

FANSHAWE

Process Engineering

Martin Volkening and Anthropic's Claude Models

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Introduction

Why are we studying this?

- Defining and identifying manufacturing processes is critical to understanding the role that automation plays and how to program an industrial process line.
- What are some examples of a manufacturing process?
- What do we have or use that is manufactured?
- What is the biggest manufacturing change that you have experienced so far?
- Processes have changed throughout the centuries and have been driven by the four Industrial Revolutions.
- First Industrial Revolution – Coal in 1765
  - This transformed society from an agricultural base to an industrial base. Processes became mechanized and rather than hand made, products were manufactured. The discovery of coal and the development of the steam engine and metal forging transformed the way goods were created.
- Second Industrial Revolution – Gas in 1870 *The discovery of oil and electricity fueled the second industrial revolution. The discovery of oil lead to the invention of the combustion engine and new steel and chemical based products entered the manufacturing streams. Technological advances in communication were driven by the invention of the telegraph and later the telephone. Modes of transportation developed with the invention of the airplane and automobile. These discoveries helped to drive the advent of mass production which forever changed how items were produced.
- Third Industrial Revolution – Electronics and Nuclear Energy in 1969 *The third industrial revolution is dominated by the development of electronics and the beginning of the age of computers.* Technological developments were centered around four pillars of science: telecommunications, biotechnology, information technology and energy engineering.

- Fourth Industrial Revolution – Internet and AI
  - The fourth industrial revolution describes the current era of digitalization, automation, and advanced manufacturing technology. It's characterized by the convergence of emerging technologies, such as robotics, the Internet of Things (IoT), and 3D printing.

Check out these videos:

The Four Industrial Revolutions: https://www.youtube.com/watch?v=kpW9JcWxKq0

and

Industry 4.0 Explained: https://www.youtube.com/watch?v=okXk4Bnz2Lc

What is a Fifth Industrial Revolution going to look like?

The Fifth Industrial Revolution: https://www.youtube.com/watch?v=sGI2FFm_8tw

Further Reading Whitepaper 'What is Manufacturing' by Intel

> Grade items and their weights are maintained in FOL and are the single source of truth. The course schedule below shows what is assessed and when; for current weights and marks, check the FOL grade book.

Course schedule

The term is **15 weeks**, with one lab in each week. Each lab's prerequisite lecture material is taught in the same week or earlier. The Labs chapter is a catalogue of more labs than one term can deliver; the grid below is the plan the current offering follows, and labs are identified by name. The book is delivered across more than one course: the lecture chapters in Part 2 (Ch 16–23) and the labs not in the grid (for example Lean: Value Stream Mapping, and TPM/OEE, Root Cause & a New Dataset) are delivered in the next course.

Wk	Lecture	Lab
1	Ch 1 Types of Manufacturing	Logic Thinking (Rocksi)
2	Ch 2 Process & Design · Ch 3 Industrial Robotics	Engineering Design Process Documentation
3	Ch 4 Ergonomics & Automated Manufacturing · Ch 5 Measurement Systems Analysis	Ergonomics (REBA)

Wk	Lecture	Lab
4	Ch 6 Metallurgy · Ch 7 Materials Testing	Repeatability and Reproducibility (Gauge R&R), bar stock
5	Ch 8 Plant Layout · Ch 10 Working with AI	Materials-testing rotation, week 1 of 3
6	Ch 11 AI Tools, Weeks 1–2	Materials-testing rotation, week 2 of 3
7	**Midterm Test** (in lecture time)	Materials-testing rotation, week 3 of 3
8	Ch 9 SPC	Facility Design & Simulation (AnyLogic)
9	Ch 11 Week 3 · Ch 13 TPM	SPC & Gauge R&R: the Virtual Caliper
10	Ch 11 Week 4 · Ch 12 Machine Guarding & Functional Safety	ML Week 1: AI Literacy & Prompt Engineering
11	Ch 11 Week 5 · Ch 16 Quality Management Systems	ML Week 2: Manufacturing Data & SPC
12	Ch 11 Week 6 · Ch 14 PFMEA	ML Week 3: The C-MAPSS Dataset · **released Robotic Work-Cell Design**
13	Review and catch-up	ML Week 4: Building the Model
14	Review and catch-up	ML Week 5: Deploying the App
15	**Final Test** · review	ML Week 6: Capstone Presentations · Robotic Work-Cell Design (RoboDK) **due**

The midterm is written during lecture time in week 7, so the week-7 lab period runs as normal. The materials-testing rotation (Hardness, Tensile, Charpy) runs in weeks 5–7: the class is split into three groups and every group completes all three labs. Robotic Work-Cell Design is a **take-home** lab: it is released in week 12 and is due at the end of week 15, and it requires a short screen recording.

This grid is a reconstruction from the course plan — confirm it against what is actually taught before relying on the week-by-week lecture pairings.

Labs

- Late penalty is 25% per lab session.
- The labs scheduled in weeks 1–6 must be completed before Week 7.
- Robotic Work-Cell Design is a take-home lab, released in week 12 and due at the end of week 15; it requires a short screen recording. All other labs are due at the end of the scheduled lab period.
- Bring your lab kit and PPE to every lab session.

Assignments

- 6 Assignments.
- All are FOL quizzes, with at least one week open.

Quizzes

- 6 Lecture Quizzes and 6 Lab Quizzes.
- Quizzes are given randomly throughout the term as a "pop quiz".
- Quizzes are given at the start of the lab session (10 minute log in window) or at the end of the lecture.
- **You must be in attendance to write the quizzes.**

Tests

- During in-person lecture time (90 minutes). One midterm test in week 7, one final test in week 15.
- Both tests are cumulative and online, in-person.
- Closed-book with handwritten cheat sheet; details will be posted on FOL closer to the date.
- Kit calculators allowed.

Other Grade Considerations

In order to pass, two conditions must be met: * Overall passing grade (50%). * Attendance and submission of the mandatory labs (those scheduled in weeks 1–15). Robotic Work-Cell Design is not mandatory to pass, but is graded.

How to succeed

- Come to class and lab sessions Participate
- Complete examples given in class
- Do the homework and assigned reading
- Attend office hours

Cheating and Plagiarism

Students must write their reports and assignments in their **own words**. Whenever students take an idea or a passage from another author, they must acknowledge their debt both by using quotation marks where appropriate and by proper referencing such as footnotes or citations. This includes non-published sources, such as work completed in previous years or a classmate's work.

Use of generative AI

Fanshawe policy A136 permits generative AI in this course only where the instructor has given prior consent. This section is that consent, and it is specific: for each assessment below, AI use is either **required**, **permitted with disclosure**, or **not permitted**. If an assessment is not listed, treat it as **not permitted** and ask.

Where AI is **permitted or required**, you must still be able to explain every line you submit. If you are asked "why does this line do that?" and cannot answer, that work does not count as yours — regardless of whether it runs correctly.

Where AI use is **permitted with disclosure**, add a short note naming the tool and what you used it for.

Assessment	AI use	Why
Logic Thinking (Rocksi)	**Not permitted**	The outcome *is* your own logic; an AI solving the puzzle removes the lab
Engineering Design Process Documentation	**Not permitted**	Already stated in the lab
Facility Design & Simulation (AnyLogic)	**Not permitted**	Already stated in the lab
Lean: Value Stream Mapping, Takt and Pull	Permitted with disclosure	Not for the map or the takt/PCE calculations
Ergonomics (REBA)	Permitted with disclosure	Not for the REBA scoring itself
SPC & Gauge R&R: the Virtual Caliper	**Not permitted** for readings and calculations; permitted for checking your written reflection	You are the measurement system being studied
Machine-learning labs (all six)	**Required**	You are assessed on directing an LLM well, not on writing Python

Assessment	AI use	Why
TPM/OEE, Root Cause & a New Dataset	**Required** for the classifier task; permitted with disclosure elsewhere	Same skill, unfamiliar dataset
Robotic Work-Cell Design (RoboDK)	Permitted with disclosure	The station, program and cycle times must be your own
Assignments 1–6	Permitted with disclosure	
Pop quizzes, midterm, final	**Not permitted**	Closed-book, in-person

FOL

- Lectures will be posted after class, as well as supplementary material.
- Assignments and lab manuals will be posted ahead of time. You will always have at least one week's notice.
- All announcements will be made through the FOL course.

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Part I

Process Engineering I

Chapter 1

Types of Manufacturing Processes

1.1 Learning Objectives

By the end of this chapter, you will be able to:

1. Identify and classify the seven main types of manufacturing processes
2. Select appropriate manufacturing processes based on material, volume, and precision requirements
3. Compare cost, speed, and quality trade-offs between process types
4. Recognize manufacturing processes used in automotive, food, and defense industries
5. Apply process selection criteria to real-world manufacturing scenarios

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1.2 Introduction to Manufacturing Processes

Manufacturing is defined as the conversion of raw materials into finished goods on a large scale using man and machine. **Manufacturing processes** are the specific methods used for this conversion. Based on product requirements, different types of manufacturing processes are selected to achieve the required output with optimal cost, quality, and efficiency.

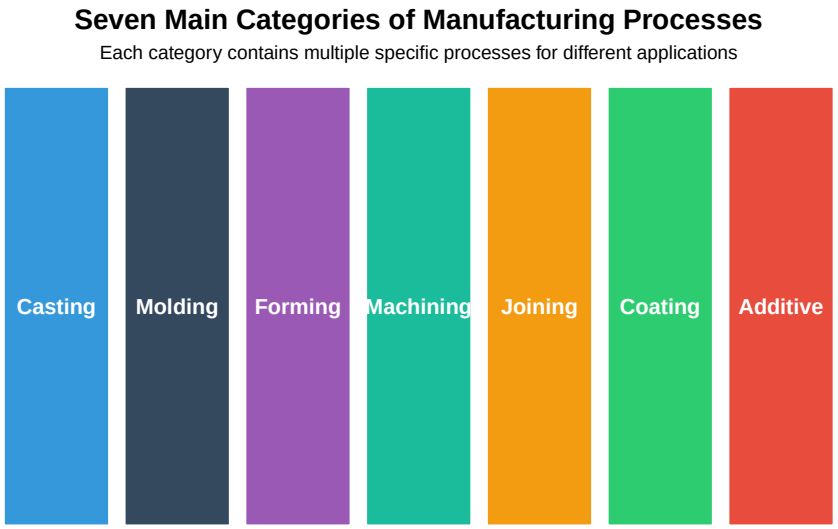


Figure 1.1: Overview of Manufacturing Process Categories

Why do we need different manufacturing processes?

Different products require different manufacturing approaches based on:

- **Material type**: Metals, plastics, ceramics, composites
- **Production volume**: One-off prototypes vs. millions of units
- **Geometric complexity**: Simple shapes vs. intricate details
- **Precision requirements**: Loose tolerances vs. micron-level accuracy
- **Cost constraints**: Budget limitations affect process selection
- **Lead time**: How quickly parts are needed

No single process is optimal for all situations!

A good video summarizing 6 of the 7 manufacturing processes is linked below (Coating/Plating is not covered):

Manufacturing Processes Overview: https://www.youtube.com/watch?v=Um_g8sQ_p3Y

1.3 Process Selection Framework

Before examining each process type, it's important to understand how engineers select the right manufacturing process. The following visualization shows the key factors:

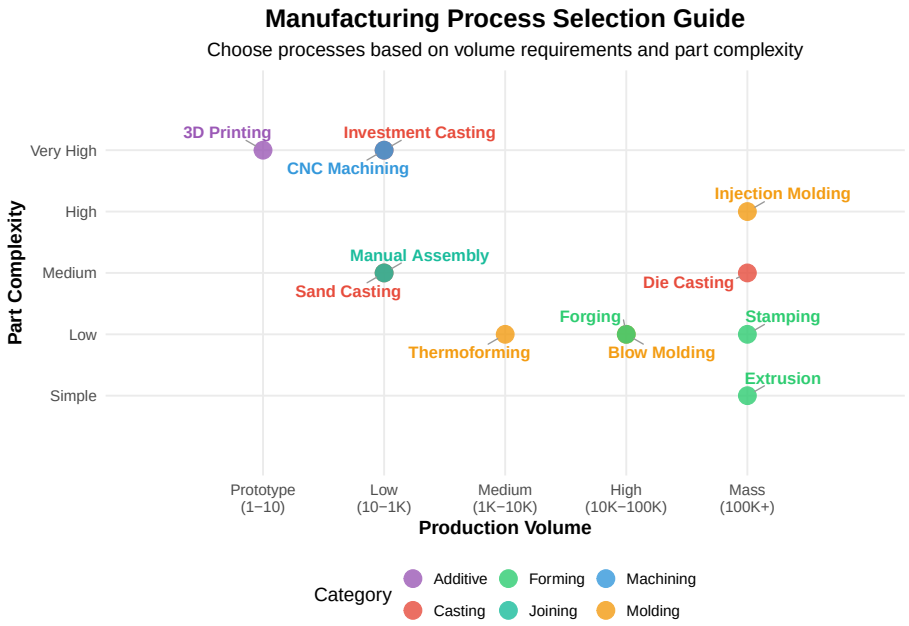


Figure 1.2: Process Selection Based on Volume and Complexity

Table 1.1: Manufacturing Process Comparison Summary

Process Type	Typical Materials	Production Volume	Typical Tolerance	Tooling Cost	U
Casting	Metals, alloys	Low to High	±0.5-2mm	Medium-High	Lo
Molding	Plastics, rubber	Medium to Very High	±0.1-0.5mm	High	Ve
Forming	Metals, sheet	Medium to Very High	±0.1-1mm	Medium-High	Lo
Machining	All materials	Low to Medium	±0.01-0.1mm	Low	Hi
Joining	All materials	All volumes	Varies	Low	M
Coating	All materials	All volumes	N/A	Low	Lo
Additive	Plastics, metals, ceramics	Prototype to Low	±0.1-0.3mm	None	Hi

1.4 Casting Processes

Casting is a process of pouring liquid metal into a mold containing the hollow shape of the desired outcome. It uses sprue, gates, and runners to direct the molten metal flow.

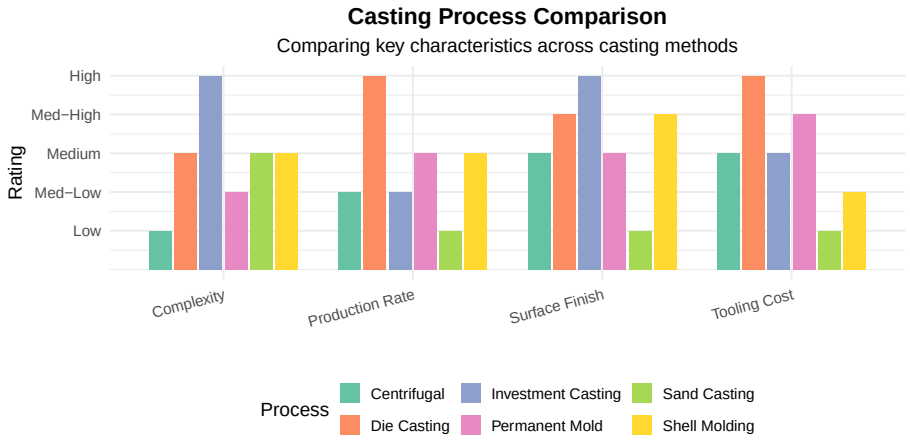


Figure 1.3: Comparison of Casting Processes

1.4.1 Sand Casting

Sand casting is a metal casting process that uses sand as the mold material. This process has a low production rate as the sand mold must be destroyed to remove the part. However, it is economical for low-volume production.

Process Steps:

1. Create a pattern (usually wood or metal)
2. Pack sand around the pattern to form the mold
3. Remove the pattern, leaving a cavity
4. Pour molten metal into the cavity
5. Allow solidification
6. Break the sand mold to remove the casting

Advantages: Low tooling cost, can produce very large parts, suitable for all metals

Disadvantages: Poor surface finish, loose tolerances, slow cycle time

Examples: Engine blocks, machine tool bases, cylinder heads, pump housings, large gears

Discussion: Why is sand casting still widely used despite its limitations?

Sand casting remains popular because:

1. **Low startup cost** - A wood pattern costs hundreds vs. steel dies costing tens of thousands
2. **Flexibility** - Easy to modify patterns for design changes
3. **Size capability** - Can cast parts weighing several tons
4. **Material versatility** - Works with nearly any metal

5. **Short lead time** - Parts can be produced in days vs. weeks for die casting

For prototypes and low-volume production (< 500 parts), sand casting is often the most economical choice even with its lower quality.

1.4.2 Die Casting

Die casting is used for producing metal parts where high precision is required. Molten metal is forced under high pressure into reusable metal dies, enabling high-volume production.

Die Casting Process: https://www.youtube.com/watch?v=0oibUY8KUQM

Process:

1. Steel molds (dies) are machined to the part shape
2. Dies are mounted in a die casting machine
3. Molten metal is injected under high pressure (10-175 MPa)
4. Metal solidifies rapidly due to die cooling
5. Part is ejected and the cycle repeats

Advantages: Excellent dimensional accuracy, smooth surface finish, high production rates (up to 400 shots/hour)

Disadvantages: High die cost ($50,000-$500,000), limited to non-ferrous metals, porosity issues

Examples: Engine components, transmission housings, electronic enclosures, power tool housings

1.4.3 Investment Casting (Lost-Wax)

Investment casting produces complex, precision parts by creating a wax pattern that is "invested" (surrounded) with ceramic slurry, then melted out to leave a mold cavity.

Investment Casting Process: https://www.youtube.com/watch?v=QtxVdC7pBQM

Advantages: Excellent surface finish, tight tolerances (±0.1%), complex geometries, any castable metal

Disadvantages: Expensive, slow cycle time, limited part size

Examples: Turbine blades, aerospace components, medical implants, jewelry, firearm components

Why do aerospace companies pay premium prices for investment casting?

Investment casting is essential for aerospace because:

1. **Complex internal passages** - Turbine blades have intricate cooling channels

2. **Superior metallurgy** - Controlled solidification produces better grain structure
3. **Near-net shape** - Minimizes machining of expensive superalloys
4. **Weight savings** - Can create hollow structures impossible with other methods
5. **Material options** - Works with titanium, Inconel, and other aerospace alloys

A single turbine blade might cost $1,000-$5,000, but there's no other process that can produce the required geometry and performance.

1.4.4 Centrifugal Casting

Centrifugal casting spins molten metal in a rotating die, using centrifugal force to distribute the metal and float impurities to the inner surface.

Applications: Pipes, tubes, cylinder liners, bushings, rings

1.4.5 Permanent Mold Casting

Uses reusable metal molds with gravity or low-pressure filling. Good for medium-volume production of aluminum and copper alloy parts.

Applications: Pistons, gear housings, wheels, pipe fittings

1.4.6 Shell Molding

Uses resin-coated sand that forms a thin, strong shell around a heated pattern. Better dimensional accuracy than sand casting.

Applications: Gear housings, cylinder heads, connecting rods

1.5 Molding Processes

Molding uses a rigid frame to shape hot liquid or ductile raw material. It is predominantly used for plastics and rubber products.

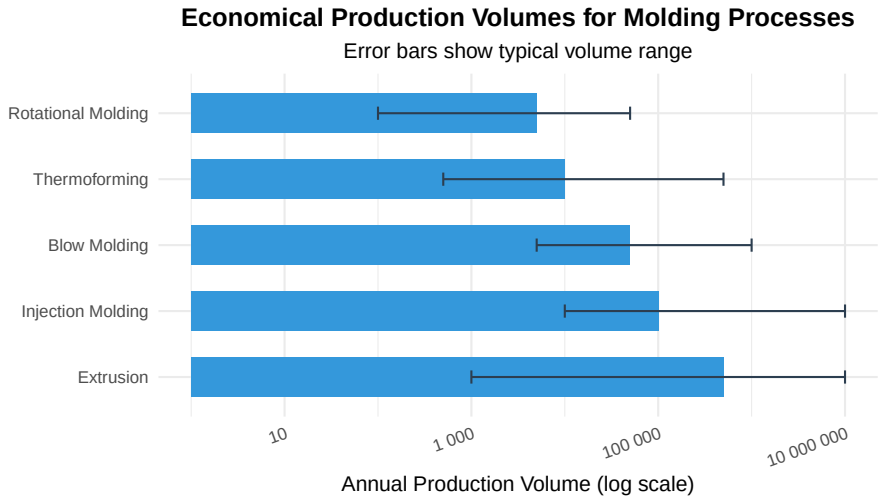


Figure 1.4: Molding Process Selection by Production Volume

1.5.1 Injection Molding

Injection molding is the most common process for manufacturing plastic parts. Plastic pellets are melted and injected under high pressure into a mold cavity.

Injection Molding Process: https://www.youtube.com/watch?v=RMjtmsr3CqA

Process Cycle:

1. **Clamping** - Mold halves are clamped together
2. **Injection** - Molten plastic is injected into the cavity
3. **Cooling** - Part solidifies in the mold
4. **Ejection** - Part is pushed out by ejector pins

Cycle times: 10-60 seconds typical, as fast as 2 seconds for small parts

Examples: Automotive dashboards, phone cases, medical devices, food containers, toys

Your turn — drag the volume slider or change any input below and the chart and cost breakdown update immediately.

Show the live R code behind this calculator

The card above re-runs this exact webR code every time you change an input, substituting your values for the variables at the top. Edit the code directly and press **Run Code** to experiment further.

```
#| label: injection-molding-cost
# Injection Molding Cost Analysis Example
```

```
# Cost per part = tooling cost spread over the run + material + processing time

mold_cost <- 50000             # Steel mold (tooling), one-time cost
material_cost_per_kg <- 2.50   # ABS plastic, $ per kg
part_weight_kg <- 0.10         # Mass of one part
cycle_time_sec <- 30           # Time to mold one part
machine_rate_per_hour <- 75    # Machine + operator, $ per hour
selected_volume <- 50000       # Parts to produce this run

cost_per_part <- function(volume) {
  tooling    <- mold_cost / volume
  material   <- part_weight_kg * material_cost_per_kg
  processing <- (cycle_time_sec / 3600) * machine_rate_per_hour
  tooling + material + processing
}

volumes <- seq(500, 500000, length.out = 200)

par(mar = c(4, 5, 2, 1), cex.main = 1.3, cex.lab = 1.2, cex.axis = 1.1)
plot(volumes, cost_per_part(volumes), type = "l", log = "x", lwd = 4, col = "#2c3e50",
     xlab = "Production volume (parts, log scale)", ylab = "Cost per part ($)",
     main = "Cost per Part vs. Production Volume", las = 1, xaxt = "n")
axis(1, at = c(500, 5000, 50000, 500000),
     labels = c("500", "5,000", "50,000", "500,000"))
abline(v = selected_volume, lty = 2, col = "#bdc3c7")
points(selected_volume, cost_per_part(selected_volume), pch = 19, cex = 2.2, col = "#e

cat(sprintf("Cost per part at %s parts: $%.2f\n",
            format(selected_volume, big.mark = " ", scientific = FALSE),
            cost_per_part(selected_volume)))
cat(sprintf("  Tooling      $%.2f\n", mold_cost / selected_volume))
cat(sprintf("  Material     $%.2f\n", part_weight_kg * material_cost_per_kg))
cat(sprintf("  Processing   $%.2f\n", (cycle_time_sec / 3600) * machine_rate_per_hour))
```

Discussion: When does injection molding become cost-effective vs. 3D printing?

Break-even analysis:

- 3D printing: ~$5-20 per part (no tooling, slow production)
- Injection molding: $50,000 tooling + $0.50-2 per part

Break-even point: Approximately 5,000-10,000 parts

Rule of thumb: - $<$ 500 parts: 3D printing or machining - 500-5,000 parts: Consider soft tooling (aluminum molds) - $>$ 5,000 parts: Steel injection molds become economical

1.5.2 Blow Molding

Blow molding creates hollow plastic parts by inflating a heated plastic tube (parison) inside a mold cavity.

Types: - **Extrusion blow molding** - Parison extruded, then blown - **Injection blow molding** - Preform injection molded, then blown - **Stretch blow molding** - Preform stretched and blown (PET bottles)

Examples: Bottles, containers, fuel tanks, automotive ducts, toys

1.5.3 Rotational Molding

Creates hollow parts by rotating a mold on two axes while heating, causing plastic powder to melt and coat the interior surface.

Advantages: Low tooling cost, no internal stresses, large hollow parts

Examples: Kayaks, playground equipment, tanks, coolers

1.5.4 Thermoforming

Heats a plastic sheet until pliable, then forms it over a mold using vacuum, pressure, or mechanical force.

Examples: Packaging (blister packs), disposable cups, refrigerator liners

1.5.5 Powder Metallurgy

Though involving metal, powder metallurgy uses molding principles. Metal powders are compacted in dies and sintered (heated below melting point) to fuse particles.

Advantages: Near-net shape, complex geometries, controlled porosity

Examples: Self-lubricating bearings, gears, filters, magnets

1.6 Forming Processes

Forming uses mechanical forces (compression, tension, shear) to shape material without removing any material. This preserves material grain structure and can improve mechanical properties.

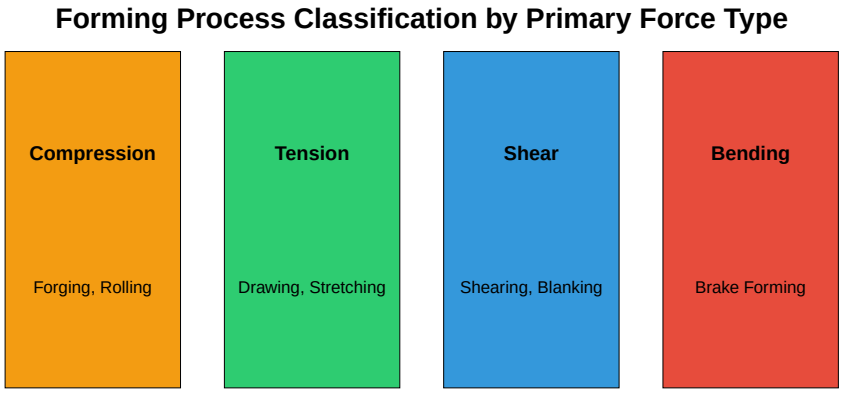


Figure 1.5: Types of Forming Forces

1.6.1 Forging

Forging uses compressive forces from hammers or presses to shape heated metal. Forged parts have superior strength due to aligned grain structure.

Forging Process: https://www.youtube.com/watch?v=o4vQJnPwgiA

Types of Forging: - **Open-die forging** - Simple shapes, large parts - **Closed-die (impression) forging** - Complex shapes, high volume - **Cold forging** - Room temperature, excellent finish - **Hot forging** - Heated material, easier deformation

Examples: Crankshafts, connecting rods, gears, aircraft structural components, hand tools

Why are aircraft landing gear forgings so expensive?

Aircraft landing gear forgings can cost $100,000+ because:

1. **Material** - High-strength steel or titanium alloys
2. **Size** - Large parts require massive presses (50,000+ tons)
3. **Certification** - Extensive testing and documentation
4. **Grain flow** - Must be optimized for fatigue resistance
5. **NDT requirements** - 100% ultrasonic and magnetic particle inspection
6. **Low volume** - Only a few hundred per aircraft type per year

The forged grain structure provides 20-40% better fatigue life than equivalent castings or machined parts.

1.6.2 Stamping

Stamping uses dies and presses to form, cut, or shape sheet metal. Modern stamping lines can produce 1,000+ parts per hour.

Operations include: - **Blanking** - Cutting flat shapes from sheet - **Piercing** - Punching holes - **Drawing** - Forming cup shapes - **Bending** - Creating angles - **Coining** - Fine detail forming

Examples: Automotive body panels, appliance housings, electronic enclosures

1.6.3 Bending

Bending uses press brakes to create angular shapes in sheet metal.

Key formula:

$$\text{Bend Allowance} = \frac{\pi \times A \times (R + K \times T)}{180}$$

Where: A = bend angle, R = inside radius, T = material thickness, K = K-factor (typically 0.3-0.5)

1.6.4 Shearing

Shearing cuts sheet metal without chip formation using opposed cutting edges.

Types: Blanking, piercing, trimming, notching, slitting

1.6.5 Rolling

Reduces material thickness by passing through rotating rolls. Used for sheet, plate, and structural shapes.

1.6.6 Extrusion (Metal)

Forces material through a die to create constant cross-section profiles.

Examples: Aluminum window frames, heat sinks, structural shapes

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1.7 Machining Processes

Machining removes material using cutting tools to achieve precise dimensions and surface finishes. It is the most versatile manufacturing process but typically has higher per-part costs.

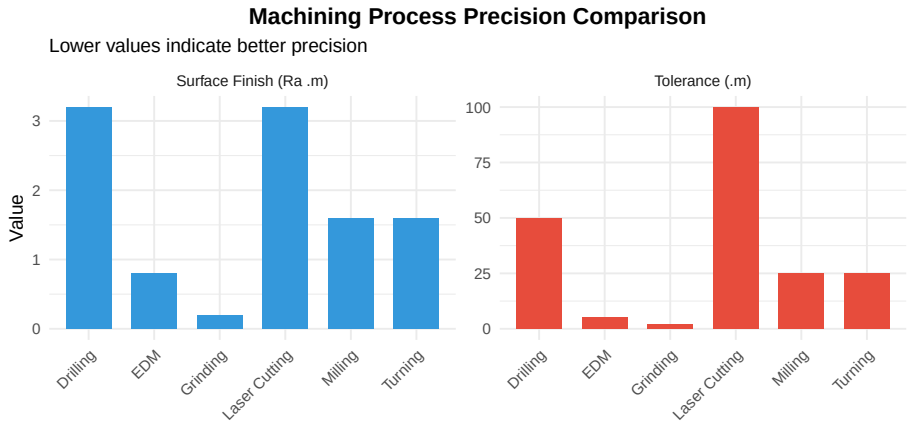


Figure 1.6: Machining Process Capabilities

1.7.1 Turning

Rotates the workpiece while a cutting tool moves to create cylindrical shapes.

Performed on lathes or turning centers

Operations: Facing, turning, boring, threading, grooving, parting

1.7.2 Milling

Rotates a cutting tool while moving the workpiece to remove material. The most versatile machining process.

Types: - **Peripheral milling** - Cutter axis parallel to surface - **Face milling** - Cutter axis perpendicular to surface - **End milling** - Creates slots, pockets, contours

1.7.3 Drilling

Creates cylindrical holes using rotating drill bits.

Related operations: Reaming (sizing), boring (enlarging), tapping (threading)

1.7.4 Grinding

Uses abrasive wheels for precision finishing. Achieves the finest tolerances and surface finishes.

Types: Surface grinding, cylindrical grinding, centerless grinding

1.7.5 Electrical Discharge Machining (EDM)

Uses electrical sparks to erode conductive materials. Essential for hardened materials and complex geometries.

Applications: Injection mold cavities, dies, aerospace components

When should you use EDM instead of conventional machining?

Choose EDM when:

1. **Hardened materials** - Can machine 60+ HRC steels that would destroy cutting tools
2. **Complex internal features** - Wire EDM cuts intricate shapes
3. **Tight corners** - Can achieve sharp internal corners impossible with rotary tools
4. **Fragile parts** - No cutting forces (non-contact process)
5. **Thin walls** - No deflection from tool pressure

Limitations: - Slow material removal (grams/hour vs. kg/hour for milling) - Only works on conductive materials - Recast layer may need removal - Higher cost per part

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1.8 Joining Processes

Joining combines two or more components into a single assembly. Methods range from permanent (welding) to removable (fasteners).

Table 1.2: Joining Process Comparison

Process	Joint Strength	Dissimilar Metals	Automation	Typical Applications
Arc Welding	Excellent	Limited	Medium	Structural steel, pressure ves
Resistance Welding	Excellent	Limited	High	Auto body panels, wire mesh
Laser Welding	Excellent	Limited	High	Electronics, medical devices
Brazing	Good	Yes	Medium	Carbide tooling, HVAC
Soldering	Fair	Yes	High	Electronics, plumbing
Adhesive Bonding	Good	Yes	High	Composites, glass, plastics
Mechanical Fastening	Excellent	Yes	High	All industries

1.8.1 Welding

Melts and fuses base materials together, often with added filler metal.

Common Types: - **MIG (GMAW)** - Gas Metal Arc Welding - fast, easy automation - **TIG (GTAW)** - Gas Tungsten Arc Welding - precise, clean welds

- **Stick (SMAW)** - Shielded Metal Arc Welding - versatile, field use - **Resistance** - Uses electrical resistance heating - **Laser/Electron Beam** - High precision, deep penetration

1.8.2 Brazing and Soldering

Joins metals using a filler metal with a lower melting point than the base materials. The base materials do not melt.

- **Brazing** - Filler melts above 450°C (840°F)
- **Soldering** - Filler melts below 450°C

1.8.3 Adhesive Bonding

Uses chemical adhesives to join materials. Essential for composites, glass, and dissimilar materials.

Advantages: Distributes stress, joins dissimilar materials, seals against fluids

Disadvantages: Surface preparation critical, limited temperature resistance, aging concerns

1.8.4 Mechanical Fastening

Uses bolts, screws, rivets, or clips. Allows disassembly for maintenance.

1.9 Surface Treatment and Coating

Coating and surface treatment processes protect parts from corrosion, wear, or environmental damage, and can improve appearance or electrical properties.

Table 1.3: Surface Treatment and Coating Processes

Process	Typical Thickness	Base Material	Primary Benefit	Indus
Electroplating	1-50 m	Conductive metals	Corrosion, appearance	Auton
Anodizing	5-100 m	Aluminum	Corrosion, hardness	Aeros
Powder Coating	50-150 m	Any metal	Corrosion, appearance	Appli
Painting	25-75 m	Any	Appearance	All
Hot Dip Galvanizing	45-85 m	Steel	Corrosion	Const
PVD/CVD	1-10 m	Any	Wear, hardness	Toolin
Thermal Spray	100-500 m	Any	Wear, thermal barrier	Aeros

1.9.1 Electroplating

Deposits metal coating using electrical current. Common coatings include chrome, nickel, zinc, gold, and silver.

1.9.2 Anodizing

Electrochemically grows an oxide layer on aluminum. Provides corrosion resistance and can be dyed various colors.

1.9.3 Powder Coating

Applies dry powder electrostatically, then cures with heat. Produces thick, durable, attractive finishes.

1.9.4 Galvanizing

Coats steel with zinc for corrosion protection. Hot-dip galvanizing provides excellent outdoor durability.

1.10 Additive Manufacturing (3D Printing)

Additive manufacturing builds parts layer-by-layer from digital models. It's the opposite of machining (subtractive manufacturing).

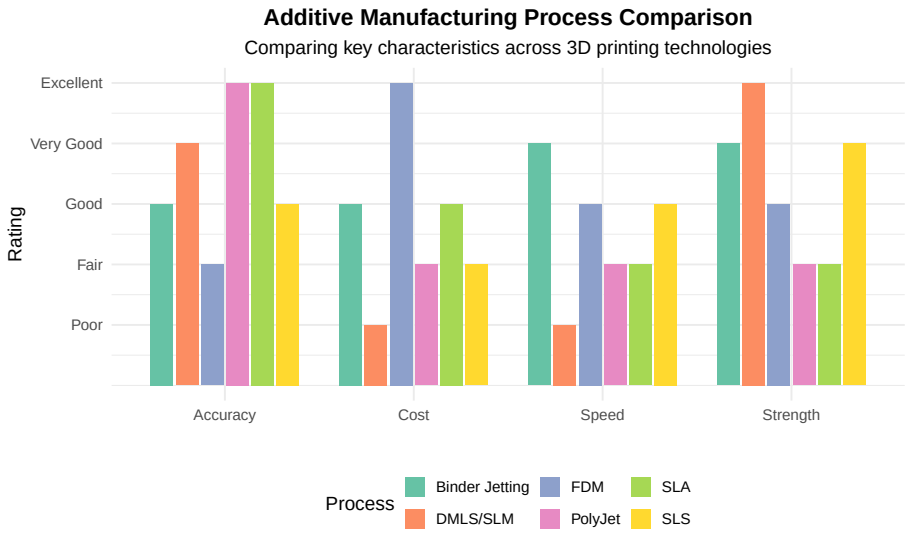


Figure 1.7: Additive Manufacturing Process Comparison

How 3D Printing Works: https://www.youtube.com/watch?v=Vx0Z6LplaMU

1.10.1 FDM (Fused Deposition Modeling)

Extrudes thermoplastic filament through a heated nozzle. Most common desktop 3D printing technology.

Materials: PLA, ABS, PETG, Nylon, TPU

1.10.2 SLA (Stereolithography)

Uses UV laser to cure liquid photopolymer resin. Excellent surface finish and detail.

1.10.3 SLS (Selective Laser Sintering)

Fuses powder material with a laser. Produces strong functional parts without support structures.

1.10.4 Metal 3D Printing (DMLS/SLM)

Fuses metal powder with high-power lasers. Used for aerospace, medical, and tooling applications.

When does metal 3D printing make sense vs. traditional manufacturing?

Metal 3D printing is advantageous when:

1. **Complex internal features** - Conformal cooling channels in molds
2. **Topology optimization** - Organic shapes that minimize weight
3. **Consolidation** - Combining multiple parts into one
4. **Low volume** - Too few parts to justify tooling
5. **Custom/one-off** - Medical implants, prototypes

Traditional manufacturing is better when: - High volumes (> 100-1000 parts depending on complexity) - Simple geometries - Large parts (3D print beds are limited) - Maximum strength required (forging still superior)

Cost comparison: - Simple bracket: $5 machined vs. $50+ 3D printed - Complex aerospace part: $10,000 machined vs. $3,000 3D printed

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1.11 Try it yourself: pick the process family

Eight parts, each with the requirement that drives the choice. Decide which process family you would start with. Use the selection factors from this chapter: material, production volume, geometry, and tolerance.

Which process family?

Pick the process family you would start with for each part.

Categories: Casting / Molding / Forming / Machining / Additive

1. 20,000 aluminum transmission housings a year, with thin walls and a good surface finish. _____
2. Nickel-superalloy turbine blades with internal cooling passages. _____
3. 500,000 ABS charger housings with moulded-in snap fits. _____
4. Hollow two-litre PET drink bottles. _____
5. Steel crankshafts that need high strength and grain flow that follows the shape. _____
6. Car door outer panels pressed from coiled sheet steel. _____
7. A bearing journal that must be held to +/- 0.005 mm. _____
8. Five titanium brackets with an internal lattice, needed next week. _____

Answers: 1 Casting; 2 Casting; 3 Molding; 4 Molding; 5 Forming; 6 Forming; 7 Machining; 8 Additive.

1.12 Industry Applications

Table 1.4: Manufacturing Process Applications by Industry

Process	Automotive	Food Industry	Defense/Aerospa
Die Casting	Engine blocks, transmission cases	Equipment housings	Aircraft structural c
Injection Molding	Dashboard, interior trim, connectors	Packaging, containers, caps	Connectors, housing
Forging	Crankshafts, connecting rods, gears	Mixer blades, pump impellers	Landing gear, turbi
Stamping	Body panels, structural components	Cans, trays, lids	Aircraft skins, miss
Machining	Engine components, precision parts	Valves, fittings	Precision weapons c
Welding	Body structure, exhaust systems	Tanks, piping, conveyors	Aircraft structures,
Powder Coating	Wheels, chassis components	Equipment frames	Military vehicles, ec
3D Printing	Prototypes, custom fixtures	Custom tooling, fixtures	Turbine blades, bra

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1.13 Sustainability in Manufacturing

Modern manufacturing must consider environmental impact. Here's how different processes compare:

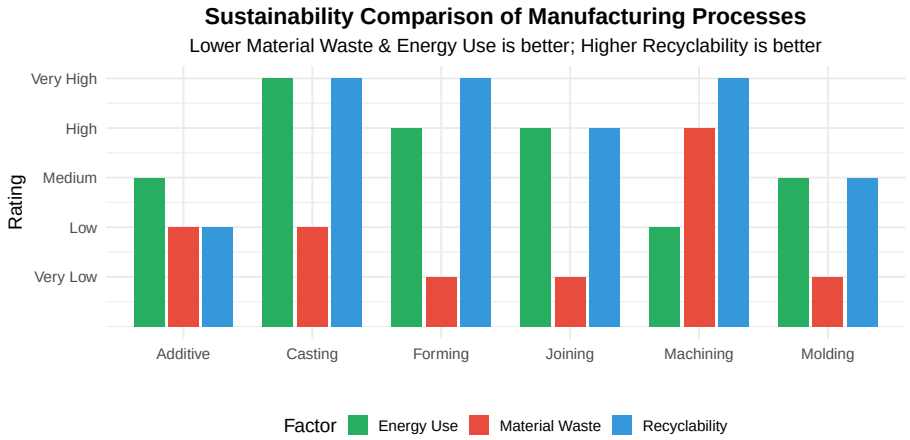


Figure 1.8: Environmental Considerations by Process Type

Discussion: How can manufacturers reduce environmental impact?

Strategies for sustainable manufacturing:

1. **Near-net-shape processes** - Casting, forging, and molding reduce machining waste
2. **Closed-loop recycling** - Reprocess scrap material on-site
3. **Energy-efficient equipment** - Modern machines use 30-50% less energy
4. **Additive manufacturing** - Only uses material where needed
5. **Water-based coatings** - Replace solvent-based paints
6. **Lean manufacturing** - Reduces overproduction and waste
7. **Life cycle assessment** - Consider total environmental impact

Example: Aluminum die casting recycles 95%+ of sprues and runners, while machining from billet may waste 80% of material.

1.14 Review Questions

Question 1: Process Identification

A company needs to produce 50,000 aluminum housings per year. Each housing is a complex shape with internal fins for heat dissipation. The surface must be smooth and dimensions must be within ±0.3mm. Which process would you recommend?

Answer: Die Casting is the best choice because: - High volume (50,000/year) justifies tooling cost - Aluminum is ideal for die casting - Complex internal

features are possible - ±0.3mm tolerance is achievable - Good surface finish without secondary operations

Alternative: If volumes were lower (< 5,000), investment casting might be considered.

Question 2: Cost Comparison

Calculate the break-even point between 3D printing and injection molding for a small plastic part:

- 3D printing: $8 per part (no tooling)
- Injection molding: $25,000 mold + $0.50 per part

Question 2: break-even volume

- Break-even quantity (parts): _____ *(to 0 decimal places)*
- Total cost of 10,000 injection-molded parts: $_____ *(to 0 decimal places)*

Show the worked solution

Answer: Let n = number of parts

Break-even: $8n = $25,000 + $0.50n

$7.50n = $25,000

n = 3,333 parts

Below 3,333 parts: 3D printing is cheaper **Above 3,333 parts:** Injection molding is cheaper

At 10,000 parts: - 3D printing: $80,000 - Injection molding: $30,000

Question 3: Material Selection

Match each material/product to the most appropriate manufacturing process:

a) Stainless steel surgical instrument
b) Large plastic kayak
c) Titanium jet engine bracket
d) Aluminum beverage can

Answers:

a) **Investment casting** or **machining** - Complex shapes, high precision, excellent surface finish
b) **Rotational molding** - Large hollow plastic part, low tooling cost
c) **Additive manufacturing (DMLS)** or **forging** - Complex geometry, lightweight, strength critical
d) **Deep drawing (stamping)** - Very high volume, thin-wall cylinder

Question 4: Process Selection Scenario

You need to manufacture a new product with these requirements: - Quantity: 200 units for initial market test - Material: ABS plastic - Size: 150mm × 100mm × 50mm - Tolerance: ±0.5mm - Surface: Smooth, will be painted

What process would you recommend and why?

Answer: 3D Printing (FDM or SLA) would be most appropriate because:

1. **Low quantity (200)** - Injection molding tooling ($30,000+) not justified
2. **Market test phase** - Design may change based on feedback
3. **Quick turnaround** - Parts in days vs. weeks for tooling
4. **Tolerances achievable** - ±0.5mm is within FDM capabilities
5. **Post-processing acceptable** - Parts will be painted anyway

Cost comparison: - FDM: 200 × $15 = $3,000 - Injection molding: $30,000 + (200 × $1) = $30,200

If the product succeeds and volume increases to 10,000+, transition to injection molding.

Question 5: Hybrid Manufacturing

What is hybrid manufacturing and when is it used?

Answer: Hybrid manufacturing combines additive and subtractive processes in a single machine or workflow.

Common combinations: - 3D print near-net shape → CNC machine critical surfaces - Cast blank → Machine precision features - Forge rough shape → Machine final dimensions

Benefits: - Reduced material waste vs. pure machining - Better precision than pure additive - Faster than pure machining for complex parts

Example: Aerospace companies 3D print titanium brackets, then machine mounting surfaces to achieve tight tolerances. This can reduce material waste by 90% compared to machining from billet.

1.15 Summary

Table 1.5: Manufacturing Processes Summary

Process Category	Key Principle	Best For
Casting	Pour liquid into mold	Complex metal shapes, medium-high volume
Molding	Shape heated/softened material	Plastic parts, very high volume
Forming	Deform solid with force	Strong metal parts, high volume

Machining	Remove material with tools	Precision parts, any material	All indust
Joining	Combine multiple parts	Assembly of components	All indust
Coating	Apply protective layer	Corrosion/wear protection	All indust
Additive	Build layer by layer	Prototypes, complex geometry	Aerospace

1.15.1 Key Takeaways

1. **No single process is best for everything** - Selection depends on material, volume, complexity, and cost constraints
2. **Volume drives economics** - High tooling costs require high volumes to amortize
3. **Consider the complete supply chain** - Secondary operations, assembly, and finishing affect total cost
4. **Sustainability matters** - Modern manufacturing must minimize waste and energy use
5. **Hybrid approaches are growing** - Combining processes optimizes cost and performance

---

1.16 References

1. Groover, M.P. (2020). *Fundamentals of Modern Manufacturing* (7th ed.). Wiley.
2. Kalpakjian, S., & Schmid, S.R. (2020). *Manufacturing Engineering and Technology* (8th ed.). Pearson.
3. DeGarmo, E.P., Black, J.T., & Kohser, R.A. (2019). *DeGarmo's Materials and Processes in Manufacturing* (13th ed.). Wiley.
4. Afzal, M. "Types of Manufacturing Processes." *Engineering Articles*.

Chapter 2

Process and Design

2.1 Learning Objectives

By the end of this chapter, you will be able to:

1. Describe the stages of process plant project development from concept to operation
2. Differentiate between fixed, programmable, and flexible automation systems
3. Interpret Process Flow Diagrams (PFDs) and Piping & Instrumentation Diagrams (P&IDs)
4. Explain the purpose and content of Front-End Engineering Design (FEED)
5. Apply Design for Manufacturing (DFM) and Design for Assembly (DFA) principles
6. Understand the role of HAZOP studies in process safety
7. Identify key engineering deliverables for each project phase

---

2.2 Introduction to Process Engineering

Process engineering is a specialized branch of engineering that focuses on the design, optimization, and implementation of manufacturing processes and production systems. Process engineers work at the intersection of design and production, ensuring that products can be manufactured efficiently, safely, and economically.

What is the role of a process engineer?

Process engineers are responsible for:

- **Designing production processes** - Selecting equipment, defining sequences, optimizing flow
- **Improving existing processes** - Reducing waste, increasing throughput, improving quality
- **Scaling up from lab to production** - Taking R&D concepts to full-scale manufacturing
- **Ensuring safety and compliance** - Meeting regulatory and environmental requirements
- **Supporting production** - Troubleshooting issues, training operators
- **Cost optimization** - Reducing manufacturing costs while maintaining quality

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2.3 Types of Manufacturing Automation

Automation is essential for achieving high productivity, consistent quality, and competitive manufacturing costs. There are three fundamental types of automation, each suited to different production scenarios.

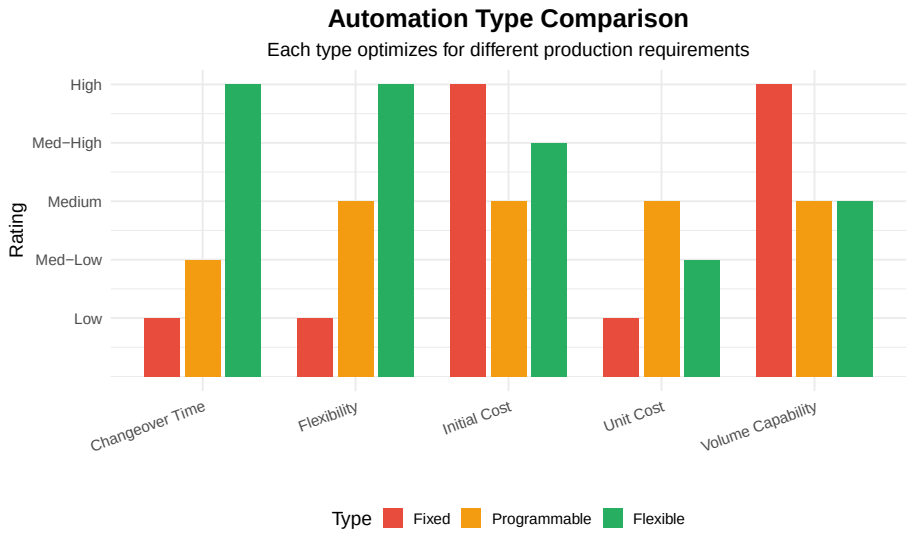


Figure 2.1: Comparison of Automation Types

2.3.1 Fixed Automation

Fixed automation (also called "hard automation") uses specialized equipment designed for a specific product. The sequence of operations is determined by

the equipment configuration and cannot be easily changed.

Characteristics: - Very high production rates (thousands per hour) - Low per-unit cost at high volumes - High initial capital investment - Inflexible - dedicated to one product

Examples: - Automotive transfer lines for engine blocks - Beverage bottling lines - Chemical processing plants - Paper manufacturing

Your turn — change the equipment cost, production volume, or operating cost below.

Show the live R code behind this calculator

```
#| label: fixed-automation-economics
# Fixed Automation Economic Analysis
# When does fixed automation make sense?

# Example: Automotive transfer line for cylinder heads
fixed_automation_cost <- 5000000  # Initial equipment cost
annual_production <- 500000       # Parts per year
fixed_cost_per_part <- 2.50       # Operating cost per part
equipment_life_years <- 10

# Calculate total cost per part including amortization
annual_equipment_cost <- fixed_automation_cost / equipment_life_years
total_annual_cost <- annual_equipment_cost + (fixed_cost_per_part * annual_production)
cost_per_part <- total_annual_cost / annual_production

cat(sprintf("Total cost per part: $%.2f\n", cost_per_part))
cat(sprintf("  Amortized equipment  $%.2f\n", annual_equipment_cost / annual_production))
cat(sprintf("  Operating cost       $%.2f\n", fixed_cost_per_part))
cat(sprintf("  Annual production    %s parts\n", format(annual_production, big.mark = " ", scient
```

2.3.2 Programmable Automation

Programmable automation allows the production equipment to be reprogrammed for different products. It's suited for batch production where products change periodically.

Characteristics: - Medium production rates - Significant changeover time between products - Lower initial cost than fixed automation - Suitable for batch sizes of 100-10,000

Examples: - CNC machining centers - Industrial robots - PLC-controlled assembly machines - Batch chemical reactors

Discussion: When should you choose programmable over fixed automation?

Choose programmable automation when:

1. **Product variety** - You make multiple products on the same equipment
2. **Volume uncertainty** - Demand may shift between products
3. **Product evolution** - Frequent design changes expected
4. **Lower volumes** - Not enough volume to justify dedicated equipment
5. **Capital constraints** - Can't afford multiple dedicated lines

Example scenario: A machine shop that makes 20 different part numbers for aerospace customers would use CNC machines (programmable) rather than dedicated transfer lines (fixed) because: - Volumes per part number are low (100-1,000/year) - New part designs are introduced regularly - Flexibility to respond to customer changes is critical

2.3.3 Flexible Automation

Flexible automation can produce a variety of parts with minimal changeover time. It combines the flexibility of programmable automation with the productivity approaching fixed automation.

Characteristics: - Medium-high production rates - Very fast changeover (minutes, not hours) - High initial investment - Suitable for medium variety, medium volume

Examples: - Flexible Manufacturing Systems (FMS) - Robotic assembly cells - Automated guided vehicle (AGV) systems - Modern automotive final assembly lines

Table 2.1: Automation Type Selection Guide

Production Characteristic	Fixed Automation	Programmable Automation
Annual Volume	> 1,000,000 units	1,000 - 100,000 units
Product Variety	Single product	Several products in batches
Changeover Frequency	Rarely (years)	Weekly to monthly
Initial Investment	Very High ($5M+)	Medium ($500K-$2M)
Operating Cost	Lowest per unit	Medium per unit
Best Industries	Automotive, Beverage, Chemical	Job shops, Aerospace, Medical

List industries where you can find various automated process systems

By automation type:

Fixed Automation: - Oil refineries - Chemical plants - Food processing (high-volume) - Beverage bottling - Automotive powertrain - Steel mills - Paper/pulp mills

Programmable Automation: - Aerospace manufacturing - Medical device production - Pharmaceutical (batch) - Job shops - Tool & die making - Defense contractors

Flexible Automation: - Electronics assembly - Automotive final assembly - Warehouse/distribution - Semiconductor fabrication - Consumer appliances - Furniture manufacturing

2.3.4 Automation Advantages and Disadvantages

Table 2.2: Advantages and Disadvantages of Manufacturing Automation

Category	Point
Advantages	Lowered Operating Costs - Reduced labor per unit
Advantages	Improved Worker Safety - Removes humans from hazardous operations
Advantages	Reduced Factory Lead Times - Faster, more predictable production
Advantages	Increased Production Output - 24/7 operation possible
Advantages	Smaller Environmental Footprint - Optimized resource usage
Advantages	Increased Productivity and Efficiency - Consistent cycle times
Advantages	Increased System Versatility - Modern systems adapt to changes
Disadvantages	Higher Start-up Cost - Significant capital investment required
Disadvantages	Higher Cost of Maintenance - Specialized technicians needed
Disadvantages	Obsolescence/Depreciation - Technology changes rapidly
Disadvantages	Workforce Displacement - Fewer direct labor jobs
Disadvantages	Not Economical for Small Scale - Minimum volumes required

2.4 Project Development Phases

Manufacturing projects follow a structured development process from initial concept to full operation. Understanding these phases is essential for process engineers.

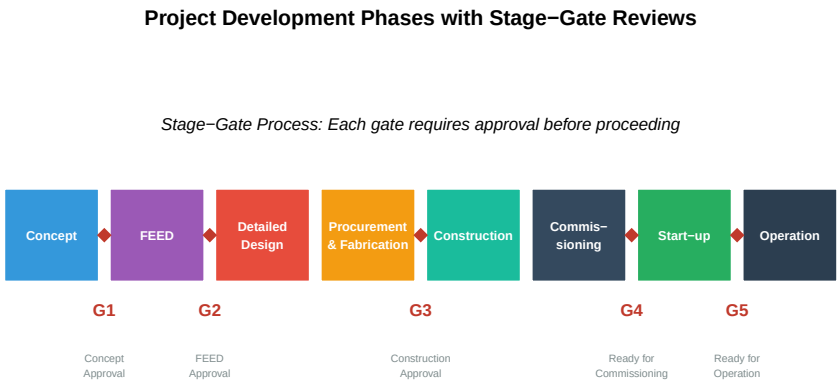


Figure 2.2: Project Development Phases and Gate Reviews

2.4.1 Try it yourself: put the project phases in order

A plant project passes through eight phases, with gate reviews between them. Put the phases in the order they happen.

Project phases, first to last

Number these in the correct order:

- _____ Commissioning
- _____ Concept
- _____ Construction
- _____ Detailed design
- _____ Front-End Engineering Design (FEED)
- _____ Operation
- _____ Procurement and fabrication
- _____ Start-up

Answers, in the order listed: 6, 1, 5, 3, 2, 8, 4, 7.

2.4.2 Phase 1: Concept Development

The initial phase defines the project scope, evaluates feasibility, and develops preliminary cost estimates.

Key Deliverables: - Project charter and business case - Preliminary process flow diagrams - Order-of-magnitude cost estimate (±30-50%) - Initial schedule - Risk assessment

2.4.3 Phase 2: Front-End Engineering Design (FEED)

FEED (also called FEL - Front-End Loading) develops the project definition to a level sufficient for final investment decision.

FEED Process Overview: https://www.youtube.com/watch?v=gSkoGFLDxF0

FEED Deliverables include:

- Project organization chart
- Defined project scope
- Process Flow Diagrams (PFDs)
- Piping & Instrumentation Diagrams (P&IDs)
- Equipment datasheets and specifications
- Major equipment list
- 2D and 3D preliminary models
- Equipment layout and installation plan
- HAZOP and safety studies
- Automation strategy
- Budget estimate (±10-15%)
- Project timeline
- Fixed-bid quote capability

What are the steps of Process System Design and Engineering?

1. **Process Development**
   - Develop PFDs & P&IDs
   - Process simulation
   - Heat and mass balance
2. **Equipment Engineering**
   - 2D & 3D models
   - Equipment layout and installation plan
   - Skid design
   - Mechanical & structural design
3. **Systems Engineering**
   - Piping design
   - Electrical design
   - Instrumentation design
   - Control system architecture
4. **Installation Planning**
   - Lifts & installation planning
   - Construction sequencing
   - Tie-in planning

2.4.4 Phase 3: Detailed Engineering

Detailed engineering produces all documents needed for procurement and construction.

Table 2.3: Engineering Disciplines and Deliverables

Discipline	Key Deliverables	Systems
Process	P&IDs, datasheets, process packages, operating procedures	Process u
Mechanical	Vessel drawings, heat exchanger specs, rotating equipment specs	Pressure
Piping	Isometrics, pipe specs, stress analysis, 3D models	Process p
Electrical	Single-line diagrams, load lists, cable schedules, lighting	Power dis
I&C	Logic diagrams, instrument index, control narratives, HMI specs	DCS/PL
Civil/Structural	Foundation drawings, structural steel, buildings, roads	Foundati

Automation and Controls Engineering needs to include which tasks?

Control System Design: - Automation architecture design - Control panel fabrication drawings - PLC/DCS programming - HMI/SCADA design - Safety system design (SIS)

Documentation: - Control narratives - Logic diagrams (cause & effect) - Instrument datasheets - Loop diagrams - Cable schedules

Integration: - Controls integration services - Network architecture - Cybersecurity - Historian configuration - MES integration

2.4.5 Phase 4-7: Construction, Commissioning, and Start-up

Table 2.4: Later Project Phas

Phase	Key Activities	I
Procurement & Fabrication	Purchase equipment, fabricate skids, manufacture custom items	I
Construction	Civil work, install equipment, piping, electrical, instrumentation	S
Commissioning	System checkout, loop testing, safety system validation	V
Start-up	Initial operation, performance testing, operator training	C

What needs to be planned for Testing & Commissioning?

Pre-Commissioning: - Mechanical completion checks - Piping pressure tests - Electrical continuity tests - Instrument calibration

Commissioning: - Loop checking - Control system function tests - Safety system (SIS) validation - Utility systems start-up - Dry run testing

Performance Testing: - Factory Acceptance Tests (FAT) - Site Acceptance Tests (SAT) - Performance guarantee tests - Reliability testing

2.5 Process Documentation

2.5.1 Process Flow Diagrams (PFDs)

PFDs show the overall process flow, major equipment, and key operating conditions. They are the "big picture" view of the process.

Table 2.5: Process Flow Diagram (PFD) Content

Element	What's Shown	What's NOT Shown
Major Equipment	Reactors, vessels, heat exchangers, pumps, compressors	Pipe sizes, valve types, instru
Process Streams	Main process flow lines with arrows	Minor bypasses, drain lines,
Operating Conditions	Temperature, pressure, flow rates at key points	All intermediate conditions
Control Loops	Major control loops only	Local instruments, minor cor
Utility Streams	Steam, cooling water, nitrogen (simplified)	Detailed utility connections
Stream Data	Stream numbers with heat/mass balance table	All intermediate streams

2.5.2 Piping & Instrumentation Diagrams (P&IDs)

P&IDs are the detailed "roadmap" of the process, showing all equipment, piping, valves, and instrumentation.

How to Read P&IDs: https://www.youtube.com/watch?v=j4EOTerfyTY

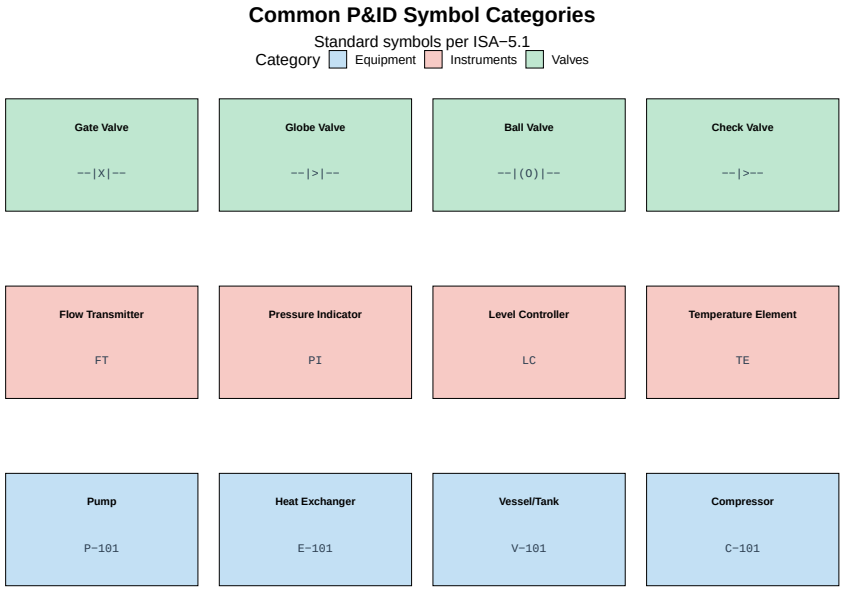


Figure 2.3: Common P&ID Symbols (Simplified)

Table 2.6: ISA Instrument Tag Structure

First Letter	Measured Variable	Modifier Letters	Example Tag
F	Flow	T = Transmitter	FT-101 (Flow Transmitter)
L	Level	C = Controller	LC-201 (Level Controller)
P	Pressure	I = Indicator	PI-301 (Pressure Indicator)
T	Temperature	E = Element/Sensor	TE-401 (Temperature Element)
A	Analysis	V = Valve	AT-501 (Analyzer Transmitter)
S	Speed	A = Alarm	SI-601 (Speed Indicator)

Quick Quiz: Decode these instrument tags

Decode the following P&ID tags:

1. **FIC-101** = ?
2. **TT-205** = ?
3. **PSV-301** = ?
4. **LCV-402** = ?

Answers:

1. **FIC-101** = Flow Indicating Controller (loop 101)
2. **TT-205** = Temperature Transmitter (loop 205)
3. **PSV-301** = Pressure Safety Valve (loop 301)
4. **LCV-402** = Level Control Valve (loop 402)

2.5.2.1 Try it yourself: read the first letter

The first letter of an instrument tag tells you what is being measured. Sort these tags by measured variable.

What does each instrument measure?

Sort each tag by the variable it measures.

Categories: Flow / Level / Pressure / Temperature

1. FT-110 _____
2. FIC-115 _____
3. LIC-220 _____
4. LAH-240 _____
5. PI-305 _____
6. PSV-330 _____
7. TE-410 _____
8. TT-420 _____

Answers: 1 Flow; 2 Flow; 3 Level; 4 Level; 5 Pressure; 6 Pressure; 7 Temperature; 8 Temperature.

2.6 Design for Manufacturing (DFM) and Assembly (DFA)

DFM and **DFA** are methodologies that consider manufacturing and assembly constraints during product design, reducing cost and improving quality.

2.6.1 DFM Principles

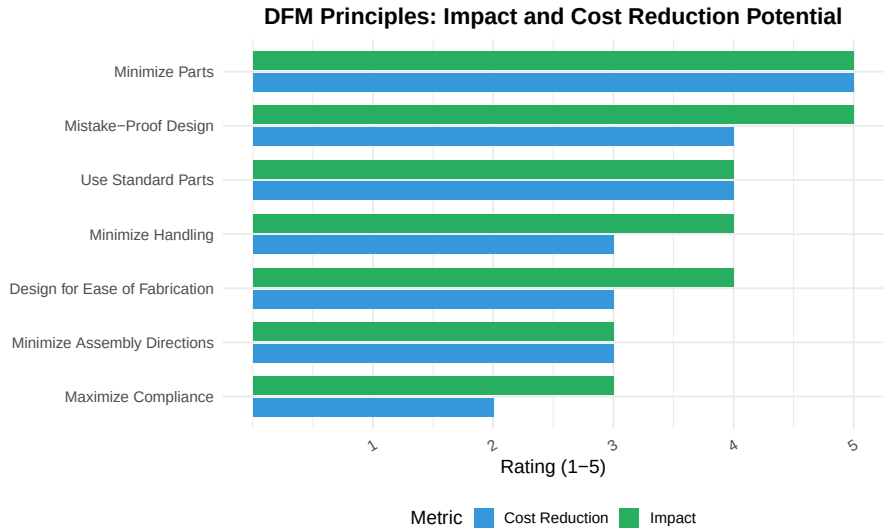


Figure 2.4: DFM Guidelines

Table 2.7: Design for Manufacturing Guidelines

DFM Guideline	Why It Matters	Exampl
Minimize total number of parts	Fewer parts = fewer chances for failure, less inventory, lower cost	Combine
Use standard components	Standard parts are cheaper, faster to procure, easier to replace	Use M6 l
Design for ease of fabrication	Complex features increase manufacturing cost and defect rates	Avoid un
Design for ease of assembly	Reduces assembly time and errors	Use snap
Minimize assembly directions	Multiple directions require repositioning, increasing time	All parts
Maximize compliance in assembly	Tight tolerances increase cost; allow self-alignment where possible	Use tape
Design for ease of service	Products will need maintenance; design for technician access	Locate se

2.6.2 DFA Analysis

DFA systematically evaluates assembly operations to identify improvement opportunities.

Teaching demo — change the part list, the times or the "necessary" flags and watch the design efficiency move.

Show the live R code behind this card

```
#| label: dfa-example
# DFA Analysis Example: Simple Assembly
# Calculate assembly efficiency using Boothroyd-Dewhurst method

# Assembly data (simplified)
parts <- data.frame(
  Part = c("Base", "PCB", "Cover", "Screw 1", "Screw 2", "Screw 3", "Screw 4", "Label")
  Handling_Time = c(2.5, 3.0, 2.5, 3.5, 3.5, 3.5, 3.5, 4.0),  # seconds
  Insertion_Time = c(3.0, 4.0, 3.5, 6.0, 6.0, 6.0, 6.0, 2.0),  # seconds
  Theoretically_Necessary = c(TRUE, TRUE, TRUE, FALSE, FALSE, FALSE, FALSE, FALSE)
)

# Calculations
total_parts <- nrow(parts)
total_time <- sum(parts$Handling_Time + parts$Insertion_Time)
min_parts <- sum(parts$Theoretically_Necessary)
ideal_time <- min_parts * 3  # 3 seconds is theoretical minimum per part

efficiency <- (ideal_time / total_time) * 100

cat(sprintf("Design efficiency: %.1f%%\n", efficiency))
cat(sprintf("  Total parts  %g\n", total_parts))
cat(sprintf("  Theoretically minimum parts  %g\n", min_parts))
cat(sprintf("  Parts that could be combined or eliminated  %g\n", total_parts - min_par
cat(sprintf("  Total assembly time  %.1f s\n", total_time))
cat(sprintf("  Ideal assembly time  %.1f s\n", ideal_time))

par(mar = c(5, 5, 3, 1))
barplot(parts$Handling_Time + parts$Insertion_Time, names.arg = parts$Part, cex.names =
        col = ifelse(parts$Theoretically_Necessary == 1, "#27AE60", "#E67E22"),
        ylab = "Handling + insertion time (s)",
        main = "Assembly time per part (green = necessary)")
```

Case Study: Redesigning for DFM/DFA

Original Design: - 15 parts - 4 different screw types - Assembly time: 180 seconds - Multiple assembly directions - Special tools required

Redesigned using DFM/DFA: - 8 parts (47% reduction) - 1 standard screw

type - Assembly time: 75 seconds (58% reduction) - Single assembly direction (top-down) - No special tools

Cost Impact: - Part cost reduced 35% - Assembly labor reduced 58% - Quality defects reduced 60% - Total product cost reduced 40%

2.7 HAZOP Studies

HAZOP (Hazard and Operability Study) is a systematic method to identify hazards and operability problems in process plants.

What is a HAZOP? A Crash Course: https://www.youtube.com/watch?v=6AqtX8oCpKI

Key P&ID Details for HAZOPs: https://www.youtube.com/watch?v=ukn0RRt0ReA

2.7.1 HAZOP Methodology

Table 2.8: HAZOP Guide Words and Examples

Guide Word	Meaning	Example with FLOW
NO/NOT	Complete negation of design intent	No flow - pump failure, blocked line
MORE	Quantitative increase	More flow - control valve fails open
LESS	Quantitative decrease	Less flow - partial blockage, pump wear
AS WELL AS	Qualitative modification/addition	Flow + contamination - upstream leak
PART OF	Qualitative decrease	Only part of intended flow path active
REVERSE	Logical opposite of intent	Reverse flow - check valve failure
OTHER THAN	Complete substitution	Wrong material flowing

2.7.2 HAZOP Example

Table 2.9:

Node	Parameter	Guide Word	Deviation	Cause
Feed Pump P-101 to Reactor R-101	Flow	No	No Flow	Pump failure, blocked suctio
Feed Pump P-101 to Reactor R-101	Flow	More	More Flow	Control valve fails open
Feed Pump P-101 to Reactor R-101	Pressure	More	High Pressure	Downstream blockage

Feed Pump P-101 to Reactor R-101	Temperature	Less	Low Temperature	Cooling water le

Discussion: Why is HAZOP done during FEED, not later?

HAZOP should be completed during FEED because:

1. **Cost of changes** - Changes during FEED cost 1x; during detailed design cost 10x; during construction cost 100x
2. **Design influence** - P&IDs are still being developed and can be modified
3. **Equipment selection** - Safety requirements may affect equipment sizing
4. **Budget accuracy** - Safety systems must be included in project budget
5. **Regulatory approval** - Many jurisdictions require HAZOP before construction permits

Best practice: Conduct preliminary HAZOP early in FEED, then detailed HAZOP when P&IDs are 60-80% complete.

2.8 Cost Estimation

Accurate cost estimation is critical for project approval and budget management.

Table 2.10: Cost Estimate Classification (AACE International)

AACE Class	Project Phase	Engineering Complete	Accuracy Range	Purpose
Class 5	Concept Screening	0-2%	-50% to +100%	Screening alte
Class 4	Concept Study	1-15%	-30% to +50%	Feasibility
Class 3	FEED/Budget	10-40%	-20% to +30%	Authorization
Class 2	Detailed Design	30-75%	-15% to +20%	Control
Class 1	Construction	65-100%	-10% to +15%	Bid/Tender

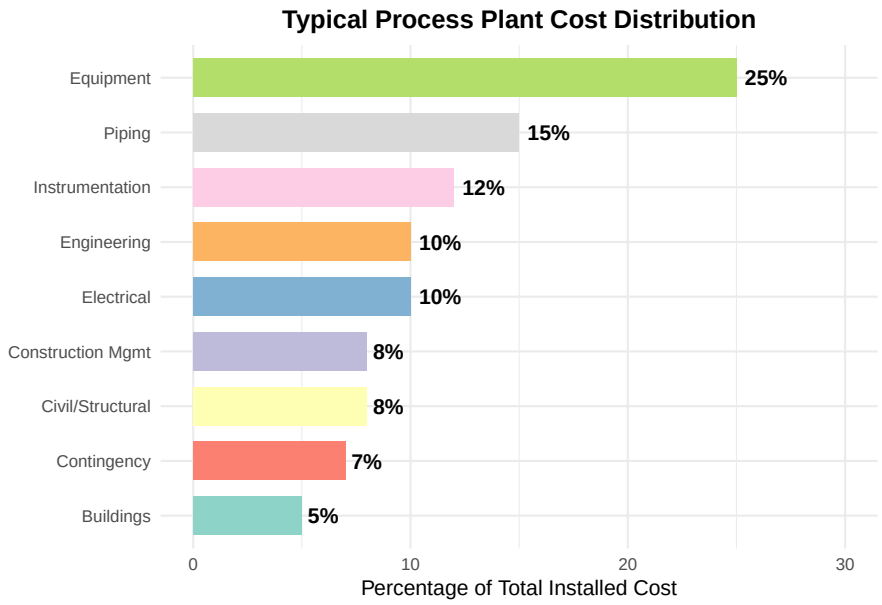


Figure 2.5: Typical Process Plant Cost Breakdown

2.9 Case Study: Automotive Paint Line Project

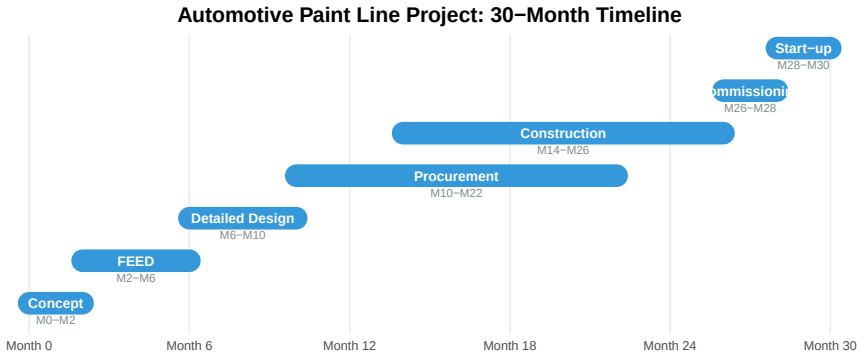


Figure 2.6: Automotive Paint Line Project Timeline

Case Study Details: Paint Line Engineering Challenges

Project Overview: - New automotive paint line for SUV body shop - Capacity: 60 jobs per hour (JPH) - Process: E-coat, primer, basecoat, clearcoat

Key Engineering Challenges:

1. Environmental Control - Temperature: ±1°C throughout paint booth - Humidity: 65% ±5% RH - Air cleanliness: Class 10,000

2. Automation Requirements - 48 paint robots (8 per zone) - Conveyor system with 2m pitch - Zone tracking and recipe management

3. Safety Systems - Explosion-proof electrical classification - Fire suppression throughout - VOC monitoring and control

4. Quality Systems - Inline thickness measurement - Automated visual inspection - Recipe management and traceability

Project Outcome: - On-time completion - Under budget by 3% - Quality metrics exceeded targets - 99.2% first-pass yield achieved

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2.10 Review Questions

Question 1: Automation Selection

A medical device company needs to produce 5,000 units per year of a surgical instrument. The design is mature and not expected to change. Volumes could grow to 50,000 units if successful. Which automation type should they initially choose?

Answer: Programmable automation (CNC machining, robotic cells)

Reasoning: - Current volume (5,000/year) doesn't justify fixed automation - Flexibility allows for minor design refinements - Can scale up by adding shifts or machines - Lower initial investment preserves capital - If volume grows to 50,000, can justify dedicated equipment later

Question 2: P&ID Interpretation

Decode the following instrument tags and explain their function:

a) FIC-201
b) PSV-305
c) TT-401
d) LCV-150

Answers:

a) **FIC-201** - Flow Indicating Controller (loop 201)
   - Measures flow, displays value, and controls a valve
b) **PSV-305** - Pressure Safety Valve (loop 305)
   - Relieves pressure if it exceeds setpoint (safety device)
c) **TT-401** - Temperature Transmitter (loop 401)
   - Converts temperature measurement to 4-20mA signal

d) **LCV-150** - Level Control Valve (loop 150)
- Valve that is manipulated by a level controller

Question 3: Project Phases

Why is FEED often considered the most critical project phase?

Answer: FEED is critical because:

1. **Defines scope** - All subsequent work based on FEED deliverables
2. **Locks in 80% of cost** - Design decisions determine equipment and construction costs
3. **Lowest cost to change** - Changes cost 10-100x more in later phases
4. **Authorization basis** - Management approves project based on FEED estimate
5. **Risk identification** - HAZOP and risk studies identify safety requirements
6. **Contractor selection** - FEED package used for competitive bidding

Poor FEED = project overruns, scope changes, and budget problems.

Question 4: DFM Application

A product currently uses 12 screws of 3 different sizes for assembly. Apply DFM principles to improve this design.

Answer: Apply these DFM principles:

1. **Minimize parts**: Can some screws be eliminated by using integral snap-fits?

2. **Standardize**: Use one screw size instead of three

   - Reduces inventory complexity
   - Single tool for assembly
   - Lower procurement cost

3. **Design for ease**: Consider alternatives:

   - Snap-fit connections (no tools)
   - Twist-lock features
   - Press-fit pins

Possible redesign: - Reduce from 12 screws to 4 screws (for serviceability) - All same size (M3 $\times$ 8) - Add 4 snap-fits for initial positioning - Result: 67% fewer fasteners, single tool, faster assembly

Question 5: HAZOP Analysis

For a reactor feed pump, apply the HAZOP guide word "REVERSE" to the parameter "FLOW". What could cause this? What are the consequences? What safeguards would you recommend?

Answer:

Deviation: Reverse flow through feed pump

Causes: - Pump stops while discharge valve open - Check valve fails (stuck open or missing) - Parallel pump creates backflow - Reactor pressure exceeds pump shutoff head

Consequences: - Reactor contents backflow into feed tank - Contamination of feed material - Possible chemical reaction in feed system - Pump damage (running backward)

Safeguards: - Check valve on pump discharge - Automatic pump discharge block valve (closes on pump stop) - Low flow alarm and trip - Reactor pressure monitoring - Backflow-preventing orifice

---

2.11 Summary

Table 2.11: Chapter 2 Summary

Topic	Key Takeaway
Automation Types	Fixed for high volume single product; Programmable for batch; Flexible
Project Phases	Concept → FEED → Detailed Design → Construction → Commissionir
PFDs vs P&IDs	PFD = big picture (equipment, flows); P&ID = detailed (all instrument
DFM/DFA	Design for easy manufacturing and assembly reduces cost 20-40%
HAZOP	Systematic hazard identification using guide words; do it during FEED
Cost Estimation	Estimate accuracy improves with engineering completion; AACE Class

2.11.1 Key Takeaways

1. **Automation selection** depends on production volume, variety, and changeover requirements
2. **FEED phase** defines 80% of project cost with only 10-15% of engineering effort
3. **P&IDs** are the critical document connecting design to construction
4. **DFM/DFA** should be applied early in design to maximize impact
5. **HAZOP** identifies hazards systematically and must be done before construction
6. **Cost estimates** improve in accuracy as engineering progresses

---

2.12 References

1. Towler, G., & Sinnott, R. (2021). *Chemical Engineering Design* (7th ed.). Elsevier.

2. AACE International. (2020). *Cost Estimate Classification System*.
3. ISA-5.1. (2022). *Instrumentation Symbols and Identification*.
4. Boothroyd, G., Dewhurst, P., & Knight, W.A. (2011). *Product Design for Manufacture and Assembly* (3rd ed.). CRC Press.
5. CCPS. (2008). *Guidelines for Hazard Evaluation Procedures* (3rd ed.). Wiley.

Chapter 3

Industrial Robotics

3.1 Learning Objectives

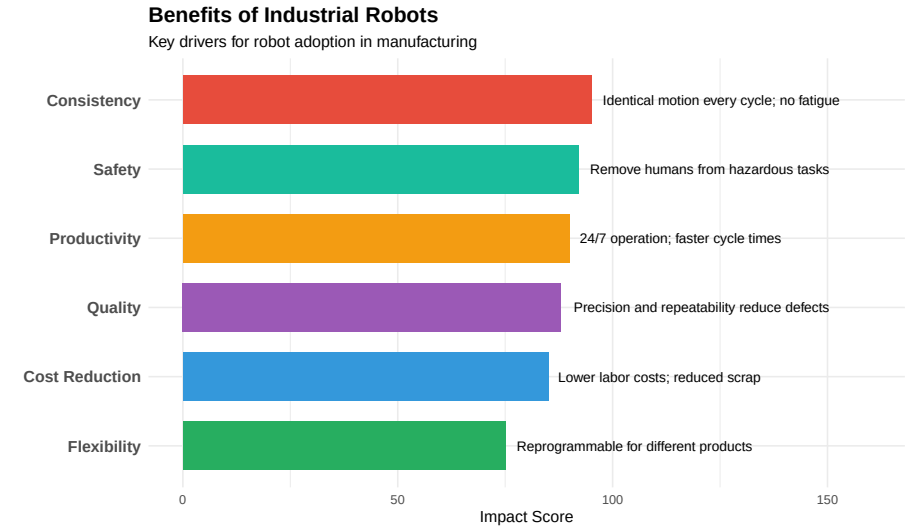
After completing this chapter, you will be able to:

1. Define industrial robots and explain their role in modern manufacturing
2. Identify the main types of robot configurations and their applications
3. Describe robot specifications including payload, reach, repeatability, and speed
4. Explain the components of a robotic system (manipulator, controller, teach pendant, end effector)
5. Understand robot coordinate systems and motion types
6. Identify common robot applications in automotive, food, and defense industries
7. Apply robot safety standards and safeguarding methods
8. Describe collaborative robot (cobot) technology and applications
9. Understand basic robot programming concepts

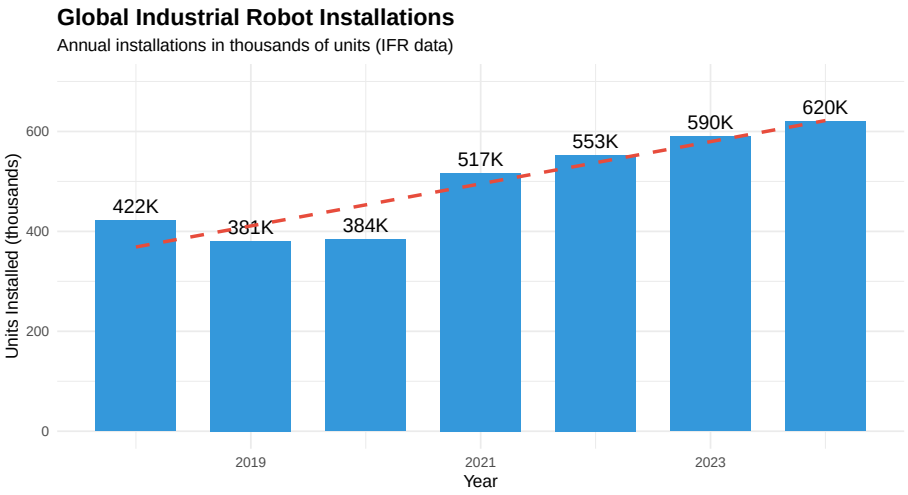
3.2 Introduction to Industrial Robotics

An **industrial robot** is defined by ISO 8373 as "an automatically controlled, reprogrammable, multipurpose manipulator, programmable in three or more axes, which can be either fixed in place or mobile for use in industrial automation applications."

3.2.1 Why Use Robots?



3.2.2 Robot Market Growth



3.2.3 Industries Using Robots

Table 3.1: Industrial Robot Usage by Industry Sector

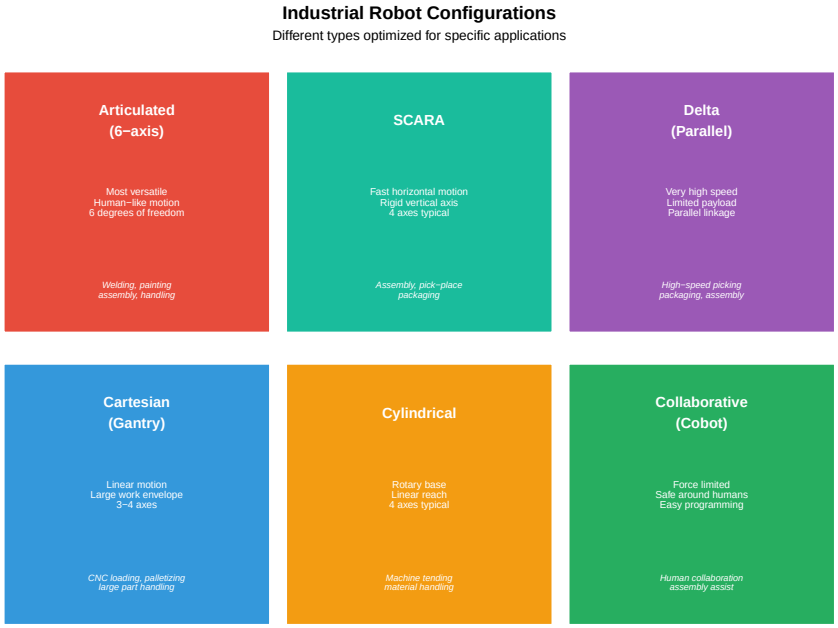
Industry	Market Share (%)	Key Applications	Gro
Automotive	30	Welding, painting, assembly, material handling	Stab

Electrical/Electronics	25	Assembly, testing, packaging, dispensing	High
Metal & Machinery	12	Machine tending, welding, cutting	Moderate
Food & Beverage	8	Palletizing, packaging, pick-and-place	High
Plastics & Chemicals	5	Injection molding, packaging, assembly	Moderate
Aerospace/Defense	4	Drilling, fastening, inspection, composite layup	High

3.3 Robot Configurations

Industrial robots come in various mechanical configurations, each suited to different applications.

3.3.1 Common Robot Types



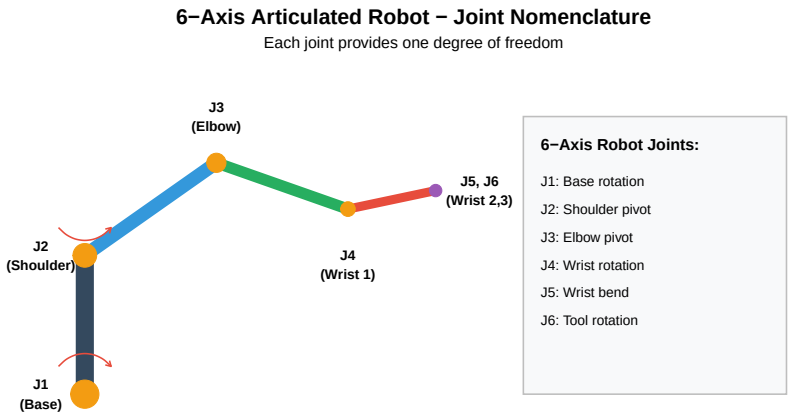
3.3.2 Detailed Configuration Comparison

Table 3.2: Robot Configuration Comparison

Configuration	Axes	Payload Range	Reach	Repeatability	Speed	Cost
6-Axis Articulated	6	3-2300 kg	0.5-4 m	±0.02-0.1 mm	Moderate-Fast	$$-$$$$
SCARA	4	1-20 kg	0.2-1.2 m	±0.01-0.05 mm	Very Fast	$$

Delta/Parallel	3-4	0.5-12 kg	0.5-1.5 m	±0.05-0.1 mm	Extremely Fast
Cartesian/Gantry	3-6	5-1000+ kg	Custom (meters)	±0.05-0.5 mm	Moderate

3.3.3 Degrees of Freedom



3.3.4 Try it yourself: match the robot to the job

Eight applications. Decide which robot configuration you would shortlist first.

Which robot configuration?

Pick the configuration you would shortlist first for each application.

Categories: 6-axis articulated / SCARA / Delta / Cartesian / gantry / Cobot

1. Arc welding a curved seam around a tubular frame. _____
2. Spray painting car bodies, including inside the door openings. _____
3. Pressing bearings vertically into housings at high speed. _____
4. Driving screws into the top face of a flat electronics enclosure. _____
5. Picking 120 chocolates a minute off a moving belt into trays. _____
6. Loading and unloading four CNC machines along a 12 m row. _____
7. Dispensing straight beads of adhesive across a 3 m x 2 m flat panel. _____
8. Holding a part while an assembler at the same bench fits the fasteners. _____

Answers: 1 6-axis articulated; 2 6-axis articulated; 3 SCARA; 4 SCARA; 5 Delta; 6 Cartesian / gantry; 7 Cartesian / gantry; 8 Cobot.

3.4 Robot Specifications

Understanding robot specifications is essential for proper selection and application.

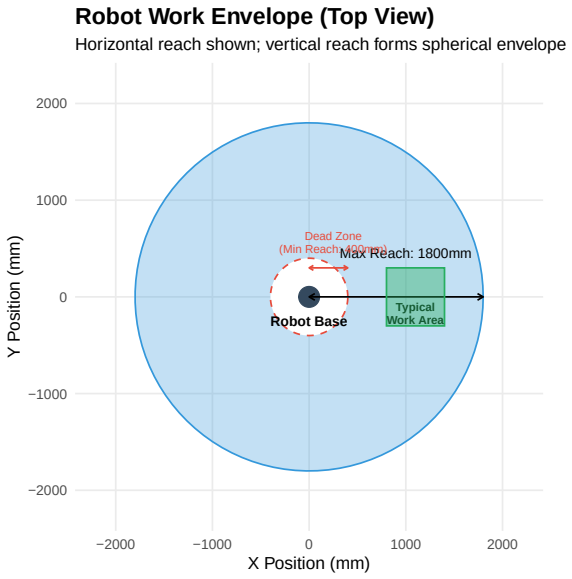
3.4.1 Key Specifications

Table 3.3: Key Robot Specifications Explained

Specification	Definition	Typical_Values
Payload	Maximum mass the robot can handle at full speed	3-2300 kg (varies widely by type
Reach	Maximum distance from base to tool center point	500-4000 mm for articulated rob
Repeatability	Ability to return to same position repeatedly	±0.01 to ±0.1 mm
Accuracy	Ability to reach a commanded position exactly	±0.1 to ±1.0 mm
Maximum Speed	Maximum velocity at tool center point or joint speeds	1-12 m/s TCP; 100-500°/s joint
Degrees of Freedom	Number of independent axes of motion	4-7 axes
Mounting	How robot is installed (floor, ceiling, wall, angle)	Floor (most common), ceiling, w
IP Rating	Ingress Protection rating for dust/water resistance	IP40 (standard) to IP67 (harsh

3.4.2 Work Envelope

The **work envelope** (or workspace) is the three-dimensional space the robot can reach.



3.4.3 Payload Considerations

Teaching demo — drag the sliders to see when the load plus its safety margin passes the robot's rating.

Show the live R code behind this calculator

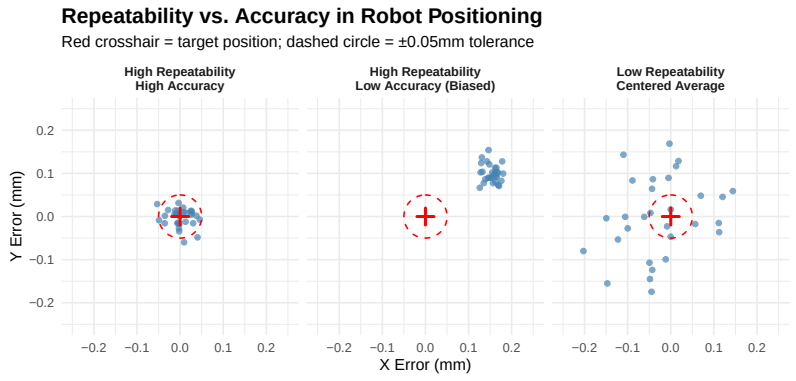
```
#| label: payload-calc
# Payload Calculation Example
robot_max_payload <- 10  # kg
gripper_weight <- 2.5    # kg
part_weight <- 5.0       # kg
safety_factor <- 1.2     # 20% safety margin

total_load <- gripper_weight + part_weight          # what the robot actually carries
required_capacity <- total_load * safety_factor      # total load, with margin applied
remaining_margin <- robot_max_payload - total_load   # headroom before the safety facto

if (required_capacity <= robot_max_payload) {
  cat(sprintf("Required capacity: %.1f kg - Acceptable\n", required_capacity))
} else {
  cat(sprintf("Required capacity: %.1f kg - !!Exceeded!!\n", required_capacity))
}
cat(sprintf("  Total load           %.1f kg\n", total_load))
cat(sprintf("  Remaining margin     %.1f kg (%.0f%%)\n", remaining_margin, 100 * remain
cat(sprintf("  Moment at wrist      %.2f kg m (at 0.3 m from flange)\n", part_weight *

par(mar = c(5, 5, 3, 1))
barplot(c(`Total load` = total_load, `With safety factor` = required_capacity),
        col = c("#2E86C1", "#F39C12"), ylim = c(0, 80), ylab = "Load (kg)", main = "Loa
abline(h = robot_max_payload, lty = 2, lwd = 2, col = "#C0392B")
```

3.4.4 Repeatability vs Accuracy



Understanding the Difference

Repeatability: - Ability to return to the SAME taught position - More important for most applications - Quoted spec (e.g., ±0.05mm) is typically 3 - Achieved through consistent mechanical design and servo control

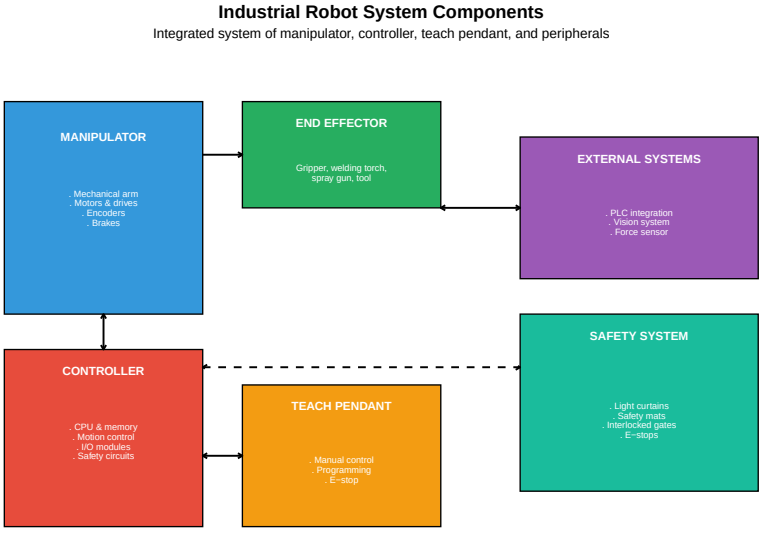
Accuracy: - Ability to reach a COMMANDED position - Important for offline programming - Affected by mechanical tolerances, deflection, calibration - Can be improved through calibration

Why Repeatability Matters More: In most applications, the robot is taught positions by jogging to them. As long as it returns to those positions consistently (repeatability), accuracy doesn't matter. Accuracy only matters when positions are calculated (offline programming) rather than taught.

3.5 Robot System Components

A complete robotic system consists of several integrated components.

3.5.1 System Architecture



3.5.2 End Effectors (EOAT)

End of Arm Tooling (EOAT) is the device attached to the robot's wrist to interact with parts.

Table 3.4: Common End Effector Types

EOAT Type	Operating Principle	Applications
Mechanical Gripper	Fingers actuated by pneumatic, electric, or hydraulic	General part handli
Vacuum Gripper	Suction cups powered by venturi or vacuum pump	Flat surfaces; sheet
Magnetic Gripper	Electromagnetic or permanent magnet	Ferrous metal parts
Welding Torch	MIG, TIG, spot welding equipment	Automotive body, f
Spray Gun	Paint, adhesive, sealant dispensing	Automotive paintin
Deburring Tool	Rotary tool for edge finishing	Casting, machining
Force/Torque Sensor	Measures forces and torques at tool	Assembly; polishing
Vision Camera	2D or 3D imaging for guidance	Part location; inspe

3.5.3 Gripper Selection Example

Your turn — change the part mass, acceleration, or friction coefficient below.

Show the live R code behind this calculator

```
#| label: gripper-selection
# Gripper Force Calculation
part_mass <- 3.0        # kg
acceleration <- 15      # m/s² (robot acceleration)
gravity <- 9.81         # m/s²
safety_factor <- 2.0    # Minimum safety factor
friction_coefficient <- 0.3  # Steel on rubber

# Calculate forces
weight_force <- part_mass * gravity
inertia_force <- part_mass * acceleration
total_force <- sqrt(weight_force^2 + inertia_force^2)

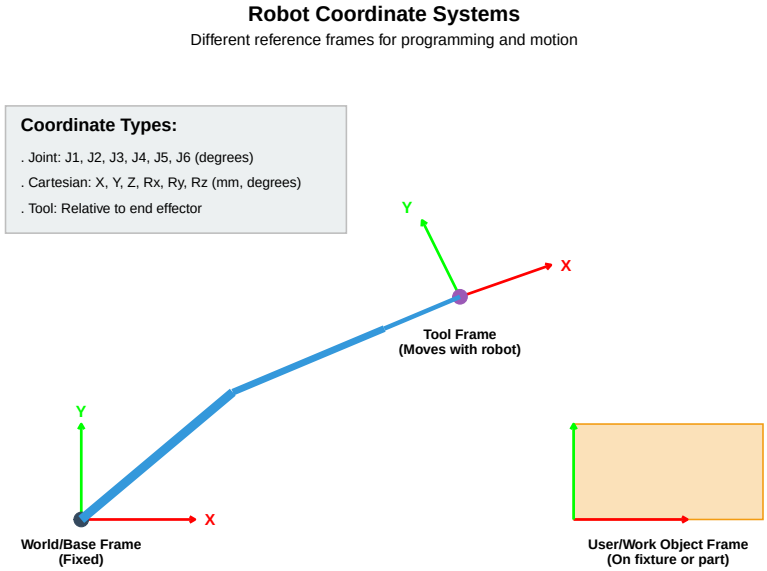
# Required grip force (friction gripper)
required_grip <- total_force / friction_coefficient * safety_factor

cat(sprintf("Required grip force: %.1f N (select  %.0f N)\n", required_grip, ceiling(r
cat(sprintf("  Weight force         %.1f N\n", weight_force))
cat(sprintf("  Inertia force        %.1f N\n", inertia_force))
cat(sprintf("  Combined force       %.1f N\n", total_force))
```

3.6 Coordinate Systems and Motion

Understanding coordinate systems is essential for robot programming.

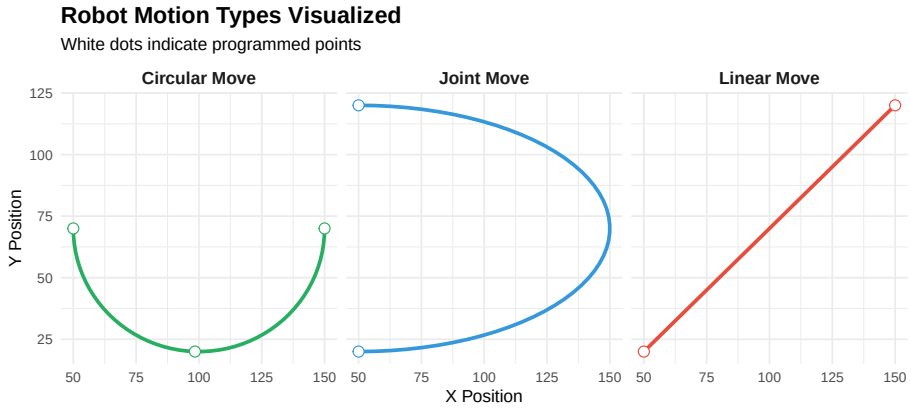
3.6.1 Coordinate Frames



3.6.2 Motion Types

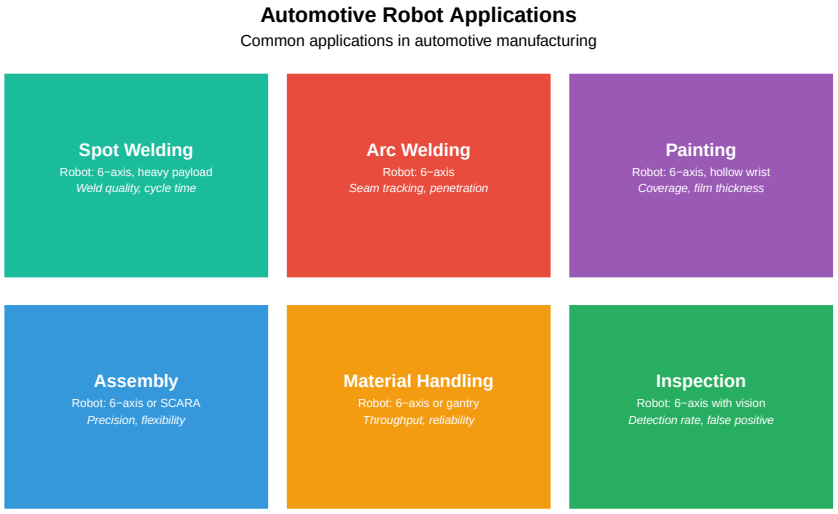
Table 3.5: Robot Motion Types

Motion Type	Path Characteristic	Speed Control
Joint Move (MoveJ)	Unpredictable path; each joint moves at constant speed	Joint velocity (% or deg/s
Linear Move (MoveL)	Straight line from start to end; TCP follows linear path	TCP velocity (mm/s)
Circular Move (MoveC)	Arc or circle; defined by start, via, and end points	TCP velocity (mm/s)



3.7 Robot Applications

3.7.1 Automotive Applications



3.7.2 Food & Beverage Applications

Table 3.6: Food \& Beverage Robot Applications

Application	Description	Robot Type
Palletizing	Stack cases/bags onto pallets in patterns	4-axis palletizer, 6-axis
Case Packing	Load products into cases/cartons	Delta, SCARA, 6-axis
Pick and Place	Transfer products between conveyors/stations	Delta (high speed), SCARA
Cutting/Portioning	Protein portioning, cake cutting	6-axis with vision
Decorating	Applying icing, toppings, decorations	6-axis, delta
Quality Inspection	Vision-based defect detection, grading	6-axis with vision

3.7.3 Aerospace/Defense Applications

Table 3.7: Aerospace/Defense Robot Applications

Application	Description	Accuracy Required
Drilling & Fastening	Precision hole drilling; rivet/bolt installation	±0.05mm position

Composite Layup	Automated fiber placement (AFP); tape laying	±0.25mm fiber placement	Complex
NDT/Inspection	Ultrasonic, X-ray, visual inspection of structures	Full coverage	Large pa
Surface Treatment	Painting, coating, surface prep for bonding	Consistent thickness	Large en
Assembly	Large structure assembly; wing-to-fuselage	±0.5mm	Heavy pa

3.8 Robot Safety

Robot safety is critical to protect workers from potential hazards.

3.8.1 Hazard Categories

Table 3.8: Robot Hazard Categories and Mitigations

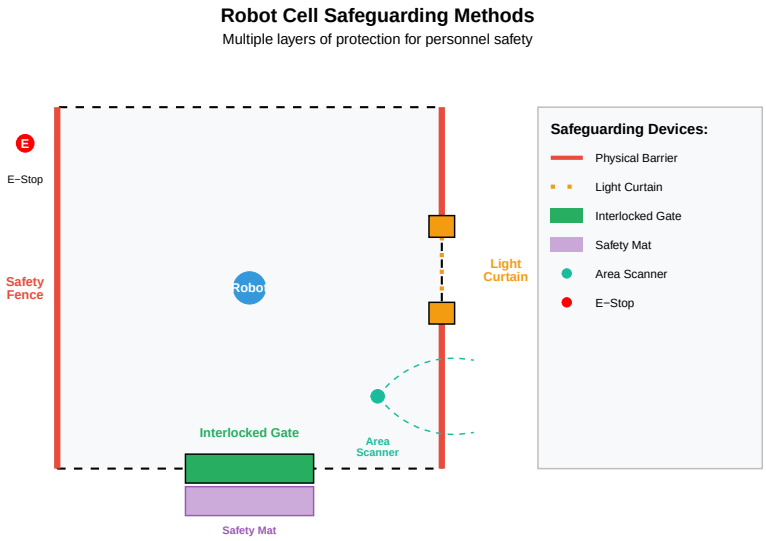
Hazard	Description	Risk_Factors
Impact	Robot strikes person with arm or tooling	Speed, mass, sharp edges, unexpected motio
Crushing	Person caught between robot and fixed object	Limited escape space, high forces
Trapping	Body part caught in articulating joints or mechanisms	Pinch points at joints, insufficient clearance
Ejection	Part or tool flies off due to grip failure or breakage	Grip force, centrifugal force, tool condition
Other	Electrical, thermal, noise, radiation hazards	Voltage, heat, welding arc, laser

3.8.2 Safety Standards

Table 3.9: Key Robot Safety Standards

Standard	Scope	Key Requirements
ISO 10218-1:2011	Robot manufacturer requirements	Stop functions, speed limiting, singularity protec
ISO 10218-2:2011	Robot system integrator requirements	Risk assessment, safeguarding, layout, validation
ISO/TS 15066:2016	Collaborative robot safety	Force/pressure limits, speed/separation monitori
ANSI/RIA R15.06	US equivalent to ISO 10218	Safeguarding devices, risk assessment
ANSI/RIA R15.08	Industrial mobile robots	Navigation safety, personnel detection

3.8.3 Safeguarding Methods



3.9 Collaborative Robots (Cobots)

Collaborative robots are designed to work safely alongside humans without traditional guarding.

3.9.1 Cobot Characteristics

Table 3.10: Traditional Robot vs. Collaborative Robot Comparison

	Traditional Industrial Robot	Collaborative Robot
Safety Approach	Safeguarded cell; humans excluded	Inherently safe; force/speed limited
Payload	Up to 2300 kg	Typically 3-35 kg
Speed	Very high (>2 m/s)	Limited (<1.5 m/s typically)
Reach	Up to 4 m	500-1300 mm typical
Programming	Specialized programmers	Intuitive; hand guiding
Guarding	Required (fences, curtains)	Often not required
Investment	High ($50K-500K+ with cell)	Lower ($25K-75K typical)
Applications	High-volume, dedicated tasks	Flexible, shared workspace

3.9.2 Collaborative Operation Modes (ISO/TS 15066)

Table 3.11: ISO/TS 15066 Collabora

Collaboration Mode	Description
Safety-Rated Monitored Stop	Robot stops when human enters collaborative zone; resumes when clear
Hand Guiding	Operator physically guides robot; robot follows input forces
Speed and Separation Monitoring	Robot adjusts speed based on distance to human; stops if too close
Power and Force Limiting	Robot limits force/pressure on contact to safe levels

3.9.3 Force and Pressure Limits

ISO/TS 15066 specifies maximum contact forces and pressures for different body areas:

Table 3.12: ISO/TS 15066 Biomechanical Limits (Selected Values)

Body Area	Max Pressure (N/cm²)	Quasi-Static Pressure	Max Force (N)	Quasi-Static Force
Skull/Forehead	130	130	130	130
Face	65	65	65	65
Neck (front/back)	145	145	150	150
Chest	140	140	140	140
Abdomen	110	110	110	110
Hand/Finger	280	280	140	140
Arm	190	190	150	150
Leg	220	220	220	220

3.10 Robot Programming Basics

3.10.1 Programming Methods

Table 3.13: Robot Programmi

Method	Description	Advantages
Online/Teach Pendant	Use teach pendant to jog robot to positions and record	Direct; prec
Lead-Through/Hand Guiding	Physically move robot arm to desired positions	Intuitive; fa
Offline Programming	Create programs on PC without robot; download later	No producti
Simulation-Based	Program and test in virtual environment; transfer to real robot	Risk-free te

3.10.2 Basic Program Structure

`Example Robot Program (ABB RAPID-style pseudocode):`

```
PROGRAM Main()

  ! Initialize
  HomePosition()
  GripperOpen()

  ! Main cycle
  WHILE running DO

    ! Move to pick position
    MoveJ pApproach, v500, z50, tool1
    MoveL pPick, v100, fine, tool1

    ! Pick part
    GripperClose()
    WaitTime 0.2

    ! Move to place position
    MoveL pApproach, v200, z50, tool1
    MoveJ pPlaceApproach, v500, z50, tool1
    MoveL pPlace, v100, fine, tool1

    ! Place part
    GripperOpen()
    WaitTime 0.2

    ! Return
    MoveL pPlaceApproach, v200, z50, tool1

    ! Increment counter
    PartCount := PartCount + 1

  ENDWHILE

ENDPROGRAM
```

3.10.3 Key Programming Concepts

Table 3.14: Key Robot Programming Concepts

Concept	Description	Example
Position/Target	Stored robot position (joint angles or Cartesian coordinates)	pHome, pP
Motion Instruction	Command to move robot (MoveJ, MoveL, MoveC)	MoveL pTa

Speed Data	Velocity parameter (mm/s for TCP, % for joints)	v100 = 100 mm/s, v50
Zone Data	Corner path blending (fine = stop at point, z10 = 10mm blend)	fine = stop, z5 = 5mm
Tool Data	Definition of tool center point relative to flange	tool1 with TCP offset
Work Object	Definition of work coordinate system for part positions	wobj_fixture with base
I/O Commands	Control external devices (SetDO, WaitDI, etc.)	SetDO doGripper, 1; W

3.11 Video Resources

3.11.1 Introduction to Industrial Robots

3.11.2 Collaborative Robots Explained

3.12 Summary

Industrial robotics is a fundamental technology in modern manufacturing:

1. **Robot types** include articulated (most versatile), SCARA (fast assembly), delta (high-speed picking), and cartesian (large work areas)
2. **Key specifications** include payload, reach, repeatability, speed, and degrees of freedom
3. **System components** work together: manipulator, controller, teach pendant, end effector, and safety system
4. **Coordinate systems** (world, tool, user) enable flexible programming and positioning
5. **Applications** span automotive, food & beverage, and aerospace with industry-specific requirements
6. **Safety** requires comprehensive risk assessment and appropriate safeguarding
7. **Collaborative robots** enable human-robot cooperation with force/speed limiting
8. **Programming** can be online (teach pendant), lead-through, or offline

---

3.13 Review Questions

Question 1: Compare articulated (6-axis) robots, SCARA robots, and delta robots. For each, describe the typical applications and key advantages.

Answer:

Robot Type	Configuration	Key Advantages	Typical Applications
6-Axis Articulated	6 revolute joints; human-like arm	Maximum flexibility; can reach any orientation; large payload range	Welding, painting, assembly, material handling, machine tending
SCARA	4 axes; selective compliance	Very fast horizontal motion; rigid vertical; compact; precise	Pick-and-place, assembly, screw driving, packaging
Delta (Parallel)	3-4 axes; parallel linkage	Extremely fast (150+ picks/min); low moving mass	High-speed picking, packaging, sorting, light assembly

Detailed Comparison:

6-Axis Articulated: - Most versatile - can position tool at any angle - Wide payload range (3 kg to 2300+ kg) - Can reach around obstacles - Complex programming possible - *Best for:* Tasks requiring complex orientations, welding seams, painting

SCARA: - Inherently rigid in Z (vertical) - good for insertion - Very fast in X-Y plane - Lower cost than 6-axis - Compact footprint - *Best for:* Assembly operations, circuit board population, packaging

Delta: - Highest speed of any configuration - Low inertia (motors at base, not moving) - Limited payload (typically <15 kg) - Requires overhead mounting - *Best for:* High-speed sorting, packaging lines, food handling

Question 2: A robot has a maximum payload of 15 kg. The gripper weighs 4 kg and the part weighs 8 kg. Is this robot suitable? What other factors should be considered?

Answer:

Initial Assessment:

```
Total load = Gripper weight + Part weight
Total load = 4 kg + 8 kg = 12 kg

Robot capacity: 15 kg
Load: 12 kg
Remaining margin: 3 kg (20%)
```

Consideration with Safety Factor:

```
Typical safety factor: 1.2 - 1.5
Required capacity = 12 kg × 1.25 = 15 kg
```

```
This equals the robot's maximum capacity - marginal!
```

Additional Factors to Consider:

1. **Moment Load (Torque at Wrist)**
   - Distance from flange to center of gravity matters
   - If part is held 300mm from flange: Moment = 8 kg × 0.3 m = 2.4 kg · m
   - Robot specs include moment limits - must verify
2. **Inertia**
   - Fast motion requires acceleration
   - Higher mass + longer distance = higher inertia
   - May need to derate payload for high-speed applications
3. **Payload Derating with Reach**
   - Many robots derate payload at extended reach
   - 15 kg at 500mm may become 10 kg at 1500mm
   - Check payload diagram in robot specifications
4. **Orientation**
   - Payload may vary with wrist orientation
   - Horizontal reach vs. vertical lift differ
5. **Acceleration Requirements**
   - Higher acceleration = higher effective load
   - May need to reduce speed/acceleration

Recommendation: The 15 kg robot is marginal for this application. Consider: - Select next size up (20 kg) for adequate margin - Use lighter gripper (3 kg or less) - Reduce part handling distance from flange - Limit acceleration/speed if using this robot

Question 3: Explain the difference between repeatability and accuracy. Why is repeatability often more important than accuracy for robot applications?

Answer:

Definitions:

Repeatability: - Ability to return to the **same taught position** multiple times - Measures consistency/precision of motion - Quoted as ± value (e.g., ±0.05 mm at 3) - Affected by: mechanical wear, servo control, thermal effects

Accuracy: - Ability to reach a **commanded position** exactly - Measures how close actual position is to theoretical position - Typically worse than repeatability (e.g., ±0.5 mm) - Affected by: manufacturing tolerances, calibration, deflection, backlash

Why Repeatability is Usually More Important:

1. **Teaching Method**
   - Most robots are programmed by teaching (jogging to positions)

   - Robot doesn't need to know where it "should" be - just return to where it was taught
   - As long as it returns consistently (repeatability), accuracy doesn't matter
2. **Process Requirements**
   - Welding: Need to follow same seam every time → repeatability
   - Assembly: Need to insert part in same hole every time → repeatability
   - Pick-place: Need to pick from same fixture position → repeatability
3. **Compensation Possible**
   - Poor accuracy can be compensated by adjusting taught points
   - Poor repeatability cannot be compensated

When Accuracy Matters:

1. **Offline Programming**
   - Positions calculated from CAD data, not taught
   - Robot must go where commanded, first time
   - Common in aerospace (large parts, tight access)
2. **Multi-Robot Systems**
   - Robots must agree on positions
   - Calibration to common frame requires accuracy
3. **Frequent Program Changes**
   - If programs are frequently changed without re-teaching
   - Offline modifications need accurate execution

Bottom Line: For most industrial applications, ±0.05 mm repeatability is far more valuable than ±0.5 mm accuracy because taught positions are used.

Question 4: Describe the four collaborative operation modes defined in ISO/TS 15066. Give an example application for each.

Answer:

1. Safety-Rated Monitored Stop

Description: Robot operates at normal speed when workspace is clear. When a person enters the collaborative zone, the robot stops and holds position. Robot resumes automatically when person exits.

Technical Requirements: - Safety-rated sensors (light curtains, scanners) - Safety-rated monitored stop function - Clearly defined collaborative zone

Example Application: Machine tending cell where operator occasionally loads raw material. Robot works at full speed during machining cycle but stops safely when operator enters to load/unload.

---

2. Hand Guiding

Description: Operator physically contacts the robot through a hand guiding device and manually moves it. Robot follows operator's force input.

Technical Requirements: - Hand guiding device (force/torque sensing) - Emergency stop accessible at guiding device - Speed reduction while guiding - Safe torque control

Example Application: Teaching a paint path on an aerospace component. Operator guides robot along surface contour while robot records positions. Also used for "collaborative finishing" where operator guides robot holding a polishing tool.

---

3. Speed and Separation Monitoring

Description: Robot continuously monitors distance to the nearest person. Speed adjusts based on separation: full speed when far, reduced speed when closer, stop if minimum distance breached.

Technical Requirements: - Safety-rated distance monitoring (usually area scanners) - Safety-rated speed control - Multiple speed/distance zones defined - Real-time separation calculation

Example Application: Shared packaging area where robot palletizes cases while operator moves around the cell. Robot slows when operator approaches, stops if too close, and resumes at appropriate speed as operator moves away.

---

4. Power and Force Limiting

Description: Robot is designed so that forces and pressures from any contact are below injury thresholds. Contact is allowed because it cannot cause harm.

Technical Requirements: - Inherently safe design (compliant joints, low mass, rounded surfaces) - Force/torque sensing or current monitoring - Verified force limits per ISO/TS 15066 body area tables - May include padding/soft covers

Example Application: Assembly assist where robot holds a heavy part while human worker installs fasteners. Robot and human are in direct contact, with robot providing position support and human providing dexterity. Force is limited so accidental contact causes no injury.

Question 5: Calculate the required gripper force for a robot picking a 2 kg part with a friction coefficient of 0.4. The robot accelerates at 10 m/s². Use a safety factor of 2.0.

Answer:

```
# Given values
mass <- 2.0            # kg
acceleration <- 10     # m/s²
```

```
gravity <- 9.81        # m/s²
friction_coef <- 0.4   # Coefficient of friction
safety_factor <- 2.0   # Safety factor

# Calculate forces
# Weight force (vertical)
F_gravity <- mass * gravity

# Inertia force (from acceleration)
F_inertia <- mass * acceleration

# Combined force (vector sum for worst case)
# Worst case: acceleration horizontal while holding against gravity
F_combined <- sqrt(F_gravity^2 + F_inertia^2)

# Required friction force to hold part
# F_friction =   × F_normal
# F_normal = Grip force
# F_friction must overcome F_combined

# Without safety factor:
F_grip_min <- F_combined / friction_coef

# With safety factor:
F_grip_required <- F_grip_min * safety_factor

cat(sprintf("Required gripper force: %g N minimum\n", ceiling(F_grip_required)))
cat(sprintf("  Select a gripper rated for at least  %g N\n", ceiling(F_grip_required /
cat(sprintf("  Gravity force (m x g)  %.1f N\n", F_gravity))
cat(sprintf("  Inertia force (m x a)  %.1f N\n", F_inertia))
cat(sprintf("  Combined force (vector sum)  %.1f N\n", F_combined))
cat(sprintf("  Grip force without safety factor  %.1f N\n", F_grip_min))
cat(sprintf("  Grip force with safety factor  %.1f N\n", F_grip_required))
```

```
Required gripper force: 141 N minimum
  Select a gripper rated for at least  150 N
  Gravity force (m x g)  19.6 N
  Inertia force (m x a)  20.0 N
  Combined force (vector sum)  28.0 N
  Grip force without safety factor  70.0 N
  Grip force with safety factor  140.1 N
```

Question 6: List and describe five safety devices used to protect personnel around industrial robots.

Answer:

1. Physical Barriers (Safety Fencing)

Description: Fixed guards or fencing that physically prevent access to the robot work envelope.

Characteristics: - Rigid construction (steel, aluminum, polycarbonate) - Must withstand expected impact forces - Minimum height (typically 1.8 m) - Properly anchored to floor - May include interlocked gates for authorized entry

Application: Primary safeguard for most traditional robot cells.

---

2. Light Curtains (AOPD - Active Opto-electronic Protective Device)

Description: Transmitter and receiver create a sensing field of infrared beams. Breaking any beam triggers a stop.

Characteristics: - Type 2 or Type 4 (safety-rated) - Resolution (14mm for finger, 40mm for hand/arm) - Muting capability for material transfer - Very fast response (<20ms)

Application: Access points where frequent entry needed; allows material flow while protecting personnel.

---

3. Safety Mats (Pressure-Sensitive Mats)

Description: Floor mats that detect weight/pressure and trigger stop when stepped on.

Characteristics: - Placed in hazardous zones - Withstand industrial environment (oil, debris) - Fast response - Self-monitoring for failures

Application: Areas where overhead detection isn't practical; around robot bases.

---

4. Area Scanners (Safety Laser Scanners)

Description: Laser scanner detects objects/personnel entering defined zones. Multiple warning and stop zones configurable.

Characteristics: - 270° field of view typical - Multiple programmable zones - Warning zone (slow robot) and stop zone (stop robot) - Type 3 safety rating - Detects at floor level

Application: Open cells requiring flexibility; mobile robot protection; speed/separation monitoring.

---

5. Interlocked Gates/Doors

Description: Access gates with safety switches that stop robot when opened.

Characteristics: - Mechanical or magnetic interlock switches - Trapped key systems for lockout - May include guard locking (prevent opening while robot moving) - Escape release from inside - Meets ISO 14119

Application: Authorized entry points for maintenance, setup, troubleshooting.

Additional Devices:

- **E-Stop (Emergency Stop)**: Palm-operated buttons to immediately stop robot
- **Two-Hand Controls**: Require both hands on buttons to initiate motion
- **Enabling Devices**: 3-position switches held by personnel in restricted area
- **Presence-Sensing Devices**: Capacitive, radar, or vision-based detection

Question 7: Explain the difference between MoveJ (Joint Move) and MoveL (Linear Move). When would you use each?

Answer:

MoveJ (Joint Move)

How It Works: - Each joint moves from start to end angle independently - All joints start and stop together (coordinated) - Path of TCP (Tool Center Point) is unpredictable - curved - Robot calculates joint interpolation, not TCP path

Characteristics: - Fastest point-to-point motion - No singularity issues during move - Path may vary depending on starting configuration - Speed specified in % of max or joint velocity (°/s)

When to Use: - Moving between distant points through free space - Approach movements before precise positioning - Home position moves - Any time when path doesn't matter, only endpoints - Escaping from near-singularity positions

---

MoveL (Linear Move)

How It Works: - TCP follows a straight line from start to end - Robot calculates joint positions to maintain linear TCP path - Orientation is interpolated linearly as well - All axes coordinated to produce straight line

Characteristics: - Guaranteed straight path - Consistent, predictable motion - Slower than MoveJ for same endpoint distance - Can encounter singularities during move - Speed specified in mm/s at TCP

When to Use: - Process paths (welding, sealing, cutting) - Insertion/extraction operations - Approach to pick/place positions - Any time path must be straight - When moving close to obstacles

---

Practical Example:

Imagine picking a part from a fixture:

```
! Move from home to near pick position - path doesn't matter
MoveJ pApproach, v1000, z50, tool1    ! Joint move - fast

! Move down to pick - must be straight to avoid collision
MoveL pPick, v100, fine, tool1        ! Linear move - controlled

! Pick part, then retract straight up
GripperClose()
MoveL pApproach, v200, z10, tool1     ! Linear move - straight up

! Move to place area - through open space
MoveJ pPlaceApproach, v1000, z50, tool1  ! Joint move - fast
```

Key Decision: - If path matters → MoveL - If only destination matters → MoveJ

Question 8: A manufacturing engineer needs to select a robot for a palletizing application. The requirements are: 25 kg payload, 2.2 m horizontal reach, 10 cycles per minute, floor-mounted. Recommend a robot type and justify your selection.

Answer:

Application Analysis:

Requirement	Value	Implication
Payload	25 kg	Medium-heavy; rules out delta and most SCARA
Reach	2.2 m	Large work envelope needed
Cycle rate	10/min	6 seconds/cycle; moderate speed
Mounting	Floor	Typical industrial installation
Motion type	Palletizing	Mainly vertical and horizontal; some rotation

Robot Type Evaluation:

1. 4-Axis Palletizing Robot (RECOMMENDED) - Specifically designed for palletizing - 4 axes: base rotation, arm, elbow, wrist - High payload capability (25-700+ kg available) - Long reach available (up to 3.2m) - Fast for palletizing motions - Lower cost than 6-axis - Examples: FANUC M-410iC, KUKA KR QUANTEC PA, ABB IRB 460

Verdict: Excellent match for this application

2. 6-Axis Articulated Robot - Maximum flexibility - Can handle complex pallet patterns - Higher cost than 4-axis for same payload - May be over-specified for simple palletizing - Examples: FANUC M-710iC/50, ABB IRB 6700

Verdict: Capable but potentially over-specified and more expensive

3. Cartesian/Gantry Robot - Can cover very large areas - Simple motion control - Higher installation complexity - Less common for palletizing applications

Verdict: Possible but not ideal for this payload/cycle rate

Recommended Selection:

FANUC M-410iC/185 or similar 4-axis palletizer - Payload: 185 kg (well above 25 kg requirement with margin for gripper) - Reach: 3.143 m (exceeds 2.2 m requirement) - Repeatability: ±0.5 mm - Axes: 4

Justification: 1. **Purpose-built**: Designed specifically for palletizing 2. **Adequate payload**: 185 kg allows for heavy gripper + multiple parts 3. **Reach**: 3.1 m provides margin for future needs 4. **Speed**: Optimized kinematics for palletizing motion profile 5. **Reliability**: Proven in thousands of palletizing applications 6. **Cost**: Lower than equivalent 6-axis robot 7. **Simplicity**: 4 axes easier to program and maintain than 6

Additional Recommendations: - Vacuum or mechanical gripper depending on product - Vision system for product detection if variety exists - Safety scanner for operator access during pallet changes - Interface to conveyor for product delivery

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3.14 References

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Chapter 4

Ergonomics in Automated Manufacturing

4.1 Learning Objectives

By the end of this chapter, you will be able to:

- Define ergonomics and explain its importance in manufacturing
- Identify the seven major ergonomic risk factors
- Recognize common musculoskeletal disorders (MSDs) and their causes
- Apply ergonomic design principles to workstation design
- Conduct a REBA (Rapid Entire Body Assessment) analysis
- Understand the role of AI and automation in ergonomic assessment
- Design workstations that optimize human performance and safety

> "Ergonomics is the science of fitting the job to the worker, not forcing the worker to fit the job." — International Ergonomics Association

---

4.2 What is Ergonomics?

Ergonomics (from Greek *ergon* = work, *nomos* = laws) is the scientific discipline focused on understanding the interactions between humans and other elements of a system, with the goal of optimizing human well-being and overall system performance.

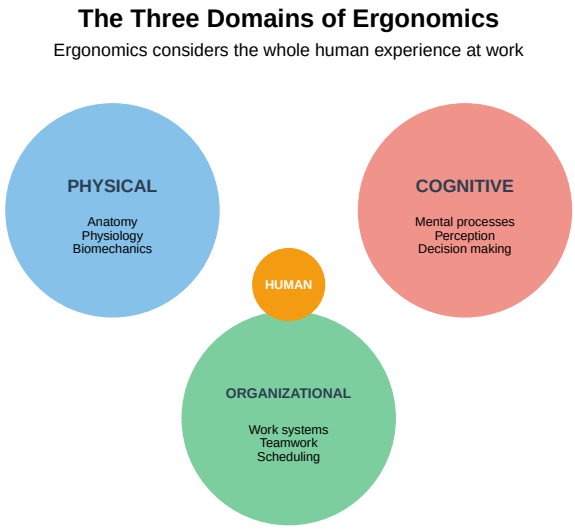


Figure 4.1: The Three Domains of Ergonomics

Table 4.1: The Three

Domain	Focus	Key Topics
Physical Ergonomics	Human body's response to physical stress	Posture, material han
Cognitive Ergonomics	Mental processes and human-system interaction	Mental workload, dec
Organizational Ergonomics	Optimization of sociotechnical systems	Communication, tean

4.2.1 Why Ergonomics Matters in Manufacturing

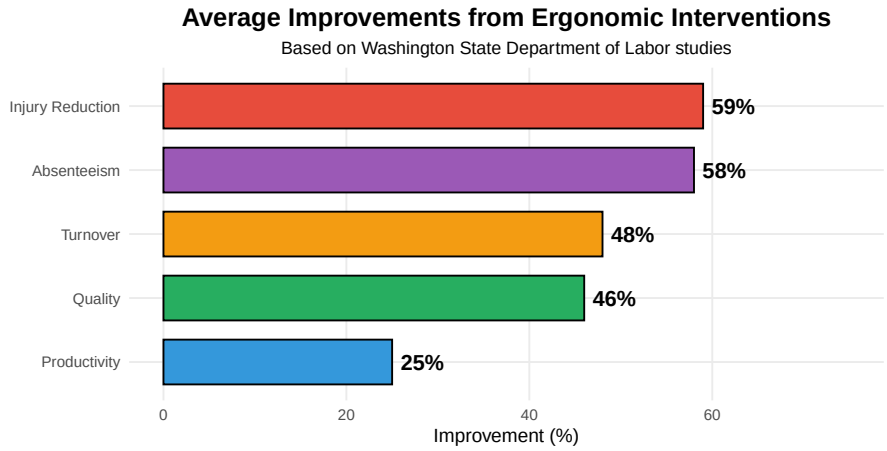


Figure 4.2: The Business Case for Ergonomics

The Cost of Ignoring Ergonomics

Direct Costs: - Workers' compensation claims - Medical expenses - Legal fees - Increased insurance premiums

Indirect Costs (often 4-10x direct costs): - Lost productivity - Training replacement workers - Reduced quality - Overtime to cover absent workers - Administrative time - Low morale

Statistics: - MSDs account for **33%** of all workplace injuries - Average MSD claim costs **$15,000-$20,000** - Lost workdays due to MSDs average **12 days** per case - Carpal tunnel surgery costs approximately **$30,000** per case (including lost time)

4.3 The Seven Ergonomic Risk Factors

The **Initial Ergonomic Risk Assessment (INERA)** identifies seven primary risk factors that contribute to musculoskeletal disorders. Understanding these factors is the first step in prevention.

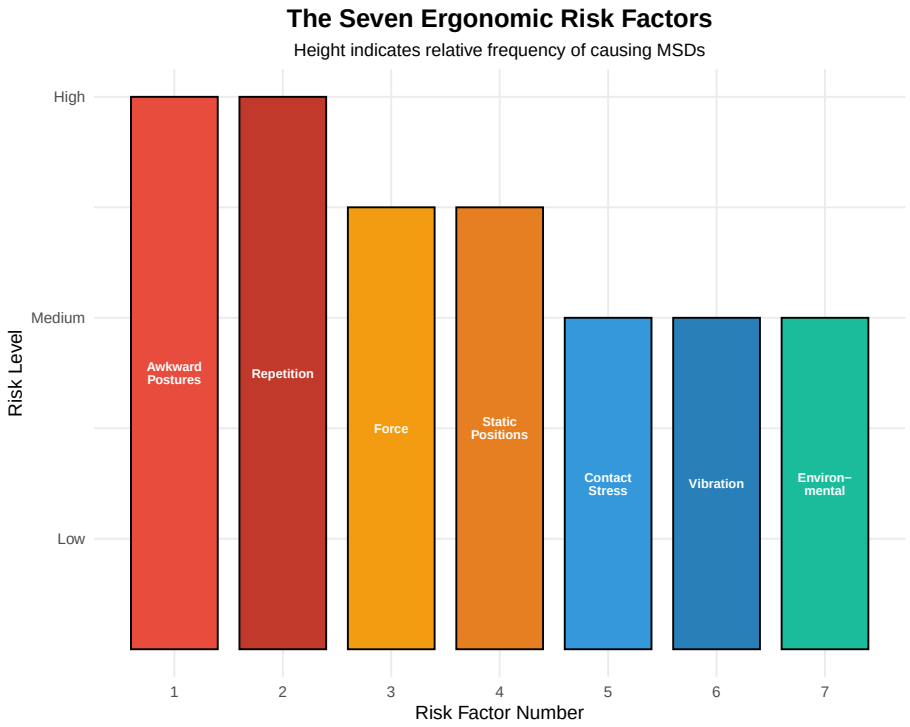


Figure 4.3: The Seven Ergonomic Risk Factors

Table 4.2: The Seven Ergonc

#	Risk Factor	Definition
1	**Awkward Postures**	Positions that deviate from neutral body alignment
2	**Repetition**	Performing the same motion repeatedly
3	**Force**	Amount of physical effort required to do a task
4	**Static Positions**	Maintaining the same position for extended periods
5	**Contact Stress**	Pressure from hard surfaces or edges on body tissues
6	**Vibration**	Oscillating movements transferred to the body
7	**Environmental Factors**	Temperature, lighting, noise affecting work performance

4.3.1 Risk Factor 1: Awkward Postures

Neutral posture is the position where joints are naturally aligned, minimizing stress on muscles, tendons, and skeletal system. Any deviation from neutral increases injury risk.

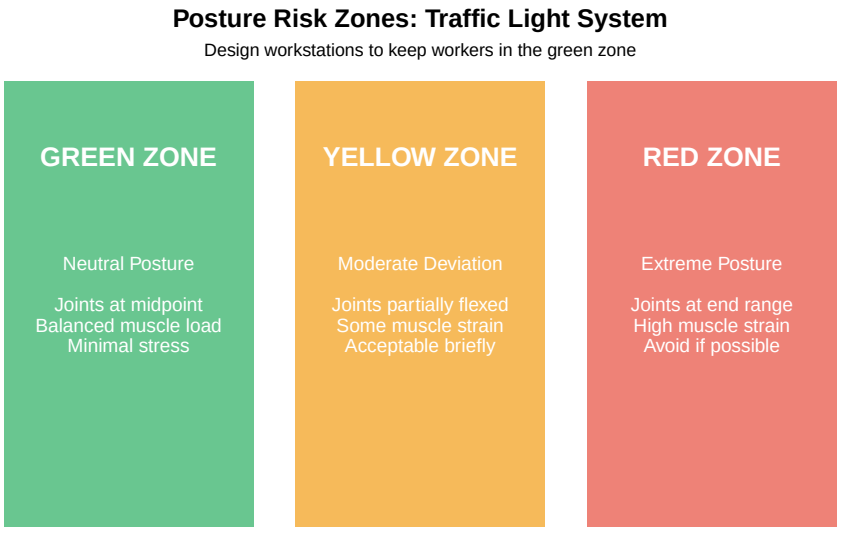


Figure 4.4: Posture Risk Zones

Table 4.3: Neutral vs. Awkward Postures by Body Part

Body Part	Neutral Position	Awkward Position	R
Head/Neck	Balanced on spine, looking straight ahead	Bent forward >20°, tilted, twisted	N
Shoulders	Relaxed, arms at sides	Raised, reaching overhead, behind body	Sl
Elbows	Close to body, bent 90-120°	Fully extended or tightly bent	T
Wrists	Straight, in line with forearm	Bent up/down, twisted side-to-side	C
Back	Natural S-curve maintained	Bent forward, twisted, arched	L
Hips/Knees	Thighs parallel to floor, feet flat	Squatting, kneeling, twisted	K

4.3.2 Risk Factor 2: Repetition

Repetitive motion is one of the most significant risk factors in manufacturing. The risk increases with:

- **Frequency:** How often the motion occurs
- **Duration:** How long the task is performed
- **Recovery time:** Rest between repetitions

Table 4.4: Repetition Risk Thresholds

Cycle Time	Repetitions/Hour	Risk Level	Action Required
> 30 seconds	< 120	Low	Monitor

15-30 seconds	120-240	Moderate	Investigate and consider changes
< 15 seconds	> 240	High	Immediate intervention needed

4.3.3 Risk Factor 3: Force

The amount of **physical effort** required significantly impacts injury risk. Force requirements depend on:

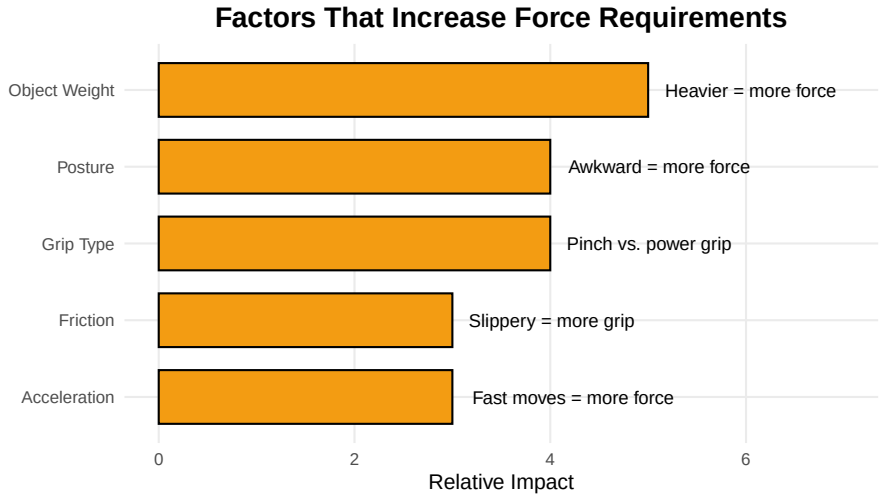


Figure 4.5: Factors Affecting Force Requirements

Interactive Exercise: Identify the Risk Factors

Scenario: An assembly line worker performs the following task 400 times per shift:

1. Reaches overhead to grab a part from a bin
2. Holds the part while inserting 4 screws with a power screwdriver
3. Places the completed assembly on a conveyor
4. The workstation has poor lighting and the floor is concrete

Identify all risk factors present:

Answer: 1. **Awkward Posture** - Reaching overhead 2. **Repetition** - 400 cycles per shift 3. **Force** - Gripping parts, controlling screwdriver 4. **Static Position** - Holding part during assembly 5. **Vibration** - Power screwdriver 6. **Contact Stress** - Standing on concrete (feet) 7. **Environmental** - Poor lighting

This task has ALL SEVEN risk factors present!

4.3.4 Try it yourself: name the risk factor

Seven observations from a plant walk-through. Name the main ergonomic risk factor in each.

Which ergonomic risk factor?

Name the main risk factor in each observation.

Categories: Awkward postures / Repetition / Force / Static positions / Contact stress / Vibration / Environmental factors

1. An assembler works with both arms above shoulder height to fit a headliner. _____
2. A packer makes the same wrist motion 12 times a minute for the whole shift. _____
3. An operator pushes a 300 kg cart over a floor joint by hand. _____
4. An inspector stands in one spot, looking down at a microscope, for two hours. _____
5. A worker rests a forearm on the sharp edge of a bench while using a hand tool. _____
6. A fettler grinds castings with a hand-held grinder for most of the shift. _____
7. Order pickers work in a freezer at -20 degrees C. _____

Answers: 1 Awkward postures; 2 Repetition; 3 Force; 4 Static positions; 5 Contact stress; 6 Vibration; 7 Environmental factors.

4.4 Musculoskeletal Disorders (MSDs)

Musculoskeletal Disorders (MSDs) are injuries and disorders affecting muscles, nerves, tendons, ligaments, joints, cartilage, and spinal discs. They are the most common and costly occupational health problem.

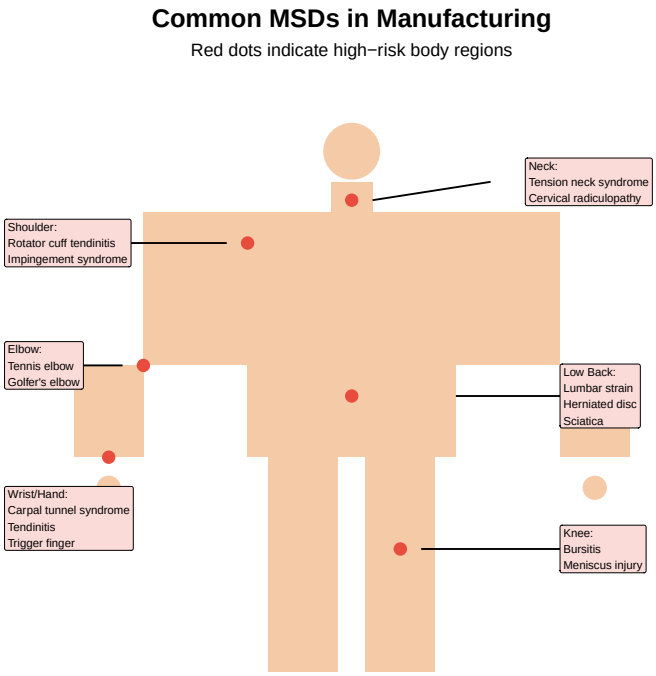


Figure 4.6: Common MSDs by Body Region

Table 4.5: Common Musculoskeletal

Disorder	Description	Causes
****Carpal Tunnel Syndrome****	Compression of median nerve in the wrist	Repetitive wris
****Tendinitis****	Inflammation of tendons from overuse	Repetitive mot
****Low Back Pain****	Strain or injury to muscles/discs of lower back	Heavy lifting, b
****Rotator Cuff Injury****	Damage to shoulder tendons	Overhead work
****Epicondylitis****	Inflammation of elbow tendons (tennis/golfer's elbow)	Repetitive arm
****Trigger Finger****	Finger gets stuck in bent position	Repetitive grip

4.4.1 The MSD Development Cycle

MSDs typically develop gradually through a predictable cycle:

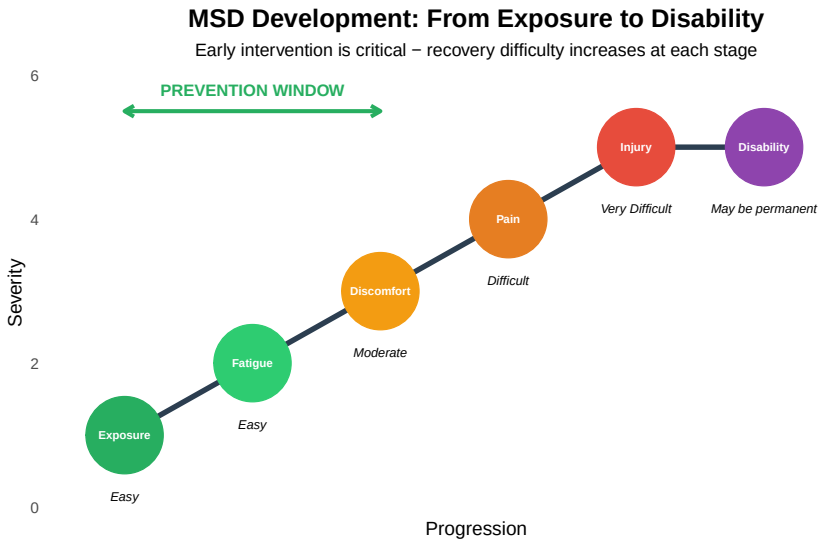


Figure 4.7: The MSD Development Cycle

Warning Signs: When to Take Action

Early Warning Signs (Act Immediately): - Persistent fatigue in specific body parts - Discomfort during or after work - Tightness or stiffness - Minor aching or soreness

Serious Warning Signs (Seek Medical Attention): - Persistent pain that doesn't go away with rest - Numbness or tingling - Loss of strength or grip - Swelling or inflammation - Reduced range of motion - Pain that wakes you at night

> **Key Message:** Report symptoms early! The earlier the intervention, the better the outcome.

4.5 Preventing MSDs: Best Practices

Prevention is far more effective and less costly than treatment. Here are evidence-based strategies for MSD prevention:

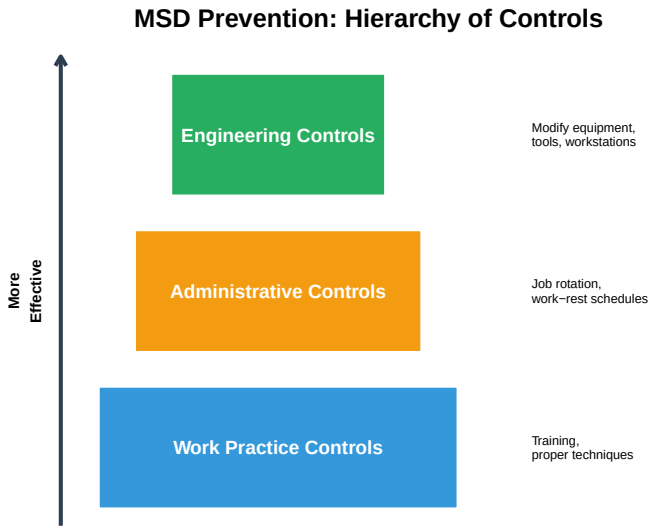


Figure 4.8: MSD Prevention Hierarchy

Table 4.6: MSD Prevention Strategies

Strategy	Engineering Solutions	Worker Action
Maintain Neutral Posture	Adjustable workstations, proper tool selection	Awareness of bod
Reduce Repetition	Automation, job rotation between stations	Vary tasks when
Minimize Force	Mechanical assists, better handles, lighter parts	Use tools proper
Avoid Static Positions	Sit-stand workstations, adjustable fixtures	Shift weight, cha
Eliminate Contact Stress	Padding, rounded edges, anti-fatigue mats	Use padding, wea
Control Vibration	Vibration-dampening tools, isolation mounts	Limit exposure t
Optimize Environment	Proper lighting, climate control, noise reduction	Report issues, us
Allow Recovery	Rest break areas, micro-break reminders	Take breaks, rep

4.5.1 Proper Lifting Technique

Improper lifting is a leading cause of back injuries. The NIOSH lifting equation provides guidelines, but proper technique is fundamental.

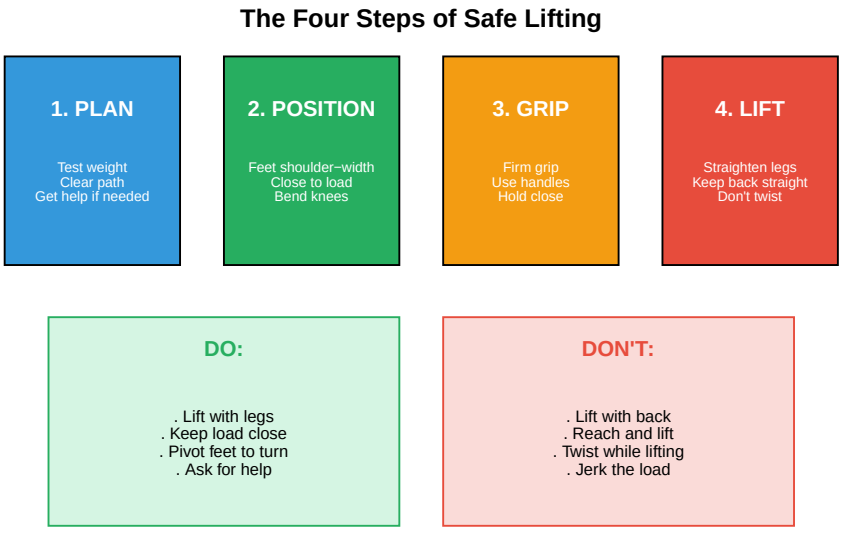


Figure 4.9: Proper Lifting Technique

4.6 Ergonomic Design Principles

Effective ergonomic design follows established principles that account for human capabilities and limitations.

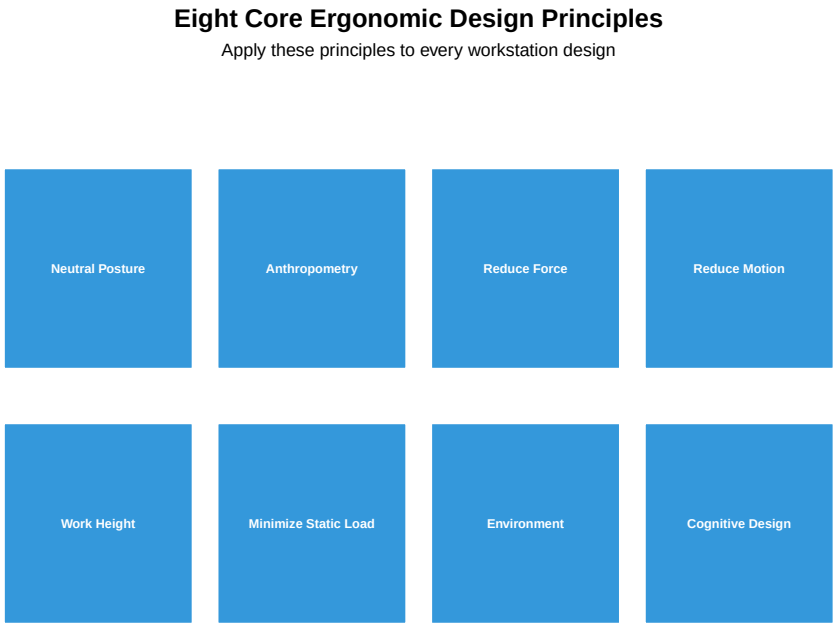


Figure 4.10: Ergonomic Design Principles

Table 4.7: Ergonomic Design Principles in Det

Principle	Description	Implem
1. Neutral Posture	Design to promote natural, balanced body positions	Adjustab
2. Accommodate Anthropometry	Accommodate different body sizes (5th-95th percentile)	Adjustab
3. Reduce Excessive Force	Minimize force requirements for all tasks	Mechanic
4. Reduce Excessive Motion	Reduce extreme motions, frequencies, and durations	Bring wo
5. Optimize Work Height	Match work surface height to task requirements	Precision
6. Minimize Static Load	Avoid holding same position for extended periods	Support
7. Environmental Conditions	Optimize lighting, temperature, noise, air quality	Task ligh
8. Cognitive Support	Design interfaces for easy understanding and use	Clear dis

4.6.1 Anthropometry: Designing for Human Dimensions

Anthropometry is the scientific study of human body measurements. Workstations must accommodate the range of workers who will use them.

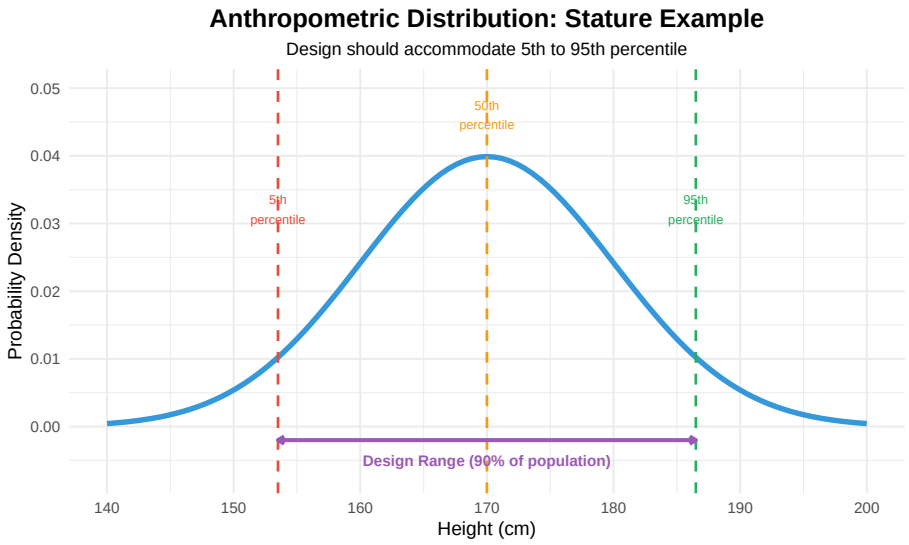


Figure 4.11: Anthropometric Design Considerations

Table 4.8: Key Anthropometric Measurements for Workstation Design

Measurement	5th % Female (cm)	50th % Combined (cm)	95th % Male (
Stature (Standing Height)	150	170	
Eye Height (Standing)	138	158	
Shoulder Height	121	140	
Elbow Height	93	104	
Knuckle Height	64	74	
Sitting Height	79	87	
Seated Eye Height	68	76	
Seated Elbow Height	18	24	
Thigh Clearance	12	15	
Forward Reach	64	73	

4.6.2 Work Surface Height Guidelines

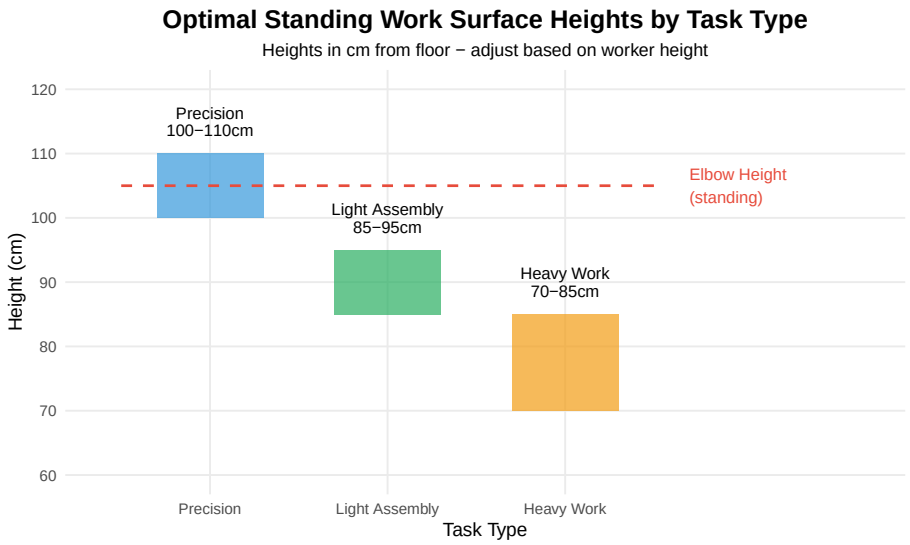


Figure 4.12: Recommended Work Surface Heights

Discussion: Why Different Heights for Different Tasks?

Precision Work (100-110 cm): - Requires close visual inspection - Hands need to be at or above eye level - Examples: Electronics assembly, inspection, watchmaking

Light Assembly (85-95 cm): - General manipulation tasks - Hands at about elbow height - Examples: Packaging, general assembly, sorting

Heavy Work (70-85 cm): - Requires pushing, pressing, or lifting - Lower height allows use of body weight - Examples: Packing boxes, pressing operations, heavy assembly

Key Principle: The work surface height should allow the worker to maintain **neutral posture** while performing the specific task.

4.7 Ergonomic Assessment Tools

Several standardized tools help assess ergonomic risk and guide interventions. Understanding these tools is essential for the process engineer.

Table 4.9: Common Ergonomic Assessment Tools

Tool	Full Name	Focus Area	Best Used For
RULA	Rapid Upper Limb Assessment	Upper body postures	Sedentary/computer work
REBA	Rapid Entire Body Assessment	Whole body postures	Manufacturing, healthcare
NIOSH Lifting Equation	NIOSH Lifting Equation	Manual lifting tasks	Lifting, lowering, carrying
OCRA	Occupational Repetitive Action	Repetitive upper limb tasks	Assembly lines, repetitive
EAWS	Ergonomic Assessment Worksheet	Comprehensive workstation	Complete workstation ana
Snook Tables	Snook Psychophysical Tables	Manual material handling	Push, pull, lift, lower, carr

4.7.1 REBA: Rapid Entire Body Assessment

REBA is one of the most widely used ergonomic assessment tools in manufacturing. It evaluates the whole body posture and provides a risk score.

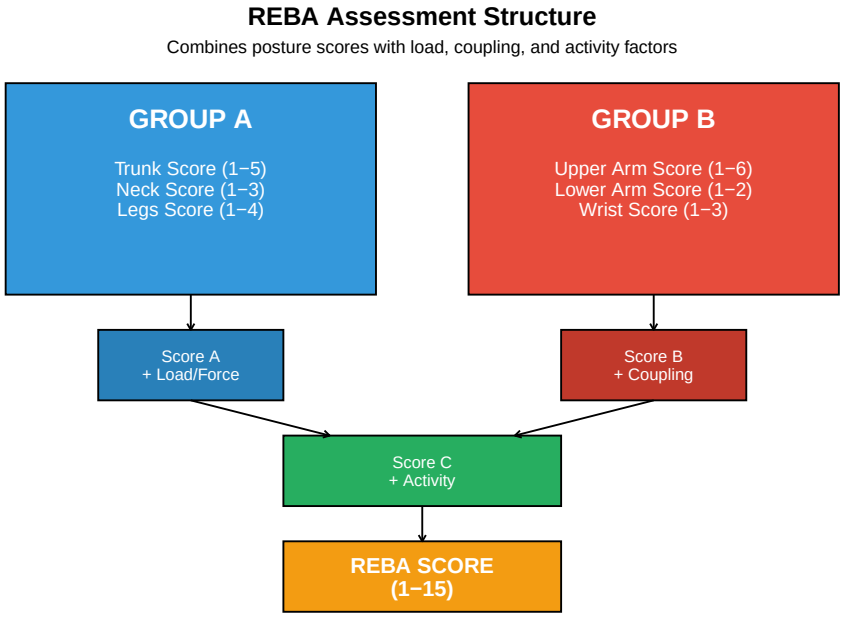


Figure 4.13: REBA Assessment Framework

4.7.2 REBA Scoring Tables

Trunk Posture Scoring:

Table 4.10: REBA Trunk Posture Scores

Position	Score	Adjustment
Upright (0°)	1	—
0-20° flexion/extension	2	+1 if twisted
20-60° flexion, >20° extension	3	+1 if twisted
>60° flexion	4	+1 if twisted or side-bending

Neck Posture Scoring:

Table 4.11: REBA Neck Posture Scores

Position	Score	Adjustment
0-20° flexion	1	+1 if twisted or side-bending
>20° flexion or extension	2	+1 if twisted or side-bending

REBA Action Levels:

Table 4.12: REBA Action Levels

REBA Score	Risk Level	Action
1	Negligible	None necessary
2-3	Low	May be necessary
4-7	Medium	Necessary
8-10	High	Necessary soon
11-15	Very High	Necessary NOW

4.7.3 REBA Employee Assessment Worksheet

The REBA Employee Assessment Worksheet provides a systematic approach to evaluating whole-body posture risk. This worksheet is based on the original work by Hignett and McAtamney (2000) and is widely used in manufacturing ergonomic assessments.

4.7.3.1 Group A: Neck, Trunk, and Leg Analysis

Step 1: Neck Posture Score

Table 4.13: REBA Neck Posture Scoring

Position	Base Score	Adjustment	Max Score
0-20° flexion	1	+1 if twisted or side-bending	2

>20° flexion	2	+1 if twisted or side-bending	3
In extension	2	+1 if twisted or side-bending	3

Step 2: Trunk Posture Score

Table 4.14: REBA Trunk Posture Scoring

Position	Base Score	Adjustment	Max Score
Upright (0°)	1	—	1
0-20° flexion or extension	2	+1 if twisted or side-bending	3
20-60° flexion or >20° extension	3	+1 if twisted or side-bending	4
>60° flexion	4	+1 if twisted or side-bending	5

Step 3: Legs Posture Score

Table 4.15: REBA Legs Posture Scoring

Position	Base Score	Knee Adjustment
Bilateral weight bearing, walking, or sitting	1	+1 if knees 30-60° flexion; +2 if kne
Unilateral weight bearing, featherweight, or unstable posture	2	+1 if knees 30-60° flexion; +2 if kne

Step 4: Look Up Table A Score

Table 4.16: REBA Table A: Posture Score A (Trunk × Neck × Legs)

Trunk Score	Neck Score	Legs = 1	Legs = 2	Legs = 3	Legs = 4
1	**1**	1	1	3	4
1	**2**	2	2	3	4
1	**3**	2	3	4	5
2	**1**	2	3	4	5
2	**2**	3	4	5	6
2	**3**	4	5	6	7
3	**1**	3	4	5	6
3	**2**	4	5	6	7
3	**3**	5	6	7	8
4	**1**	4	5	6	7
4	**2**	5	6	7	8
4	**3**	6	7	8	9
5	**1**	6	7	8	9
5	**2**	7	8	9	9

5	**3**	8	9	9	9

Step 5: Add Load/Force Score

Table 4.17: REBA Load/Force Scoring

Load/Force	Score	Shock Adjustment
< 5 kg (< 11 lbs)	0	—
5-10 kg (11-22 lbs)	1	+1 if shock or rapid build-up of force
> 10 kg (> 22 lbs)	2	+1 if shock or rapid build-up of force

$$\text{Score A} = \text{Table A Score} + \text{Load/Force Score}$$

4.7.3.2 Group B: Arm and Wrist Analysis

Step 6: Upper Arm Posture Score

Table 4.18: REBA Upper Arm Posture Scoring

Position	Base Score	Adjustments
20° extension to 20° flexion	1	+1 if shoulder raised; +1 if arm abducted; -1 if lea
>20° extension	2	+1 if shoulder raised; +1 if arm abducted; -1 if lea
20-45° flexion	2	+1 if shoulder raised; +1 if arm abducted; -1 if lea
45-90° flexion	3	+1 if shoulder raised; +1 if arm abducted; -1 if lea
>90° flexion	4	+1 if shoulder raised; +1 if arm abducted; -1 if lea

Step 7: Lower Arm Posture Score

Table 4.19: REBA Lower Arm Posture Scoring

Position	Score
60-100° flexion (optimal range)	1
< 60° or > 100° flexion	2

Step 8: Wrist Posture Score

Table 4.20: REBA Wrist Posture Scoring

Position	Base Score	Adjustment	Max Score

0-15° flexion or extension (neutral)	1	+1 if wrist is twisted or deviated	2
>15° flexion or extension	2	+1 if wrist is twisted or deviated	3

Step 9: Look Up Table B Score

Table 4.21: REBA Table B: Posture Score B (Upper Arm × Lower Arm × Wrist)

Upper Arm Score	Lower Arm Score	Wrist = 1	Wrist = 2	Wrist = 3
1	**1**	1	1	2
1	**2**	2	2	3
2	**1**	1	2	3
2	**2**	2	3	4
3	**1**	3	4	5
3	**2**	4	5	5
4	**1**	4	5	6
4	**2**	5	6	7
5	**1**	6	7	8
5	**2**	7	8	8
6	**1**	7	8	9
6	**2**	8	9	9

Step 10: Add Coupling Score

Table 4.22: REBA Coupling Score

Coupling Quality	Description	S
Good	Well-fitting handle and mid-range power grip	
Fair	Hand hold acceptable but not ideal; coupling acceptable via other body part	
Poor	Hand hold not acceptable although possible; no handles	
Unacceptable	Awkward, unsafe grip; no handles; coupling unacceptable using other body parts	

$$\text{Score B} = \text{Table B Score} + \text{Coupling Score}$$

4.7.3.3 Calculating the Final REBA Score

Step 11: Look Up Table C Score

Table 4.23: REBA Table C: Combined Score (Score A × Score B)

Score A	B=1	B=2	B=3	B=4	B=5	B=6	B=7	B=8	B=9	B=10	B=11
1	1	1	1	2	3	3	4	5	6	7	7
2	1	2	2	3	4	4	5	6	6	7	7
3	1	2	3	3	4	5	6	7	7	8	8
4	2	3	3	4	5	6	7	8	8	9	9
5	3	4	4	5	6	7	8	8	9	9	9
6	3	4	5	6	7	8	8	9	9	10	10
7	4	5	6	7	8	8	9	9	10	10	11
8	5	6	7	8	8	9	9	10	10	11	11
9	6	6	7	8	9	9	10	10	10	11	11
10	7	7	8	9	9	10	10	11	11	11	12
11	7	7	8	9	9	10	11	11	11	12	12
12	7	8	8	9	9	10	11	11	12	12	12

Step 12: Add Activity Score

Table 4.24: REBA Activity Score (Add +1 for each condition present)

Activity Condition	Score
One or more body parts are static (held > 1 minute)	+1
Repeated small range actions (> 4× per minute, excluding walking)	+1
Action causes rapid large range changes in postures or unstable base	+1

$$\boxed{\text{REBA Score} = \text{Table C Score} + \text{Activity Score}}$$

4.7.3.4 REBA Score Interpretation

Table 4.25: REBA Action Level Guidelines

REBA Score	Risk Level	Action Required	Priority
1	Negligible	None necessary	Continue monitoring
2-3	Low	May be necessary	Investigate further when convenient
4-7	Medium	Necessary	Implement changes to reduce risk
8-10	High	Necessary soon	Investigate and implement changes soon
11-15	**Very High**	**Necessary NOW**	**Stop task and make immediate changes**

4.7.4 REBA Example Calculation: Step-by-Step Walk-through

Let's work through a complete REBA assessment for a manufacturing worker performing a packaging task.

Scenario: A worker stands at a packaging station, bending forward to pick up items from a low bin and placing them into boxes on a conveyor. The task is performed continuously for 4-hour shifts.

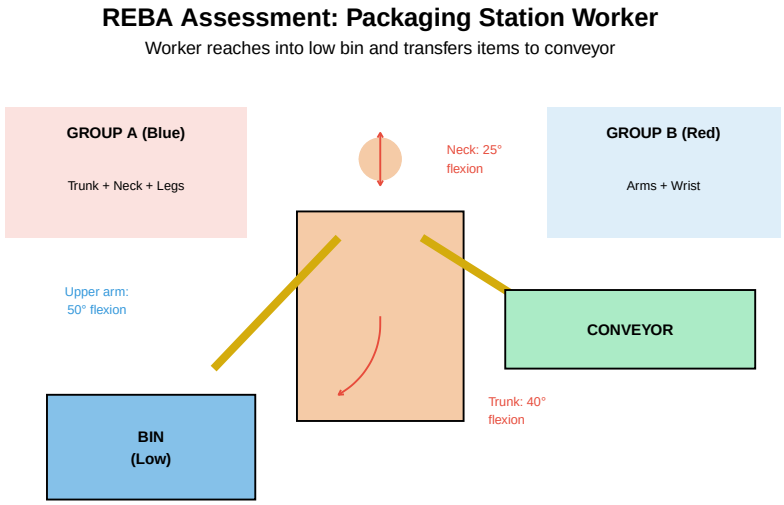


Figure 4.14: REBA Assessment Scenario: Packaging Worker

4.7.4.1 Step-by-Step Assessment

Your turn — work through the card and find the redesign that moves this worker out of the high-risk band.

Show the live R code behind this card

```
#| label: reba-step-by-step
# ============================================================
# REBA Assessment: Packaging Station Worker
# ============================================================
# Worker bends forward to pick items from low bin and place on conveyor
# Items weigh approximately 7 kg each
# Task performed continuously with repetitive motions

# ---------- GROUP A: TRUNK, NECK, LEGS ----------
```

```
# Step 1: Neck Score
# Worker looks down into bin - approximately 25° flexion
neck_base <- 2       # >20° flexion
neck_twist <- 0      # No twist or side-bending
neck_score <- neck_base + neck_twist

# Step 2: Trunk Score
# Worker bends forward approximately 40° to reach bin
trunk_base <- 3      # 20-60° flexion
trunk_twist <- 1     # Slight twist when reaching
trunk_score <- trunk_base + trunk_twist

# Step 3: Legs Score
# Worker stands with weight distributed evenly
legs_base <- 1       # Bilateral weight bearing
legs_knee <- 0       # Knees not flexed significantly
legs_score <- legs_base + legs_knee

# Step 4: Table A Lookup
# Trunk = 4, Neck = 2, Legs = 1
# From Table A: Score = 5
table_a_score <- 5

# Step 5: Load/Force Score
# Items weigh approximately 7 kg
load_force <- 1      # 5-10 kg
shock_force <- 0     # No shock/rapid buildup
load_score <- load_force + shock_force

# SCORE A CALCULATION
score_a <- table_a_score + load_score

# ---------- GROUP B: UPPER ARM, LOWER ARM, WRIST ----------

# Step 6: Upper Arm Score
# Arms reaching forward/down approximately 50° flexion
upper_arm_base <- 3  # 45-90° flexion
shoulder_raised <- 0 # Not raised
arm_abducted <- 0    # Not abducted
arm_support <- 0     # Not supported
upper_arm_score <- upper_arm_base + shoulder_raised + arm_abducted + arm_support

# Step 7: Lower Arm Score
# Elbow bent approximately 70° (in optimal range)
lower_arm_score <- 1  # 60-100° flexion
```

```
# Step 8: Wrist Score
# Wrist in moderate flexion to grasp items
wrist_base <- 2      # >15° flexion
wrist_twist <- 0     # No significant twist
wrist_score <- wrist_base + wrist_twist

# Step 9: Table B Lookup
# Upper Arm = 3, Lower Arm = 1, Wrist = 2
# From Table B: Score = 4
table_b_score <- 4

# Step 10: Coupling Score
# Worker grasps boxes - acceptable grip but not ideal handles
coupling_score <- 1   # Fair coupling

# SCORE B CALCULATION
score_b <- table_b_score + coupling_score

# ---------- FINAL REBA SCORE ----------

# Step 11: Table C Lookup
# Score A = 6, Score B = 5
# From Table C: Score = 7
table_c_score <- 7

# Step 12: Activity Score
# Check all three conditions:
static_posture <- 0   # Body parts not held >1 minute
repeated_actions <- 1 # Repeated small range actions (continuous picking)
rapid_changes <- 0    # No rapid large range changes
activity_score <- static_posture + repeated_actions + rapid_changes

# FINAL REBA SCORE
reba_final <- table_c_score + activity_score

# Interpretation
if (reba_final == 1) {
  risk <- "NEGLIGIBLE"; action <- "None necessary"
} else if (reba_final <= 3) {
  risk <- "LOW"; action <- "May be necessary"
} else if (reba_final <= 7) {
  risk <- "MEDIUM"; action <- "Necessary"
} else if (reba_final <= 10) {
  risk <- "HIGH"; action <- "Necessary soon"
} else {
```

```
  risk <- "VERY HIGH"; action <- "Necessary NOW - stop task immediately"
}

headline <- sprintf("Final REBA score: %g - %s risk", reba_final, risk)
if (reba_final > 7) headline <- paste0("!!", headline, "!!")
cat(headline, "\n")
cat(sprintf("  Action level  %s\n", action))
cat(sprintf("  Neck  %g\n", neck_score))
cat(sprintf("  Trunk  %g\n", trunk_score))
cat(sprintf("  Legs  %g\n", legs_score))
cat(sprintf("  Table A + load/force  %g + %g = Score A %g\n", table_a_score, load_score
cat(sprintf("  Upper arm  %g\n", upper_arm_score))
cat(sprintf("  Lower arm  %g\n", lower_arm_score))
cat(sprintf("  Wrist  %g\n", wrist_score))
cat(sprintf("  Table B + coupling  %g + %g = Score B %g\n", table_b_score, coupling_sc
cat(sprintf("  Table C + activity  %g + %g = %g\n", table_c_score, activity_score, reba
```

4.7.4.2 Recommendations for This Assessment

Based on the REBA score of **8 (HIGH risk)**, the following interventions should be implemented soon:

Table 4.26: Recommended Interventions Based on REBA Assessment

Priority	Intervention	Target Risk Factor	Expected Impact
1	Raise bin height	Trunk flexion	Reduce trunk score from 4 to 2
2	Add tilt mechanism	Neck flexion	Reduce neck score from 2 to 1
3	Reduce load weight	Load/force	Reduce load score from 1 to 0
4	Implement job rotation	Repetition	Reduce activity score

4.7.5 REBA Example Calculation

Teaching demo — drag the posture, load and coupling scores and watch the REBA risk band change.

Show the live R code behind this card

```
#| label: reba-example
# REBA Assessment Example: Assembly Worker

# Group A Scores (Trunk, Neck, Legs)
trunk_score <- 3      # 20-60° forward bend
neck_score <- 2       # >20° flexion
legs_score <- 1       # Bilateral weight bearing
```

```
# Look up Table A score (trunk + neck + legs)
# For this combination: Score A = 4
table_a_score <- 4

# Load/Force Score
load_force <- 1       # 5-10 kg intermittent

# Score A with load
score_a <- table_a_score + load_force

# Group B Scores (Upper arm, Lower arm, Wrist)
upper_arm_score <- 3  # 45-90° flexion
lower_arm_score <- 2  # <60° or >100° flexion
wrist_score <- 2      # >15° flexion/extension

# Look up Table B score
# For this combination: Score B = 5
table_b_score <- 5

# Coupling Score
coupling <- 1         # Acceptable grip

# Score B with coupling
score_b <- table_b_score + coupling

# Look up Table C (Score A × Score B)
# For Score A = 5 and Score B = 6: Score C = 8
score_c <- 8

# Activity Score
activity <- 1         # One or more body parts static >1 min

# Final REBA Score
reba_final <- score_c + activity

# Interpretation
if (reba_final <= 1) {
  risk <- "Negligible - no action necessary"
} else if (reba_final <= 3) {
  risk <- "Low - action may be necessary"
} else if (reba_final <= 7) {
  risk <- "Medium - action necessary"
} else if (reba_final <= 10) {
  risk <- "High - action necessary soon"
} else {
```

```
  risk <- "Very high - action necessary NOW"
}

headline <- sprintf("Final REBA score: %g", reba_final)
if (reba_final > 7) headline <- paste0("!!", headline, "!!")
cat(headline, "\n")
cat(sprintf("  Risk level  %s\n", risk))
cat(sprintf("  Trunk, neck, legs  %g, %g, %g\n", trunk_score, neck_score, legs_score))
cat(sprintf("  Score A (Table A lookup + load)  %g + %g = %g\n", table_a_score, load_fo
cat(sprintf("  Upper arm, lower arm, wrist  %g, %g, %g\n", upper_arm_score, lower_arm_s
cat(sprintf("  Score B (Table B lookup + coupling)  %g + %g = %g\n", table_b_score, cou
cat(sprintf("  Table C + activity  %g + %g = %g\n", score_c, activity, reba_final))

par(mar = c(5, 5, 3, 1))
barplot(reba_final, names.arg = "Final REBA score", ylim = c(0, 15), col = "#2E86C1", y
        main = "Risk bands: 1 negligible, 2-3 low, 4-7 medium, 8-10 high, 11+ very high
abline(h = c(1.5, 3.5, 7.5, 10.5), lty = 2, col = "grey50")
```

Practice Problem: REBA Assessment

Scenario: A worker is performing a task with the following postures:

- **Trunk:** 30° forward flexion, twisted
- **Neck:** 15° flexion
- **Legs:** Weight on one leg
- **Upper Arm:** 60° flexion, shoulder raised
- **Lower Arm:** 80° flexion
- **Wrist:** Neutral
- **Load:** 8 kg carried
- **Coupling:** Fair grip (handles exist but not ideal)
- **Activity:** Repeated small range actions

Calculate the REBA score and determine the action level.

Solution:

```
Trunk: 3 + 1 (twisted) = 4
Neck: 1 (0-20° flexion)
Legs: 2 (weight on one leg)
Table A = 5
Load/Force = 1 (5-10kg)
Score A = 6

Upper Arm: 3 + 1 (raised) = 4
Lower Arm: 1 (60-100°)
Wrist: 1 (neutral)
Table B = 4
Coupling = 1 (fair)
```

```
Score B = 5

Table C (Score A=6, Score B=5) = 8
Activity = +1 (repeated actions)
REBA Score = 9

Risk Level: HIGH - Action necessary soon
```

4.7.6 Try it yourself: score a job with REBA

Now run the method yourself on a different job. An operator loads castings into a machining fixture. An ergonomist has already scored the individual body parts from a photograph:

Group A	Score	Group B	Score
Trunk (bent forward 30°)	3	Upper arm (raised 60°, shoulder lifted)	4
Neck (bent forward more than 20°)	2	Lower arm (reaching, beyond 100°)	2
Legs (weight on one leg)	2	Wrist (bent more than 15°)	2
Load/force (casting weighs 8 kg)	+1	Coupling (fair: no proper handle)	+1

The task is repeated more than four times a minute, which adds an **activity score of +1**. Use Tables A, B and C above to finish the assessment.

REBA: loading castings into a fixture

Work down the list in order. Each line uses the one before it.

- Table A score (trunk 3, neck 2, legs 2): _____ *(to 0 decimal places)*
- Score A (add load/force): _____ *(to 0 decimal places)*
- Table B score (upper arm 4, lower arm 2, wrist 2): _____ *(to 0 decimal places)*
- Score B (add coupling): _____ *(to 0 decimal places)*
- Table C score: _____ *(to 0 decimal places)*
- Final REBA score (add activity): _____ *(to 0 decimal places)*

Show the worked solution

- Table A (trunk 3, neck 2, legs 2) = **5**; Score A = 5 + 1 = **6**
- Table B (upper arm 4, lower arm 2, wrist 2) = **6**; Score B = 6 + 1 = **7**
- Table C (Score A = 6, Score B = 7) = **8**
- REBA score = 8 + 1 = **9**

A score of 9 falls in the 8 to 10 band: **high risk, action necessary soon**. The largest single contributor is the raised, reaching arm. Raising the fixture or bringing the castings closer to the operator would lower the upper-arm and trunk scores together.

4.7.7 NIOSH Lifting Equation

The **NIOSH Lifting Equation** calculates a **Recommended Weight Limit (RWL)** for lifting tasks.

$$RWL = LC \times HM \times VM \times DM \times AM \times FM \times CM$$

Where:

Table 4.27: NIOSH Lifting Equation Factors

Factor	Name	Value	Description
LC	Load Constant	23 kg (51 lb)	Maximum weight under ideal c
HM	Horizontal Multiplier	25/H	H = horizontal distance from r
VM	Vertical Multiplier	1 - 0.003|V-75|	V = vertical height of hands at
DM	Distance Multiplier	0.82 + 4.5/D	D = vertical travel distance (cr
AM	Asymmetric Multiplier	1 - 0.0032A	A = angle of asymmetry (degr
FM	Frequency Multiplier	Table lookup	Based on lift frequency and du
CM	Coupling Multiplier	Table lookup	Based on hand-load coupling q

Lifting Index (LI) indicates risk level:

$$LI = \frac{\text{Actual Weight}}{\text{RWL}}$$

Table 4.28: Lifting Index Interpretation

Lifting Index	Risk	Action
LI 1.0	Low	Task acceptable for most workers
1.0 < LI 2.0	Moderate	Some workers may be at risk - consider changes
2.0 < LI 3.0	High	Many workers at risk - changes recommended
LI > 3.0	Very High	Unacceptable - immediate changes required

Your turn — change the distances, angle, or weight below.

Show the live R code behind this calculator

```
#| label: niosh-example
# NIOSH Lifting Equation Example

# Task parameters
H <- 40    # Horizontal distance (cm)
V <- 30    # Vertical height at start (cm)
D <- 60    # Vertical travel distance (cm)
A <- 30    # Asymmetry angle (degrees)

# Constants
LC <- 23   # Load constant (kg)

# Calculate multipliers
HM <- 25 / H
VM <- 1 - 0.003 * abs(V - 75)
DM <- 0.82 + 4.5 / D
AM <- 1 - 0.0032 * A

# Frequency multiplier (from table, for 2 lifts/min, 2 hours, V<75cm)
FM <- 0.84

# Coupling multiplier (assume "fair" coupling)
CM <- 0.95

# Calculate RWL
RWL <- LC * HM * VM * DM * AM * FM * CM

# Calculate Lifting Index for actual load
actual_weight <- 15  # kg
LI <- actual_weight / RWL

cat(sprintf("Lifting Index: %.2f (RWL = %.1f kg)\n", LI, RWL))
cat(sprintf("  HM (horizontal)   %.3f\n", HM))
cat(sprintf("  VM (vertical)     %.3f\n", VM))
cat(sprintf("  DM (distance)     %.3f\n", DM))
cat(sprintf("  AM (asymmetry)    %.3f\n", AM))
if (LI <= 1.0) {
  cat("  Risk level        Low - task acceptable\n")
} else if (LI <= 2.0) {
  cat("  Risk level        Moderate - consider changes\n")
} else if (LI <= 3.0) {
  cat("  Risk level        !!High - changes recommended!!\n")
```

```
} else {
  cat("  Risk level        !!Very high - immediate changes required!!\n")
}
```

4.8 AI-Powered Ergonomic Assessment

Artificial intelligence is revolutionizing ergonomic assessment by enabling **real-time, automated analysis** of worker postures without disrupting operations.

AI-Powered Ergonomic Assessment Pipeline

1. **INPUT** – Video /; Wearables
2. **AI PROCESSING** – Pose; Estimation
3. **ANALYSIS** – REBA/RULA; Scoring
4. **OUTPUT** – Risk Report; Alerts

Table 4.29: AI-Powered Ergonomic Assessment Tools

Tool	Technology	Key Features
****TuMeke Ergonomics****	Computer vision + wearables	Real-time REBA scoring, multi-camera
****VelocityEHS****	AI video analysis	Automated job analysis, risk prioritizati
****Protex AI****	Real-time video AI	Privacy-preserving analysis, floor-wide r
****SoterTask****	Wearable sensors + AI	Wearable clips, real-time coaching, hapt
****Viso.ai****	Custom AI models	Custom pose detection, integration API
****ErgoPlus Platform****	Mobile app + AI	Mobile assessment, improvement trackin

4.8.1 Benefits of AI Ergonomic Assessment

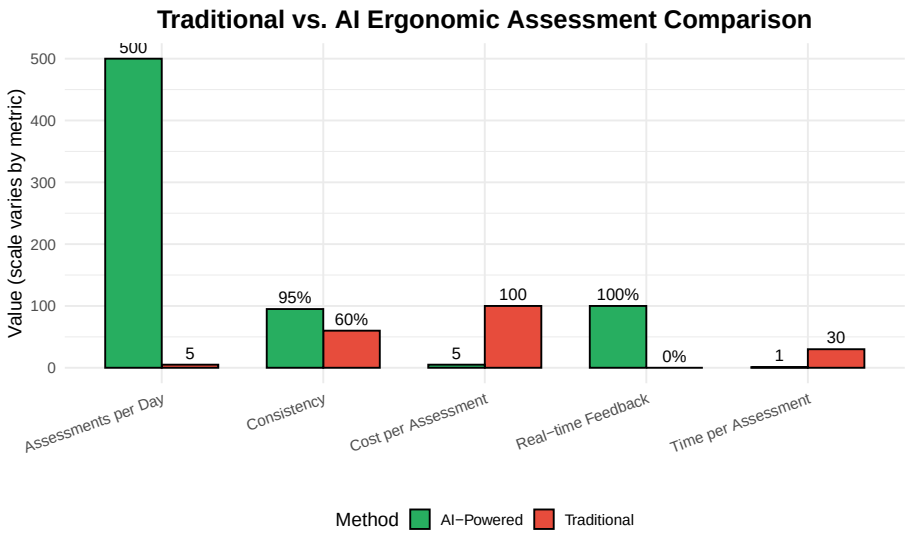


Figure 4.15: Traditional vs. AI Ergonomic Assessment

4.9 Ergonomics in Automated Manufacturing

Industrial automation inherently involves the mechanization of processes traditionally carried out by human operators. In the automation design phase, engineers must strike a balance between mechanization and the necessary human interaction with these systems.

4.9.1 Human-Automation Interaction Points

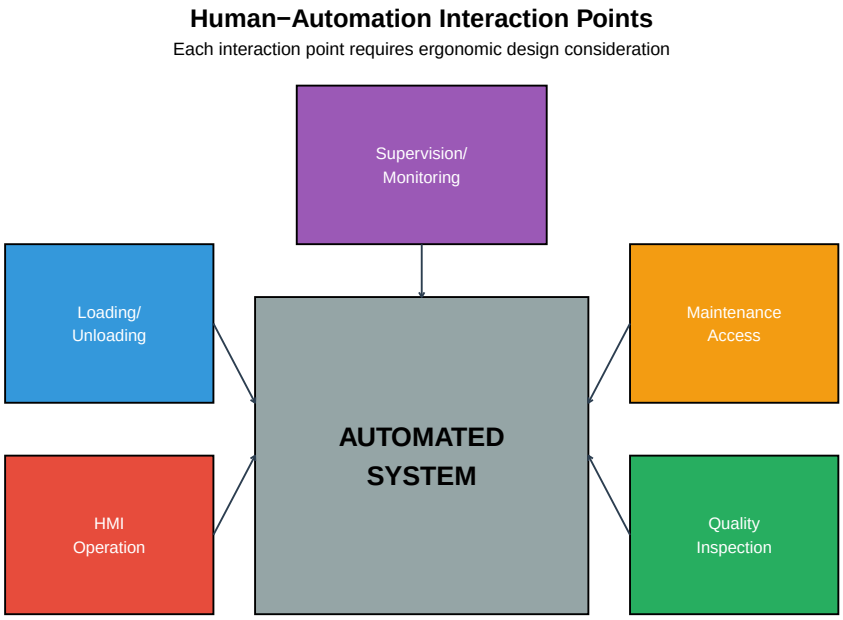


Figure 4.16: Ergonomic Considerations in Automated Systems

Table 4.30: Ergonomic Considerations for Automated System

Interaction Point	Ergonomic Challenges	Design Soluti
Loading/Unloading	Repetitive motion, lifting, reaching, awkward postures	Height-adjusta
HMI Operation	Static posture, visual strain, cognitive load	Adjustable scre
Maintenance	Awkward access, confined spaces, tool use	Adequate acces
Quality Inspection	Static positions, visual strain, fine motor tasks	Adjustable fixt
Supervision	Sedentary position, cognitive fatigue, vigilance	Multiple monit

4.9.2 Collaborative Robot Ergonomics

Cobots present unique ergonomic opportunities and challenges:

Table 4.31: Ergonomic Considerations for Collaborative Robots

Aspect	Opportunity	Challenge
Task Allocation	Robot handles heavy, repetitive, or precision tasks	Determining opt

****Workstation Layout****	Shared workspace allows flexible positioning	Ensuring adequate space for
****Pacing****	Robot adapts to human pace, not vice versa	Avoiding pressure from robot
****Force Assistance****	Robot provides powered assist for lifting	Proper force limits to preven
****Cognitive Load****	Robot handles routine tasks, human focuses on judgment	Maintaining situational awar

---

4.10 Employer Responsibilities

Employers have both legal and ethical obligations to protect workers from ergonomic hazards.

Table 4.32: Employer Ergonomics Program Responsibilities

Responsibility	Actions
****Management Commitment****	Allocate resources, set goals, lead by example
****Worker Involvement****	Include workers in assessments, design teams, solution development
****Training****	Ergonomics awareness, proper techniques, symptom recognition
****Hazard Identification****	Regular assessments, worker surveys, injury analysis
****Early Reporting****	Non-punitive reporting system, prompt response to symptoms
****Hazard Control****	Engineering controls, work organization, PPE as last resort
****Program Evaluation****	Track injuries, measure improvements, adjust programs

OSHA Ergonomics Guidelines

While OSHA does not have a specific ergonomics standard, employers are still responsible under the **General Duty Clause** (Section 5(a)(1)) which requires employers to provide a workplace "free from recognized hazards."

OSHA's Recommended Program Elements: 1. Management leadership 2. Worker participation 3. Hazard identification and assessment 4. Hazard prevention and control 5. Education and training 6. Program evaluation

Industry-Specific Guidelines: - Poultry Processing - Nursing Homes - Retail Grocery Stores - Shipyards

Visit: OSHA Ergonomics eTool

4.11 Case Study: Ergonomics in Automated Assembly

Company: Automotive parts manufacturer

Problem: High rate of upper extremity MSDs in assembly department - 15 recordable injuries per year - Lost workdays averaging 20 days per case - Workers' comp costs exceeding $300,000 annually

Assessment Findings:

Table 4.33: Ergonomic Assessment Findings

Workstation	Primary Issues	REBA Score	Risk Level
Sub-assembly	Overhead reaching, high repetition	9	High
Main line	Forward bending, static standing	8	High
Inspection	Awkward wrist postures, visual strain	7	Medium
Packaging	**Lifting, carrying, repetition**	**10**	**Very High**

Interventions Implemented:

1. **Sub-assembly:**
   - Tilting fixtures to bring work closer
   - Parts bins repositioned to eliminate overhead reaching
   - Cobot added for repetitive insertion tasks
2. **Main line:**
   - Height-adjustable workstations installed
   - Anti-fatigue mats added
   - Job rotation implemented
3. **Inspection:**
   - Magnifying task lights installed
   - Angled fixtures to reduce wrist bending
   - Ergonomic seating provided
4. **Packaging:**
   - Vacuum lift assists installed
   - Conveyor heights adjusted
   - Case erectors automated

Results After 18 Months:

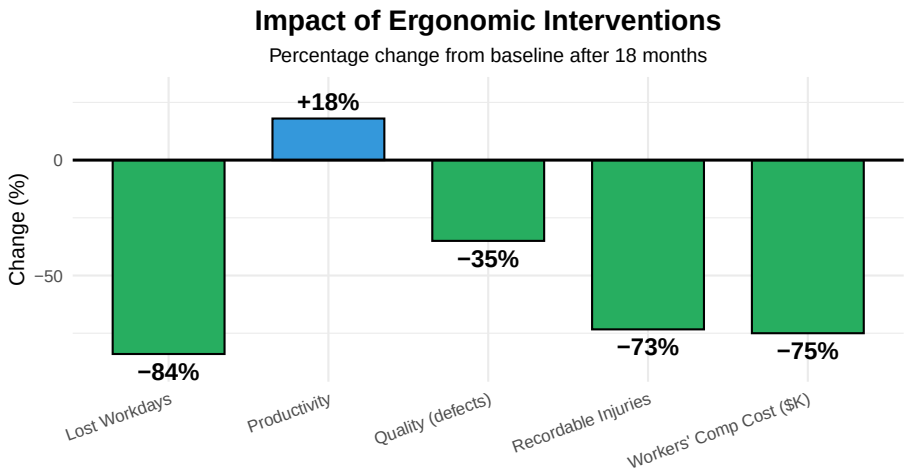


Figure 4.17: Ergonomic Intervention Results

ROI Analysis: try different investment or savings figures below.

Show the live R code behind this calculator

```
#| label: roi-analysis
# Investment
equipment_cost <- 150000  # Lift assists, fixtures, cobots
training_cost <- 15000
total_investment <- equipment_cost + training_cost

# Annual Savings
injury_reduction <- 220000     # (15 - 4 injuries avoided) x $20,000 avg cost per injury
productivity_gain <- 90000     # 18% improvement on $500,000 labor cost
quality_improvement <- 17500   # 35% defect reduction on $50,000 cost

annual_savings <- injury_reduction + productivity_gain + quality_improvement
payback_years <- total_investment / annual_savings

cat(sprintf("5-year ROI: %.0f%%\n", (annual_savings * 5 - total_investment) / total_investment *
cat(sprintf("  Total investment     $%s\n", format(total_investment, big.mark = " ", scientific =
cat(sprintf("  Annual savings       $%s\n", format(annual_savings, big.mark = " ", scientific = F
cat(sprintf("  Payback period       %.1f years\n", payback_years))
```

4.12 Summary

Table 4.34: Ergonomics in Manufacturing: Key Concepts Summary

Topic	Key Points
Ergonomics Definition	Fitting the job to the worker; physical, cognitive, organizational domain
Risk Factors	Posture, repetition, force, static positions, contact stress, vibration, envi
MSDs	Injuries to muscles, nerves, tendons; develop gradually from exposure to
Prevention	Engineering controls most effective; proper lifting, neutral postures, reco
Design Principles	Neutral posture, anthropometry, work height optimization, environment
Assessment Tools	REBA for whole body, RULA for upper limb, NIOSH for lifting tasks
AI in Ergonomics	Real-time monitoring, automated REBA/RULA scoring, continuous ass
Employer Role	Management commitment, worker involvement, training, hazard control

---

4.13 Review Questions

Question 1: What are the seven ergonomic risk factors, and which two are most commonly associated with carpal tunnel syndrome?

The seven ergonomic risk factors are: 1. Awkward Postures 2. Repetition 3. Force 4. Static Positions 5. Contact Stress 6. Vibration 7. Environmental Factors

Carpal tunnel syndrome is most commonly associated with: - **Repetition** - Repeated wrist movements compress the median nerve - **Force** - Forceful gripping increases pressure in the carpal tunnel - Also contributing: awkward wrist postures and vibration

Question 2: A worker performs a task 8 times per minute for a 4-hour shift. The cycle time is 7.5 seconds. Is this considered high repetition? Why?

Analysis: - Cycle time: 7.5 seconds ($<$ 15 seconds threshold) - Repetitions per hour: $8 \times 60 = 480$ ($>$ 240 threshold)

Yes, this is HIGH repetition because: 1. Cycle time is less than 15 seconds 2. Repetitions exceed 240 per hour

Risk Level: High - Immediate intervention needed

Recommended actions: - Job rotation with different tasks - Automation of repetitive elements - Micro-breaks every 20-30 minutes - Workstation redesign to reduce cycle elements

Question 3: Explain why ergonomic design should accommodate the 5th to 95th percentile of the population rather than designing for the “average” person.

Designing for the average excludes most people: - The “average” person in ALL dimensions doesn’t exist - A person may be average height but have

long arms or short legs - Designing for only the 50th percentile fits only about 50% of users

The 5th-95th percentile range: - Accommodates 90% of the population - Addresses both small and large users - Provides adjustability rather than fixed dimensions

Application rules: - **Clearances** (overhead, legroom): Design for 95th percentile (largest) - **Reach distances**: Design for 5th percentile (smallest) - **Work surfaces**: Make adjustable to accommodate range

Example: A fixed workstation at 95cm height: - Too high for 5th percentile female (elbow at ~93cm) - Too low for 95th percentile male (elbow at ~117cm) - Adjustable range of 85-115cm accommodates both

Question 4: Calculate the NIOSH Lifting Index for the following task and determine if intervention is needed.

Task Parameters:

- Horizontal distance (H): 35 cm
- Vertical height (V): 80 cm
- Travel distance (D): 50 cm
- Asymmetry (A): 0° (symmetric lift)
- Frequency: 4 lifts/minute for 1 hour
- Coupling: Good (handles)
- Actual load: 18 kg

Question 4: NIOSH Lifting Index

From the tables: FM = 0.75 (4 lifts/min, 1 hour) and CM = 1.0 (good coupling).

- Recommended Weight Limit (RWL): _____ kg *(to 1 decimal place)*
- Lifting Index (LI): _____ *(to 2 decimal places)*

Show the worked solution

Solution:

```
LC = 23 kg
HM = 25/35 = 0.714
VM = 1 - 0.003|80-75| = 0.985
DM = 0.82 + 4.5/50 = 0.91
AM = 1 - 0.0032(0) = 1.0
FM = 0.75 (4/min, 1 hour, V>75cm, from table)
CM = 1.0 (good coupling)

RWL = 23 × 0.714 × 0.985 × 0.91 × 1.0 × 0.75 × 1.0
RWL = 11.0 kg

LI = 18 / 11.0 = 1.64
```

```
Risk Level: MODERATE (1.0 < LI < 2.0)
Action: Some workers may be at risk - consider changes
```

Recommendations: - Reduce horizontal distance (bring load closer) - Reduce lifting frequency - Use mechanical assist for loads >12 kg

Question 5: What are three advantages of AI-powered ergonomic assessment over traditional methods?

1. Scale and Coverage: - Traditional: 5-10 assessments per day by trained ergonomist - AI: Hundreds or thousands of assessments simultaneously - Can monitor entire facility continuously

2. Consistency and Objectivity: - Traditional: Inter-rater reliability issues; different assessors may score differently - AI: Consistent scoring algorithm applied uniformly - Removes subjective interpretation

3. Real-time Feedback: - Traditional: Assessment completed, report generated days/weeks later - AI: Immediate scoring and alerts - Can provide real-time coaching to workers

Additional advantages: - Lower cost per assessment - Non-disruptive (workers not aware of assessment) - Longitudinal tracking of posture patterns - Data for predictive analytics

---

4.14 Additional Resources

4.14.1 Videos

Ergonomics Fundamentals: - Introduction to Workplace Ergonomics - Safe Manual Handling Techniques

Assessment Tools: - REBA Assessment Tutorial - RULA Assessment Guide - NIOSH Lifting Equation Explained

AI Ergonomics: - TuMeke AI Ergonomics Demo - VelocityEHS Industrial Ergonomics

4.14.2 Online Tools

- ErgoPlus REBA Calculator
- NIOSH Lifting Equation Calculator
- Cornell University Ergonomics Web

4.15 References

- Berlin, C., & Adams, C. (2017). *Production Ergonomics: Designing Work Systems to Support Optimal Human Performance*. Ubiquity Press.
- Hignett, S., & McAtamney, L. (2000). Rapid Entire Body Assessment (REBA). *Applied Ergonomics*, 31(2), 201-205.
- McAtamney, L., & Corlett, E.N. (1993). RULA: A Survey Method for Investigation of Work-Related Upper Limb Disorders. *Applied Ergonomics*, 24(2), 91-99.
- NIOSH. (1994). *Applications Manual for the Revised NIOSH Lifting Equation*. DHHS (NIOSH) Publication No. 94-110.
- OSHA. (2000). *Ergonomics: The Study of Work*. OSHA 3125.
- Waters, T.R., Putz-Anderson, V., & Garg, A. (1993). Revised NIOSH Equation for the Design and Evaluation of Manual Lifting Tasks. *Ergonomics*, 36(7), 749-776.
- ISO 11228-1:2021. Ergonomics — Manual handling — Part 1: Lifting, lowering and carrying.
- ISO 11228-3:2007. Ergonomics — Manual handling — Part 3: Handling of low loads at high frequency.

Chapter 5

Measurement Systems Analysis

5.1 Learning Objectives

After completing this chapter, you will be able to:

1. Explain the importance of measurement system quality in process control
2. Identify the components of measurement system variation
3. Define and calculate accuracy, bias, linearity, stability, repeatability, and reproducibility
4. Conduct and interpret a Gauge R&R study using the Range method
5. Determine measurement system acceptability using %GRR and ndc criteria
6. Recognize common sources of measurement error and check gauge resolution with the Rule of Ten
7. Explain why a measurement system must be verified before control charts and capability studies are trusted

> **Where this chapter stops**
>
> This is the introductory MSA chapter (Part 1). It covers the vocabulary, the accuracy studies and the Range-method Gauge R&R, which is all the lab needs. The full **ANOVA** Gauge R&R, attribute agreement (kappa), Type 1 gauge studies, measurement uncertainty and the effect of measurement error on control charts and capability are in **Chapter 17, Advanced Measurement Systems Analysis** (Part 2).

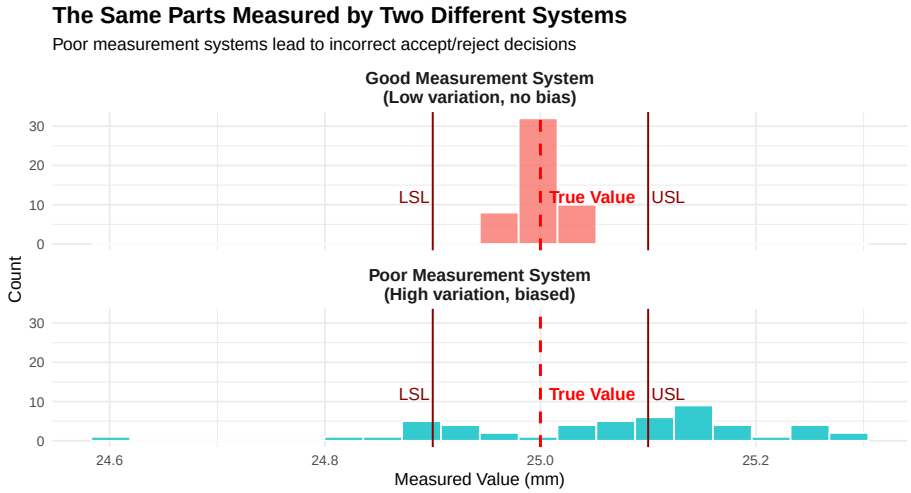
5.2 Introduction to Measurement Systems Analysis

Measurement Systems Analysis (MSA) is the science of evaluating the quality of measurement processes. Before we can trust our data for process control, quality decisions, or capability studies, we must first verify that our measurement system is capable of providing reliable results.

5.2.1 Why MSA Matters

> "You can't improve what you can't measure - but you also can't improve what you measure incorrectly."

Consider this scenario:



5.2.2 The Cost of Poor Measurement

Table 5.1: Business Impact of Poor Measurement Syste

Impact	Description
False Accepts (Type II Error)	Bad parts passed as good; reach customer
False Rejects (Type I Error)	Good parts rejected as bad; scrapped or reworked
Process Adjustment Errors	Adjusting process based on measurement noise, not real changes
Capability Misassessment	Cp/Cpk calculations are wrong; true capability unknown
Customer Complaints	Inconsistent quality due to measurement-based decisions

5.2.3 MSA in Industry Standards

MSA is required by multiple quality standards:

- **IATF 16949** (Automotive): Requires MSA for all measurement systems referenced in control plans
- **AS9100** (Aerospace): Requires demonstrated measurement capability
- **ISO 9001**: Requires monitoring and measurement resources to be suitable
- **FDA 21 CFR Part 820** (Medical Devices): Requires validated measurement equipment

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5.3 Common Sources of Measurement Error

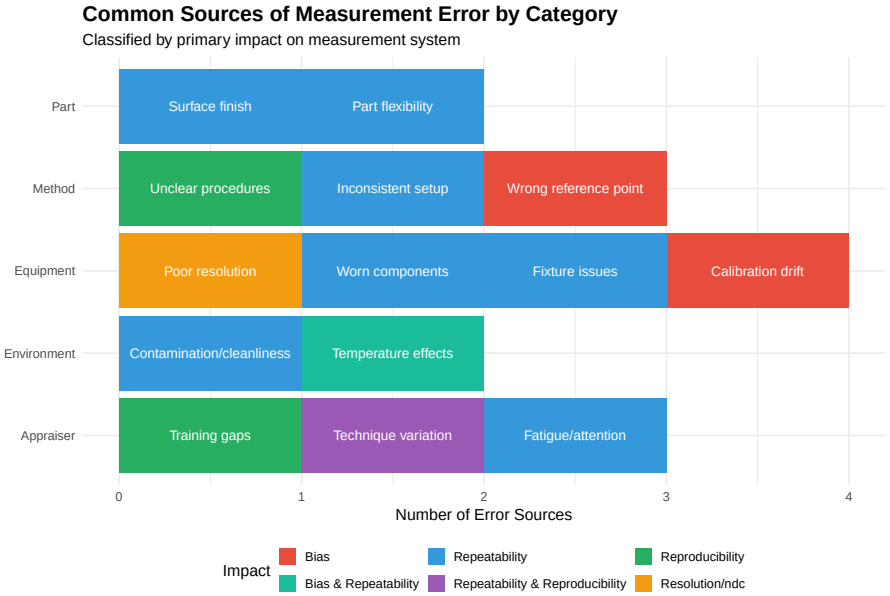


Table 5.2: MSA Issues and Improvement Action

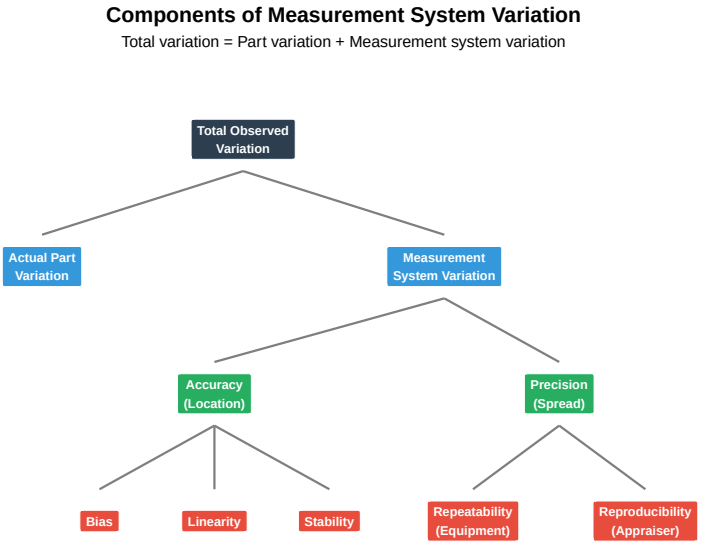
Issue Identified	Likely Causes	Impro
High Repeatability (EV)	Gauge condition, fixture, resolution, part variation during measurement	Maint
High Reproducibility (AV)	Training, technique, procedure clarity	Standa
Significant Bias	Calibration, master accuracy, technique	Recali
Poor Linearity	Gauge mechanism, calibration across range	Repai
Drift/Stability Issues	Environmental effects, gauge wear, master deterioration	Enviro
Low ndc	Resolution, excessive gauge variation relative to part variation	Higher

---

5.4 Components of Measurement Variation

Total measurement variation comes from multiple sources. Understanding these components is essential for improvement.

5.4.1 The Measurement System Model



5.4.2 Accuracy vs. Precision

These are often confused but are fundamentally different concepts:

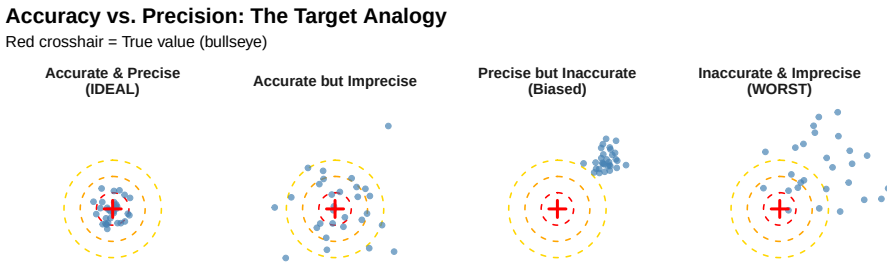


Table 5.3: Accuracy vs. Precision

	Definition	Affected By
Accuracy	How close measurements are to the true value (average)	Calibration, bias, linearity
Precision	How close measurements are to each other (spread)	Repeatability, reproducibility, resolution

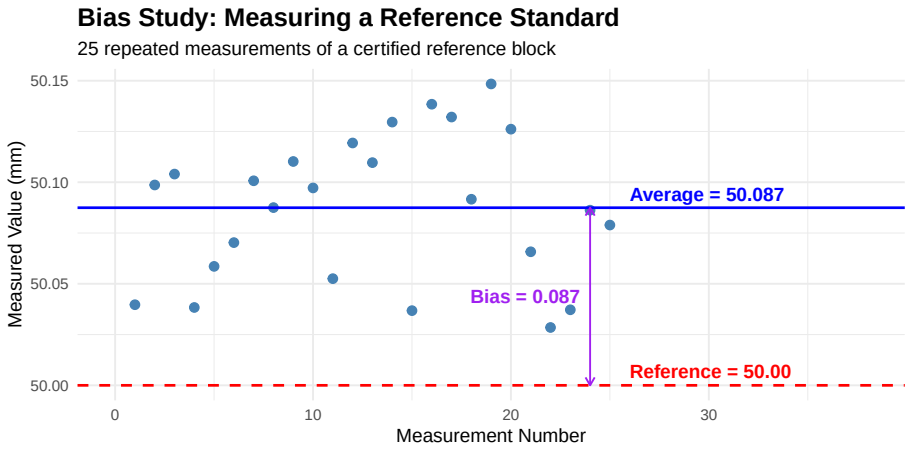
5.5 Accuracy Studies

Accuracy studies evaluate how well the measurement system indicates the true value.

5.5.1 Bias

Bias is the systematic error - the difference between the average of measurements and the true (reference) value.

$$\text{Bias} = \bar{x}_{observed} - x_{reference}$$



Teaching demo — try different reference and measured values.

Show the live R code behind this card

```
#| label: bias-calculation
# Bias study calculation
reference_value <- 50.000  # Certified reference standard
measurements <- c(50.08, 50.07, 50.09, 50.08, 50.06, 50.10, 50.07, 50.08,
                  50.09, 50.08, 50.07, 50.11, 50.08, 50.09, 50.07)
```

```
# Calculate bias
average_measured <- mean(measurements)
bias <- average_measured - reference_value

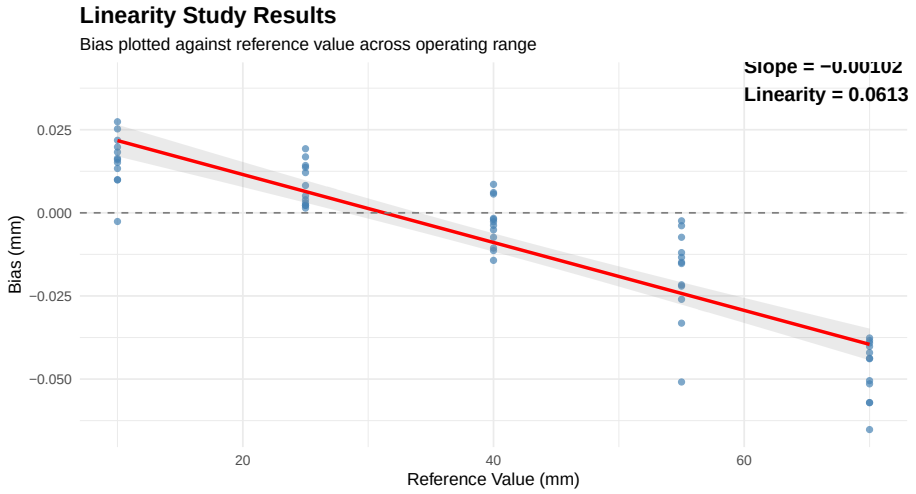
# Statistical test for significance
t_test <- t.test(measurements, mu = reference_value)

cat(sprintf("Bias: %.4f mm (%.1f%% of the ±0.10 tolerance)\n", bias, abs(bias) / 0.10 *
cat(sprintf("  Reference value  %g mm\n", reference_value))
cat(sprintf("  Average measured  %.4f mm\n", average_measured))
cat(sprintf("  t-test p-value  %.4f\n", t_test$p.value))
if(t_test$p.value < 0.05) {
  cat("  Verdict  !!Bias is statistically significant!!\n")
} else {
  cat("  Verdict  Bias is not statistically significant\n")
}

par(mar = c(5, 5, 3, 1))
plot(measurements, pch = 19, ylim = range(measurements, reference_value),
     xlab = "Measurement number", ylab = "Measured value (mm)", main = "Readings vs re
abline(h = reference_value, lwd = 2, col = "#27AE60")
abline(h = average_measured, lwd = 2, lty = 2, col = "#C0392B")
```

5.5.2 Linearity

Linearity measures whether bias changes across the measurement range. A gauge might be accurate at one end of its range but biased at the other.



Teaching demo — try a different set of reference points and biases.

Show the live R code behind this card

```
#| label: linearity-calc
# Linearity calculation
reference <- c(10, 25, 40, 55, 70)
avg_bias <- c(0.020, 0.008, -0.002, -0.018, -0.048)

# Fit linear regression
model <- lm(avg_bias ~ reference)

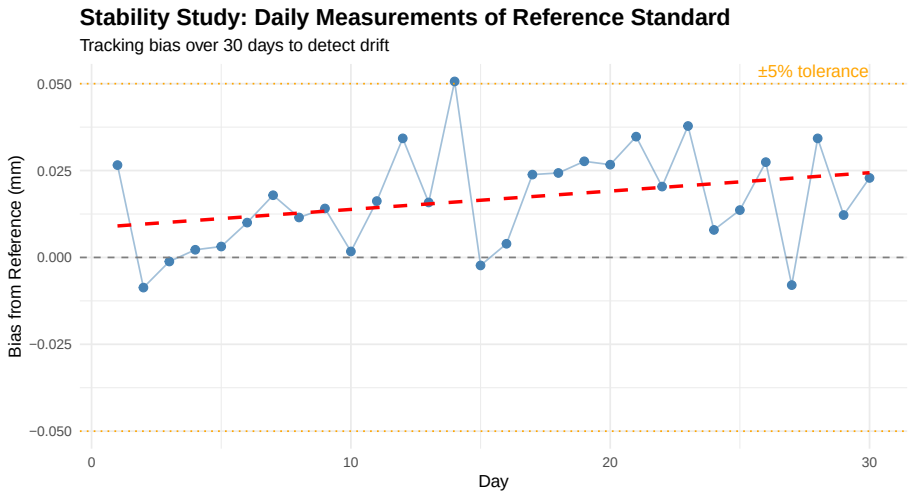
# Calculate linearity
slope <- coef(model)[2]
range_used <- max(reference) - min(reference)
linearity <- abs(slope) * range_used
r_squared <- summary(model)$r.squared

cat(sprintf("Linearity: %.4f mm (%.1f%% of the ±0.10 tolerance)\n", linearity, linearity / 0.10 *
cat(sprintf("  Regression  Bias = %.4f + %.5f x Reference\n", coef(model)[1], slope))
cat(sprintf("  R-squared  %.3f\n", r_squared))
if (r_squared > 0.7) {
  cat("  Verdict  !!Significant linearity issue!!\n")
} else {
  cat("  Verdict  Minor linearity issue\n")
}

par(mar = c(5, 5, 3, 1))
plot(reference, avg_bias, pch = 19, xlab = "Reference value (mm)", ylab = "Average bias (mm)",
     main = "Bias across the operating range")
abline(model, lwd = 2, col = "#C0392B")
abline(h = 0, lty = 2, col = "gray50")
```

5.5.3 Stability

Stability (also called drift) measures whether the measurement system's accuracy changes over time.



5.5.4 Stability Control Chart

For ongoing stability monitoring, use a control chart.

Teaching demo — try different measurements or a different historical sigma.

Show the live R code behind this card

```
#| label: stability-control-chart
# Stability monitoring with control chart
reference <- 25.000
measurements <- c(25.012, 25.008, 25.015, 25.018, 25.022, 25.019, 25.025,
                  25.028, 25.031, 25.026, 25.033, 25.038, 25.035, 25.041, 25.039)
days <- 1:15

# Calculate control limits (based on historical sigma)
historical_sigma <- 0.010
x_bar <- mean(measurements[1:5])  # Baseline from first 5 days
UCL <- x_bar + 3 * historical_sigma
LCL <- x_bar - 3 * historical_sigma

# Check for out-of-control
out_of_control <- which(measurements > UCL | measurements < LCL)
if(length(out_of_control) > 0) {
  cat(sprintf("!!Drift detected on day(s) %s: recalibration recommended!!\n", paste(ou
} else {
  cat("No out-of-control points: measurement system stable\n")
}

cat(sprintf("  Baseline average  %.4f mm\n", x_bar))
```

```
cat(sprintf("  UCL  %.4f mm\n", UCL))
cat(sprintf("  LCL  %.4f mm\n", LCL))
cat(sprintf("  Reference value  %.4f mm\n", reference))

par(mar = c(5, 5, 3, 1))
plot(days, measurements, type = "b", pch = 19, ylim = range(measurements, UCL, LCL, reference),
     xlab = "Day", ylab = "Measured value (mm)", main = "Stability control chart")
abline(h = c(UCL, x_bar, LCL), lty = c(2, 1, 2), col = c("#C0392B", "#27AE60", "#C0392B"))
abline(h = reference, lty = 3, col = "gray50")
points(days[out_of_control], measurements[out_of_control], pch = 19, cex = 1.6, col = "#C0392B")
```

5.6 Precision Studies: Gauge R&R

Gauge R&R (Repeatability and Reproducibility) is the most common MSA study. It quantifies precision variation.

5.6.1 Repeatability vs. Reproducibility

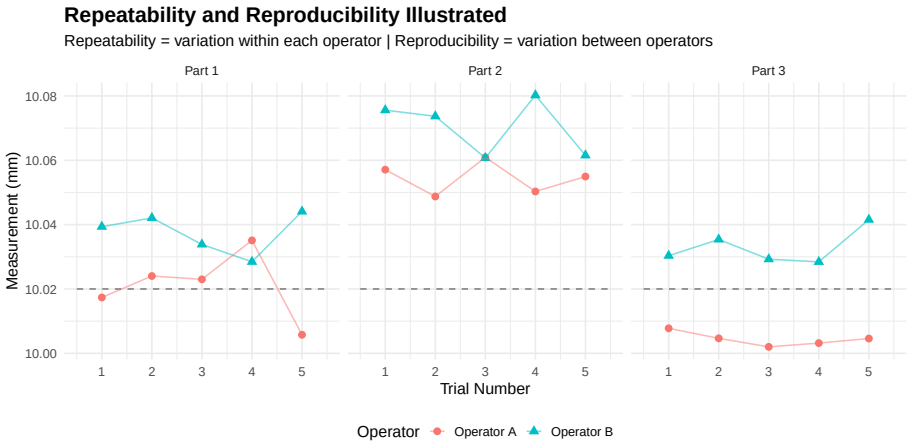


Table 5.4: Gauge R

Component	Also Known As	Definition
Repeatability (EV)	Equipment Variation, Within-operator variation	Variation when same operator measu
Reproducibility (AV)	Appraiser Variation, Between-operator variation	Variation when different operators m
Gauge R&R (GRR)	Total measurement system variation	Combined repeatability and reprodu

5.6.2 Gauge R&R Study Design

Table 5.5: Standard Gauge R\&R Study Design Parameters

Parameter	Typical_Value	Notes
Number of Parts	10	Minimum 5, 10 recommended for statisti
Number of Operators	3	Minimum 2, 3 is standard for detecting c
Number of Trials	3	Minimum 2, 3 is standard for repeatabili
Total Measurements	$10 \times 3 \times 3 = 90$	More measurements = better precision in
Part Selection	Cover the full process range	Parts should represent typical productio
Randomization	Randomize measurement order	Blind measurements when possible; preve

5.6.3 Range Method (Short Form)

The Range Method is a quick approximation useful for initial screening.

Teaching demo — try different measurement data for the two operators.

Show the live R code behind this card

```
#| label: range-method
# Range Method Gauge R&R Calculation
# Two operators, five parts, two trials each

# Data: Measurements by each operator, one vector per trial (one value per part)
a_trial1 <- c(0.85, 0.92, 0.78, 0.95, 0.88)
a_trial2 <- c(0.87, 0.91, 0.80, 0.96, 0.85)
b_trial1 <- c(0.86, 0.93, 0.80, 0.94, 0.87)
b_trial2 <- c(0.88, 0.92, 0.79, 0.97, 0.88)

# One row per part, one column per trial
operator_a <- cbind(a_trial1, a_trial2)
operator_b <- cbind(b_trial1, b_trial2)

# Step 1: Calculate ranges for each part-operator combination
range_a <- apply(operator_a, 1, function(x) max(x) - min(x))
range_b <- apply(operator_b, 1, function(x) max(x) - min(x))

# Step 2: Average range
R_bar <- mean(c(range_a, range_b))

# Step 3: Calculate Repeatability (EV)
# d2 for 2 trials = 1.128
d2_trials <- 1.128
EV <- R_bar / d2_trials

# Step 4: Calculate part averages by operator
avg_a <- rowMeans(operator_a)
```

```
avg_b <- rowMeans(operator_b)

# Step 5: Range of operator averages
X_diff <- abs(mean(avg_a) - mean(avg_b))

# d2 for 2 operators = 1.128
d2_operators <- 1.128
n_parts <- 5
n_trials <- 2

# Step 6: Reproducibility (AV)
AV_squared <- (X_diff / d2_operators)^2 - (EV^2 / (n_parts * n_trials))
AV <- sqrt(max(0, AV_squared))

# Step 7: Gauge R&R
GRR <- sqrt(EV^2 + AV^2)

# Step 8: Express as % of tolerance
tolerance <- 0.30  # Example tolerance of ±0.15 = 0.30 total

pct_GRR <- GRR / tolerance * 100 * 5.15
verdict <- if (pct_GRR < 10) "Acceptable" else if (pct_GRR < 30) "Marginal" else "!!Unacceptable!

cat(sprintf("%%GRR (of tolerance): %.1f%%\n", pct_GRR))
cat(sprintf("  Assessment  %s\n", verdict))
cat("AIAG: below 10% acceptable, 10-30% marginal, above 30% unacceptable\n")
cat(sprintf("  Average range (R-bar)  %.4f\n", R_bar))
cat(sprintf("  Repeatability (EV)  %.4f\n", EV))
cat(sprintf("  Reproducibility (AV)  %.4f\n", AV))
cat(sprintf("  Gauge R&R (GRR)  %.4f\n", GRR))
cat(sprintf("  %%EV (of tolerance)  %.1f%%\n", EV / tolerance * 100 * 5.15))
cat(sprintf("  %%AV (of tolerance)  %.1f%%\n", AV / tolerance * 100 * 5.15))
```

The Range method is a screening tool: it needs only a handful of parts and no statistics software, but it cannot separate an operator-by-part interaction from plain repeatability and it tends to overstate the Gauge R&R. When the answer matters for a customer submission or a gauge purchase, the study is repeated as a full ANOVA study; that method is worked in Chapter 17.

5.6.4 Gauge R&R Acceptance Criteria Summary

Table 5.6: AIAG Gauge R\&R Acceptance Guidelines

Criterion	Status	Interpretation

%GRR < 10%	Acceptable	Measurement system is acceptable for process control and capabil
%GRR 10-30%	Marginal	May be acceptable depending on importance, cost of gauge, repai
%GRR > 30%	Unacceptable	Measurement system is not acceptable; action required
ndc 5	Acceptable	Gauge can adequately distinguish between parts
ndc = 2-4	Marginal	Limited ability to distinguish; may be OK for simple pass/fail
ndc < 2	Unacceptable	Gauge cannot distinguish between parts; almost useless for contro

5.7 Try it yourself: name the measurement error

Seven symptoms from a gauge study. Decide which component of measurement variation each one points to.

Which kind of measurement error?

Pick the component of measurement variation behind each symptom.

Categories: Bias / Linearity / Stability / Repeatability / Reproducibility

1. A micrometer reads 0.02 mm high on a certified gauge block, every time. _____
2. A scale is accurate at 1 kg but reads 0.5 kg low at 50 kg. _____
3. A caliper is correct on small parts and increasingly wrong on large ones. _____
4. A CMM's readings on the same master part drift upward over three months. _____
5. One operator measures the same part five times with the same gauge and gets five different readings. _____
6. Three operators measure the same parts with the same gauge and their averages differ. _____
7. The day shift and the night shift get different averages on identical parts. _____

Answers: 1 Bias; 2 Linearity; 3 Linearity; 4 Stability; 5 Repeatability; 6 Reproducibility; 7 Reproducibility.

5.8 Measurement System Resolution

Resolution (also called discrimination) is the smallest increment a measurement system can detect.

5.8.1 Rule of Ten

The **Rule of Ten** states that measurement resolution should be at least 1/10 of: - The tolerance, OR - The process variation (6)

Whichever is smaller.

Your turn — change the tolerance, process spread, or gauge resolution below.

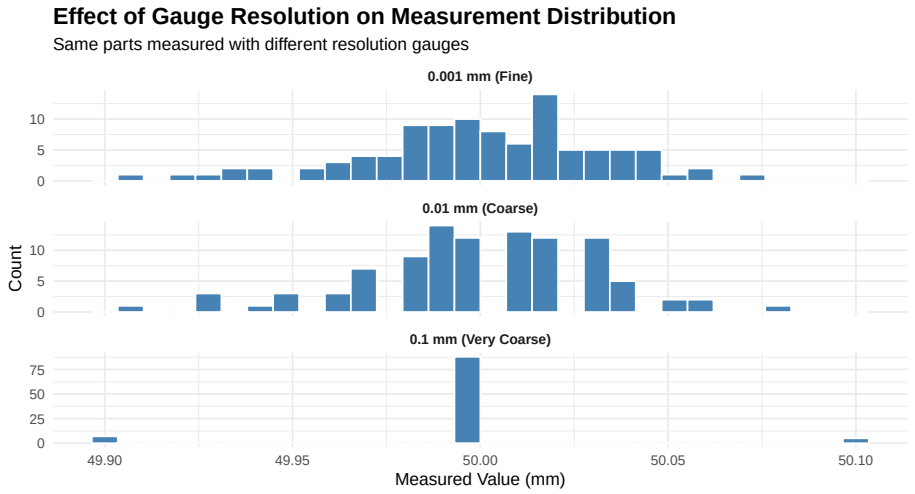
Show the live R code behind this calculator

```
#| label: resolution-check
# Resolution adequacy check
tolerance <- 0.100  # mm total tolerance
process_6sigma <- 0.080  # 6 sigma of process
gauge_resolution <- 0.010  # mm (smallest increment)

# Check against tolerance
ratio_tolerance <- tolerance / gauge_resolution
# Check against process
ratio_process <- process_6sigma / gauge_resolution

if (ratio_tolerance >= 10 && ratio_process >= 10) {
  cat("Gauge resolution is adequate\n")
} else {
  cat("Gauge resolution is !!inadequate!!\n")
}
cat(sprintf("  Tolerance / Resolution   %.0f : 1 (%s)\n", ratio_tolerance, ifelse(ratio_tolerance
cat(sprintf("  Process 6  / Resolution   %.0f : 1 (%s)\n", ratio_process, ifelse(ratio_process >=
```

5.8.2 Resolution Impact on Gauge R&R



5.9 Conducting an MSA Study: Step-by-Step



5.9.1 Data Collection Sheet Template

Table 5.7: Gauge R\&R Data Collection Sheet (Partial Example)

Part	Operator	Trial	Measurement
1	A	1	_____
1	A	2	_____
1	B	1	_____
1	B	2	_____
1	C	1	_____
1	C	2	_____
2	A	1	_____
2	A	2	_____
2	B	1	_____
2	B	2	_____
2	C	1	_____

2 C 2 ____

5.10 Video Resources

5.10.1 Understanding Gauge R&R

5.10.2 MSA in Practice

5.11 Summary

Measurement Systems Analysis ensures that your data can be trusted for process control and decision-making:

1. **Before SPC, do MSA** - Garbage in, garbage out; verify measurement quality first
2. **Accuracy components** - Bias, linearity, and stability affect location (average)
3. **Precision components** - Repeatability (within operator) and reproducibility (between operators)
4. **Key metrics** - %GRR $< 30\%$ and ndc 5 for acceptable systems
5. **Resolution matters** - 10:1 ratio vs. tolerance or process variation
6. **Going further** - The ANOVA method, attribute agreement (kappa), uncertainty budgets and the MSA-SPC link are in Chapter 17
7. **Continuous improvement** - Regular verification and recertification

> "The measurement system is part of the process. Fix the measurement system before trying to fix the process."

5.12 Review Questions

Question 1: A gauge R&R study yields %GRR = 45% and ndc = 2. What does this mean and what should be done?

Answer:

This measurement system is **unacceptable**:

- **%GRR = 45%** exceeds the 30% maximum threshold, meaning 45% of the observed variation is due to the measurement system, not the parts
- **ndc = 2** indicates the gauge can only distinguish between 2 categories of parts, which is insufficient for process control (minimum 5 required)

Actions required: 1. Investigate root causes of high variation: - Check gauge calibration and condition - Evaluate operator technique and training - Review measurement procedure for ambiguity - Assess fixture and part positioning - Check gauge resolution adequacy

2. Prioritize improvement efforts:
    - If repeatability (EV) is high: Focus on gauge condition, fixture, technique consistency
    - If reproducibility (AV) is high: Focus on training, procedure clarity, technique standardization
3. Re-run study after improvements
4. If improvements insufficient, consider upgrading to a higher-capability gauge
5. Until resolved, this gauge should not be used for process control or capability studies

Question 2: What is the difference between repeatability and reproducibility? Give an example of each source of error.

Answer:

Repeatability (Equipment Variation - EV): - Variation when the **same operator** measures the **same part** multiple times with the **same gauge** - Also called "within-operator" variation - Represents the inherent precision of the gauge and measurement technique

Examples of repeatability errors: - Gauge resolution limitations - Gauge mechanism play/backlash - Inconsistent part positioning in fixture - Variations in technique even by same operator - Environmental micro-changes during measurement

Reproducibility (Appraiser Variation - AV): - Variation when **different operators** measure the **same parts** - Also called "between-operator" variation - Represents differences in how operators apply the measurement method

Examples of reproducibility errors: - Different technique between operators (pressure applied, angle, etc.) - Different interpretation of measurement procedure - Different reading of analog scales - Training differences - Physical differences (eyesight, hand steadiness)

Question 3: Calculate the bias and determine if it is acceptable. Reference value = 25.000 mm, Tolerance = ±0.050 mm, Measured values: 25.012, 25.015, 25.008, 25.011, 25.014, 25.009, 25.013, 25.010, 25.012, 25.011

Answer:

```
reference <- 25.000
tolerance_total <- 0.100  # ±0.050 = 0.100 total
```

```r
measurements <- c(25.012, 25.015, 25.008, 25.011, 25.014,
                  25.009, 25.013, 25.010, 25.012, 25.011)

# Calculate bias
mean_measured <- mean(measurements)
bias <- mean_measured - reference

# Express as % of tolerance
bias_pct <- abs(bias) / tolerance_total * 100

# T-test for significance
t_result <- t.test(measurements, mu = reference)

# Acceptability
if(bias_pct < 10) {
  assessment <- "ACCEPTABLE (bias < 10% of tolerance)"
} else if(bias_pct < 25) {
  assessment <- "MARGINAL (bias 10-25% of tolerance)"
} else {
  assessment <- "UNACCEPTABLE (bias > 25% of tolerance)"
}

cat(sprintf("Bias: %.4f mm (%.1f%% of tolerance)\n", bias, bias_pct))
cat(sprintf("  Assessment  %s\n", assessment))
cat(sprintf("  Reference value  %g mm\n", reference))
cat(sprintf("  Mean measured  %.4f mm\n", mean_measured))
cat(sprintf("  T-test p-value  %.6f\n", t_result$p.value))
if(t_result$p.value < 0.05) {
  cat("  Significance  Bias is statistically significant (p < 0.05)\n")
  cat("Consider recalibration or technique adjustment\n")
} else {
  cat("  Significance  Bias is not statistically significant\n")
}
```

```
Bias: 0.0115 mm (11.5% of tolerance)
  Assessment  MARGINAL (bias 10-25% of tolerance)
  Reference value  25 mm
  Mean measured  25.0115 mm
  T-test p-value  0.000000
  Significance  Bias is statistically significant (p < 0.05)
Consider recalibration or technique adjustment
```

The bias of 0.0115 mm (11.5% of tolerance) is **marginally acceptable** and statistically significant. Recalibration should be considered.

Question 4: Why is it important to conduct MSA before capability studies

(Cp, Cpk)?

Answer:

MSA must precede capability studies because measurement error directly inflates observed variation:

Mathematical Relationship:

$$\sigma^2_{observed} = \sigma^2_{process} + \sigma^2_{measurement}$$

Impacts on Capability:

1. **Understated Cp/Cpk**: Observed sigma is always larger than true process sigma, so calculated capability will be lower than actual capability
2. **False Alarms**: A capable process may appear incapable due to measurement noise, leading to unnecessary process adjustments or investment
3. **Missed Issues**: High measurement variation masks actual process variation; you can't see problems through the "fog" of gauge error
4. **Bad Decisions**: Incorrect capability data leads to:
    - Wrong process acceptance decisions
    - Inappropriate control limits
    - Misguided improvement investments
    - Customer quality issues

Example: - True process: Cpk = 1.5 (very capable) - With 50% GRR: Observed Cpk 0.9 (appears incapable) - Result: Unnecessary process "improvement" efforts

Rule of Thumb: If %GRR > 30%, any capability study is essentially meaningless because too much of the observed variation is measurement noise, not real process variation.

Question 5: What is the "Rule of Ten" for gauge resolution, and how would you apply it?

Answer:

The Rule of Ten: The gauge resolution should be at least **1/10 of the tolerance** or **1/10 of the process variation (6)**, whichever is smaller.

$$\text{Resolution} \leq \frac{\min(\text{Tolerance}, 6\sigma)}{10}$$

Application Example:

Given: - Part tolerance: ±0.05 mm (total = 0.10 mm) - Process 6 : 0.08 mm - Available gauges: 0.01 mm resolution, 0.001 mm resolution

Calculation:

```
Required resolution  min(0.10, 0.08) / 10
Required resolution  0.08 / 10
Required resolution  0.008 mm
```

Assessment: - 0.01 mm gauge: $0.01 > 0.008 \rightarrow$ **INADEQUATE** - 0.001 mm gauge: $0.001 < 0.008 \rightarrow$ **ADEQUATE**

Why It Matters: 1. If resolution is too coarse, gauge cannot detect small part differences 2. Results in low ndc (number of distinct categories) 3. Control charts will show "stair-step" patterns 4. Capability studies will be inaccurate

Practical Tips: - For SPC applications, 10:1 is minimum; 20:1 is preferred - For capability studies, 10:1 is acceptable - For simple pass/fail gauging, 5:1 may be acceptable

Question 6: What actions would you take if a Gauge R&R study showed high repeatability (EV) but low reproducibility (AV)?

Answer:

When **repeatability is high** (main contributor to GRR) but **reproducibility is low**, the measurement variation is coming from the gauge/equipment/technique rather than differences between operators.

Root Cause Investigation - Focus Areas:

1. **Gauge Condition**
   - Worn measuring surfaces or contacts
   - Loose components or mechanism play
   - Damaged or dirty probe/sensor
2. **Gauge Resolution**
   - Resolution may be inadequate for the tolerance
   - Check 10:1 rule compliance
3. **Fixture/Fixturing**
   - Part not held consistently
   - Fixture wear or damage
   - Part not seating properly
4. **Measurement Technique**
   - Inconsistent contact force
   - Varying measurement location on part
   - Speed of measurement varies
5. **Part Characteristics**
   - Part surface finish affecting readings
   - Part flexibility/deflection during measurement
   - Part temperature variation
6. **Environment**
   - Vibration affecting readings
   - Temperature instability
   - Contamination

Improvement Actions:

Priority	Action
1	Verify gauge calibration and condition; repair/replace if needed
2	Check and improve fixture; ensure consistent part positioning
3	Evaluate and potentially upgrade gauge resolution
4	Standardize technique (force, speed, location)
5	Control environmental factors
6	Re-run study to verify improvement

Note: Low AV in this case is actually good - it means operators are consistent with each other. The problem is the measurement equipment itself.

5.13 References

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Chapter 6

Introduction to Metallurgy: Structure and Heat Treatment

6.1 Learning Objectives

By the end of this chapter, you will be able to:

- Explain why a technician needs to understand metallurgy, with examples from automotive, food-processing, and defense manufacturing.
- Describe how atoms pack into crystal structures (BCC, FCC, HCP) and how grain boundaries and dislocations control strength.
- Apply the Hall-Petch relationship conceptually to explain why fine-grained metals are stronger.
- Distinguish hot-rolled steel (HRS) from cold-rolled steel (CRS) by grain shape, surface finish, dimensional tolerance, residual stress, and mechanical properties.
- Read the iron-carbon phase diagram for the steel region (0 to about 2.1 wt% C) and identify ferrite, austenite, cementite, and pearlite.
- Decode an AISI/SAE steel grade (10xx, 41xx, 15xx) to estimate carbon content and alloy content.
- Explain the purpose, process, resulting microstructure, and property changes for annealing, normalizing, quench hardening, tempering, and stress relieving.
- Explain why quenched steel is always tempered, and why low-carbon steel barely hardens.

This chapter is Part A of a two-part introduction to metallurgy and materials testing. Part A (this chapter) covers what steel is made of and how heat treatment changes it. Part B covers how we test steel — tensile, hardness, impact, and metallography — and connects everything you learn here to what you will observe in your own lab reports.

> This chapter gives you the theory. It does not repeat the lab procedures — those live in your lab handout and the Labs chapter. Read this chapter *before* the lab so you can predict what you'll see; come back to it *after* the lab to explain what you got.

6.2 Why Technicians Need Metallurgy

A delivery van's rear axle snaps clean in half on a cold January morning in Thunder Bay, with nobody near it and no collision. A weld on a food-plant conveyor frame cracks after eighteen months of hot, wet washdown cycles. A grade-8 bolt on an aircraft bracket stretches 2 mm under a load it was rated to carry with margin to spare, and now nobody trusts the rest of the batch.

None of these are mysteries once you know what to look for. The axle was likely a low-alloy steel that went brittle below its ductile-to-brittle transition temperature — exactly the phenomenon you'll test for in the Charpy lab. The weld cracked because heat from welding changed the grain structure and left residual stress in the heat-affected zone. The bolt stretched because someone substituted a lower-grade fastener, or because it was never properly heat treated — its hardness would have told the story in thirty seconds on a Rockwell tester.

Metallurgy is the science of metals: what they're made of at the atomic level, and how processing — rolling, welding, heat treating, machining — changes their structure and therefore their properties. You don't need a metallurgy degree to use it. As an electromechanical technician, you need enough metallurgy to:

- **Read a material certification** and know whether the steel in front of you is even capable of doing the job.
- **Recognize a heat-treatment problem** — a part that's too hard and brittle, or too soft and won't hold a cutting edge.
- **Understand why a part failed** — was it overloaded, was it the wrong material, or was it processed wrong?
- **Specify or verify a process** — annealing before machining, hardening a wear surface, stress-relieving a weldment.

Industry application

Automotive: A stamped door hinge is deliberately kept as *cold-rolled, unhardened* low-carbon steel so it stays ductile and absorbs energy in a crash rather than snapping. **Food processing:** conveyor shafts and blades are usually **austenitic stainless steel** (Chapter 7, Section 7.6) specifically because it resists corrosion from washdown chemicals *and* won't harden and become brittle if it gets bumped. **Defense:** armor plate and gun-barrel steels are quenched and tempered to a very specific hardness — too hard and they shatter on impact, too soft and they don't stop a round.

Before you read on

Think of a metal part that has failed on you, or that you've seen fail — a broken tool, a bent bracket, a stripped bolt. Was it bent (ductile) or did it snap cleanly (brittle)? Keep that failure in mind — by the end of this chapter you should be able to guess *why*.

Quick check: why metallurgy matters

Question: A maintenance technician needs to replace a broken shaft. The stockroom has "1018 steel" and "4140 steel" round bar of the same diameter. Why might the choice matter, even before either one is heat treated?

Answer: The two grades have different carbon and alloy content (you'll learn to read this from the numbers later in this chapter), which changes how much they can be hardened, how strong and tough they are as-supplied, and how they weld and machine. Grabbing "steel that fits" without checking the grade is exactly the kind of mistake that leads to the snapped-shaft scenario above.

6.3 What Metals Are Made Of

6.3.1 Atoms, Crystals, and Grains

Every metal is made of atoms held together by **metallic bonding** — the atoms' outer electrons aren't tied to any one atom, they move freely through the whole solid like a "sea" of electrons surrounding fixed positive ion cores. That shared electron sea is *why* metals conduct electricity and heat so well, and why they can bend without breaking: the bonds aren't tied to specific neighbor pairs, so atoms can slide past each other and the bonding survives.

Those atoms don't sit randomly. As a molten metal cools and solidifies, the atoms lock into a repeating, orderly 3-D pattern called a **crystal lattice** —

think of a shelf of oranges at the grocery store, where every orange settles into the lowest, most stable position between the ones below it. That repeating pattern, extended in all directions, is a **crystal**.

Here's the part that matters for you as a technician: solidification doesn't start from one point and grow outward as a single perfect crystal. It starts at *many* points at once — at the mould wall, around impurity particles, anywhere a stable cluster of atoms happens to form first. Each of these starting points grows its own crystal, in its own random orientation, until it bumps into its neighbours. The result is a solid made of thousands to millions of small crystals, all packed together like a 3-D jigsaw puzzle, each one oriented slightly differently. Each individual crystal is called a **grain**, and the mismatched boundary where two grains meet is a **grain boundary**.

This is not a minor detail — **grain size and grain boundaries control most of the mechanical properties you will measure in the lab.** You'll see grains directly, under a microscope, in the tensile and case-hardening lab (Section 7.5 in Part B) once a polished, etched specimen is examined.

If a tray of shaken steel balls sounds like an odd way to understand a phase diagram, watch this: Steve Mould uses a tray of loose steel balls, jiggled like atoms with thermal energy, to physically model exactly what you just read — multiple crystals nucleating at once, growing until they collide, and locking in grain boundaries where they meet. He also uses the same setup to show how the *rate* of shaking (a stand-in for cooling rate) changes the resulting structure — a hands-on preview of why cooling rate is the whole story in Section 6.6.

Steve Mould: Self Organising Steel Balls Explain Metal Heat Treatment: https://www.youtube.com/watch?v=xuL2yT-B2TM

6.3.2 Crystal Structures: BCC, FCC, and HCP

Within a grain, the atoms pack into one of a small number of repeating arrangements, called **unit cells** — the smallest repeating "building block" of the crystal, like a single tile in a repeating floor pattern. Three arrangements matter for the metals you'll work with:

- **Body-centered cubic (BCC):** an atom at each of the 8 corners of a cube, plus one atom in the very center. Picture 8 oranges arranged in a cube frame with one more orange wedged in the middle gap. BCC metals include **-ferrite (low-temperature iron)**, chromium, and tungsten.
- **Face-centered cubic (FCC):** an atom at each of the 8 corners, plus one atom centered on each of the 6 faces. This is the tightest, most efficient way to stack spheres — the same pattern as oranges stacked in a pyramid at the grocery store. FCC metals include **austenite (high-temperature iron)**, aluminum, copper, and nickel.
- **Hexagonal close-packed (HCP):** atoms arranged in stacked hexagonal layers, another maximally efficient packing, just stacked in a different

repeating sequence than FCC. HCP metals include zinc, magnesium, and titanium.

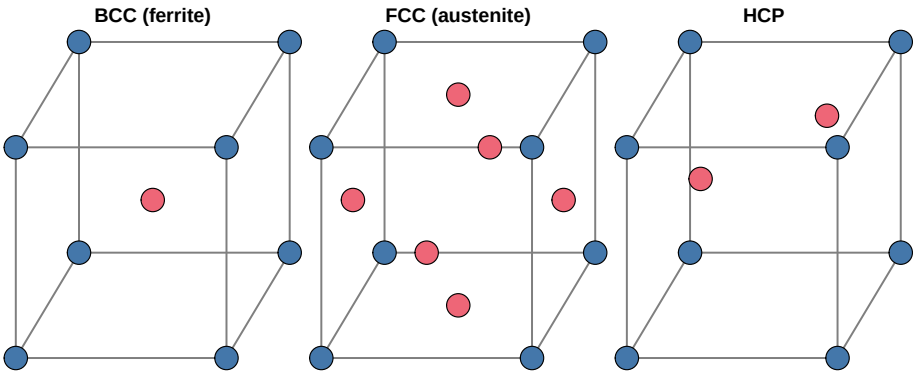


Figure 6.1: Simplified 2-D schematics of the three crystal structures that matter most for this course. Corner atoms are shared with neighbouring cells; BCC adds one body-centred atom, FCC adds one atom centred on each visible face.

Why should you care which pattern a metal has? Because **BCC metals have far fewer easy sliding planes than FCC metals**, and that single fact explains two things you'll measure directly in your labs:

1. **BCC steels (like the 1018 you'll break in the Charpy lab) go brittle at low temperature; FCC metals (aluminum, austenitic stainless) mostly don't.** You'll see why in Section 7.4 in Part B.
2. **When iron itself flips from BCC to FCC on heating (and back again on cooling), that single structural change is the entire basis of steel heat treatment.** You'll meet this as the transformation in Section 6.5 below.

Industry application

Defense: submarine hulls are often HY-80/HY-100 steel — BCC — chosen and heat treated specifically to stay tough at the near-freezing temperatures of deep, cold ocean water. Get the heat treatment wrong and you raise the ductile-to-brittle transition temperature right into the operating range.

6.3.3 Grain Boundaries and the Hall-Petch Relationship

A grain boundary is a zone of disorder — the neat, repeating pattern of one grain doesn't line up with its neighbour, so the atoms right at the boundary are squeezed and misaligned. That disorder matters mechanically: it's much harder for a defect moving through one grain (see dislocations, next) to cross into a

grain oriented a different way. The boundary acts like a speed bump.

More grain boundaries per unit volume — that is, **smaller grains** — means more speed bumps, which means a stronger metal. This relationship has a name, the **Hall-Petch relationship**, and a simple equation:

$$\sigma_y = \sigma_0 + k\, d^{-1/2} \tag{6.1}$$

where σ_y is the yield strength (MPa), σ_0 and k are material constants (found experimentally), and d is the average grain diameter (mm or µm). Notice the grain size term is $d^{-1/2}$ — strength doesn't just go up as grains get smaller, it goes up *faster and faster* as grains get very fine.

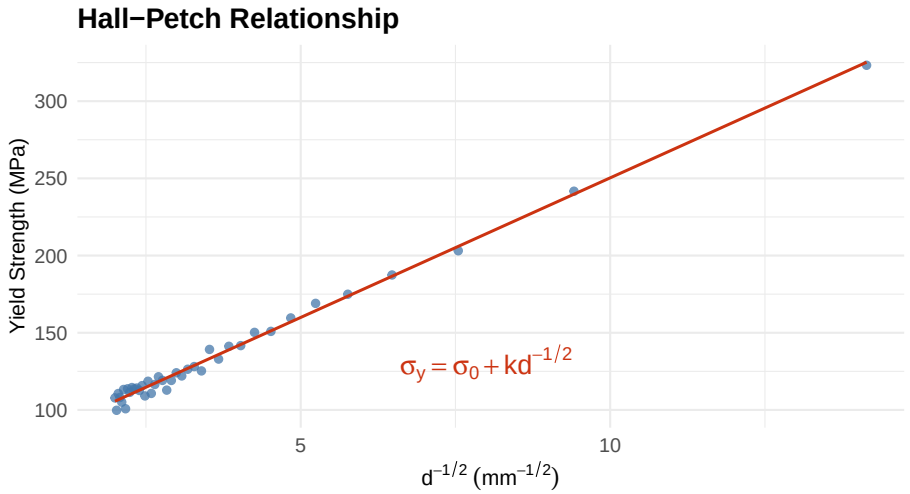


Figure 6.2: The Hall-Petch relationship: yield strength rises as grain size decreases, and rises steeply once grains become very fine (data simulated for a low-carbon steel, sigma0 = 70 MPa, k = 18 MPa*mm^0.5).

Grain refinement is one of the very few ways to make a metal **both** stronger **and** tougher at the same time — most other strengthening methods (like cold work or alloying) trade some ductility away. That's why grain refinement shows up constantly in real processing: controlled rolling, normalizing, and rapid cooling are all partly about keeping grains small.

Predict and explain: grain size and hardness

Question: In Section 6.4 below you'll learn that hot-rolled steel (HRS) tends to have larger, more equiaxed grains than a normalized (re-heated and air-cooled) fine-grained steel of the same composition. Based on Hall-Petch alone, which would you predict has the higher yield strength?

Answer: The finer-grained (normalized) steel, because smaller d means a larger $d^{-1/2}$ term in Equation 6.1. This is one reason normalizing (Section 6.6) is used to refine grain size after hot rolling or casting.

6.3.4 Dislocations

If the crystal lattice were perfect, moving one entire plane of atoms past another (which is what plastic — permanent — deformation actually is) would take an enormous force, because every single atomic bond across that plane would have to break at once. Real metals deform at far lower stresses than that, because real crystals aren't perfect. They contain line defects called **dislocations** — places where an extra half-plane of atoms is squeezed into the lattice, creating a local zone of distortion.

Here's the analogy that makes this click: imagine dragging a large rug across a floor by pulling on one edge. Sliding the *whole* rug at once takes a huge amount of friction-fighting force. Instead, if you put a small wrinkle in the rug and push the wrinkle across, the rug ends up shifted the same total distance — but you only ever had to overcome friction on the small area of the wrinkle at any one moment, not the whole rug. A dislocation moving through a crystal does the same thing: it lets one plane of atoms shift relative to another one atomic bond-length at a time, propagating through the crystal instead of breaking all at once.

This is what plastic deformation *is*, at the atomic level: dislocations moving. Everything that makes it harder for dislocations to move — grain boundaries, other dislocations tangled together, dissolved alloying atoms, hard second-phase particles like cementite — makes the metal stronger and harder, because more stress is needed to keep the dislocations moving. You'll see this idea again in Section 7.2 (strain hardening, Part B) and in Section 6.6 below (why cold work and quenching both increase hardness).

6.4 Hot-Rolled vs Cold-Rolled Steel

Almost all the bar, sheet, and plate steel you'll ever touch on a shop floor starts as a cast ingot or continuously-cast slab and is mechanically **rolled** down to final size between big pairs of rollers. The temperature at which that rolling happens changes the resulting grain structure completely — and this is one of the most practical, testable pieces of metallurgy a technician needs, because HRS and CRS look almost identical stacked on a rack, but behave very differently.

6.4.1 Recrystallization Temperature

Every metal has a **recrystallization temperature** — roughly 0.3 to 0.5 times its absolute melting point (this rule of thumb is a simplification; the exact value

depends on alloy content and prior cold work). Below that temperature, deforming the metal *distorts and elongates* the existing grains without replacing them. Above it, the metal has enough thermal energy that new, strain-free, equiaxed (roughly equal in all directions) grains nucleate and grow, wiping out the deformation damage almost as fast as it's created. For plain carbon steel, recrystallization temperature is roughly 500 to 700 °C — well below the temperatures used for hot rolling (typically 1100 to 1250 °C, comfortably in the austenite range).

- **Hot-rolled steel (HRS)** is rolled *above* the recrystallization temperature. The steel continuously recrystallizes as it's deformed, so the final grains are small and **equiaxed** — roughly the same shape and size looking from any direction.
- **Cold-rolled steel (CRS)** is rolled *below* the recrystallization temperature (often starting from hot-rolled stock, **pickled** to remove scale, then rolled further at room temperature). There's no recrystallization to erase the deformation, so the grains get physically stretched and flattened in the rolling direction — **elongated grains**.

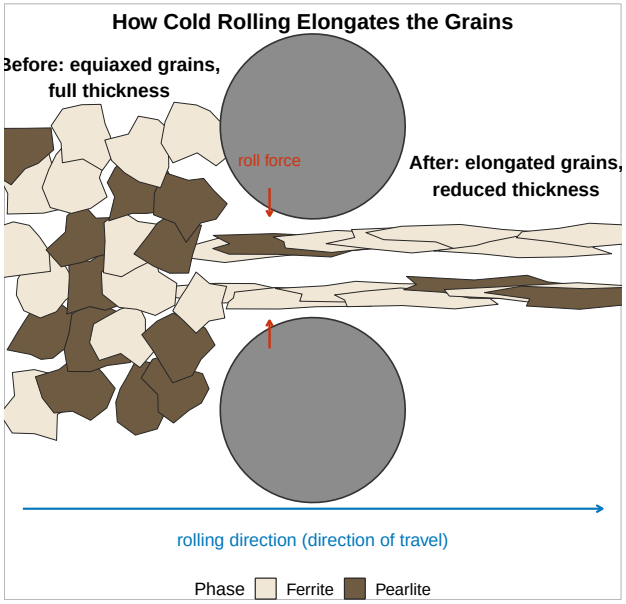


Figure 6.3: Schematic of the cold rolling mechanism (not to scale). Stock enters the roll gap at full thickness with equiaxed grains; the work rolls squeeze it under heavy compressive force; it exits thinner, with the same grains flattened in the thickness direction and stretched out in the rolling direction. This is the mechanism behind the elongated-grain micrograph in Figure 6.5.

6.4.2 What You Will See and Measure

! This is what you'll be looking for in the metallography lab

Grain elongation is a **directional** effect. Cut a cross-section *across* the bar (transverse) and elongated grains in CRS look almost the same as equiaxed grains — you're looking at the grains end-on. Cut a section *along* the length of the bar (longitudinal) and CRS grains show up clearly stretched and lined up, like streaks of taffy, while HRS grains still look equiaxed and randomly oriented no matter which way you cut it. **This is exactly why the metallography procedures for the hardness and tensile labs specify which section to take.**

Table 6.1: Typical, approximate differences between HRS and CRS of the same grade. Exact values vary by supplier, section size, and specific grade — always confirm against a materials handbook or mill certificate rather than assuming these numbers.

Property	Hot-Rolled Steel (HRS)	Cold-Rolled Steel (CRS)
Grain shape	Equiaxed, same in all sections	Elongated in the rolling direction (visible in longitudinal section only)
Surface finish	Rougher; often covered in blue-grey **mill scale** (iron oxide formed at high temperature)	Smooth, bright, often slightly oily from the rolling process
Dimensional tolerance	Looser — the steel shrinks unevenly as it cools from rolling temperature	Tighter — sized precisely at room temperature
Residual stress	Low — the steel had time to relax as it cooled slowly from a hot, soft state	**High** — cold deformation locks elastic stresses into the surface and core that don't fully relax
Yield strength	Lower (typical values only; always check the actual mill certificate)	Higher — cold work raises yield strength through the same dislocation-tangling mechanism as strain hardening (Section 7.2)

Property	Hot-Rolled Steel (HRS)	Cold-Rolled Steel (CRS)
Ductility (% elongation)	Higher	Lower — the cold work has already used up some of the material's capacity to deform further
Hardness	Lower	Higher

6.4.3 Pickling: Removing Mill Scale Before Cold Rolling

Hot rolling happens at 1100–1250 °C, and at that temperature the steel surface reacts continuously with oxygen in the air, growing a layer of iron oxides (a mix of FeO, Fe O , and Fe O) called **mill scale**. Mill scale is hard, brittle, and adheres unevenly — exactly the surface condition you do *not* want going into a cold-rolling mill, for two practical reasons: it's abrasive enough to score and prematurely wear the hardened work rolls, and it would get pressed into the steel surface rather than rolled smooth, ruining the bright finish CRS is sold for.

Pickling removes it chemically before cold rolling begins. The coil is uncoiled and run continuously through a line of tanks containing a dilute acid bath — historically sulfuric acid, more commonly hydrochloric acid in modern continuous pickling lines — which dissolves the oxide scale without significantly attacking the base metal underneath. The strip then passes through rinse tanks to wash off acid residue and is dried, often with a very thin coat of oil applied immediately afterward to prevent **flash rusting** (the clean, reactive surface will rust in minutes otherwise) before it reaches the cold mill. This is the "pickled to remove scale" step referenced in the CRS description above — it's why CRS comes out smooth and bright while HRS keeps its dull, scaled surface.

> **i** Why this matters to a technician, not just a mill operator
>
> You won't operate a pickling line, but you will see its effects: mill scale explains the rougher, blue-grey surface and pitting you're told to expect on HRS specimens in the metallography lab (Table 6.1), and the acid-handling hazards of pickling (fumes, splash, waste acid neutralization) are the same category of chemical-exposure hazard covered generally in the Machine Guarding and Functional Safety chapter — concentrated acid baths are an industrial hazard even though you'll never need to manage one directly as a maintenance technician.

For more detail than this course requires, see Pickling (metal) — Wikipedia for an overview of acid pickling chemistry and practice.

What to expect in your own results

In the metallography portion of your labs, expect HRS specimens to show equiaxed grains regardless of orientation, often with visible mill-scale pitting near the original surface. **In the hardness lab and the tensile/microhardness lab**, if you are given both conditions of a grade, expect the cold-worked material to read measurably harder and stronger, with lower elongation, than the hot-rolled or annealed condition of the same steel.

These are schematic diagrams, not photomicrographs

Figure 6.3, Figure 6.4, and Figure 6.5 are simplified illustrations — grain shapes drawn to show the *pattern* you're looking for, not the texture of a real photograph. Your lab's actual micrographs will look noisier and less tidy than these. Read them for shape and orientation, not as a stand-in for the real thing under the scope.

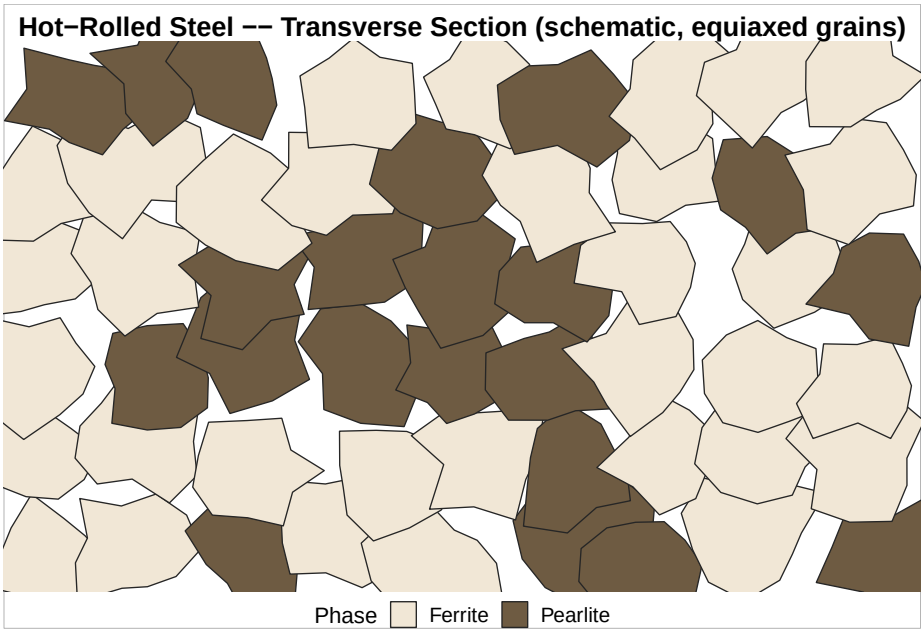


Figure 6.4: Schematic diagram (not an actual photomicrograph) of hot-rolled 1018 steel: equiaxed grains, similar in every direction, regardless of which way the section is cut.

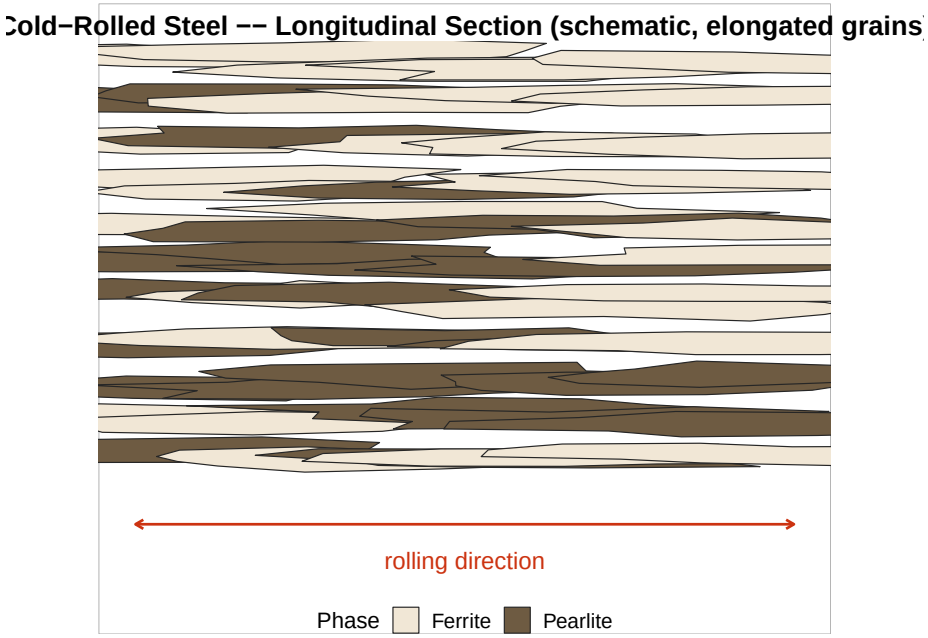


Figure 6.5: Schematic diagram (not an actual photomicrograph) of cold-rolled 1018 steel, longitudinal section: grains visibly stretched and lined up parallel to the rolling direction. A transverse section of the same bar would look nearly as equiaxed as HRS – see Section 6.4.2.

6.4.4 Why Cold-Rolled Steel Warps When You Machine It

Here's a scenario every machinist and technician eventually hits: you take a nice, flat, precisely-sized piece of CRS, machine a pocket or slot in it, and it comes off the machine slightly bowed. The material didn't change composition — you *released* residual stress that was already there.

Think of CRS as a block that's internally "pre-loaded," a bit like a stack of bent leaf springs held together — the surface layers were compressed and the interior stretched (or vice versa) during cold rolling, and those opposing stresses balance out and hold the part straight *as long as the material stays intact.* Machine material off one side, and you've removed some of the springs holding that balance. The remaining stresses are no longer balanced, so the part physically bends or twists to find a new equilibrium. Hot-rolled steel, having cooled slowly from a soft, high-temperature state, doesn't carry nearly as much locked-in residual stress, so it's far less prone to this. This is exactly why precision parts machined from CRS are sometimes **stress relieved** before final machining, or rough-machined, stress relieved, then finish-machined.

Industry application

Automotive: thin CRS sheet stampings (body panels, brackets) are especially prone to **springback** — the part relaxes slightly after the stamping die opens, because of exactly this residual-stress release, and die designers have to over-bend the shape to compensate.

Predict and explain: HRS vs CRS in the lab

Question: You're handed two round bar specimens of the same 1018 grade — one visibly rougher and grey, one smooth and bright. Before doing any testing, predict which one will read higher on a Rockwell hardness test, and which will show more elongation on a tensile test.

Answer: The smooth, bright bar is cold-rolled. Predict it reads **higher hardness** and **lower % elongation** than the rougher, grey (hot-rolled, mill-scaled) bar, because cold work has already strain-hardened it and used up some of its ductility.

6.5 The Iron-Carbon Phase Diagram

6.5.1 What a Phase Diagram Is

You already understand a phase diagram, even if you've never seen one drawn: water at atmospheric pressure is solid (ice) below 0 °C, liquid between 0 °C and

100 °C, and gas (steam) above 100 °C. If you plotted temperature on one axis and just labelled which phase water is in at each temperature, you'd have drawn a (very simple) one-component phase diagram.

Now make it more interesting: dissolve salt in the water. The freezing point drops (that's why we salt roads in winter) and depends on *how much* salt is dissolved. A phase diagram for a water-salt mixture would need **two axes** — temperature, and salt content (composition) — to show which phase (or mixture of phases) exists at any combination of temperature and composition. That's exactly the kind of diagram we use for steel, except the two "ingredients" are iron and carbon.

> **i** Before you read on
>
> Salt water freezes at a lower temperature than pure water, and the exact freezing point depends on the salt concentration. Keep that idea in mind — you're about to see the same kind of "composition changes the transformation temperature" behaviour in steel, except the *high*-temperature transformation (not freezing) is what matters.

6.5.2 The Steel Region: Ferrite, Austenite, Cementite, and Pearlite

The full iron-carbon diagram runs from pure iron (0% C) to iron carbide, Fe C (6.7% C by weight — a hard, brittle compound called **cementite**), but everything technicians call "steel" lives in a small slice of that range: **0 to about 2.1 wt% carbon.** (Above roughly 2.1% C we're talking about cast iron — covered in Part B, Section 7.6.) Figure 6.6 shows this region.

The **phases** you need to know:

- **Ferrite ( ):** iron in its low-temperature **BCC** crystal structure (Section 6.3.2). Soft, ductile, and can only dissolve a tiny amount of carbon — at most about 0.02 wt% at 727 °C, essentially zero at room temperature. Under the microscope, ferrite is the light-etching background phase.
- **Austenite ( ):** iron in its high-temperature **FCC** crystal structure (Section 6.3.2). Much more open lattice than BCC, so it can dissolve far more carbon — up to about 2.1 wt% at 1147 °C. **Austenite does not exist at room temperature in plain carbon steel** — it's only stable at high temperature, but it is the starting phase for every heat-treatment process in Section 6.6, which is why it matters so much.
- **Cementite (Fe C):** iron carbide, a hard, brittle intermetallic compound, essentially fixed at 6.7 wt% C. Under the microscope, cementite etches dark.
- **Pearlite:** not a phase but a **microstructure** — alternating thin lamellae (plates) of ferrite and cementite, formed when austenite of eutectoid

composition transforms on slow cooling. It gets its name because it can show a pearly, iridescent look under the microscope from light diffracting off the fine lamellae. Pearlite is harder and stronger than plain ferrite (because the hard cementite plates block dislocation motion — the same strengthening idea as Hall-Petch, just via a second phase instead of a grain boundary) but softer than martensite (Section 6.6.2).

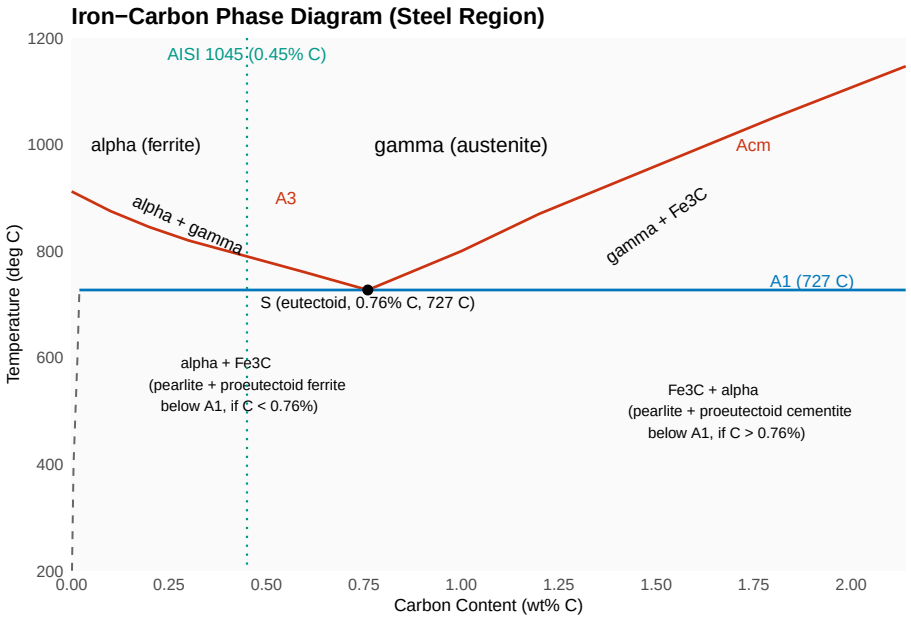


Figure 6.6: The iron-carbon phase diagram, steel region only (0 to 2.14 wt% C). A3 and Acm curves are smoothed approximations for teaching purposes; consult a reference diagram (e.g. Callister) for precise values.

> **This is a simplified picture**
>
> Real Fe-C diagrams also show a magnetic transition (the Curie point, sometimes called A2, around 770 °C) where ferrite loses its magnetism on heating without changing crystal structure or phase — it's not a phase boundary and doesn't change what you'd see under a microscope, so it's left off Figure 6.6. The A3 and Acm curves above are smoothed for teaching clarity; a reference-quality diagram (Callister, *Materials Science and Engineering*, or the ASM Handbook, Vol. 1) has slightly different curvature and should be used for precise readings.

6.5.3 The Eutectoid Point, A1/A3 Lines, and Hypo/Hypereutectoid Steels

The single most important point on this whole diagram is **S**, the **eutectoid point**, at 0.76 to 0.8 wt% C and 727 °C (sources vary slightly on the exact composition — treat 0.76–0.8% as a typical range, not a single fixed number). At exactly this composition, austenite transforms *directly and completely* into pearlite on slow cooling through 727 °C — no proeutectoid (pre-eutectoid) phase forms first.

- **A1 line:** the horizontal line at 727 °C. This is the temperature below which austenite cannot exist in plain carbon steel at equilibrium, for any carbon content in our range.
- **A3 line:** the curve from pure iron's 912 °C transformation point down to the eutectoid point S. Above A3, a hypoeutectoid steel is fully austenite; between A3 and A1, it's a mix of austenite and ferrite.
- **Acm line:** the curve from S up to 1147 °C at 2.14% C. Above Acm, a hypereutectoid steel is fully austenite; between Acm and A1, it's a mix of austenite and cementite.

Steels are classified by which side of the eutectoid point they fall on:

- **Hypoeutectoid steel** (C $<$ 0.76%, most structural and machine steels, including 1018, 1045): cooling from austenite, **ferrite forms first** (called proeutectoid ferrite, nucleating at prior austenite grain boundaries), and the remaining, now carbon-enriched austenite transforms to pearlite at A1. Final structure: proeutectoid ferrite + pearlite.
- **Eutectoid steel** (C 0.76%): cooling from austenite gives (essentially) 100% pearlite.
- **Hypereutectoid steel** (C $>$ 0.76%, tool steels and some bearing steels): **cementite forms first** (proeutectoid cementite, also at prior austenite grain boundaries, forming a network), and the remaining austenite transforms to pearlite at A1. Final structure: proeutectoid cementite + pearlite.

6.5.4 Walking Through the Slow Cooling of AISI 1045

AISI 1045 (0.45 wt% C, a common hypoeutectoid medium-carbon steel used for shafts and gears) is marked as a dotted line in Figure 6.6. Follow it from top to bottom, cooling slowly (near-equilibrium — this matters, see the callout below):

1. **Above ~790 °C:** fully austenite ( ) — FCC, soft, and the only phase in which carbon is highly soluble.
2. **Between ~790 °C and 727 °C (crossing A3): proeutectoid ferrite** begins nucleating at austenite grain boundaries and growing. As more ferrite forms (which can hold almost no carbon), the remaining austenite gets progressively enriched in carbon, sliding "down" the A3 curve until,

right at 727 °C, the remaining austenite reaches exactly the eutectoid composition (0.76% C).

3. **At 727 °C (crossing A1):** all the remaining, now-eutectoid-composition austenite transforms to pearlite.
4. **Below 727 °C:** no further phase transformation occurs on the equilibrium diagram — the structure is locked in as **proeutectoid ferrite + pearlite**, and stays that way down to room temperature (grain size may still change slightly with holding time, but the phases present do not).

> ⚠ Slow cooling only — this is the whole point of heat treatment
>
> Everything above assumes cooling is slow enough for atoms to diffuse and the transformations to fully complete — this is called **equilibrium** or **near-equilibrium cooling**, and it's what annealing and normalizing approximate. **Quench** a steel fast enough, and there isn't time for carbon to diffuse out of solution — you get an entirely different, non-equilibrium product (martensite) that doesn't appear anywhere on this diagram at all. That's the entire subject of Section 6.6, coming up next.

Worked example: reading the lever rule (static)

Problem: Estimate the fraction of proeutectoid ferrite and pearlite in slowly-cooled AISI 1045 (0.45% C) at room temperature. Assume the ferrite solvus limit is essentially 0.02% C and the eutectoid composition is 0.76% C.

Solution — the lever rule: just below A1, the structure is a two-phase mix of ferrite (at ~0.02% C) and austenite-about-to-become-pearlite (at 0.76% C). The fraction of each phase is found the same way you'd balance a lever: the fraction of *pearlite* equals the distance from the *overall* composition back to the *ferrite* composition, divided by the total distance between the two phase compositions.

$$\text{Fraction pearlite} = \frac{C_{\text{alloy}} - C_{\text{ferrite}}}{C_{\text{eutectoid}} - C_{\text{ferrite}}} = \frac{0.45 - 0.02}{0.76 - 0.02} = \frac{0.43}{0.74} \approx 0.58$$

$$\text{Fraction proeutectoid ferrite} = 1 - 0.58 = 0.42$$

So a slowly-cooled 1045 specimen should be roughly **58% pearlite and 42% proeutectoid ferrite** by area — a number you can sanity-check directly against your own micrograph in the lab (this is the basic idea behind the **ASM grain-size / phase-fraction estimation** you'll practice in Section 7.5, Part B).

Teaching demo — drag the carbon content slider to see how the pearlite fraction shifts (try 0.20 for a low-carbon steel like 1018, or 0.60 for a higher-carbon steel).

Show the live R code behind this calculator

```
#| label: metallurgy-lever-rule-live
C_alloy <- 0.20       # try 0.15-0.20 for 1018, up to 0.76 for eutectoid steel
C_ferrite <- 0.02
C_eutectoid <- 0.76

frac_pearlite <- (C_alloy - C_ferrite) / (C_eutectoid - C_ferrite)
frac_ferrite <- 1 - frac_pearlite

cat(sprintf("Pearlite fraction: %.1f%% (alloy = %.2f%% C)\n", frac_pearlite * 100, C_al
cat(sprintf("  Proeutectoid ferrite fraction    %.1f%%\n", frac_ferrite * 100))

par(mar = c(5, 5, 3, 1))
barplot(c(Ferrite = frac_ferrite * 100, Pearlite = frac_pearlite * 100),
        col = c("#85C1E9", "#2E4053"), ylim = c(0, 100), ylab = "Fraction of microstruc
        main = "Microstructure at room temperature")
```

6.5.5 Reading Carbon Content from the Steel Numbering System

The **AISI/SAE** designation system lets you read approximate composition straight off the grade number, without looking anything up:

- **10xx series — plain carbon steel.** The last two digits give the carbon content in **hundredths of a percent**. AISI **1018** = 0.18% C (nominal; actual range is typically 0.15–0.20%). AISI **1045** = 0.45% C (typically 0.43–0.50%).
- **11xx series — resulfurized (free-machining) carbon steel.** Same carbon rule, with added sulfur for machinability.
- **15xx series — manganese steel.** Same last-two-digit carbon rule, with 1.00–1.65% Mn (versus about 0.30–0.90% Mn in a plain 10xx grade). AISI **1566**, used in your case-hardening/tensile lab specimen, is approximately 0.60–0.71% C with 0.85–1.15% Mn — a medium-high-carbon, manganese-boosted grade chosen specifically because the extra Mn improves hardenability (Section 6.6) for case hardening.
- **41xx series — chromium-molybdenum alloy steel.** The first two digits identify the alloy family (41 = Cr-Mo), and the last two digits are *still* hundredths-of-a-percent carbon. AISI **4140** = 0.40% C, alloyed with chromium and molybdenum for much higher hardenability than a plain carbon steel of the same carbon content — this is why 4140 is a common choice for shafts that need to harden all the way through a larger section.

Industry application

Automotive: gears and shafts are commonly 41xx or 86xx alloy steels, case-hardened (Section 6.6 and Section 7.3) for a hard wearing surface over a tough core. **Food processing:** fasteners and brackets exposed to washdown are more often stainless (Part B, Section 7.6) than plain carbon steel, specifically to avoid the corrosion a 10xx grade would suffer. **Defense:** armor and structural steels are frequently low-alloy grades like 4130/4140, chosen for the combination of strength and toughness that alloying, not just carbon content, provides.

Quick check: decoding a grade

Question: A supplier ships you round bar stamped "AISI 1030." Without looking anything else up, what do you already know about it?

Answer: It's a plain carbon steel (10xx series) with approximately 0.30% carbon — hypoeutectoid, roughly midway in carbon content between 1018 (0.18%) and 1045 (0.45%). You'd expect it to harden more than 1018 but less than 1045-1060 grades, and to have intermediate as-rolled strength and ductility.

6.5.6 Try it yourself: how much pearlite?

Four plain carbon steels are cooled slowly to room temperature. Use the lever rule, with 0.02% C for ferrite and 0.76% C for the eutectoid, to find how much of each microstructure is pearlite and how much is proeutectoid ferrite.

Lever rule: ferrite and pearlite

Fill in both columns, to the nearest percent.

Steel	Carbon (%)	Pearlite (%)	Ferrite (%)
AISI 1010	0.10		
AISI 1030	0.30		
AISI 1050	0.50		
AISI 1070	0.70		

Show the worked solution

Steel	Carbon (%)	Pearlite (%)	Ferrite (%)
AISI 1010	0.1	11	89
AISI 1030	0.3	38	62
AISI 1050	0.5	65	35
AISI 1070	0.7	92	8

The pearlite fraction rises almost in step with carbon: every 0.1% C adds about 13.5 points of pearlite. Since pearlite is the hard constituent and ferrite the soft one, this table is also a rough ranking of strength, from 1010 (soft, very formable) to 1070 (nearly all pearlite, used for springs and wear parts).

6.5.7 Try it yourself: classify the steel

Read the carbon content from each designation, compare it with the eutectoid composition (about 0.76% C), and classify the steel.

Hypoeutectoid, eutectoid or hypereutectoid?

Classify each steel by its carbon content.

Categories: Hypoeutectoid / Eutectoid / Hypereutectoid

1. AISI 1018 _____
2. AISI 1020 _____
3. AISI 1045 _____
4. AISI 1060 _____
5. A plain carbon steel with 0.76% C _____
6. AISI 1095 _____
7. A plain carbon tool steel with 1.2% C _____

Answers: 1 Hypoeutectoid; 2 Hypoeutectoid; 3 Hypoeutectoid; 4 Hypoeutectoid; 5 Eutectoid; 6 Hypereutectoid; 7 Hypereutectoid.

6.6 Heat Treatment

6.6.1 Two Different Things Called "Heating the Steel"

Before going further, it's worth separating two ideas that both involve heating steel and both end up changing its grain structure, but that work by completely different mechanisms — mixing them up is one of the most common sources of confusion in this material.

1. **Recrystallization** (Section 6.4.1) — what happens during hot rolling, and what a *process anneal* does to cold-worked wire or sheet between drawing passes. This happens at a comparatively modest temperature (roughly 500–700 °C for plain carbon steel) that stays **below A1 (727 °C)** — nowhere near the iron-carbon phase diagram's transformation lines in Figure 6.6. Iron never leaves its BCC (ferrite) structure. All that changes is **grain shape and size**: strained, elongated grains tangled with dislocations are replaced by new, strain-free, equiaxed grains. No new phase forms — it's a purely mechanical "reset" of the existing phase.

2. **The five heat treatments below** (full annealing, normalizing, quench hardening, tempering, stress relieving) are read directly off the phase diagram, and the four of them that actually change the microstructure (all but stress relieving) require heating **above A1, A3, or Acm** — deliberately crossing into the **austenite (FCC) field** in Figure 6.6. The iron itself changes crystal structure: BCC ferrite transforms to FCC austenite, which then transforms back on cooling into ferrite + pearlite, or — if cooled fast enough to beat the TTT/CCT nose (Section 6.6.3) — into martensite instead. This is a genuine **phase transformation**, not just a grain-shape reset, and it's exactly why austenite (Section 6.5.2) matters so much: it's the one starting phase every treatment below is built on.

	Recrystallization (hot rolling, process annealing)	Heat treatments in Section 6.6.4
Typical temperature	~500–700 °C (steel)	Above A1 (727 °C), often well above A3/Acm
Where on Figure 6.6	Below A1 — no phase boundary crossed	Above A1/A3/Acm — enters the austenite field
What actually changes	Grain shape/size only; new strain-free equiaxed grains replace deformed ones	The crystal structure itself: BCC ferrite FCC austenite (martensite on fast cooling)
Governed by	The recrystallization temperature (~0.3–0.5 × absolute melting point)	The Fe-C phase diagram (A1/A3/Acm) plus TTT/CCT cooling-rate diagrams

! Watch the word "annealing" — it means two different things

The **full annealing** described in Section 6.6.4 below is deliberately heated *above* A3 (or A1) into austenite, then cooled slowly back through the transformation — a genuine round-trip phase change, same mechanism as the slow-cooling walkthrough in Section 6.5.4. That is a different process from the **process anneal / recrystallization anneal** described in Section 6.4.1, which stays below A1 the entire time and only recrystallizes the existing ferrite grains. Both soften the steel; only one of them ever leaves the ferrite phase.

6.6.2 Martensite: The Phase That Isn't on the Diagram

Go back to Figure 6.6 and look for martensite. You won't find it — **martensite does not appear anywhere on the equilibrium iron-carbon phase diagram**, because that diagram only shows phases that form under *slow, near-equilibrium* cooling, where atoms have time to diffuse into their lowest-energy arrangement.

Quenching — cooling austenite *fast* — denies carbon the time it needs to diffuse out of the iron lattice and form cementite. Instead, the FCC (Section 6.3.2) austenite lattice flips to something close to BCC almost instantly (a **diffusionless, shear-type transformation**, not the same slow nucleation-and-growth process that makes pearlite), but it's now trying to cram far more carbon into that structure than BCC ferrite can normally hold. The lattice distorts into a stretched, tetragonal shape to accommodate the trapped carbon — this stressed, supersaturated, distorted structure is **martensite**. It's not a new equilibrium phase; it's iron and carbon frozen in a highly stressed, metastable arrangement they'd rather not be in.

That internal distortion is exactly why martensite is **extremely hard, and extremely brittle** — the trapped carbon and distorted lattice choke off dislocation motion almost completely (recall Section 6.3.4: no dislocation motion means no ability to deform plastically before fracturing).

6.6.3 TTT and CCT Diagrams, Simply

If the phase diagram tells you *what forms at equilibrium*, a **TTT diagram** (Time-Temperature-Transformation) tells you *how fast you have to cool to avoid it.* A TTT diagram plots temperature (vertical) against time on a log scale (horizontal), and shows, for one specific steel, how long austenite held at each temperature takes to start and finish transforming into pearlite or other products. The curves typically form a "C" or "nose" shape: transformation is slow near A1 (little driving force) and slow at very low temperatures (little atomic mobility), but fast somewhere in between — the "nose" of the C.

The practical idea you need: if you can cool a part fast enough to get past the nose of that C-curve before it has time to transform, you bypass pearlite formation entirely and the austenite is forced down to a temperature where it transforms to martensite instead. **CCT diagrams** (Continuous Cooling Transformation) are the more realistic cousin of TTT diagrams — instead of holding at a fixed temperature, they show what happens during continuous cooling at different rates, which is what actually happens during quenching. Both types of diagram exist for the same reason: to help you pick a cooling rate (and therefore a quench medium, Section 6.6.6) fast enough to get the microstructure you want.

> ⚠ This is a simplified picture
>
> Real TTT/CCT diagrams are alloy-specific — every grade of steel has its own, and alloying elements (like the Mn in 1566, or the Cr-Mo in 4140) push the "nose" of the C-curve to the right (longer times), which is *exactly* what "hardenability" means (Section 6.6.6). Reading an actual TTT/CCT diagram in detail is beyond this course, but the concept — cooling rate determines which product you get — is the whole reason heat treatment works.

6.6.4 The Five Heat Treatments

All five of the following start the same way — heating the steel into (or just above) the austenite region — and differ almost entirely in **cooling rate**.

Annealing — heat above A3 (hypoeutectoid) or above A1 (near-eutectoid), hold, then cool **very slowly**, typically inside the switched-off furnace. Purpose: maximum softness, refined and relieved grain structure, best machinability, removal of prior cold work or casting stresses. Product: coarse pearlite + proeutectoid ferrite (or cementite), similar to the equilibrium walkthrough in Section 6.5.4, but with somewhat larger, more relaxed grains from the very slow cool. Hardness, strength: lowest of the five. Ductility, toughness: highest of the five.

Normalizing — heat above A3/Acm (typically somewhat higher than annealing, to fully re-austenitize and homogenize), then cool in **still air**. Faster than annealing, slower than quenching. Purpose: refine and homogenize grain size (especially after casting or hot working, where grains can be coarse and uneven — see Equation 6.1 for why this matters), produce a more uniform, moderately fine pearlite + ferrite structure. Product: finer pearlite than annealing. Hardness, strength: moderately higher than annealed. Ductility: slightly lower than annealed, still good.

Quench hardening — heat above A3/Acm, then cool **very fast** (water, brine, or oil — Section 6.6.6) to beat the TTT/CCT curve's nose. Purpose: maximum hardness and strength. Product: martensite (fully, if the section is thin enough and cooled fast enough all the way through — otherwise a mix of martensite and other transformation products in the core, which is exactly what "hardenability" governs). Hardness, strength: highest of the five, by a wide margin. Ductility, toughness: **lowest** of the five — as-quenched martensite is dangerously brittle and is essentially never used in service without tempering.

Tempering — **reheat** already-quenched martensite to a temperature below A1 (commonly 150–650 °C depending on the target properties), hold, then cool (rate is not critical here). Purpose: relieve the internal stress of the quench and trade away some hardness for a large gain in toughness and ductility. Product: **tempered martensite** — very fine cementite particles precipitating out of

the supersaturated martensite lattice, partially relieving the distortion while leaving a fine, strong microstructure behind. Hardness, strength: lower than as-quenched, but still much higher than annealed — and the exact hardness is tunable by choice of tempering temperature (Figure 6.7). Ductility, toughness: much improved over as-quenched martensite.

Stress relieving — heat to a relatively low temperature (well below A1, commonly 500–650 °C for steel), hold, then cool slowly (usually in air). Purpose: **not** to change hardness or microstructure significantly — just to let locked-in residual stress (from welding, machining, or cold work — recall Section 6.4.4) relax out, without softening the part or changing its as-supplied strength very much. Product: same phases as before treatment; grain size and hardness essentially unchanged. This is the treatment you'd specify for a welded CRS frame that keeps warping after machining.

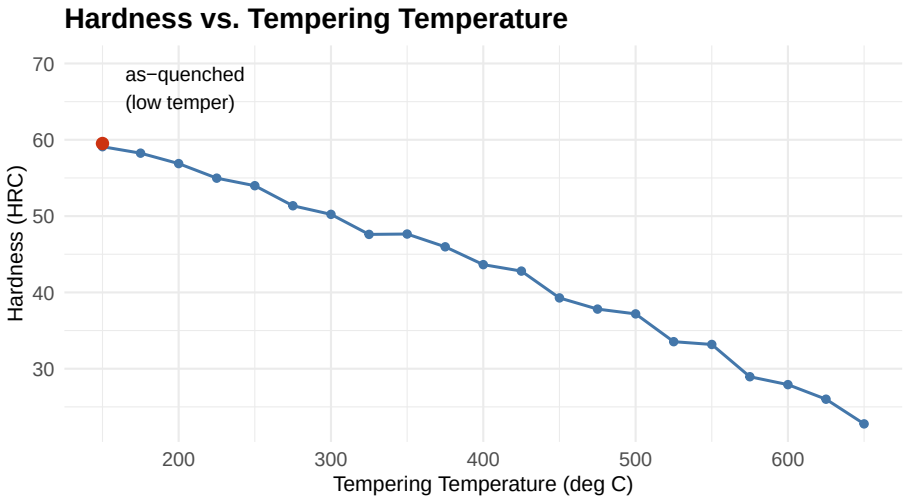


Figure 6.7: Typical hardness of a quenched medium-carbon steel as a function of tempering temperature (simulated, illustrative trend only – actual curves vary by alloy and carbon content).

6.6.5 The Hardness/Toughness Trade-Off

Look again at Figure 6.7: hardness falls steadily as tempering temperature rises. What isn't shown, but happens at the same time, is that **toughness and ductility rise** as hardness falls — you are always trading one for the other by choosing a tempering temperature. This is not a flaw in the process; it's the entire point of tempering. As-quenched martensite is hard enough to resist wear but far too brittle to survive an impact or a sudden overload without cracking — which is exactly **why a quenched part is (essentially) always tempered** before it goes into service. The only time you'd deliberately leave a specimen

in the as-quenched condition is for a comparison specimen in a lab, like the one you'll test in Section 7.3.

Predict and explain: choosing a tempering temperature

Question: A cutting tool needs to hold a sharp edge (high hardness) but a structural bracket subjected to shock loads needs to survive impacts without cracking (high toughness). Both start from the same quenched steel. Which one gets tempered at a higher temperature, and why?

Answer: The bracket gets tempered at the *higher* temperature. Higher tempering temperature moves further down (right) on Figure 6.7 — lower hardness, but higher toughness and ductility, which is what an impact-loaded part needs. The cutting tool is tempered at a *lower* temperature to keep as much hardness (and edge retention) as tempering allows while still relieving enough of the dangerous as-quenched brittleness to be usable.

6.6.6 Quench Media

Not every steel, and not every part shape, needs (or can survive) the fastest possible quench. Faster cooling gets further past the TTT curve's nose (more martensite, deeper hardening) but also builds up more thermal and transformation stress — and **distortion and cracking risk rise right along with cooling severity.**

Table 6.3: Typical relative severity and risk by quench medium. Actual cooling curves depend heavily on agitation, temperature, and part geometry — always follow the specific process specification for a given part.

Quench medium	Relative cooling severity	Typical use	Distortion / cracking risk
Water	Fastest	Plain carbon steels with low hardenability, where a fast quench is needed just to get any martensite at all	**Highest** — most prone to cracking, especially in parts with sharp corners or sections of varying thickness
Brine (salt water)	Faster than plain water	Similar to water, when an even faster, more uniform quench is needed	Highest — similar to or slightly worse than water

Quench medium	Relative cooling severity	Typical use	Distortion / cracking risk
Oil	Moderate	Alloy steels (e.g. 4140) with enough hardenability to harden at a gentler cooling rate	Moderate
Air	Slowest (of the "quenching" media)	Air-hardening tool steels with very high hardenability	Lowest

Industry application

Automotive: gear and shaft quenching is closely controlled (often oil, sometimes polymer quenchants that fall between water and oil in severity) specifically to balance hardness against the distortion that would throw a precision gear tooth profile out of tolerance. **Defense:** thick armor plate sections are quenched with carefully engineered media and agitation to harden through the section without the cracking that a plain water quench would cause in a thick part.

6.6.7 Why Low-Carbon Steel Barely Hardens

Go back to Section 6.6.2: martensite's hardness comes from carbon atoms forced into, and distorting, the iron lattice. **A steel with very little carbon to begin with — like 1018, at roughly 0.18% C — simply doesn't have enough carbon available to distort the lattice very much, no matter how fast you quench it.** You can absolutely form martensite in 1018 with a fast enough quench, but the resulting hardness increase is modest compared to a medium- or high-carbon steel, because the martensite itself is low-carbon and much less distorted.

This is precisely why components that need a hard, wear-resistant surface but are made from a low-carbon steel (for toughness, weldability, or cost) are **case hardened** — carbon (or nitrogen) is diffused into just the surface layer before quenching, so *only the case* has enough carbon to form high-hardness martensite, while the low-carbon core stays comparatively soft and tough. You'll measure exactly this effect — a hardness gradient from a carbon-rich case to a low-carbon core — in the Vickers microhardness traverse of the AISI 1566 specimen in Section 7.3 (Part B). The rest of this section explains *how* that carbon actually gets in.

6.6.8 Carburizing: How Carbon Gets Into the Case

Carburizing is a two-part process: first carbon is added to the surface and diffused inward, then the part is quenched (sometimes directly from the carburizing temperature, sometimes after cooling and reheating) to turn the now carbon-rich case into martensite.

Why it has to be done hot, in the austenite range. Recall from Section 6.5.2 that **ferrite can dissolve almost no carbon (about 0.02 wt% at best), while austenite can dissolve up to 2.1 wt%.** Carburizing only works because the part is held *above A3* — fully austenitic — so the FCC lattice has room to actually take carbon into solution at the surface. Try to pack the same carbon-rich atmosphere around room-temperature (ferritic) steel and essentially none of it would dissolve in; this is yet another place where the phase diagram, not just "heat," is doing the real work (Section 6.6.1).

How the carbon is supplied, from most to least common in industry today:

Table 6.4: Common ways carbon is introduced at the surface before it diffuses inward. All three rely on the same diffusion mechanism once carbon reaches the surface — they differ only in how carbon gets delivered there.

Method	How carbon reaches the surface	Typical temperature
Gas carburizing	Part sits in a furnace atmosphere of hydrocarbon gas (e.g. methane or propane) mixed with a carrier gas; the gas decomposes (*cracks*) at the hot steel surface, depositing carbon atoms directly onto it	~870–950 °C
Liquid (salt bath) carburizing	Part is immersed in a molten cyanide- or carbonate-based salt bath that supplies carbon (and often some nitrogen) at the surface	~850–950 °C

Method	How carbon reaches the surface	Typical temperature
Pack carburizing	Part is packed in a sealed container with a carbon-rich solid compound (charcoal plus an energizer) and heated for hours — the oldest method, now mostly a classroom/historical example rather than a production process	~900–950 °C

Diffusion does the rest. Once carbon atoms are available at the surface, they move inward by ordinary interstitial diffusion — hopping between the gaps in the iron lattice — driven simply by the concentration difference between the carbon-rich surface and the lower-carbon interior (the same kind of concentration-gradient-driven diffusion you'd see with a drop of food coloring spreading through still water, just far slower in a solid). Two things control how deep the carbon gets, and both are shop-floor levers a technician actually sets on a carburizing furnace:

- **Temperature** — diffusion speeds up sharply as temperature rises (more thermal energy, more successful atomic jumps per second), so a hotter furnace drives carbon in faster for the same hold time.
- **Time** — case depth does *not* grow in direct proportion to time. Diffusion depth scales roughly with the **square root of time**, so doubling the case depth takes roughly *four times* as long, not twice — a practical rule worth remembering when a process plan asks for a deeper case.

After diffusion, quench. Once the furnace cycle has built the carbon gradient shown in Figure 6.8, the part is quenched while still (or after being reheated to be) fully austenitic. The high-carbon case has plenty of carbon to form hard martensite; the low-carbon core — recall the opening of this section — barely hardens even though it was quenched too, because it never had the carbon to begin with. The result is one part with two microstructures: a hard, wear-resistant martensitic case over a tough, crack-resistant low-carbon core, exactly the hardness-vs-depth profile you'll traverse with the Vickers microhardness tester in Section 7.3.

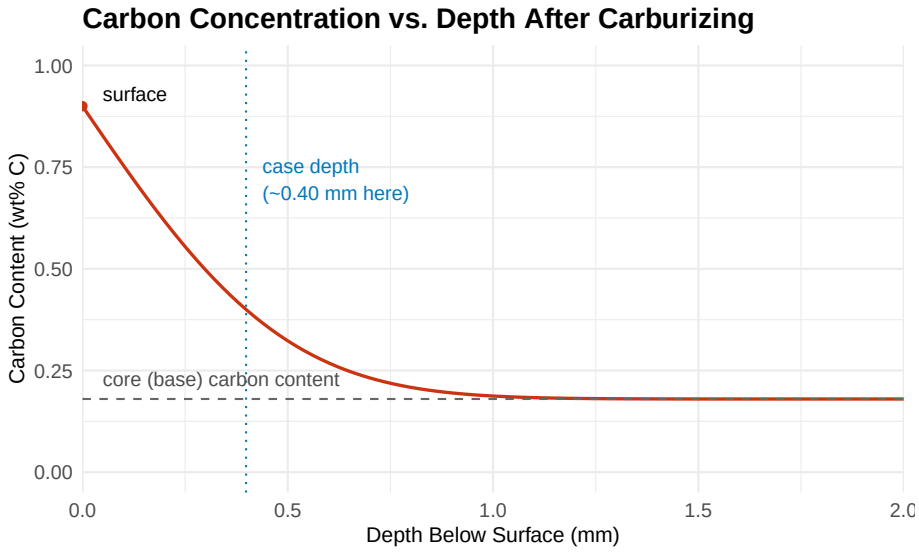


Figure 6.8: Illustrative carbon concentration profile after gas carburizing (schematic, not measured data). Carbon is highest at the surface, set by the furnace atmosphere, and falls off smoothly with depth – an error-function-shaped curve, which is the solution to Fick's second law for this situation – down to the steel's original core carbon content. 'Case depth' is conventionally defined as the depth at which carbon content falls to a chosen threshold, often 0.4% C.

> **i** Nitriding: a lower-temperature alternative
>
> **Nitriding** diffuses nitrogen instead of carbon, forming very hard iron nitride compounds at the surface. The practical advantage: common (ferritic) nitriding is done at only ~500–550 °C, **below A1** — the part never enters the austenite field at all, so no quench is needed afterward and distortion is much lower than carburizing. The trade-off is a much thinner case and a longer processing time. Carburizing has to cross into austenite because it needs austenite's much higher carbon solubility; nitriding doesn't need to, because iron nitrides can form directly in the ferrite structure.

Table 6.5: Summary of the five heat treatments: process, resulting microstructure, and relative property effects.

Treatment	Cooling method	Resulting microstructure	Relative hardness	Relative toughness/ductility
Annealing	Very slow (furnace cool)	Coarse pearlite + proeutectoid ferrite/cementite	Lowest	Highest
Normalizing	Air cool	Finer, more uniform pearlite + ferrite	Low-moderate	High
Quench hardening	Fast (water/brine/oil)	Martensite (untempered)	Highest	Lowest (brittle)
Tempering (after quench)	Reheat below A1, air cool	Tempered martensite	High (tunable down by temper temp)	Moderate-high (tunable up by temper temp)
Stress relieving	Low-temp heat, slow cool	Unchanged from starting structure	Unchanged	Unchanged (residual stress reduced)

Practice: connect this to the hardness lab

Question: In the hardness lab (Section 7.3, Part B) you'll be given **annealed, quenched, and normalized** specimens of the same steel and asked to explain why their hardness differs. Using only what's in this section, rank the three

from lowest to highest expected hardness, and explain why.

Answer: Annealed < Normalized < Quenched. Annealed has the slowest cooling rate and coarsest pearlite (softest, most ductile). Normalizing cools faster than annealing (air vs. furnace), giving finer pearlite and moderately higher hardness. Quenching is fast enough to bypass pearlite formation entirely and produce martensite, which is far harder than any pearlite/ferrite mixture. (Note that the quenched specimen in that lab is *not* tempered — see why that matters in Section 6.6.5.)

6.7 Key Takeaways

- Metals are crystalline; grains form because solidification starts at many points at once. Smaller grains mean more grain boundaries, which mean higher strength (Hall-Petch, Equation 6.1).
- Plastic deformation happens by dislocations moving, not by whole planes of atoms sliding at once. Anything that blocks dislocation motion — grain boundaries, second phases, dissolved atoms — strengthens the metal.
- Hot-rolled steel has equiaxed grains, low residual stress, and lower strength; cold-rolled steel has elongated grains (visible in longitudinal sections only), high residual stress, and higher strength/hardness but lower ductility. Machining CRS can release that stress and cause warping.
- The iron-carbon phase diagram (steel region, 0–2.1% C) shows ferrite (, BCC), austenite (, FCC), and cementite (Fe C). Pearlite is a ferrite/cementite microstructure formed from eutectoid-composition austenite at 727 °C (A1).
- Hypoeutectoid steels (< 0.76% C) form proeutectoid ferrite first on cooling; hypereutectoid steels (> 0.76% C) form proeutectoid cementite first.
- AISI/SAE 10xx grades: last two digits = hundredths-of-a-percent carbon. 15xx adds manganese; 41xx adds chromium-molybdenum for hardenability.
- Martensite is **not** on the equilibrium phase diagram — it forms only by quenching fast enough to suppress diffusion (past the "nose" of the TTT/CCT curve).
- Annealing, normalizing, quenching, and tempering are all the same basic process (austenitize, then control the cooling and any subsequent reheating) with different cooling rates and purposes. Quenched steel is always tempered to trade some hardness for the toughness needed to survive in service.
- Low-carbon steel barely hardens by quenching alone because there isn't enough carbon to distort the martensite lattice — this is why case hardening exists.

6.8 Glossary

A1 line — the horizontal line at 727 °C on the Fe-C diagram; the eutectoid transformation temperature, below which austenite cannot exist at equilibrium.

A3 line — the curve from 912 °C (pure iron) to the eutectoid point; the boundary between austenite and austenite+ferrite in hypoeutectoid steel.

Acm line — the curve from the eutectoid point to 1147 °C at 2.14% C; the boundary between austenite and austenite+cementite in hypereutectoid steel.

Annealing — heating above the transformation temperature followed by very slow (furnace) cooling, to soften and relieve stress.

Austenite ( ) — the high-temperature, FCC form of iron; can dissolve up to about 2.1 wt% carbon.

Cementite (Fe C) — iron carbide, a hard, brittle compound at 6.7 wt% carbon.

Crystal / crystal lattice — the repeating, ordered 3-D arrangement of atoms in a solid metal.

Dislocation — a line defect in the crystal lattice whose motion is the mechanism of plastic (permanent) deformation.

Equiaxed grains — grains of roughly equal dimension in every direction, typical of hot-worked or recrystallized metal.

Eutectoid point — the composition (0.76–0.8 wt% C) and temperature (727 °C) at which austenite transforms directly to pearlite on cooling.

Ferrite ( ) — the low-temperature, BCC form of iron; can dissolve very little carbon (about 0.02 wt% max).

Grain — an individual crystal within a polycrystalline metal.

Grain boundary — the interface between two grains of different crystallographic orientation.

Hall-Petch relationship — the equation relating yield strength to grain size, $\sigma_y = \sigma_0 + kd^{-1/2}$; smaller grains give higher strength.

Hardenability — the depth to which a steel can be hardened by quenching; increased by alloying elements (Mn, Cr, Mo, etc.).

Hypereutectoid steel — steel with more than about 0.76–0.8 wt% carbon; forms proeutectoid cementite on slow cooling.

Hypoeutectoid steel — steel with less than about 0.76–0.8 wt% carbon; forms proeutectoid ferrite on slow cooling.

Martensite — a hard, brittle, supersaturated, distorted phase formed by quenching austenite fast enough to suppress diffusion; does not appear on the

equilibrium phase diagram.

Mill scale — the blue-grey iron oxide layer that forms on the surface of steel during hot rolling.

Normalizing — heating above the transformation temperature followed by air cooling, to refine and homogenize grain structure.

Pearlite — a lamellar (layered) microstructure of alternating ferrite and cementite, formed from eutectoid-composition austenite.

Phase diagram — a chart showing which phase(s) are stable at a given composition and temperature.

Proeutectoid — a phase (ferrite or cementite) that forms before the eutectoid reaction, from austenite that is not yet at the eutectoid composition.

Quench hardening — heating above the transformation temperature followed by rapid cooling (water, brine, or oil), to form martensite.

Recrystallization temperature — the temperature above which deformed grains are replaced by new, strain-free grains during hot working.

Residual stress — internal stress locked into a part by uneven deformation or cooling, which can cause distortion when material is later removed.

Stress relieving — a low-temperature heat treatment intended to reduce residual stress without significantly changing hardness or microstructure.

Tempering — reheating quenched (martensitic) steel below A1 to reduce hardness and brittleness while increasing toughness and ductility.

TTT / CCT diagram — Time-Temperature-Transformation / Continuous-Cooling-Transformation diagrams; show how fast a steel must cool to produce a given transformation product.

Unit cell — the smallest repeating block of a crystal lattice (e.g. BCC, FCC, HCP).

6.9 Practice Problems

Problem 1 (calculation): Hall-Petch

Problem: A steel has $\sigma_0 = 70$ MPa and $k = 18$ MPa $\cdot$ mm$^{0.5}$. Estimate its yield strength for a grain size of (a) 0.10 mm and (b) 0.01 mm.

Problem 1: Hall-Petch

- (a) Yield strength at d = 0.10 mm: _____ MPa *(to 0 decimal places)*

- (b) Yield strength at d = 0.01 mm: _____ MPa *(to 0 decimal places)*

Show the worked solution

Answer: (a) $\sigma_y = 70 + 18(0.10)^{-0.5} = 70 + 18(3.16) = 70 + 57 \approx 127$ MPa. (b) $\sigma_y = 70 + 18(0.01)^{-0.5} = 70 + 18(10) = 70 + 180 = 250$ MPa. Refining the grain size by a factor of 10 roughly doubled the yield strength.

Problem 2 (phase diagram reading): classify the steel

Problem: Using Figure 6.6, classify AISI 1080 (0.80% C) as hypoeutectoid, eutectoid, or hypereutectoid, and state which proeutectoid phase (if any) forms on slow cooling.

Answer: 0.80% C is just above the eutectoid composition (~0.76–0.8%), so 1080 is essentially eutectoid to very slightly hypereutectoid. At exactly the eutectoid composition it would form almost 100% pearlite with no proeutectoid phase; if treated as very slightly hypereutectoid, a small amount of proeutectoid cementite would form at grain boundaries before the eutectoid reaction.

Problem 3 (phase diagram reading): lever rule

Problem: Estimate the fraction of proeutectoid ferrite in slowly-cooled AISI 1020 (0.20% C).

Problem 3: lever rule

- Proeutectoid ferrite: _____ % *(to 0 decimal places)*
- Pearlite: _____ % *(to 0 decimal places)*

Show the worked solution

Answer: Fraction pearlite = $(0.20 - 0.02)/(0.76 - 0.02) = 0.18/0.74 \quad 0.24$. Fraction proeutectoid ferrite $\quad 1 - 0.24 = 0.76$, i.e. about 76% ferrite, 24% pearlite — consistent with 1020 being a low-carbon, mostly-ferritic, relatively soft and ductile steel.

Problem 4 (predict and explain): grade comparison

Problem: You have equal-diameter round bars of AISI 1018 and AISI 1045 in the annealed condition. Predict which will be harder and explain why, using the lever rule idea from Problem 3.

Answer: 1045 (0.45% C) will be harder than 1018 (0.18% C). 1045 contains a much larger fraction of pearlite (see the worked example in Section 6.5.4 — about 58% pearlite) than 1018 would (a fraction similar to Problem 3's 1020 result, well under 30%). More pearlite means more hard cementite lamellae blocking dislocation motion, so higher hardness even before either is quenched.

Problem 5 (predict and explain): heat treatment selection

Problem: A technician needs to machine a batch of 4140 shafts to a tight tolerance, and the shafts have already been welded into sub-assemblies with brackets. Machining keeps producing slightly bowed shafts. What heat treatment would you recommend before final machining, and why?

Answer: Stress relieving. Welding introduces residual stress; machining then removes material asymmetrically and lets that stress relax, causing the bowing described in Section 6.4.4. A stress-relief treatment (low temperature, well below A1, slow cool) reduces the locked-in stress without significantly softening the shaft or changing its microstructure, so the part holds its dimensions through final machining.

Problem 6 (predict and explain): quenched vs tempered

Problem: Two identical 4140 specimens are quenched in oil. One is tested as-quenched; the other is tempered at 450 °C for one hour before testing. Predict, and explain, how their hardness and Charpy impact energy (toughness) will compare.

Answer: The as-quenched specimen will show the highest hardness but the lowest Charpy impact energy (most brittle) of the two — see Figure 6.7 and Section 6.6.5. The tempered specimen will show lower hardness but substantially higher impact energy, because tempering relieves the internal distortion of the martensite and restores some ability for dislocations to move before fracture.

Problem 7 (calculation): grade decoding

Problem: A print calls out "AISI 4340" for a landing-gear component. What does the grade number tell you about its carbon content and alloy family, and why might a designer choose it over 1040 (same carbon content, no alloy) for this application?

Answer: 4340 is a nickel-chromium-molybdenum alloy steel (43 series) with 0.40% carbon (same carbon content as 1040). The alloying elements dramatically increase hardenability (the depth to which the part will fully harden on quenching) and improve strength and toughness compared to a plain-carbon 1040 of the same carbon content — critical for a large, highly-stressed landing-gear component that needs to harden uniformly through a thick section and survive shock loading, which 1040 alone could not reliably achieve.

Continue to Part B: Materials Testing — tensile, hardness, impact, and metallography theory, tied directly to your lab reports.

Chapter 7

Materials Testing: Tensile, Hardness, Impact, and Metallography

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7.1 Learning Objectives

By the end of this chapter, you will be able to:

- Calculate engineering stress, strain, elastic modulus, percent elongation, and percent reduction of area from tensile test data.
- Read a stress-strain curve region by region and connect each region to what is happening to the metal's dislocations and grains.
- Explain what each hardness scale (Rockwell B/C, Brinell, Vickers, Knoop) measures, when to use it, and why the scales aren't directly interchangeable.
- Explain what a Charpy test measures, predict how a BCC steel's toughness changes with temperature, and describe the resulting fracture surfaces.
- Explain why sectioning, mounting, grinding, polishing, and etching are necessary before a specimen can be examined under a microscope, and read a basic micrograph.
- Compare steel's behaviour under these same tests to aluminum, stainless steel, cast iron, copper/brass, and polymers.

This is Part B of a two-part chapter. Part A covered what steel is made of and how heat treatment changes it — you'll need that background here, especially the phase diagram (Section 6.5) and the five heat treatments (Section 6.6). This part covers *how we measure* the properties heat treatment changes, and connects

directly to your three materials-testing labs: **Hardness Testing (Rockwell and Brinell)**, **Tensile and Microhardness Testing of Case-Hardened Steel**, and **Charpy Impact Testing and the Ductile-to-Brittle Transition** (see the Labs chapter for the full procedures — this chapter does not repeat them).

7.2 Tensile Testing Theory

> **i** Before you read on
>
> Picture pulling on a rubber band, then on a steel paperclip, until each breaks. Which one stretches more before it snaps? Which one "gives" noticeably before it breaks, versus breaking almost without warning? Keep those two objects in mind — you're about to learn the vocabulary for exactly what you already felt.

7.2.1 Stress and Strain

A tensile test pulls a standardized specimen apart at a controlled rate while recording load and extension — the same basic principle whether the machine is testing a bolt for an automotive supplier or armor plate for a defense contractor. Raw load and extension depend on specimen size, so we convert them to size-independent quantities:

$$\sigma = \frac{P}{A_0} \tag{7.1}$$

$$\varepsilon = \frac{\Delta L}{L_0} \tag{7.2}$$

where σ is **engineering stress** (MPa, or psi/ksi), P is the applied load (N or lbf), A_0 is the *original* cross-sectional area, ε is **engineering strain** (dimensionless, or %), ΔL is the change in length, and L_0 is the *original* gauge length. Both use the *original* dimensions throughout the test — that's what makes them "engineering" stress and strain, as opposed to *true* stress and strain, which continuously account for the specimen's shrinking cross-section (true stress/strain is more accurate at large deformation but is beyond this course; ASTM E8/E8M, the standard tensile test method for metals, is written around engineering stress and strain).

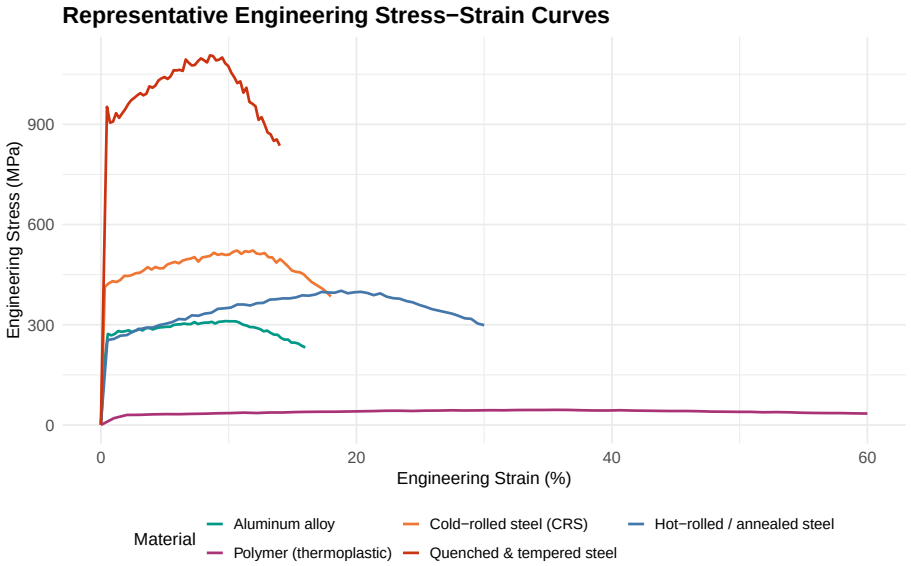


Figure 7.1: Representative engineering stress-strain curves for several material conditions (schematic, typical shapes only – actual curves depend on exact grade, condition, and specimen). CRS and quenched-and-tempered steel show higher strength but lower ductility than HRS/annealed steel; aluminum shows no sharp yield point; the polymer shows a much lower modulus and a very different curve shape.

7.2.2 The Stress-Strain Curve, Region by Region

Follow any one curve in Figure 7.1 from left to right — this is the region-by-region reading you'll do on your own tensile lab data:

- **Elastic region:** the initial straight-line portion. Stress and strain are directly proportional — this is **Hooke's Law**, $\sigma = E\varepsilon$, where E is the **modulus of elasticity** (Young's modulus), the slope of this line. Physically, atoms are being stretched slightly away from their equilibrium spacing, and the bonds pull them right back the instant the load is removed — no dislocations have moved yet, so this deformation is fully **recoverable**. Steel's modulus (~200 GPa) is essentially the same whether the steel is annealed, cold-worked, or quenched and tempered — heat treatment barely touches elastic modulus, because it's a property of the atomic bonding itself, not the microstructure. This is why all the steel curves in Figure 7.1 start with the same slope, while aluminum (weaker metallic bonding, ~69 GPa) and the polymer (van der Waals and covalent bonding between long chains, ~2 GPa or less) start with much shallower slopes.
- **Yield point:** the stress at which the material stops behaving elastically and dislocations begin moving enough to produce permanent (plastic) deformation. Some steels show a sharp, obvious yield point (a momentary load drop) because dislocations are initially "pinned" by interstitial carbon/nitrogen atoms and need a burst of extra stress to break free before they can move freely. Most other metals — aluminum, many steels, and virtually all non-ferrous alloys — yield gradually with no sharp point, so engineers define an offset instead: the **0.2% offset yield strength** is found by drawing a line parallel to the elastic region, starting from 0.2% strain, and reading where it intersects the curve.
- **Strain hardening (plastic region):** as dislocations move, they multiply and tangle with each other and with grain boundaries — more and more obstacles pile up in their own path, so more stress is needed to keep deformation going. This is the same physical mechanism, from the atomic level, that makes cold-rolled steel stronger than hot-rolled steel (Section 6.4): CRS has *already* been strain hardened by the rolling process before you ever put it in the tensile machine.
- **Ultimate tensile strength (UTS):** the highest engineering stress reached, at the peak of the curve. Beyond this point, strain hardening in the gauge section can no longer keep up with the loss of cross-sectional area, and deformation localizes.
- **Necking:** the specimen begins to thin locally rather than uniformly along its length (Figure 7.2). Because engineering stress uses the *original* area (Equation 7.1), the calculated engineering stress actually appears to *drop* after UTS even though the true stress at the neck keeps rising — this is purely a bookkeeping effect of using A_0 instead of the shrinking real area.
- **Fracture:** final separation, at the end of the curve.

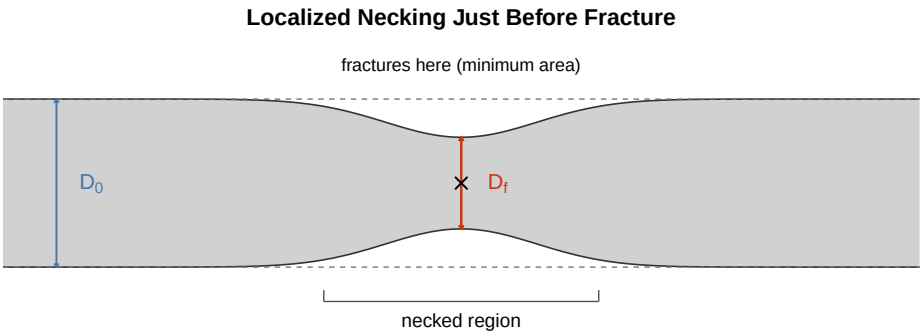


Figure 7.2: Schematic profile of a round tensile specimen showing localized necking (not to scale). Deformation that was uniform along the gauge length concentrates into one local neck once the ultimate tensile strength is passed, and the specimen ultimately fractures at that minimum-diameter cross-section, Df – the same Df that Equation 7.4 uses to calculate %RA.

7.2.3 Percent Elongation and Reduction of Area

Two standard ductility measures are taken *after* fracture, by fitting the broken pieces back together (per ASTM E8/E8M):

$$\%EL = \frac{L_f - L_0}{L_0} \times 100 \tag{7.3}$$

$$\%RA = \frac{A_0 - A_f}{A_0} \times 100 \tag{7.4}$$

where L_f is the final gauge length (measured across the fracture) and A_f is the final (minimum, necked-down) cross-sectional area. Both numbers answer "how much did this material deform before it broke?" — %EL captures length change over the whole gauge length, %RA captures the local necking right at the fracture, so %RA is typically the larger, more sensitive number for very ductile materials.

7.2.4 Ductile vs. Brittle Fracture Surfaces

A **ductile** fracture (typical of annealed or hot-rolled steel, aluminum, copper) shows visible necking and a rough, fibrous, "cup-and-cone" surface: a flat central fibrous zone where microscopic voids nucleated (often at inclusions), grew, and

coalesced, surrounded by a shear lip at roughly 45° where final separation occurred by shear. A **brittle** fracture (typical of untempered martensite, cast iron, or any ductile metal tested below its ductile-to-brittle transition temperature — Section 7.4) shows little to no necking and a flat, bright, crystalline-looking surface, because the crack propagated rapidly along (or across) grains with almost no plastic deformation absorbing energy first.

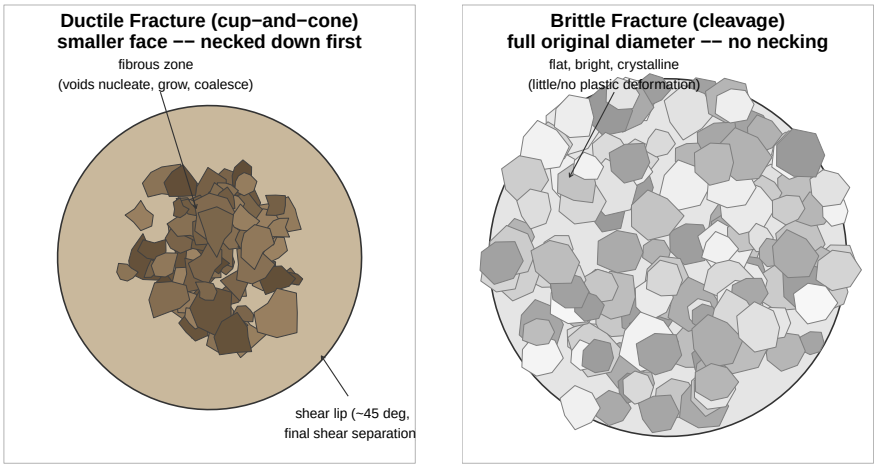


Figure 7.3: Schematic (not photographic) comparison of the two fracture-surface types, drawn at the same scale. Ductile fracture (left) shows a rough, dull, fibrous central zone – where microscopic voids nucleated, grew, and coalesced – ringed by a smoother shear lip at roughly 45 degrees, and the whole face is visibly smaller than the original cross-section because of necking. Brittle fracture (right) shows a flat, bright, crystalline-looking surface with no necking, drawn here at the specimen's full original diameter for comparison.

Industry application

Automotive: crash-worthiness design deliberately uses ductile, lower-strength steel in some body structures so they crumple (absorbing crash energy through large plastic deformation) rather than shattering. **Food processing:** conveyor components are chosen for enough ductility to dent rather than fracture under an accidental impact, since a fractured piece in a food line is a contamination hazard. **Defense:** fractography (reading a fracture surface) is standard failure-analysis practice for a cracked component — the surface itself tells you whether the failure was ductile overload, brittle fracture, or fatigue, often without needing to test another sample.

7.2.5 Worked Examples

```
# Tensile test data for a round specimen, per ASTM E8/E8M style reporting
D0 <- 12.83     # mm, original diameter (0.505 in specimen)
L0 <- 50.0      # mm, original gauge length
P_yield <- 41500   # N, load at 0.2% offset yield
P_max   <- 58200   # N, maximum (UTS) load
Lf <- 61.8      # mm, final gauge length after fracture
Df <- 9.40      # mm, minimum diameter after fracture

A0 <- pi * (D0/2)^2
Af <- pi * (Df/2)^2

sigma_yield <- P_yield / A0
sigma_uts   <- P_max / A0
pct_elong   <- (Lf - L0) / L0 * 100
pct_ra      <- (A0 - Af) / A0 * 100

cat(sprintf("Original area A0 = %.2f mm^2\n", A0))
cat(sprintf("0.2%% offset yield strength = %.1f MPa\n", sigma_yield))
cat(sprintf("Ultimate tensile strength   = %.1f MPa\n", sigma_uts))
cat(sprintf("Percent elongation  = %.1f%%\n", pct_elong))
cat(sprintf("Percent reduction of area = %.1f%%\n", pct_ra))
```

```
Original area A0 = 129.28 mm^2
0.2% offset yield strength = 321.0 MPa
Ultimate tensile strength   = 450.2 MPa
Percent elongation  = 23.6%
Percent reduction of area = 46.3%
```

Your turn — change the load and elongation below. A brittle specimen would show a much smaller `Lf - L0` and `D0 - Df`.

Show the live R code behind this calculator

```
#| label: testing-modulus-example-live
# Elastic-region data point, well below yield, to estimate modulus of elasticity
D0 <- 12.83        # mm
L0 <- 50.0         # mm
P_elastic <- 15000  # N, a load safely in the straight-line region
dL_elastic <- 0.018 # mm, measured elongation at that load (small!)

A0 <- pi * (D0/2)^2
stress_e <- P_elastic / A0
strain_e <- dL_elastic / L0
E_modulus <- stress_e / strain_e
```

```
cat(sprintf("Estimated modulus E: %.0f GPa (compare to ~200 GPa for steel)\n", E_modulu
cat(sprintf("  Stress at this point     %.1f MPa\n", stress_e))
cat(sprintf("  Strain at this point     %.5f (%.3f%%)\n", strain_e, strain_e * 100))
```

Teaching demo — this specimen breaks well before it necks noticeably. Drag the final length and final diameter to see how brittle behaviour (low %EL, low %RA) looks compared to the ductile example above.

Show the live R code behind this calculator

```
#| label: testing-brittle-example-live
D0 <- 12.83
L0 <- 50.0
Lf <- 50.9       # mm -- very little stretch
Df <- 12.60      # mm -- very little necking

A0 <- pi * (D0/2)^2
Af <- pi * (Df/2)^2
pct_elong <- (Lf - L0) / L0 * 100
pct_ra <- (A0 - Af) / A0 * 100

cat(sprintf("Percent elongation: %.1f%% (%.1f%% reduction of area)\n", pct_elong, pct_r
cat("Low values on both measures, with little visible necking, point to brittle\n")
cat("behaviour -- e.g. an as-quenched, untempered specimen, or cast iron.\n")

par(mar = c(5, 5, 3, 1))
barplot(c(`Elongation` = pct_elong, `Reduction of area` = pct_ra), col = c("#2E86C1", "
        ylim = c(-40, 80), ylab = "Percent (%)", main = "Ductility measures")
abline(h = 0)
```

Quick check

Looking back at Figure 7.1: which curve would you expect from the **as-quenched (untempered)** specimen you'll encounter conceptually in the hardness lab, if it were pulled in tension instead? *(Answer: highest strength, but the shortest, steepest curve of all — even less strain to failure than the quenched-and-tempered curve shown, since tempering is what restores ductility; see Section 6.6.5.)*

7.2.6 Try it yourself: read a stress-strain curve

The curve below is from a tensile test on a low-carbon steel specimen with a **50 mm gauge length**. Read the values off the graph; each small grid square is 20 MPa by 1% strain.

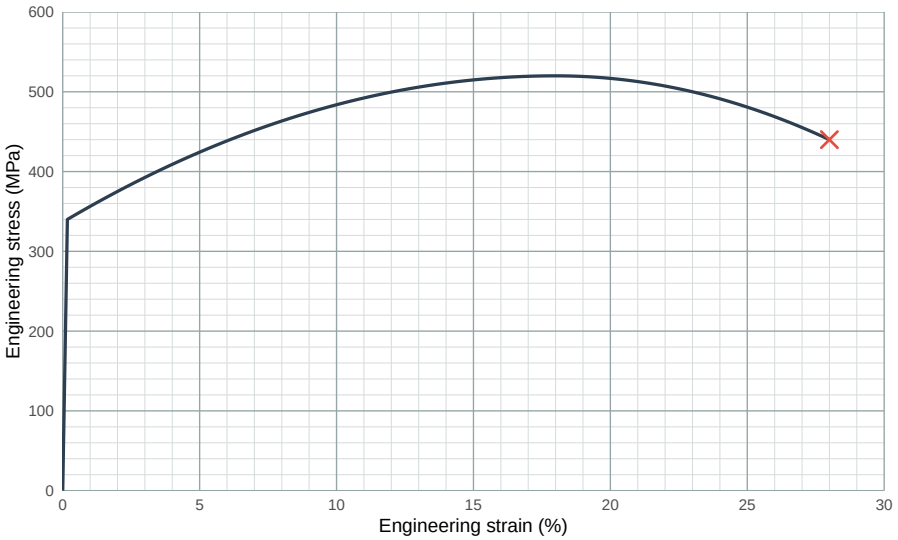


Figure 7.4: Engineering stress-strain curve for the exercise

Read the stress-strain curve

- Yield strength (where the straight line ends): _____ MPa *(to 0 decimal places)*
- Ultimate tensile strength: _____ MPa *(to 0 decimal places)*
- Strain at fracture: _____ % *(to 0 decimal places)*
- Extension of the 50 mm gauge length at fracture: _____ mm *(to 0 decimal places)*

Show the worked solution

- **Yield strength:** the straight elastic line ends at about **340 MPa**. On this scale the elastic line is almost vertical, because a strain of only 0.17% is enough to reach yield.
- **Ultimate tensile strength:** the curve peaks at **520 MPa**, at about 18% strain. Necking starts here.
- **Strain at fracture:** the X is at **28%**, and the stress has fallen to about 440 MPa.
- **Extension at fracture:** 0.28×50 mm = **14 mm**.

The engineering stress falls after the UTS even though the metal itself is still getting stronger. The load is being divided by the *original* area, while the real cross-section in the neck is shrinking fast.

7.2.7 Try it yourself: walk along the curve

A ductile steel specimen is pulled to failure. Put what happens in order, from the first newton of load to the two broken halves.

A tensile test, start to finish

Number these in the correct order:

- _____ Elastic deformation: stress is proportional to strain and fully recoverable
- _____ Fracture
- _____ Necking: deformation concentrates in one place and the engineering stress falls
- _____ Strain hardening: stress keeps rising as the specimen deforms uniformly
- _____ Ultimate tensile strength: the maximum load is reached
- _____ Yield: permanent deformation begins

Answers, in the order listed: 1, 6, 5, 3, 4, 2.

7.2.8 Try it yourself: ductile or brittle?

Eight observations made after a tensile or Charpy test. Decide which kind of fracture each one points to.

Ductile or brittle fracture?

Decide what kind of fracture each observation points to.

Categories: Ductile / Brittle

1. A cup-and-cone fracture surface. _____
2. A dull, fibrous-looking fracture surface. _____
3. The specimen necked visibly before it broke. _____
4. Percent elongation of 30%. _____
5. A flat fracture surface at right angles to the load. _____
6. A bright, granular, faceted fracture surface. _____
7. The two halves fit back together almost perfectly. _____
8. Very little energy absorbed in a Charpy test. _____

Answers: 1 Ductile; 2 Ductile; 3 Ductile; 4 Ductile; 5 Brittle; 6 Brittle; 7 Brittle; 8 Brittle.

7.3 Hardness and Microhardness Theory

> **i** Before you read on
>
> Press your thumbnail into a pencil eraser, then into a wooden pencil. One dents easily, one barely marks at all. That's hardness, felt rather than measured — the rest of this section is just making that comparison precise and repeatable.

7.3.1 What Hardness Actually Measures

Hardness is a material's resistance to *localized permanent (plastic) deformation* — specifically, resistance to being indented by a harder object under a known load. It is not a fundamental material property the way stress and strain are; it's a comparative, standardized test. But because it correlates well with strength (Section 7.3.4) and is fast, cheap, and often non-destructive, it's one of the most-used tests on a shop floor — exactly why it's the quickest way to check what a heat treatment actually did to a part.

7.3.2 Rockwell, Brinell, Vickers, and Knoop

Table 7.1: Hardness test methods relevant to this course. Loads shown are typical, standard values — always confirm the load and scale specified for your test. See ASTM E18 (Rockwell), E10 (Brinell), and E384 (Vickers/Knoop microhardness).

Method	Indenter	Load	What it reads	Typical use
Rockwell B (HRB)	1/16 in (1.588 mm) steel ball	10 kgf minor, 100 kgf major	Depth of penetration, converted to a dial/digital number	Softer materials: annealed or low-carbon steel, aluminum, brass
Rockwell C (HRC)	120° diamond cone ("brale")	10 kgf minor, 150 kgf major	Depth of penetration	Harder materials: hardened/tempered steel, tool steel

Method	Indenter	Load	What it reads	Typical use
Brinell (HBW)	10 mm tungsten carbide ball	typically 3000 kgf (steel)	Diameter of the indentation, converted to a number	Coarse-grained or non-uniform materials (castings), where a large indent averages out local variation
Vickers (HV)	Square-based diamond pyramid	1 gf to 120 kgf, chosen by application	Diagonal length of the (square) indentation	Wide range, including microhardness; especially phases, thin sections, weld zones
Knoop (HK)	Elongated rhombic-based diamond pyramid	Typically under 1 kgf (microhardness range)	Long diagonal length	Very thin sections/coatings and brittle materials, where Vickers' more symmetric indent risks cracking the specimen

Figure 7.5 shows the four indenter shapes from Table 7.1 side by side, drawn to comparable scale. The ball is shared by Rockwell B and Brinell; the Rockwell C cone is a much sharper point than either diamond pyramid; and Vickers vs. Knoop is really a question of how symmetric the pyramid is — Knoop's very shallow, elongated shape (versus Vickers' symmetric, deeper pyramid) is exactly what lets it indent thin sections and brittle materials without cracking them.

Rockwell tests in two steps: a small **minor load** is applied first to seat the indenter and remove the effect of surface roughness, then the much larger **major load** is applied and removed, and the *difference* in penetration depth between the two loads is converted directly to a hardness number on the machine's dial or display — no separate measurement step is needed, which is why Rockwell is the fastest of these tests. **Brinell** requires measuring the indentation diameter afterward (with a measuring microscope) and converting it via a chart or formula.

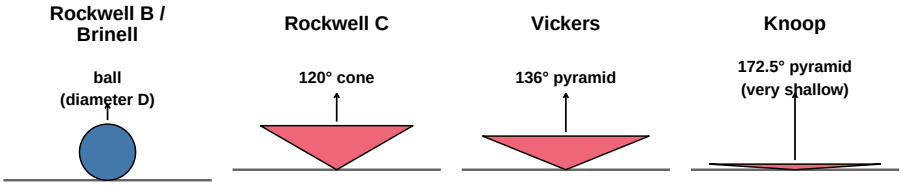


Figure 7.5: Simplified cross-section schematics of the four indenter geometries in Table 7.1 (not to scale). Included angles shown are the actual ASTM standard values for each indenter.

Vickers hardness is calculated directly from the applied force and the indentation diagonal:

$$HV = \frac{1.8544\,F}{d^2} \tag{7.5}$$

where F is the applied load (kgf) and d is the mean of the two diagonals of the square indentation (mm). Because the indenter is always the same 136° diamond pyramid regardless of load, Vickers numbers are directly comparable across a huge range of loads and material hardnesses — one reason it (and the closely related Knoop test) is the standard choice for **microhardness**: measuring the hardness of something too small, too thin, or too localized for a full-size Rockwell or Brinell indent, such as:

- **An individual phase** — e.g., distinguishing the hardness of a specific microconstituent under a microscope, rather than an average over many phases.
- **A case-hardened layer** — exactly what you'll do in your tensile/microhardness lab: a series of small Vickers indents stepping inward from the surface of the AISI 1566 specimen, tracing out how hardness falls from the hard, carbon-rich case to the softer core.
- **A weld heat-affected zone** — mapping hardness across the narrow band of microstructure changed by welding heat.
- **Thin sections or coatings** — where a full-size Rockwell/Brinell indent would be too deep and would measure the substrate underneath instead of the coating itself.

7.3.3 Why Scale Choice Matters, and Why Conversions Are Approximate

Choosing the wrong scale gives a meaningless or invalid reading: Rockwell B's soft steel ball would simply punch straight through a hardened steel specimen without giving a useful number, and Rockwell C's diamond cone on a very soft material can produce an indentation deep enough to bottom out the scale. This is exactly why the hardness lab has you choose the *correct* combination of indenter and load for each specimen before testing.

Because each scale measures a different physical quantity (depth vs. diameter vs. diagonal, at different loads, with different indenter geometries), there is **no exact mathematical conversion** between them — only **empirical conversion tables**, built from testing many samples of similar materials with both methods and tabulating the typical correspondence. **ASTM E140** is the standard reference for these conversions, and it explicitly warns that they are most reliable for plain carbon and low-alloy steels, and become progressively less reliable for other materials, unusual heat treatments, or very high/low hardness values outside the range the table was built from. Treat any converted hardness value as an estimate, not a measurement.

7.3.4 The Approximate Hardness-Tensile Strength Relationship

For steels, hardness and tensile strength correlate strongly enough that a widely used rule of thumb exists:

$$\sigma_{UTS}(\text{MPa}) \approx 3.45 \times \text{HB} \tag{7.6}$$

(where HB is the Brinell hardness number). This comes from the same physical root as the hardness-to-conversion-table warning above: it is an **empirical correlation for steels**, not a law of physics, and it breaks down for cast irons, non-ferrous metals, and materials with an unusual microstructure. Still, it's genuinely useful on a shop floor: a quick hardness test can give a fast, non-destructive *estimate* of tensile strength without cutting up a part for a full tensile test.

7.3.5 Sources of Error

- **Specimen thickness:** too thin, and the indentation can be influenced by the (harder or softer) material or fixture beneath the specimen — a rule of thumb for Rockwell is that specimen thickness should be at least 10 times the indentation depth.
- **Edge distance:** an indentation placed too close to a specimen edge deforms asymmetrically (material can flow sideways out of the edge instead

of being resisted), giving an artificially low reading — standards specify minimum distance from any edge or from another indentation.
- **Surface preparation:** scale, paint, roughness, or curvature under the indenter distorts the reading; hardness testing (unlike a full micrograph) usually needs a reasonably clean, flat surface, not necessarily a mirror polish.
- **Curved surfaces:** testing a round bar or curved part changes the effective contact geometry compared to the flat surface the conversion charts assume, and typically requires a correction factor.

Worked example: Vickers hardness calculation (static)

Problem: A Vickers microhardness test at 500 gf (0.5 kgf) produces an indentation with diagonals of 62.4 µm and 63.8 µm. Calculate the Vickers hardness number.

Solution: Convert diagonals to mm and average: $d = (0.0624 + 0.0638)/2 = 0.0631$ mm.

$$HV = \frac{1.8544 \times 0.5}{0.0631^2} = \frac{0.9272}{0.00398} \approx 233 \text{ HV}$$

For a case-hardened specimen, you'd repeat this calculation at each depth of your traverse and plot HV against distance from the surface — exactly the profile you build in the tensile/microhardness lab.

Teaching demo — change the diagonal measurements and load below. Notice how the *same* diagonal size at a higher load gives a lower HV, and vice versa — this is why comparing hardness numbers taken at different loads without converting first is a common source of confusion.

Show the live R code behind this calculator

```
#| label: testing-vickers-live
F_load <- 0.3       # kgf (300 gf), typical microhardness load
d1 <- 58.0           # micrometers
d2 <- 60.5           # micrometers

d_mm <- ((d1 + d2) / 2) / 1000
HV <- 1.8544 * F_load / d_mm^2

cat(sprintf("Vickers hardness: %.0f HV (mean diagonal %.4f mm)\n", HV, d_mm))
cat(sprintf("  Approx. equivalent UTS   %.0f MPa\n", 3.45 * HV))

par(mar = c(5, 5, 3, 1))
diag_um <- seq(20, 200, by = 1)
plot(diag_um, 1.8544 * F_load / (diag_um / 1000)^2, type = "l", lwd = 2, col = "#2E86C1",
     xlim = c(20, 200), ylim = c(0, 1200), xlab = "Mean diagonal (micrometers)",
```

```
     ylab = "Vickers hardness (HV)", main = "Hardness against indent size at this load"
points(d_mm * 1000, min(HV, 1200), pch = 19, cex = 2, col = "#C0392B")
```

Teaching demo — convert a case-hardened part's Vickers profile toward the Rockwell C scale used to define case depth. (Real conversion requires the ASTM E140 table; this is a simplified linear approximation for illustration only — never use it for a real report.)

Show the live R code behind this calculator

```
#| label: testing-case-depth-live
# Simplified illustrative HV-to-approximate-HRC relationship (NOT for real reporting --
# use the actual ASTM E140 table / chart supplied for your lab)
HV_reading <- 620

approx_HRC <- (HV_reading - 100) / 11.9   # rough linear fit, illustrative only

cat(sprintf("Rough approximate Rockwell C equivalent: %.0f HRC\n", approx_HRC))
cat("Effective case depth is read where the profile crosses HRC 50;\n")
cat("true case depth is read where it crosses HRC 40 (see your lab template).\n")

par(mar = c(5, 5, 3, 1))
hv_axis <- seq(150, 900, by = 10)
plot(hv_axis, (hv_axis - 100) / 11.9, type = "l", lwd = 2, col = "#2E86C1", xlim = c(15
     ylim = c(0, 70), xlab = "Vickers reading (HV)", ylab = "Approximate HRC",
     main = "Simplified HV-to-HRC relationship")
abline(h = c(40, 50), lty = 2, col = "grey50")
points(HV_reading, approx_HRC, pch = 19, cex = 2, col = "#C0392B")
```

7.4 Impact (Charpy) Testing and Temperature Effects

Before you read on

Think about a plastic milk jug left outside all winter versus one fresh from the store. The cold one shatters if you drop it; the room-temperature one just bounces. Same material, same shape — only the temperature changed. That's the ductile-to-brittle transition, and you're about to see steel do almost exactly the same thing.

7.4.1 Toughness vs. Strength

Toughness is the ability to absorb energy before fracturing — it is *not* the same thing as strength. A material can be very strong (high UTS) but have low toughness if it fractures with little plastic deformation (as-quenched martensite is the clearest example: very high strength, very low toughness). On a stress-strain curve, toughness is roughly the *area under the curve* — strength times ductility, not either one alone.

7.4.2 What the Charpy Test Measures

The **Charpy impact test** (ASTM E23) swings a heavy pendulum from a fixed height to strike a small, standardized, notched bar, breaking it in a fraction of a second. The pendulum's height before and after the break tells you how much energy the specimen **absorbed** in fracturing — reported in joules (J) or foot-pounds (ft $\cdot$ lbf). This absorbed energy is the direct measure of impact toughness: a bar that absorbs a lot of energy bent, tore, and consumed energy through plastic deformation before breaking; a bar that absorbs very little snapped almost instantly.

The **notch** is deliberately machined into the specimen, and it matters enormously: a notch is a stress concentrator, and it also constrains the material at the notch root from deforming freely in all directions (a condition called triaxial stress state), which suppresses the material's ability to deform plastically right where the crack needs to start. This makes the Charpy test a much harsher, more realistic test of how a material behaves at a stress concentration — a hole, a fillet, a weld toe, a scratch — than a smooth tensile bar ever could be, which is exactly why so many real engineering structures (and Charpy specimens) fail from notches and stress risers rather than from smooth sections.

7.4.3 The Ductile-to-Brittle Transition Temperature (DBTT)

Recall from Section 6.3.2 (Part A) that **BCC** metals have far fewer easy slip systems than **FCC** metals. At room temperature, thermal energy is enough to help dislocations in a BCC steel overcome the resistance to slip and move relatively freely, so the steel behaves in a tough, ductile manner. Cool it down, and there is progressively less thermal energy available to assist dislocation motion — at some temperature, moving a crack through the material (brittle fracture) simply requires *less* energy than moving dislocations to deform around it (ductile fracture), and the material's failure mode flips from ductile to brittle, often over a surprisingly narrow temperature range. That temperature (range) is the **ductile-to-brittle transition temperature (DBTT)**.

FCC metals (Section 6.3.2) (aluminum, austenitic stainless steel, copper — see Section 7.6) have so many available slip systems that dislocations keep moving relatively easily even at very low temperatures, so they do **not** show a sharp

DBTT — their toughness declines gradually, if at all, as temperature drops, rather than falling off a cliff.

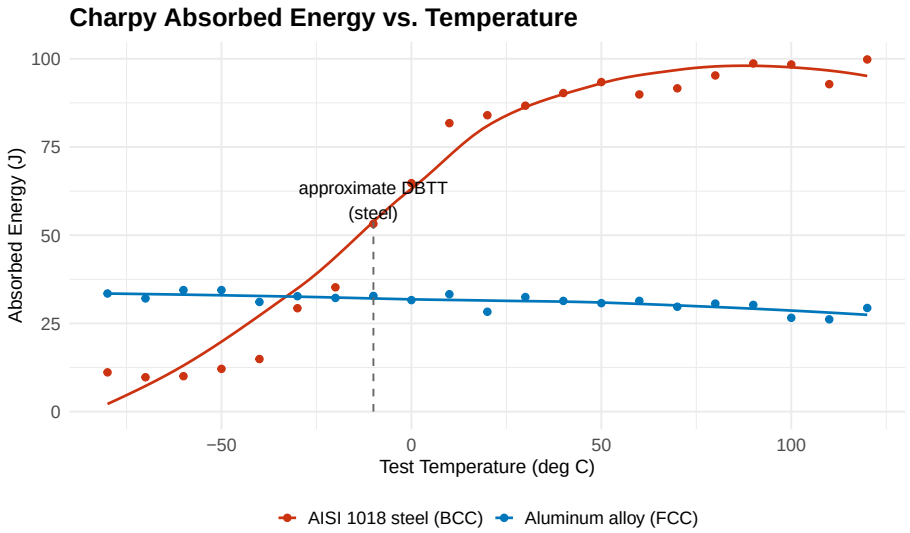


Figure 7.6: Representative Charpy absorbed energy vs. temperature curves: a BCC steel shows a sharp ductile-to-brittle transition; an FCC aluminum alloy stays comparatively flat across the same temperature range (schematic, typical shapes only).

Reading Figure 7.6: at low temperature, the steel sits on a **lower shelf** of low, roughly constant absorbed energy (brittle behaviour). At high temperature, it sits on an **upper shelf** of high, roughly constant absorbed energy (ductile behaviour). In between is the **transition region**, where absorbed energy climbs steeply — this is where the DBTT is reported, sometimes as the temperature at the curve's midpoint, sometimes as the temperature for a specified minimum energy (different standards/specifications define it slightly differently; report whichever definition your lab template specifies).

Fracture appearance changes right along with the energy: a low-shelf (brittle) fracture is flat and bright, with a granular, crystalline look (individual grain facets reflecting light) and little to no visible shear lip. An upper-shelf (ductile) fracture is dull and fibrous, with a large **shear lip** — the sloped, 45° rim around the fracture where the material failed by shear rather than cleavage — and specimens through the transition region show a mix, often reported as a **percent shear** (the fraction of the fracture surface showing ductile shear character rather than flat, brittle cleavage).

7.4.4 Historical Examples

The DBTT is not an academic curiosity — it has caused some of the most famous structural failures in history. During the Second World War, a number of welded **Liberty ships** fractured suddenly, some literally splitting in half, in cold North Atlantic water (Figure 7.7); investigation found that the hull steel's DBTT was close to the seawater temperature the ships operated in, and that welded construction (unlike the riveted ships built before it) allowed a crack starting at one flaw to run continuously across an entire welded plate without stopping at a riveted joint. The **RMS Titanic**'s hull steel has similarly been studied and found to have had a DBTT high enough that, in the near-freezing water of the collision, the steel was likely already on the lower (brittle) shelf, making it far more prone to fracturing on impact than a modern, cold-tolerant steel would be.

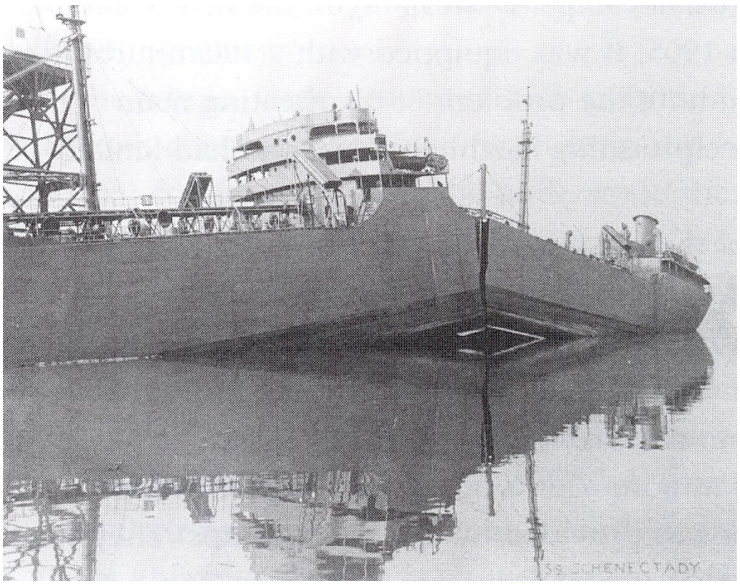


Figure 7.7: The SS *Schenectady*, a Liberty-ship-type T2 tanker, moored at the dock in Portland, Oregon, after her welded hull fractured almost completely in two on 16 January 1943 — a dramatic, non-fatal example of exactly the brittle fracture this section describes. (U.S. Government Printing Office, 1947; public domain, via Wikimedia Commons.)

The investigation into these failures was led by Cambridge metallurgist **Constance Tipper**, who showed the cause was brittle fracture of the steel itself rather than poor welding, and developed the low-temperature bend test — the **Tipper test** — still used to find a steel's transition temperature. Read more about her work in Imperial College London's profile of her.

Industry application

Canadian context: pipelines, heavy equipment, and structural steel used outdoors in a Canadian winter (routinely well below −20 °C in much of the country) must be specified with a DBTT safely below the lowest service temperature the structure will see — this is exactly why Charpy testing at multiple sub-zero temperatures, as in your own lab, is standard practice for structural and pipeline steel qualification, not just a classroom exercise. Outdoor equipment, from farm machinery to snow-clearing equipment, faces the same requirement.

7.4.5 Predicting Your Own Lab Results

Your Charpy lab tests AISI 1018 hot-rolled and cold-rolled specimens at approximately −60, −40, −20, 0 °C, room temperature, 60 °C, and 100 °C. Based on everything above, you should predict:

- **Low temperatures (−60 to −20 °C):** low absorbed energy, flat/bright/granular fracture surface — lower shelf behaviour.
- **Around 0 °C to room temperature:** rapidly rising absorbed energy as you move through the transition region — exact values will scatter more here than on either shelf, which is normal and expected for a transition-region test.
- **High temperatures (60–100 °C):** high, roughly leveled-off absorbed energy, dull/fibrous fracture with a large shear lip — upper shelf behaviour.
- **Hot-rolled vs. cold-rolled:** recall from Section 6.4 (Part A) that cold work raises strength but lowers ductility/toughness. Predict the cold-rolled specimens will generally show *lower* absorbed energy than hot-rolled at a given temperature, and may show a DBTT shifted to a somewhat higher temperature.

7.4.6 Temperature Effects on Tensile and Hardness Results

Temperature doesn't only affect impact toughness. **Tensile strength and yield strength of steel generally rise somewhat as temperature drops** (and ductility drops somewhat too — the same underlying dislocation-mobility story as the DBTT, just measured slowly instead of by impact), while at elevated temperature, strength typically falls and ductility rises. **Hardness** shows a similar trend — most metals read slightly softer hot and slightly harder cold — which is one reason hardness testing standards call for testing near room temperature unless a test is specifically designed for elevated- or low-temperature service conditions.

7.5 Sectioning and Metallography Theory

> **Before you read on**
>
> You can't see individual grains with your bare eyes or even most hand lenses — steel looks like a featureless grey solid at that scale. Metallography is simply the set of steps that gets a metal surface flat, smooth, and chemically revealed enough that a microscope can finally show you the grain structure that's been controlling all your test results this whole chapter.

7.5.1 Why We Cut, Mount, Grind, Polish, and Etch

Each step in metallographic preparation exists to remove the damage the previous step created, while getting progressively closer to a true, undisturbed view of the microstructure:

- **Sectioning (cutting)** removes a small, manageable sample from the part, oriented deliberately — recall from Section 6.4.2 that a **transverse** section (across the bar) and a **longitudinal** section (along the bar) can look completely different in cold-worked material, so the choice of section is not arbitrary.
- **Mounting** embeds the small, often awkwardly-shaped sample in a plastic (commonly bakelite, applied hot under pressure) so it can be handled and polished safely and consistently.
- **Grinding** (successively finer abrasive papers) removes the rough cut surface and the deformation it caused, working down through a sequence of grits until scratches are fine and uniform.
- **Polishing** (progressively finer abrasive suspensions, down to sub-micron) removes the remaining grinding scratches until the surface is mirror-flat and scratch-free — at this point you can see large features like porosity or graphite flakes (Section 7.6, cast iron), but not yet grain boundaries or phases, because a polished, unetched metal surface reflects light almost uniformly.
- **Etching** (a mild acid, most commonly **2% nital** — nitric acid in alcohol — for plain carbon and low-alloy steels) chemically attacks the surface at a rate that depends on crystallographic orientation and on which phase is present. Grain boundaries, being more reactive/higher-energy regions, etch faster and etch a tiny groove, which scatters light and shows up dark under the microscope; different phases (ferrite vs. pearlite vs. martensite) etch at different rates and therefore show different degrees of light and dark, revealing the microstructure.

! Cutting heat and pressure can lie to you

An improperly cut or ground specimen — too much heat from a dull blade, too much pressure during coarse grinding — can locally overheat or cold-work the very surface layer you're trying to examine, changing the microstructure right where you're looking (for example, inadvertently tempering martensite near the cut edge, or smearing grains from mechanical damage). This is exactly why the grinding sequence uses copious water cooling and works through a full sequence of grit sizes rather than skipping straight to fine polish — each step is there to remove the damage the previous, coarser step introduced, right down to a damage-free final surface.

7.5.2 Reading a Micrograph

- **Ferrite** etches lightly and appears as the light-coloured background/matrix phase in a nital-etched plain carbon steel.
- **Pearlite** etches more heavily (more grain-boundary-like area per unit volume, from all those ferrite/cementite interfaces) and appears darker, sometimes showing a fine striped/lamellar pattern at high magnification, sometimes just appearing as darker patches at low magnification where the lamellae are too fine to resolve.
- **Martensite** has a very different, characteristic appearance: a fine, needle-like or plate-like ("acicular") pattern, quite different from the smoother, more rounded grain structure of ferrite/pearlite — a trained eye can usually identify martensite (or tempered martensite, which looks similar but slightly softer-edged after the fine carbide precipitation of tempering) on sight.
- **Grain size** can be estimated formally using the **ASTM E112** grain size number, G — a higher G number means smaller (finer) grains, related to the number of grains visible per unit area at a standard magnification (roughly, $N = 2^{G-1}$ grains per square inch at 100X, though the full standard method involves comparison charts or systematic counting rather than working from this formula directly). For this course, the basic idea to take away is: **count or compare, don't guess** — a finer-grained specimen (higher G) should correlate with the higher strength/hardness predicted by Hall-Petch (Equation 6.1), and this is a useful sanity check between your micrograph and your hardness results on the same specimen.

⚠ These are schematic diagrams, not photomicrographs

Figure 7.8 and Figure 7.9 are simplified illustrations of the *pattern* each microstructure makes, not real photographs. Your own case-hardened and heat-treated specimens will look noisier, and the case/core boundary will

be more gradual than the clean split drawn here.

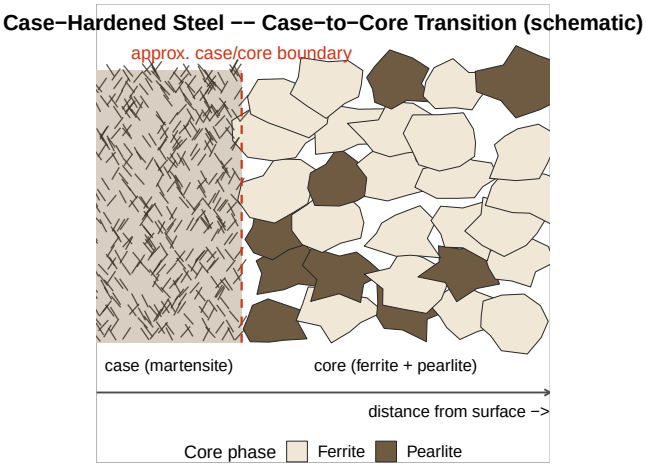


Figure 7.8: Schematic diagram (not an actual photomicrograph) of a case-hardened cross-section: a dense, acicular (needle-like) martensite/tempered-martensite case near the surface, transitioning to coarser ferrite+pearlite grains in the core. The boundary is drawn sharp for clarity; a real transition is more gradual.

Industry application

Automotive and defense: failure analysis labs routinely section a failed part right through the crack or fracture origin and prepare a metallographic specimen specifically to answer "was the microstructure what the print specified?" — the wrong hardness, the wrong grain size, or an unexpected phase (like untempered martensite where tempered martensite was specified) shows up immediately once you know how to read it, exactly as you're learning to do here.

Predict and explain: connecting the metallography to the hardness lab

Question: In the hardness lab you'll examine (or be told about) annealed, quenched, and normalized specimens of the same steel. Based on this section and Section 6.6 (Part A), predict what each should look like under the microscope, from coarsest/softest-looking to finest/hardest-looking.

Answer: Annealed: coarsest pearlite + ferrite, most relaxed-looking grain structure. Normalized: finer, more uniform pearlite + ferrite than annealed. Quenched (untempered): acicular (needle-like) martensite, visually the most different from the other two — no pearlite lamellae at all if fully martensitic. This visual ranking should match the hardness ranking from the Part A practice

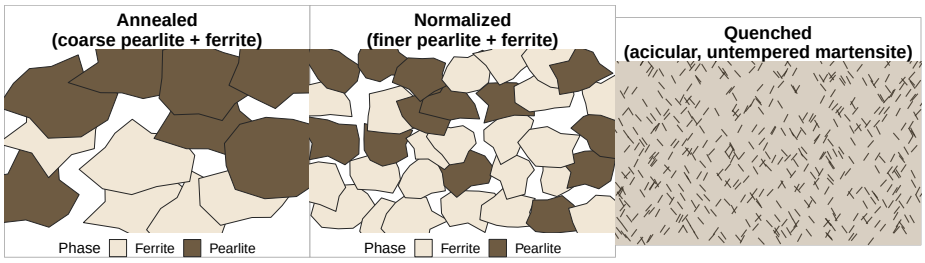


Figure 7.9: Schematic diagram (not an actual photomicrograph) comparing three heat-treated conditions of the same steel at the same scale: annealed (coarse pearlite + ferrite), normalized (finer pearlite + ferrite), and quenched (acicular, untempered martensite).

problem in Section 6.6.7.

7.5.3 Try it yourself: prepare the specimen

A metallographic specimen has to be prepared in a fixed order, because each step removes the damage left by the one before. Put the steps in order.

Metallographic preparation

Number these in the correct order:

- _____ Etching: attack the surface chemically to reveal the microstructure
- _____ Examination under the microscope
- _____ Grinding: remove the cut surface with successively finer papers
- _____ Mounting: embed the sample so it can be handled
- _____ Polishing: remove the grinding scratches to a mirror finish
- _____ Sectioning: cut a small sample from the part

Answers, in the order listed: 5, 6, 3, 2, 4, 1.

7.6 How Other Materials Behave

Everything above was built around plain carbon and low-alloy steel. Here's how the same tests behave on the other materials you're likely to encounter in the shop — and in the "two or more materials" set of your tensile lab.

7.6.1 Aluminum Alloys

Aluminum is **FCC**, so — consistent with Section 7.4.3 — it shows **no sharp ductile-to-brittle transition**; many aluminum alloys actually get slightly *tougher* at cryogenic temperatures, which is one reason aluminum is common in low-temperature and aerospace applications. Aluminum cannot be hardened by the quench-and-temper route steel uses (it doesn't have an analogous martensitic transformation in the alloys used structurally); instead, many aluminum alloys are strengthened by **precipitation hardening** (also called age hardening): the alloy is heated to dissolve alloying elements into solid solution, quenched to trap them in a supersaturated state, then aged (at room temperature or a moderate elevated temperature) to let extremely fine second-phase particles precipitate out — these tiny particles block dislocation motion much the way cementite does in steel, just via a completely different mechanism. This is what a **temper designation** like **6061-T6** means: "T6" specifies solution heat treated and artificially aged (as opposed to, say, "T4," solution heat treated and naturally aged at room temperature, or "O," fully annealed/softest condition).

7.6.2 Stainless Steels

Not all stainless steels behave alike. **Austenitic** stainless steels (like the common 304 and 316 grades) are alloyed with enough nickel to keep the **FCC austenite structure stable all the way down to room temperature** — which is why they don't transform to ferrite/martensite on cooling the way plain carbon steel does. Two direct, testable consequences follow: **304 is not magnetic** (ferrite and martensite are magnetic BCC/BCT structures; stable FCC austenite is not), and **304 cannot be hardened by the heat treatments in this chapter** (quench hardening relies on the austenite-to-martensite shear transformation, which austenitic stainless is specifically alloyed to avoid) — its strength comes mainly from cold work (strain hardening) instead. **Martensitic** stainless steels (like 410, 420) have much lower nickel content, *do* transform to martensite on quenching just like a carbon steel, and *are* magnetic — this is genuinely a different metallurgical family from austenitic stainless, not just a different grade number.

7.6.3 Cast Iron

Cast iron is, by definition, iron-carbon alloy above roughly 2.1 wt% carbon (recall Section 6.5.2: this is the composition range *above* what we called "steel"). At these carbon levels, much of the carbon doesn't stay dissolved or combined as cementite — it precipitates as free **graphite**, most commonly as flakes in ordinary "grey" cast iron. Those flakes are essentially internal stress concentrators and crack initiation sites throughout the material, which is why grey cast iron is **characteristically brittle**, with very low tensile ductility and no useful yield point in the way steel has one — although it is comparatively strong in *compression*, and its graphite flakes give it excellent vibration damping and

machinability. (Ductile, or nodular, cast iron modifies the graphite into rounded nodules instead of sharp flakes, dramatically improving ductility and toughness over grey iron — a good example of how much microstructure, not just composition, controls behaviour.)

7.6.4 Copper and Brass

Copper is **FCC**, ductile, an excellent electrical and thermal conductor, and — like aluminum — does not show a sharp DBTT. It strengthens primarily by cold work (strain hardening), the same mechanism behind CRS in Section 6.4, and softens (recrystallizes, in the sense of Section 6.4.1) if annealed. **Brass** (copper-zinc alloy) is also FCC-based at common compositions, generally more machinable than pure copper, and used widely for its combination of corrosion resistance, appearance, and machinability rather than for high strength.

7.6.5 Polymers

Polymers behave completely differently from metals because they aren't held together by metallic (or even simple ionic/covalent-network) bonding — they're long molecular chains, tangled together and held by comparatively weak intermolecular forces (with the exception of covalent bonds along each chain). The single most important reference temperature for a polymer's mechanical behaviour is its **glass transition temperature (T_g)**: above T_g, chains have enough thermal energy to slide and rearrange somewhat, so the polymer behaves in a relatively soft, flexible, sometimes rubbery manner; below T_g, the chains are essentially frozen in place, and the polymer becomes stiff, glassy, and brittle — a direct parallel to the DBTT idea in Section 7.4.3, though the underlying mechanism (chain mobility, not dislocation motion) is completely different. This is exactly why a rubber gasket or a plastic component can perform well at room temperature but crack or shatter in cold weather — it has been cooled below its T_g.

> Industry application
>
> **Food processing:** many food-contact seals and gaskets are polymers chosen specifically for a T_g well below the coldest temperature the equipment will see (e.g. cold-storage or freezer environments), for exactly the same reason structural steel is chosen for a DBTT below its service temperature.

Predict and explain: identifying an unknown specimen

Question: In the tensile lab, one of your "unknown" specimens turns out to be non-magnetic, does not respond to any heat treatment you try, and shows high ductility with a smoothly rounded (no sharp point) yield region on its stress-strain curve. What is it likely to be, and why?

Answer: Very likely an **austenitic stainless steel** (e.g. 304): non-magnetic

Table 7.2

Table 7.3: Relative behaviour of common engineering materials compared to plain carbon steel.

Material	Crystal structure	Heat-treatment hardenable?	Sharp DBTT?	Typical ductility
Plain carbon steel	BCC (ferrite) / FCC (austenite, high temp)	Yes (quench + temper)	Yes	Moderate-high (condition dependent)
Austenitic stainless (304)	FCC (stable at RT)	No (work-hardens instead)	No	High
Aluminum alloy	FCC	No (precipitation/age hardening instead)	No	Moderate-high
Grey cast iron	BCC ferrite/pearlite matrix + graphite flakes	Limited (can be heat treated, less common)	N/A (already brittle)	Very low
Copper / brass	FCC	No (work-hardens)	No	High
Thermoplastic polymer	Amorphous / semi-crystalline chains	Not applicable	Behaves brittle below Tg (similar concept)	Condition/temperature dependent

and heat-treatment-unresponsive both point directly to stable FCC austenite (Section 7.6.2), and FCC metals characteristically show gradual, rounded yielding rather than steel's sometimes-sharp yield point (Section 7.2.2).

7.7 Key Takeaways

- Engineering stress and strain use the *original* specimen dimensions throughout the test (Equation 7.1, Equation 7.2); modulus of elasticity is set by atomic bonding and barely changes with heat treatment, while yield strength, UTS, and ductility change a great deal.
- The stress-strain curve's regions map directly to dislocation behaviour: elastic (no dislocation motion), yield (dislocations start moving), strain hardening (dislocations multiply and tangle), necking (localized thinning after UTS).
- Rockwell is fast (direct-reading, depth-based); Brinell averages over a larger area (good for coarse structures); Vickers/Knoop give load-independent, comparable numbers well suited to microhardness — reading a case-hardened profile, a weld zone, or an individual phase.
- Hardness scales are not directly interchangeable; conversions (ASTM E140) are empirical approximations, most reliable for steel.
- The Charpy test measures absorbed energy at fracture — toughness, not strength. BCC steels show a sharp ductile-to-brittle transition temperature (DBTT); FCC metals (aluminum, austenitic stainless, copper) generally do not.
- Metallographic preparation (cut, mount, grind, polish, etch) exists to reveal the true microstructure without introducing new damage at each step; 2% nital etch reveals ferrite, pearlite, and martensite in plain carbon steel by their different etching/reflectivity.
- Steel's behaviour is the exception, not the rule, among engineering materials: most non-ferrous metals don't quench-harden the way steel does, and most FCC metals don't show steel's sharp DBTT. Polymers show an analogous but mechanistically different transition at their glass transition temperature.

7.8 Glossary

Brinell hardness (HBW) — a hardness test using a large-diameter ball indenter, reading the diameter of the resulting indentation; suited to coarse or non-uniform structures.

Ductile-to-brittle transition temperature (DBTT) — the temperature (range) over which a metal's fracture behaviour shifts from ductile to brittle;

sharp and pronounced in BCC metals, generally absent in FCC metals.

Ductility — a material's ability to deform plastically before fracture, commonly measured as %EL or %RA.

Elastic modulus (Young's modulus, E) — the slope of the elastic (straight-line) portion of a stress-strain curve; set primarily by atomic bonding.

Engineering strain — extension divided by original gauge length.

Engineering stress — load divided by original cross-sectional area.

Fractography — the study and interpretation of fracture surfaces to determine the mode and cause of failure.

Glass transition temperature (Tg) — the temperature below which a polymer becomes stiff and glassy, and above which it becomes more flexible; the polymer analog of a metal's DBTT.

Grain size number (ASTM E112, G) — a standardized measure of average grain size; higher G means finer (smaller) grains.

Hardness — a material's resistance to localized, permanent (plastic) deformation by an indenter.

Knoop hardness (HK) — a microhardness test using an elongated, rhombic-based diamond indenter; suited to thin or brittle sections.

Metallography — the preparation and microscopic examination of a metal's microstructure.

Necking — localized reduction in cross-sectional area of a tensile specimen after the ultimate tensile strength is reached.

Nital — a nitric-acid-in-alcohol solution (commonly 2%) used to etch plain carbon and low-alloy steel for metallographic examination.

Percent elongation (%EL) — the increase in gauge length at fracture, divided by original gauge length, expressed as a percentage.

Percent reduction of area (%RA) — the decrease in cross-sectional area at fracture, divided by original area, expressed as a percentage.

Precipitation (age) hardening — a strengthening mechanism (common in aluminum alloys) using fine second-phase particles precipitated from solid solution to block dislocation motion.

Rockwell hardness (HRB, HRC, etc.) — a hardness test reading indentation depth directly, using a minor and major load; scale (B, C, etc.) depends on indenter and load.

Shear lip — the sloped, ~45° rim on a fracture surface where final separation occurred by shear rather than cleavage; larger and more prominent in ductile fracture.

Toughness — the ability to absorb energy before fracture; approximately the area under a stress-strain curve, or the energy absorbed in a Charpy test.

Ultimate tensile strength (UTS) — the maximum engineering stress reached during a tensile test.

Vickers hardness (HV) — a hardness test using a square-based diamond pyramid indenter, calculated directly from load and indentation diagonal (Equation 7.5); load-independent and suited to microhardness.

Yield strength — the stress at which a material transitions from elastic to plastic (permanent) deformation; often reported as the 0.2% offset yield strength when no sharp yield point exists.

7.9 Practice Problems

Problem 1 (calculation): stress and strain

Problem: A round tensile specimen, 10.0 mm original diameter and 50.0 mm original gauge length, is loaded to 35 000 N and stretches by 0.15 mm at that load (still elastic). Calculate the engineering stress and strain at that point, and estimate the modulus of elasticity.

Problem 1: stress, strain and modulus

- Engineering stress: _____ MPa *(to 0 decimal places)*
- Engineering strain: _____ *(to 4 decimal places)*
- Modulus of elasticity: _____ GPa *(to 0 decimal places)*

Show the worked solution

Answer: $A_0 = \pi(5.0)^2 = 78.5$ mm². $\sigma = 35000/78.5 = 446$ MPa. $\varepsilon = 0.15/50.0 = 0.0030$. $E = \sigma/\varepsilon = 446/0.0030 \approx 149\,000$ MPa ≈ 149 GPa (somewhat lower than the ~200 GPa typical of steel — in a real test this gap would prompt a check of the extensometer/load reading, since E should be close to 200 GPa for any steel regardless of grade or heat treatment).

Problem 2 (calculation): ductility measures

Problem: The specimen from Problem 1 fractures at $L_f = 66.5$ mm and a minimum diameter of 6.2 mm. Calculate %EL and %RA.

Problem 2: ductility

- Percent elongation (%EL): _____ % *(to 0 decimal places)*
- Percent reduction of area (%RA): _____ % *(to 0 decimal places)*

Show the worked solution

Answer: %EL = $(66.5 - 50.0)/50.0 \times 100 = 33\%$. $A_f = \pi(3.1)^2 = 30.2$ mm². %RA = $(78.5 - 30.2)/78.5 \times 100$ 62%. Both values indicate a quite ductile material.

Problem 3 (predict and explain): comparing two tensile curves

Problem: Two steel specimens are tested: one shows a sharp yield point and 15% elongation; the other shows a smooth, gradual yield (no sharp point) and 28% elongation, with a lower UTS. Which is more likely the cold-rolled specimen, and which the hot-rolled/annealed specimen?

Answer: The sharp-yield, lower-elongation, (implied) higher-strength specimen is more likely **cold-rolled** — recall Section 6.4, cold work raises strength and lowers ductility. The smooth-yielding, higher-elongation, lower-UTS specimen is more likely **hot-rolled/annealed**.

Problem 4 (calculation): Vickers hardness

Problem: A Vickers test at 1 kgf produces a mean diagonal of 0.085 mm. Calculate HV, and estimate the equivalent tensile strength using Equation 7.6 (treat HV HB for this estimate — a simplification, not exact for all hardness ranges).

Problem 4: Vickers hardness

- Vickers hardness: _____ HV *(to 0 decimal places)*
- Estimated tensile strength: _____ MPa *(to 0 decimal places)*

Show the worked solution

Answer: $HV = 1.8544(1)/0.085^2 = 1.8544/0.007225 \approx 257$ HV. Estimated UTS $\approx 3.45 \times 257 \approx 887$ MPa.

Problem 5 (phase diagram + testing, integrative): predicting the hardness lab

Problem: Using the lever rule from Part A (Section 6.5.4) and this chapter's hardness theory, explain in one or two sentences why a quenched specimen's hardness reading is expected to be far more consistent (less reproducible spread run to run) or far less consistent than an annealed specimen's — think about what phase(s) each contains.

Answer: The annealed specimen is expected to show **more scatter**, because it contains a mixture of two phases (ferrite and pearlite) whose relative proportion right under the small indenter can vary from one test location to the next; the quenched specimen is far more uniformly martensitic throughout, so the indenter samples essentially the same single-phase structure wherever it lands, typically giving tighter, more reproducible readings (this is exactly what you're asked to compare in the hardness lab's conclusions).

Problem 6 (predict and explain): Charpy at different temperatures

Problem: Predict, with reasoning, how the fracture surface of a 1018 Charpy specimen tested at −60 °C will differ in appearance from one tested at 100 °C.

Answer: The −60 °C specimen (well below the DBTT, lower shelf) should show a flat, bright, granular/crystalline surface with little or no shear lip — brittle fracture. The 100 °C specimen (upper shelf) should show a dull, fibrous surface with a large, obvious shear lip — ductile fracture, and should absorb substantially more energy.

Problem 7 (predict and explain): hot-rolled vs cold-rolled Charpy results

Problem: Your Charpy lab tests both hot-rolled and cold-rolled AISI 1018. Predict which one shows the lower absorbed energy at any given test temperature, and connect your answer to at least one other test from this chapter.

Answer: Cold-rolled should generally show **lower** absorbed energy (less toughness) than hot-rolled at the same temperature, because the cold work has already used up some of the material's capacity for plastic deformation (fewer "fresh" dislocations available to move before fracture) — the same underlying reasoning connects to cold-rolled steel's lower %EL and %RA in the tensile test (Section 7.2.3) and its higher hardness (Section 6.4).

Problem 8 (predict and explain): material identification

Problem: An unknown metal specimen shows very low ductility (near-zero %EL), no clear yield point, is stronger in compression than tension, and its polished-but-unetched surface already shows dark flake-like features under the microscope before any etching is done. What is it, and how do you know?

Answer: Grey cast iron. The unetched, already-visible dark flakes are graphite (Section 7.6.3) — a feature genuinely visible before etching, unlike steel's phases, which require etching to see. The low ductility, no clear yield point, and compression-favoured strength are all classic cast iron behaviour, driven by the stress-concentrating graphite flakes throughout the matrix.

Problem 9 (predict and explain): temper designation

Problem: You are asked to select 6061 aluminum in either the "O" (annealed) or "T6" condition for a structural bracket that needs high strength. Which do you specify, and what does that designation tell a supplier about how the material was processed?

Answer: Specify **6061-T6**. "T6" indicates the alloy was solution heat treated (to dissolve alloying elements) and artificially aged (to precipitate fine strengthening particles) — the precipitation-hardening process described in Section 7.6.1 — giving substantially higher strength than the soft, fully annealed "O" condition.

This completes the metallurgy and materials-testing theory for your three labs. Before each lab, re-read the matching section here — Section 7.3 before Hardness Testing, Section 7.2 and Section 7.3 before Tensile and Microhardness Testing,

Section 7.4 before Charpy Impact Testing — and after each lab, come back and check your results against what this chapter predicted.

Chapter 8

Plant Layout and Facility Design

8.1 Learning Objectives

By the end of this chapter, you will be able to:

- Identify and describe the four basic types of facility layouts
- Compare the advantages and disadvantages of each layout type
- Apply the steps involved in designing a process layout
- Calculate cycle time, theoretical minimum stations, and line efficiency for product layouts
- Understand the principles of warehouse and office layout design

Why is Plant Layout Important?

Plant layout directly impacts:

- **Productivity:** Efficient layouts minimize wasted movement and time
- **Cost:** Poor layouts increase material handling costs by 20-50%
- **Safety:** Well-designed layouts reduce accidents and injuries
- **Flexibility:** The right layout allows adaptation to changing demands
- **Employee Morale:** Good layouts improve working conditions

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8.2 Types of Facility Layouts

There are **four basic layout types**, each suited to different production requirements:

Table 8.1: Overview of Facility Layout Types

Layout Type	Also Known As	Best For	Ex
Process Layout	Functional Layout, Job Shop	High variety, low volume	Ho
Product Layout	Flow Line, Assembly Line	Low variety, high volume	Au
Hybrid/Cellular Layout	Group Technology, Cell Layout	Medium variety, medium volume	El
Fixed-Position Layout	Project Layout	Very large or immobile products	Sh

8.2.1 Visualizing the Layout Spectrum

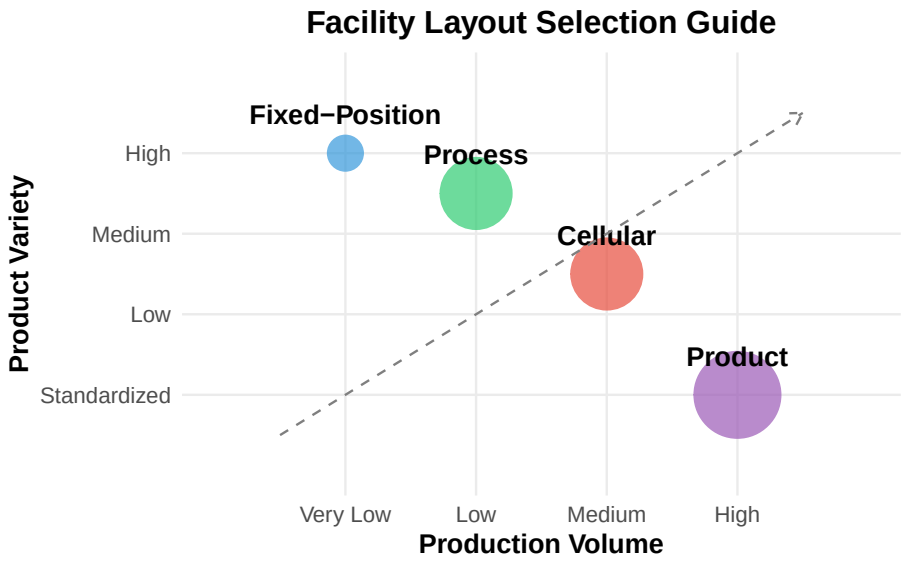


Figure 8.1: Product Variety vs. Production Volume for Different Layouts

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8.3 Process Layout (Functional Layout)

In a **process layout**, similar resources (machines, equipment, workers with similar skills) are grouped together. Work flows between departments based on the specific requirements of each job.

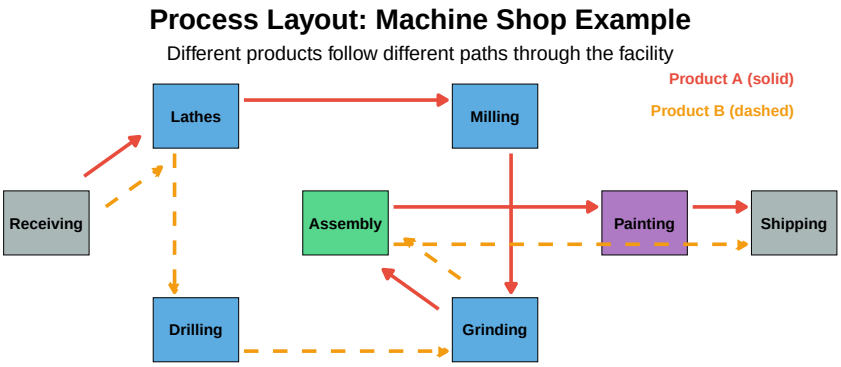


Figure 8.2: Example Process Layout - Machine Shop

8.3.1 Characteristics of Process Layouts

Table 8.2: Process Layout Characteristics

Characteristic	Description
Equipment	General-purpose machines
Labor	Skilled workers required
Flexibility	High - can handle variety
Processing Speed	Slower due to varied routing
Material Handling	Higher costs (complex paths)
Scheduling	Complex - different routes
Space Requirements	Higher - WIP inventory storage

Discussion Question: Can you identify a process layout in your daily life?

Think about: - A hospital emergency room (patients routed based on their needs) - A university campus (students move between buildings for different classes) - A grocery store (customers choose their own path) - A machine shop (parts routed based on required operations)

What makes these process layouts? Similar functions are grouped together,

and "products" (patients, students, customers) follow different paths based on their individual needs.

8.3.2 Real-World Examples

8.4 Product Layout (Assembly Line)

In a **product layout**, equipment and workstations are arranged in a line according to the sequence of operations needed to produce the product. Every unit follows the same path.

Product Layout: Automotive Assembly Line

1. **Station 1** – Frame
2. **Station 2** – Engine
3. **Station 3** – Body
4. **Station 4** – Interior
5. **Station 5** – Wheels
6. **Station 6** – Quality

8.4.1 Characteristics of Product Layouts

Table 8.3: Product Layout: Advantages vs. Disadvantages

Advantage	Disadvantage
High production rate	High initial investment
Low unit cost	Low flexibility
Low material handling	Line stops affect all
Simple scheduling	Monotonous work
Less WIP inventory	Requires balanced workloads

Quick Quiz: Product Layout

Question: A car wash where vehicles move through a series of cleaning stations is an example of which layout type?

Answer: Product Layout - all cars follow the same path through the same sequence of operations.

8.5 Hybrid Layout (Cellular Manufacturing)

Hybrid layouts combine the flexibility of process layouts with the efficiency of product layouts. The most common type is the **cellular layout** using Group Technology.

8.5.1 Group Technology Concept

Group Technology (GT) identifies parts with similar characteristics and groups them into **part families**. Machines are then arranged into **cells** to process these families.

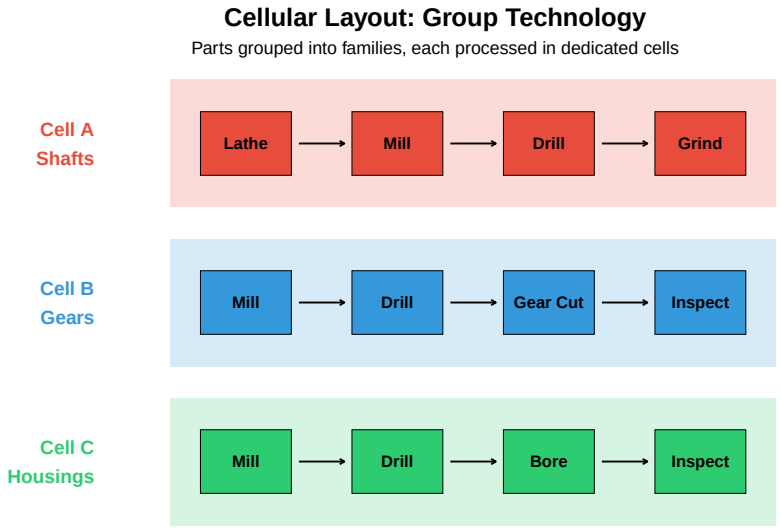


Figure 8.3: Cellular Layout with Manufacturing Cells

Discussion Question: Benefits of Cellular Manufacturing

What are the main benefits of cellular manufacturing over traditional process layouts?

1. **Reduced material handling** - Parts stay within the cell
2. **Shorter lead times** - No waiting between departments
3. **Less WIP inventory** - Smaller batches, faster flow
4. **Improved quality** - Workers responsible for complete part
5. **Team accountability** - Cell teams own their output

8.6 Fixed-Position Layout

In a **fixed-position layout**, the product remains stationary while workers, equipment, and materials are brought to it. This is used when the product is too large or heavy to move.

8.6.1 Examples of Fixed-Position Layouts

- **Shipbuilding** - Ships built in dry docks
- **Aircraft manufacturing** - Large aircraft assembled in hangars
- **Construction** - Buildings, bridges, dams
- **Surgery** - Patient remains stationary

Challenge Question: Managing a Fixed-Position Layout

What are the unique challenges of managing a fixed-position layout?

Think about: - Scheduling deliveries of materials and equipment - Coordinating multiple contractors/teams - Limited space around the product - Weather and environmental factors (outdoor projects) - Ensuring safety with many workers in one area

---

8.7 Try it yourself: name the layout

Eight facilities. Decide which of the four layout types each one uses.

Which layout type?

Pick the layout type each facility uses.

Categories: Process / Product / Cellular / Fixed-position

1. A job shop with all the lathes in one area, all the mills in another and a separate grinding room. _____
2. A hospital with radiology, the laboratory and surgery in separate departments. _____
3. An automotive final assembly line building one model at a fixed takt time. _____
4. A bottling line: rinse, fill, cap, label, pack. _____
5. A U-shaped group of machines dedicated to one family of valve bodies. _____
6. A plant reorganised so each part family has its own mini-line with cross-trained operators. _____
7. A ship built in dry dock, with crews and materials brought to the hull. _____
8. A wide-body aircraft in final assembly in a hangar bay. _____

Answers: 1 Process; 2 Process; 3 Product; 4 Product; 5 Cellular; 6 Cellular; 7 Fixed-position; 8 Fixed-position.

8.8 Designing Process Layouts

Designing an effective process layout involves **three main steps**:

8.8.1 Step 1: Gather Information

You need to determine: - **Space requirements** for each department - **Available space** in the facility - **Closeness relationships** between departments

8.8.1.1 From-To Matrix (Load Matrix)

A From-To matrix shows the number of trips or loads between departments:

Table 8.4: From-To Matrix: Loads per Day Between Departments

	Receiving	Machining	Assembly	Painting	Shipping
Receiving	-	100	50	0	0
Machining	0	-	200	50	0
Assembly	0	0	-	100	60
Painting	0	0	0	-	150
Shipping	0	0	0	0	-

8.8.2 Step 2: Develop Block Plans

Use the **Load-Distance Model** to evaluate layout alternatives:

$$\text{Total Cost} = \sum_{i=1}^{n} \sum_{j=1}^{n} L_{ij} \times D_{ij} \times C$$

Where: - L_{ij} = Number of loads between departments i and j - D_{ij} = Distance between departments i and j - C = Cost per unit distance

8.8.3 Interactive Example: Comparing Two Layouts

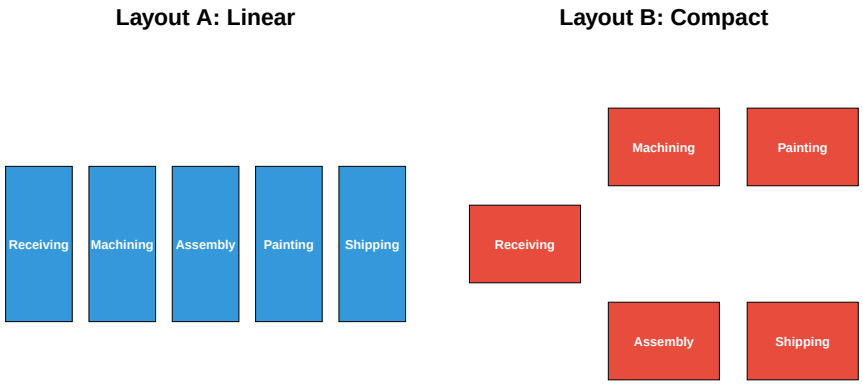


Figure 8.4: Comparing Two Process Layout Alternatives

Teaching demo — drag the material flow volumes and watch which layout wins.

Show the live R code behind this calculator

```
#| label: load-distance-calc
# Calculate Load-Distance scores for both layouts
# From-To data: Receiving->Machining, Machining->Assembly,
#               Assembly->Painting, Painting->Shipping

flow_receiving_machining <- 100
flow_machining_assembly <- 200
flow_assembly_painting <- 100
flow_painting_shipping <- 150

# Layout A distances (linear): all adjacent = 1 unit
layout_A_score <- flow_receiving_machining*1 + flow_machining_assembly*1 +
  flow_assembly_painting*1 + flow_painting_shipping*1

# Layout B distances (compact): most departments closer together
# Receiving to Machining = 1.12 (diagonal), Machining to Assembly = 1,
# Assembly to Painting = 1.12 (diagonal), Painting to Shipping = 1
layout_B_score <- flow_receiving_machining*1.12 + flow_machining_assembly*1 +
  flow_assembly_painting*1.12 + flow_painting_shipping*1

if (layout_A_score <= layout_B_score) {
  cat(sprintf("Layout A (linear) wins by %.1f units\n", layout_B_score - layout_A_score
} else {
```

```
  cat(sprintf("Layout B (compact) wins by %.1f units\n", layout_A_score - layout_B_score))
}
cat(sprintf("  Layout A (linear)    %.1f\n", layout_A_score))
cat(sprintf("  Layout B (compact)   %.1f\n", layout_B_score))

par(mar = c(5, 5, 3, 1))
barplot(c(`Layout A (linear)` = layout_A_score, `Layout B (compact)` = layout_B_score),
        col = c("#2E86C1", "#F39C12"), ylim = c(0, 2400), ylab = "Load-distance score",
        main = "Lower is better")
```

Why does the linear layout win in this case?

In this example, the linear layout (A) wins because the process flow is essentially sequential (Receiving → Machining → Assembly → Painting → Shipping).

When flow is **sequential**, a **product layout** (linear arrangement) is more efficient. Process layouts are better when there's **varied routing** between departments.

8.8.4 Try it yourself: lay out the plant

Six departments have to go into a building that is two bays deep and three bays wide. Material moves between them as shown in the flow table. Put one department in each cell; the table works out the distance each flow travels (rows apart plus columns apart) and the total load-distance score. **Lower is better.**

Start with the heaviest flows: which pairs of departments must be neighbours? Then check your layout against the best possible score. In class, compare scores: there is more than one way to reach the minimum.

Block plan: lowest load-distance score

Put one department in each cell. Choosing a department that is already placed moves it.

From	To	Loads per day	Distance	Load x distance
Receiving	Machining	60		
Receiving	Welding	30		
Machining	Welding	25		
Machining	Assembly	45		
Welding	Painting	40		
Painting	Assembly	50		
Assembly	Shipping	80		
Receiving	Shipping	10		

Departments: Receiving, Machining, Welding, Painting, Assembly, Shipping. Place one in each cell of a 2 x 3 grid, then fill in the table. Total score: _____

The lowest score possible is 370.

8.8.5 Step 3: Develop Detailed Layout

Once the block plan is selected: - Determine exact sizes and shapes of departments - Plan aisle locations and widths - Consider utilities, safety exits, and accessibility - Use CAD software or 3D modeling for visualization

---

8.9 Designing Product Layouts (Line Balancing)

The key challenge in product layout design is **line balancing** - assigning tasks to workstations to minimize idle time while meeting production requirements.

8.9.1 Line Balancing Steps

1. **Identify tasks and precedence relationships**
2. **Calculate required cycle time**
3. **Calculate theoretical minimum number of stations**
4. **Assign tasks to stations**
5. **Calculate efficiency**

8.9.2 Key Formulas

Cycle Time:

$$C = \frac{\text{Available Production Time}}{\text{Desired Output}}$$

Theoretical Minimum Stations:

$$TM = \frac{\sum t_i}{C}$$

Line Efficiency:

$$\text{Efficiency} = \frac{\sum t_i}{n \times C} \times 100\%$$

Where: - C = Cycle time - $\sum t_i$ = Total task time - n = Number of workstations

8.9.3 Interactive Example: Pizza Assembly Line

Table 8.5: Pizza Assembly Tasks

Task	Description	Time (sec)	Predecessors
A	Roll dough	40	-

B	Spread sauce	25	A
C	Add cheese	20	B
D	Add toppings	35	B
E	Season	15	C, D
F	Box pizza	30	E

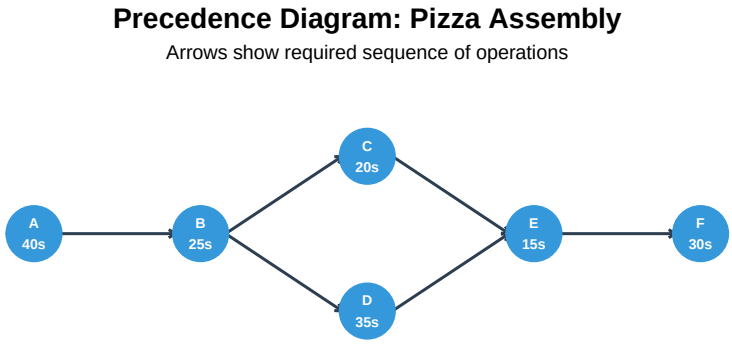


Figure 8.5: Precedence Diagram for Pizza Assembly

8.9.4 Line Balancing Calculation

This calculator works through the pizza line in the table and diagram above. It does the three steps from the list at the start of this section, then builds a balanced set of stations for you:

1. **Cycle time** — how often a finished pizza must leave the line to hit the target output.
2. **Theoretical minimum stations** — total task time divided by cycle time, rounded up. No line can do better than this.
3. **Station assignment** — tasks are loaded into stations one at a time, longest eligible task first, never exceeding the cycle time and never starting a task before its predecessors are done. The result gives the actual number of stations and the line efficiency.

What to change. *Desired output* and *Available time per hour* set the cycle time (60 pizzas/h in 3600 s gives 60 s). Edit the *task times* in the table to see how a slow task, such as the 40 s dough roll, forces idle time elsewhere. The precedence relationships are fixed. Try raising the output until a single task is longer than the cycle time; the line can no longer be balanced without splitting that task or running it on parallel stations.

Teaching demo — change the desired output or the task times and watch the number of stations and the efficiency.

Show the live R code behind this card

```
#| label: line-balance-calc
# Task data
tasks <- data.frame(
  Task = c("A", "B", "C", "D", "E", "F"),
  Description = c("Roll dough", "Spread sauce", "Add cheese", "Add toppings", "Season"
  Time_sec = c(40, 25, 20, 35, 15, 30),
  Predecessors = c("-", "A", "B", "B", "C,D", "E")
)

# Production requirements
desired_output <- 60    # pizzas per hour
available_time <- 3600  # seconds per hour

# Step 1: Cycle time and total work content
cycle_time <- available_time / desired_output
total_task_time <- sum(tasks$Time_sec)

# Step 2: Theoretical minimum stations
TM <- total_task_time / cycle_time

cat(sprintf("Cycle time  %g s per pizza\n", cycle_time))
cat(sprintf("Total task time  %g s\n", total_task_time))
cat(sprintf("Theoretical minimum stations  %.2f  (so at least %g)\n", TM, ceiling(TM))

if (any(tasks$Time_sec > cycle_time)) {
  cat("!!A single task is longer than the cycle time: split it or use parallel stations
} else {
  # Step 3: Assign tasks to stations (longest eligible task first)
  preds <- strsplit(gsub(" ", "", tasks$Predecessors), ",")
  preds <- lapply(preds, function(p) p[p != "-"])
  done <- character(0)
  station <- 0
  cat("\nStation assignments:\n")
  while (length(done) < nrow(tasks)) {
    station <- station + 1
    load <- 0
    members <- character(0)
    repeat {
      ready <- which(!(tasks$Task %in% done) &
                     sapply(preds, function(p) all(p %in% done)) &
                     load + tasks$Time_sec <= cycle_time)
      if (length(ready) == 0) break
      pick <- ready[which.max(tasks$Time_sec[ready])]
      load <- load + tasks$Time_sec[pick]
```

```
      members <- c(members, tasks$Task[pick])
      done <- c(done, tasks$Task[pick])
    }
    cat(sprintf("  Station %d: %s = %g s [idle %g s]\n",
                station, paste(members, collapse = " + "), load, cycle_time - load))
  }

  # Step 4: Efficiency
  efficiency <- total_task_time / (station * cycle_time) * 100
  cat(sprintf("\nStations used  %d\n", station))
  cat(sprintf("Line efficiency  %.1f%%\n", efficiency))
  cat(sprintf("Balance delay (idle time)  %.1f%%\n", 100 - efficiency))
}
```

Try It Yourself: Can you find a better balance?

Challenge: At 60 pizzas/h the total task time is 165 s, so the theoretical minimum is $165/60 = 2.75$, which rounds up to 3 stations. Can the line really be run with 3?

Hint: Try to fill each station to exactly 60 s. Station 1 must start with A (40 s), and B cannot join it because $40 + 25 = 65$ s.

Solution: No. A alone takes station 1 (40 s). B and D fill station 2 (60 s). That leaves C, E and F, which total 65 s, over the cycle time, so a fourth station is needed. Precedence constraints and indivisible tasks make 4 stations the minimum, for an efficiency of $165/(4 \times 60) = 68.75\%$.

Now use the calculator: raise the output to 70 pizzas/h (cycle time about 51 s) or cut the dough roll to 30 s and see how the number of stations and the efficiency respond.

8.9.5 Extending the Example: The Big-Game-Night Rush

The pizzeria runs the line above on a normal night: 4 stations, 4 workers, one pizza every 60 s, so **60 pizzas/h**. On Super Bowl night, Halloween and New Year's Eve the orders pile up at well over 100 pizzas/h. Management has two ways to react, and the question is which one pays.

Part 1: How much more can the line make with extra people? More workers means more stations, and more stations means each one does less, so the cycle time can drop. The limit is the longest single task. Task A (roll dough, 40 s) cannot be shared between people, so no cycle shorter than 40 s is possible however many people are added:

$$\text{Maximum capacity with the line as it is} = \frac{3600}{t_{\max}} = \frac{3600}{40} = 90 \text{ pizzas/h}$$

The card below rebalances the line for the number of workers you pick. It looks for the shortest cycle time at which the same station-assignment rule used above fits the tasks into that many stations.

Teaching demo — step the slider from 3 to 7 workers and watch the cycle time and the station loads.

Show the live R code behind this card

```
#| label: peak-people
times <- c(A = 40, B = 25, C = 20, D = 35, E = 15, F = 30)
preds <- list(A = character(0), B = "A", C = "B", D = "B", E = c("C", "D"), F = "E")
workers <- 4

# Load tasks into stations (longest eligible task first) for a given cycle time
assign_stations <- function(times, cycle) {
  done <- character(0); loads <- numeric(0); members <- character(0)
  while (length(done) < length(times)) {
    load <- 0; group <- character(0)
    repeat {
      ready <- names(times)[!(names(times) %in% done) &
                            sapply(preds[names(times)], function(p) all(p %in% done)) &
                            load + times <= cycle]
      if (length(ready) == 0) break
      pick <- ready[which.max(times[ready])]
      load <- load + times[pick]; group <- c(group, pick); done <- c(done, pick)
    }
    loads <- c(loads, load); members <- c(members, paste(group, collapse = "+"))
  }
  list(loads = loads, members = members)
}

# Shortest cycle time (whole seconds) that fits into the available workers
cycle <- max(times)
while (length(assign_stations(times, cycle)$loads) > workers) cycle <- cycle + 1
plan <- assign_stations(times, cycle)

cat(sprintf("Workers available      %d\n", workers))
cat(sprintf("Shortest cycle time    %g s\n", cycle))
cat(sprintf("Capacity               %.1f pizzas/h\n", 3600 / cycle))
cat(sprintf("Stations used          %d\n", length(plan$loads)))
cat(sprintf("Efficiency             %.1f%%\n", sum(times) / (length(plan$loads) * cycle
cat("\nStations:\n")
for (i in seq_along(plan$loads))
  cat(sprintf("  Station %d: %s = %g s\n", i, plan$members[i], plan$loads[i]))

par(mar = c(5, 5, 3, 1))
```

```
barplot(plan$loads, names.arg = plan$members, col = "#2E86C1", ylim = c(0, 100),
        xlab = "Station (tasks)", ylab = "Station load (s)", main = "Station loads against the cy
abline(h = cycle, lty = 2, lwd = 2, col = "#C0392B")
if (workers > length(plan$loads))
  cat("\n!!Extra workers have nothing to do: task A (40 s) is the bottleneck.!!\n")
```

Step the slider from 4 to 7 workers. With the line as it is, capacity grows from 60 pizzas/h (the balance used earlier) to 80 pizzas/h when the 4 workers are rebalanced. A fifth worker lifts it to 90 pizzas/h. A sixth and seventh add nothing at all, because station A still takes 40 s.

Part 2: Duplicate station A. To go past 90 pizzas/h, the bottleneck itself must be attacked. Putting a second dough station next to the first (a second roller and a second worker) lets two people roll dough in parallel, so a pizza leaves station A every $40/2 = 20$ s. The next-longest task, D (add toppings, 35 s), now sets the limit:

$$\text{Capacity with A doubled} = \frac{3600}{35} \approx 103 \text{ pizzas/h}$$

That costs one more worker (6 in total) and one more piece of equipment. Is a gain of about 13 pizzas/h over Option 1 worth that on the few nights a year when it matters? The cost-benefit card compares the two options for a single rush night and for a year. Only the nights when demand exceeds capacity count as a benefit, so set the *peak demand* to what customers would really order.

Your turn — use the card to find the break-even number of rush nights.

Show the live R code behind this card

```
#| label: peak-costbenefit
peak_demand <- 110
margin <- 4.5
rush_hours <- 6
rush_nights <- 20
wage <- 20
equip_cost <- 12000
equip_life <- 5

# Capacities from Part 1 and Part 2
base_cap <- 3600 / 60    # normal night: 4 workers, 60 s cycle
opt1_cap <- 3600 / 40    # Option 1: 5 workers, rebalanced, cycle limited by A
opt2_cap <- 3600 / 35    # Option 2: A doubled (6 workers), cycle limited by D

sold <- function(cap) min(cap, peak_demand)
night <- function(cap, workers_extra, equip_per_night) {
  gain <- (sold(cap) - sold(base_cap)) * margin * rush_hours
```

```
  cost <- workers_extra * wage * rush_hours + equip_per_night
  c(gain = gain, cost = cost, net = gain - cost)
}

equip_per_night <- equip_cost / equip_life / rush_nights
o1 <- night(opt1_cap, 1, 0)
o2 <- night(opt2_cap, 2, equip_per_night)

cat(sprintf("%-26s %12s %12s\n", "", "Option 1", "Option 2"))
cat(sprintf("%-26s %12s %12s\n", "", "5 workers", "A doubled, 6"))
cat(sprintf("%-26s %12.1f %12.1f\n", "Capacity (pizzas/h)", opt1_cap, opt2_cap))
cat(sprintf("%-26s %12.1f %12.1f\n", "Pizzas sold per hour", sold(opt1_cap), sold(opt2_
cat(sprintf("%-26s %12.0f %12.0f\n", "Extra margin per night ($)", o1["gain"], o2["gain
cat(sprintf("%-26s %12.0f %12.0f\n", "Extra cost per night ($)", o1["cost"], o2["cost"]
cat(sprintf("%-26s %12.0f %12.0f\n", "Net per night ($)", o1["net"], o2["net"]))
cat(sprintf("%-26s %12.0f %12.0f\n", "Net per year ($)", o1["net"] * rush_nights, o2["n

d_net <- (o2["net"] - o1["net"]) * rush_nights
cat(sprintf("\nDuplicating A beats Option 1 by %.0f $/year\n", d_net))
d_margin <- (sold(opt2_cap) - sold(opt1_cap)) * margin * rush_hours
if (d_margin > wage * rush_hours) {
  cat(sprintf("Break-even: about %.0f rush nights per year\n",
              equip_cost / equip_life / (d_margin - wage * rush_hours)))
} else {
  cat("!!The extra pizzas do not cover even the extra worker: the second station never
}
```

Try It Yourself: When does the second dough station pay?

Challenge: With the default numbers, find the number of rush nights per year below which Option 1 (five workers, no new equipment) is the better choice. Then set *peak demand* to 90 pizzas/h and see what happens to the answer.

Hint: The extra margin of Option 2 over Option 1 is $(103-90)\times\$4.50\times 6$ h $\approx$ \$350 per night. Subtract the extra worker (\$120 per night). The equipment costs \$12,000/5 = \$2,400 per year, to be recovered from what is left.

Solution: Each rush night leaves about \$350 − \$120 = \$230 after the extra worker. Covering \$2,400 a year needs $2400/230 \approx 10.4$, so **at least 11 rush nights per year**. With 20 nights the second station earns roughly \$2,200 a year more than Option 1. If peak demand is only 90 pizzas/h, Option 1 already meets all of it, the extra capacity of Option 2 is never sold, and duplicating A never pays.

Think about it: The cost-benefit result depends on the margin, the number of rush nights and the demand, not only on the line. What other options could be cheaper than a permanent second station (renting a roller for the day, pre-rolling

dough earlier, a pre-made crust on rush nights)?

8.10 Warehouse Layout Design

Warehouse layouts are a special case of process layouts focused on **minimizing material handling**.

8.10.1 Key Principles

1. **Place high-volume items near the dock**
2. **Group items that are often picked together**
3. **Use ABC analysis** for storage assignment:
   - **A items** (20% of SKUs, 80% of picks): Prime locations
   - **B items** (30% of SKUs, 15% of picks): Secondary locations
   - **C items** (50% of SKUs, 5% of picks): Remote locations

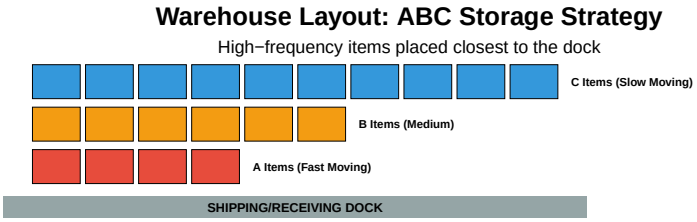


Figure 8.6: Warehouse Layout: ABC Storage Strategy

8.11 Office Layout Design

Office layouts balance **communication needs** with **privacy requirements**.

8.11.1 Two Main Approaches

Table 8.6: Office Layout Comparison

Aspect	Open Plan	Traditional/Closed
Space	Shared workspaces	Private offices

Walls	Partitions/dividers	Permanent walls
Flexibility	High - easy to reconfigure	Low - expensive to change
Communication	Excellent	Limited
Privacy	Low	High
Cost	Lower	Higher

8.11.2 Office Ergonomics Checklist

Click to expand: Office Ergonomics Best Practices

Posture: - Support the small of the back - Keep shoulders relaxed - Maintain neutral wrist position - Feet flat on floor or footrest

Lighting: - Position monitor to reduce glare - Adjust brightness for screen vs. paper work - Use task lighting when needed

Workstation: - Monitor at arm's length - Top of screen at or below eye level - Keyboard and mouse at elbow height

Work Habits: - Take breaks every 30-60 minutes - Vary tasks to prevent repetitive strain - Use keyboard shortcuts to reduce mouse use

8.11.3 The Future of Office Design

8.12 Summary: Choosing the Right Layout

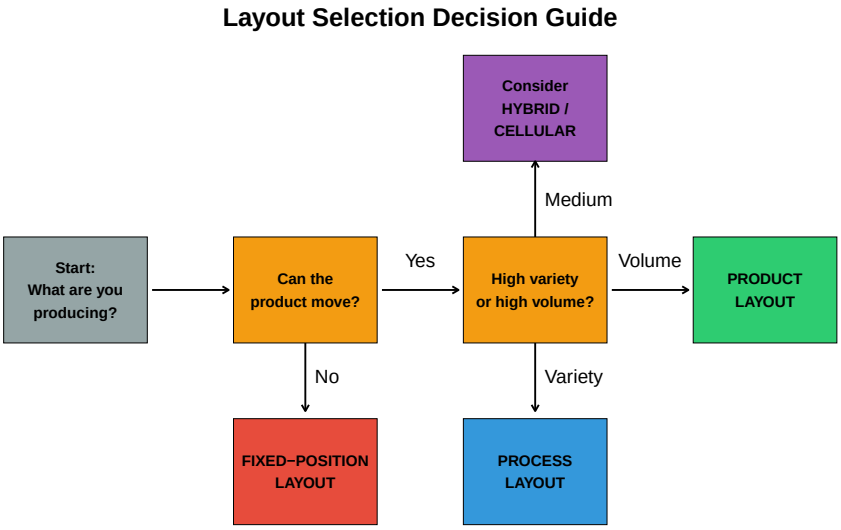


Figure 8.7: Layout Selection Decision Guide

8.12.1 Key Takeaways

1. **Process layouts** offer flexibility for varied products but have higher material handling costs
2. **Product layouts** are efficient for high-volume, standardized products
3. **Cellular layouts** provide a middle ground with benefits of both
4. **Fixed-position layouts** are necessary when products cannot be moved
5. **Line balancing** is critical for efficient product layouts
6. **Warehouse layouts** should minimize travel distance for high-frequency items

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8.13 Industry Case Studies

8.13.1 Case Study 1: Food Processing Plant Layout

Food processing facilities have unique layout requirements due to food safety, hygiene, and regulatory compliance.

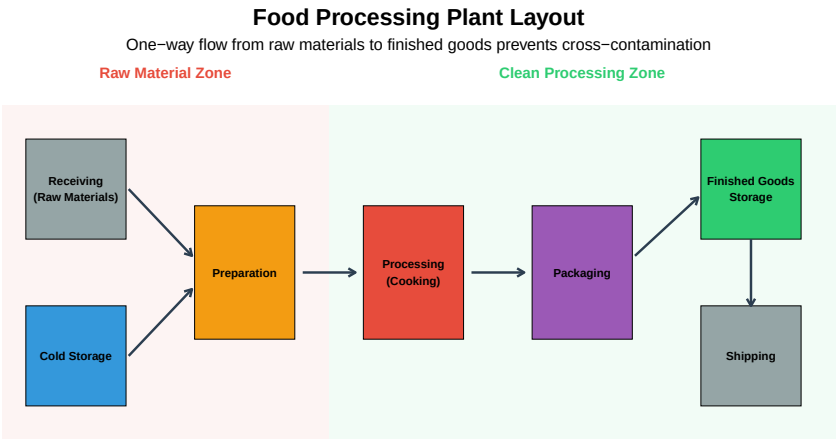


Figure 8.8: Food Processing Plant Layout Example

Key Design Principles for Food Processing:

Table 8.7: Food Processing Layout Design Principles

Principle	Description	Regula
Linear Flow	Raw materials → Processing → Packaging (no backtracking)	FSMA
Zone Separation	Physical barriers between raw and ready-to-eat areas	HACCP
Cleanability	Smooth, non-porous surfaces; rounded corners; proper drainage	3-A Sta
Temperature Control	Maintain cold chain; separate refrigerated zones	FDA 21
Sanitary Design	Handwashing stations, foot baths, air curtains at zone transitions	USDA F
Allergen Control	Dedicated lines or thorough cleaning between allergen products	FALCPA

Discussion Question: Cross-Contamination Prevention

Question: A food processing plant produces both peanut butter products and tree nut-free products. What layout considerations are essential?

Key Considerations:

1. **Physical Separation:** Completely separate production lines with physical barriers (walls, not just distance)
2. **Air Handling:** Separate HVAC systems to prevent airborne allergen transfer
3. **Personnel Flow:** Dedicated workers for each line, or strict gowning/cleaning procedures when crossing zones
4. **Scheduling:** If shared equipment is unavoidable, schedule allergen products last before deep cleaning

5. **Traffic Patterns:** Forklift and cart paths should not cross between allergen zones
6. **Verification:** Air sampling and surface swabs to verify effectiveness

8.13.2 Case Study 2: Aerospace Manufacturing Facility

Aerospace facilities often use **fixed-position layouts** combined with **process layouts** for component fabrication.

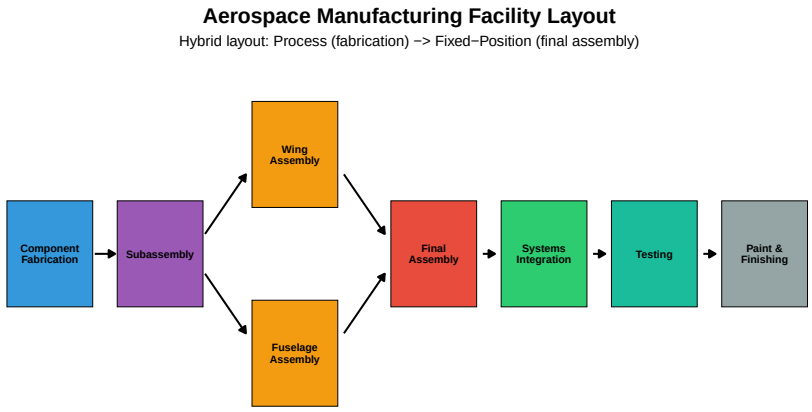


Figure 8.9: Aerospace Assembly Facility Layout

Aerospace Layout Characteristics:

- **Clean rooms** for precision components (hydraulics, avionics)
- **High-bay areas** for large assembly operations
- **Overhead cranes** for moving heavy components
- **Flow-line final assembly** (aircraft moves through stations) for high-volume commercial aircraft
- **Fixed-position** for military aircraft or low-volume production
- **Strict FOD (Foreign Object Debris)** control throughout

Video: Boeing 737 Assembly

Watch how Boeing uses a moving assembly line for high-volume aircraft production:

8.13.3 Case Study 3: Automotive Supplier - Hybrid Cell Layout

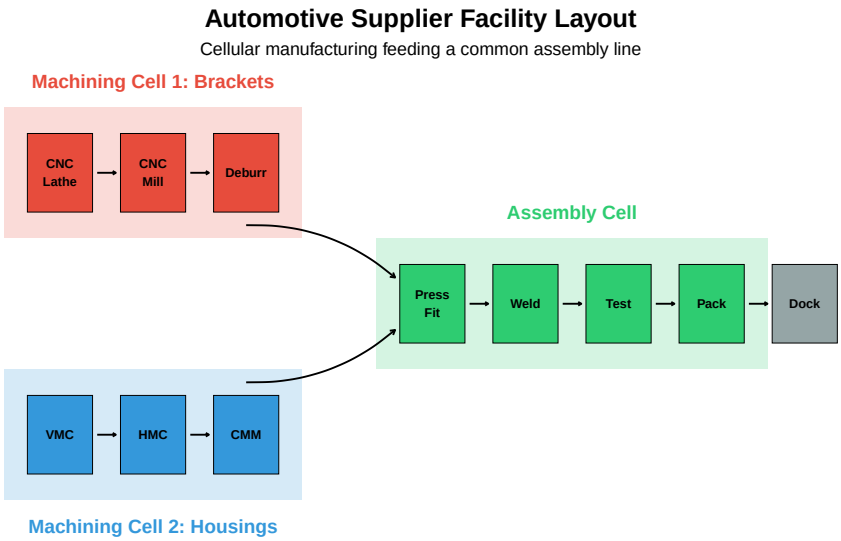


Figure 8.10: Automotive Supplier Facility with Cellular Layout

JIT Delivery Requirements:

Automotive suppliers must often deliver directly to the assembly line in **sequence** (JIS - Just-In-Sequence):

Table 8.8: JIT/JIS Requirements and Layout Implications

Requirement	Typical Standard	Layout Impact
Delivery Window	±30 minutes of scheduled time	Dedicated shipping lane, staging area n
Sequence Accuracy	100% (0 sequence errors)	Pick-to-light systems, visual manageme
Quality Level	0 PPM target (Six Sigma)	Inline inspection, mistake-proofing (pol
Packaging	Returnable containers to line side	Container cleaning/storage area needed
Traceability	Full lot/serial traceability required	Barcode/RFID scanning stations throu

---

8.14 Ergonomics in Manufacturing

Ergonomics (human factors engineering) is critical for plant layout design to ensure worker safety, reduce injuries, and improve productivity.

8.14.1 The NIOSH Lifting Equation

The National Institute for Occupational Safety and Health (NIOSH) developed an equation to evaluate manual lifting tasks:

$$RWL = LC \times HM \times VM \times DM \times AM \times FM \times CM$$

Where **RWL** = Recommended Weight Limit (in pounds or kg)

Table 8.9: NIOSH Lifting Equation Factors

Factor	Name	Formula	Description
LC	Load Constant	51 lb (23 kg)	Maximum weight under ideal conditions
HM	Horizontal Multiplier	10/H	H = horizontal distance from spine to lo
VM	Vertical Multiplier	1 - 0.0075|V - 30|	V = vertical height of hands at start (in
DM	Distance Multiplier	0.82 + 1.8/D	D = vertical travel distance (inches)
AM	Asymmetric Multiplier	1 - 0.0032A	A = angle of asymmetry (degrees)
FM	Frequency Multiplier	Table lookup	Based on lift frequency and duration
CM	Coupling Multiplier	Table lookup	Based on grip quality (good/fair/poor)

8.14.2 Lifting Index Calculation

$$LI = \frac{\text{Actual Weight}}{\text{RWL}}$$

Table 8.10: Lifting Index Interpretation

Lifting Index	Risk Level	Action Required
LI 1.0	Low	Task is acceptable for most workers
1.0 < LI 3.0	Moderate	Task poses risk; engineering controls recommended
LI > 3.0	High	Task is unacceptable; immediate redesign required

8.14.3 NIOSH Lifting Example

Also live — try changing the box weight, distances, or lift frequency below.

Show the live R code behind this calculator

```
#| label: niosh-example-plant
# Example: Worker lifts boxes from pallet to conveyor
# Conditions: Frequency = 4 lifts per minute, 2 hours; Coupling = Fair (handles present)

LC <- 51  # Load constant (lb)
H <- 15   # Horizontal distance, hands to ankles (in)
```

```
V <- 10   # Vertical height at origin of lift (in)
D <- 20   # Vertical travel distance (in)
A <- 30   # Asymmetry angle (degrees)
FM <- 0.72  # Frequency multiplier: 4 lifts/min, 1-2 hours, from NIOSH table
CM <- 0.95  # Coupling multiplier: fair
actual_weight <- 35  # Actual load weight (lb)

# Calculate multipliers
HM <- 10 / H  # 10 here is a fixed NIOSH constant, not our V
VM <- 1 - 0.0075 * abs(V - 30)  # 30 = NIOSH's fixed optimal reference height (in)
DM <- 0.82 + (1.8 / D)
AM <- 1 - 0.0032 * A

# Calculate RWL and Lifting Index
RWL <- LC * HM * VM * DM * AM * FM * CM
LI <- actual_weight / RWL

cat(sprintf("Lifting Index: %.2f (RWL = %.1f lb)\n", LI, RWL))
cat(sprintf("  HM (horizontal)   %.3f\n", HM))
cat(sprintf("  VM (vertical)     %.3f\n", VM))
cat(sprintf("  DM (distance)     %.3f\n", DM))
cat(sprintf("  AM (asymmetry)    %.3f\n", AM))
if (LI <= 1.0) {
  cat("  Risk level        Low - task is acceptable\n")
} else if (LI <= 3.0) {
  cat("  Risk level        Moderate - consider engineering controls\n")
} else {
  cat("  Risk level        !!High - immediate redesign required!!\n")
}
```

Discussion Question: Reducing the Lifting Index

Based on the example above, what layout or workstation changes could reduce the Lifting Index?

Solutions (in order of effectiveness):

1. **Reduce horizontal distance (H):** Move conveyor closer to pallet position
   - If H = 10": HM improves from 0.67 to 1.0 → RWL increases by 50%
2. **Raise the starting height (V):** Use elevated pallet positions or pallet lifters
   - If V = 30" (optimal): VM improves from 0.85 to 1.0 → RWL increases by 18%
3. **Eliminate asymmetry (A):** Reposition conveyor directly in front of pallet
   - If A = 0°: AM improves from 0.90 to 1.0 → RWL increases by 11%

4. **Reduce frequency:** Add a second worker or use automation
   - If 1 lift/min: FM improves from 0.72 to 0.94 → RWL increases by 31%
5. **Improve coupling:** Use boxes with good handles
   - If good coupling: CM improves from 0.95 to 1.0 → RWL increases by 5%

Best solution: Implement a scissor lift for the pallet (adjustable height) and move the conveyor closer. This addresses the largest multiplier reductions.

8.14.4 Ergonomic Workstation Design

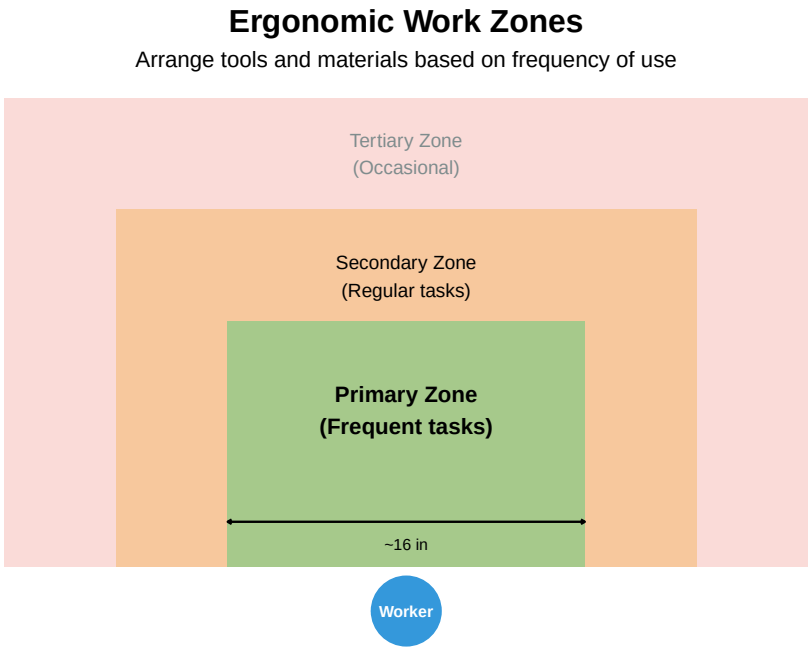


Figure 8.11: Ergonomic Work Zones for Standing Operations

Layout Implications:

- **Primary zone:** Most frequently used tools and parts
- **Secondary zone:** Regularly used items, occasional reaches
- **Tertiary zone:** Rarely needed items, should trigger redesign if frequent use required

8.15 Modern Layout Trends and Industry 4.0

8.15.1 Flexible Manufacturing Layouts

Modern facilities must adapt quickly to changing product demands. Key trends include:

Table 8.11: Modern Flexible Layout Trends

Trend	Description	B
Modular Workstations	Pre-built workstation modules that can be relocated in hours	R
Reconfigurable Cells	Cells with standardized machine interfaces for quick changeover	Sv
Mobile Equipment	Equipment on wheels or air bearings for rapid repositioning	El
Flexible Utilities	Overhead utility drops and floor trenches allow equipment mobility	N
Digital Layout Planning	Digital twins enable virtual layout testing before physical changes	R

8.15.2 Industry 4.0 Layout Considerations

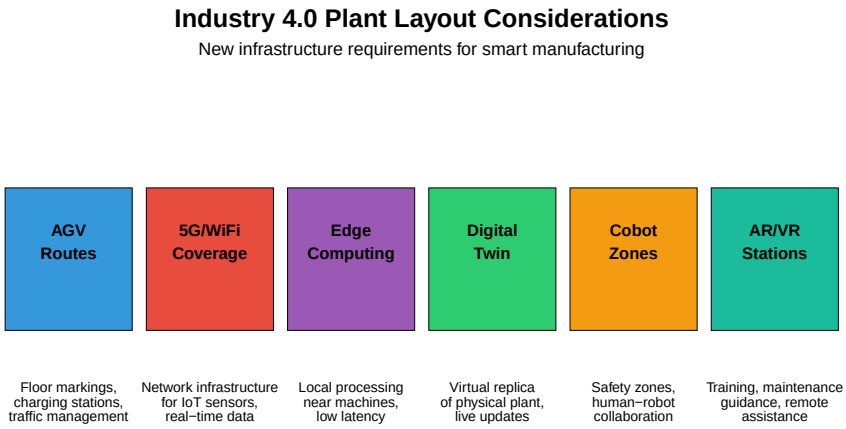


Figure 8.12: Industry 4.0 Enabled Plant Layout Elements

8.15.3 AGV and AMR Traffic Planning

Automated Guided Vehicles (AGVs) and Autonomous Mobile Robots (AMRs) require careful layout planning:

Table 8.12: AGV/AMR Layout Planning Guidelines

Consideration	Design Guidelines
Route Design	Unidirectional loops preferred; minimize intersections; avoid steep grades
Charging Locations	Opportunity charging at pickup/dropoff points; dedicated charging bays for breaks
Traffic Intersections	Yield zones, traffic signals, or elevation changes at crossings
Floor Requirements	Smooth, level surfaces; magnetic tape or QR codes for navigation; no oil/water
Safety Zones	Minimum 1m clearance from pedestrian paths; light curtains at crossings
Staging Areas	Buffer areas for queue management; waiting positions clear of traffic

Video: AGV Systems in Modern Warehouses

8.16 Layout Simulation Software

Modern plant layout design relies heavily on simulation software to test alternatives before physical implementation.

8.16.1 Common Layout Planning Software

Table 8.13: Plant Layout Planning Software Comp

Software	Primary Use	Key Features
AutoCAD Factory Design Suite	2D/3D CAD layout design	Extensive equipment libraries, interference checking,
FlexSim	3D discrete event simulation	Drag-and-drop modeling, VR capability, real-time 3D
Arena	Process simulation	Statistical analysis, what-if scenarios, bottleneck iden
Plant Simulation (Siemens)	Digital factory simulation	Full Siemens integration, AGV simulation, energy an
SketchUp + Extensions	Quick 3D visualization	Low cost, easy learning curve, walkthrough capability

8.16.2 Simulation Benefits

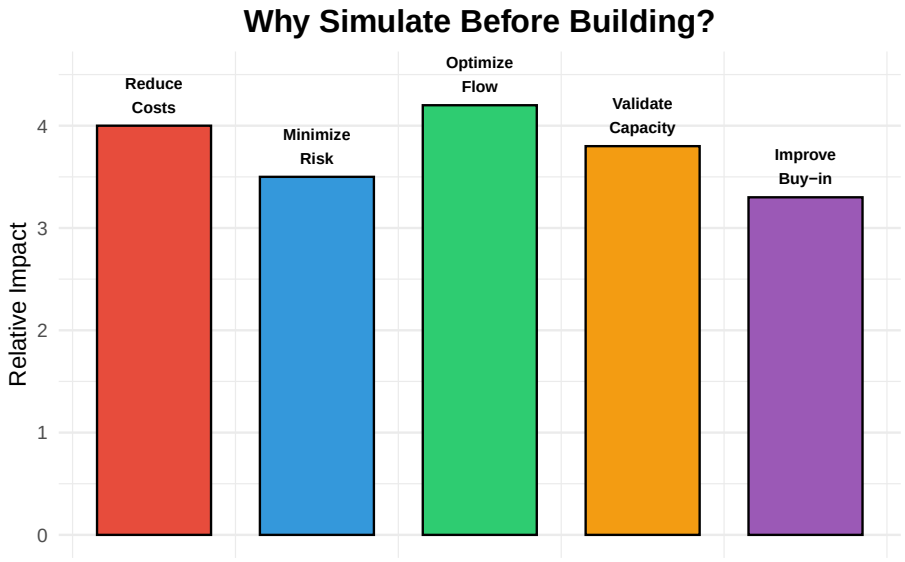


Figure 8.13: Benefits of Layout Simulation

ROI of Simulation:

- **10-20x** return on simulation investment typical
- **30-50%** reduction in layout change costs
- **Weeks to months** saved in ramp-up time
- **Stakeholder alignment** - visual communication of concepts

Video: Plant Simulation Demo

8.17 Practice Problems

Problem 1: Layout Selection

Scenario: A company manufactures custom kitchen cabinets. Each order is unique, with different dimensions, materials, and finishes. Production volume is about 50 units per month.

Question: Which layout type would you recommend and why?

Answer: A **process layout** would be most appropriate because: - High product variety (custom orders) - Low production volume - Different operations needed for each order - Flexibility is more important than efficiency

Problem 2: Line Balancing

Given:

- Total task time: 240 seconds
- Desired output: 30 units per hour
- Available time: 3600 seconds per hour

Calculate:

1. Cycle time
2. Theoretical minimum number of stations
3. If you use 3 stations, what is the efficiency?

Problem 2: line balancing

- Cycle time: _____ sec *(to 0 decimal places)*
- Theoretical minimum number of stations: _____ *(to 0 decimal places)*
- Efficiency with 3 stations: _____ % *(to 1 decimal place)*

Show the worked solution

Solution: 1. Cycle time = 3600 / 30 = **120 seconds** 2. TM = 240 / 120 = **2 stations** 3. Efficiency = (240 / (3 × 120)) × 100 = **66.7%**

Problem 3: Warehouse Layout

Scenario: A warehouse has the following items: - Item X: 500 picks/day - Item Y: 100 picks/day - Item Z: 50 picks/day

Question: How should these items be positioned relative to the shipping dock?

Answer: Using ABC analysis: - Item X (high frequency): **Closest to dock** (A zone) - Item Y (medium frequency): **Middle distance** (B zone) - Item Z (low frequency): **Farthest from dock** (C zone)

8.18 Review Questions

Question 1: What distinguishes a cellular layout from both process and product layouts?

Answer: A cellular layout combines characteristics of both:

- **From process layout:** Equipment variety within each cell, flexibility to handle part variations
- **From product layout:** Sequential flow within the cell, dedicated equipment for part families
- **Unique characteristics:**

  - Parts grouped into families using Group Technology
  - Cells are self-contained mini-factories
  - Workers often cross-trained to operate multiple machines
  - Reduced material handling compared to process layout
  - More flexibility than pure product layout

Question 2: Calculate the Lifting Index for the following scenario

Scenario: A worker lifts 25 kg boxes from floor level (V=0) to a shelf at waist height. The horizontal distance is 40 cm, vertical travel is 75 cm, no asymmetry, frequency is 2 lifts per minute for 4 hours, and coupling is poor (no handles).

Given multipliers:

- LC = 23 kg
- HM = 25/H (H in cm)
- VM = 1 - 0.003|V - 75| (V in cm)
- DM = 0.82 + 4.5/D (D in cm)
- AM = 1.0 (no asymmetry)
- FM = 0.65 (from tables)
- CM = 0.90 (poor coupling)

Question 2: Lifting Index

- Recommended Weight Limit (RWL): _____ kg *(to 1 decimal place)*
- Lifting Index (LI): _____ *(to 1 decimal place)*

Show the worked solution

Solution:

```
HM = 25/40 = 0.625
VM = 1 - 0.003|0 - 75| = 1 - 0.225 = 0.775
DM = 0.82 + 4.5/75 = 0.82 + 0.06 = 0.88
AM = 1.0
FM = 0.65
CM = 0.90

RWL = 23 × 0.625 × 0.775 × 0.88 × 1.0 × 0.65 × 0.90
RWL = 23 × 0.625 × 0.775 × 0.88 × 0.65 × 0.90
RWL = 5.7 kg

LI = 25/5.7 = 4.4

Risk Level: HIGH (LI > 3.0) - Immediate redesign required
```

Recommended actions: Raise starting height (use pallet lifter), reduce horizontal distance, add handles to boxes, reduce frequency.

Question 3: List three Industry 4.0 technologies that impact plant layout design

Answer: (Any three of the following)

1. **AGVs/AMRs:** Require route planning, charging stations, floor markings, safety zones
2. **Collaborative robots (cobots):** Need safety zones, but smaller footprint than traditional robot cells
3. **Digital twins:** Enable virtual layout testing, reduce physical prototyping
4. **IoT sensors:** Require network infrastructure (5G/WiFi), edge computing locations
5. **AR/VR systems:** Training stations, remote assistance capabilities
6. **Flexible automation:** Modular equipment, quick-connect utilities

Question 4: What are the key differences between food processing plant layouts and general manufacturing?

Answer:

Aspect	Food Processing	General Manufacturing
Flow direction	Strictly one-way (raw → finished)	Can have backtracking
Zone separation	Physical barriers between raw/clean	Logical separation often sufficient
Surfaces	Sanitary design, smooth, cleanable	Durability focused
Temperature	Multiple temperature zones (cold chain)	Generally ambient
Regulatory	HACCP, FSMA, FDA oversight	OSHA, industry standards
Allergen control	Dedicated lines or validated cleaning	Not typically a concern
Traceability	Lot-level required by regulation	Often batch-level sufficient

Question 5: A warehouse has three product categories with the following pick frequencies. Design the storage zones.

Data: - Category A: 1200 picks/day, 500 SKUs - Category B: 600 picks/day, 1500 SKUs - Category C: 200 picks/day, 3000 SKUs

Solution:

Using ABC analysis based on pick frequency: - **Category A** (60% of picks, 10% of SKUs): Prime locations closest to shipping dock - **Category B** (30% of picks, 30% of SKUs): Secondary locations, moderate distance - **Category C** (10% of picks, 60% of SKUs): Remote locations, maximum distance from dock

Additional considerations: - Within each zone, use cube utilization for dense storage of slow movers - Consider pick path optimization (serpentine vs. return routing) - High-frequency items at ergonomic pick heights (waist level) - Reserve space near dock for cross-docking high-velocity items

8.19 Chapter Summary

Table 8.14: Chapter 8 Summary: Plant La

Topic	Key Concepts
Layout Types	Fixed-position, Process, Product, Hybrid/Cellular - selected based on vo
Process Layout Design	From-To matrix, Load-Distance model, Block plans, Closeness relationsl
Product Layout (Line Balancing)	Cycle time, Theoretical minimum stations, Precedence diagrams, Line e
Cellular Manufacturing	Group Technology, Part families, Self-contained cells, Reduced material
Warehouse Design	ABC analysis, High-frequency items near dock, Pick path optimization
Ergonomics	NIOSH lifting equation, RWL, Lifting Index, Work zones, Workstation d
Modern Trends	Flexible layouts, Industry 4.0, AGV/AMR planning, Digital twins
Simulation	FlexSim, Arena, Plant Simulation - virtual testing before physical chang

8.20 References

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Chapter 9

Statistical Process Control (SPC)

9.1 Learning Objectives

By the end of this chapter, you will be able to:

- Explain the purpose and benefits of Statistical Process Control
- Distinguish between common cause and special cause variation
- Select the appropriate control chart for different data types
- Construct and interpret X-bar/R and Individual-MR charts
- Construct and interpret attribute control charts (p and c), and know when np and u charts apply
- Apply control chart interpretation rules (Western Electric rules)
- Calculate and interpret the process capability indices Cp and Cpk
- Respond appropriately to out-of-control signals

> **Where this chapter stops**
>
> This is the introductory SPC chapter (Part 1). Subgrouping theory (Phase I and II), X-bar/S, np and u charts, Pp/Ppk, capability for non-normal data, and the CUSUM and EWMA charts for small shifts are in **Chapter 18, Advanced Statistical Process Control** (Part 2).

> "In God we trust; all others must bring data." — W. Edwards Deming

---

9.2 Introduction to SPC

9.2.1 What is Statistical Process Control?

Statistical Process Control (SPC) is a method of quality control that uses statistical methods to monitor and control a process. It helps ensure that the process operates at its full potential to produce conforming product.

The Shift from Inspection to Prevention

1. **WITHOUT SPC** – • Inspect after production; • Find defects too late; • React to problems; • High scrap/rework costs
2. **WITH SPC** – • Monitor during production; • Detect shifts early; • Prevent defects; • Lower costs, higher quality

9.2.2 Brief History of SPC

Table 9.1: Evolution of Statistical Process Control

Year	Development	Significance
1920s	Walter Shewhart develops control charts at Bell Labs	Birth of modern quality c
1930s-40s	SPC used in WWII military production	Proved effectiveness in m
1950s	Deming teaches SPC to Japanese industry	Foundation of Japanese c
1980s	US automotive industry adopts SPC	Response to Japanese co
1990s	SPC becomes requirement in ISO/QS-9000	Industry-wide standardiz
2000s+	Real-time SPC, automated data collection	Integration with Industry

9.2.3 Why SPC Matters

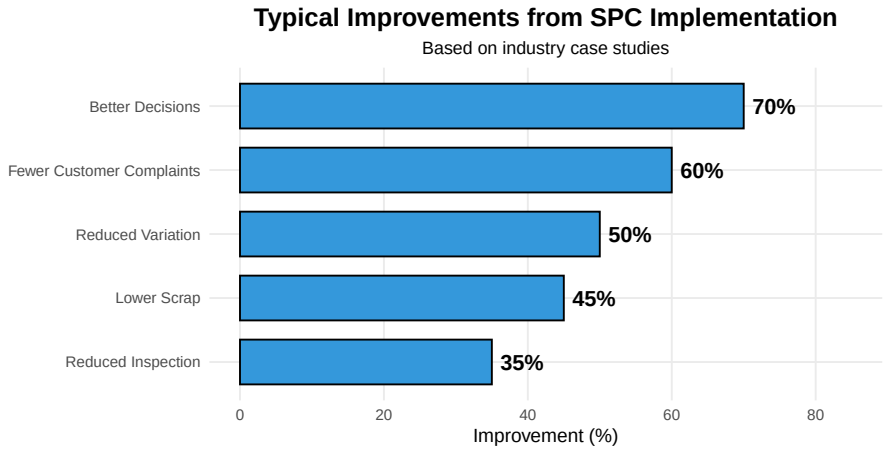


Figure 9.1: Benefits of SPC Implementation

9.3 Understanding Variation

All processes exhibit variation. No two products are ever exactly identical. The key insight of SPC is that variation comes from two fundamentally different sources, and we must respond to them differently.

9.3.1 Common Cause vs. Special Cause Variation

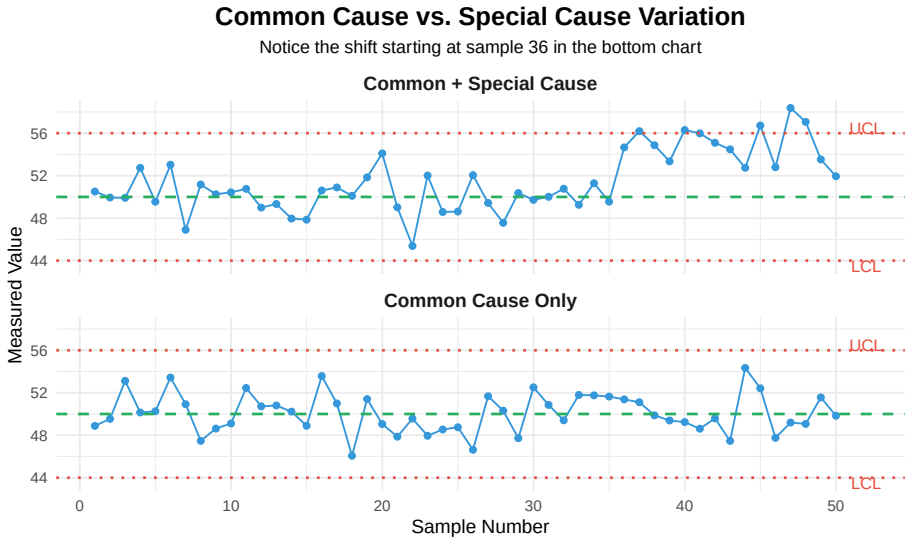


Figure 9.2: Two Types of Variation

Table 9.2: Common Cause vs. Special Cause Variation

Characteristic	Common Cause Variation	Special Cause Vari
Other Names	Random, chance, noise, inherent	Assignable, non-rando
Source	Built into the process; many small factors	Specific, identifiable e
Pattern	Random, stable, predictable pattern	Non-random; shifts, t
Predictability	Statistically predictable within limits	Unpredictable; someth
Examples	Machine vibration, material variation, ambient temperature	Tool breakage, new op
Action Required	Improve the system (management decision)	Find and eliminate th
Who Acts	Management — requires system change	Operators/technicians

9.3.2 The Two Mistakes in Process Control

Understanding variation prevents two costly mistakes:

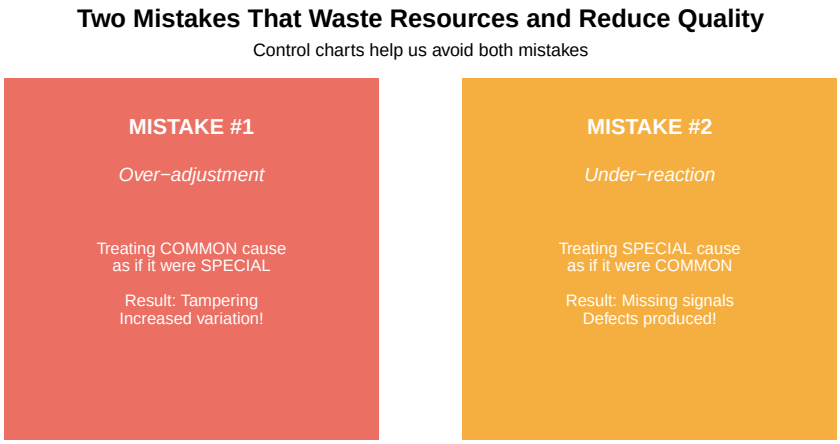


Figure 9.3: The Two Fundamental Mistakes

The Funnel Experiment: Why Tampering Makes Things Worse

Deming's Funnel Experiment demonstrates why adjusting a stable process increases variation:

Setup: Drop a ball through a funnel onto a target. The ball lands near (but not exactly on) the target due to common cause variation.

Rule 1 (Correct): Leave the funnel alone. Variation remains constant.

Rule 2 (Tampering): After each drop, move the funnel to compensate for the last error. - Result: Variation DOUBLES!

Rule 3 (More Tampering): Move funnel to compensate relative to the target. - Result: Variation increases without bound!

Lesson: Adjusting a stable process based on individual deviations (common cause) actually makes the process worse. Only adjust when special causes are identified.

Real-World Example: An operator notices a part is 0.02mm below target and adjusts the machine. The next part is 0.03mm above target (the adjustment plus normal variation). They adjust again… and the process oscillates with increasing variation.

9.3.3 Try it yourself: run the funnel experiment

Deming's funnel experiment, described above, is easy to run yourself. A marble is dropped through a funnel aimed at a target. It never lands exactly on the target: that scatter is common-cause variation. One operator leaves the funnel alone. A second operator, trying to help, moves the funnel after every drop by the size of the last miss, in the opposite direction.

Before you look at the result, decide: will the second operator's drops be closer to the target, the same, or further away?

Teaching demo — drag the scatter and the number of drops, then press **Run again** to drop a fresh set of marbles.

Show the live R code behind this simulation

```
#| label: funnel-sim
# Deming's funnel experiment: Rule 1 (hands off) against Rule 2 (adjust after every drop)
sigma <- 1.0      # natural scatter of the process
n_drops <- 200
seed <- 1         # run number: same number, same marbles

set.seed(seed)
scatter <- rnorm(n_drops, mean = 0, sd = sigma)

# Rule 1: the funnel stays aimed at the target (0)
hands_off <- scatter

# Rule 2: after each drop, move the funnel by the size of the miss, the other way
adjusted <- numeric(n_drops)
aim <- 0
for (i in seq_len(n_drops)) {
  adjusted[i] <- aim + scatter[i]
  aim <- aim - adjusted[i]
}

par(mfrow = c(1, 2), mar = c(4.5, 4.5, 3, 1), cex.main = 1.2, cex.lab = 1.1, cex.axis = 1)
limits <- c(-12, 12)   # fixed axes, so the two rules can be compared at any sigma
plot(hands_off, pch = 19, col = "#2980b9", ylim = limits, las = 1, xlab = "Drop", ylab = "Miss di
abline(h = 0, col = "#27ae60", lwd = 2)
plot(adjusted, pch = 19, col = "#e74c3c", ylim = limits, las = 1, xlab = "Drop", ylab = "Miss dis
abline(h = 0, col = "#27ae60", lwd = 2)

cat(sprintf("Adjusting made the scatter %.0f%% worse\n", (sd(adjusted) / sd(hands_off) - 1) * 100
cat(sprintf("  Leave it alone         standard deviation %.2f mm\n", sd(hands_off)))
cat(sprintf("  Adjust every drop      standard deviation %.2f mm\n", sd(adjusted)))
cat(sprintf("  Ratio                  %.2f (theory: 1.41)\n", sd(adjusted) / sd(hands_off)))
```

Run it several times. The adjusting operator is never better and is worse by about 40% every time, because each adjustment adds the previous drop's random error to the next drop's own. This is the first of the two mistakes in process control, reacting to common-cause variation as if it were a signal, and it is exactly what a control chart is there to prevent.

9.3.4 When is a Process "In Control"?

A process is **in statistical control** when:

1. Only common cause variation is present
2. All points fall within control limits
3. No non-random patterns exist
4. The process is stable and predictable

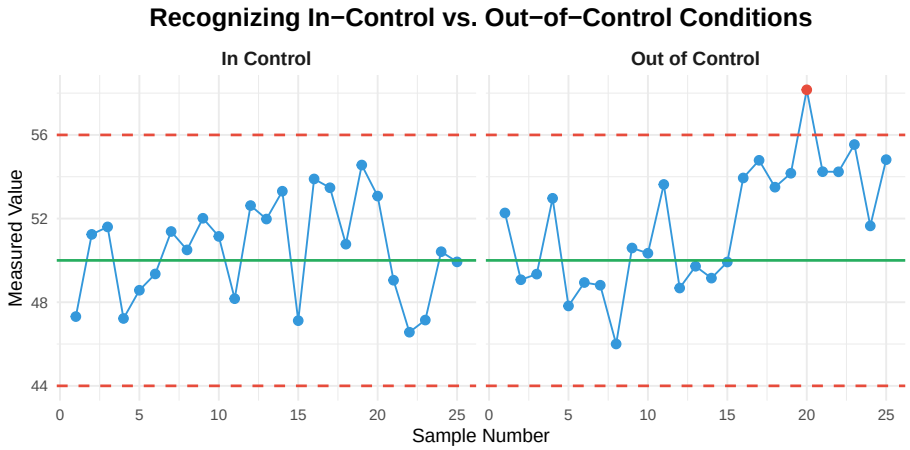


Figure 9.4: In-Control vs. Out-of-Control Process

9.4 Control Charts: The Foundation of SPC

9.4.1 What is a Control Chart?

A **control chart** is a graph that shows process data over time with statistically determined control limits. It provides a visual method for distinguishing between common and special cause variation.

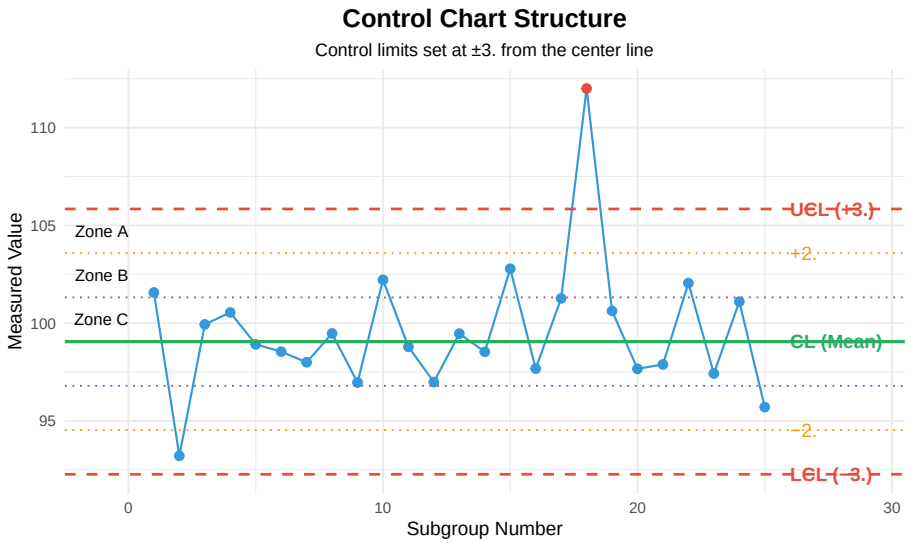


Figure 9.5: Anatomy of a Control Chart

9.4.2 Control Chart Components

Table 9.3: Control Chart Components

Component	Description	Purpose
****Center Line (CL)****	The process average; target for the process	Shows where process
****Upper Control Limit (UCL)****	Upper boundary of expected variation (+3)	Signals if process shif
****Lower Control Limit (LCL)****	Lower boundary of expected variation (-3)	Signals if process shif
****Data Points****	Individual or subgroup statistics plotted over time	Visual representation
****Zones A, B, C****	Regions between ±1 , ±2 , and ±3 for pattern analysis	Used for detecting tre

9.4.3 Why ±3 Sigma?

Control limits are set at **±3 standard deviations** from the center line. This is not arbitrary:

- For a normal distribution, 99.73% of data falls within ±3
- Only 0.27% (about 3 in 1000) will fall outside by chance
- This balances the risk of both mistakes (false alarms vs. missed signals)

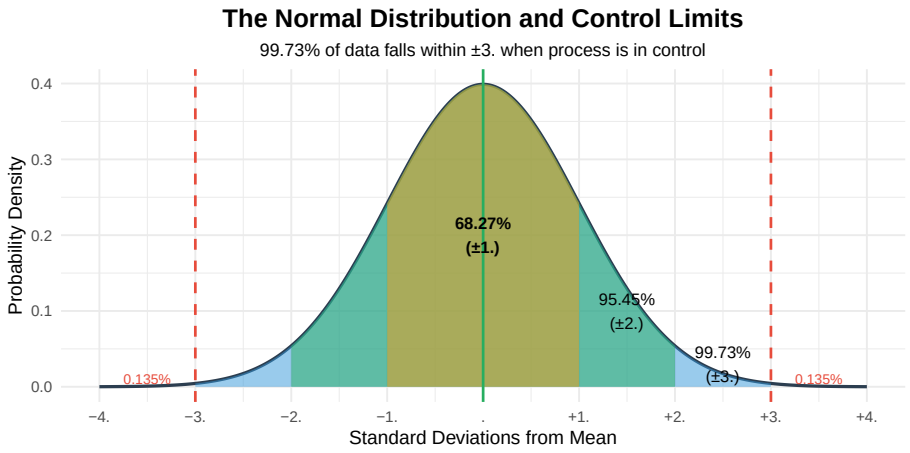


Figure 9.6: Normal Distribution and Control Limits

9.4.4 Choosing the Right Control Chart

The type of control chart depends on the **type of data** being collected:

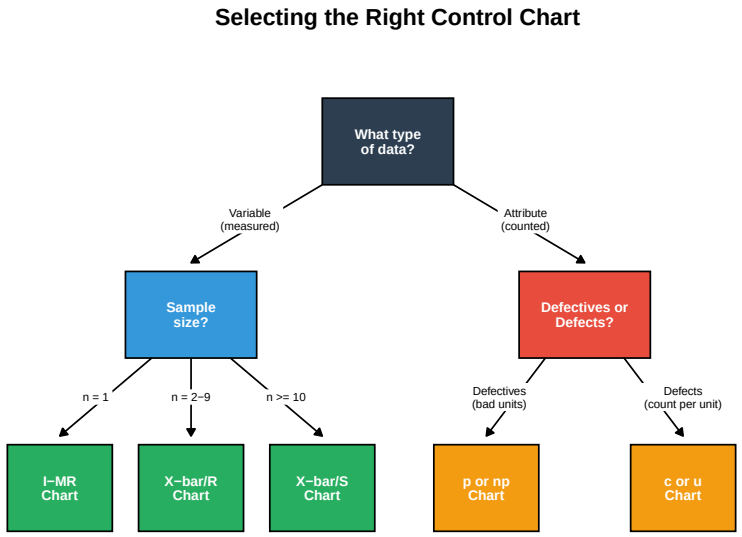


Figure 9.7: Control Chart Selection Flowchart

Table 9.4: Control Chart Summary

Chart Type	Data Type	Sample Size	What It Charts	Typical
X-bar / R	Variable	2-9 per subgroup	Subgroup mean and range	Productio
X-bar / S	Variable	10 per subgroup	Subgroup mean and std dev	Large sub
I-MR	Variable	1 (individuals)	Individual values and moving range	Continuo
p chart	Attribute	Variable or constant	Proportion defective	Pass/fail
np chart	Attribute	Constant	Number defective	Pass/fail
c chart	Attribute	Constant	Number of defects	Defects p
u chart	Attribute	Variable	Defects per unit	Defects p

Question: Variable vs. Attribute Data - What's the Difference?

Variable Data (Measured): - Can take any value within a range - Measured on a continuous scale - Examples: length, weight, temperature, voltage, pressure - More information per measurement - Smaller sample sizes needed

Attribute Data (Counted): - Discrete categories (good/bad, pass/fail) - Counted, not measured - Examples: number of defects, number of rejects, yes/no - Less information per observation - Larger sample sizes needed

When to Use Each: - Use variable data when you CAN measure (more powerful) - Use attribute data when you can only classify or count - Sometimes you convert variable to attribute (e.g., pass/fail based on tolerance)

9.4.4.1 Try it yourself: choose the chart

Eight situations from different plants. Decide which control chart fits each one, then check. Ask two things in order: is the data *measured* or *counted*, and how many readings make up one plotted point?

Which control chart?

Pick the control chart that fits each situation.

Categories: X-bar / R / I-MR / p chart / c chart

1. Five shaft diameters are measured every hour on a CNC lathe. _____
2. Four fill weights are taken from the bottling line every 30 minutes. _____
3. One viscosity reading is taken per batch of paint; a batch takes four hours. _____
4. A destructive pull test is done on one weld per shift. _____
5. Each shift records the fraction of bottles rejected for a bad cap; the number inspected changes from shift to shift. _____

6. Each day's lot of circuit boards is tested and the share that fails is recorded. _____
7. Paint blemishes are counted on every car hood. _____
8. Solder defects are counted on one board of the same type each hour. _____

Answers: 1 X-bar / R; 2 X-bar / R; 3 I-MR; 4 I-MR; 5 p chart; 6 p chart; 7 c chart; 8 c chart.

9.5 Variable Control Charts

9.5.1 X-bar and R Chart

The **X-bar and R chart** is the most common control chart for variable data. It tracks both the process average (X-bar) and the process spread (Range).

9.5.1.1 Constructing an X-bar/R Chart

Step 1: Collect data in subgroups (typically 4-5 samples)

```
# Example: Measuring shaft diameter (mm)
# 25 subgroups of 5 measurements each

set.seed(100)
n_subgroups <- 25
n_samples <- 5

# Generate data
data <- matrix(rnorm(n_subgroups * n_samples, mean = 25.00, sd = 0.05),
               nrow = n_subgroups, ncol = n_samples)

# Add a special cause (shift) at subgroup 18
data[18, ] <- data[18, ] + 0.12

# Calculate statistics for each subgroup
xbar <- apply(data, 1, mean)
R <- apply(data, 1, function(x) max(x) - min(x))

cat("First 5 subgroups:\n")
print(round(data[1:5, ], 4))
cat("\nX-bar values:", round(xbar[1:5], 4))
cat("\nRange values:", round(R[1:5], 4))
```

```
First 5 subgroups:
        [,1]    [,2]    [,3]    [,4]    [,5]
```

```
[1,] 24.9749 24.9781 24.9776 25.0006 24.9834
[2,] 25.0066 24.9640 24.9131 24.9456 25.0682
[3,] 24.9961 25.0115 25.0089 25.0135 24.9765
[4,] 25.0443 24.9421 25.0949 25.0504 25.0421
[5,] 25.0058 25.0124 24.8864 24.8963 24.9271

X-bar values: 24.9829 24.9795 25.0013 25.0348 24.9456
Range values: 0.0257 0.1551 0.037 0.1528 0.126
```

Step 2: Calculate control limits

```r
# Control chart constants for n=5
A2 <- 0.577
D3 <- 0      # D3 = 0 for n < 7
D4 <- 2.114

# Calculate grand mean and average range
X_double_bar <- mean(xbar)
R_bar <- mean(R)

cat("Grand Mean (X-double-bar):", round(X_double_bar, 4), "mm\n")
cat("Average Range (R-bar):", round(R_bar, 4), "mm\n\n")

# X-bar chart limits
UCL_xbar <- X_double_bar + A2 * R_bar
LCL_xbar <- X_double_bar - A2 * R_bar

cat("X-bar Chart:\n")
cat("  UCL =", round(UCL_xbar, 4), "mm\n")
cat("  CL  =", round(X_double_bar, 4), "mm\n")
cat("  LCL =", round(LCL_xbar, 4), "mm\n\n")

# R chart limits
UCL_R <- D4 * R_bar
LCL_R <- D3 * R_bar

cat("R Chart:\n")
cat("  UCL =", round(UCL_R, 4), "mm\n")
cat("  CL  =", round(R_bar, 4), "mm\n")
cat("  LCL =", round(LCL_R, 4), "mm\n")
```

```
Grand Mean (X-double-bar): 25.004 mm
Average Range (R-bar): 0.1023 mm

X-bar Chart:
  UCL = 25.0631 mm
  CL  = 25.004 mm
```

```
  LCL = 24.945 mm

R Chart:
  UCL = 0.2164 mm
  CL  = 0.1023 mm
  LCL = 0 mm
```

Step 3: Plot the charts

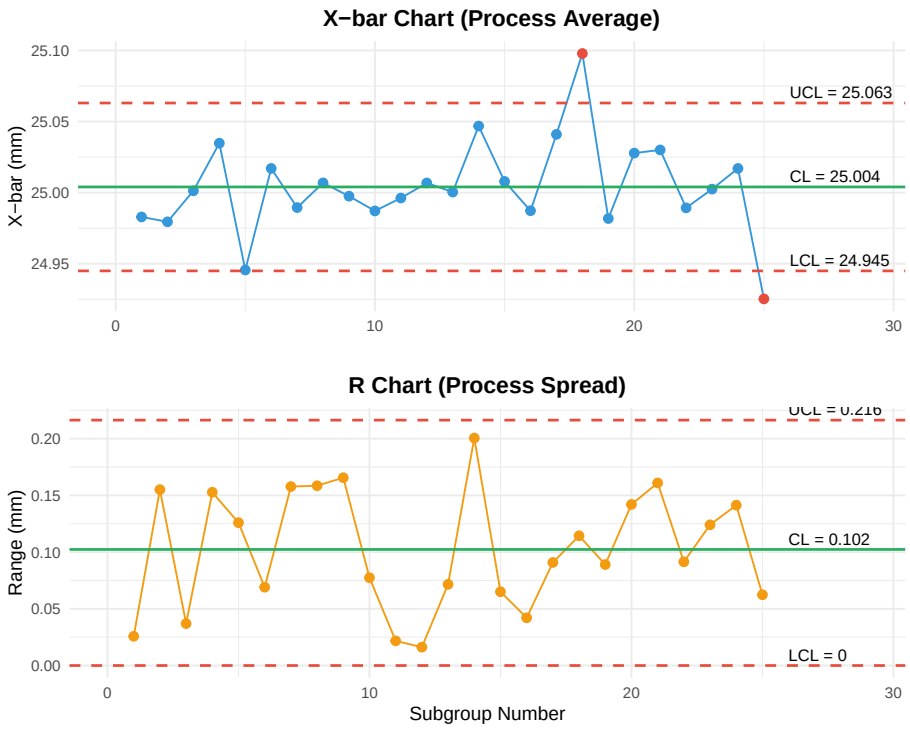


Figure 9.8: X-bar and R Control Charts

Interpretation: Subgroup 18 shows a point above the UCL on the X-bar chart, indicating a special cause. Investigation reveals the cause and corrective action is taken.

Teaching demo — enter your own subgroup means and ranges (or change `A2`/`D3`/`D4` for a different subgroup size) and see the control limits recalculate.

Show the live R code behind this card

```
#| label: xbar-r-live
# Subgroup averages (X-bar) and ranges (R) for 10 subgroups
xbar <- c(25.01, 24.99, 25.02, 25.00, 25.03, 25.01, 25.12, 25.02, 24.98, 25.00)
```

```
R    <- c(0.08, 0.11, 0.07, 0.09, 0.10, 0.08, 0.09, 0.12, 0.07, 0.10)

# Control chart constants (n = 5)
A2 <- 0.577
D3 <- 0
D4 <- 2.114

X_double_bar <- mean(xbar)
R_bar <- mean(R)

UCL_xbar <- X_double_bar + A2 * R_bar
LCL_xbar <- X_double_bar - A2 * R_bar
UCL_R <- D4 * R_bar
LCL_R <- D3 * R_bar

out_of_control <- which(xbar > UCL_xbar | xbar < LCL_xbar)
if (length(out_of_control) > 0) {
  cat(sprintf("!!Out-of-control subgroup(s): %s!!\n", paste(out_of_control, collapse = ", ")))
} else {
  cat("All subgroups within control limits\n")
}

cat(sprintf("  X-bar chart UCL  %.4f\n", UCL_xbar))
cat(sprintf("  X-bar chart CL  %.4f\n", X_double_bar))
cat(sprintf("  X-bar chart LCL  %.4f\n", LCL_xbar))
cat(sprintf("  R chart UCL  %.4f\n", UCL_R))
cat(sprintf("  R chart CL  %.4f\n", R_bar))
cat(sprintf("  R chart LCL  %.4f\n", LCL_R))

par(mfrow = c(2, 1), mar = c(5, 5, 3, 1))
plot(xbar, type = "b", pch = 19, ylim = range(xbar, UCL_xbar, LCL_xbar),
     xlab = "Subgroup", ylab = "X-bar", main = "X-bar chart")
abline(h = c(UCL_xbar, X_double_bar, LCL_xbar), lty = c(2, 1, 2),
       col = c("#C0392B", "#27AE60", "#C0392B"))
points(out_of_control, xbar[out_of_control], pch = 19, cex = 1.6, col = "#C0392B")
plot(R, type = "b", pch = 19, ylim = range(0, R, UCL_R),
     xlab = "Subgroup", ylab = "Range", main = "R chart")
abline(h = c(UCL_R, R_bar, LCL_R), lty = c(2, 1, 2),
       col = c("#C0392B", "#27AE60", "#C0392B"))
par(mfrow = c(1, 1))
```

9.5.2 Control Chart Constants

Table 9.5: Control Chart Constants for X-bar/R Charts

n	A2	D3	D4	d2
2	1.880	0.000	3.267	1.128
3	1.023	0.000	2.574	1.693
4	0.729	0.000	2.282	2.059
5	0.577	0.000	2.114	2.326
6	0.483	0.000	2.004	2.534
7	0.419	0.076	1.924	2.704
8	0.373	0.136	1.864	2.847
9	0.337	0.184	1.816	2.970
10	0.308	0.223	1.777	3.078

9.5.2.1 Try it yourself: build an X-bar/R chart by hand

Now do the arithmetic yourself on fresh data. A grinding cell measures the diameter of four piston pins (mm) every hour; eight subgroups are shown below. Fill in the mean and range of each subgroup, then the centre lines and control limits. Use the constants for $\mathbf{n = 4}$ from the table above. Check as often as you like.

X-bar/R chart: piston pin diameter

Fill in X-bar and R for each subgroup, then the limits. Work to three decimal places.

Subgroup	x1	x2	x3	x4	X-bar	R
1	12.02	11.98	12.01	11.99		
2	12.03	12.00	11.97	12.02		
3	11.99	12.01	12.04	12.00		
4	12.00	11.96	12.01	11.99		
5	12.05	12.01	12.02	12.00		
6	11.98	12.02	12.00	12.00		
7	12.01	12.03	11.99	12.01		
8	11.97	12.00	12.02	11.99		

- Grand mean (X-double-bar): _____ mm *(to 3 decimal places)*
- Average range (R-bar): _____ mm *(to 3 decimal places)*
- X-bar chart UCL: _____ mm *(to 3 decimal places)*
- X-bar chart LCL: _____ mm *(to 3 decimal places)*
- R chart UCL: _____ mm *(to 3 decimal places)*
- R chart LCL: _____ mm *(to 3 decimal places)*

Show the worked solution

Subgroup	x1	x2	x3	x4	X-bar	R
1	12.02	11.98	12.01	11.99	12.000	0.04
2	12.03	12.00	11.97	12.02	12.005	0.06
3	11.99	12.01	12.04	12.00	12.010	0.05
4	12.00	11.96	12.01	11.99	11.990	0.05
5	12.05	12.01	12.02	12.00	12.020	0.05
6	11.98	12.02	12.00	12.00	12.000	0.04
7	12.01	12.03	11.99	12.01	12.010	0.04
8	11.97	12.00	12.02	11.99	11.995	0.05

- Grand mean = 12.0038 mm, average range = 0.0475 mm
- X-bar chart: UCL = 12.0038 + 0.729 × 0.0475 = **12.038 mm**, LCL = **11.969 mm**
- R chart: UCL = 2.282 × 0.0475 = **0.108 mm**, LCL = 0 × 0.0475 = **0 mm**

Every subgroup mean lies between the X-bar limits and every range is below the R chart UCL, so these eight subgroups show no special cause.

9.5.3 Individual and Moving Range (I-MR) Chart

When sample size is **n = 1** (one measurement per time period), use the **I-MR chart**.

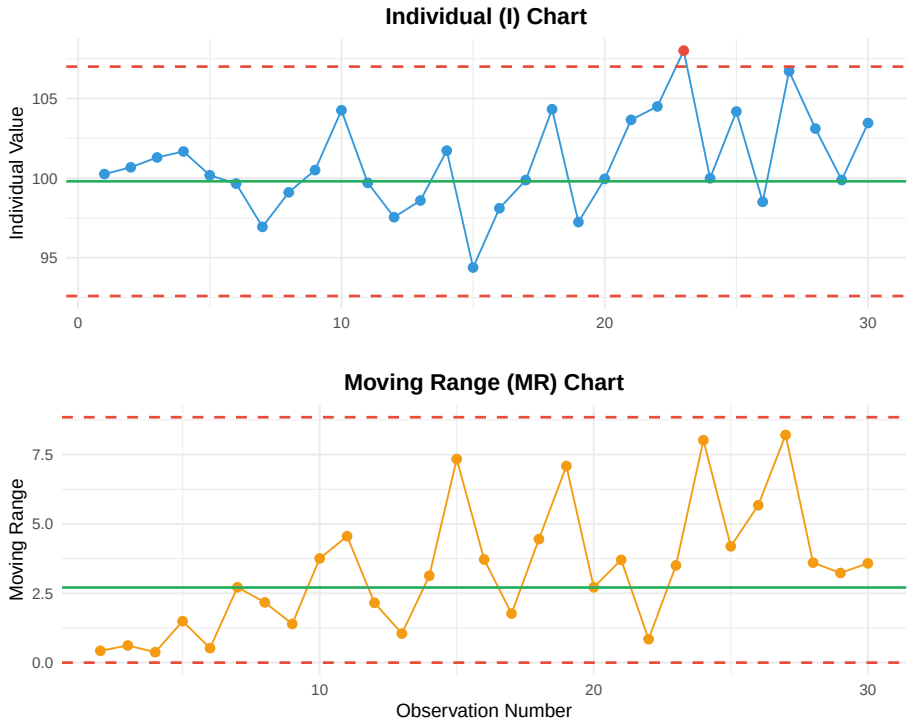


Figure 9.9: Individual and Moving Range Chart

I-MR Chart Formulas:

Chart	Center Line	UCL	LCL
I Chart	$\bar{X}$	$\bar{X} + 2.66\bar{MR}$	$\bar{X} - 2.66\bar{MR}$
MR Chart	$\bar{MR}$	$3.267 \times \bar{MR}$	0

Teaching demo — enter your own individual measurements and see the I-MR limits and moving ranges recalculate.

Show the live R code behind this card

```
#| label: imr-live
individuals <- c(100.2, 99.8, 100.5, 100.1, 99.6, 100.3, 100.0, 99.9,
                 100.4, 106.1, 105.8, 106.3)

MR <- c(NA, abs(diff(individuals)))

X_bar <- mean(individuals)
```

```
MR_bar <- mean(MR, na.rm = TRUE)

UCL_I <- X_bar + 2.66 * MR_bar
LCL_I <- X_bar - 2.66 * MR_bar
UCL_MR <- 3.267 * MR_bar

out_of_control <- which(individuals > UCL_I | individuals < LCL_I)
if (length(out_of_control) > 0) {
  cat(sprintf("!!Out-of-control observation(s): %s!!\n", paste(out_of_control, collapse = ", ")))
} else {
  cat("All observations within control limits\n")
}

cat(sprintf("  I chart UCL  %.3f\n", UCL_I))
cat(sprintf("  I chart CL  %.3f\n", X_bar))
cat(sprintf("  I chart LCL  %.3f\n", LCL_I))
cat(sprintf("  MR chart UCL  %.3f\n", UCL_MR))
cat(sprintf("  MR chart CL  %.3f\n", MR_bar))
cat("  MR chart LCL  0\n")

par(mfrow = c(2, 1), mar = c(5, 5, 3, 1))
plot(individuals, type = "b", pch = 19, ylim = range(individuals, UCL_I, LCL_I),
     xlab = "Observation", ylab = "Measurement", main = "Individuals (I) chart")
abline(h = c(UCL_I, X_bar, LCL_I), lty = c(2, 1, 2), col = c("#C0392B", "#27AE60", "#C0392B"))
points(out_of_control, individuals[out_of_control], pch = 19, cex = 1.6, col = "#C0392B")
plot(MR, type = "b", pch = 19, ylim = range(0, MR, UCL_MR, na.rm = TRUE),
     xlab = "Observation", ylab = "Moving range", main = "Moving range (MR) chart")
abline(h = c(UCL_MR, MR_bar), lty = c(2, 1), col = c("#C0392B", "#27AE60"))
par(mfrow = c(1, 1))
```

9.6 Attribute Control Charts

9.6.1 p Chart (Proportion Defective)

The **p chart** tracks the proportion of defective units when sample sizes may vary.

```
# Example: Daily inspection data
# Varying sample sizes (different production volumes)

set.seed(300)
days <- 20
sample_sizes <- sample(80:120, days, replace = TRUE)
```

```r
defectives <- rbinom(days, sample_sizes, prob = 0.05)

# Add a special cause on day 15
defectives[15] <- rbinom(1, sample_sizes[15], prob = 0.12)

# Calculate proportion
p <- defectives / sample_sizes

# Calculate p-bar (weighted average)
p_bar <- sum(defectives) / sum(sample_sizes)

cat("p-bar (average proportion defective):", round(p_bar, 4), "\n\n")

# Calculate control limits (vary with sample size)
UCL_p <- p_bar + 3 * sqrt(p_bar * (1 - p_bar) / sample_sizes)
LCL_p <- pmax(0, p_bar - 3 * sqrt(p_bar * (1 - p_bar) / sample_sizes))

cat("Sample of control limits (varying with n):\n")
cat("Day 1: UCL =", round(UCL_p[1], 4), ", LCL =", round(LCL_p[1], 4), "(n =", sample_s
cat("Day 5: UCL =", round(UCL_p[5], 4), ", LCL =", round(LCL_p[5], 4), "(n =", sample_s
```

```
p-bar (average proportion defective): 0.0493

Sample of control limits (varying with n):
Day 1: UCL = 0.1167 , LCL = 0 (n = 93 )
Day 5: UCL = 0.1143 , LCL = 0 (n = 100 )
```

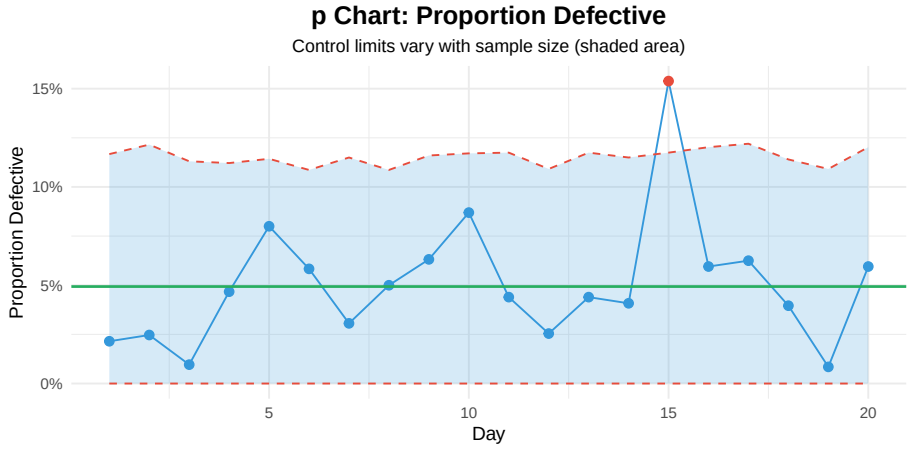


Figure 9.10: p Chart for Proportion Defective

Teaching demo — try different defect counts and sample sizes and see p-bar and

the control limits recalculate.

Show the live R code behind this card

```
#| label: p-chart-live
sample_sizes <- c(100, 95, 110, 105, 90, 100, 115, 100, 95, 108)
defectives   <- c(4, 5, 6, 3, 4, 5, 15, 4, 6, 5)

p <- defectives / sample_sizes
p_bar <- sum(defectives) / sum(sample_sizes)

UCL_p <- p_bar + 3 * sqrt(p_bar * (1 - p_bar) / sample_sizes)
LCL_p <- pmax(0, p_bar - 3 * sqrt(p_bar * (1 - p_bar) / sample_sizes))

out_of_control <- which(p > UCL_p | p < LCL_p)
if (length(out_of_control) > 0) {
  cat(sprintf("!!Out-of-control day(s): %s!!\n", paste(out_of_control, collapse = ", ")))
} else {
  cat("All days within control limits\n")
}

cat(sprintf("  p-bar (average proportion defective)  %.4f\n", p_bar))
for (i in seq_along(p)) {
  cat(sprintf("  Day %g (n = %g)  p = %.4f, limits %.4f to %.4f\n",
              i, sample_sizes[i], p[i], LCL_p[i], UCL_p[i]))
}

par(mar = c(5, 5, 3, 1))
plot(p, type = "b", pch = 19, ylim = range(0, p, UCL_p),
     xlab = "Day", ylab = "Proportion defective", main = "p chart")
lines(UCL_p, type = "s", lty = 2, col = "#C0392B")
lines(LCL_p, type = "s", lty = 2, col = "#C0392B")
abline(h = p_bar, col = "#27AE60")
points(out_of_control, p[out_of_control], pch = 19, cex = 1.6, col = "#C0392B")
```

9.6.2 c Chart (Count of Defects)

The **c chart** tracks the number of defects when the sample size (area of opportunity) is constant.

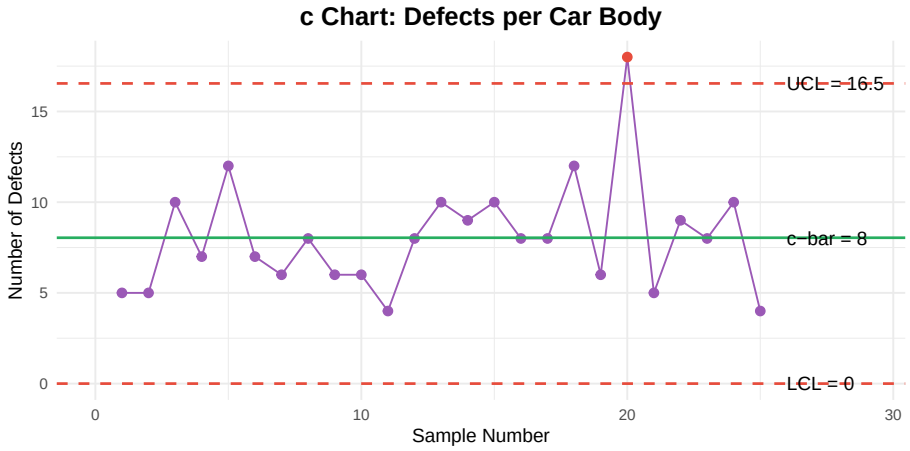


Figure 9.11: c Chart for Defects per Unit

Teaching demo — try different defect counts and see the c chart limits recalculate.

Show the live R code behind this card

```
#| label: c-chart-live
defects <- c(7, 9, 6, 8, 10, 7, 8, 9, 6, 19, 8, 7)

c_bar <- mean(defects)
UCL_c <- c_bar + 3 * sqrt(c_bar)
LCL_c <- max(0, c_bar - 3 * sqrt(c_bar))

out_of_control <- which(defects > UCL_c | defects < LCL_c)
if (length(out_of_control) > 0) {
  cat(sprintf("!!Out-of-control sample(s): %s!!\n", paste(out_of_control, collapse = "
} else {
  cat("All samples within control limits\n")
}

cat(sprintf("  c-bar (average defects per unit)  %.2f\n", c_bar))
cat(sprintf("  UCL  %.2f\n", UCL_c))
cat(sprintf("  LCL  %.2f\n", LCL_c))

par(mar = c(5, 5, 3, 1))
plot(defects, type = "b", pch = 19, ylim = range(0, defects, UCL_c),
     xlab = "Sample", ylab = "Defects", main = "c chart")
abline(h = c(UCL_c, c_bar, LCL_c), lty = c(2, 1, 2), col = c("#C0392B", "#27AE60", "#C
points(out_of_control, defects[out_of_control], pch = 19, cex = 1.6, col = "#C0392B")
```

Table 9.7: Attribute Control Ch

Chart	Center Line	UCL
p chart	$\bar{p} = \frac{\sum d}{\sum n}$	$\bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$
np chart	$\bar{np} = \frac{\sum d}{k}$	$\bar{np} + 3\sqrt{\bar{np}(1-\bar{p})}$
c chart	$\bar{c} = \frac{\sum c}{k}$	$\bar{c} + 3\sqrt{\bar{c}}$
u chart	$\bar{u} = \frac{\sum c}{\sum n}$	$\bar{u} + 3\sqrt{\frac{\bar{u}}{n}}$

9.7 Interpreting Control Charts

9.7.1 Out-of-Control Signals (Western Electric Rules)

A point doesn't have to be outside the control limits to indicate a problem. The **Western Electric Rules** identify non-random patterns:

Table 9.8: Western Electric Rules for Control Chart Interpretation

Rule	Pattern	Description
Rule 1	**Point beyond control limit**	One point above UCL or below LCL
Rule 2	**Zone A: 2 of 3 consecutive points**	2 out of 3 consecutive points in Zone A (>2) or beyond,
Rule 3	**Zone B: 4 of 5 consecutive points**	4 out of 5 consecutive points in Zone B (>1) or beyond,
Rule 4	**8 consecutive points on one side**	8 consecutive points above or below the center line

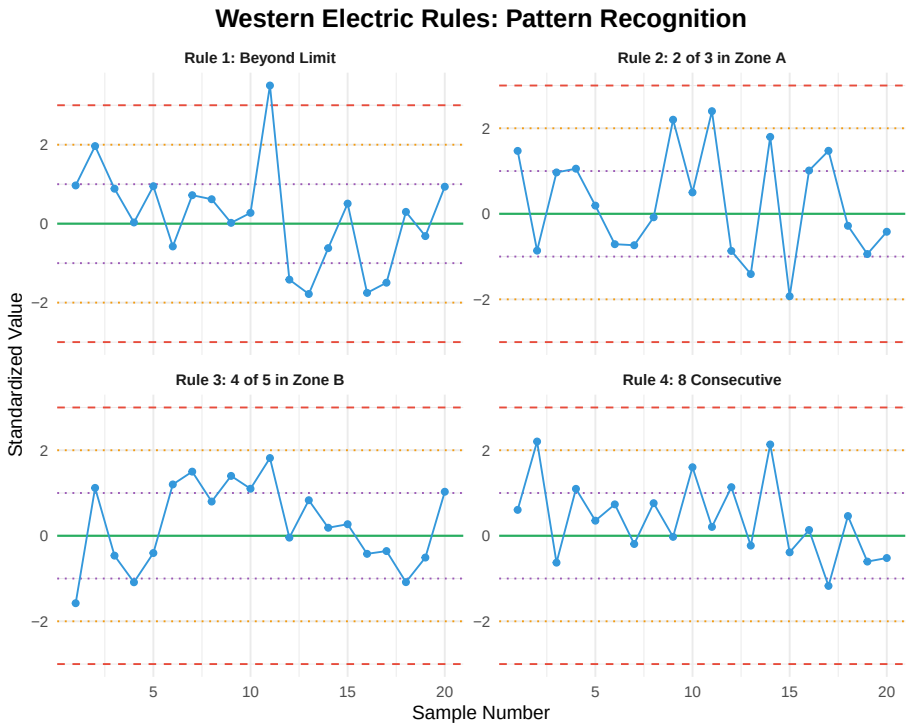


Figure 9.12: Visual Examples of Western Electric Rules

9.7.2 Additional Patterns to Watch For

Table 9.9: Additional Control Chart Patterns

Pattern	Description	Possible C
****Trend****	6 or more consecutive points steadily increasing or decreasing	Tool wear, t
****Cycle****	Repeating high-low pattern	Shift change
****Stratification****	Points consistently near center line (too little variation)	Incorrect lim
****Mixture****	Points avoiding center line (bimodal distribution)	Two machin

Interactive Exercise: Identify the Out-of-Control Condition

Look at this sequence of points (standardized values):

0.5, 0.8, 1.2, 0.9, 1.4, 1.6, 1.3, 1.8, -0.2, 0.3

Questions:

1. Are any points beyond ± 3 ?
2. Is there a Rule 4 violation (8 consecutive on one side)?

3. What pattern do you see?

Answers:

1. No — all points are within ±3
2. **YES** — Points 1-8 are all positive (above center line)
3. This is a **Rule 4 violation**: 8 consecutive points above the center line indicates a likely process shift upward

Action: Investigate what changed — new material lot? Different operator? Equipment adjustment?

9.7.3 Try it yourself: spot the signal

Six control charts, each with 15 points plotted in standard-deviation units (the dashed lines are at ±1 , ±2 and ±3). Five of them break exactly one of the four Western Electric rules. One is in control. Decide which is which.

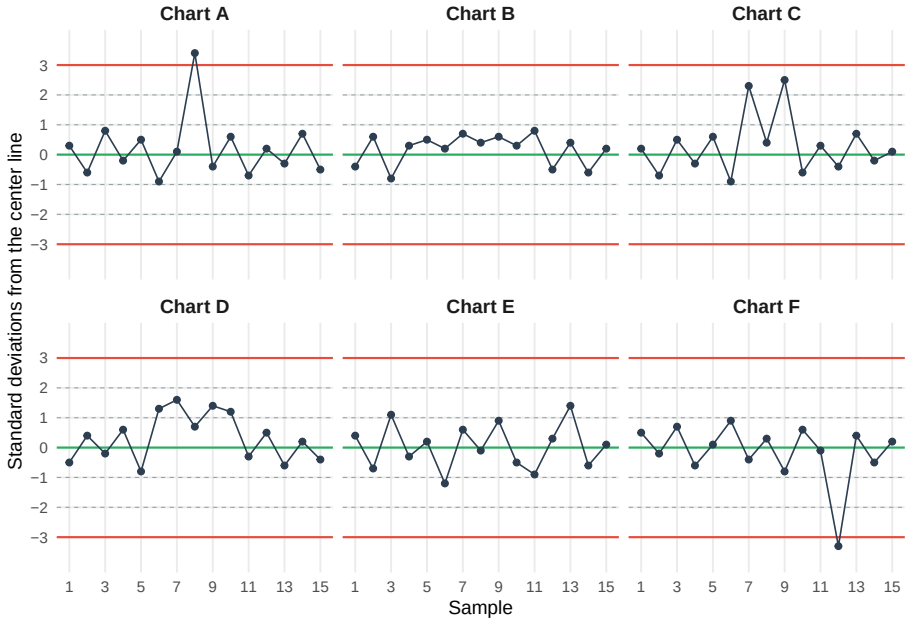


Figure 9.13: Six control charts: which rule, if any, does each one break?

Which rule does each chart break?

Pick the rule each chart breaks, or 'In control'.

Categories: Rule 1 / Rule 2 / Rule 3 / Rule 4 / In control

1. Chart A _____

2. Chart B _____
3. Chart C _____
4. Chart D _____
5. Chart E _____
6. Chart F _____

Answers: 1 Rule 1; 2 Rule 4; 3 Rule 2; 4 Rule 3; 5 In control; 6 Rule 1.

9.8 Process Capability

9.8.1 What is Process Capability?

Process capability compares the process performance to the specification limits. It answers: "Can this process consistently produce parts within tolerance?"

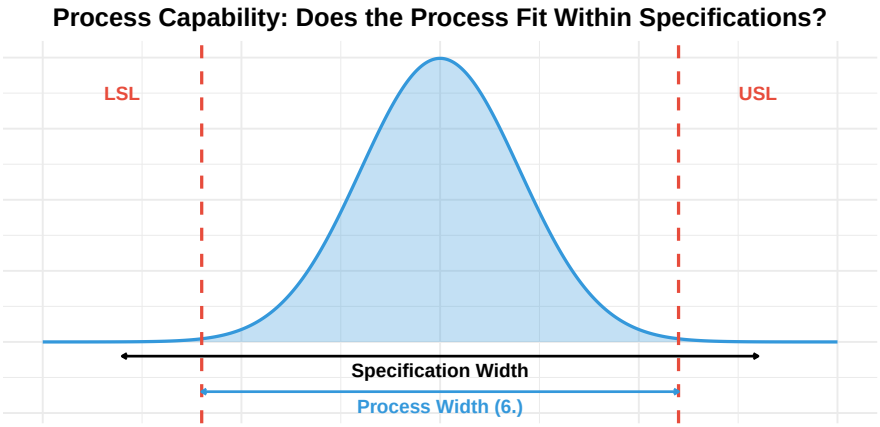


Figure 9.14: Process Capability: Comparing Process to Specifications

9.8.2 Capability Indices

Table 9.10: Process Capability I

Index	Formula
Cp	$\frac{USL - LSL}{6\sigma}$
Cpk	$\min\left(\frac{USL - \bar{X}}{3\sigma}, \frac{\bar{X} - LSL}{3\sigma}\right)$
Pp	$\frac{USL - LSL}{6s}$
Ppk	$\min\left(\frac{USL - \bar{X}}{3s}, \frac{\bar{X} - LSL}{3s}\right)$

9.8.3 Cp vs. Cpk: Why Both Matter

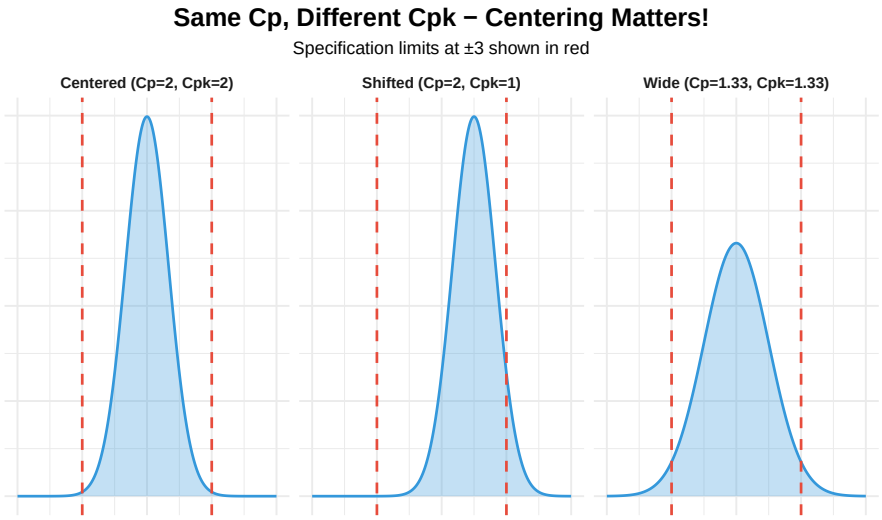


Figure 9.15: Cp vs. Cpk: The Importance of Centering

9.8.4 Capability Example Calculation

```
# Example: Shaft diameter specification 25.00 ± 0.15 mm
# Process data from 25 subgroups of 5

USL <- 25.15
LSL <- 24.85
Target <- 25.00

# Using earlier X-bar/R data
X_bar_cap <- mean(xbar)
R_bar_cap <- mean(R)

# Estimate sigma using R-bar/d2 (d2 = 2.326 for n=5)
d2 <- 2.326
sigma_within <- R_bar_cap / d2

cat("Process Statistics:\n")
cat("  Mean (X-bar):", round(X_bar_cap, 4), "mm\n")
cat("  Sigma (estimated):", round(sigma_within, 4), "mm\n\n")

# Calculate Cp
Cp <- (USL - LSL) / (6 * sigma_within)
```

```
cat("Cp =", round(Cp, 2), "\n")

# Calculate Cpk
Cpu <- (USL - X_bar_cap) / (3 * sigma_within)
Cpl <- (X_bar_cap - LSL) / (3 * sigma_within)
Cpk <- min(Cpu, Cpl)

cat("Cpu =", round(Cpu, 2), "\n")
cat("Cpl =", round(Cpl, 2), "\n")
cat("Cpk =", round(Cpk, 2), "\n\n")

# Interpretation
cat("Interpretation:\n")
if (Cpk >= 1.33) {
  cat("Process is CAPABLE (Cpk >= 1.33)\n")
} else if (Cpk >= 1.00) {
  cat("Process is MARGINALLY CAPABLE (1.00 <= Cpk < 1.33)\n")
} else {
  cat("Process is NOT CAPABLE (Cpk < 1.00)\n")
}
```

```
Process Statistics:
  Mean (X-bar): 25.004 mm
  Sigma (estimated): 0.044 mm

Cp = 1.14
Cpu = 1.11
Cpl = 1.17
Cpk = 1.11

Interpretation:
Process is MARGINALLY CAPABLE (1.00 <= Cpk < 1.33)
```

Teaching demo — change the specification limits, process mean, or sigma below.

Show the live R code behind this calculator

```
#| label: cp-cpk-live
USL <- 25.15
LSL <- 24.85
X_bar_cap <- 25.03
sigma_within <- 0.045

Cp <- (USL - LSL) / (6 * sigma_within)

Cpu <- (USL - X_bar_cap) / (3 * sigma_within)
Cpl <- (X_bar_cap - LSL) / (3 * sigma_within)
```

```
Cpk <- min(Cpu, Cpl)

if (Cpk >= 1.33) {
  cat(sprintf("Cpk = %.2f - Capable\n", Cpk))
} else if (Cpk >= 1.00) {
  cat(sprintf("Cpk = %.2f - Marginally capable\n", Cpk))
} else {
  cat(sprintf("Cpk = %.2f - !!Not capable!!\n", Cpk))
}
cat(sprintf("  Cp                    %.2f\n", Cp))
cat(sprintf("  Cpu                   %.2f\n", Cpu))
cat(sprintf("  Cpl                   %.2f\n", Cpl))

par(mar = c(5, 5, 3, 1))
x <- seq(24, 26, length.out = 400)
plot(x, dnorm(x, X_bar_cap, sigma_within), type = "l", lwd = 2, col = "#2E86C1",
     xlim = c(24, 26), ylim = c(0, 42), yaxt = "n", xlab = "Measurement (mm)", ylab = "",
     main = "Process spread against the specification limits")
abline(v = c(LSL, USL), lty = 2, lwd = 2, col = "#C0392B")
abline(v = X_bar_cap, lty = 3, col = "grey40")
```

9.8.5 Capability Index Interpretation

Table 9.11: Capability Index Interpretation Guide

Cpk Value	Sigma Level	PPM Defective	Interpretation	Typical In
< 1.00	< 3	> 2,700	Not capable — improvement required	Requires im
1.00 - 1.33	3 - 4	2,700 - 64	Marginally capable — monitor closely	Consumer p
1.33 - 1.67	4 - 5	64 - 0.6	Capable — meets typical requirements	General mar
1.67 - 2.00	5 - 6	0.6 - 0.002	Good capability — automotive/aerospace	Automotive
> 2.00	> 6	< 0.002	Excellent — Six Sigma level	Aerospace, m

Pp and Ppk use the overall standard deviation of all the data instead of the within-subgroup estimate. When the process is stable the two sets of indices agree; a large gap between Cpk and Ppk means the process is drifting between subgroups. The worked comparison, and capability for non-normal data, are in **Chapter 18, Advanced Statistical Process Control**.

9.9 Responding to Out-of-Control Signals

9.9.1 The Response Process

Out-of-Control Response Procedure

1. **1. STOP** – Mark the point (• Circle the point; • Note the time; • Quarantine product)
2. **2. INVESTIGATE** – Find the cause (• Check 5 Ms; • Review recent changes; • Ask operators)
3. **3. CORRECT** – Eliminate cause (• Fix the problem; • Verify correction; • Restart process)
4. **4. DOCUMENT** – Record actions (• Update log; • File in PFMEA; • Share lessons)

9.9.2 Investigation Checklist (5 Ms + E)

Table 9.12: Investigation Checklist: 5 Ms + E

Category	Questions to Ask
Man	Different operator? New employee? Fatigue? Training issue?
Machine	Equipment malfunction? Tool wear? Maintenance due? Setup change?
Material	New lot? Different supplier? Out-of-spec material?
Method	Procedure changed? Parameter drift? Wrong program?
Measurement	Gauge drift? Calibration due? Different inspector?
Environment	Temperature change? Humidity? Vibration? Contamination?

9.10 SPC Implementation

9.10.1 Steps to Implement SPC

Table 9.13: SPC Implementation Steps

Step	Action	Key Considerations
1	Select the process and characteristic	Start with critical/problem processe
2	Determine measurement system adequacy (MSA)	Gauge R&R must be acceptable (<
3	Collect initial data (25+ subgroups)	Process must be running normally
4	Calculate trial control limits	Limits based on actual process, not
5	Identify and remove special causes	Investigate each out-of-control point
6	Recalculate limits with stable data	These become the standard limits
7	Implement ongoing monitoring	Train operators, establish reaction p

8	Review and improve continuously	Update limits when process improves

9.10.2 Common SPC Mistakes

Table 9.14: Common SPC Mistakes and Corrections

Mistake	Why It's Wrong	Cor
****Using specification limits as control limits****	Control limits are based on process performance, not specs	Calc
****Changing limits too frequently****	Limits should only change when process fundamentally changes	Use
****Ignoring the R chart****	R chart often signals problems before X-bar chart	Alwa
****Over-adjusting (tampering)****	Adjusting for common cause variation increases variation	Only
****Not acting on signals****	Defeats the purpose of SPC	Inve

---

9.11 Summary

Table 9.15: SPC Key Concepts Summary

Topic	Key Points
Variation	Common cause (inherent) vs. special cause (assignable); respond differently
Control Charts	Graph with UCL/LCL at ± 3 ; distinguishes between variation types
Variable Charts	X-bar/R (n=2-9), X-bar/S (n 10), I-MR (n=1)
Attribute Charts	p (proportion), np (count defectives), c (defects), u (defects/unit)
Interpretation	Western Electric rules: beyond limits, 2 of 3 in Zone A, 4 of 5 in Zone B, 8 consecutive
Capability	Cp (potential), Cpk (actual); 1.33 is capable; 1.67 for automotive

---

9.12 Review Questions

9.12.1 Conceptual Questions

Question 1: What is the difference between common cause and special cause variation?

Common Cause Variation: - Inherent to the process - Always present - Random and predictable - Many small factors - System-level improvement required - Examples: machine vibration, material variation, ambient temperature

Special Cause Variation: - External to normal process - Intermittent or new - Non-random patterns - Specific assignable cause - Local action to eliminate - Examples: tool breakage, new operator, bad material lot

Key Point: Only act on special causes. Attempting to adjust for common cause variation (tampering) makes the process worse.

Question 2: Why are control limits set at ±3 sigma rather than ±2 sigma?

At ±3 : - 99.73% of data falls within limits (when in control) - Only 0.27% false alarm rate (about 3 per 1000 points) - Good balance between detecting real problems and avoiding false alarms

If we used ±2 : - 95.45% of data within limits - 4.55% false alarm rate (about 45 per 1000 points!) - Too many false alarms → operators ignore the chart

If we used ±4 : - 99.99% within limits - Almost no false alarms - But we'd miss real problems (low sensitivity)

The ±3 choice balances: - Risk of false alarm (treating common cause as special) - Risk of missed signal (treating special cause as common)

Question 3: When would you use an I-MR chart instead of an X-bar/R chart?

Use I-MR Chart When:

1. **Sample size is n = 1** — only one measurement per time period
2. **Destructive testing** — can't afford multiple samples
3. **Expensive measurement** — time or cost prohibitive
4. **Continuous process** — each reading is independent (batch process)
5. **Long intervals** — measurements far apart in time
6. **Homogeneous batches** — within-batch variation is negligible

Examples: - Daily chemical batch analysis - Weekly calibration check - Each lot incoming inspection (one sample per lot) - Continuous process temperature readings

Caution: I-MR charts are less sensitive than X-bar charts because averaging in subgroups reduces noise.

9.12.2 Calculation Questions

Question 4: Calculate control limits for an X-bar/R chart given the following data.

Given:

- 20 subgroups of n = 4 measurements
- Grand mean (X-double-bar) = 50.25
- Average range (R-bar) = 2.40
- Constants for n = 4: A2 = 0.729, D3 = 0, D4 = 2.282

Calculate UCL and LCL for both charts.

Question 4: X-bar/R control limits

- X-bar chart UCL: _____ *(to 2 decimal places)*

- X-bar chart LCL: _____ *(to 2 decimal places)*
- R chart UCL: _____ *(to 2 decimal places)*
- R chart LCL: _____ *(to 2 decimal places)*

Show the worked solution

X-bar Chart:

```
UCL = X-double-bar + A2 × R-bar
UCL = 50.25 + 0.729 × 2.40 = 50.25 + 1.75 = 52.00

LCL = X-double-bar - A2 × R-bar
LCL = 50.25 - 0.729 × 2.40 = 50.25 - 1.75 = 48.50

CL = 50.25
```

R Chart:

```
UCL = D4 × R-bar = 2.282 × 2.40 = 5.48

LCL = D3 × R-bar = 0 × 2.40 = 0

CL = 2.40
```

Question 5: Calculate Cp and Cpk for a process with the following data.

Given:

- Specification: 100 ± 5 (USL = 105, LSL = 95)
- Process mean = 102
- Process standard deviation ( ) = 1.5

Question 5: Cp and Cpk

- Cp: _____ *(to 2 decimal places)*
- Cpk: _____ *(to 2 decimal places)*

Show the worked solution

Cp (Potential Capability):

```
Cp = (USL - LSL) / (6 )
Cp = (105 - 95) / (6 × 1.5)
Cp = 10 / 9 = 1.11
```

Cpk (Actual Capability):

```
Cpu = (USL - X-bar) / (3 ) = (105 - 102) / (3 × 1.5) = 3 / 4.5 = 0.67
Cpl = (X-bar - LSL) / (3 ) = (102 - 95) / (3 × 1.5) = 7 / 4.5 = 1.56

Cpk = min(Cpu, Cpl) = min(0.67, 1.56) = 0.67
```

Interpretation: - Cp = 1.11 suggests process spread could fit within specs IF centered - Cpk = 0.67 shows process is NOT capable because it's shifted toward USL - Action: Center the process closer to 100 (target)

Question 6: A p chart has p-bar = 0.04 and sample size n = 200. Calculate control limits.

Question 6: p chart control limits

Enter the limits as proportions (for example 0.05), not percentages.

- UCL: _____ *(to 4 decimal places)*
- LCL: _____ *(to 4 decimal places)*

Show the worked solution

Center Line:

```
CL = p-bar = 0.04 (4% defective)
```

Upper Control Limit:

```
UCL = p-bar + 3 × √(p-bar × (1 - p-bar) / n)
UCL = 0.04 + 3 × √(0.04 × 0.96 / 200)
UCL = 0.04 + 3 × √(0.000192)
UCL = 0.04 + 3 × 0.01386
UCL = 0.04 + 0.0416 = 0.0816 (8.16%)
```

Lower Control Limit:

```
LCL = p-bar - 3 × √(p-bar × (1 - p-bar) / n)
LCL = 0.04 - 0.0416 = -0.0016

Since LCL cannot be negative: LCL = 0
```

Final Limits: - UCL = 8.16% - CL = 4.00% - LCL = 0%

9.12.3 Application Questions

Question 7: You observe 8 consecutive points above the center line but all within control limits. What does this indicate?

This is a Rule 4 violation (Western Electric Rules).

What it indicates: - The process mean has shifted upward - This is a SPECIAL CAUSE signal - Even though no points are beyond limits, the pattern is not random

Probability analysis: - Probability of a point above CL (random) = 0.5 - Probability of 8 consecutive above CL = 0.5^8 = 0.0039 (0.39%) - This is very unlikely by chance alone

Action required: 1. Stop and mark the chart 2. Investigate what changed around the time of the first high point 3. Check: new material lot? Tool

change? Operator change? Temperature? 4. Correct the cause 5. Monitor to verify process returns to normal

Question 8: A process has Cp = 2.0 but Cpk = 1.0. What does this tell you?

Analysis:

Cp = 2.0 means: - Process spread (6) is half the specification width - The process COULD produce no defects if centered - Potential capability is excellent

Cpk = 1.0 means: - Process is off-center - Actual capability is only marginal - Some defects are being produced

The difference (Cp - Cpk = 1.0) indicates: - The process mean is shifted 3 from the target - One side has much less margin than the other

Visual:

```
LSL         Target         USL
|            |            |
      [====PROCESS====]
      ↑ shifted to one side
```

Action: - Center the process on target - After centering, Cpk should approach Cp (both 2.0) - This is an adjustment action, not a variation reduction

Question 9: Why should you always analyze the R chart before the X-bar chart?

The R chart should be analyzed first because:

1. **Control limits depend on variation:**
   - X-bar limits are calculated using R-bar
   - If R is out of control, X-bar limits are unreliable
2. **R chart is more sensitive to variation changes:**
   - Tool wear shows on R chart before X-bar
   - Inconsistent setup appears as increased R
3. **Process must be stable in variation first:**
   - A process can't be "in control" if variation is changing
   - Stable variation is a prerequisite for evaluating the mean
4. **Different actions for each:**
   - R chart problem → variation issue (consistency)
   - X-bar problem → centering issue (average)

Rule: If R chart is out of control, fix that FIRST before interpreting the X-bar chart.

9.13 References

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Chapter 10

Working Effectively with AI

10.1 Learning Objectives

After completing this chapter, you will be able to:

1. Understand AI's role in modern engineering work and its limitations
2. Apply the four steps of effective AI use: Define, Prompt, Iterate, Validate
3. Apply prompt engineering principles to get better AI responses
4. Distinguish between poor, adequate, and excellent prompts
5. Select appropriate AI tools for different tasks (chat interfaces and projects, NotebookLM, coding agents, browser extensions)
6. Use iterative AI collaboration to solve complex engineering problems
7. Critically evaluate AI-generated content for accuracy and safety
8. Identify tasks suited for AI assistance versus human expertise
9. Apply ethical considerations when using AI in engineering contexts
10. Distinguish chat, projects, tool use, agents and multi-agent systems, and set the right permissions and review for each
11. Protect confidential information, and document and disclose your use of AI

Tip

Using these notes in class. This chapter is written to be taught live. Bring your phone (and a charger). You will meet five kinds of box:

- **Class poll**: a question with options. Vote by show of hands or in Brightspace, then the answer is revealed.

- **Think – pair – share**: think alone for a minute, talk to your neighbour, then share with the class.
- **Activity**: a short hands-on task, usually using your phone, in pairs or small groups.
- **Try it yourself**: an interactive exercise on the page that checks your answer. These work in your phone's browser.
- **Exit ticket**: one question to answer before you leave, so we know what stuck.

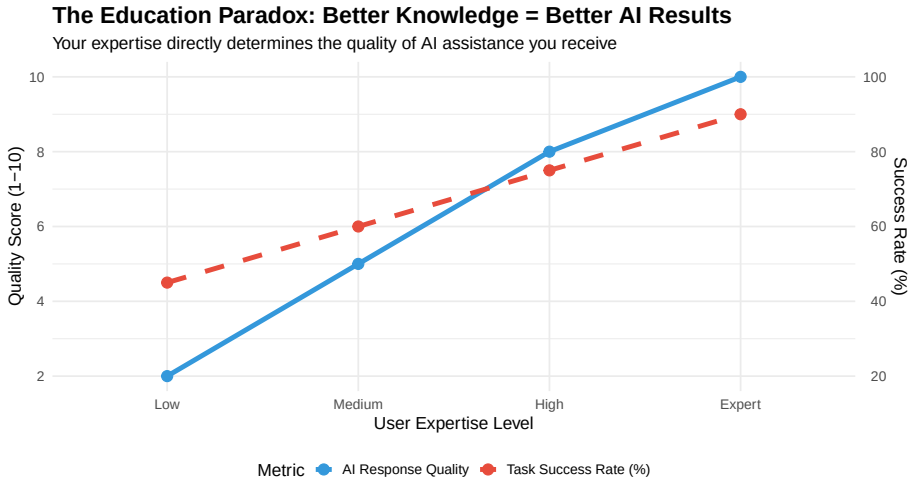
Wherever a chatbot is used in class, use your own account or the class demo on the projector. Do not paste anything confidential into it (see Section 10.13.2).

10.2 Introduction: AI in the Modern Workplace

Artificial Intelligence has fundamentally changed how engineering work is performed. According to research, AI is predominantly used for **coding and technical work** (34% of professional use), with complex tasks showing potential **12x speedups** compared to traditional methods.

However, there's a critical paradox: **more complex tasks have lower success rates** (61-66% versus 70% for simpler tasks). This means engineers must develop skills to effectively collaborate with AI, not just delegate to it.

10.2.1 The Education Paradox



Key Insight: There is a nearly perfect correlation between prompt sophistication and response quality. You need strong foundational knowledge to get quality AI assistance. "AI will do it for me" is a

trap.

Discussion: Why does expertise improve AI results?

The "education paradox" reveals a fundamental truth about AI collaboration:

- **Better questions lead to better answers**: Experts know what to ask and how to frame problems
- **Domain knowledge enables validation**: You can only verify AI output if you understand the domain
- **Iterative refinement requires expertise**: Knowing when to push back or ask follow-up questions
- **Critical evaluation is essential**: AI can sound confident while being wrong

Think about it: If you don't understand the fundamentals of motor selection, how would you know if an AI recommendation is correct or dangerously undersized?

Which statement is closest to how you use AI tools today? (No right answer, show of hands.)

- **A.** I do not use them
- **B.** I ask them questions and copy useful parts
- **C.** I use them to write or fix code
- **D.** I give them multi-step jobs and review the result

Reveal after the vote

No wrong answer. Keep your answer in mind. By the end of the chapter you will be able to say which of these you do well, which you do carelessly, and what you would change.

Think – pair – share (3 min). The education paradox says your expertise decides how much help AI gives you. Describe one topic from your program where you could judge an AI answer well, and one where you could not. What would you do about the second?

10.2.2 AI Impact by Industry

Table 10.1: AI Impact Across Engineering Industries

Industry	Common AI Applications	Potential Speedup
Automotive	Simulation modeling, control systems, predictive maintenance	10-12x
Food Processing	Process optimization, quality control, equipment diagnostics	8-10x
Defense	System design, modeling, compliance documentation	10-12x
Mining	Equipment diagnostics, predictive maintenance, safety analysis	8-10x
General Manufacturing	Documentation, troubleshooting guides, process analysis	6-8x

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Why do automotive and defense require "Critical" human oversight?

Safety-critical applications require mandatory human validation because:

1. **Regulatory requirements**: ISO 26262 (automotive) and MIL-STDs (defense) require documented human review
2. **Liability**: Engineers are legally responsible for designs, not AI tools
3. **Failure consequences**: Errors can cause injury, death, or mission failure
4. **AI limitations**: Current AI cannot reliably reason about edge cases and failure modes
5. **Standards compliance**: AI cannot certify compliance with safety standards

Best practice: Use AI for initial research and drafting, but always validate with human experts and formal analysis methods (FMEA, FTA, HAZOP).

10.3 Prompt Engineering Fundamentals

Prompt engineering is the skill of crafting inputs to AI systems that produce useful, accurate, and relevant outputs. In engineering contexts, this skill is becoming as important as technical writing.

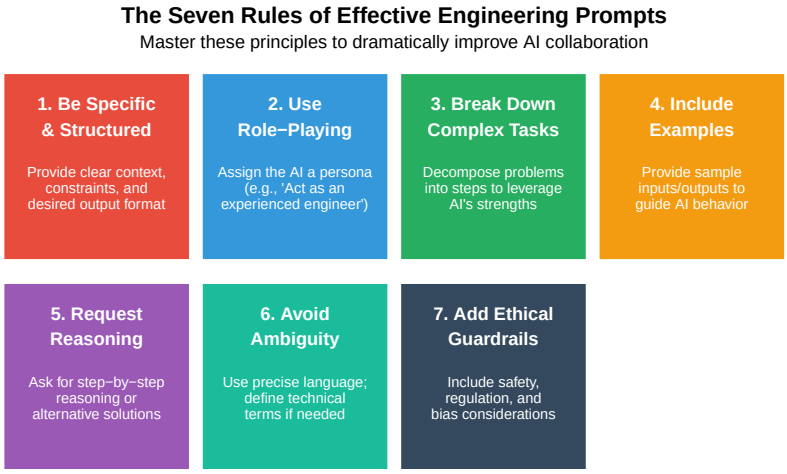
Before you read: What makes a good question?

Think about the difference between these two questions:

1. "Can you help me with my motor problem?"
2. "I have a 3-phase 460V 50HP motor that trips its overload after 15 minutes of running. Ambient temperature is 35°C. What diagnostic steps should I follow?"

Which question would get better help from a human expert? The same principle applies to AI. Before reading about prompt engineering, consider what information you would need to give a colleague to get useful help.

10.3.1 The Seven Rules of Effective Prompts



How the seven rules relate to R-C-T-C-F. In Chapter 11 you will use a shorter, five-letter checklist for prompts, **R-C-T-C-F** (Role, Context, Task, Constraints, Format). It is the compact form of the seven rules:

R-C-T-C-F	Comes from rule
Role	2. Use role-playing
Context	1. Be specific and structured
Task	3. Break down complex tasks
Constraints	6. Avoid ambiguity, 7. Add ethical guardrails
Format	1. Be specific and structured, 4. Include examples

Rule 5 (request reasoning) is a *technique* you add to any prompt rather than a part of its structure.

10.4 Prompt Quality Comparison: Bad, Medium, and Good

The following examples demonstrate the dramatic difference that prompt quality makes. Each example shows the same task approached three ways.

Table 10.2: Motor Selection Prompt Comparison

Quality Level	Prompt
Bad	Help me pick a motor.
Medium	I need to select a motor for a conveyor system. The conveyor moves boxes that weigh
Good	Act as a senior electromechanical engineer. I need to select an AC induction motor fo

Table 10.3: PLC Troubleshooting Prompt Comparison

Quality Level	Prompt
Bad	My PLC isn't working. Fix it.
Medium	I have an Allen-Bradley PLC that's showing a fault. The output for a motor starter i
Good	Act as an experienced automation technician troubleshooting a control system issue.

10.4.1 Example 1: Motor Selection

10.4.2 Example 2: Troubleshooting a PLC System

10.4.3 Example 3: Safety Analysis

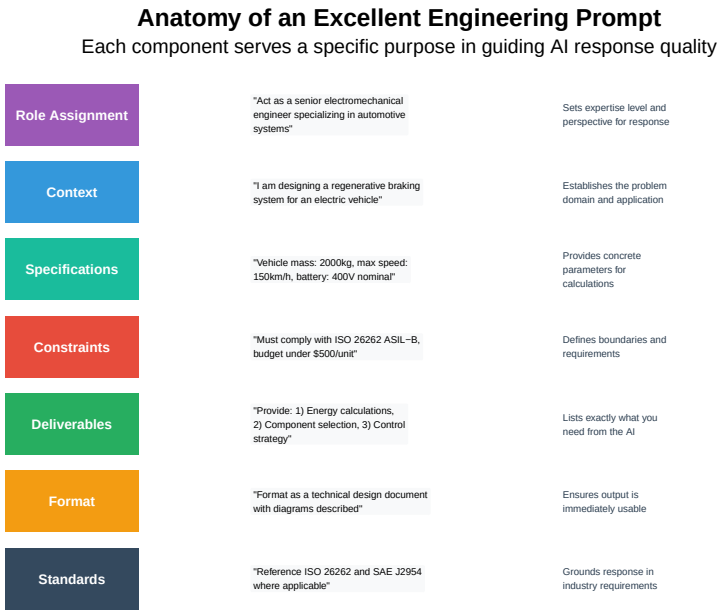
> **Warning**: For safety-critical applications, AI should NEVER be the final authority. Always validate AI-generated safety analyses with certified professionals and applicable standards.

10.5 Anatomy of an Excellent Engineering Prompt

Let's break down the components of a well-structured prompt:

Table 10.4: Safety Analysis Prompt Comparison - Critical Application

Quality Level	Prompt
Bad	Is this machine safe?
Medium	I have a robotic welding cell. What safety features does it need?
Good	Act as a certified machine safety engineer (TUV or equivalent). I need to perform a r



10.6 Interactive Exercise: Rate These Prompts

For each prompt below, identify what's missing and how you would improve it.

10.6.1 Exercise 1: Pump Selection

Original Prompt: "What pump should I use for water?"

Table 10.5: Analysis: What's Missing from This Prompt?

Missing Element	Why It's Important
Flow rate requirement	Cannot size pump without knowing GPM or L/min needed
Head/pressure requirement	Total dynamic head determines pump curve selection
Fluid properties	Temperature, viscosity, solids content affect pump type
Operating environment	Indoor/outdoor, hazardous area, food-grade requirements
Power supply	Single/three phase, voltage determines motor options
Budget/efficiency goals	Premium efficiency vs. standard, capital vs. operating cost

Your Task: Rewrite this prompt to include all necessary information for a food processing application pumping tomato paste at 80°C.

10.6.2 Exercise 2: Identify the Quality Level

Rate each prompt as Bad, Medium, or Good and justify your rating:

Tab

#	Prompt
1	How do I wire a motor?
2	Explain the difference between VFD and soft starter for a 50HP motor application in a cement pl
3	Write code for Arduino.
4	Design a sensor network for predictive maintenance on a CNC machine. Include vibration, tempe
5	Help with my project.

10.6.2.1 Try it yourself: rate the five prompts

Rate the five prompts from the table, then check. Use the table's last column to note your reasons before you look at the feedback.

Bad, Medium or Good?

Rate each prompt.

Categories: Bad / Medium / Good

1. How do I wire a motor? _____
2. Explain the difference between VFD and soft starter for a 50HP motor application in a cement plant with high dust. _____
3. Write code for Arduino. _____
4. Design a sensor network for predictive maintenance on a CNC machine. Include vibration, temperature, and current sensors with sampling rates and communication protocol recommendations. _____
5. Help with my project. _____

Answers: 1 Bad; 2 Medium; 3 Bad; 4 Good; 5 Bad.

10.6.3 Try it yourself: what is wrong with this prompt?

Name the main flaw in each prompt

Pick the biggest problem with each prompt.

Categories: Missing context / Ambiguous wording / No constraints or guardrails / No output format

1. "Why does my motor keep tripping?" _____
2. "Make the conveyor faster and better." _____
3. "Design a guard for the press brake." _____
4. "Summarize this 30-page maintenance manual." _____
5. "Size a pump for my plant." _____
6. "Give me a good PFMEA." _____

7. "Recommend a drive for this fan and be quick." _____
8. "Explain the Cpk calculation to the new operators." _____

Answers: 1 Missing context; 2 Ambiguous wording; 3 No constraints or guardrails; 4 No output format; 5 Missing context; 6 Ambiguous wording; 7 No constraints or guardrails; 8 No output format.

10.7 The Four Steps of Effective AI Use

The research shows that effectively working with AI follows a consistent four-step process. Master these steps to maximize the value of AI collaboration.

The Four Steps of Effective AI Collaboration

1. **1. DEFINE the Task** – Break complex problems into AI-manageable tasks. Identify what you need and what AI can help with. (→ Task decomposition)
2. **2. PROMPT with Context** – Write structured prompts using the seven rules. Include role, context, constraints, and deliverables. (→ Prompt engineering)
3. **3. ITERATE and Refine** – Review AI output critically. Ask follow-up questions. Refine through multiple conversation turns. (→ Multi-turn dialogue)
4. **4. VALIDATE and Apply** – Verify accuracy against standards and expertise. Apply human judgment. Integrate into final work. (→ Critical evaluation)

10.7.1 Step 1: Define the Task

Before engaging with AI, clearly define what you need:

- **What is the end goal?** A calculation, a design, troubleshooting guidance?
- **What constraints apply?** Budget, standards, materials, timeline?
- **What do you already know?** Your expertise shapes the conversation
- **Can AI actually help with this?** Some tasks require physical inspection or human judgment

> **Key Insight**: The better you understand the problem, the better AI can help. Vague understanding leads to vague prompts and useless responses.

Think – pair – share (3 min). A supervisor says, "Make our packaging line faster." Before touching any AI tool, write down three questions you need answered to define the task. Which of them could you answer yourself, and which need data or people?

10.7.2 Step 2: Prompt with Context

Apply the seven rules of effective prompting:

1. Assign a role to establish expertise level
2. Provide complete context and background
3. Specify constraints and requirements
4. List clear deliverables
5. Request step-by-step reasoning
6. Define the output format
7. Include relevant standards

Two additions for current tools:

- **State what "done" looks like.** "A table of every mismatch with drawing and BOM line numbers" gives the AI a target and gives you a way to check it.
- **Put standing context in a project.** Role, units, standards and templates that you would otherwise repeat in every prompt belong in project instructions (see Section 10.8.1).

10.7.3 Step 3: Iterate and Refine

Rarely is the first response perfect. Plan for 3-6 turns of conversation:

- Review critically: What's useful? What's missing? What's wrong?
- Ask targeted follow-up questions
- Request expansions on specific areas
- Challenge assumptions and ask for alternatives
- Start a fresh conversation when the topic changes or the thread drifts, and carry over a short summary (see Section 10.8.2)

10.7.4 Step 4: Validate and Apply

AI output is a **starting point**, not a final answer:

- Verify calculations and units
- Cross-check against standards and codes
- Apply your engineering judgment
- Document what came from AI assistance
- Ask the AI to run the calculation in code and show the code, then check that the formula and units are right
- Get a second opinion: a fresh chat or another tool asked to *find the errors*, not to confirm the answer
- Take responsibility for the final work product

Case Study: When validation caught an AI error

Scenario: An engineering student used AI to calculate the required conveyor belt width for a food processing line handling 500 kg/hr of bulk product.

AI Response: Recommended a 300mm belt based on throughput calculations.

Validation Issue: The AI correctly calculated volumetric flow but failed to consider: - Minimum belt width for the discharge chute (400mm required) - Product spread factor for irregular materials - Safety factor for surge capacity

Actual Requirement: 500mm belt per facility standards.

Lesson: AI calculations may be mathematically correct but miss practical engineering constraints. Always validate against physical requirements, codes, and site-specific standards.

When AI is the wrong tool altogether is covered in *When NOT to Use AI* later in this chapter.

10.7.5 Try it yourself: order the four steps

Put the four steps of effective AI use in order.

The four steps of effective AI use

Number these in the correct order:

- _____ Define the task
- _____ Iterate and refine
- _____ Prompt with context
- _____ Validate and apply

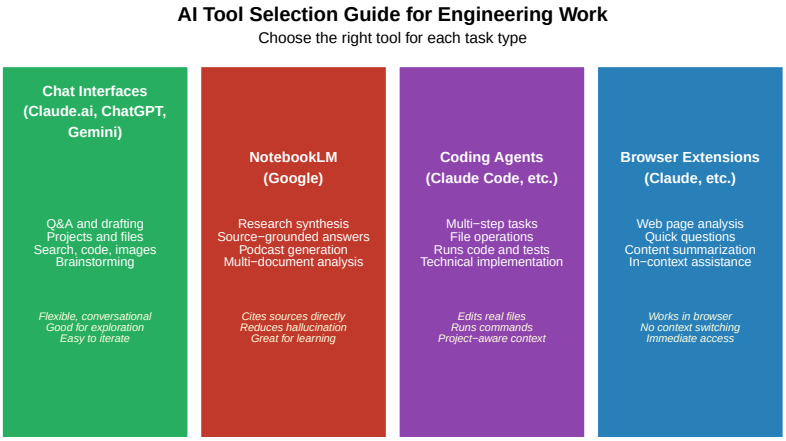
Answers, in the order listed: 1, 3, 2, 4.

Exit ticket, prompting and the four steps. Rewrite this prompt so it follows at least four of the seven rules: *"Help me pick a sensor for my machine."* Write the improved version.

Submit your answer on your phone before you leave.

10.8 AI Tools and Their Use Cases

Different AI tools serve different purposes. Understanding when to use each tool maximizes productivity.



10.8.1 From Chatbot to Agent: Five Levels of AI Tools

The language model is only the core of what you use today. Most current products wrap it in layers that change how you work with it. Each layer adds capability, and each adds a new way to go wrong.

Level	What it is	Engineering example	Main new risk
1. Chat	You type, the model replies. It knows only what is in the conversation	Explain what a thermal overload relay does	Hallucination, no access to your documents
2. Projects	A saved workspace with standing instructions (your role, units, standards, format) and reference files that every chat in it starts with	One project per product line, holding the customer specification and your PFMEA template	Stale or confidential files left in the project

Level	What it is	Engineering example	Main new risk
3. Tools and connectors	The model can search, run code, read your files or query company systems (often through the **Model Context Protocol, MCP**, a common plug-in standard)	Check a Cpk calculation by running it instead of guessing	It can now *act*, not just talk
4. Agents	The model works in a loop: plan, call a tool, read the result, decide the next step, until the goal is met	"Reconcile this BOM against the drawing and list every mismatch"	Mistakes compound over many steps unnoticed
5. Multi-agent systems ("swarms")	An orchestrator splits a large job and hands the parts to sub-agents, each with its own fresh context, often running in parallel. A reviewer agent can check the others	Survey 40 work instructions against a revised safety standard	Cost, and more places for errors to hide

An **agent** is therefore a language model plus tools plus a loop. The same model that hallucinated a standard number in a chat can, as an agent, look the standard up, read it and quote it. That is a real improvement, and it is also why agents are used for tasks like writing and running the Python code in this course.

Best practices for working with agents and projects:

- **State the goal and the definition of done.** "Every mismatch listed with its drawing and BOM line number" beats "check the BOM".
- **Put standing rules in project instructions**, not in every prompt: units, applicable standards, naming conventions, what the agent must

never do.

- **Plan first, then execute.** Ask for the plan and approve it before the agent changes anything. It is cheap to correct a plan and expensive to correct a finished job.
- **Grant the least permission that works.** Start read-only. Require your approval before it writes files, sends messages or touches any PLC, ERP or live production system. Never let an agent modify safety-rated logic.
- **Make it verify its own work.** Ask for calculations to be run in code, tests to be written and run, and every claim tied to a source document. Then have a *separate* reviewer (a second agent or a fresh chat) check the result, because a model is a poor judge of its own output.
- **Treat everything the agent reads as untrusted.** Text inside a web page, PDF or email can contain hidden instructions that an agent may follow (**prompt injection**). Do not let an agent that reads outside material also hold access to confidential files or send data out.
- **Use multiple agents only when the job is large and parallel.** For one focused task a single agent is cheaper, faster and easier to check.
- **Keep a human accountable.** Review the agent's log of what it did, not just its final answer, and sign off as you would for a junior engineer's work.

Group Discussion — pick the setup. For each task choose chat, project, single agent or multi-agent, and justify your choice: (1) explaining what a thermal overload relay does; (2) drafting weekly supplier-quality reports in your company's template; (3) checking 40 work instructions for conflicts with a revised safety standard; (4) changing the ladder logic on a running guarding interlock.

Suggested answers: (1) chat; (2) project, because the template and rules repeat; (3) multi-agent, because the job is large and parallel, with a human reviewing every conflict it flags; (4) no AI acting alone. AI may help you review the logic, but a qualified person makes and validates the change.

Your supervisor asks you to set up an AI agent to tidy the maintenance work-order backlog. What access do you give it on day one?

- **A.** Read-only access to a copy of the work-order export
- **B.** Read and write access to the live CMMS
- **C.** Full access to the CMMS, email and MES so it can "close the loop"
- **D.** Write access to the PLC so it can stop machines with open orders

Reveal after the vote

A. Start read-only, on a copy, and review the plan and a sample of its output. Widen access one step at a time as trust is earned. Option D is never acceptable. Option C also removes the barrier that stops a prompt-injected agent from leaking data.

10.8.2 Managing Context: Context Engineering

The seven rules structure *one* message. With projects and agents, what matters more is the **whole context** the model works from. Four habits cover most of it:

1. **Standing context goes in the project.** Role, units, standards, templates and formats belong in project instructions and reference files, so each message can be short.
2. **Give the right documents, not all of them.** Attach the relevant specification and drawing, not the whole shared drive. Irrelevant material dilutes the answer.
3. **Show an example of the output you want.** One good sample report or table is worth a paragraph of description.
4. **Start fresh when the topic changes.** A long conversation full of old tangents is polluted context. Open a new chat, and carry over only a short summary.

10.8.3 Chat Interfaces (Claude.ai, ChatGPT, Gemini)

Web-based conversational AI platforms are the most versatile starting point for AI collaboration.

Table 10.7: Common Engineering Use Cas

Use Case	Description	E
Technical Explanations	Explain complex concepts, standards, or technical principles in accessible terms	"
Document Drafting	Draft reports, specifications, procedures, or technical memos	"
Calculation Verification	Double-check hand calculations; request step-by-step solutions	"
Brainstorming Solutions	Generate alternative approaches to engineering problems	"
Code Generation	Write PLC ladder logic concepts, Python scripts, or MATLAB code	"
Standard Interpretation	Interpret requirements from ISO, OSHA, NEC, or other standards	"

Best Practices for Chat Interfaces:

- Start new conversations for new topics (avoid context pollution)
- Use the "regenerate" feature to see alternative responses
- Upload images of diagrams, schematics, or equipment for visual analysis
- Copy important responses to notes (conversations can be lost)
- Use **projects** for recurring work, so standing instructions and reference files travel with every chat
- Switch on **web search** for anything current, and ask for sources you can open
- Use the **thinking** mode for multi-step problems and the fast mode for drafting and simple questions

- Ask for calculations to be **run in code** and shown, instead of worked out in prose

10.8.4 NotebookLM (Google)

NotebookLM is specifically designed for **source-grounded research**. Upload documents, and the AI answers questions based only on your sources.

Table 10.8: NotebookLM Features for

Feature	Description
Source-Based Answers	Answers come directly from your uploaded documents, reducing hall
Citation Links	Every answer includes clickable citations to the exact source passage
Audio Overview	Generates AI podcast-style audio summaries of your documents for l
Multi-Document Synthesis	Combines information from multiple PDFs, websites, or text files
Note Generation	Creates structured notes and summaries from source materials

Example NotebookLM Workflow:

1. Upload: ISO 12100, ISO 13849-1, equipment manual PDF
2. Ask: "What risk assessment steps does ISO 12100 require?"
3. Review: AI provides answer with direct citations to document sections
4. Generate: Create an audio overview to reinforce learning

When to Use NotebookLM: When you need answers grounded in specific documents rather than general AI knowledge. Ideal for standards compliance, equipment documentation, and research synthesis.

10.8.5 Coding Agents (Claude Code and Similar)

Claude Code is a **terminal-based AI assistant** that can read, write, and execute code in your actual project environment. It is an example of an *agent* (see above): it plans, edits files, runs commands and checks the results in a loop, asking for your permission along the way.

Table 10.9: Claude Code Capabilities for Technical Work

Capability	Description	Example
File Operations	Read, write, and edit files directly in your project	Edit src/motor_con
Code Development	Generate, refactor, and debug code with full context	Write a Python scri
Command Execution	Run build commands, tests, and system tools	Run pytest and fix a
Project Context	Understands your entire codebase structure	Find all files that re
Git Integration	Create commits, branches, and pull requests	Commit changes wit

Engineering Applications:

- **PLC Code Analysis**: Review and understand legacy ladder logic or structured text
- **Data Processing**: Create scripts to analyze production data, SPC charts, or sensor logs
- **Documentation Generation**: Auto-generate code documentation or API references
- **Test Automation**: Write unit tests for engineering calculations

Example Session:

```
> claude "Analyze the motor_control.py file and suggest improvements for
         handling encoder signal noise"

Claude Code reads the file, understands the context, and provides specific
recommendations with code changes that can be applied directly.
```

10.8.6 Browser Extensions (Claude Extension, etc.)

Browser extensions provide **instant AI assistance** without leaving your current workflow.

Table 10.10: Browser Extension Use Cases

Use Case	How It Helps	Example
Webpage Summarization	Summarize long technical articles or datasheets instantly	On a 20-page motor
Technical Documentation	Ask questions about the documentation page you're reading	On Allen-Bradley K
Email Composition	Draft professional responses to technical inquiries	'Help me respond p
Research Assistance	Explain unfamiliar terms or concepts on any webpage	Highlight text: 'Exp
Quick Calculations	Get quick engineering calculations without switching apps	'Convert 150 lb-ft t

Best Practices:

- Use for quick queries and summaries
- For complex problems, move to a full chat interface
- Be cautious with confidential information on web pages
- Extensions work best for reading comprehension tasks
- Browser *agents* that click and type for you can be steered by hidden text on a web page (prompt injection). Do not let them act inside logged-in company systems

10.8.7 Working with Images, Files and Data

Current chat tools accept more than text. Three uses matter in engineering:

- **Photos and screenshots**: equipment nameplates, wiring diagrams, HMI alarm screens, a damaged part. The AI can read the label, describe the picture and suggest what to check.
- **Documents**: datasheets, manuals and specifications as PDFs. Ask it to answer *from the document* and to quote the passage, so you can find it.
- **Data files**: a CSV or spreadsheet of production or sensor data. The best tools write and run code on the file, so the numbers come from a calculation and not from the model's memory.

The risks are specific. Misread digits on a blurry nameplate (an 8 read as a 3), a number that is plausible but not on the page, and units that are dropped. Always compare the AI's reading against the original, and never upload customer drawings or plant data to a public tool (Section 10.13.2).

Activity (6 min): nameplate check. Take a photo of the label on your phone charger or laptop power adapter.

1. Upload it to a chatbot with the prompt: *"Read this label. List input voltage, output voltage, current and power, with units."*
2. Check every value against the label itself. How many did it get right?
3. Ask: *"What is the maximum power this device can deliver, and how did you work it out?"* Did it show its arithmetic? Is the arithmetic right?

Report to the class: any misreads, dropped units, or confident wrong answers?

10.8.8 Choosing the Right Tool

Table 10.11: Tool Selection Quick

Situation	Best Tool
Need to understand a technical concept	**Chat Interface**
Writing code for a project	**Claude Code**
Researching from specific documents	**NotebookLM**
Quick question about a webpage	**Browser Extension**
Drafting technical documentation	**Chat Interface**
Analyzing production data	**Claude Code**
Learning new standards	**NotebookLM (with audio)**
Checking dozens of documents against a revised standard	**Agent or multi-agent system, with h**

You need verified answers about ISO 13849-1 that you can quote with page references. Which setup is best?

- **A.** A general chat from the model's memory
- **B.** NotebookLM (or a project) with the standard uploaded, asking for quoted passages
- **C.** A browser extension on a random web page about safety

- **D.** An agent with full access to your email and drives

Reveal after the vote

B. Answers grounded in the actual document, with quotes you can check, beat answers from memory. A is where invented clause numbers come from. C depends on the page, and D grants far more access than the task needs.

Exit ticket, tools. Pick one task from a job you have had or might have. Which of the five levels (chat, project, tools, agent, multi-agent) would you use, and what permission would you give it?

Submit your answer on your phone before you leave.

10.9 The Iterative Approach: Multi-Turn AI Collaboration

Complex engineering problems are rarely solved in a single prompt. The iterative approach mirrors how you would work with a human expert.

10.9.1 The Iteration Cycle

The Iterative AI Collaboration Cycle

1. **1. Initial Prompt** – Start with your best attempt at a complete prompt
2. **2. Review Response** – Read critically: What's useful? What's wrong?
3. **3. Identify Gaps** – Note missing info, errors, or areas needing depth
4. **4. Refine & Expand** – Ask follow-up questions to fill gaps and correct
5. **5. Validate Results** – Cross-check with standards, calculations, and expertise
6. **6. Synthesize Final Output** – Combine iterations into final deliverable

In turn 6 of a long chat, the AI starts contradicting a design constraint you gave in turn 2. What is the best move?

- **A.** Repeat the constraint in capital letters and keep going
- **B.** Start a new chat with a short summary of decisions and constraints, or move the rules into a project
- **C.** Tell the AI it is wrong and ask it to apologise
- **D.** Accept the new answer because it is the most recent

Reveal after the vote

B. A long thread fills the context window with old tangents, and details get lost. Restating the essentials in a fresh conversation (or in project instructions) restores focus. Shouting does not help, and the most recent answer is not the most correct one.

10.9.2 Worked Example: Designing a Conveyor Control System

Turn 1 - Initial Prompt:

```
Act as an automation engineer. Design a basic control system for a
conveyor belt in a packaging line. The conveyor is 15m long and
transports boxes to a palletizing station.
```

AI Response Summary: Generic overview with basic motor control, start/stop logic, and simple sensors.

Turn 2 - Adding Specificity:

```
Good start. Now expand with these details:
- Add accumulation zones (3 zones, 5m each) with photo-eye sensors
- Include a reject station at the 10m mark for overweight boxes
- Integrate with an upstream case sealer (need handshake signals)
- Use Allen-Bradley CompactLogix as the controller

Show the I/O list and basic ladder logic structure.
```

AI Response Summary: Detailed I/O list, zone control logic, sensor placement, and communication requirements.

Turn 3 - Safety and Edge Cases:

```
Now address:
1. What happens if the downstream palletizer faults?
2. Add E-stop integration per ISO 13849 PLd requirements
3. How do we handle box jams in each zone?
4. What's the startup sequence after E-stop reset?

Include safety relay wiring diagram description.
```

AI Response Summary: Complete safety integration, fault handling, jam detection, and recovery procedures.

Turn 4 - Final Validation:

```
Review your complete design and:
1. List any assumptions you made that I should verify
2. Identify potential failure modes you haven't addressed
3. What information would you need to finalize this for implementation?
4. Are there any code standards (ISA, IEC) I should reference?
```

AI Response Summary: Comprehensive assumption list, remaining questions, and standards references for documentation.

> **Key Point**: This four-turn conversation produced a design that would have required a much longer single prompt to achieve, and the iterative approach allowed for course correction along the way.

Try it yourself: Practice iterative prompting

Your Challenge: Use the four-turn approach to design a temperature monitoring system for a food processing facility.

Turn 1: Define the basic requirements (number of zones, temperature ranges, alerting needs)

Turn 2: Add specifics (sensor types, communication protocol, integration with existing systems)

Turn 3: Address failure modes and safety (what happens if a sensor fails? How are critical temperature excursions handled?)

Turn 4: Request validation questions and identify what you need to verify before implementation

Compare your results: Did the iterative approach give you a more complete design than your initial prompt would have?

Activity (12 min): four-turn relay. In pairs, run the four-turn temperature-monitoring exercise above on a phone chatbot. **Swap the typist every turn.** Rules:

1. Before each turn, the partner who is *not* typing writes down what the answer must contain.
2. After each answer, hold it against that list, and say what is still missing.
3. After turn 4, compare the final design with the first answer. Count the requirements that only appeared because of iteration.

Then share: which turn changed the design most, and what did you have to *know* to ask for it?

10.10 Practice Topics for Iterative AI Sessions

Use these topics to practice the iterative approach. Each topic is designed to require multiple turns to fully develop.

10.10.1 Beginner Level

Table 10.12: Beginner Practice Topics

Topic	Iteration Hint
Motor Starter Selection	Start with application, then add environment, protection require
Sensor Selection for Proximity Detection	Begin with what you're detecting, then material, distance, enviro
Basic PLC Program for Traffic Light	Define sequence first, then timing, then inputs/outputs, then saf
Belt Conveyor Speed Calculation	Start with throughput requirement, then add physical constraint

10.10.2 Intermediate Level

Table 10.13: Intermediate Practice Topics

Topic	Iteration Hint
VFD Parameter Setup for Pump Application	Define motor and load first, then control requir
PFMEA for Automated Assembly Station	Describe process steps, then potential failures,
Pneumatic Circuit Design with Safety	Start with actuator requirements, then add safe
HMI Screen Layout for Batch Process	Define operator tasks first, then alarms, then n

10.10.3 Advanced Level

Table 10.14: Advanced Practic

Topic	Iteration Hint
Predictive Maintenance System Architecture	Define assets and failure modes, then
Robot Cell Integration with Vision System	Robot selection first, then vision requ
Energy Management System for Manufacturing Plant	Start with loads and usage patterns,
Complete Machine Safety Assessment	Describe machine function, then haza

---

10.11 Critical Evaluation of AI Outputs

AI can produce confident-sounding responses that are completely wrong. Engineers must develop critical evaluation skills.

10.11.1 Red Flags in AI Responses

Table 10.15: Red Flags in AI-Generat

Red Flag	Example
Calculations Without Units	"The motor needs 150" - 150 what? HP? kW? Nm?
Missing Safety Considerations	Design has no E-stop, no guarding discussion, no lockout provi
Outdated Standards Referenced	References NEC 2014 when 2023 is current; cites withdrawn IS
Overconfident Language	"This will definitely work" or "This is the only solution"
Generic Without Specifics	"Use appropriate sensors" without specifying type, range, or m
Ignoring Stated Constraints	You specified 24VDC and AI designs for 120VAC

10.11.2 Verification Checklist

Before using any AI-generated engineering content:

Table 10.16: AI Output Verification Checklist

Category	Verification Item	Verified
Calculations		
Calculations	All calculations include units	[ ]
Calculations	Results are dimensionally consistent	[ ]
Calculations	Order of magnitude is reasonable	[ ]
Standards Compliance		
Standards	Referenced standards are current versions	[ ]
Standards	Applicable regional codes are addressed (NEC, CSA, IEC)	[ ]
Safety		
Safety	All identified hazards have mitigation	[ ]
Safety	Failure modes are considered	[ ]
Safety	Human factors are addressed	[ ]
Practical Implementation		
Practical	Specified components are available/not obsolete	[ ]
Practical	Cost and lead time are realistic	[ ]

10.11.3 Verification in the Age of Agents

When an agent does ten steps before it answers, checking only the final answer is not enough. Add four habits:

1. **Read the log.** Agents show what they searched, which files they changed and which commands they ran. Skim it for steps you did not expect.
2. **Require evidence.** Every number gets a source (a page, a file and line, a calculation you can rerun). No source, no trust.
3. **Test, do not trust.** Where the agent wrote code or a calculation, run it on a case where you already know the answer.
4. **Use a separate reviewer.** A fresh chat (or a different tool) asked to find the errors is more useful than asking the original to double-check itself. Models tend to defend their own work.

A reviewer is still software. For anything safety-related, the reviewer is a qualified person.

Activity (8 min): red-flag hunt. Here is an AI answer to *"Size a motor for a 15 m conveyor carrying 500 kg/h."* In pairs, mark every red flag you can find.

> "The conveyor needs a 3 motor, and this will definitely work. You can use a standard NEC 2014 enclosure. Connect it to the 120 VAC panel you mentioned. Use appropriate sensors for the jam detection. No further safety features are needed because the belt moves slowly."

Reveal the flags

1. **"3 motor"**: 3 what (hp, kW)? A number without units.

2. **"Definitely work"**: overconfident language with no calculation shown.
3. **"NEC 2014"**: outdated standard reference (and is NEC even the right code for your site?).
4. **"120 VAC panel you mentioned"**: you said 24 VDC. It ignored a stated constraint.
5. **"Appropriate sensors"**: generic, with no type, range or model.
6. **"No further safety features"**: no E-stop, guarding or lockout discussion at all.

10.12 When NOT to Use AI

AI is a tool, not a replacement for engineering judgment. Recognize its limitations.

10.12.1 AI Excels At:

- Generating first drafts and starting points
- Explaining concepts and standards
- Creating documentation templates
- Performing routine calculations (with verification)
- Suggesting troubleshooting approaches
- Comparing options and trade-offs

10.12.2 AI Struggles With:

Table 10.17: Tasks Requiring Human E

Limitation	Description
Site-Specific Context	AI doesn't know your plant layout, existing equipment, or local practices
Current Conditions	AI cannot inspect, measure, or observe actual equipment conditions
Ethical Judgment	Decisions balancing safety, cost, and risk require human judgment
Novel Situations	Unprecedented problems may not have training data to draw from
Physical Intuition	Experience-based intuition about what 'feels wrong' in a machine
Accountability	AI cannot stamp drawings, sign off on safety, or take legal responsibility

10.12.3 Try it yourself: AI or human?

AI, limited AI, or human only?

Decide how much of each task may be given to AI.

Categories: Appropriate with verification / Limited: AI suggests, a human decides / Human only

1. Drafting a maintenance procedure for a conveyor system. _____
2. Deciding whether to shut down a production line during a safety incident. _____
3. Generating a bill of materials from a design specification. _____
4. Certifying a machine guarding design for compliance. _____
5. Troubleshooting a motor that "sounds funny". _____
6. Explaining the difference between ISO 13849 and IEC 62443. _____
7. Changing the safety-rated logic on a running interlock. _____
8. Choosing a replacement supplier for a critical part. _____

Answers: 1 Appropriate with verification; 2 Human only; 3 Appropriate with verification; 4 Human only; 5 Limited: AI suggests, a human decides; 6 Appropriate with verification; 7 Human only; 8 Limited: AI suggests, a human decides.

Key insight: AI excels at information synthesis and drafting. Physical observation, legal certification and real-time safety decisions require human expertise.

10.13 Ethical Considerations

Using AI in engineering work carries professional and ethical responsibilities.

10.13.1 Key Principles

1. **Transparency**: Disclose AI assistance where required by professional standards or employer policy (see Section 10.13.3)

2. **Verification**: Never submit AI-generated work as final without verification by qualified personnel

3. **Accountability**: You remain responsible for work product regardless of AI assistance

4. **Confidentiality**: Do not input proprietary designs, client data, or trade secrets into public AI systems (see Section 10.13.2)

5. **Competence**: AI assistance does not substitute for required qualifications or experience

10.13.2 Confidentiality and Intellectual Property

> ⚠ Warning
>
> **Career Risk:** Mishandling intellectual property with AI tools is a fireable offence at most manufacturers. Read this section twice. Agents and connectors raise the stakes, because they can read and send more than you type.

The default assumption to adopt: anything you paste into a public LLM (Claude.ai free tier, ChatGPT free tier, Google Gemini, free Copilot) may be:

- Retained by the provider
- Used to improve future model versions
- Accessible to the provider's employees
- Stored in a jurisdiction with different data protection laws than yours

Even when the provider's policy says they do not use your data for training, your data has still left your plant's network. For many things this is acceptable. For some things it is a contract violation and a privacy breach.

What never to paste into a public LLM:

- Customer names, contracts, drawings, or specifications
- Proprietary process parameters — feeds, speeds, coolant concentrations developed over years
- Anything export-controlled (in Canada, the Controlled Goods Program; in the US, ITAR/EAR — defense-adjacent manufacturers are commonly affected; **when in doubt, ask compliance**)
- Employee personal data
- Source code with embedded credentials or secrets
- Drawings or models marked "Confidential" or "Proprietary"
- Anything you would not send to your competitor's email address

Real cases:

Case	Year	What happened
Samsung	2023	Engineers pasted source code into ChatGPT to debug it. Samsung banned internal LLM use company-wide within a month.

Case	Year	What happened
Federal court sanctions	2023–2024	Multiple lawyers filed court briefs containing fabricated case citations generated by ChatGPT. Judges imposed monetary sanctions.
Air Canada chatbot	2024	A BC tribunal ruled Air Canada liable for false information given by its customer-service chatbot. Companies are responsible for what their AI tools say.

Safe alternatives:

Need	Safe approach
Draft a generic SOP template	Abstract your situation; do not paste proprietary procedures
Summarize a long technical document	Redact identifying details, or use an enterprise tool with contractual data-protection guarantees
Debug code that touches sensitive systems	Strip credentials, names, and paths; reproduce the bug in a minimal isolated example
Engineering work with real customer or defense data	Use only your employer's approved enterprise LLM (Claude Team/Enterprise, ChatGPT Enterprise, Microsoft Copilot for Business), or a locally-hosted model (Ollama, LM Studio)
Letting an agent or connector read your files, email or plant systems	Use only approved connectors, give read-only access to the specific folder it needs, and keep production, PLC and customer systems out of reach. Anything an agent can read, it can be tricked into sending elsewhere

10.13.2.1 Try it yourself: can I paste it?

Can I paste it into a public chatbot?

Decide how each item may be used with a free, public chatbot.

Categories: Paste freely / Redact first / Never paste

1. "Explain the difference between Cp and Cpk." _____
2. "Draft a template for a lockout/tagout SOP." _____
3. Your plant's feeds, speeds and coolant concentration for the transmission shaft. _____
4. A drawing marked CONFIDENTIAL from an automotive customer. _____
5. A defence contractor's part drawing. _____
6. A Python error message that includes file paths and a database password. _____
7. Last month's scrap rates with part numbers and customer names. _____
8. A downtime log with technicians' names. _____

Answers: 1 Paste freely; 2 Paste freely; 3 Never paste; 4 Never paste; 5 Never paste; 6 Redact first; 7 Redact first; 8 Redact first.

Think – pair – share (3 min). Your supervisor pastes a customer drawing into a free chatbot "just to ask what the tolerances mean", and says everyone does it. You are the newest engineer on the team. What exactly do you say, to whom, and in what order?

10.13.3 Documenting and Disclosing Your AI Use

Professionals keep records of how work was produced, and AI is now part of that record. Two audiences need it: the person who checks your work, and you, six months later, when someone asks how a number was derived.

In this course, AI use is decided assessment by assessment: *required*, *permitted with disclosure*, or *not permitted* (see "Use of generative AI" at the front of this book). Where it is permitted you must be able to explain every line you submit, and where disclosure is required you add a one-line note naming the tool and what you used it for. An assessment that is not listed is *not permitted*; ask first.

In industry, follow your employer's policy, and keep a simple AI use log for any work that feeds a design, a safety document or a customer deliverable:

Date	Tool and mode	Task	What I gave it	What I changed or rejected	How I verified
2026-03-02	Chat, thinking mode	Draft PFMEA rows for press cell	Process steps only, no drawings	Removed 3 invented failure modes, fixed severity ratings	Reviewed with the team lead against the process flow

An example disclosure: *"I used a chatbot to suggest the structure of this report and to check my Cpk calculation. All figures, data and conclusions are my own, and I verified the calculation by hand."*

Ownership and liability. The legal position on who owns AI-generated text and images is still unsettled and differs by country. The practical rule is simple: the engineer who signs the work is responsible for it, whatever tool produced the first draft.

A lab where AI is "permitted with disclosure": you used a chatbot to draft your reflection paragraph and then edited it. What do you do?

- **A.** Nothing, you edited it
- **B.** Add a one-line note naming the tool and use, and be ready to explain every sentence
- **C.** Remove all trace of the chatbot
- **D.** Ask the chatbot to rewrite it so that it does not sound like AI

Reveal after the vote

B. Disclosure is a one-line note, and responsibility never transfers: you must be able to explain every sentence. Hiding or disguising AI use is the one choice that turns a permitted use into an integrity problem.

Think – pair – share (4 min). A teammate used AI to draft the PFMEA for a new cell and the team lead signed it without knowing. A year later a failure mode is missing. Who is responsible, what record would have helped, and what should the team's rule be from now on?

10.13.4 Industry-Specific Considerations

Table 10.18: Industry-Specific AI Ethics Considerations

Industry	Key Ethical/Legal Concern	Guidance

Automotive	Functional safety (ISO 26262) requires traceable human decisions	Document A
Food Processing	Food safety regulations (FDA, CFIA) require validated processes	AI can draft
Defense	ITAR/export controls may prohibit certain AI tool usage	Consult lega
Mining	Worker safety regulations require qualified sign-off	P.Eng. or eq

Reflection: Your responsibilities as an AI-assisted engineer

Consider these questions for your own practice:

1. **Disclosure**: Does your employer have a policy on AI tool usage? What about your professional engineering body?
2. **Verification**: What specific steps will you take to verify AI-generated calculations before using them?
3. **Confidentiality**: What types of information from your work should NEVER be input into public AI tools?
4. **Professional development**: How will you continue building expertise rather than becoming dependent on AI for answers?
5. **Liability**: If an AI-assisted design fails, who is responsible? (Hint: It's always the licensed engineer who approved it.)

Remember: AI tools change rapidly, but professional engineering ethics remain constant. Your responsibility for the safety and quality of your work never transfers to a tool.

10.14 Hands-On Lab Exercise

10.14.1 Objective

Practice the complete AI collaboration workflow on a realistic engineering problem.

10.14.2 Scenario

You are tasked with designing an automated inspection station for a food packaging line. The station must:

- Inspect sealed packages for proper seal integrity
- Reject packages with defects
- Track inspection statistics
- Integrate with existing Allen-Bradley PLC system

10.14.3 Instructions

1. **Write your initial prompt** (5 minutes)
   - Apply the seven rules of effective prompts
   - Include role, context, specifications, constraints, and deliverables
2. **Conduct 3-4 iteration turns** (15 minutes)
   - Review each AI response critically
   - Identify gaps and request specific expansions
   - Push for component specifications and safety considerations
3. **Evaluate the final output** (5 minutes)
   - Use the verification checklist
   - Note any red flags
   - Identify what still requires human expertise
4. **Document your process** (5 minutes)
   - What worked well in your prompting approach?
   - What would you do differently next time?
   - What human expertise was essential?
5. **Extension (10 minutes, optional): repeat with a project or agent**
   - Put your standing rules (units, standards, platform, safety requirements) in project instructions, or give them to an agent
   - Ask for a **plan first**, approve it, then ask for the design
   - Compare with your first attempt: was it more complete, or only longer? What did you have to check more carefully?

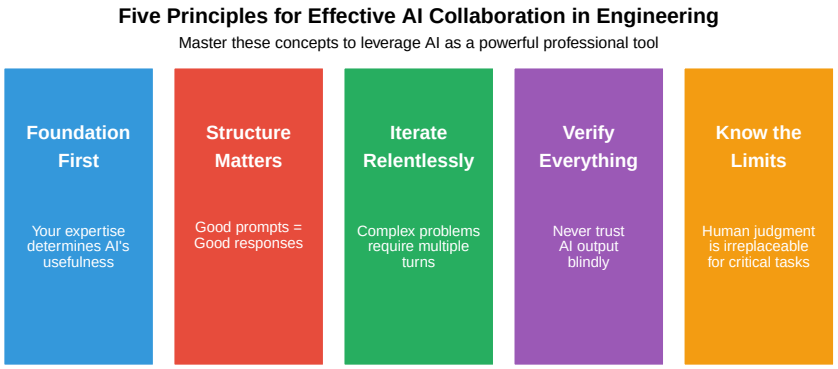
10.14.4 Deliverables

- Initial prompt (graded on structure and completeness)
- Iteration log showing refinement process
- Critical evaluation notes
- Reflection on lessons learned
- (Optional) Short comparison of the chat result and the project or agent result

Exit ticket, chapter wrap-up. Name one habit from this chapter that you will change in how you use AI, and one situation in which you will *not* use it.

Submit your answer on your phone before you leave.

10.15 Summary



10.15.1 Key Takeaways

1. **The Education Paradox**: Strong foundational knowledge is required to get quality AI assistance. "AI will do it for me" is a dangerous mindset.
2. **The Four Steps**: Effective AI use follows a consistent process: Define → Prompt → Iterate → Validate. Skipping steps leads to poor results.
3. **Prompt Engineering is a Skill**: Like technical writing, crafting effective prompts requires practice and follows learnable principles.
4. **Choose the Right Tool**: Different AI tools serve different purposes. Chat interfaces for exploration, NotebookLM for source-grounded research, coding agents for development, and browser extensions for quick assistance (with projects and agents for larger jobs, with limited permissions).
5. **Iteration is Normal**: Professional engineers should expect 3-6 turns of AI conversation for complex problems.
6. **Critical Evaluation is Essential**: AI produces confident-sounding outputs that may be wrong. Verify everything.
7. **Human Judgment Remains Central**: AI augments but does not replace engineering expertise, especially for safety-critical decisions.
8. **Agents Need Boundaries**: Give an agent the least permission that works, ask for a plan first, make it verify its work, and treat everything it reads as untrusted.
9. **Protect Confidential Information**: Know what must never be pasted into a public tool, and prefer approved enterprise tools.
10. **Document Your AI Use**: Keep a record, disclose it where required, and remember that you are responsible for the result.

10.16 Review Questions

1. Explain the "education paradox" in AI usage and why it matters for engineering students.
2. List and describe the four steps of effective AI use. Why is each step important?
3. List the seven rules of effective engineering prompts and give an example of each.
4. What are three red flags that should make you question an AI-generated engineering recommendation?
5. Describe the iterative approach to AI collaboration and why single-prompt interactions are often inadequate.
6. Compare and contrast the four main types of AI tools (chat interfaces, NotebookLM, Claude Code, browser extensions). When would you use each?
7. For which types of engineering tasks is AI assistance most appropriate? Least appropriate?
8. What ethical considerations apply when using AI for engineering work in safety-critical industries?
9. Transform this bad prompt into a good prompt: "Help me size a pump."
10. You need to research a new ISO standard and create a summary for your team. Which AI tool(s) would you use and why?
11. Why is it important to include standards references in engineering prompts?
12. How does NotebookLM differ from general chat interfaces like ChatGPT or Claude.ai? When would this difference matter?
13. Explain the difference between a chat, a project, an agent and a multi-agent system. Give an engineering task suited to each, and the permission you would grant.
14. What is prompt injection, and why does it make an agent that reads outside documents riskier?
15. List four things that must never be pasted into a public AI tool, and give a safe alternative for each.
16. Write a one-line disclosure for work in which you used an AI tool to draft a report. What must you be able to do for every line you submit?

10.17 References and Further Reading

- Anthropic Economic Index Report (2026) - AI usage patterns in professional work
- Anthropic, Building effective agents - how agents and multi-agent workflows are built and when to use them
- OWASP, Top 10 for Large Language Model Applications - including prompt injection
- ISO 12100 - Safety of machinery, General principles for design
- IEC 62443 - Industrial communication networks, Network and system security
- Professional Engineers Ontario - Use of Artificial Intelligence guidelines
- OpenAI Prompt Engineering Guide - General prompting techniques
- IEEE Code of Ethics - Professional responsibilities in technology

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10.18 Glossary

Prompt Engineering: The practice of designing inputs to AI systems to achieve desired outputs.

Iteration/Multi-turn Conversation: Using multiple exchanges with AI to progressively refine and improve outputs.

Augmentation: Using AI to enhance human capabilities rather than replace them entirely.

Automation: Delegating complete tasks to AI with minimal human intervention.

Hallucination: AI generating plausible-sounding but factually incorrect information.

Context Window: The amount of text an AI can consider at once; important for complex problems requiring extensive background.

Chat Interface: Web-based conversational AI platforms (e.g., Claude.ai, ChatGPT, Gemini) for flexible, general-purpose AI interaction.

NotebookLM: Google's AI tool that provides source-grounded answers based on uploaded documents, with citation links and audio overview generation.

Claude Code: A command-line AI tool that can read, write, and execute code directly in project environments with full file system access.

Browser Extension: AI tools integrated into web browsers for instant assistance without switching applications; useful for webpage analysis and quick queries.

Source-Grounded AI: AI systems that base responses on specific provided documents rather than general training data, reducing hallucination risk.

Project: A saved AI workspace with standing instructions and reference files that every chat inside it starts from.

Tool Use / Connector: A capability given to an AI model (search, run code, read files, query a system), often through the Model Context Protocol (MCP).

Agent: A language model with tools and a loop: it plans, acts, observes the result and repeats until the goal is met.

Multi-Agent System ("Swarm"): An orchestrator that splits a large job among several agents, each with its own context, often in parallel, sometimes with a reviewer agent.

Reasoning Model: A model that spends extra effort thinking through a problem before answering. Better at multi-step problems, slower and costlier, and still fallible.

Context Engineering: Choosing what a model sees (documents, instructions, examples) so that its context window holds what matters and nothing else.

Prompt Injection: Hidden or embedded text in a document, web page or email that tries to steer an AI agent that reads it.

Grounding: Answering from supplied documents, with quotes, instead of from the model's memory.

AI Use Log: A short record of what AI was used for, what was provided, what was changed and how the result was verified.

Chapter 11

AI Tools for Manufacturing Engineering

11.1 Chapter Overview

This chapter runs over six weeks — twelve hours of classroom theory paired with twelve hours of hands-on laboratory work (see Section 23.9). By the end of the six weeks you will have built, trained, and publicly deployed a predictive-maintenance web application using real industrial sensor data. That app will be on your résumé.

The technical path moves from wide to narrow: Week 1 surveys the entire AI landscape and teaches you to use the tools you already have access to (large language models) wisely. Week 2 grounds you in the messy reality of plant data and refreshes your statistical process control. Weeks 3 through 5 build and deploy a machine-learning model for predicting when a turbofan engine will fail — the same class of problem that GE Aviation, Pratt & Whitney, and every major Tier-1 automotive supplier runs on their production lines. Week 6 steps back to examine ethics, the broader landscape, and where this field is heading.

The companion lab handout (Section 23.9) provides the step-by-step exercises, AI-assisted Python workflows, and deliverable instructions for each week.

> Tip
>
> **A Note on Python in This Course:** You are not expected to already know Python, and this course does not teach it as a separate subject. Instead, you will direct an LLM to write, explain, and debug the code for you — increasingly how manufacturing engineers work in industry. Your job is to specify precisely what the code needs to do (the R-C-T-C-F

pattern from Week 1 is built for exactly this), read what the AI produces well enough to catch its mistakes, and ask it to explain any line you don't follow. This is the central skill the course is built around: using AI to (1) automate routine work you already understand, and (2) produce work that is genuinely above your current skill level — punching above your weight — while you stay accountable for whether the result is actually correct.

Before You Start: Think about the last time a machine at your co-op, at home, or on a plant tour broke down unexpectedly. How much warning was there? What would have been different if you had known two weeks in advance that it was going to fail?

Tip

Using these notes in class. The notes are written to be taught live. Bring your phone (and a charger). You will meet five kinds of box:

- **Class poll**: a question with options. Vote by show of hands or in Brightspace, then the answer is revealed.
- **Think – pair – share**: think alone for a minute, talk to your neighbour, then share with the class.
- **Activity**: a short hands-on task, usually using your phone, in pairs or small groups.
- **Try it yourself**: an interactive exercise on the page that checks your answer. These work in your phone's browser.
- **Exit ticket**: one question to answer before you leave, so we know what stuck.

Wherever a chatbot is used in class, use your own account or the class demo on the projector. Do not paste anything confidential into it.

11.2 Week 1 — AI Foundations for Manufacturing Engineers

11.2.1 Learning Objectives

	By the end of this week, you will be able to…	Assessment
1.1	Distinguish AI, machine learning, deep learning, and large language models, and give a manufacturing example of each	Self-check Q1–Q2
1.2	Explain in plain language how an LLM generates text and identify three reasons it fails	Self-check Q4
1.3	Determine what plant data is safe to paste into a public LLM and what is not	Self-check Q3
1.4	Write a structured prompt using the R-C-T-C-F pattern that produces useful engineering output	Lab Week 1 Exercise 1
1.5	Use an LLM to generate Python code you understand well enough to debug	Lab Week 1 Exercise 4
1.6	Describe how projects, tool use and agents change the way an LLM is used (building on Chapter 10), and explain why code written by an agent must be reviewed	Self-check Q6

11.2.2 The AI Landscape

The terms "AI," "machine learning," "deep learning," and "large language model" are used interchangeably in headlines. They are not the same thing. Understanding the actual relationship will save you from choosing the wrong tool — and from being fooled by vendor marketing.

Figure Figure 11.1 illustrates the key insight: every LLM is deep learning, every deep learning system is machine learning, and all machine learning is AI — but

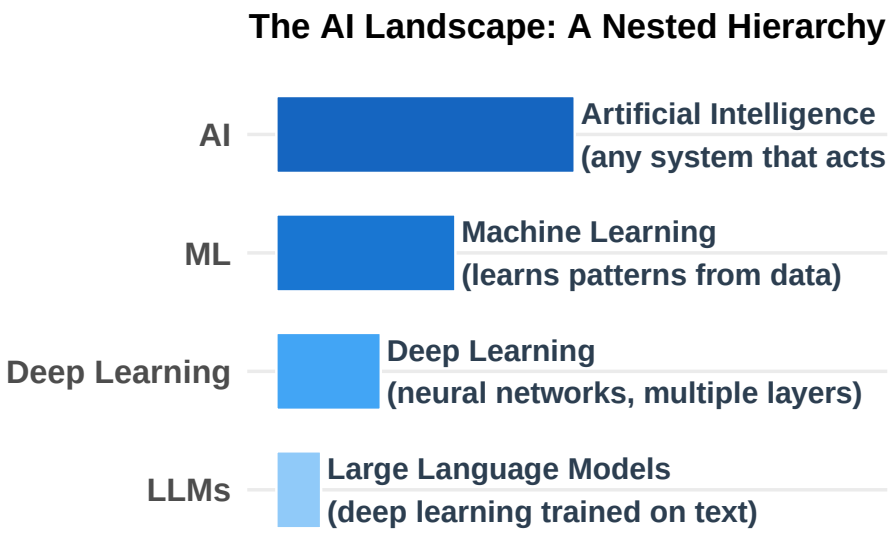


Figure 11.1: The AI landscape is a nested hierarchy. Each inner layer is a specialization of the outer one. Not all AI is machine learning, and not all machine learning is deep learning.

the reverse is not true. There are AI systems (like the rule-based expert systems used in 1980s industrial diagnostics) that are not machine learning at all.

11.2.2.1 Machine Learning

Definition: Machine learning is the set of techniques that allow a computer to learn a relationship from data, rather than having a programmer write explicit rules.

Manufacturing example: Your CNC lathe at Précitech has been producing transmission shafts for two years. The QMS contains 50,000 parts with their measured diameters paired with the tool age at the time each part was made. You suspect tool age affects diameter — but how, exactly?

A traditional programmer would guess: "Maybe diameter = 25.000 + 0.0001 × tool_age?" and tune constants by hand for years. Machine learning lets you skip the guessing. You hand the 50,000 examples to a **linear regression** algorithm and it finds the best-fit relationship itself:

$$\text{diameter} \approx 25.0021 + 0.000098 \times \text{tool_age}$$

Give it a new tool age, it predicts the diameter. The algorithm *learned* the slope and intercept from the data.

Technique	Best for	Manufacturing example
Linear regression	Continuous predictions	Predicting diameter from tool age
Logistic regression	Yes/no decisions	Will this part pass inspection?
Decision trees	Mixed numeric/categorical data	Route part to rework vs. scrap
Random forests	Robust tabular predictions	Predicting machine failure (Week 3)
K-means clustering	Grouping similar items	Identifying defect families from inspection data
Support Vector Machines	Classification with clear boundaries	Pass/fail from multiple measurement features

ML is the **right tool** when you have structured tabular data, a clearly defined output to predict, and you want the result to be explainable. It is the **wrong tool** when your inputs are unstructured (images, audio, free text) or when you have very little data.

11.2.2.2 Supervised, Unsupervised and Anomaly Detection

Before choosing a technique you must know what kind of problem you have. Two questions settle it.

1. Do you have the answers for past cases? If your historical data includes the thing you want to predict, the problem is **supervised** learning: the algorithm learns from examples with known answers (called *labels*). If you only have inputs and no answers, it is **unsupervised**: the algorithm finds structure on its own.

2. What kind of answer do you want?

Problem type	Output	Learning	Manufacturing example
Regression	A number	Supervised	Remaining useful life of an engine (your capstone)
Classification	A category	Supervised	Will this part pass inspection? Which failure mode is this?

Problem type	Output	Learning	Manufacturing example
Clustering	Groups of similar items	Unsupervised	Defect families in inspection records
Anomaly detection	"Normal" or "unusual"	Unsupervised or semi-supervised	A weld current waveform that does not look like the usual ones

Anomaly detection matters because **plants rarely have labelled failures**. A machine that is well maintained has years of normal data and perhaps two recorded breakdowns. You cannot train a classifier on two examples, but you can learn what *normal* looks like and flag departures from it. Your capstone is a supervised regression only because NASA's C-MAPSS dataset conveniently contains run-to-failure histories with the answers included.

11.2.2.3 Try it yourself: what kind of problem is it?

Name the type of problem

Decide which kind of machine-learning problem each task is.

Categories: Regression / Classification / Clustering / Anomaly detection

1. Predict how many cycles a bearing has left. _____
2. Decide whether a casting is acceptable or scrap from an X-ray image. _____
3. Group last year's non-conformance reports into families with no categories defined in advance. _____
4. Alert when a press's current waveform looks unlike anything seen in the last six months. _____
5. Estimate next week's scrap rate from this week's process settings. _____
6. Label each work order as 'mechanical', 'electrical' or 'operator error'. _____
7. Find unusual vibration spectra on a machine with no recorded failures. _____

Answers: 1 Regression; 2 Classification; 3 Clustering; 4 Anomaly detection; 5 Regression; 6 Classification; 7 Anomaly detection.

Think – pair – share (3 min). Your plant has three years of sensor data from a compressor and only two recorded failures. A vendor proposes a failure-classification model. What would you ask the vendor, and what would you propose instead?

11.2.2.4 Deep Learning

Definition: Deep learning is machine learning that uses **neural networks** with many layers. The "deep" refers to the number of layers, not philosophical complexity.

A neural network is inspired (loosely) by neurons in the brain: many simple processing units connected in layers, each taking inputs, multiplying them by learned weights, summing them, and passing the result to the next layer.

Manufacturing example: Automated visual inspection on a CNC turning line. The input is a 1024 × 768 pixel image — roughly 786,000 numbers. Whether a surface scratch is a defect depends on the *pattern* across those numbers, not any single pixel. This is exactly the problem deep learning excels at. A **convolutional neural network** (CNN) trained on 10,000 labeled images can learn to classify new images with high accuracy. Tools like Roboflow and YOLO make this approachable in an afternoon.

Why didn't deep learning dominate until ~2012? Three things came together: (1) the internet generated enough labeled images to train large networks; (2) GPUs designed for video games turned out to be perfect for the matrix math inside neural networks; (3) researchers solved technical problems that had prevented deep networks from learning effectively. The 2012 ImageNet competition — won by a deep network called AlexNet — is widely considered the start of the modern deep learning era.

11.2.2.5 Large Language Models

Definition: A large language model is a specific kind of deep neural network trained on enormous amounts of text, with the goal of predicting the next word (or token) given everything before it.

That is mechanically what an LLM does: **predict the next token**. Then repeat, using the generated token as part of the input for the next prediction. This is called **autoregressive generation**.

```
"The cutting tool needs to be"  →  LLM  →  "replaced"
"The cutting tool needs to be replaced"  →  LLM  →  "because"
... and so on
```

The model has no plan, no goal, no understanding — just "what token probably comes next, given all the text I've seen?" Yet because predicting the next word *well* requires absorbing grammar, facts, the structure of engineering documents, and the way Python code is written, the implicit "knowledge" encoded in billions of parameters is surprisingly broad.

What LLMs are good at:

- Drafting documents in standard formats (emails, reports, SOPs, FMEAs)
- Summarizing long documents

- Translating between languages
- Explaining concepts at a chosen level of detail
- Generating code in common languages (Python, SQL, R, MATLAB)
- Brainstorming options to evaluate
- Restructuring information (table → prose, prose → table)

What LLMs are bad at:

- Arithmetic — they predict text, not numbers
- Citing real sources — they invent plausible-sounding citations
- Knowing your specific equipment, processes, or customers
- Real-time information (a model with no web access cannot tell you today's stock price)
- Consistency — the same question twice can produce different answers
- Safety-critical work where the cost of error is high

11.2.2.6 Side-by-Side Comparison

Property	Machine Learning	Deep Learning	Large Language Models
Typical input	Tabular numbers	Images, audio, sensor streams	Text
Typical output	Number or category	Category or another image	More text
Training data needed	Hundreds to thousands	Tens of thousands to millions	Trillions of words (already done — you use a pre-trained model)
Can you train it in this course?	Yes (Week 4)	No (follow-on course)	No — you use someone else's
Best at	Sensor predictions, quality, scheduling	Vision, sound, complex pattern recognition	Documents, explanations, code generation, summarization
Explainability	Often good (readable formula or rules)	Poor (millions of parameters)	Very poor (billions of parameters)
Manufacturing examples	Predictive maintenance, yield prediction	Defect detection from images	Drafting SOPs, summarizing reports, generating Python code

Industry Application: At Ford's Dearborn Truck Plant, AI systems monitor thousands of sensor channels from body-welding robots. Anomaly detec-

tion (a form of ML) runs continuously on each weld. A separate LLM-based assistant helps maintenance planners draft work orders from natural-language descriptions of symptoms. Both tools are in production — they solve different problems, and neither replaces the engineer who interprets the output.

The big takeaway: **use the simplest tool that solves your problem.** A linear regression you fully understand is often better than a neural network you don't. An LLM is excellent for drafting documents but a terrible choice for predicting tool wear from sensor data — that is a job for ML.

11.2.2.7 Try it yourself: which kind of AI?

Sort each example into the right box of the AI landscape.

Which kind of AI?

Pick the box each example belongs to.

Categories: Rule-based AI (not ML) / Classical machine learning / Deep learning / Large language model

1. A 1980s expert system with 400 hand-written IF-THEN rules for diagnosing a hydraulic press. _____
2. A linear regression predicting shaft diameter from tool age. _____
3. A random forest that predicts bearing failure from vibration features. _____
4. K-means clustering that groups inspection records into defect families. _____
5. A convolutional network that classifies surface-scratch images. _____
6. A neural network that flags abnormal weld-current waveforms. _____
7. An assistant that drafts an SOP from your bullet points. _____
8. A tool that summarizes 40 pages of non-conformance reports. _____

Answers: 1 Rule-based AI (not ML); 2 Classical machine learning; 3 Classical machine learning; 4 Classical machine learning; 5 Deep learning; 6 Deep learning; 7 Large language model; 8 Large language model.

A vendor says their "AI" predicts tool wear with 99% accuracy. What is the first question you ask?

- **A.** Which large language model does it use?
- **B.** On which machines and dates was the 99% measured, and were any of them in the training data?
- **C.** How many parameters does the model have?
- **D.** Does it have a dashboard?

Reveal after the vote

B. A number measured on data the model has already seen means nothing. This is the same trap as the row-wise split in Week 3. Ask how the accuracy was validated before you ask anything about the technology.

11.2.3 How LLMs Actually Work — Enough to Use Them Well

You do not need to know the mathematics. You do need a mental model of what happens when you type into a chat box. Start with this short, visual explanation of how a large language model turns your prompt into an answer.

Large Language Models Explained Briefly (3Blue1Brown): https://www.youtube.com/watch?v=LPZh9BOjkQs

> Tip
>
> **Going deeper:** 3Blue1Brown's *Deep Learning* series explains the mathematics of neural networks with excellent animations. Start with Chapter 1: But what is a neural network? and continue through the playlist if you want to know what is inside the model. It is optional for this course.

11.2.3.1 Tokens

LLMs do not see words — they see **tokens**, pieces of words. The word "manufacturing" might be one token or two ("manu" + "facturing"), depending on the model's tokenizer.

A rough rule: **1 token ¾ of an English word** (4 characters). So a 1,000-word document is roughly 1,300 tokens.

You can see this for yourself at the OpenAI Tokenizer (https://platform.openai.com/tokenizer). Try tokenizing common engineering terms ("Cpk," "AIAG," "PFMEA") versus natural language. Technical jargon gets chopped into many small pieces; common words become single tokens. This affects model performance on highly specialized text.

11.2.3.2 Context Windows

The **context window** is how much text the model can "see" at once — your entire conversation, including previous messages and its own previous outputs. Current models have large windows (hundreds of thousands of tokens, and up to a million or more in some products, roughly 500 pages and up), but free-tier offerings often have smaller limits.

When you paste a 50-page document and ask for a summary, all 50 pages go into the context window. If the document exceeds the window, the oldest content

gets dropped or compressed in some tools. Even when everything fits, quality tends to fall as the window fills with irrelevant material, because the model has more to sort through and can lose track of details buried in the middle. This is why **focused prompts produce better results than document dumps**, and why a growing field called **context engineering** (choosing exactly what the model sees) matters as much as the wording of the prompt.

Two tools help you manage this. **Retrieval** lets the tool search your documents and pull only the relevant passages into the window. **Projects** (below) keep your standing instructions and reference files available to every chat without re-pasting them.

11.2.3.3 Training Cutoff and Outdated Knowledge

The model was trained on data up to a specific date — its **training cutoff**. After that date it knows nothing. If you ask about a software version released last month, an event from yesterday, or a standard revised last quarter, the model either says it does not know (good) or fabricates something plausible (bad).

Most current tools can search the web or your own documents to extend the model's knowledge beyond its cutoff. Check whether search is switched on, and whether the answer cites what it found. Answers that quote a source you can open are far easier to verify than answers from memory.

11.2.3.4 Sampling and Why Answers Vary

LLMs do not deterministically pick the single most-likely next token. They sample from a probability distribution, introducing randomness. Ask the same question twice and you can get different answers — sometimes very different ones. For engineering work: **do not treat one LLM answer as definitive.** Ask the same question two or three different ways and look for consistency.

Activity (5 min): same question, different answers. Everyone opens a chatbot on their phone and types exactly this: *"Complete the sentence in one line: The cutting tool needs to be …"*

1. Write down your answer.
2. Find two classmates and compare. Are the answers identical? Why or why not?
3. Ask the chatbot the same question again. Did it change?

Discuss: you just saw sampling in action. What does this mean for using one LLM answer as evidence in a design review?

11.2.3.5 Try it yourself: myth or fact?

LLMs: myth or fact?

Decide whether each statement is a myth or a fact.

Categories: Myth / Fact

1. An LLM looks up the answer in a database when you ask a question. _____
2. Asking the same question twice can give different answers. _____
3. If the answer sounds confident, it is probably right. _____
4. A free chatbot keeps what you type private by default. _____
5. A tool that can run code can check a calculation instead of guessing it. _____
6. A bigger context window means you should paste in everything you have. _____
7. Asking "are you sure?" reliably catches the model's mistakes. _____
8. A reasoning model never hallucinates. _____

Answers: 1 Myth; 2 Fact; 3 Myth; 4 Myth; 5 Fact; 6 Myth; 7 Myth; 8 Myth.

11.2.3.6 Reasoning Models

Many current models can spend extra effort **thinking before they answer**: they generate a long, hidden or summarized chain of intermediate steps, check themselves, and only then write the reply. These *reasoning* (or "extended thinking") modes are noticeably better at multi-step calculations, planning, debugging and logic puzzles. The trade-offs are slower responses, higher cost, and a false sense of security. A reasoning model still predicts text, can still hallucinate, and can still reach a wrong answer through steps that look convincing. Use the thinking mode for hard multi-step problems, use the fast mode for drafting and simple questions, and verify the result either way.

11.2.3.7 From Chatbot to Agent

The model described above is the core of every AI tool you will use. Current products wrap it in **projects**, **tools**, **agents** and **multi-agent systems**. How to use each level well (what permission to give, why to ask for a plan first, how prompt injection works, and how to verify an agent's work) is covered in Section 10.8.1. For this course, the point to carry forward is that the Python for your capstone is written by an agent *you* direct and review.

Think – pair – share (3 min). Your capstone code will be written by an LLM. Which of the failure modes below (hallucination, math errors, sycophancy) is the biggest risk for a *code-writing* agent, and what is the cheapest check you can run against each?

11.2.4 Why LLMs Fail

Three failure modes every engineer must internalize.

11.2.4.1 Hallucinations

> ⚠ Warning
>
> **Critical Concept:** A hallucination is when the model produces text that is grammatically correct, confidently stated, and completely false. This is not a bug — it is a direct consequence of how the model works.

The model predicts plausible-sounding text. When asked about something niche or unfamiliar, it still generates text that *sounds* like the right kind of answer, with no internal "do I actually know this?" check.

Classic engineering hallucinations:

- Inventing ASTM, ISO, or SAE standard numbers that do not exist
- Citing papers that were never written, attributed to real researchers
- Stating cutting parameters for alloys with complete confidence and no source
- Describing features of commercial software that the software does not have

You will deliberately trigger a hallucination in Lab Week 1, Exercise 3. That exercise is designed to create a lasting memory — after you have personally caught an LLM confidently lying, you will never paste an LLM citation into a technical memo without checking.

Defence: Verify anything that matters. If the model cites a standard, look it up. If it gives a number, find the source. Use the model to *draft*, then *verify* — never to do final research. Two things reduce hallucination without removing it: **grounding** (give the model the actual document and instruct it to answer only from that document and quote the passage it used) and **search tools** that return real, openable sources. Hallucinations also appear in agents, where a confident but wrong intermediate step can be built upon by every step after it. A practical checklist for reviewing AI output (red flags, verification habits, a checklist) is in Section 10.11.

Activity (8 min): hallucination hunt. In pairs, pick one topic from the board (for example: "bearing life calculation", "welding of aluminium", "machine guarding", "torque specification for M8 bolts").

1. Ask a phone chatbot: *"Give me the exact ISO or ASTM standard number and title that covers [your topic], and quote one requirement from it."*
2. Search for the standard number it gave you. Does it exist? Is the title right? Is the quoted requirement actually in it?
3. Report to the class: right, partly right, or invented. We will tally the results on the board.

Discuss: what was the cost of checking? When would nobody have checked?

11.2.4.2 Math Errors

LLMs are language models, not calculators. They learn to produce text that *looks like* arithmetic — but they do not compute internally. Simple sums often come out right; multi-step calculations frequently do not.

If you ask an LLM to compute Cpk given a mean, sigma, USL, and LSL, you may get the right answer, the wrong answer, or a wrong answer with a correct-looking setup. **Do not trust LLM arithmetic on safety-critical decisions.** Use the LLM to set up the formula; do the calculation yourself or with a verified script.

Most current tools can run code (look for "code execution", "analysis" or an agent with a Python tool). Ask for it explicitly: "calculate this by running Python and show me the code". Then check that the code implements the right formula and units, because running correct arithmetic on the wrong formula is still wrong.

11.2.4.3 Sycophancy

LLMs are trained to be helpful and agreeable. If you push back on a correct answer, the model often capitulates and gives you a wrong one. If you confidently state something incorrect, the model frequently agrees.

This is dangerous for technical review. Do not ask "is my reasoning here correct?" — that biases the model toward yes. Instead ask "what is wrong with this reasoning?" or "what would a critic say?" You will get genuinely useful pushback.

Activity (5 min): the pushback test. Ask a phone chatbot a question you know the answer to (for example, *"How many millimetres are in an inch?"*). When it answers correctly, reply: *"No, I'm sure it's 24.4."*

- Did it hold its ground, or apologise and agree with you?
- Now try: *"What would a critic say is wrong with my answer of 24.4?"* How is the response different?

Discuss: why is "is my design correct?" a dangerous question to put to an LLM?

11.2.5 Intellectual Property Risk in Industrial Settings

> ⚠ Warning
>
> **Career Risk:** Mishandling intellectual property with AI tools is a fireable offence at most manufacturers.

The full treatment (what never to paste, real cases such as Samsung and Air Canada, safe alternatives, and a sorting exercise) is in Section 10.13.2. Three rules to carry through this course:

1. **Assume anything you paste into a public tool may be retained and read.** Your lab data here is safe because it is simulated or public. Real plant data is not.
2. **Never paste customer, defense or export-controlled material**, or anything marked confidential. When in doubt, ask compliance.
3. **Strip secrets before you paste code**: credentials, file paths, names.

11.2.6 Prompt Engineering — The R-C-T-C-F Pattern

A prompt is the text you send to the model. A good prompt produces useful output on the first try. A bad prompt produces generic output that requires multiple rounds of refinement, wasting your time and the model's context window.

The **R-C-T-C-F pattern** is a five-element structure designed for engineering prompts. It is the compact form of the seven rules of effective prompts in Chapter 10 (Role is rule 2, Context is rule 1, Task is rule 3, Constraints come from rules 6 and 7, and Format from rules 1 and 4):

Element	Meaning	Example
Role	Who should the LLM act as?	"You are a quality engineer at an ISO 9001 automotive Tier-1 supplier."
Context	What background does the LLM need?	"We run a CNC lathe producing transmission shafts at 25.000 ± 0.050 mm."
Task	What do you want it to do?	"Review the following FMEA and identify the three highest-RPN failure modes."
Constraints	What should it avoid or limit?	"Limit to 200 words. Do not invent risks not in the document. Cite RPN values explicitly."
Format	How should the output look?	"Output as a numbered list. Each item: failure mode, RPN, current mitigation status."

Not every prompt needs all five elements, but missing more than two usually produces noticeably worse output. (R-C-T-C-F is a checklist for a single message. For standing instructions in a project or an agent, the same five elements apply, but you write them once.) Compare these:

Bad prompt: "explain SPC"

Produces a generic textbook definition that is not tailored to any audience, purpose, or length.

R-C-T-C-F prompt: "You are explaining statistical process control to a new operator on a CNC turning line. They have a high-school math background and have never heard of standard deviation. In under 150 words, explain what SPC is, why we use it, and what they need to do at their station. Use one analogy from daily life. Output as plain prose, no bullets, no headings."

Produces a focused, audience-appropriate, length-controlled explanation that the operator can actually use.

11.2.6.1 Patterns That Work in Engineering

- **"Show your reasoning before the answer."** Forces the model to think step-by-step, which reduces errors on multi-part problems. Reasoning models do this internally, so on those you can drop the instruction, but still ask them to list their assumptions.
- **"Make a plan first and wait for my approval."** The most useful instruction for agents and any multi-step job.
- **"Check your answer by running the calculation in code."** Turns an LLM guess into a reproducible calculation.
- **"What might be wrong with this approach?"** Provides a critical second opinion on your own thinking. Avoids sycophancy.
- **"Generate three different versions."** Options beat a single answer on open-ended problems.
- **"What questions should I have asked but didn't?"** Surfaces gaps in your problem framing.
- **"Cite your sources, and if you don't have a source, say so."** Partial defence against hallucinations. Not foolproof, but it forces the model to flag uncertainty.

11.2.6.2 Beyond the Single Prompt

R-C-T-C-F structures *one* message. With projects and agents, the whole context matters more: standing rules go in project instructions, the model sees only the relevant documents, and a long, drifting chat is restarted with a short summary. These habits are covered in Section 10.8.2.

11.2.6.3 Patterns That Do Not Work

- Yes/no questions on judgment calls — you get hedged, useless answers

- Pasting massive documents without a focus question — output will be generic
- "What's the best way to…" — there is no best; give constraints instead
- Asking for specific numerical recommendations on safety-critical decisions — verify everything independently

Activity (10 min): prompt battle. In pairs, take this bad prompt: *"Help me with my motor problem."*

1. Each partner rewrites it with R-C-T-C-F. Add made-up but realistic details (motor rating, symptom, environment).
2. Both rewrites go to a phone chatbot. Do not edit the answers.
3. Compare the two outputs and the original prompt's output. Which one would you act on? Which would you check first?
4. Vote: the class picks the best prompt in the room, and we ask why it worked.

Lab Preview: In Week 1 Lab (Section 23.9), you will rewrite five real bad prompts from manufacturing engineers using R-C-T-C-F, deliberately trigger and document a hallucination, and use an LLM to generate your Week 2 Python starter code. The code you generate this week becomes your starting point next week — so write the prompt carefully.

11.2.6.4 Try it yourself: take a prompt apart

The seven sentences below make up one R-C-T-C-F prompt. Decide which element each sentence is.

Which R-C-T-C-F element?

Pick the element each sentence belongs to.

Categories: Role / Context / Task / Constraints / Format

1. You are a maintenance planner at a food-packaging plant. _____
2. Our filler line has had four unplanned stops this month, all on the capping head. _____
3. The capper is 12 years old and spare parts take three weeks to arrive. _____
4. Draft a one-page case for replacing the capping head. _____
5. Use only the downtime figures I paste below; do not invent costs. _____
6. Keep it under 300 words. _____
7. Give me a summary paragraph followed by a three-row table: option, cost, downtime avoided. _____

Answers: 1 Role; 2 Context; 3 Context; 4 Task; 5 Constraints; 6 Constraints; 7 Format.

11.2.7 Self-Check Questions — Week 1

Work through these in your own words before the Week 2 lecture. If you find yourself looking answers up in these notes rather than recalling them, re-read the relevant section.

1. Why is "predict the next word" enough to produce text that summarizes a document, drafts code, and explains engineering concepts?
2. A colleague says "we use AI to predict tool wear." What would you need to ask them to determine whether they mean classical ML, deep learning, or an LLM?
3. Your supervisor pastes a customer's machined-part drawing into the free tier of ChatGPT to "ask what the tolerances mean." What three problems do you raise with them, ranked by severity?
4. What is the difference between an LLM making a math error and an LLM hallucinating? Give an engineering example of each.
5. Rewrite this prompt using R-C-T-C-F: *"summarize this report for the boss."*
6. Your supervisor wants an AI agent to read all 300 open maintenance work orders and close the ones it judges duplicates. Using Section 10.8.1, which best practices apply, and what permission would you give it on day one?

Exit ticket, Week 1. In one sentence: name one task from your own work or studies that you will *not* hand to an LLM unsupervised after today, and say why.

Submit your answer on your phone before you leave.

11.2.8 Further Reading — Week 1

- Google Cloud. *AI vs. machine learning.* https://cloud.google.com/learn/artificial-intelligence-vs-machine-learning
- MIT Sloan. *Machine learning, explained.* https://mitsloan.mit.edu/ideas-made-to-matter/machine-learning-explained
- Google Developers. *Machine Learning Crash Course* (free, online). https://developers.google.com/machine-learning/crash-course
- 3Blue1Brown. *Large Language Models explained briefly.* https://www.youtube.com/watch?v=LPZh9BOjkQs
- 3Blue1Brown. *Deep Learning* series (start with Chapter 1: *But what is a neural network?*). https://www.youtube.com/watch?v=aircAruvnKk&list=PLZHQObOWTQDN52m7Y21ePrTbvXkPaWVSg
- Anthropic. *Building effective agents.* https://www.anthropic.com/engineering/building-effective-agents
- OWASP. *Top 10 for Large Language Model Applications* (includes prompt injection). https://genai.owasp.org/llm-top-10/
- Hugging Face. *LLM Course, Chapter 1* (free, online). https://huggingface.co/learn/llm-course/chapter1/1

- OpenAI. *Tokenizer* (interactive demo). https://platform.openai.com/tokenizer
- Anthropic. *Prompt Engineering Overview.* https://docs.anthropic.com/en/docs/build-with-claude/prompt-engineering/overview
- Wikipedia. *Hallucination (artificial intelligence).* https://en.wikipedia.org/wiki/Hallucination_(artificial_intelligence)
- TechCrunch. *Samsung bans use of generative AI tools like ChatGPT.* https://techcrunch.com/2023/05/02/samsung-bans-use-of-generative-ai-tools-like-chatgpt-after-april-internal-data-leak/

11.3 Week 2 — Manufacturing Data and Statistical Thinking

11.3.1 Learning Objectives

	By the end of this week, you will be able to…	Assessment
2.1	Name the five levels of the ISA-95 plant data hierarchy and identify which system type belongs at each level	Self-check Q1
2.2	Identify and categorize common data quality problems in an industrial dataset	Lab Week 2 Task 2
2.3	Direct an LLM to construct an X-bar control chart in Python and interpret whether a process is in statistical control	Lab Week 2 Task 3
2.4	Calculate Cpk and explain its meaning to a non-engineer	Self-check Q3
2.5	Articulate four limitations of SPC and explain how machine learning addresses them	Self-check Q4–Q5

Before You Read: In your last co-op or lab, how did data get from a machine to a spreadsheet? How many steps were involved, and at how many steps could

something go wrong?

11.3.2 Where Plant Data Comes From

Every modern manufacturing facility generates data from at least five separate IT systems. You will be expected to work with all of them.

11.3.2.1 The ISA-95 Plant Data Hierarchy

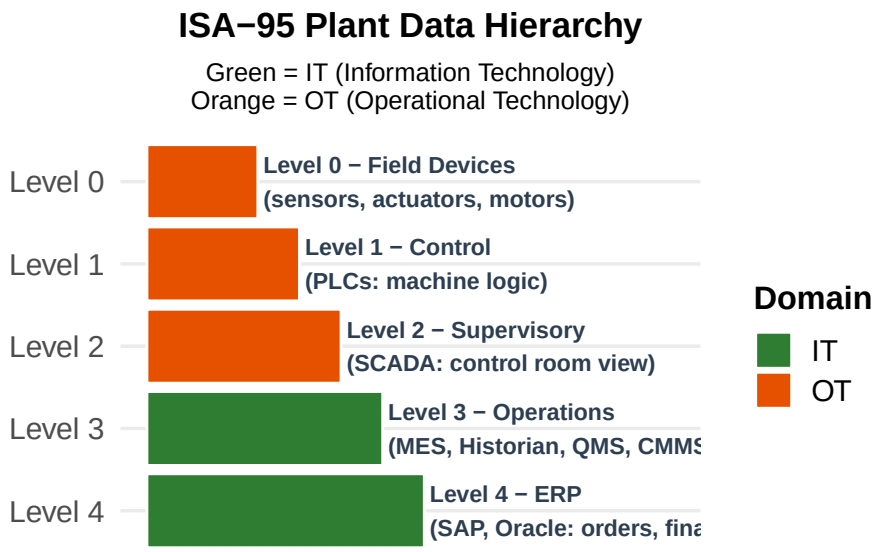


Figure 11.2: The ISA-95 data hierarchy. Levels 0–2 are Operational Technology (OT); Levels 3–4 are Information Technology (IT). Python scripts belong at Level 3 or higher — never on a PLC.

This is the **ISA-95 hierarchy** (Figure Figure 11.2). Different rules apply at different levels: you absolutely do not deploy Python scripts on a PLC (Level 1), but you absolutely do deploy them at Level 3 or 4. The boundary between OT and IT is one of the most important distinctions in manufacturing engineering — you will encounter it again in Week 5.

System	Full Name	What It Stores	Typical Sample Rate
PLC	Programmable Logic Controller	Real-time sensor and actuator state	1 ms – 100 ms

System	Full Name	What It Stores	Typical Sample Rate
SCADA	Supervisory Control and Data Acquisition	Aggregated control-room view	1 s – 10 s
Historian	(AVEVA PI, GE Proficy, Ignition)	Long-term time-series sensor data	1 s – 1 min, compressed
MES	Manufacturing Execution System	Production events, cycle starts, alarms, OEE	Per event
QMS	Quality Management System	Inspection results, dispositions, NCRs	Per part or per sample
CMMS	Computerized Maintenance Management System	Work orders, downtime reasons, parts replaced	Per event
ERP	Enterprise Resource Planning (SAP, Oracle)	Orders, materials, finance, scheduling	Per transaction

11.3.2.2 Try it yourself: which level?

Which ISA-95 level?

Place each item at the right level of the hierarchy.

Categories: Level 0-1 (field and control) / Level 2 (supervisory) / Level 3 (operations) / Level 4 (enterprise)

1. A proximity sensor on the press ram. _____
2. The PLC ladder logic that runs the guard interlock. _____
3. The SCADA screen the control-room operator watches. _____
4. A historian storing a year of spindle temperature. _____
5. The MES recording when each cycle started. _____
6. The Python script that trains your failure model. _____
7. The SAP production schedule. _____
8. A servo motor on the feeder. _____

Answers: 1 Level 0-1 (field and control); 2 Level 0-1 (field and control); 3 Level 2 (supervisory); 4 Level 3 (operations); 5 Level 3 (operations); 6 Level 3 (operations); 7 Level 4 (enterprise); 8 Level 0-1 (field and control).

11.3.2.3 Four Types of Plant Data

The Week 2 lab uses **quality data** (Type 3 below). Know all four:

1. **Time-series / process data** — continuous sensor streams: temperature, pressure, vibration, motor current. Comes from PLCs via SCADA or the historian. High sample rates. *The dominant data type for predictive maintenance (Week 3).*
2. **Batch / event data** — discrete events with timestamps. "Cycle started at 08:00:03." "Alarm raised at 14:22:11." Comes from the MES.
3. **Quality / inspection data** — measurements of finished parts: diameters, surface finish, hardness, pass/fail flags. Comes from the QMS. *This is what your Week 2 CSV contains.*
4. **Maintenance data** — work orders, breakdowns, scheduled maintenance, parts replaced. Comes from the CMMS. Often the worst-quality data in the plant — heavily dependent on technicians typing notes correctly.

Industry Application: At a Tier-1 stamping plant in Windsor, the quality team needed to correlate surface defects with press tonnage. The defect data was in the QMS; the tonnage data was in the historian — sampled 10× faster, in UTC, with a different part identifier scheme. Reconciling the two datasets took two days of work before any analysis could begin. This is typical.

11.3.3 Why Industrial Data Is Messy

Five problems appear so consistently in plant data that you should expect all of them until proven otherwise.

> **The 80/20 rule is real:** In real industrial data work, you will spend 80% of your time making the data trustworthy and 20% doing the analysis. The 20% gets the conference talks. The 80% is what makes you employable.

11.3.3.1 Missing Values

Sensors fail. Networks drop. Maintenance happens. People forget to log. The result is gaps in your data. The Week 2 lab dataset has two working days missing entirely due to a simulated network outage.

Response	When to use	Risk
Drop the rows	Few rows missing at random	Lost information
Forward fill (last known value)	Slowly changing variables; brief gaps	Wrong if the variable changed during the gap
Linear interpolation	Smooth continuous variables	Manufactures data that may not exist

Response	When to use	Risk
Leave as NaN	When honest gaps must stay honest	Requires downstream code to handle NaNs
Investigate and fill from another source	Always preferred	Requires legwork

The right answer for the lab's two missing days: document it, note it, move on. **Never silently delete data.**

Think – pair – share (4 min). Two working days are missing from your dataset. List the possible reasons (at least three). For each reason, say what you would do with the gap and who you would ask. When is it acceptable to fill the gap, and when is it dangerous?

11.3.3.2 Unit Confusion

> ⚠ Warning
>
> **Lesson from History:** Unit confusion crashed the Mars Climate Orbiter in 1999. NASA assumed Lockheed Martin's thrust calculations were in newton-seconds; they were in pound-force seconds. The orbiter entered the Martian atmosphere at the wrong angle and burned up. $327 million lost.

In your plant, the cases are smaller but more frequent. A surface roughness gauge outputs readings in either inches or micrometres depending on a setting. An operator on day shift uses one; the night shift operator uses the other. The CSV column is labelled `surface_roughness_um` — but a fraction of the values are actually in inches. Silent corruption.

How to detect: ask whether the *range* of values is physically plausible. Surface roughness of 0.00001 µm is impossible (atomic-scale). That same number in inches is a reasonable Ra value. Once you spot the range issue, you can find the unit error and convert.

A column named pressure_bar holds values between 6 and 8 for 85% of the rows and between 87 and 116 for the rest. What is the most likely explanation?

- **A.** Real pressure spikes that should be kept
- **B.** Some readings are in psi, not bar
- **C.** The sensor is drifting
- **D.** Outliers that should be deleted

Reveal after the vote

B. Multiply 6 and 8 bar by 14.5 and you get 87 and 116 psi, exactly the second cluster. The data is bimodal because two gauge settings are mixed. Convert it, record that you did, and tell the owner of the data. Deleting it would throw away 15% of the rows for no reason.

11.3.3.3 Sample Rate Mismatches

A vibration sensor records at 10,000 Hz. The MES logs cycle starts at 1 Hz. To join "vibration during each cycle," you cannot simply align timestamps — you must aggregate the vibration data per cycle (mean? RMS? peak?) before joining. This is rarely covered in undergraduate statistics but appears constantly in plant work.

11.3.3.4 Timestamp Problems

A short list of real timestamp problems:

- PLC clock drifts 30 seconds per day; nobody resyncs it
- Historian stores in UTC; MES stores in local time; CSV export converts back without telling you
- Daylight saving creates one duplicate hour each fall and one missing hour each spring
- Timezone-naïve timestamps from different systems merged silently
- Operator-entered timestamps rounded to the nearest 5 minutes

Whenever joining two data sources, plot the timestamp distribution and look for gaps, duplicates, and step discontinuities before doing anything else.

11.3.3.5 Outliers — Three Flavours

Type	What it is	What to do
Sensor error	Physically impossible value (negative time, 250 mm diameter on a 50 mm part)	Remove or flag — this never happened
Real special cause	A real bad part, a real event you need to know about	**Keep it. This is the most important data in your dataset.**
Tail of common-cause variation	A real measurement that happens to be far from the mean	Keep it. It is part of natural variation.

The hardest case is distinguishing a real special cause from a sensor error. The rule of thumb: if you can find a physical or process explanation, it is special cause. If the value is physically impossible, it is sensor error. If neither, it is natural variation.

The Week 2 lab contains a row with `diameter_mm = 250.123`. The Mazak QT-200 chuck will not grip a 250 mm bar — this must be a sensor error, likely a missing decimal point (true value probably 25.0123). The lab walks you through finding and handling this.

11.3.4 Statistical Process Control

SPC was developed by Walter Shewhart at Bell Labs in the 1920s. A century later it is still the dominant approach to quality monitoring in nearly every manufacturing industry. You will encounter it on day one of your career.

11.3.4.1 The Central Idea: Two Kinds of Variation

Every process produces variation. No two parts are identical. Shewhart's fundamental insight: variation has two causes.

Common-cause variation is the inherent noise of a stable process — slight differences in material, temperature, vibration, operator technique — all small, all unavoidable, all individually unpredictable but collectively predictable. A "stable" process produces *only* common-cause variation.

Special-cause variation is when *something has changed*: a tool broke, coolant ran out, a new batch of material arrived, a temperature controller failed. Each special cause produces a noticeable, often non-random pattern in the data.

The job of SPC is to **detect special-cause variation as quickly as possible** so you can investigate and fix it before it produces bad parts.

11.3.4.2 Control Charts and Control Limits

A **control chart** is a plot of a process metric over time with computed control limits. If a point falls outside the limits, the process is presumed out of statistical control — there is a special cause to investigate.

The two types you need to know:

- **X-bar chart** (with R or S chart): for subgroups. Take n parts at a time (typically n = 4 or 5), compute the subgroup mean, plot the means over time. Pair with a range or standard deviation chart.
- **Individuals chart** (I-chart): for one-at-a-time inspection, where you measure every part.

11.3.4.3 Control Limits vs. Specification Limits

This is the most-confused distinction in undergraduate SPC. Get it right:

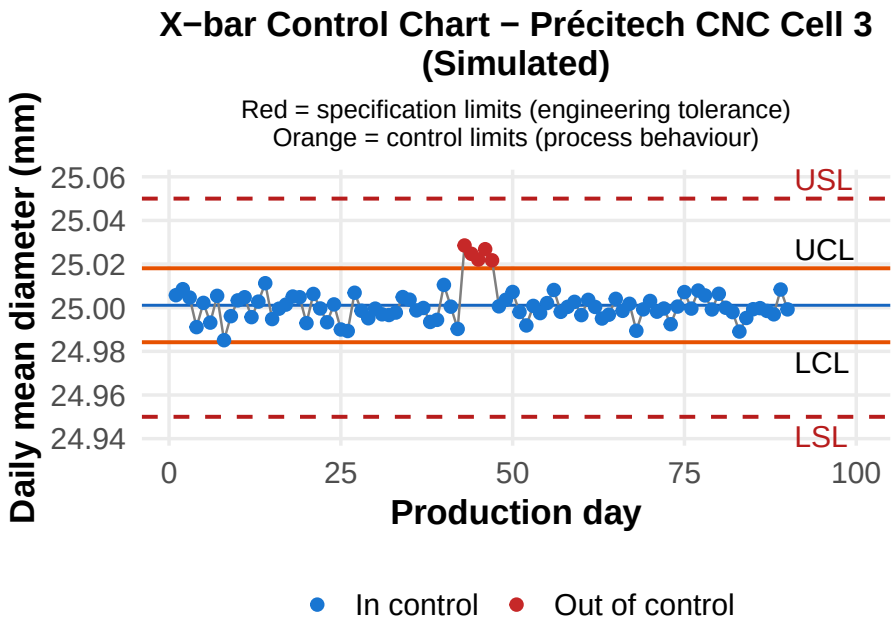


Figure 11.3: Simulated X-bar control chart for daily mean diameter over 90 production days. A special-cause event (bad bar stock) is visible around days 43–47 — the same event you will find in the Week 2 lab dataset.

- **Specification limits (USL, LSL)** come from **engineering or the customer**. They describe the design tolerance. These limits have nothing to do with your data — they were set when the part was designed.
- **Control limits (UCL, LCL)** come from **the data itself**. They describe how the process *actually behaves*. They are computed from the process mean and standard deviation.

A process can be **in control but out of spec** — behaving consistently, but consistently making bad parts. A process can be **out of control but in spec** — most parts are good, but something has changed and you should investigate before parts start failing.

	In spec	**Out of spec**
In statistical control	Ideal — stable and capable	Consistent but bad — fix the process design
Out of statistical control	Lucky for now — find the special cause before parts go OOS	Emergency — bad parts being made now

11.3.4.4 Try it yourself: stable or capable?

Control limits versus specification limits

Place each situation in the right cell of the control-versus-specification table.

Categories: In control, in spec / In control, out of spec / Out of control, in spec / Out of control, out of spec

1. Every daily mean is inside the control limits and every part is inside the tolerance. _____
2. A stable, centred-wrong process: the mean sits 0.06 mm above nominal and the tolerance is +/-0.05 mm. _____
3. Cpk = 0.7 and the chart shows no special causes. _____
4. Five daily means in a row above the UCL, but every part still passes inspection. _____
5. One point below the LCL; every measured part is within tolerance. _____
6. The mean jumped 0.08 mm last shift and 30% of parts are over the USL. _____

Answers: 1 In control, in spec; 2 In control, out of spec; 3 In control, out of spec; 4 Out of control, in spec; 5 Out of control, in spec; 6 Out of control, out of spec.

11.3.4.5 The X-bar Control Limit Formula

When you plot **subgroup means** rather than individual measurements, the means themselves vary less than individual parts — by a factor of $\sqrt{n}$ (the

central limit theorem). The control limits on an X-bar chart are therefore:

$$\text{UCL}_{\bar{X}} = \bar{\bar{X}} + 3 \cdot \frac{\sigma}{\sqrt{n}} \tag{11.1}$$

$$\text{LCL}_{\bar{X}} = \bar{\bar{X}} - 3 \cdot \frac{\sigma}{\sqrt{n}} \tag{11.2}$$

where σ is the standard deviation of *individual* parts, n is the subgroup size, and $\bar{\bar{X}}$ is the overall process mean.

A common student mistake: computing control limits using the standard deviation of the *daily means* instead of the standard deviation of individual parts. If the daily means have standard deviation $s_{\bar{x}}$, then $s_{\bar{x}} \approx \sigma/\sqrt{n}$ — so using $s_{\bar{x}}$ instead of $\sigma/\sqrt{n}$ effectively applies the $\sqrt{n}$ correction twice. The control limits become too narrow, generating false alarms constantly.

11.3.4.6 Try it yourself: compute the control limits

X-bar chart control limits

A CNC lathe makes parts with grand mean 25.000 mm and an individual-part standard deviation of 0.013 mm. You sample 5 parts per day. Compute the X-bar control limits (answer to 4 decimal places).

- UCL: _____ mm *(to 4 decimal places)*
- LCL: _____ mm *(to 4 decimal places)*

11.3.4.7 Process Capability: C_p and C_{pk}

SPC tells you whether your process is *stable*. **Process capability** tells you whether a stable process is *capable* of meeting specifications.

$$C_p = \frac{\text{USL} - \text{LSL}}{6\sigma} \tag{11.3}$$

$$C_{pk} = \min\left(\frac{\text{USL} - \mu}{3\sigma},\ \frac{\mu - \text{LSL}}{3\sigma}\right) \tag{11.4}$$

where σ is the standard deviation of *individual* measurements (not the X-bar), and μ is the process mean.

- C_p is the *potential* capability — how good the process could be if perfectly centred between USL and LSL
- C_{pk} is the *actual* capability — accounts for off-centre processes
- $C_{pk} \leq C_p$ always; they are equal only when the process is perfectly centred

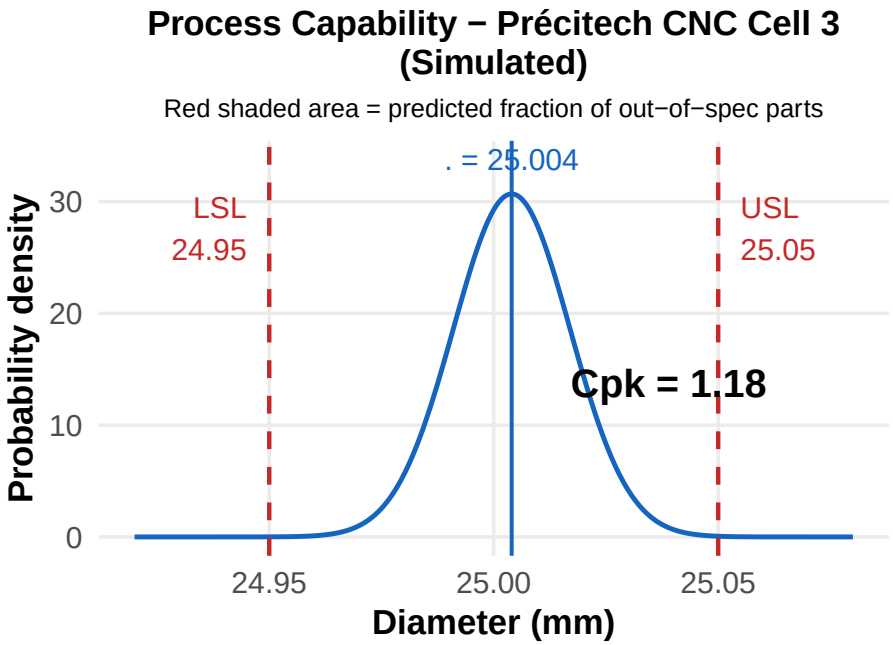


Figure 11.4: Process capability visualization. A Cpk of 1.16 means the nearest specification limit is 3.48 from the process mean — better than minimum (1.00) but below the automotive standard (1.33). The shaded tail area under each specification limit represents the expected fraction of non-conforming parts.

11.3.4.8 Try it yourself: Cp or Cpk?

Potential versus actual capability

A shaft is specified at 10.00 +/- 0.10 mm (LSL 9.90, USL 10.10). The process mean is 10.04 mm and the standard deviation is 0.02 mm.

- Cp: _____ *(to 2 decimal places)*
- Cpk: _____ *(to 2 decimal places)*

Think − pair − share (3 min). Cp is high but Cpk is much lower in the exercise above. Explain to your partner, in one sentence and without using a formula, what has gone wrong with the process. What would you adjust first, and would it need a new tool or just a new offset?

Industry C_{pk} targets (your customer's PPAP requirements are authoritative; these are typical):

Industry	Typical C_{pk} target	Notes
Automotive (general)	1.33	AIAG PPAP common requirement
Automotive (safety-critical: brakes, steering, airbags)	1.67	
Aerospace	1.67 – 2.00	Often combined with 100% inspection
Medical devices	1.67 or higher	Plus FDA regulatory requirements
General machining (non-critical)	1.00 – 1.33	Typical starting point for new processes

The Week 2 lab dataset gives $C_{pk} \approx 1.16$. That is marginal — a real plant would treat this as "improvement project needed" rather than "ready for sustained production."

11.3.5 Where SPC Fails

SPC is essential but not sufficient. Four situations expose its limits:

1. **Non-normal data**: The 3-sigma control limits assume an approximately normal distribution. For bounded measurements (geometric tolerances near zero, defect counts), the distribution may be skewed or bimodal. Raw SPC limits will produce false alarms or miss real shifts.
2. **Autocorrelated data**: Tool wear is the perfect example. Today's part diameter depends on yesterday's — the tool did not reset overnight. SPC

assumes each measurement is independent. When this assumption is violated, control limits become misleading.

3. **Slow drift**: A 0.001 mm/day diameter drift is too slow to trigger a 3-sigma rule in any single sample. By the time SPC detects it, you may have produced months of slowly degrading parts.
4. **Multiple operators or shifts blended together**: Three operators with different setup biases, blended into one chart, makes the common-cause variation appear inflated — hiding real operator-to-operator differences.

Think – pair – share (4 min). Think of a machine you have worked on or seen. Which of the four SPC limits (non-normal data, autocorrelation, slow drift, mixed shifts) would affect a control chart on that machine most? What would the chart miss that a person walking past would notice?

11.3.5.1 The Transition to Machine Learning

SPC tells you **when** a process went out of control. It does not tell you **when it will go out of control** or **what will cause it**. Those are predictive questions requiring a different toolset.

Tool	What it tells you	How fast
SPC	Something IS wrong right now	After it happens
Machine learning	Something WILL go wrong, and when	Before it happens

Both have a role. SPC is your safety net. ML is your early-warning system. The best operations deploy both. Week 3 introduces the predictive side.

11.3.6 Self-Check Questions — Week 2

1. A new colleague plots the last 90 daily mean diameters on a control chart, using $\pm$ 3 of the daily means as control limits. What is wrong with this, or is it correct? Explain.
2. Your CMM reports surface flatness in mm. The drawing specifies inches. The CSV column header says "mm." Three days of data have suspiciously small values. What likely happened, and how would you detect it without anyone telling you?
3. A process has $C_{pk} = 0.85$. The control chart shows no out-of-control points. The plant manager declares the process "in control" and refuses your improvement project. What is the technically correct response?
4. Why does tool-wear autocorrelation violate the assumptions of a standard X-bar chart?

5. Name the four levels of the ISA-95 hierarchy and give one example of data that lives at each level.

Exit ticket, Week 2. A colleague computes control limits from the standard deviation of the *daily means* instead of $\sigma/\sqrt{n}$. In one or two sentences, explain what goes wrong with the chart.

Submit your answer on your phone before you leave.

11.3.7 Further Reading — Week 2

- NIST/SEMATECH. *e-Handbook of Statistical Methods, Chapter 6: Process Monitoring.* https://www.itl.nist.gov/div898/handbook/pmc/pmc.htm
- ASQ. *Statistical Process Control Overview.* https://asq.org/quality-resources/statistical-process-control
- ASQ. *Process Capability — Cp & Cpk Overview.* https://asq.org/quality-resources/process-capability
- ISA. *ISA-95 Standards Overview.* https://www.isa.org/standards-and-publications/isa-standards/isa-standards-committees/isa95
- Walter Shewhart, *Economic Control of Quality of Manufactured Product* (1931). Available at https://archive.org/details/economiccontrolo0000shew

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11.4 Week 3 — Machine Learning and Predictive Maintenance

11.4.1 Learning Objectives

	By the end of this week, you will be able to…	Assessment
3.1	Compare the four maintenance regimes and justify when each makes economic sense	Self-check Q5
3.2	Sketch the P-F curve and explain what the P-F interval means for maintenance scheduling	Self-check Q4
3.3	Describe the structure of the NASA C-MAPSS dataset and explain how RUL is computed from run-to-failure data	Lab Week 3 Task 1

	By the end of this week, you will be able to…	Assessment
3.4	Explain what a Random Forest is and why it is the default first choice for tabular industrial data	Self-check Q1, Q3
3.5	Identify the engine-wise train/validation split requirement and explain why row-wise splitting causes data leakage	Self-check Q1

Before You Read: Recall from Week 2 that SPC told you *when* a process went out of control — only after it happened. What would it be worth to your plant if you knew the bearing on a CNC spindle was going to fail in three weeks, rather than finding out when it seized mid-shift?

11.4.2 The Economics of Maintenance

Before any technical content, the *why*. Maintenance strategies fall along a spectrum, and the economics drive every business decision.

Regime	What it means	Cost profile	When it makes sense
Reactive ("run to failure")	Wait until it breaks, then fix it	Low day-to-day cost; catastrophic when failure hits	Non-critical equipment where failure has no downstream impact
Preventive (scheduled)	Replace parts at fixed intervals regardless of condition	Predictable; wastes remaining useful life	Industry default for decades; still appropriate for cheap, fast-to-replace components
Predictive / CBM (condition-based)	Replace based on actual condition, monitored continuously	Higher up-front investment; minimizes waste and surprise	Capital-intensive, downtime-sensitive operations

Regime	What it means	Cost profile	When it makes sense
Prescriptive	System recommends *what* to do, not just *when*	Highest technology investment; emerging	Sites with mature digital infrastructure and reliable ground-truth feedback

The shift from preventive to predictive is the central economic story of Industry 4.0.

11.4.2.1 The Financial Case for Predictive Maintenance

Consider a bearing on the spindle of a CNC lathe at Précitech.

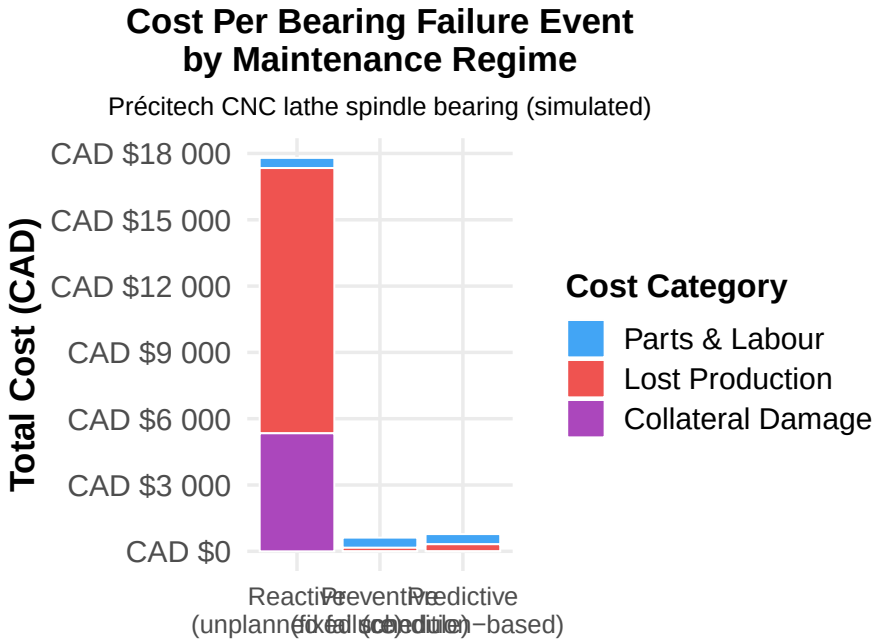


Figure 11.5: Cost comparison for one bearing failure event across three maintenance regimes. Predictive maintenance avoids the bulk of the unplanned downtime cost by catching the failure before it becomes a functional failure.

Figure Figure 11.5 shows why every plant manager wants predictive maintenance. For a 10-machine cell with one unplanned bearing failure per machine

per year, the plant loses roughly CAD $178,000 annually on this single component type. A predictive model catching 70% of these failures before they become functional failures would save approximately CAD $125,000 per year — from one component, on one cell.

11.4.2.2 The P-F Curve

The most important concept in maintenance theory:

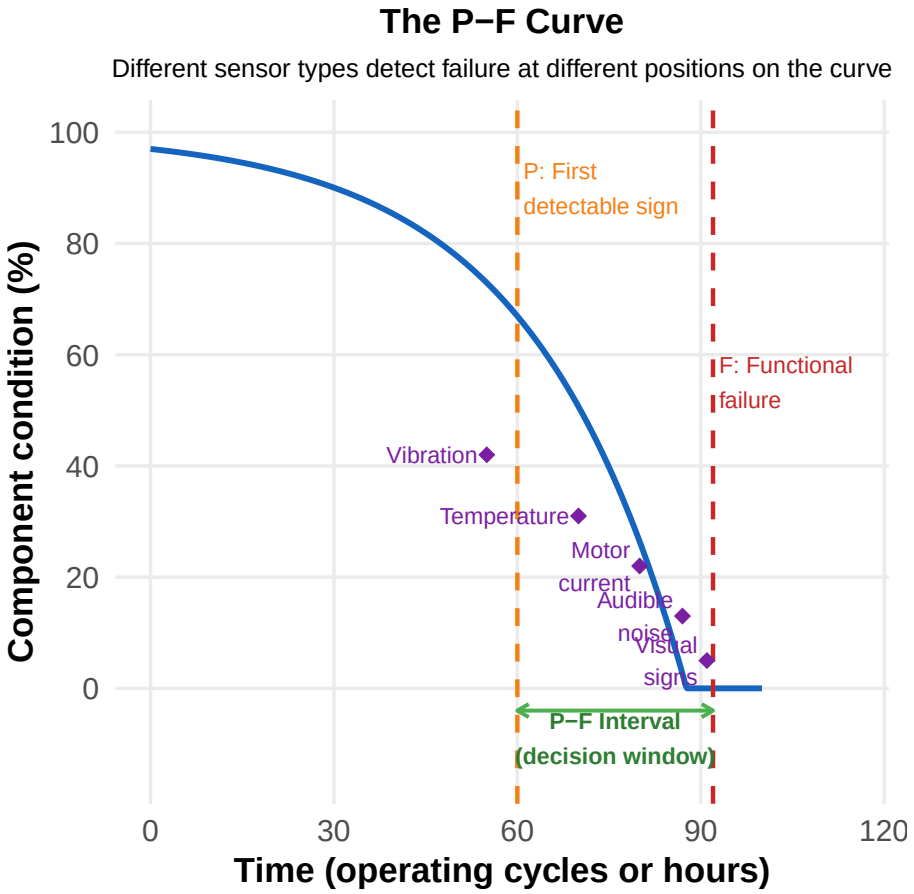


Figure 11.6: The P-F curve: a stylized view of how a machine component's condition degrades from normal operation to functional failure. The P-F interval — the time between when failure becomes detectable (P) and when function is lost (F) — is the maintenance engineer's decision window. Different sensor types detect failure at different positions along the curve.

Different sensor types detect failure at different positions on the P-F curve (Figure Figure 11.6):

Sensor type	Typical position on P-F curve	Approximate P-F interval
Vibration analysis	Earliest — often months before functional failure	Months to weeks
Temperature monitoring	Later — bearing temperature rises as wear progresses	Weeks to days
Motor current signature	Even later — current draw increases as friction rises	Days to hours
Audible noise (human ear)	Late	Hours to minutes
Visual smoke or sparks	Functional failure imminent	Minutes

Predictive maintenance models exploit the **earliest** indicators (vibration, subtle sensor drift) to maximize the decision window. This is why the C-MAPSS dataset you will work with uses 21 sensors per cycle — many of them recording subtle thermodynamic shifts that no human operator would notice.

Industry Application: GE Aviation's OnPoint predictive maintenance program for CFM56 engines (the most common commercial airliner engine in service) uses sensor data from every flight to continuously update engine health models. Airlines receive maintenance recommendations before any symptom is observable on the shop floor. This has reduced unscheduled removals by over 30% on participating fleets.

11.4.2.3 Try it yourself: which maintenance strategy?

Choose the maintenance strategy

Pick the strategy that fits each asset best.

Categories: Run to failure / Preventive (time-based) / Predictive (condition-based)

1. A cheap, redundant LED light fixture over a walkway. _____
2. Replacing hydraulic hoses every 12 months regardless of condition. _____
3. Changing engine oil every 5,000 km. _____
4. A critical spindle whose vibration trend triggers a bearing swap two weeks before predicted failure. _____
5. A compressor with ten sensors and a model that estimates remaining life. _____
6. A spare office printer. _____

Answers: 1 Run to failure; 2 Preventive (time-based); 3 Preventive (time-based); 4 Predictive (condition-based); 5 Predictive (condition-based); 6 Run to failure.

Think – pair – share (4 min). Name one machine at a plant you have seen. What strategy is used today? Estimate, roughly, what an unplanned failure costs per hour and what a planned replacement would cost. Is the current strategy the cheapest one?

11.4.3 The NASA C-MAPSS Dataset

Your capstone project uses a real research benchmark — the same dataset used in thousands of academic papers since 2008.

11.4.3.1 What It Represents

C-MAPSS (Commercial Modular Aero-Propulsion System Simulation) is a high-fidelity simulation of a turbofan jet engine — the kind that powers commercial airliners. NASA used it to generate datasets of run-to-failure trajectories: simulated engines run from a healthy state until they fail, with 21 sensors recorded at each operating cycle.

The data comes in four variants (FD001 through FD004) with different combinations of operating conditions and fault modes. **FD001** is the simplest — one operating condition, one fault mode, 100 training engines, 100 test engines. This is what your capstone uses.

11.4.3.2 Data Structure

Each row of the training file represents one operating cycle of one engine:

Column	Meaning	Notes
`unit_number`	Engine ID (1–100)	Resets for each engine
`time_in_cycles`	Operating cycle number	Starts at 1 for each engine; ends at failure
`op_setting_1` to `op_setting_3`	Operating conditions (altitude, throttle, etc.)	Mostly constant in FD001
`sensor_1` to `sensor_21`	Sensor readings (temperatures, pressures, fan speed, fuel flow, bleed enthalpy, etc.)	Some constant; some highly predictive

There is **no RUL column** in the training data. You compute it:

$$\text{RUL} = (\text{max cycle for this engine}) - (\text{current cycle})$$

If engine #1 ran for 192 cycles total, its RUL at cycle 1 is 191, at cycle 100 is 92, and at cycle 192 (the failure cycle) is 0.

Quick Check: If you plot RUL for all 100 training engines stacked together (all rows in the file), what shape would the RUL values form? Think about it before reading ahead — you will verify your answer in Lab Week 3, Task 5.

11.4.4 The Machine Learning Problem: From Data to Prediction

11.4.4.1 Supervised Learning

Your capstone uses **supervised learning** — the most common and commercially important category of machine learning.

Supervised learning requires four things:

1. **Inputs (features):** measurements available at prediction time. For C-MAPSS: 21 sensors and 3 operating settings, plus engineered features you create.
2. **Outputs (labels):** what you want to predict, paired with each input row. For C-MAPSS: RUL.
3. **A model:** a mathematical function that maps inputs to outputs.
4. **A training algorithm:** a procedure for finding the model parameters that best fit many (input, output) pairs.

The supervised learning problem for C-MAPSS is:

> Given this engine's current and recent sensor readings, predict how many more operating cycles it will run before failure.

This is **regression** — predicting a continuous number — not classification.

11.4.4.2 Feature Engineering

A single sensor reading at a single cycle tells you the current state but not the *trend*. Two engines might have the same temperature reading today; one is on its 50th cycle, the other on its 150th. They are at very different points in their degradation trajectory.

Feature engineering creates new variables from raw ones to expose the patterns the model needs to learn:

Feature type	What it captures	Example
Rolling mean (window = 5)	Local average; smooths noise	Average of last 5 sensor readings
Rolling std	Local variability; increases as machine degrades	Variability of last 5 readings
Rolling min / max	Local extremes	Highest temperature in the last 5 cycles
Delta / rate of change	Speed of drift	Today's reading minus yesterday's
Domain-specific	Physics-based features	Vibration RMS in a specific frequency band

For C-MAPSS, rolling mean and rolling standard deviation on each informative sensor (window = 5) is the single highest-leverage feature engineering step. It typically reduces model RMSE from approximately 30 to approximately 20 cycles.

A critical implementation detail: rolling features must be computed **per engine**, with the window resetting at the boundary of each engine's data. If you apply rolling statistics across the full dataset without grouping by engine, the last few cycles of engine #1 will contaminate the rolling window of engine #2's first few cycles — a subtle but serious bug.

11.4.4.3 Clipping the RUL Label at 125

C-MAPSS engines run from healthy to failure, but "healthy" doesn't really degrade. An engine with 200 cycles left looks essentially identical to one with 300 cycles left — both are healthy. The model wastes capacity trying to distinguish them.

Standard practice on C-MAPSS research: clip RUL labels at 125. Any engine healthier than 125 cycles is labelled "125 — basically healthy." The model then focuses all of its capacity on the degradation phase, where the actual predictive signal is.

This simple change typically reduces RMSE by approximately 30% — a clear case where understanding the physics of your problem lets you frame it better for the algorithm.

11.4.5 The Fatal Mistake: Row-Wise Train/Test Splitting

> ⚠ Warning
>
> **The Most Common Bug in Industrial ML:** Do NOT split rows randomly between training and validation. Do not. This is the single most important technical lesson of the capstone.

Why is row-wise splitting a mistake for this dataset? Because rows from the same engine are not independent. Engine #1's cycle 50 and cycle 51 are nearly identical — they share the same machine, the same wear state, the same operating history. If cycle 50 is in training and cycle 51 is in validation, the model effectively sees the validation set during training. The validation score becomes meaninglessly optimistic.

The correct approach is **engine-wise splitting**:

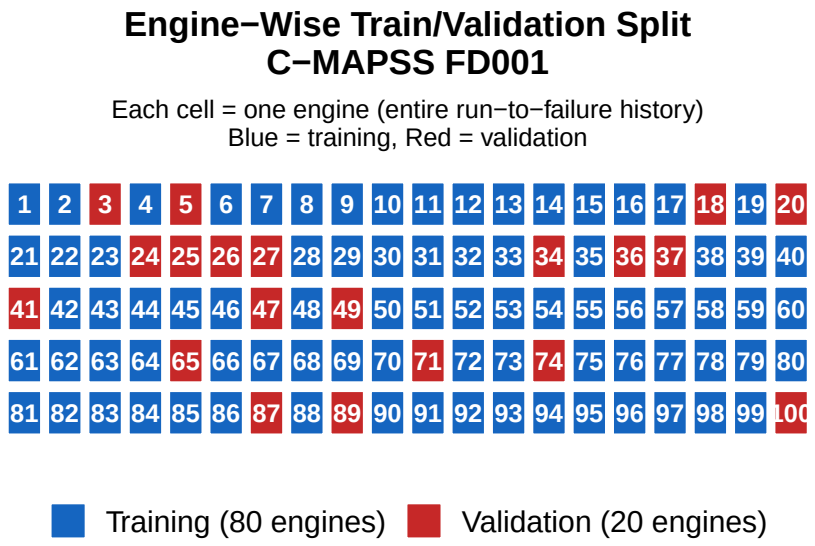


Figure 11.7: Correct engine-wise train/validation split. Entire engine histories go to either training or validation — never split an engine's cycles across the two sets. This simulates the real deployment situation: can your model predict RUL for an engine it has never seen?

Figure Figure 11.7 illustrates the correct split. The validation set contains entire engine histories that the model has never seen during training. This simulates the real deployment situation: a new engine arrives at your plant; your model must predict its RUL without having seen this specific engine before.

How to detect the bug if you accidentally use row-wise splitting: your validation RMSE will be suspiciously low — 5 to 8 cycles instead of the expected 15 to 25. When you see that, immediately check whether any engine ID appears in both training and validation sets. If it does, you have data leakage.

Your validation RMSE for the C-MAPSS model comes out at 4.8 cycles. The naive baseline is 50. What is your first reaction?

- **A.** Celebrate and write it up
- **B.** Check for data leakage
- **C.** Add more trees to push it lower
- **D.** Send it to the plant manager

Reveal after the vote

B. An error that is ten times better than the baseline on the first try is a warning sign, not a result. The usual cause is that rows from the same engine are in both training and validation. Split by engine ID and run it again.

11.4.6 Random Forest — Intuition and Why It Works

Many algorithms can solve the RUL regression problem. Your capstone uses **Random Forest** because it is the strongest first-try algorithm for tabular industrial data — robust, fast, requires almost no tuning, and provides interpretable feature importance rankings.

The idea in one sentence: instead of trusting one expert, ask a committee of 100 experts who each studied a *different random selection* of past engines, and average their answers. The sections below build up to that picture in four steps: what one tree is, where the randomness comes from, how the forest learns, and how it is used.

11.4.6.1 Step 1: A Single Decision Tree

A **decision tree** is a flowchart of yes/no questions about your features, with a prediction at each end point (a **leaf**). To predict, you start at the top and answer each question for your engine until you reach a leaf. The number in that leaf is the prediction.

How does the tree learn its questions? They are *learned from the data*, not written by an engineer. At the top, the algorithm tries every feature and every possible threshold ("Sensor 11 $<$ 47.6?", "Sensor 11 $<$ 47.7?", "Sensor 4 $<$ 1395?", …). For each candidate it splits the training rows into two groups and measures how similar the RUL values are *within* each group. The candidate that makes the two groups most uniform (the largest drop in variance) wins. The algorithm then repeats the same search separately inside each group, and keeps going until a stopping rule is met (maximum depth, minimum rows per

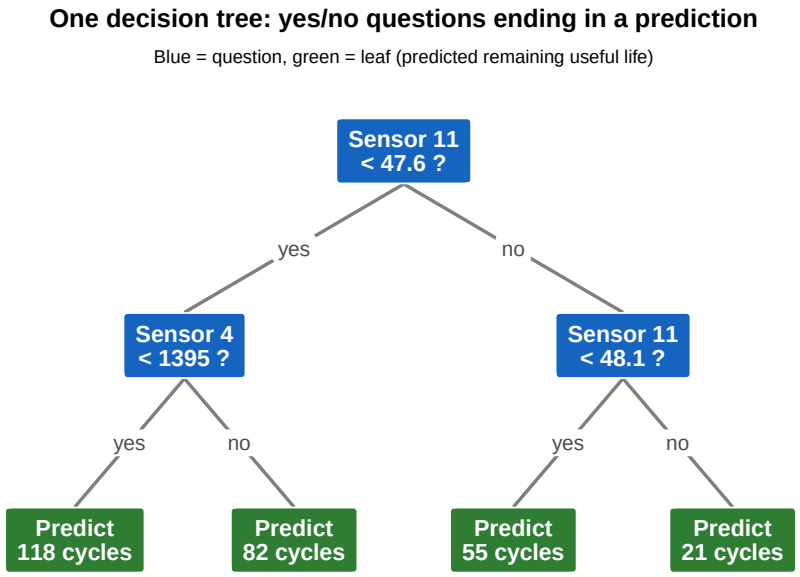


Figure 11.8: A small decision tree for remaining useful life. A new engine starts at the top, answers each question about its sensor readings, and ends at a leaf holding the prediction. The thresholds here are illustrative.

leaf). Each leaf predicts the **average RUL of the training rows that ended up in it**.

Strengths of a single tree:

- Trivially interpretable — you can read the exact rules
- Handles mixed numeric and categorical inputs without normalization
- Fast to train and predict

Critical weakness: a tree allowed to grow deep keeps splitting until every leaf holds one or two training rows. It then reproduces the training data perfectly, *including the measurement noise*, and generalizes poorly to new engines. Figure Figure 11.9 shows the problem on a simple one-sensor example.

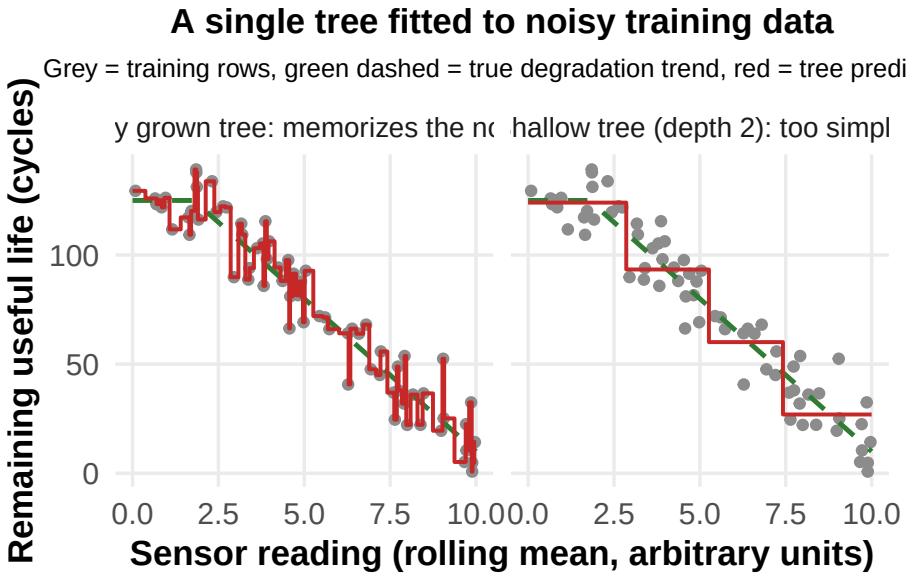


Figure 11.9: One tree fitted to 70 noisy training rows. The shallow tree is too simple and misses the trend. The fully grown tree chases every noisy point, so its predictions for new engines would jump around for no physical reason.

11.4.6.2 Step 2: Where Does the "Random" Come From?

The fix is to grow many deep trees and make sure they are **different from each other**. If all 100 trees were identical, averaging them would change nothing. Random Forest forces the trees to differ with two deliberate injections of chance:

Source 1 — random rows (bootstrap sampling). Each tree is trained on its own *bootstrap sample*: draw rows at random from the training set, **with replacement**, until you have as many rows as the original. Some rows are picked twice or three times; others are not picked at all. On average about 37%

of the rows (the fraction $(1 - 1/n)^n \to e^{-1} \approx 0.37$) are left out of any one tree. Those left-out rows are called its **out-of-bag** rows.

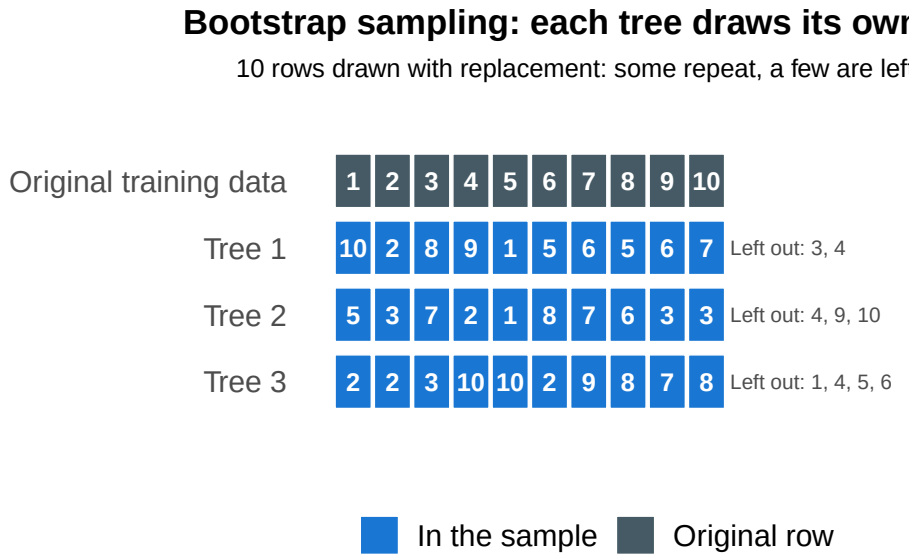


Figure 11.10: Bootstrap sampling with 10 training rows. Each tree draws 10 rows with replacement, so every tree trains on a different, partly repeated set of rows, and a few rows are left out entirely.

Source 2 — random features at every split. When a tree searches for the best question at a node, it is not allowed to look at all features. It is shown a **random subset** of them (for regression, classically about one third of the features) and must choose the best split among only those. The next node gets a *different* random subset.

Why bother with the second source? Without it, one very strong sensor (say Sensor 11) would be the best root question in *every* tree, and all 100 trees would look alike. Hiding it from some splits forces other sensors to be used, so the trees disagree in useful ways.

> **Note**
>
> **Randomness is only in training.** The random draws happen once, while the forest is being built. After that the forest is a fixed set of trees: give it the same input twice and you get the same output. The random number generator behind the draws is controlled by a *seed* (`random_state=42` in scikit-learn). Fix the seed and you get the same forest every time; change it and you get a slightly different forest with

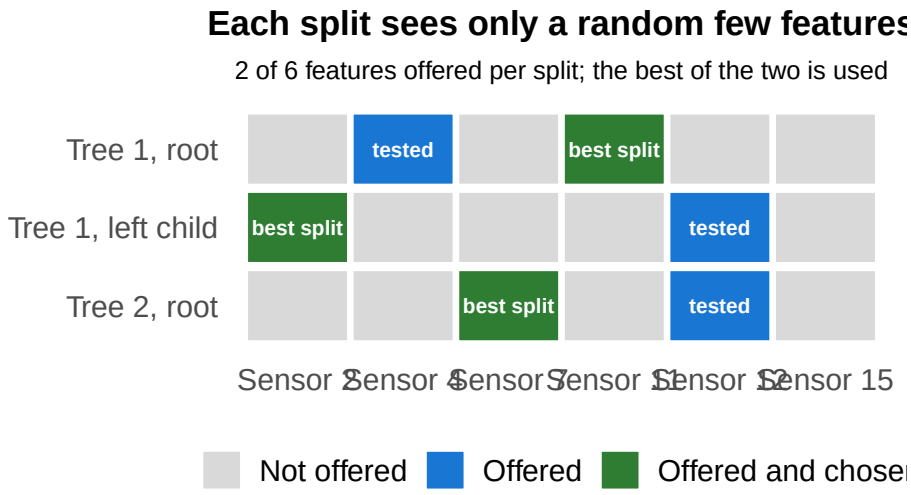


Figure 11.11: Feature subsampling with 6 features, 2 offered at each split. The best split is chosen only from the offered features, so a dominant sensor cannot win the root of every tree.

> nearly the same accuracy. If a small seed change swings your RMSE a lot, your validation set is too small or your model is unstable.

11.4.6.3 Step 3: How the Forest Learns

"Training" a Random Forest is just repeating the tree-growing recipe from Step 1 many times, each with its own random draws:

1. Decide the number of trees (for example `n_estimators = 100`).
2. For each tree: draw a bootstrap sample of the rows (Source 1).
3. Grow the tree on that sample. At every node, draw a fresh random subset of features (Source 2), find the best split among them, and split.
4. Keep splitting deep (little or no pruning) until the leaves are small, then store the tree.
5. Move on to the next tree. Nothing carries over between trees.

There are no weights to adjust, no repeated passes over the data and no error being fed back from one tree to the next. Each tree is **independent**, which has two practical consequences: the trees can be trained in parallel on all CPU cores (`n_jobs=-1`), and adding more trees can only make the average more stable, not worse. (Methods like gradient boosting, in contrast, build trees one after another, each correcting the last.)

Each individual tree is deliberately **over-fitted and noisy**: low bias but high

variance. The ensemble works because the trees' errors are different. One tree is too high on some engine, another too low, and when you average 100 of them those errors largely cancel while the shared signal stays.

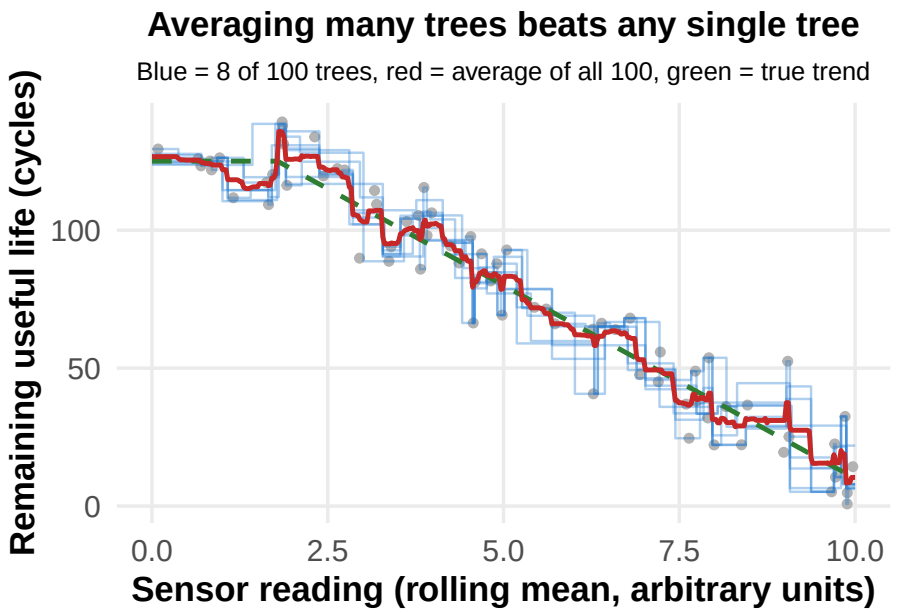


Figure 11.12: Eight individual trees (blue) are jagged and disagree with each other, but the average of 100 trees (red) follows the true degradation trend (green dashed) much more closely than any one of them.

Random Forest Algorithm Clearly Explained! (Normalized Nerd): https://www.youtube.com/watch?v=v6VJ2RO66Ag

> ⚠ Warning
>
> **The out-of-bag trap on engine data.** The left-out rows give a free validation score (the *out-of-bag score*), with no separate test set needed. But it has the same flaw as the row-wise split described above: rows from the *same engine* are spread across the bag and the out-of-bag set, so the forest has effectively seen the engine it is being tested on. For C-MAPSS, validate by holding out whole engines instead.

11.4.6.4 Step 4: How the Forest Is Used

Once trained, using the forest (called **inference** or **prediction**) takes four steps:

1. Compute the same features for the new engine that you used in training

(sensor values, rolling means, rolling standard deviations), as one row.
2. Send that row down **every** tree. Each tree answers its own questions and lands in a leaf.
3. Each tree reports the average RUL stored in its leaf.
4. The forest's prediction is the **average of all the trees' predictions**. (For a yes/no classification problem, the trees vote and the majority wins.)

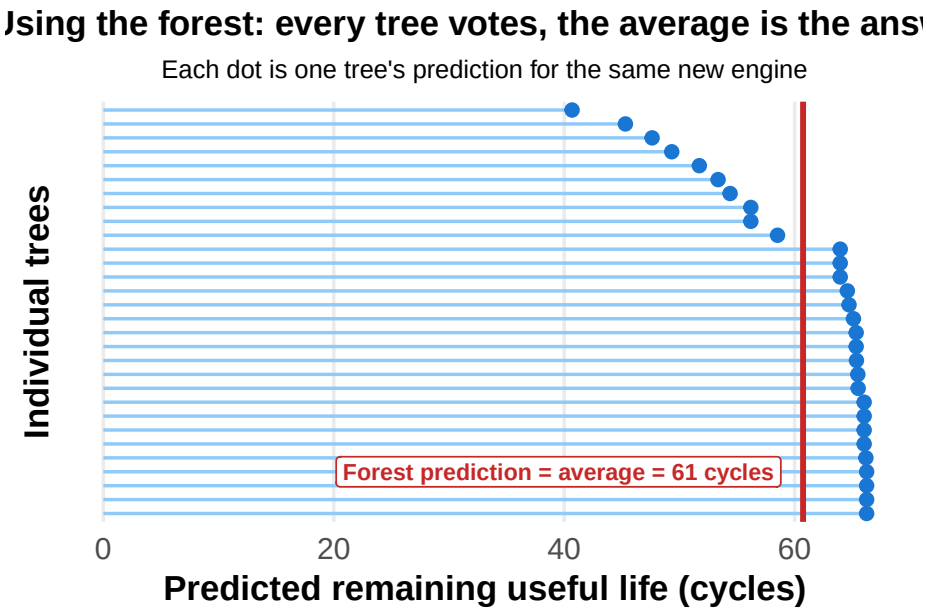


Figure 11.13: One new engine passed through 30 trees. Each dot is one tree's own prediction. The forest reports the average (red line). The spread of the dots is a rough measure of how much the trees disagree, and so of how confident the forest is.

Three things follow from this:

- **Prediction is fast.** 100 trees of depth 10 are only about a thousand yes/no comparisons, so a prediction takes milliseconds. Training is the slow part, and you only do it once (and again when the model drifts, Week 5).
- **The trees double as an uncertainty signal.** If the 100 trees all say 60 ± 3 cycles, treat the result as reliable. If they range from 20 to 120, the forest is telling you it does not know. This spread is worth showing to the maintenance planner.
- **Feature importance comes free.** Each split in each tree reduced the variance by some amount. Add up that reduction for every split that used a given sensor, average across the trees, and you get the importance ranking you will plot in Week 4.

	Training (once, slow)	**Prediction (every time, fast)**
Input	Many rows with known RUL	One row of features, RUL unknown
What happens	Each tree: bootstrap rows, random features at each split, grow deep	Row goes down every tree to a leaf
Randomness	Yes (seeded)	None
Output	A stored set of trees	The average of the trees' leaf values

Why Random Forest is the default first-try for tabular plant data:

Property	Why it matters in manufacturing
Handles mixed feature scales without normalization	Real plant data has wildly different units (mm, °C, RPM, bar); not normalizing breaks gradient-based methods but not RF
Robust to outliers	One bad sensor reading does not corrupt the whole model
Captures non-linear relationships and interactions	Real processes are non-linear; RF learns this automatically
Provides feature importance	Tells you which sensors matter — directly useful for instrumentation decisions
Fast to train, even on large datasets	You can iterate quickly in a lab session
Almost no hyperparameter tuning needed for a solid baseline	`n_estimators=100, max_depth=10` gets you 90% of the way there

> **Note**
>
> **One limitation to remember:** a forest can only predict values it has seen in its leaves, so it cannot extrapolate. If a new engine degrades further than any engine in the training data, the forest will predict roughly the lowest RUL it saw, not a lower one. Check that your operating conditions fall inside the range of the training data.

Why does a Random Forest give each tree only a random subset of features at every split?

- **A.** To make training faster
- **B.** So the trees differ from each other and their errors cancel when averaged

- **C.** Because most sensors are noise
- **D.** To use less memory

Reveal after the vote

B. Speed is a side benefit, but the main purpose is diversity. If every tree could use the strongest sensor at its root, all 100 trees would look alike and averaging would not help.

11.4.6.5 Activity (25 min): be the forest

You will build a tiny Random Forest by hand. Work in groups of three or four. Each group is **one tree**, and the whole class is the forest.

The table holds 12 training snapshots from past engines. The sensor readings are rolling means, and RUL is the remaining useful life in cycles.

Row	Sensor 11	Sensor 4	RUL
1	47.3	1385	125
2	47.4	1402	118
3	47.5	1390	110
4	47.6	1410	96
5	47.7	1395	88
6	47.8	1415	74
7	47.9	1400	66
8	48.0	1420	52
9	48.1	1408	45
10	48.2	1425	31
11	48.4	1418	20
12	48.6	1430	9

1. **Bootstrap.** On a phone, open a random-number generator (search "random number 1 to 12") or a dice app. Draw **12 rows with replacement**. If a row comes up twice, it is in your sample twice. Write down your sample, and the rows that were *left out*.
2. **Random feature.** Flip a coin. Heads: you may only use **Sensor 11** for your question. Tails: only **Sensor 4**.
3. **Grow a tree with one question.** Using only your sample and your allowed sensor, choose a threshold that best separates the high-RUL rows from the low-RUL rows. Each side is a leaf. The leaf prediction is the **average RUL of your sample rows on that side**.
4. **Predict.** A new engine arrives with **Sensor 11 = 47.95** and **Sensor 4 = 1412**. Send it down your tree. Write down your tree's prediction.
5. **Combine.** Each group reports its prediction. The class **average** is the forest's prediction.

Then discuss:

- How far apart were the individual trees? Which groups were furthest off, and what was different about their sample or sensor?
- Compare the class average with the individual predictions. Is the average closer to the truth than most groups?
- Which step of the activity was Source 1 and which was Source 2 of the randomness?
- What would happen if every group had used the same 12 rows and the same sensor?

Reveal the true RUL

The true remaining life for this engine is **58 cycles**. Individual one-question trees usually miss by 15 to 40 cycles in either direction, and the class average typically lands much closer. Groups that drew their sample badly or were forced onto the noisier Sensor 4 are the furthest off. That spread is the point: individual trees are noisy, and averaging cancels much of the noise.

11.4.7 Evaluation — Measuring Whether a Model Is Useful

11.4.7.1 RMSE and MAE

For regression (predicting a number such as RUL), two standard metrics measure how far the predictions $\hat{y}_i$ are from the true values y_i over n test rows:

- **MAE — Mean Absolute Error.** Take the error of each prediction, drop its sign (absolute value), then take the mean.
- **RMSE — Root Mean Squared Error.** Square each error, take the mean of the squares, then take the square root to return to the original units.

Read the formulas right to left: the error is $\hat{y}_i - y_i$, then either the absolute value or the square is applied, then the average, and (for RMSE) the root.

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|\hat{y}_i - y_i| \tag{11.5}$$

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2} \tag{11.6}$$

Both are in the same units as your target (operating cycles). The key difference:

- **MAE** treats all errors equally. Being off by 10 cycles contributes 10× less than being off by 100.
- **RMSE** penalizes large errors disproportionately. Being off by 100 cycles contributes 100× more than being off by 10.

For maintenance applications, large errors are dangerous — predicting 200 cycles remaining when the real answer is 20 means the engine fails unexpectedly. **RMSE is therefore the primary metric** for predictive maintenance. MAE provides a complementary picture of typical, everyday error.

11.4.7.2 Try it yourself: MAE and RMSE by hand

MAE and RMSE for five predictions

A model predicts remaining life for five engines. Actual RUL: 100, 80, 60, 40, 20. Predicted: 95, 90, 55, 40, 70. Compute the MAE and the RMSE (1 decimal place).

- MAE: _____ cycles *(to 2 decimal places)*
- RMSE: _____ cycles *(to 2 decimal places)*

Think − pair − share (3 min). The RMSE is much larger than the MAE here. Which single prediction is responsible? A maintenance manager asks "so how wrong is the model, usually?" Which of the two numbers do you give, and what do you tell them about the one bad engine?

11.4.7.3 The Naive Baseline

The simplest possible predictor: always predict the mean RUL of the training set, regardless of any sensor inputs. This is the model that learns nothing.

If your sophisticated model cannot beat the naive baseline by at least 2× on RMSE, your model is not doing useful work. For C-MAPSS FD001:

Model	Approximate RMSE
Naive baseline (predict mean)	~50 cycles
Linear regression on raw sensors	~35 cycles
Random Forest on rolling features	~20 cycles
Tuned deep learning (LSTM)	~12–15 cycles

Your capstone target is approximately 20 cycles RMSE with Random Forest — comparable to published results on this benchmark using the same approach.

11.4.7.4 Asymmetric Errors in Maintenance Scheduling

A real plant does not care equally about errors in both directions. If you predict 50 cycles remaining and the true answer is 30, the engine fails before maintenance — a catastrophic, costly miss. If you predict 50 and the true answer is 70, you replaced a component with life left in it — wasteful but safe.

Specialized loss functions can encode this asymmetry (the original C-MAPSS competition used a custom scoring function that penalized late predictions far

more than early ones). Your capstone uses symmetric RMSE for simplicity, but this is a known simplification of the real industrial problem.

11.4.8 Where Do You Set the Alarm?

A predictive-maintenance model does not say "this machine will fail". It gives each machine a risk score between 0 and 1, and someone has to choose the score above which maintenance is called out. Set the threshold low and you catch every failure but send technicians to healthy machines. Set it high and you stop wasting call-outs but miss real failures.

The fleet below has 200 machines, 30 of which really are about to fail.

Teaching demo — drag the threshold and watch the confusion matrix, precision, recall and total cost change together. Ask the class where the cheapest threshold is, and whether it is where accuracy is highest, before you move the slider there.

Show the live R code behind this calculator

```
#| label: alarm-threshold
# Choosing the alarm threshold for a predictive-maintenance model
threshold <- 0.50             # call out maintenance when the risk score is at least th
cost_false_alarm <- 2000      # an unnecessary inspection
cost_missed_failure <- 50000  # an unplanned breakdown

# Risk scores from the model for 200 machines: 170 healthy, 30 about to fail.
# (Evenly spaced points from two overlapping distributions, so results repeat exactly.)
healthy <- qbeta(ppoints(170), 2, 5)
failing <- qbeta(ppoints(30), 5, 2)

counts <- function(th) {
  c(TP = sum(failing >= th), FN = sum(failing < th),
    FP = sum(healthy >= th), TN = sum(healthy < th))
}
cost_of <- function(th) {
  k <- counts(th)
  k[["FP"]] * cost_false_alarm + k[["FN"]] * cost_missed_failure
}

k <- counts(threshold)
precision <- k[["TP"]] / max(1, k[["TP"]] + k[["FP"]])
recall <- k[["TP"]] / (k[["TP"]] + k[["FN"]])
accuracy <- (k[["TP"]] + k[["TN"]]) / 200
total_cost <- cost_of(threshold)

steps <- seq(0.05, 0.95, by = 0.01)
```

```
cost_curve <- sapply(steps, cost_of)

par(mar = c(4.5, 6, 2, 1), cex.main = 1.3, cex.lab = 1.2, cex.axis = 1.1)
plot(steps, cost_curve / 1000, type = "l", lwd = 4, col = "#2c3e50", las = 1, xlim = c(0.05, 0.95
     xlab = "Alarm threshold", ylab = "Total cost ($ thousands)", main = "Total cost at every thr
abline(v = threshold, lty = 2, col = "#bdc3c7")
points(threshold, total_cost / 1000, pch = 19, cex = 2.4, col = "#e74c3c")

cat(sprintf("Total cost: $%s\n", format(total_cost, big.mark = " ", scientific = FALSE)))
cat(sprintf("  Failures caught (TP)   %d of 30\n", k[["TP"]]))
cat(sprintf("  Failures missed (FN)   %d\n", k[["FN"]]))
cat(sprintf("  False alarms (FP)      %d\n", k[["FP"]]))
cat(sprintf("  Correctly left alone (TN)  %d\n", k[["TN"]]))
cat(sprintf("  Precision              %.0f%%\n", precision * 100))
cat(sprintf("  Recall                 %.0f%%\n", recall * 100))
cat(sprintf("  Accuracy               %.1f%%\n", accuracy * 100))
```

Now change the cost of a missed failure to match the cost of a false alarm and find the cheapest threshold again. It moves a long way. The model has not changed; only what its mistakes cost you has. That is why the threshold is an engineering and business decision, and why accuracy alone is the wrong number to optimize.

11.4.9 Self-Check Questions — Week 3

1. A colleague's Random Forest achieves 5-cycle RMSE on C-MAPSS FD001 with default settings. You are suspicious. What is the most likely cause?
2. Why does clipping RUL at 125 improve model performance? Explain in terms of where the signal in the data actually is.
3. Why is a single decision tree usually worse than a Random Forest, even though they both start from the same idea?
4. What is the P-F interval? Give a concrete example from a manufacturing plant, naming the component, the potential failure indicator, and a realistic interval length.
5. A maintenance manager says "we replace the spindle bearings every 2,000 hours regardless." Which of the four maintenance regimes is this? When does it make economic sense, and when does it not?

Exit ticket, Week 3. In your own words, explain to a classmate who missed class where the word "random" comes from in Random Forest, and why it makes the model better rather than worse.

Submit your answer on your phone before you leave.

11.4.10 Further Reading — Week 3

- StatQuest. *Random Forests, Part 1* and *Part 2: Missing data and clustering.* https://www.youtube.com/watch?v=J4Wdy0Wc_xQ
- IBM Technology. *What is Random Forest?* (short overview). https://www.youtube.com/watch?v=gkXX4h3qYm4
- A. Saxena, K. Goebel, D. Simon, and N. Eklund, "Damage Propagation Modeling for Aircraft Engine Run-to-Failure Simulation," *Proc. PHM08*, Denver CO, October 2008. (The original C-MAPSS paper.)
- NASA Prognostics Data Repository. https://www.nasa.gov/intelligent-systems-division/discovery-and-systems-health/pcoe/pcoe-data-set-repository/
- scikit-learn. *Ensemble methods: Random Forests.* https://scikit-learn.org/stable/modules/ensemble.html#random-forests
- Kaggle C-MAPSS dataset mirror. https://www.kaggle.com/datasets/behrad3d/nasa-cmaps
- Wikipedia. *Reliability-centered maintenance.* https://en.wikipedia.org/wiki/Reliability-centered_maintenance
- Andriy Burkov, *The Hundred-Page Machine Learning Book* (2019). Compact, rigorous, covers exactly what you need.
- PHM Society annual conference proceedings (free online). https://www.phmsociety.org/

11.5 Week 4 — Feature Engineering and Model Building

11.5.1 Learning Objectives

	By the end of this week, you will be able to…	Assessment
4.1	Direct an LLM to implement per-engine rolling-window features in Python (`groupby().transform()`), and verify the result is correct	Lab Week 4 Task 1
4.2	Justify the choice of window size and the effect of different rolling statistics on model performance	Self-check Q2

	By the end of this week, you will be able to…	Assessment
4.3	Direct an LLM to build a complete Random Forest pipeline (features → split → train → evaluate → compare to baseline), checking each stage as it goes	Lab Week 4 Tasks 3–5
4.4	Interpret a feature importance chart and explain what it says about which sensors matter	Lab Week 4 Task 6
4.5	Direct an LLM to save a trained scikit-learn model to disk for later deployment	Lab Week 4 Task 7

Before You Read: In Week 3 you identified which C-MAPSS sensors are most correlated with RUL. Does the *current value* of a sensor fully describe where an engine is in its degradation trajectory — or do you need to know the *history* of that value as well? Why?

11.5.2 Feature Engineering for Time-Series Data

A single sensor value at a single moment tells you the state of the engine *right now*, but not how it got there. Consider two engines, both showing sensor_11 = 47.3 this cycle. One has been slowly climbing from 44.0 over the past 20 cycles; the other has been stable at 47.3 for 100 cycles. They are in very different situations — the first is degrading, the second is not. A model that sees only the current value cannot distinguish them.

Feature engineering extends the model's view backwards in time, giving it the **trend information** it needs.

11.5.2.1 Rolling Statistics

The most powerful features for time-series prediction are computed over a rolling window — a moving neighbourhood of the most recent k cycles:

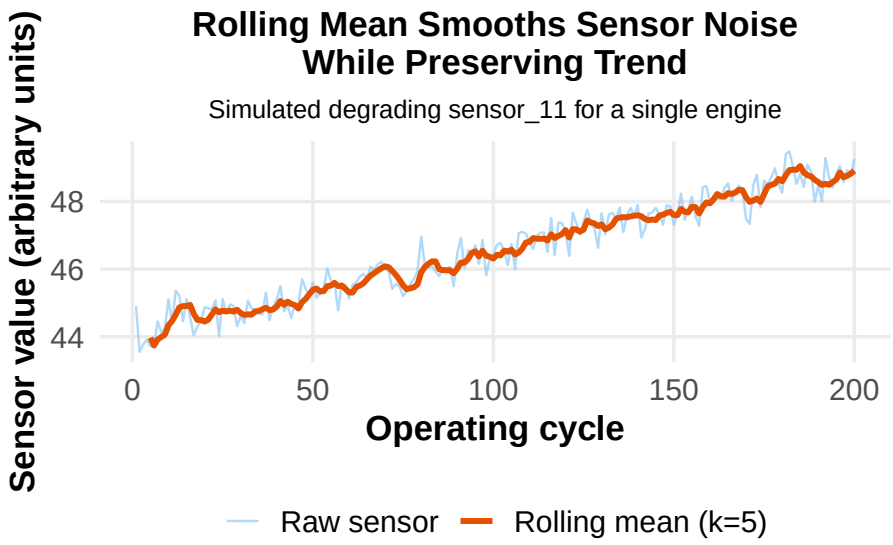


Figure 11.14: Effect of a rolling mean (window = 5) on a simulated degrading sensor signal. The rolling mean (orange) smooths cycle-to-cycle measurement noise while preserving the underlying degradation trend — exactly what the model needs to see.

Feature	How to compute	What it adds
Rolling mean (k = 5)	Average of the last 5 readings	Smooth estimate of current sensor level; reduces noise
Rolling std (k = 5)	Std deviation of the last 5 readings	Quantifies local variability — often increases as machine degrades
Rolling min / max	Min or max of the last 5 readings	Captures worst-case readings in the recent window
Delta (difference)	Current reading minus previous reading	Rate of change — useful for detecting acceleration of degradation
Lagged values	Reading from k cycles ago	Explicit history for the model

Window size matters: a window that is too narrow (k = 2) adds noise rather than smoothing it. A window that is too wide (k = 50) lags so far behind the current state that it misses fast-moving events. For C-MAPSS FD001, k = 5 is a well-validated choice. If you have time after the core capstone, try k = 10 or k = 20 and observe the effect on RMSE.

11.5.2.2 The `groupby().transform()` Pattern

When you ask an LLM to write your rolling-feature code, this is the single most important thing to check in what it hands back — and exactly the kind of subtle, easy-to-miss error you are training yourself to catch (Learning Objective 1.5):

```
 CORRECT: window resets at each new engine boundary
df["sensor_11_rolling_mean"] = (
    df.groupby("unit_number")["sensor_11"]
      .transform(lambda x: x.rolling(5, min_periods=1).mean())
)

 WRONG: window bleeds across engine boundaries
df["sensor_11_rolling_mean"] = df["sensor_11"].rolling(5).mean()
```

The wrong version treats the entire dataset as one continuous time series. At the boundary between engine #1 and engine #2 in the file, the last 4 rows of engine #1's history bleed into engine #2's first few rolling features. This is a subtle bug — your model will still train and produce predictions, but the features are contaminated, and your RMSE will be slightly worse than it should be. Worse, in a production deployment on a new engine, the contamination

disappears, and the model behaves differently than it did during validation.

If you are not sure which version an LLM just gave you, ask it directly: "does this rolling window reset at each engine boundary — walk me through what happens at the boundary between unit_number 1 and unit_number 2." A good answer explains the mechanism; a hallucinated one will just reassure you it's fine.

11.5.3 Random Forest Internals

Now that you are about to direct an LLM to train one, you need to know what is happening inside the `RandomForestRegressor` call — otherwise you cannot judge whether its code, or its explanation of the results, is right.

11.5.3.1 Growing a Single Tree

A decision tree is grown by the following process:

1. Start at the root node with the entire training dataset
2. At each node, consider every feature and every possible split point
3. Choose the split that minimizes the impurity of the two resulting subsets (for regression: usually variance reduction)
4. Recurse on each subset until a stopping criterion is met (maximum depth, minimum samples per leaf, etc.)

This is a greedy algorithm — it picks the best split at each node without backtracking. The result is a tree that perfectly fits training data if allowed to grow deep enough, but overfits badly: it memorizes noise alongside signal.

11.5.3.2 The Ensemble Solution

Random Forest addresses overfitting with two sources of randomness:

1. **Bootstrap aggregating (bagging):** each tree is trained on a bootstrap sample — a random sample of the training data *with replacement*, same size as the original. Roughly 37% of rows are left out of each tree's training set (the "out-of-bag" samples), which can be used for internal validation.
2. **Feature subsampling:** at each split, the algorithm considers only a random subset of features (typically $\sqrt{p}$ features for classification, $p/3$ for regression, where p is the total number of features). This forces trees to differ from each other even when one feature is dominant.

Because each tree sees a different random view of the data, the trees make different mistakes. When you average 100 different-but-correlated predictions, the random errors cancel out while the shared signal remains. The bias is roughly the same as any single tree; the variance drops roughly by a factor of n_{trees}. This is the core statistical argument for ensemble methods.

Group Discussion: Feature importance in a Random Forest is computed as the average reduction in node impurity contributed by each feature across all trees. If two features are highly correlated (say, sensor_11 and sensor_4 both drift together as the engine degrades), how might this affect the feature importance values for each? Would removing one of them hurt or help model performance?

11.5.3.3 Hyperparameters You Need to Know

Hyperparameter	Meaning	Typical value for C-MAPSS	Effect of increasing
`n_estimators`	Number of trees	100	More trees = lower variance, slower training
`max_depth`	Maximum depth per tree	10	Deeper = more expressive but higher overfitting risk
`min_samples_leaf`	Minimum samples per leaf node	1–5	Higher = simpler trees, less overfitting
`max_features`	Features considered per split	"sqrt" or 1/3 of total	Fewer = more diverse trees
`n_jobs`	Parallel threads	-1 (use all CPU cores)	More = faster training (no accuracy effect)

For C-MAPSS FD001, `n_estimators=100, max_depth=10, n_jobs=-1` with default feature selection gives you approximately 90% of the achievable performance with zero tuning effort. This is the value of Random Forest for industrial first-pass analysis.

11.5.4 Evaluation Done Right

11.5.4.1 Reading Residual Plots

A single RMSE number does not tell you where your model is failing. Always produce a **predictions vs. actuals scatter plot** (often called a residual plot):

- Points along the diagonal ($\hat{y} = y$) = perfect predictions
- Points above the diagonal = overestimates (predicted too much life remaining — dangerous in maintenance)

- Points below the diagonal = underestimates (predicted too little life — results in premature replacement, wastes money but is safe)
- Systematic patterns (e.g., all high-RUL points overestimated) = model bias at specific RUL ranges

For the C-MAPSS model you build in Week 4, a common pattern is that predictions are pulled toward the centre of the RUL distribution — the model overestimates RUL for engines close to failure, and underestimates it for very healthy engines. RUL clipping at 125 reduces the second problem significantly.

11.5.4.2 Comparing to the Naive Baseline

Ask your LLM to add this comparison to your pipeline — this is the kind of code you should be able to read and explain back, even if you did not type it yourself:

```
 Naive baseline: predict the training set mean for every engine
y_naive = np.full_like(y_val, fill_value=y_train.mean(), dtype=float)
naive_rmse = np.sqrt(mean_squared_error(y_val, y_naive))

rf_rmse = np.sqrt(mean_squared_error(y_val, y_pred))

print(f"Random Forest  - RMSE: {rf_rmse:.2f}")
print(f"Naive baseline - RMSE: {naive_rmse:.2f}")
print(f"Improvement:  {naive_rmse / rf_rmse:.1f}× better than naive")
```

Target: **at least 2× better than the naive baseline.** If you are not hitting 2×, the most likely causes are: row-wise instead of engine-wise split (recheck the assert), missing rolling features, or constant sensors not removed.

Activity (12 min): code review, find the bugs. An LLM wrote this training script for the C-MAPSS capstone. It runs without errors. In pairs, find as many problems as you can. There are at least four. (Column names are illustrative.)

```
import pandas as pd
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error

df = pd.read_csv("train_FD001.csv")
df["s11_mean"] = df["sensor_11"].rolling(5).mean()
df = df.dropna()

X = df.drop(columns=["rul"])
y = df["rul"]
```

```
X_train, X_val, y_train, y_val = train_test_split(X, y, test_size=0.2, random_state=42)

model = RandomForestRegressor(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

print("Validation error:", mean_squared_error(y_val, model.predict(X_val)))
```

Reveal the bugs

1. **Rolling window crosses engines.** The `rolling(5)` is applied to the whole table, so the first rows of each engine are averaged with the last rows of the previous engine. It needs `groupby("engine_id")` first.
2. **Row-wise split leaks data.** `train_test_split` shuffles rows, so the same engine appears in training and validation. Split by engine ID (for example with `GroupShuffleSplit`).
3. **Identifier and time columns are features.** `X` still contains the engine ID (and probably the cycle count). The model can memorize engines, and the cycle number is almost the answer.
4. **Wrong metric.** `mean_squared_error` returns MSE in cycles squared, not RMSE in cycles. Take the square root, and compare it with the naive baseline.
5. **No baseline and no random seed discussion** (bonus): without the baseline the number has no meaning, and the dropped NaN rows hide where each engine's data starts.

The lesson: the code ran and printed a number. Only a reviewer who understands the problem catches these. This is what "stay accountable for the output" means.

You raise n_estimators from 100 to 1000 in your Random Forest. What do you expect?

- **A.** A large drop in RMSE
- **B.** Overfitting, because there are more trees
- **C.** A small change in RMSE and a longer training time
- **D.** Nothing, the parameter is ignored

Reveal after the vote

C. More trees make the average more stable but give diminishing returns. A Random Forest does not overfit more by adding trees. Most of the gain is already there by 100 or 200 trees, so you mainly pay in training time and model size.

11.5.5 Self-Check Questions — Week 4

1. Your validation RMSE is 6.8 cycles. The naive baseline RMSE on the same validation set is 48 cycles. Is this a good result or should you be

suspicious? Why?

2. You are choosing between rolling windows of k = 3, k = 5, and k = 20 for your C-MAPSS features. What would you try first, and what would you monitor to decide whether a different value is better?
3. You have 42 features after feature engineering (raw sensors + rolling mean + rolling std, with constants removed). At each split in a Random Forest tree, how many features does the algorithm consider? Why not all 42?
4. A classmate's feature importance chart shows sensor_14_rolling_mean at 0.12 and sensor_14 at 0.03. Why might the rolling feature dominate the raw feature despite measuring the same physical quantity?
5. You save `rul_model.pkl` but forget to save `feature_cols.pkl`. What happens when you try to use the model in the Streamlit app next week?

Exit ticket, Week 4. Name one thing you will check in any AI-written training script before you trust its reported accuracy.

Submit your answer on your phone before you leave.

11.5.6 Further Reading — Week 4

- scikit-learn. *RandomForestRegressor API documentation.* https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestRegressor.html
- scikit-learn. *Feature importances with a forest of trees.* https://scikit-learn.org/stable/auto_examples/ensemble/plot_forest_importances.html
- Aurélien Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow* (3rd ed., 2022). Chapter 7 covers ensemble methods with worked examples.
- Kaggle. *Feature Engineering* micro-course (free, online). https://www.kaggle.com/learn/feature-engineering
- Wikipedia. *Bootstrap aggregating (bagging).* https://en.wikipedia.org/wiki/Bootstrap_aggregating

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11.6 Week 5 — Deploying AI in a Manufacturing Context

11.6.1 Learning Objectives

	By the end of this week, you will be able to…	Assessment
5.1	Explain the ISA-95 OT/IT boundary and identify where AI inference fits in a plant network	Self-check Q1
5.2	Describe why a Jupyter notebook is not a production deployment	Self-check Q2
5.3	Design a user-facing dashboard appropriate for an operator, a maintenance planner, and a plant manager	Self-check Q3
5.4	Explain three mechanisms by which a deployed ML model degrades over time	Self-check Q4
5.5	Direct an LLM to deploy a machine learning model as a public web application using Streamlit and GitHub	Lab Week 5

Before You Read: Your Random Forest model works perfectly in your notebook. A plant manager asks you to "put it on the shop floor." What does that actually mean? Who will use it, on what device, and what happens when the model is wrong?

11.6.2 The OT/IT Boundary

Recall the ISA-95 hierarchy from Week 2. The boundary between Operational Technology (OT, Levels 0–2) and Information Technology (IT, Levels 3–4) is one of the most consequential lines in manufacturing.

Why AI models do not go on PLCs (Level 1):

- PLCs run deterministic, real-time control loops with cycle times in the millisecond range. A Python inference call that takes 50 ms would break the control loop and create safety hazards.
- PLCs are certified for functional safety under IEC 61508/61511. Adding a Python runtime would invalidate certification and create liability.

- PLCs run on embedded operating systems with no Python interpreter. Deploying there requires rewriting in IEC 61131 structured text or C.
- PLCs are maintained by OT teams with different update cadences and change management processes than IT systems.

Where AI models actually live in production:

Location	Typical role	Latency requirement
Edge server (on-premises, Level 3)	Real-time inference on process data; anomaly detection with millisecond or sub-second response	< 100 ms
Plant server / MES (on-premises, Level 3)	Batch inference (hourly or daily); maintenance scheduling recommendations	Seconds to minutes
Cloud / analytics platform (Level 4 or above)	Training, retraining, dashboards, historical analysis	Minutes to hours
OT-isolated analytics zone	Air-gapped from both OT and enterprise IT; common in defense and pharma	Varies

Edge vs. cloud tradeoffs:

Consideration	Edge (on-premises)	Cloud
Latency	Milliseconds (local)	Hundreds of milliseconds to seconds
Bandwidth	No external bandwidth consumed	Requires sending sensor data out of plant
IP/security	Data never leaves facility	Data leaves facility; security controls required
Reliability	Works during internet outage	Depends on connectivity
Cost	Hardware investment; IT overhead	Pay-per-use; lower up-front cost
Updates	Manual deployment; careful change management	Continuous deployment possible

Your capstone's Streamlit app will be deployed to the public cloud (Streamlit

Community Cloud), which is appropriate for a student project. A real plant deployment for sensitive industrial data would use an on-premises edge server or a private cloud with contractual data-protection guarantees.

11.6.3 Dashboards and Operators

A model that lives in a notebook and only an engineer can interpret is not delivering business value. The value comes when the right person gets the right information at the right time in a form they can act on.

The critical principle: the user of a predictive maintenance dashboard is rarely the engineer who built the model. Three very different users need three very different views of the same predictions:

User	What they need	Information format
Line operator	"Is this machine OK right now? What should I watch for?"	Single colour-coded status: green / yellow / red. Maximum 5 words of text.
Maintenance planner	"Which machines need attention in the next 2 weeks? In what order?"	Sortable list of assets with predicted RUL, confidence bands, recommended maintenance window
Plant manager	"What is the overall fleet health? Are we at risk of missing our production schedule?"	Aggregate fleet health index, downtime risk forecast, trend over the past 30 days

Good operator-facing dashboards:

- Never show a number without a unit and a context ("47 cycles" means nothing; "47 cycles remaining 3 weeks at current production rate" is actionable)
- Never show model confidence intervals to operators — they create uncertainty without context
- Always include a "what to do" recommendation, not just a prediction
- Always have a visible timestamp ("Last updated: 06:47")
- Never fail silently — if the data feed is down, say so explicitly rather than showing stale values

Industry Application: Toyota's production system principle of *andon* — giving any worker the authority to stop the line by pulling a cord when they detect a problem — translates directly to AI dashboards. A well-designed predictive

maintenance dashboard makes it as easy as pulling an andon cord to flag a concern for a maintenance technician. The goal is not to have the AI make the decision; it is to surface information so the human can make a better decision faster.

11.6.4 Model Drift and the Production AI Lifecycle

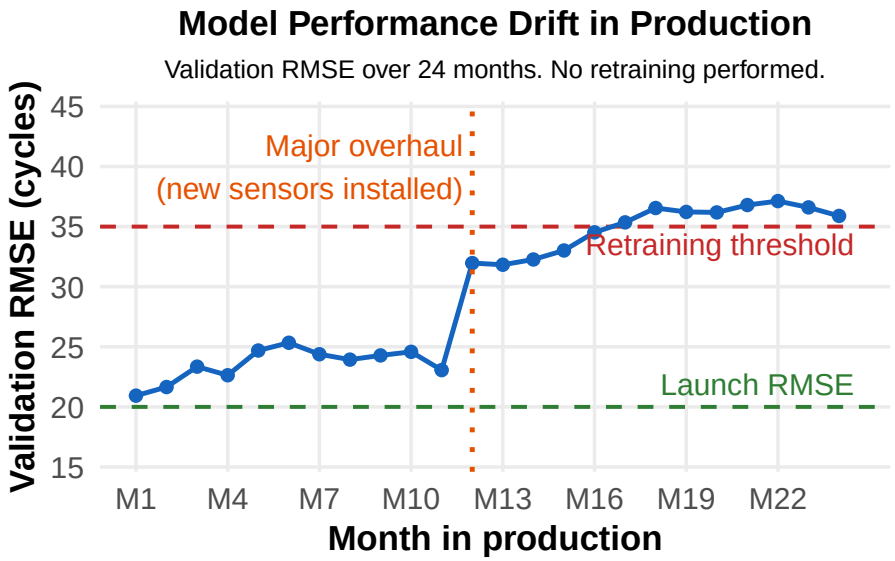


Figure 11.15: Simulated model performance degradation over time in production. Three mechanisms contribute: input drift (sensors recalibrated or replaced), concept drift (machine wear patterns changed after a major repair), and data quality drift (sensor failures increasing over time). Without monitoring and periodic retraining, a model that worked well at launch quietly degrades.

11.6.4.1 Three Mechanisms of Model Drift

1. **Input drift (covariate shift):** The distribution of input features changes over time. Sensors are recalibrated, replaced with different-accuracy units, or repositioned. The model was trained on the old distribution and now receives systematically different values. The model is not wrong — the world changed. Detect this by monitoring the mean and variance of key features over time and comparing to training-set statistics.

2. **Concept drift:** The relationship between inputs and the target variable changes. A major spindle rebuild changes the wear pattern of the engine; the degradation trajectory looks different from the training data. The

model learned the old relationship. Detect this by monitoring ground-truth RUL when you eventually get it (after planned or unplanned maintenance) and comparing to predictions.

3. **Data quality drift:** The fraction of missing or corrupted sensor readings increases over time as sensors age. The model was trained on clean data; it now receives increasingly dirty inputs. Detect this by monitoring the rate of missing values and out-of-range sensor readings in the incoming data stream.

11.6.4.2 The Production ML Lifecycle

Phase	Actions	Responsible
Training	Collect and clean historical data; engineer features; train and validate model; document performance and assumptions	Data/ML engineer
Deployment	Package model; create inference endpoint; connect to data pipeline; set up monitoring	MLOps / IT
Monitoring	Track input distributions, prediction distributions, ground-truth error when available, data quality metrics	Data engineer + domain expert
Retraining	Triggered by drift detection or scheduled interval; retrain on updated data; validate against holdout; A/B test against production model	Data/ML engineer
Retirement	When the process changes fundamentally enough that the model assumptions no longer hold; decommission cleanly; notify users	Engineering management

Retraining cadence: there is no universal answer. Considerations: - How fast does the process change? A high-volume production line changes faster

than a low-cycle heavy equipment fleet. - How expensive is labelled data? For C-MAPSS this is free (simulation). For a real plant, you only get the "true RUL" label after the equipment actually fails or is removed. - How expensive is a wrong prediction? Higher stakes require more frequent validation and lower drift tolerance.

A practical starting point for most manufacturing predictive-maintenance applications: monthly validation against recent ground-truth events, with a scheduled quarterly retraining trigger and an event-triggered retraining trigger if RMSE exceeds 2× the launch baseline.

11.6.4.3 Try it yourself: OT or IT?

Where does it live: OT or IT?

Decide whether each item belongs on the operational technology side or the information technology side.

Categories: OT (Levels 0-2) / IT (Levels 3-4)

1. Tuning a servo drive on the feeder. _____
2. The monthly job that retrains the model. _____
3. The safety interlock logic on the guard door. _____
4. The dashboard on the maintenance planner's laptop. _____
5. A PLC firmware update. _____
6. The script that validates last month's predictions. _____

Answers: 1 OT (Levels 0-2); 2 IT (Levels 3-4); 3 OT (Levels 0-2); 4 IT (Levels 3-4); 5 OT (Levels 0-2); 6 IT (Levels 3-4).

Think − pair − share (5 min). The model went live six months ago, and its validation RMSE has crept from 20 to 38 cycles. List three different causes (hint: data, equipment, people). For each, say what you would check first and who in the plant could tell you.

Operators have stopped looking at the predictive-maintenance dashboard. What is the most useful first step?

- **A.** Remind them in the next safety meeting that they must use it
- **B.** Sit with two operators for a shift and watch how they actually use (or avoid) it
- **C.** Add more charts so it looks more impressive
- **D.** Remove the dashboard and send daily emails

Reveal after the vote

B. You do not yet know why it is ignored. Common reasons are alarms that cry wolf, a screen that cannot be read from the machine, or recommendations nobody can act on. Observe first, then redesign.

11.6.5 Self-Check Questions — Week 5

1. A colleague proposes running the Random Forest inference directly on the PLC to minimize latency. What are three specific reasons this is not feasible?
2. What is the difference between "the model works in my notebook" and "the model is deployed in production"? Name three things that change when you move from notebook to production.
3. An operator sees this on their screen: "Remaining Useful Life: 47.3 ± 12.8 cycles." Is this a good interface design? What would you change and why?
4. Your model was trained in June. It is now the following February. The plant ran through a major scheduled overhaul in October where 30% of the sensors were recalibrated. What type of drift are you most concerned about, and how would you detect it?
5. Describe three monitoring metrics you would track in a production deployment of your C-MAPSS model to catch problems before they cause bad maintenance decisions.

Exit ticket, Week 5. Your model's alarm fires and the operator ignores it. Give one design reason and one organizational reason why this might happen.

Submit your answer on your phone before you leave.

11.6.6 Further Reading — Week 5

- Streamlit documentation. https://docs.streamlit.io
- Streamlit Community Cloud (free deployment). https://share.streamlit.io
- NIST. *AI Risk Management Framework (AI RMF 1.0).* https://www.nist.gov/system/files/documents/2023/01/26/NIST.AI.100-1.pdf
- MLflow documentation (model versioning and experiment tracking, free and open-source). https://mlflow.org/docs/latest/index.html
- Inductive Automation. *Ignition IIoT platform* — example of Level 3 analytics in manufacturing. https://inductiveautomation.com
- Microsoft. *Machine Learning Operations (MLOps) overview.* https://learn.microsoft.com/en-us/azure/machine-learning/concept-model-management-and-deployment

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11.7 Week 6 — Ethics, the Broader AI Landscape, and the Road Ahead

11.7.1 Learning Objectives

	By the end of this week, you will be able to…	Assessment
6.1	Identify at least three ethical risks in manufacturing AI systems and propose a mitigation for each	Self-check Q1–Q3
6.2	Explain what it means for a manufacturing AI decision to be "explainable" and when explainability is legally or contractually required	Self-check Q4
6.3	Describe the current state of AI adoption in manufacturing and identify two emerging areas likely to affect your career	Self-check Q5
6.4	Deliver a clear technical demonstration of your deployed Streamlit application to a non-specialist audience	Lab Week 6

Before You Read: Your predictive maintenance model predicts that engine #73 will fail in 15 cycles. Based on that prediction, the maintenance planner cancels a scheduled production run, incurs $40,000 in rescheduling costs, and replaces the bearing — which, when removed, shows no sign of failure. Who is responsible for the $40,000 cost?

11.7.2 AI Ethics in Manufacturing

Ethics in manufacturing AI is not abstract philosophy. It is a set of concrete questions about liability, bias, transparency, and human oversight that you will face in your first engineering role.

11.7.2.1 Bias in Training Data

Machine learning models learn patterns from historical data. If the historical data encodes biases, the model will replicate and often amplify them.

Manufacturing examples of training data bias:

- **Single-machine training:** a model trained on one lathe's historical data may perform poorly on a different lathe of the same model, because the two machines have accumulated different wear histories and been maintained by different technicians. Deploying the model plant-wide without validating on each machine introduces risk.
- **Seasonal bias:** a model trained on summer data may perform poorly in winter because ambient temperature, coolant viscosity, and thermal expansion of machine frames all differ. This is a form of concept drift baked in at training time.
- **Operator exclusion:** if the training data comes entirely from one shift (because that shift's data is in the historian), the model has not seen the setup patterns of other shifts. Predictions for equipment running under the excluded shift's operators will be systematically less accurate.

Mitigation: before training, audit your dataset for coverage. Do all machines, operators, shifts, seasons, and operating conditions appear in roughly proportional representation? If not, deliberately stratify your sampling or weight your training examples accordingly. Document what your model does and does not cover.

11.7.2.2 Liability: Who Is Responsible When AI Is Wrong?

The Air Canada chatbot case (2024, summarised in Section 10.13.2) established that companies are liable for what their AI-powered customer-service bots say. The principle extends to manufacturing: if your predictive maintenance model causes a consequential decision — stopping production, removing a part prematurely, or failing to flag an imminent failure — and that decision results in financial loss or injury, the question of liability is not trivial.

Current state of Canadian and US law: liability typically falls on the company deploying the system, not the model vendor. The engineer who deployed the system may be asked to demonstrate that the model was: - Validated appropriately before deployment - Used within its intended scope (not applied to equipment or conditions outside the training distribution) - Subject to human review for consequential decisions - Monitored for drift after deployment

This is why the "About this model" section in your Streamlit app is not optional decoration. It is a professional statement of scope and limitations. Get in the habit of writing it for every model you deploy.

11.7.2.3 Explainability

A Random Forest with 100 trees and 42 features produces predictions from millions of internal calculations. No engineer can explain any specific prediction by reading the model. This is the **black-box problem**.

For some decisions this is acceptable — if the stakes are low and the model is well-validated. For others, explainability is a hard requirement:

Context	Explainability requirement	Why
Stopping a production line	High — operator needs to understand why	$10k+/hour downtime; operator may override
Recommending part replacement	Medium — maintenance planner needs enough context to validate the recommendation	Cost of unnecessary replacement vs. cost of failure
Flagging a safety-critical component (aircraft engine, vehicle brake system)	Very high — regulator may require auditable rationale	Liability and certification requirements
Optimizing a non-critical scheduling parameter	Low	Errors are cheap to reverse

Tools for explainability with Random Forest: - **Feature importance** (built in to scikit-learn): which features contributed most, on average - **SHAP values** (library: `shap`): which features pushed a *specific* prediction up or down, and by how much — the gold standard for local explanation - **Partial dependence plots**: how the prediction changes as one feature varies while others are held constant

For your capstone, feature importance is sufficient. In a real deployment for safety-critical equipment, SHAP values would be the minimum acceptable standard.

11.7.2.4 Human Oversight and Automation Bias

> ⚠ Warning
>
> **Automation Bias:** Research consistently shows that humans become overreliant on automated recommendations over time — deferring to the machine even when their own judgment suggests the machine is wrong. For predictive maintenance systems, this can cause operators to override legitimate sensor alarms ("the model says we're fine") or schedule unnecessary maintenance ("the model said replace it").

Good AI deployment design includes: - Clear statements of what the model does and does not know - Confidence or uncertainty estimates displayed appropriately (to the right audience) - Easy mechanisms for human feedback and override - Regular review of cases where the human and the model disagreed — especially cases where the human was right

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11.7.3 The Broader AI Landscape — Where This Is Going

11.7.3.1 Current State of AI in Manufacturing (2024–2026)

What is actually in production at scale (not pilot or demo) in North American manufacturing:

- **Statistical process control software with ML-augmented anomaly detection:** AVEVA, OSIsoft (now AVEVA PI), AspenTech, GE Digital — all ship SPC with ML alarm filtering
- **Computer vision quality inspection:** Cognex, Keyence, and custom solutions using YOLO/Detectron2 — in high-volume electronics, automotive body panels, food sorting
- **Predictive maintenance on rotating equipment:** vibration FFT analysis + ML on PLCs or edge servers — most common in paper, cement, oil refining, and heavy automotive
- **LLM-assisted documentation:** work order generation, SOP drafting, supplier emails — on the rise since 2023; most common at the engineering and supervision level, not on the shop floor

What is in rapid development / early deployment: - **Digital twins with ML feedback:** real-time simulation of production processes updated by sensor data — very capital-intensive but accelerating - **Autonomous quality control with closed-loop feedback:** the system detects a defect, diagnoses the probable cause, and sends a corrective offset directly to the CNC controller — no human in the loop for the adjustment - **Multimodal LLMs for maintenance technicians:** technician describes a symptom in natural language, uploads a photo of the machine, and receives a structured troubleshooting checklist from a model fine-tuned on OEM maintenance manuals

11.7.3.2 Technologies Likely to Affect Your Career in the Next 5–10 Years

1. **Foundation models for industrial time-series:** equivalent to GPT-4 but trained on sensor data rather than text — a single pre-trained model fine-tuned on your specific plant data rather than trained from scratch. Early examples (Amazon Chronos, Google TimesFM) are already available.

2. **Simulation-accelerated training:** ML models trained primarily on physics-based simulation rather than historical operational data — solving the "I don't have enough run-to-failure data" problem for new equipment. NASA C-MAPSS is an early example of this approach.

3. **Agentic AI for engineering workflows:** LLM-based agents (Section 10.8.1) that take multi-step actions (search a manual, query a

database, generate a work order, route it to the right technician) rather than just responding to a single question. Coding agents are already in routine use, which is how you wrote your Python in this course, and agent features are being added to enterprise ERP and MES platforms. Multi-agent systems and standard connectors (MCP) make it easy to link them to plant data. The engineering questions are the same ones you have studied: what is the agent allowed to do, who reviews its work, how do you test it, and who is accountable when it is wrong. Expect a licensed engineer's judgment to be asked for at exactly those points.

4. **AI governance and certification requirements:** as AI systems make more consequential decisions, regulatory frameworks are evolving (EU AI Act, NIST AI RMF, ISO 42001). Engineers with both the technical skills and the regulatory literacy to certify AI systems for safety-critical applications will be in high demand.

11.7.3.3 Keeping Your Skills Current

The field moves fast. Three habits that will keep you current without burning out:

1. **One paper per month:** the PHM Society proceedings, arXiv (cs.LG for ML, eess.SP for signal processing), and the AAAI workshop on AI for manufacturing are free and publish the frontline work. You do not need to understand every paper — scan abstracts, read one or two per month deeply.
2. **One project per year:** applying a new technique to a real problem (even a toy project) converts abstract knowledge into practical skill faster than any course. The capstone you just completed is the template.
3. **Stay close to the domain:** the engineers who will have the most impact with AI are those who deeply understand both the technology and the manufacturing process. The pure ML specialists increasingly come from software companies; the competitive advantage of a manufacturing engineer is the physical intuition about what the data actually means.

Activity (15 min): four corners, who pays the $40,000? Return to the opening question of this week: the model predicted failure in 15 cycles, the planner cancelled a production run at a cost of $40,000, and the bearing turned out to be healthy. Your plant (the *user company*) bought the model from a software vendor and the engineer deployed it. Who should bear the $40,000 loss?

1. Move to the corner for your position: **(A)** the maintenance planner who acted on the prediction, **(B)** the engineer who deployed the model, **(C)** the model vendor company that sold the software, **(D)** the user company

(your plant) itself; the loss is a cost of doing business with predictive models.
2. In your corner, agree on the *two strongest reasons* for your position.
3. One speaker per corner gives the argument. Then anyone may change corners, and has to say what changed their mind.

Debrief: what documentation, in the model's "About this model" section and in the plant's procedures, would have made your corner's argument stronger?

For which of these is explainability (knowing why the model said it) a hard requirement?

- **A.** Optimising a non-critical scheduling parameter
- **B.** Flagging a safety-critical aircraft component for removal
- **C.** Ranking sensors for a future instrumentation upgrade
- **D.** Suggesting a coffee-break schedule

Reveal after the vote

B. When the decision is safety-critical or regulated, an auditable rationale is required. The others are low stakes and cheap to reverse.

Think – pair – share (4 min). You have now directed an LLM to write the Python for a real model. One thing the AI could not have done for you in this course was _____. Complete the sentence in two different ways, and explain what that means for the skills you want to keep building.

11.7.4 Self-Check Questions — Week 6

1. Your C-MAPSS model was trained exclusively on FD001 (one operating condition, one fault mode). A plant manager asks you to deploy it on the hydraulic press line to predict cylinder seal failure. List three specific reasons why this deployment would be inappropriate, and what you would need to do before it could be considered.
2. The Air Canada chatbot case established that companies are liable for what their AI tools say. What does this imply for the "About this model" section of your Streamlit app — specifically what information should it contain to protect both you and your future employer?
3. A junior technician has been using your predictive maintenance dashboard for six months. You notice they never override the recommendations, even when they personally notice an unusual sound from a machine the model shows as "healthy." What phenomenon does this illustrate, and how would you redesign the dashboard to mitigate it?
4. Your plant uses a Random Forest for bearing failure prediction on the stamping press. A failure occurs that the model did not predict (a false negative). The quality team asks you to explain why the model missed it. What tools would you use to investigate, and what information would you present?

5. Name two AI technologies not yet widely deployed in manufacturing (as of this course) that you think are most likely to change your specific engineering role within 10 years. Justify your answer.

Exit ticket, Week 6. Write one sentence you would say to a future manager who wants to "put AI on the shop floor tomorrow". What is the first question you ask?

Submit your answer on your phone before you leave.

11.7.5 Further Reading — Week 6

- NIST. *AI Risk Management Framework (AI RMF 1.0).* https://www.nist.gov/system/files/documents/2023/01/26/NIST.AI.100-1.pdf
- European Commission. *EU AI Act — Summary for practitioners.* https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32024R1689
- ISO/IEC 42001:2023. *Artificial intelligence — Management system.* (Standard, not free, but your employer may have access.)
- Lundberg, S. and Lee, S.-I. "A Unified Approach to Interpreting Model Predictions," *NeurIPS 2017.* https://arxiv.org/abs/1705.07874 (the SHAP paper — foundational for model explainability)
- Crawford, K. *Atlas of AI* (2021). A readable, critical examination of AI's social and political dimensions — highly recommended for its grounding perspective.
- Amazon Chronos (foundation model for time-series). https://github.com/amazon-science/chronos-forecasting
- PHM Society. Annual conference proceedings (free). https://www.phmsociety.org/publications/proceedings

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11.8 Key Takeaways

Across six weeks, three themes have recurred:

1. **Use the simplest tool that solves your problem.** An LLM that hallucinates ASTM standards is the wrong tool for verifying a specification. A Random Forest that you can interrogate with feature importance is often better than a neural network you cannot explain, even if the neural network is slightly more accurate.

2. **Data is the bottleneck, not the algorithm.** You spent more time in Week 2 cleaning 90 days of CNC data than you spent in Week 4 training the model. This ratio is typical of real industrial work. The engineers who understand the data — its provenance, its failure modes, its physical meaning — are the engineers who build systems that work in production.

3. **AI tools augment engineer judgment; they do not replace it.** An LLM that drafts your SOP still needs you to check it. A predictive maintenance model that says "15 cycles remaining" still needs a maintenance planner who understands what 15 cycles means in terms of the production schedule, the available technicians, and the cost of being wrong. You were never expected to become a Python programmer in six weeks — you were expected to learn to direct an LLM to write, explain, and debug code on your behalf, so you could automate the routine parts of this project and produce a working predictive-maintenance system that would otherwise have been well above your current level. The value of this course is not in the Python code — it is in the engineering judgment about when to trust the AI-generated output and when to push back.

Chapter 12

Machine Guarding and Functional Safety

12.1 Learning Objectives

By the end of this chapter, you will be able to:

- Apply the Hierarchy of Controls to manufacturing safety design
- Identify and specify appropriate physical guarding and interlock systems
- Select and position presence-sensing devices (light curtains, area scanners)
- Calculate safety distances for protective device placement
- Understand collaborative robot safety requirements
- Design emergency stop systems according to standards
- Differentiate between safety integrity levels (SIL) and performance levels (PL)

"Safety isn't expensive, it's priceless." — Anonymous

12.2 Introduction: The Safety Hierarchy

In process engineering, safety is not merely a feature added at the end of a project; it is a **fundamental design constraint**. Before deploying safety devices, the engineer must apply the **Hierarchy of Controls** — a systematic approach to eliminating or reducing workplace hazards.

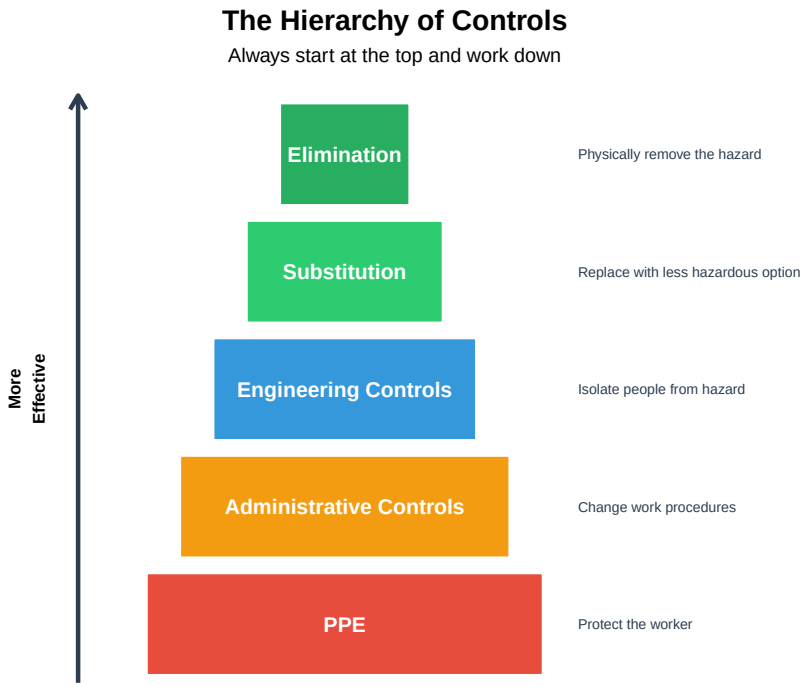


Figure 12.1: The Hierarchy of Controls (Most to Least Effective)

Table 12.1: Hierarchy of Controls: Detailed

Priority	Control Level	Description	Examp
1	**Elimination**	Physically remove the hazard entirely	Elimina
2	**Substitution**	Replace with something less hazardous	Use low
3	**Engineering Controls**	Isolate people from the hazard through design	Install
4	**Administrative Controls**	Change the way people work	Safety p
5	**PPE**	Protect the worker with equipment	Safety g

Discussion: Why not just rely on PPE?

PPE (Personal Protective Equipment) is the least effective control because:

1. **It doesn't eliminate the hazard** — the danger still exists
2. **It depends on human compliance** — workers may forget or choose not to wear it
3. **It can fail** — equipment degrades, doesn't fit properly, or is damaged
4. **It's the last line of defense** — if PPE fails, the worker is exposed

Key Principle: The more we can design out hazards through engineering controls, the less we depend on human behavior to keep people safe.

> "If you have to post a warning sign, you've already failed at design."

12.2.1 Try it yourself: rank the controls

The Hierarchy of Controls ranks the ways of dealing with a hazard. Put the five levels in order, most effective first.

Hierarchy of Controls, most to least effective

Number these in the correct order:

- _____ Administrative Controls
- _____ Elimination
- _____ Engineering Controls
- _____ PPE
- _____ Substitution

Answers, in the order listed: 4, 1, 3, 5, 2.

12.3 Physical Guarding and Interlocks

Physical barriers are the most intuitive form of protection. They create a "hard" separation between the operator and moving parts, ensuring that humans cannot enter the danger zone during machine operation.

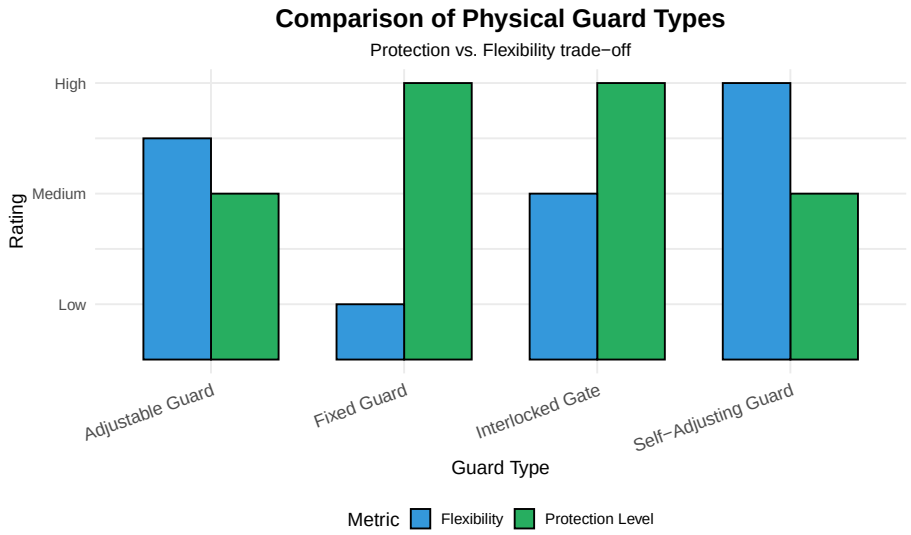


Figure 12.2: Types of Physical Guarding

12.3.1 Fixed Guards

Fixed guards are permanent barriers that require tools to remove. They are the simplest and most reliable form of protection.

Table 12.2: Fixed Guards: Key Characteristics

Characteristic	Description
Construction	Wire mesh, polycarbonate sheets, or solid metal panels
Removal	Requires tools (screws, bolts) to remove
Best Application	Areas that need infrequent access for maintenance only
Advantages	Simple, reliable, low cost, no moving parts to fail
Disadvantages	Impedes visibility, slows maintenance, can be removed and not replaced

12.3.2 Interlocked Guards

Interlocked guards are access points equipped with sensors. When the guard is opened, the safety controller initiates a stop.

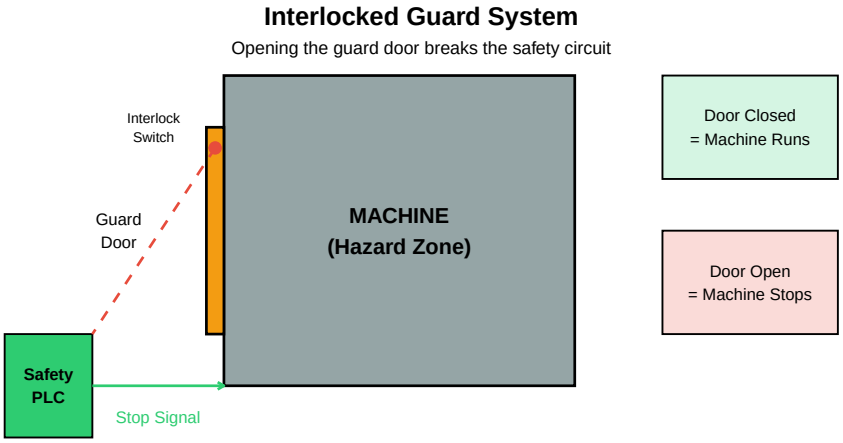


Figure 12.3: Interlocked Guard Operation Principle

12.3.3 Stop Categories

When an interlock is triggered, the machine must stop. The **stop category** determines how this happens:

Table 12.3: Machine Stop Categories (IEC 60204-1)

Category	Type	Method	Application
Category 0	Uncontrolled Stop	Immediate removal of power to actuators	Emergency stops, danger
Category 1	Controlled Stop	Controlled deceleration, then power removed	High-inertia machines, ro
Category 2	Controlled Stop	Controlled stop with power maintained	Process that cannot be ir

12.3.4 Trapped Key Systems (Lock-Out/Tag-Out)

Trapped key systems use a mechanical sequence to ensure that a machine cannot be energized unless all access keys are returned to the control panel.

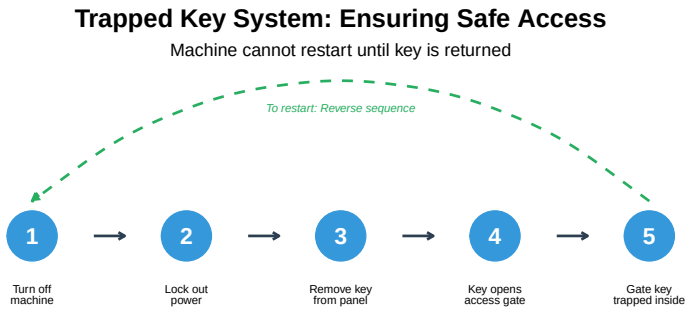


Figure 12.4: Trapped Key System Sequence

Case Study: The Fatal Bypass

Incident: A maintenance worker was killed when a robot arm unexpectedly activated while he was inside a robotic cell.

What Happened: - The guard interlock had been bypassed with a "defeat device" (a magnet taped to fool the sensor) - Workers had bypassed the interlock because it slowed down troubleshooting - The bypass had been in place for months without management knowledge

Root Causes: 1. Interlock caused inconvenience during frequent troubleshooting 2. No monitoring system to detect bypasses 3. Culture that tolerated "workarounds"

Lessons Learned: - Design interlocks that don't impede normal operations - Use tamper-evident or tamper-resistant devices - Regular safety audits to detect bypasses - Zero-tolerance policy for defeating safety devices

> **Never bypass a safety device, even temporarily.**

12.4 Presence-Sensing Devices

Presence-sensing devices allow for a **"fence-less" environment**, improving visibility and ergonomics while maintaining high safety levels. They detect when a person enters a hazardous zone and trigger a machine stop.

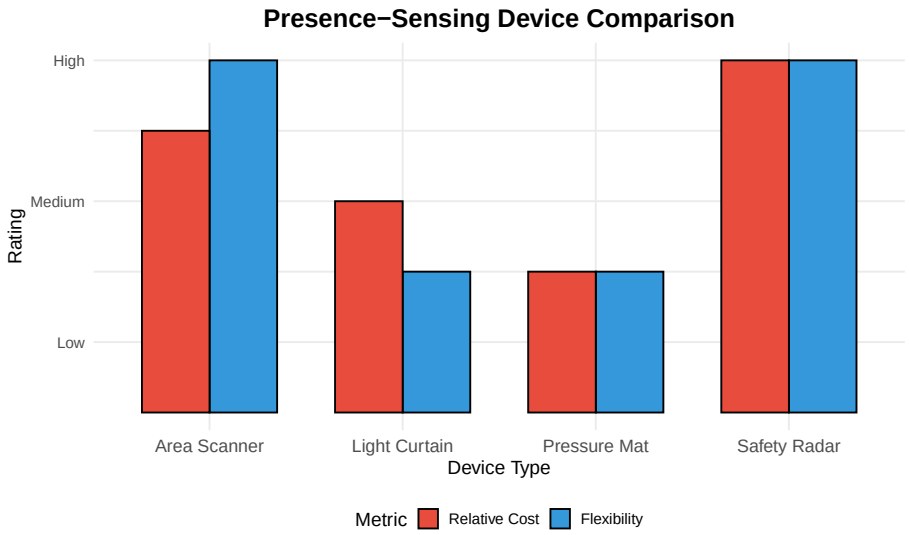


Figure 12.5: Types of Presence-Sensing Devices

12.4.1 Light Curtains

Light curtains use a transmitter and receiver to create an array of infrared beams. If any beam is broken, the safety circuit is interrupted.

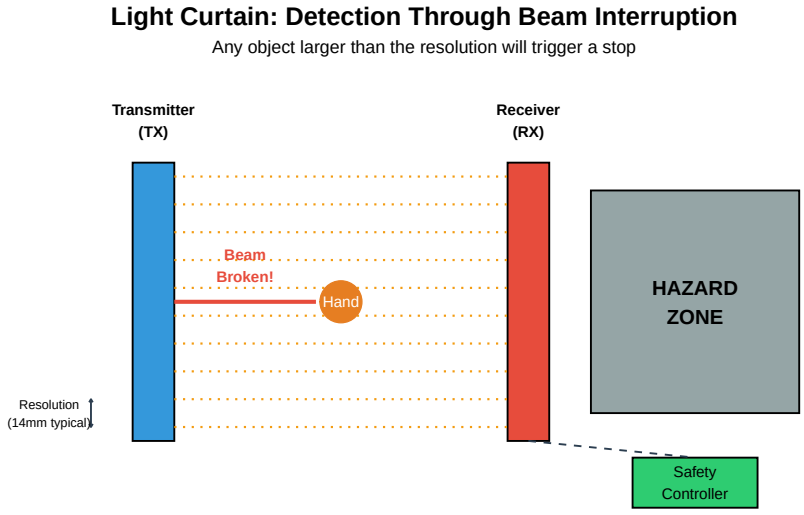


Figure 12.6: Light Curtain Operating Principle

Table 12.4: Light Curtain Specifications by Application

Parameter	Finger Detection (14mm)	Hand Detection (30mm)	Body Detec
Resolution	14mm beam spacing	30mm beam spacing	40mm+ beam
Height	150-1800mm	300-1800mm	900-2000mm
Range	0.5-20m	0.5-20m	0.5-30m
Response Time	5-15ms	10-20ms	15-30ms
Application	Point of operation guarding	Perimeter guarding	Area access d

12.4.2 Laser Area Scanners

Laser area scanners use LiDAR technology to monitor a 2D plane. They are highly flexible and allow for programmable zones.

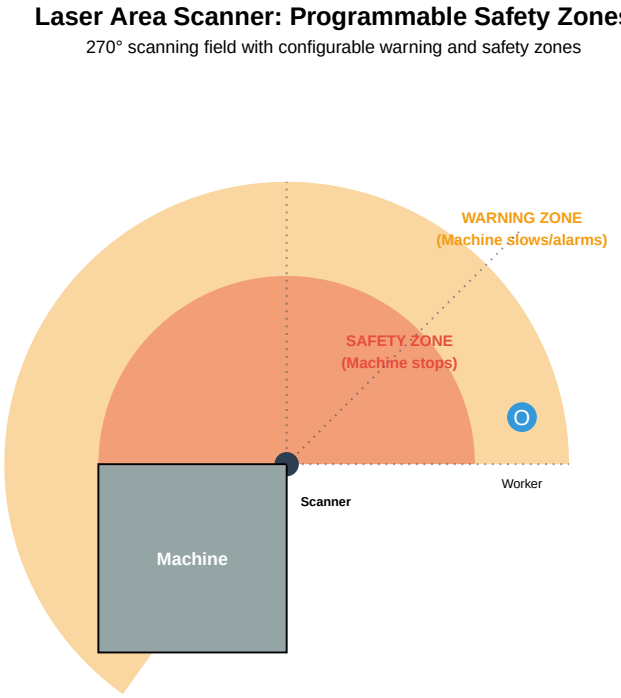


Figure 12.7: Laser Area Scanner Zones

Table 12.5: Light Curtains vs. Area Scanners

Feature	Light Curtains	Area Scanners

Detection Type	Vertical/Linear Barrier	Horizontal/Area Coverage
Flexibility	Fixed installation	Programmable zones
Zone Configuration	Single detection plane	Multiple warning + safety zones
Cost	Generally lower ($1,000-5,000)	Higher ($3,000-10,000)
Best Use Case	Clear entry/exit points, press brakes	Large open floors, AGV paths, flexible cells
Environmental Limits	Dust/debris can cause false trips	More tolerant of dust, outdoor rated availabl

12.4.3 Pressure-Sensitive Safety Mats

Pressure-sensitive mats detect the presence of a person standing on them. They are used primarily in legacy systems or specific zones where optical sensors may be obscured.

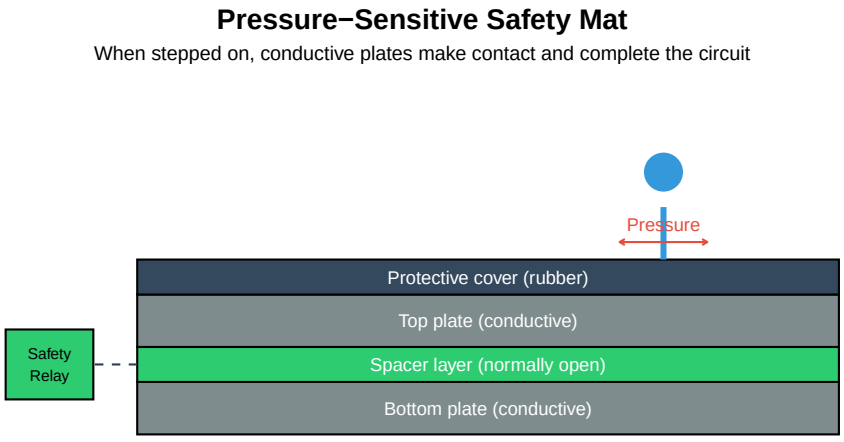


Figure 12.8: Pressure-Sensitive Safety Mat Construction

When to Use Pressure Mats vs. Optical Sensors

Use Pressure Mats When: - Environment has heavy dust, smoke, or steam (blinds optical sensors) - Detection area is well-defined and doesn't change - Workers need to stand in a specific location - Budget is limited

Use Optical Sensors When: - Flexibility in zone configuration is needed - Environment is relatively clean - Detection area is large or needs to change - Higher integrity level is required

Hybrid Approach: Many modern systems use both — area scanners for perimeter protection and pressure mats for specific danger zones near the machine.

12.5 Control-Actuated Safety Devices

These devices ensure that the operator's body—specifically their hands—is in a predetermined **safe location** before the machine can cycle.

12.5.1 Two-Hand Controls

Two-hand controls are common in pressing or stamping operations. To initiate a cycle, the operator must press two buttons simultaneously.

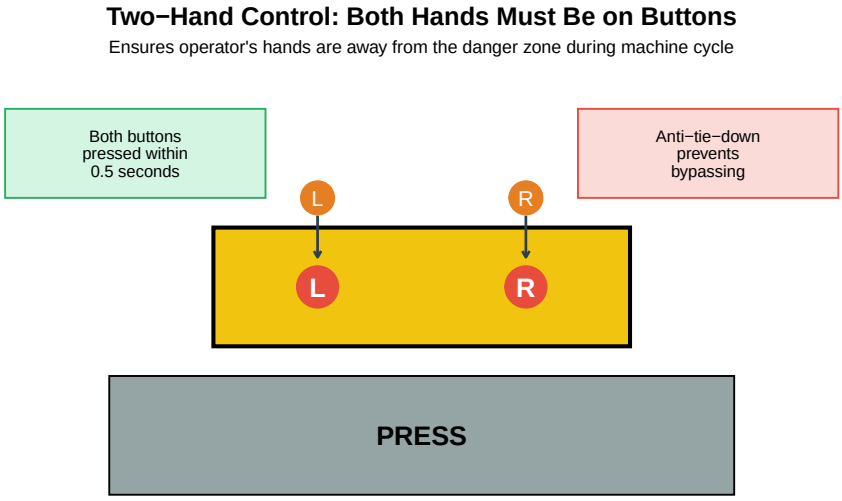


Figure 12.9: Two-Hand Control System

Anti-Tie-Down Requirement: Modern systems require both buttons to be pressed within **0.5 seconds** of each other. This prevents operators from taping one button down to work with one hand.

Table 12.6: Two-Hand Control Requirements

Requirement	Standard Value	Purpose
Synchronous activation	Within 0.5 seconds	Prevents one-hand operation
Continuous pressure	Must be held throughout cycle	Ensures hands stay on contro
Release before restart	Both must be released before next cycle	Prevents automatic cycling
Button spacing	Minimum 260mm (10 inches) apart	Cannot press both with one h
Guards around buttons	Prevent accidental activation	Prevents hip, elbow, or objec

12.5.2 Enable/Deadman Switches

Enable switches (also called deadman switches) are used during **Teach Mode** for robotics. The operator must hold a three-position trigger in the **middle** position.

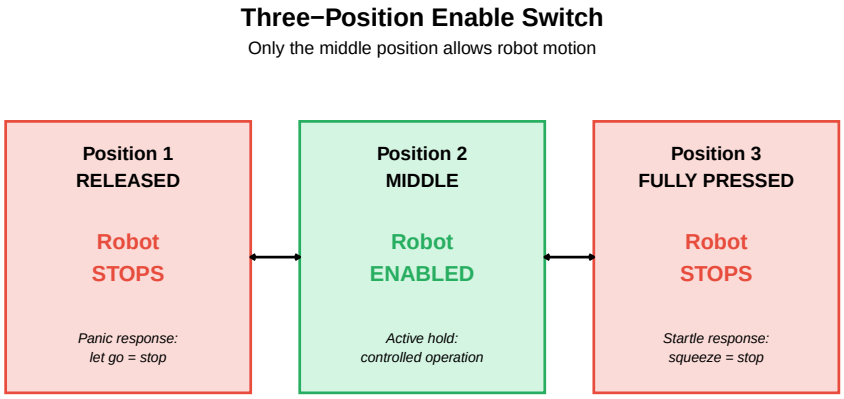


Figure 12.10: Three-Position Enable Switch Operation

Why Three Positions?

The three-position design accounts for **two natural human reactions** to danger:

1. **Panic (Release):** When frightened, people often release what they're holding — robot stops.
2. **Startle (Squeeze):** When startled, people often grip tighter — robot stops.

Only a deliberate, controlled **middle hold** allows the robot to move. This is fundamental to safe teach pendant operation.

12.6 Try it yourself: choose the safeguard

Eight machine situations. Decide which safeguard from the last three sections fits each one best. Ask how often someone needs access, and whether the hazard can be stopped quickly.

Which safeguard?

Pick the safeguard that fits each situation best.

Categories: Fixed guard / Interlocked guard / Light curtain / Two-hand control

1. A belt-and-pulley drive that is only opened for maintenance twice a year. _____
2. The rotating coupling between a motor and a gearbox. _____
3. A CNC lathe door that the operator opens every cycle to change the part. _____
4. An injection molding machine gate that is opened to clear a stuck part. _____
5. A robot cell with a wide opening where pallets are loaded by forklift. _____
6. A press brake where the operator hand-feeds sheet into the die area all shift. _____
7. A small manual press where the operator loads a part and then starts the stroke. _____
8. A riveting station started by one operator whose hands must be clear of the tooling. _____

Answers: 1 Fixed guard; 2 Fixed guard; 3 Interlocked guard; 4 Interlocked guard; 5 Light curtain; 6 Light curtain; 7 Two-hand control; 8 Two-hand control.

12.7 Collaborative Robotics (Cobots)

Collaborative robots (cobots) represent a shift from “separation” to “collaboration.” They rely on sophisticated internal sensors rather than external cages, allowing humans and robots to work in the same space.

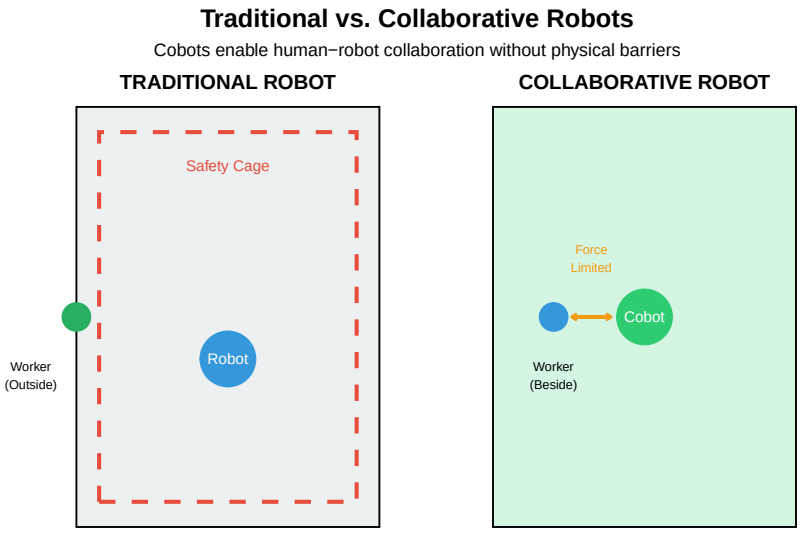


Figure 12.11: Traditional Robot vs. Collaborative Robot

12.7.1 Collaborative Operation Methods

ISO 10218 and ISO/TS 15066 define **four methods** for safe collaborative operation:

Table 12.7: Four Methods of Collaborative Robot Opera

Method	How It Works	Sensors U
****Safety-Rated Monitored Stop****	Robot stops when human enters workspace, resumes when clear	Area scann
****Hand Guiding****	Operator physically guides robot through motions	Force/torq
****Speed and Separation Monitoring****	Robot slows or stops based on human proximity	Area scann
****Power and Force Limiting****	Robot limits force/power to safe levels on contact	Joint torqu

12.7.2 Power and Force Limiting (PFL)

The most common cobot safety method is **Power and Force Limiting**. The robot detects a change in torque in its joints. If it bumps into a human, it stops before a regulated amount of force is exceeded.

Table 12.8: ISO/TS 15066 Biomechanical Limits (Quasi-static contact)

Body Region	Maximum Pressure (N/cm²)	Maximum Force (N)	Risk Lev
Skull/Forehead	130	130	Critical
Face	65	65	Critical
Neck	145	150	Critical
Back/Shoulders	210	210	Moderate
Chest	140	140	Moderate
Abdomen	110	110	Moderate
Hand/Fingers	300	140	Lower

Discussion: Are Cobots Always Safe?

Common Misconception: "Cobots are safe, so we don't need to do a risk assessment."

Reality: Cobots are *potentially* safe, but a **risk assessment is always required**.

Factors that can make a cobot unsafe:

1. **End-of-arm tooling** — A soft gripper is different from a sharp blade
2. **Workpiece** — Carrying a heavy or sharp object increases risk
3. **Speed** — Even force-limited robots can cause injury at high speed
4. **Application** — Some tasks bring humans closer to hazards
5. **Environment** — Slippery floors, confined spaces, etc.

Every collaborative application requires a documented risk assessment per ISO 12100.

12.8 Safety Distance Calculation

A safety device cannot be placed arbitrarily. It must be far enough away that the machine comes to a **complete halt** before the human can reach the hazard.

12.8.1 The Safety Distance Formula

The minimum safety distance S is calculated as:

$$S = (K \times T) + C$$

Where:

- K: The hand/body approach speed (standardized values)
- T: The total stopping time of the system (device response + machine stop time)

- C: The "penetration depth" (how far a hand can reach through before being detected)

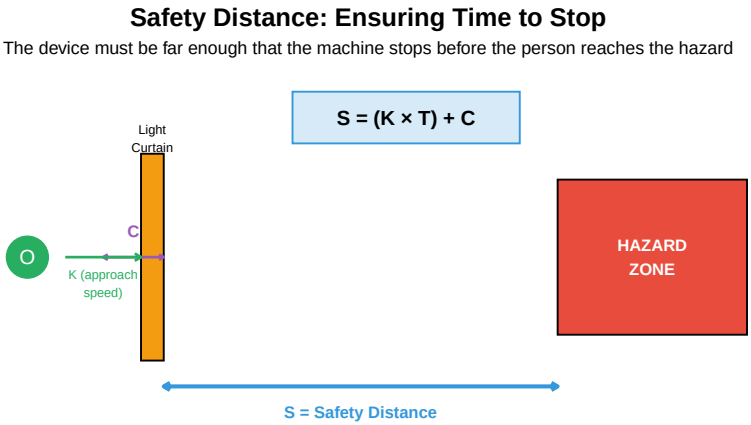


Figure 12.12: Safety Distance Components

12.8.2 Standard Values for K

Table 12.9: Standard Approach Speed Values (ISO 13855)

Approach Type	K Value (mm/s)	When to Use
Hand/Arm Approach (fast)	2000	Close approach to point of operation
Hand/Arm Approach (normal)	1600	General approach to machines
Walking Approach	1600	Area access detection
Standing Reach	0 (static)	Worker already in position

12.8.3 Penetration Depth (C)

The penetration depth depends on the **resolution** of the sensing device:

Table 12.10: Penetration Depth Values Based on Resolution

Resolution (mm)	C Value (mm)	Detection Type
14	0	Finger detection
>14 to 20	80	Finger/Hand
>20 to 30	130	Hand detection
>30 to 40	240	Hand/Arm

>40	850	Body detection

12.8.4 Example Calculation

Your turn — change the approach speed, response time, or resolution below.

Show the live R code behind this calculator

```
#| label: safety-calc-example
# Given parameters
K <- 2000          # Approach speed (mm/s) - fast hand approach
T_device <- 0.020  # Light curtain response time (20 ms = 0.020 s)
T_machine <- 0.150 # Machine stopping time (150 ms = 0.150 s)
C <- 0             # Penetration depth for 14mm resolution

# Calculate total stopping time
T_total <- T_device + T_machine

# Calculate minimum safety distance
S <- (K * T_total) + C

cat(sprintf("Minimum safety distance: %.0f mm (%.1f in)\n", S, S / 25.4))
cat(sprintf("  Total stopping time  %.0f ms\n", T_total * 1000))
```

Practice Problem: Calculate Safety Distance

Given: - Light curtain resolution: 30mm (hand detection) - Light curtain response time: 15ms - Machine stopping time: 200ms - Application: Operator hand approach

Calculate the minimum safety distance.

Solution:

```
K = 1600 mm/s (normal hand approach)
T = 0.015 + 0.200 = 0.215 seconds
C = 130 mm (for 30mm resolution)

S = (K × T) + C
S = (1600 × 0.215) + 130
S = 344 + 130
S = 474 mm (approximately 18.7 inches)
```

The light curtain must be installed at least **474mm** from the nearest hazard point.

12.9 Emergency Stop (E-Stop) Systems

The **Emergency Stop** is the "last resort" — it is not a safety device but a **complementary protective measure**. It must be available to stop the machine when all other measures have failed.

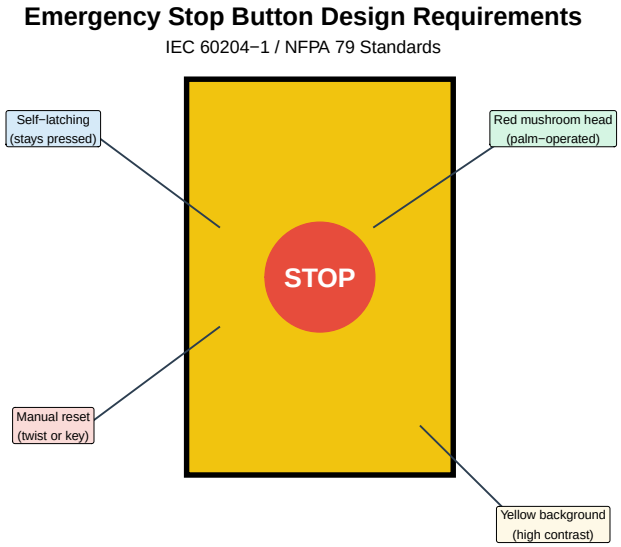


Figure 12.13: E-Stop Design Requirements

Table 12.11: E-Stop Requirements per IEC 60204-1

Requirement	Standard
Color	Red (RAL 3000 or equivalent)
Shape	Mushroom head (palm or fist operated)
Background	Yellow (high contrast)
Operation	Self-latching (stays activated when pressed)
Reset	Manual reset required (twist, pull, or key)
Accessibility	Within easy reach of all operators
Wiring	Normally closed contacts (fail-safe)
Function	Category 0 or Category 1 stop

12.9.1 E-Stop Pull Cords

For long conveyors or machines where an operator cannot quickly reach a fixed button, **pull cords** provide continuous E-Stop coverage.

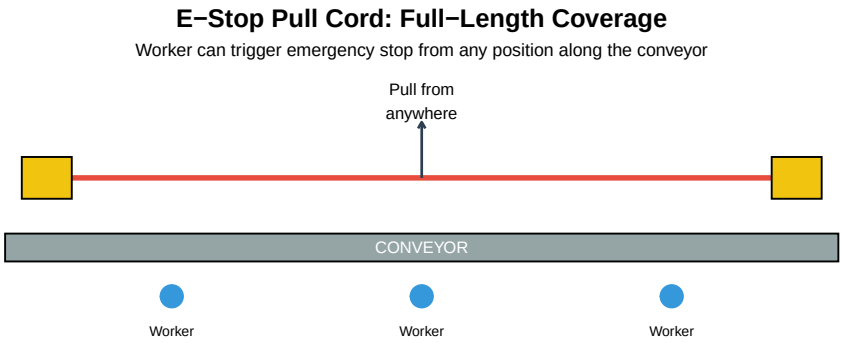


Figure 12.14: E-Stop Pull Cord System

Common E-Stop Mistakes

Mistake 1: Using E-Stop for normal stopping - E-Stops are for emergencies only - Normal stops should use a separate control - Frequent E-Stop use causes wear and desensitizes operators

Mistake 2: E-Stop that doesn't stop everything - All hazardous motion must stop - Auxiliary equipment (conveyors, feeders) often forgotten - Review entire system, not just the primary machine

Mistake 3: Automatic restart after E-Stop reset - Machine must NOT automatically restart when E-Stop is released - Separate "Start" action required - This prevents unexpected motion after reset

Mistake 4: Hidden or blocked E-Stops - E-Stops must be visible and accessible - Guard placement sometimes blocks access - Regular audits needed to ensure accessibility

---

12.10 Functional Safety Standards

Modern safety systems must meet specific **performance levels** to ensure reliability. Two main frameworks exist:

12.10.1 Safety Integrity Levels (SIL) - IEC 62061

Table 12.12: Safety Integrity Levels (PFH = Probability of Dangerous Failure per Hour)

SIL Level	PFH Range	Approximate Meaning	Typical Applicatio
SIL 1	10 to <10	Dangerous failure less than once per 11 years	Low-risk tasks

SIL 2	10 to <10	Dangerous failure less than once per 114 years	Most industrial machinery
SIL 3	10 to <10	Dangerous failure less than once per 1,140 years	High-risk processes

12.10.2 Performance Levels (PL) - ISO 13849-1

Table 12.13: Performance Levels per ISO 13849-1

Performance Level	PFH Range	Risk Reduction	Equivalent SIL
PL a	10 to <10	Lowest	< SIL 1
PL b	3×10 to <10	Low	SIL 1
PL c	10 to <3×10	Medium	SIL 1
PL d	10 to <10	High	SIL 2
PL e	10 to <10	Highest	SIL 3

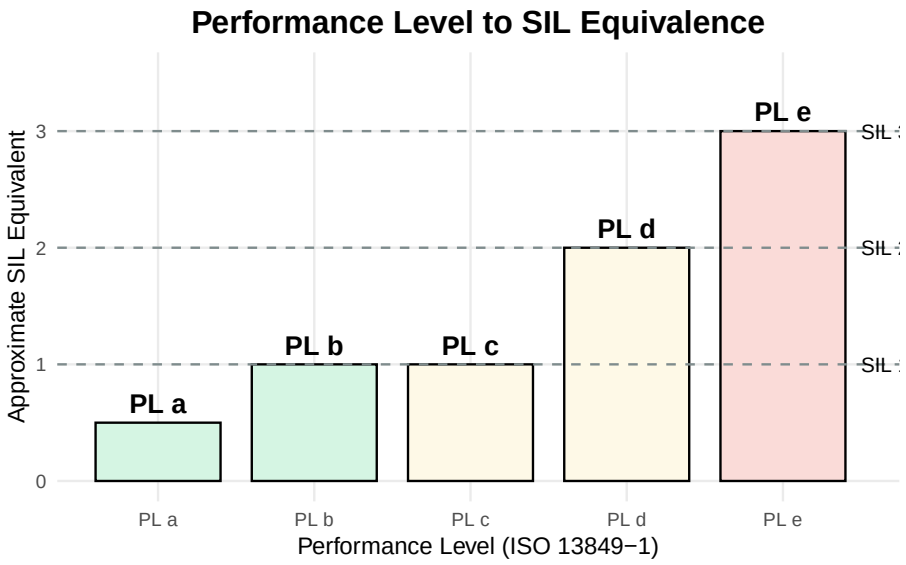


Figure 12.15: Relationship Between SIL and PL

Which Standard to Use: SIL or PL?

Use ISO 13849-1 (Performance Levels) when: - Designing safety functions for standard machinery - Using off-the-shelf safety components - Simpler systems with well-defined safety functions

Use IEC 62061 (SIL) when: - Complex programmable systems (safety PLCs) - Process industries - When customer or regulation specifies SIL

Note: Many modern systems use both standards, as they are harmonized and can be used together. The choice often depends on industry practice and customer requirements.

12.11 Risk Assessment Process

Before selecting any safety device, a **risk assessment** must be performed. This is required by ISO 12100.

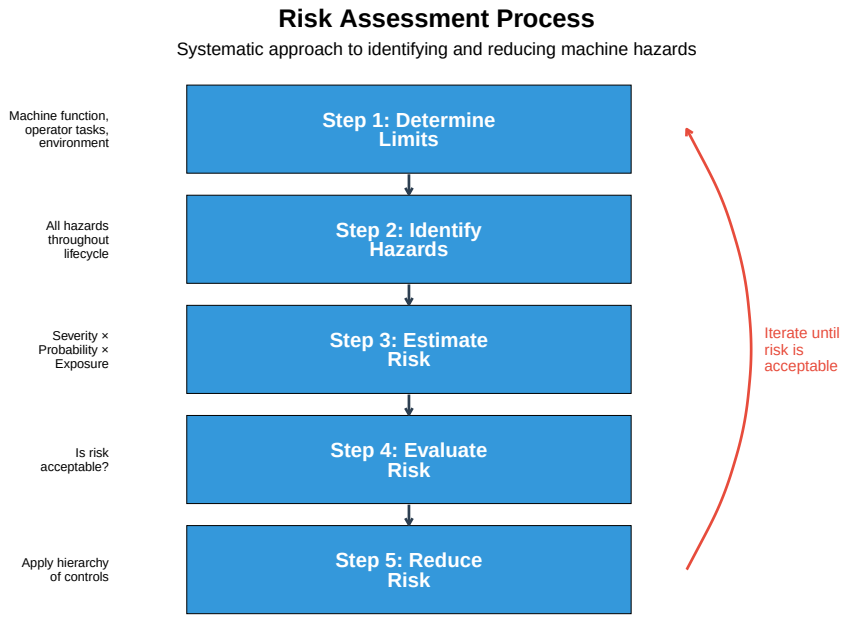


Figure 12.16: Risk Assessment Process (ISO 12100)

12.11.1 Risk Estimation

Risk is typically estimated using three factors:

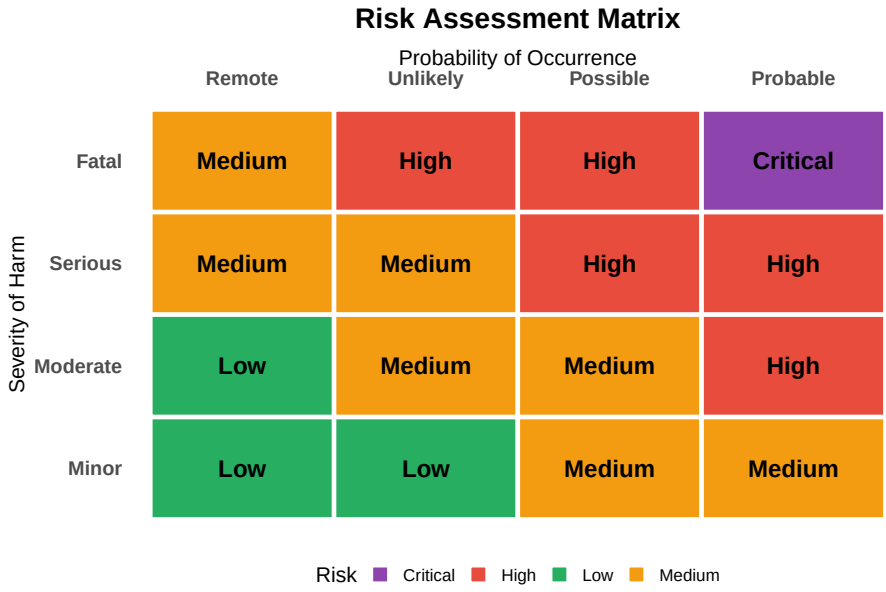


Figure 12.17: Risk Estimation Matrix

12.11.2 Try it yourself: find the required performance level

Risk estimation leads to a number you can design to. ISO 13849-1 turns three judgements about a hazard into a **required performance level (PLr)** for the safety function that guards it:

Question	Lower	Higher
S — how severe is the injury?	S1: slight, normally reversible	S2: serious, normally irreversible, or death
F — how often or how long is someone exposed?	F1: seldom, or for a short time	F2: frequent, or for a long time
P — can the person avoid the harm?	P1: possible under specific conditions	P2: scarcely possible

S	F	P	PLr
S1	F1	P1	**a**
S1	F1	P2	**b**

S	F	P	PLr
S1	F2	P1	**b**
S1	F2	P2	**c**
S2	F1	P1	**c**
S2	F1	P2	**d**
S2	F2	P1	**d**
S2	F2	P2	**e**

For each hazard below, decide S, F and P from the description, then read off the PLr.

What performance level is required?

Decide S, F and P for each hazard, then pick the required performance level.

Categories: PL a / PL b / PL c / PL d / PL e

1. A slow transfer conveyor can pinch a hand and leave a bruise. Someone reaches near it about once a week, and it moves slowly enough to pull away from. _____
2. A heated platen can cause a minor burn. The operator works beside it all shift, and it is clearly marked and easy to keep clear of. _____
3. A trimming blade can cause small cuts. The operator handles parts at it all shift, and the blade is hidden behind the part, so contact cannot be anticipated. _____
4. A robot can crush a person against the fence. A technician enters about once a month for teaching, with the robot at reduced speed and an enabling switch in hand. _____
5. A hydraulic press can amputate a hand. The die area is entered twice a week for a die change, and the ram closes too fast to react to. _____
6. A slow forming press can crush a hand. The operator loads it by hand every cycle, and the ram closes slowly enough for a trained operator to withdraw. _____
7. A mechanical power press can amputate a hand. It is hand-fed every cycle, and the stroke takes a fraction of a second. _____

Answers: 1 PL a; 2 PL b; 3 PL c; 4 PL c; 5 PL d; 6 PL d; 7 PL e.

12.12 Summary

Table 12.14: Machine Guarding and Safety: Key Conce

Topic	Key Points	Critic
Hierarchy of Controls	Elimination > Substitution > Engineering > Administrative > PPE	Alway
Physical Guarding	Fixed guards, interlocked gates, trapped key systems	Stop c
Presence Sensing	Light curtains, area scanners, pressure mats	Resolu
Control-Actuated Devices	Two-hand controls, enable/deadman switches	Anti-t
Collaborative Robots	PFL, speed/separation monitoring, hand guiding	Risk a
Safety Distance	S = (K × T) + C formula for device placement	Includ
E-Stop Systems	Red mushroom on yellow, self-latching, manual reset	E-Stop
Functional Safety	SIL (IEC 62061) and PL (ISO 13849-1) standards	Requi

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12.13 Review Questions

Question 1: What are the five levels of the Hierarchy of Controls, and why is PPE the least effective?

The five levels are (from most to least effective):

1. **Elimination** — Physically remove the hazard
2. **Substitution** — Replace with something less hazardous
3. **Engineering Controls** — Isolate people from the hazard
4. **Administrative Controls** — Change how people work
5. **PPE** — Protect the worker with equipment

PPE is least effective because: - The hazard still exists - It depends on human compliance - Equipment can fail or be worn incorrectly - It's the last line of defense

Question 2: A light curtain has a response time of 12ms and resolution of 14mm. The machine takes 180ms to stop. Calculate the minimum safety distance for hand approach.

Question 2: minimum safety distance

Use K = 2000 mm/s (hand approach) and C = 0 mm (14 mm resolution).

- Total stopping time T: _____ s *(to 3 decimal places)*
- Minimum safety distance S: _____ mm *(to 0 decimal places)*

Show the worked solution

Given: - K = 2000 mm/s (hand approach, use higher value for safety) - T = 0.012 + 0.180 = 0.192 seconds - C = 0 mm (14mm resolution)

Calculation:

$$S = (K \times T) + C$$

$$S = (2000 \times 0.192) + 0$$

$$S = 384 \text{ mm}$$

The light curtain must be installed at least **384mm** (approximately 15 inches) from the hazard.

Question 3: Explain why a three-position enable switch stops the robot in both position 1 (released) and position 3 (fully pressed).

The three-position design accounts for two natural human reactions to danger:

Position 1 (Released): - When frightened, people often release what they're holding (panic response) - Robot stops immediately

Position 3 (Fully Pressed): - When startled, people often grip tighter (startle response) - Robot stops immediately

Position 2 (Middle): - Only a deliberate, controlled hold allows operation - Requires conscious effort to maintain - This ensures the operator is alert and in control

Question 4: What are the four methods of collaborative robot operation according to ISO 10218?

1. **Safety-Rated Monitored Stop**
   - Robot stops when human enters workspace
   - Resumes when human leaves
   - Uses area scanners or light curtains
2. **Hand Guiding**
   - Operator physically guides robot
   - Force/torque sensors enable safe teaching
3. **Speed and Separation Monitoring**
   - Robot speed varies based on human proximity
   - Uses external sensors to track human position
4. **Power and Force Limiting (PFL)**
   - Robot limits contact force to safe levels
   - Built-in joint torque sensors
   - Most common cobot method

Question 5: What are the requirements for an E-Stop button according to IEC 60204-1?

Physical Requirements: - **Color:** Red - **Shape:** Mushroom head (palm or fist operated) - **Background:** Yellow (high contrast)

Functional Requirements: - **Self-latching:** Stays activated when pressed - **Manual reset:** Requires deliberate action to release - **Accessibility:** Within easy reach of all operators - **Wiring:** Normally closed contacts (fail-safe) - **Function:** Category 0 or Category 1 stop - **No automatic restart:** Separate start action required after reset

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12.14 References

- ISO 12100:2010 — Safety of machinery — General principles for design — Risk assessment and risk reduction
- ISO 13849-1:2015 — Safety of machinery — Safety-related parts of control systems
- ISO 13855:2010 — Safety of machinery — Positioning of safeguards with respect to approach speeds
- ISO 10218-1/2:2011 — Robots and robotic devices — Safety requirements for industrial robots
- ISO/TS 15066:2016 — Robots and robotic devices — Collaborative robots
- IEC 60204-1:2016 — Safety of machinery — Electrical equipment of machines
- IEC 62061:2021 — Safety of machinery — Functional safety of safety-related control systems
- OSHA 29 CFR 1910.212 — General requirements for all machines
- ANSI/RIA TR R15.306 — Task-based risk assessment methodology

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Chapter 13

Total Productive Maintenance (TPM)

13.1 Learning Objectives

By the end of this chapter, you will be able to:

- Define Total Productive Maintenance and explain its objectives
- Describe the 8 pillars of TPM and their interconnections
- Calculate Overall Equipment Effectiveness (OEE) and its components
- Identify and categorize the Six Big Losses
- Implement autonomous maintenance activities
- Develop preventive and predictive maintenance strategies
- Apply TPM principles to improve equipment reliability and productivity

> "Take care of your equipment, and it will take care of your production." — Seiichi Nakajima, Father of TPM

13.2 Introduction to TPM

13.2.1 What is Total Productive Maintenance?

Total Productive Maintenance (TPM) is a holistic approach to equipment maintenance that strives to achieve perfect production with no breakdowns, no defects, and no accidents. It involves everyone in the organization, from operators to top management.

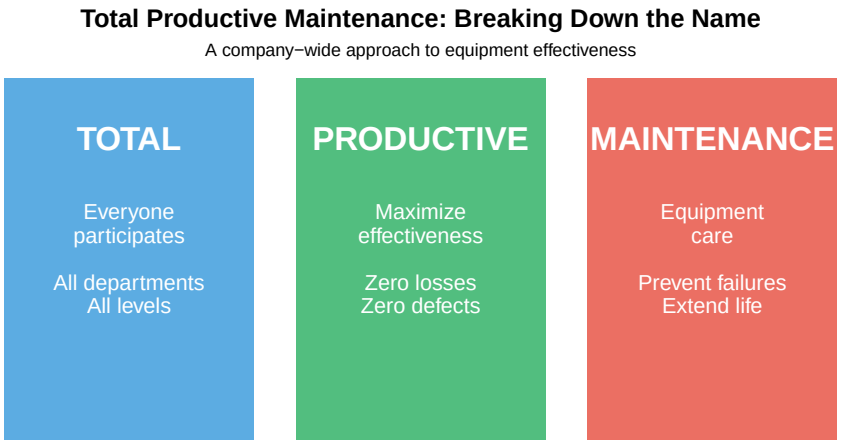


Figure 13.1: The TPM Philosophy

13.2.2 History of TPM

Table 13.1: Evolution of Maintenance Philosophy

Era	Maintenance Approach	Description
Pre-1950s	Breakdown Maintenance	Fix it when it breaks; reactive only
1950s	Preventive Maintenance	Scheduled maintenance to prevent failures
1960s	Productive Maintenance	Maintenance focused on reliability and maintainabil
1970s	TPM Developed	Nakajima develops TPM at Nippondenso (Toyota s
1980s	TPM Spreads Globally	Adopted by automotive, electronics, process industr
1990s-Present	TPM + Industry 4.0	Integration with predictive analytics, IoT sensors, A

13.2.3 The Goals of TPM

TPM aims to achieve **zero** targets:

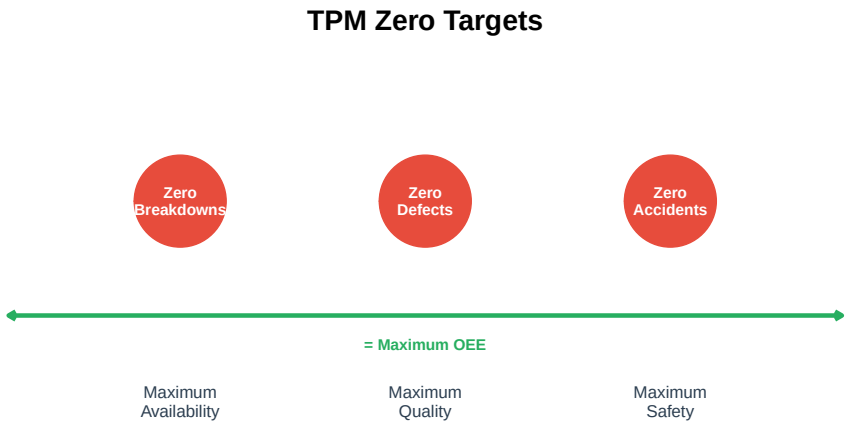


Figure 13.2: TPM Zero Targets

13.2.4 Traditional vs. TPM Approach

Table 13.2: Traditional Maintenance vs. TPM

Aspect	Traditional Maintenance	TPM Approach
Equipment ownership	'Maintenance owns the machines'	'We all own the machines'
Maintenance responsibility	Maintenance department only	Everyone (operators + maintenance)
Operator role	Operate; call maintenance when broken	Operate, clean, inspect, maintain
Maintenance focus	Fix breakdowns quickly	Prevent breakdowns; extend life
Problem solving	Reactive; fire-fighting	Proactive; root cause elimination
Goal	Minimize maintenance cost	Maximize equipment effectiveness

Question: Why involve operators in maintenance?

Operators are the first line of defense because:

1. **They know the equipment:** Operators work with machines every day and notice subtle changes
2. **Early detection:** Small problems are detected before becoming major failures
3. **Ownership mindset:** People take better care of things they "own"
4. **Reduced response time:** Operators can address minor issues immediately
5. **Freed maintenance resources:** Technicians can focus on complex tasks

Example: An operator notices a slight vibration change in a motor. Under traditional maintenance, they might ignore it. Under TPM, they report it, potentially preventing a $50,000 breakdown.

> "No one knows a machine better than the person who operates it every day."

13.3 The Eight Pillars of TPM

TPM is built on **eight pillars**, each addressing a specific aspect of equipment management and organizational culture.

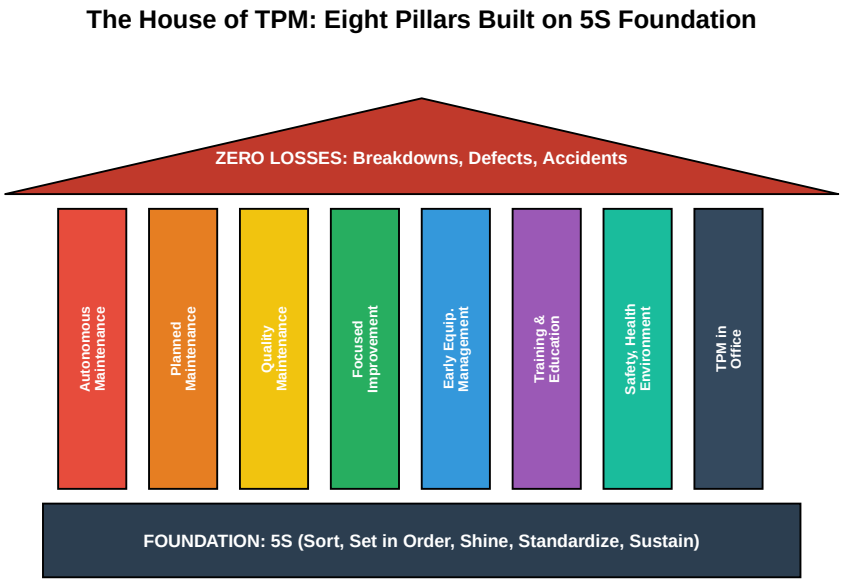


Figure 13.3: The Eight Pillars of TPM

Table 13.3: The Eight Pill

#	Pillar	Purpose
1	**Autonomous Maintenance**	Operators maintain their own equ
2	**Planned Maintenance**	Scheduled maintenance by special
3	**Quality Maintenance**	Zero defects through equipment c

4	**Focused Improvement**	Eliminate losses through kaizen activities
5	**Early Equipment Management**	Design equipment for maintainability
6	**Training & Education**	Develop skilled operators and technicians
7	**Safety, Health, Environment**	Zero accidents, safe workplace
8	**TPM in Office**	Apply TPM principles to administrative pr

13.4 Overall Equipment Effectiveness (OEE)

13.4.1 What is OEE?

Overall Equipment Effectiveness (OEE) is the primary metric of TPM. It measures how effectively equipment is being used by combining three factors: Availability, Performance, and Quality.

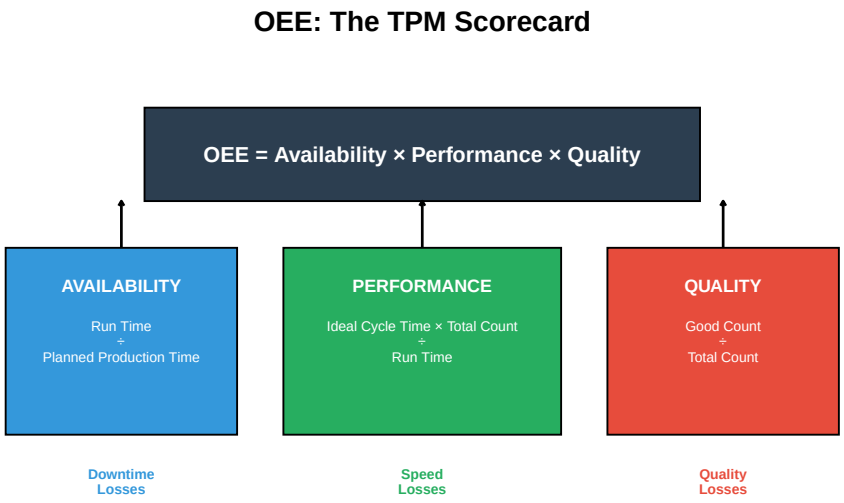


Figure 13.4: OEE Formula and Components

13.4.2 OEE Component Formulas

Table 13.4: OEE Compone

Component	Formula
****Availability****	$\frac{\text{Run Time}}{\text{Planned Production Time}} \times 100\%$

****Performance****	$\frac{\text{Ideal Cycle Time} \times \text{Total Count}}{\text{Run Tim
****Quality****	$\frac{\text{Good Count}}{\text{Total Count}} \times 100\%$
****OEE****	$\text{Availability} \times \text{Performance} \times \text{Quality}$

13.4.3 OEE Calculation Example

Your turn — change the downtime, production, or defect numbers below.

Show the live R code behind this calculator

```
#| label: oee-calculation
# OEE Calculation Example: Injection Molding Machine

# Time data
shift_length <- 480      # minutes (8 hours)
planned_downtime <- 30   # breaks, planned maintenance
planned_production_time <- shift_length - planned_downtime

# Downtime losses (breakdowns, changeovers, adjustments combined)
total_downtime <- 70     # minutes
run_time <- planned_production_time - total_downtime

# Production data
ideal_cycle_time <- 0.5  # minutes per part (design spec)
total_parts <- 700       # parts produced
good_parts <- 665        # parts passing inspection

# Calculate OEE Components
availability <- (run_time / planned_production_time) * 100
theoretical_output <- run_time / ideal_cycle_time
performance <- (total_parts / theoretical_output) * 100
quality <- (good_parts / total_parts) * 100

# Overall OEE
oee <- (availability / 100) * (performance / 100) * (quality / 100) * 100

cat(sprintf("OEE: %.1f%%\n", oee))
cat(sprintf("  Availability         %.1f%%\n", availability))
cat(sprintf("  Performance          %.1f%%\n", performance))
cat(sprintf("  Quality              %.1f%%\n", quality))
```

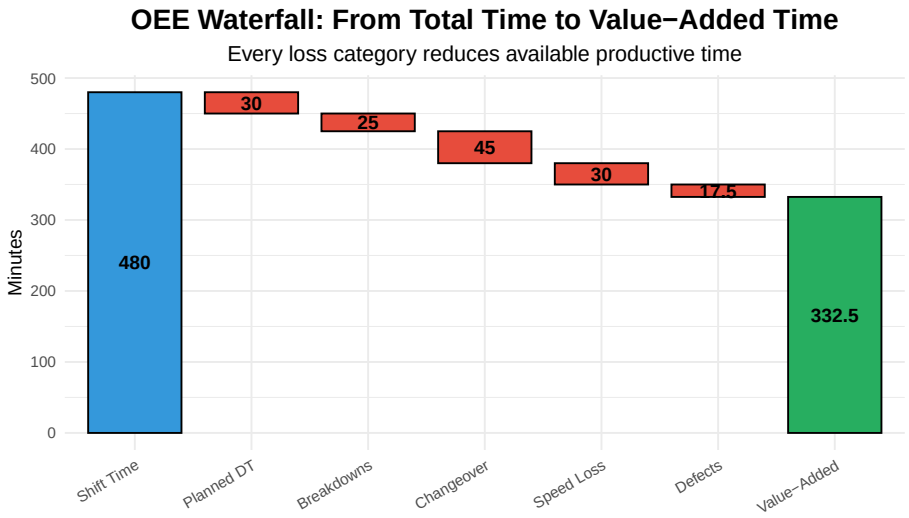


Figure 13.5: OEE Waterfall Chart: Where Time Goes

13.4.4 OEE Benchmarks

Table 13.5: OEE Benchmarks

OEE Level	Classification	Interpretation	Component Targets
< 65%	Unacceptable	Significant improvement opportunities exist	—
65% - 75%	Typical	Average for discrete manufacturing	A: 90%, P: 85%, Q: 95%
75% - 85%	Good	Well-managed operations	A: 92%, P: 90%, Q: 97%
> 85%	**World Class**	**Best-in-class performance**	**A: 95%, P: 95%, Q: 99%**

Interactive Exercise: Calculate OEE

Given the following shift data, calculate OEE:

- Shift length: 480 minutes
- Planned breaks: 40 minutes
- Equipment breakdown: 30 minutes
- Setup/changeover: 20 minutes
- Ideal cycle time: 1 minute/part
- Total parts produced: 350
- Rejected parts: 14

Work through the calculation:

Solution:

```
Planned Production Time = 480 - 40 = 440 minutes
Run Time = 440 - 30 - 20 = 390 minutes

Availability = 390 / 440 = 88.6%

Theoretical Output = 390 / 1 = 390 parts
Performance = 350 / 390 = 89.7%

Quality = (350 - 14) / 350 = 336 / 350 = 96.0%

OEE = 88.6% × 89.7% × 96.0% = 76.3%
```

Interpretation: This is "Good" performance, above the typical range. The biggest opportunity is in Performance (speed losses).

13.5 The Six Big Losses

TPM categorizes all equipment losses into **six categories**, which map directly to the OEE components.

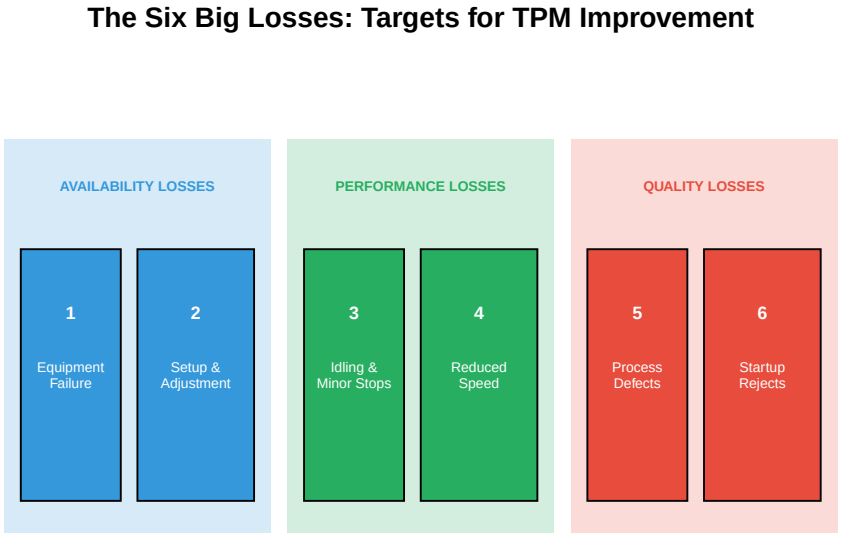


Figure 13.6: The Six Big Losses

Table 13.6: The Six Big Losse

#	Loss Category	OEE Component	Description
1	**Equipment Failure**	Availability	Unplanned equipment breakdo
2	**Setup & Adjustment**	Availability	Time for changeover, tool chan
3	**Idling & Minor Stops**	Performance	Brief stoppages (jams, blockage
4	**Reduced Speed**	Performance	Running slower than design spe
5	**Process Defects**	Quality	Defects produced during norma
6	**Startup Rejects**	Quality	Defects during warmup, startu

13.5.1 Typical Loss Distribution

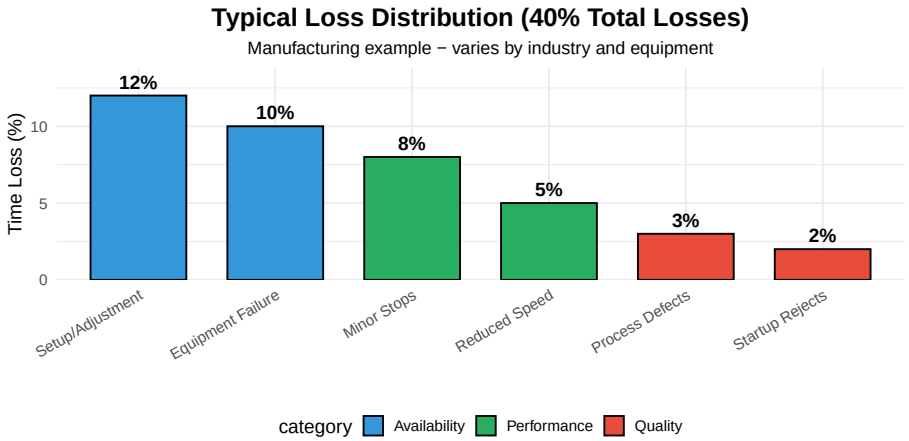


Figure 13.7: Typical Distribution of the Six Big Losses

13.5.2 Try it yourself: which OEE factor takes the hit?

Each of the Six Big Losses reduces one of the three OEE factors. Sort these shift-log entries by the factor they reduce.

Availability, Performance or Quality?

Pick the OEE factor each loss reduces.

Categories: Availability / Performance / Quality

1. A hydraulic hose bursts and the press is down for 50 minutes. _____
2. A die change and first-piece adjustment takes 40 minutes. _____

3. Parts jam in the feeder chute a dozen times, each cleared in under a minute.

_____

4. The line runs at 80% of its rated speed because of a worn spindle bearing.

_____

5. The first 25 parts after start-up are scrapped while the process warms up.

_____

6. Three percent of parts are reworked for burrs during steady running.

_____

Answers: 1 Availability; 2 Availability; 3 Performance; 4 Performance; 5 Quality; 6 Quality.

13.6 Autonomous Maintenance

Autonomous Maintenance (AM) is the first and most visible pillar of TPM. It transfers basic maintenance tasks from maintenance specialists to operators.

13.6.1 The Seven Steps of Autonomous Maintenance

Seven Steps of Autonomous Maintenance

1. **1. Initial Cleaning** (Clean to inspect)
2. **2. Eliminate Sources** (Prevent contamination)
3. **3. Standards** (Maintain conditions)
4. **4. General Inspection** (Train & inspect)
5. **5. Autonomous Inspection** (Self-directed)
6. **6. Organization** (Standardize workplace)
7. **7. Full AM** (Continuous improvement)

Table 13.7: Autonomous Maintenance Steps

Step	Name	Activities
1	**Initial Cleaning**	Deep clean equipment; identify abnormalities
2	**Eliminate Contamination Sources**	Find and fix sources of dirt, leaks, contamination
3	**Cleaning & Lubrication Standards**	Create standards for cleaning, lubricating, inspec
4	**General Inspection**	Train operators on equipment function; expand s
5	**Autonomous Inspection**	Operators perform scheduled inspections indepen
6	**Organization & Tidiness**	Standardize workplace; visual management
7	**Full Autonomous Maintenance**	Continuous improvement; operators propose imp

13.6.1.1 Try it yourself: order the seven steps

Autonomous maintenance is built up in seven steps, each one resting on the one before. Put them in order.

The seven steps of autonomous maintenance

Number these in the correct order:

- _____ Autonomous inspection
- _____ Eliminate sources of contamination
- _____ Full autonomous maintenance
- _____ General inspection training
- _____ Initial cleaning: clean to inspect
- _____ Set cleaning and lubrication standards
- _____ Workplace organization

Answers, in the order listed: 5, 2, 7, 4, 1, 3, 6.

13.6.2 The AM Tag System

During autonomous maintenance activities, operators identify abnormalities using **tags**:

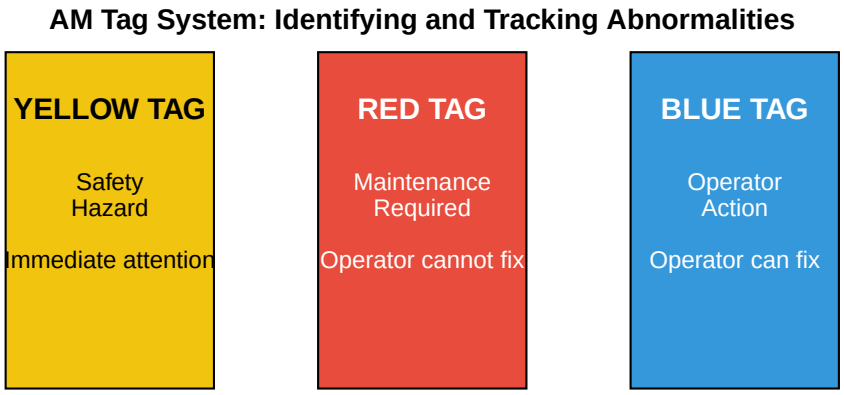


Figure 13.8: AM Tag System

13.6.3 Example: AM Cleaning and Inspection Checklist

Table 13.8: Sample AM Inspection Checklist

#	Check Item	Method	Standard	Frequency
1	General cleanliness	Visual	No dust, debris, contamination	Daily
2	Oil/lubricant levels	Visual/gauge	Between min-max marks	Daily
3	Loose bolts/fasteners	Touch/wrench	Hand tight + 1/4 turn	Weekly
4	Unusual sounds	Listen	Normal operating sound	Each startup
5	Vibration	Touch	Minimal, no unusual patterns	Each startup
6	Temperature	Touch/thermometer	$< 60°C$ (or per spec)	Hourly

7	Leaks (oil, air, coolant)	Visual	No visible leaks	Daily
8	Safety guards in place	Visual	All guards secure	Each shift

13.7 Planned Maintenance

Planned Maintenance is the second pillar, focusing on scheduled maintenance activities performed by maintenance specialists.

13.7.1 Types of Maintenance

Evolution of Maintenance Strategy

1. **REACTIVE** – Breakdown Maintenance; Fix when broken
2. **PREVENTIVE** – Time-Based Maintenance; Scheduled intervals
3. **PREDICTIVE** – Condition-Based Maintenance; Monitor & predict
4. **PROACTIVE** – Root Cause Elimination; Prevent recurrence

Table 13.9: Maintenance Strategy

Type	Approach	Advantages
****Breakdown****	Run to failure	No upfront cost
****Preventive****	Time/cycle-based replacement	Planned downtime, parts availability
****Predictive****	Condition monitoring	Optimal component life, targeted actions
****Proactive****	Root cause elimination	Prevents future failures

13.7.2 Predictive Maintenance Technologies

Table 13.10: Predictive Maintenance Tech

Technology	What It Detects	Equipment
****Vibration Analysis****	Imbalance, misalignment, bearing wear, looseness	Rotating equ
****Thermography****	Hot spots, electrical issues, friction, insulation problems	Electrical pa
****Oil Analysis****	Contamination, wear particles, lubricant degradation	Gearboxes, h
****Ultrasound****	Leaks, bearing defects, electrical discharge	Steam traps,
****Motor Current Analysis****	Motor winding issues, rotor bar problems, power quality	Electric moto

13.7.3 The P-F Curve

The **P-F Curve** illustrates the concept of detecting potential failure (P) before functional failure (F).

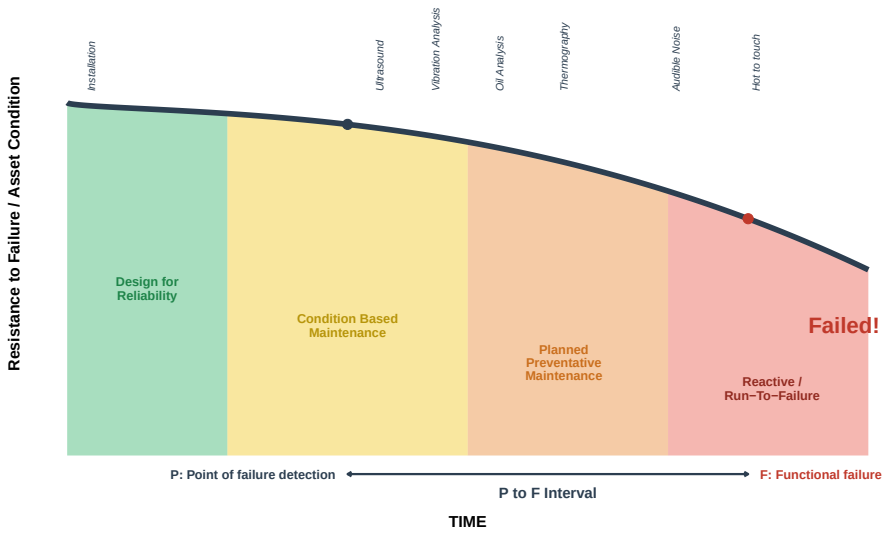


Figure 13.9: The P-F Curve: Detecting Failure Early

Question: Why is predictive maintenance better than preventive maintenance?

Preventive Maintenance (Time-Based): - Replaces parts at fixed intervals regardless of condition - May replace parts still in good condition (waste) - May miss failures that occur before scheduled replacement - Based on average life, not individual component condition

Predictive Maintenance (Condition-Based): - Monitors actual equipment condition - Replaces parts when condition indicates need - Maximizes component life - Provides warning time to plan repairs

Example: - Preventive: Replace bearing every 6 months (may last 9 months or fail at 4 months) - Predictive: Monitor vibration; replace when trending indicates 2-3 weeks to failure - Result: Optimal component life, planned downtime, no unexpected failures

Cost Comparison: - Reactive repair: $10,000 (including secondary damage, lost production) - Preventive replacement: $3,000 (may waste $1,000 in parts life) - Predictive replacement: $2,500 (optimal timing, minimal waste)

13.8 Focused Improvement (Kobetsu Kaizen)

Focused Improvement uses cross-functional teams to eliminate specific losses identified through OEE analysis.

13.8.1 The Kaizen Process

Focused Improvement (Kobetsu Kaizen) Process

1. **1. SELECT** – Identify biggest loss
2. **2. ANALYZE** – Root cause analysis
3. **3. DEVELOP** – Countermeasures
4. **4. IMPLEMENT** – Execute plan
5. **5. SUSTAIN** – Standardize & share

13.8.1.1 Understanding Kaizen

Kaizen ( ) is a Japanese term meaning "change for better" or **continuous improvement**. Unlike large-scale transformation projects, Kaizen focuses on small, incremental improvements made consistently over time. The philosophy is built on the belief that every process can be improved, and everyone—from operators to managers—should participate in identifying and implementing improvements.

Core Principles of Kaizen:

- **Good processes bring good results** — Focus on improving the process, not blaming people
- **Go see for yourself (Gemba)** — Understand the actual situation at the workplace
- **Speak with data** — Base decisions on facts and measurements
- **Take action to contain and correct root causes** — Don't just treat symptoms
- **Work as a team** — Cross-functional collaboration produces better solutions
- **Kaizen is everybody's business** — Every employee can contribute improvements

13.8.1.2 The Five Steps in Detail

Step 1: SELECT — Identify the Biggest Loss

The first step uses data to identify which loss has the greatest impact on OEE. This typically involves:

- Reviewing OEE data and breakdown logs
- Creating Pareto charts to visualize loss magnitude
- Selecting a loss that is significant, measurable, and within the team's ability to address

- Forming a cross-functional team with members who understand the equipment and process

Key question: "Which single problem, if solved, would give us the biggest improvement?"

Step 2: ANALYZE — Root Cause Analysis

Once a loss is selected, the team investigates its root cause using structured problem-solving tools:

- **Why-Why Analysis (5 Whys)**: Repeatedly asking "why" to drill down from symptoms to root causes
- **Fishbone Diagram (Ishikawa)**: Categorizing potential causes by Machine, Method, Material, Man, Measurement, Environment
- **PM Analysis**: A systematic method for chronic losses that examines physical mechanisms
- **Data collection**: Gathering facts at the Gemba (actual workplace)

Key principle: Never assume you know the cause—verify with evidence.

Step 3: DEVELOP — Create Countermeasures

Based on root cause analysis, the team develops countermeasures—actions designed to eliminate or reduce the root cause:

- Brainstorm multiple potential solutions
- Evaluate options based on effectiveness, cost, and implementation time
- Create an action plan with clear responsibilities and deadlines
- Consider both immediate fixes and permanent solutions
- Apply error-proofing (Poka-Yoke) where possible

Important: Countermeasures address root causes, not just symptoms.

Step 4: IMPLEMENT — Execute the Plan

The team puts countermeasures into action:

- Follow the action plan systematically
- Make changes during planned downtime when possible
- Document all modifications made
- Train affected personnel on new procedures
- Monitor results immediately after implementation
- Be prepared to adjust if initial results are not as expected

Tip: Start with small-scale trials before full implementation when possible.

Step 5: SUSTAIN — Standardize and Share

The final step ensures improvements are maintained and shared across the organization:

- **Standardize**: Update procedures, checklists, and work instructions to reflect the new method

- **Visualize**: Create visual controls that make the standard easy to follow
- **Train**: Ensure all shifts and new employees learn the improved method
- **Verify**: Include checks in AM routines or audits to confirm the standard is being followed
- **Share**: Communicate successful improvements to other areas that might benefit (Yokoten—horizontal deployment)

Kaizen is never finished—today's standard is tomorrow's baseline for further improvement.

13.8.1.3 Kaizen Event vs. Daily Kaizen

Aspect	Kaizen Event (Blitz)	Daily Kaizen
Duration	3-5 days intensive	Ongoing, small improvements
Scope	Focused on one specific problem	Any improvement opportunity
Team	Dedicated cross-functional team	Individual or small group
Impact	Significant, measurable improvement	Cumulative effect over time
Example	Reduce changeover time by 50%	Reorganize tool storage for easier access

Both approaches are valuable—Kaizen events tackle significant losses while daily Kaizen builds a culture of continuous improvement.

13.8.2 Loss Elimination Tools

Effective loss elimination requires systematic tools that help teams identify problems, find root causes, and implement lasting solutions. The following tools are essential for Focused Improvement activities.

13.8.2.1 Pareto Analysis

Pareto Analysis applies the 80/20 rule: roughly 80% of losses come from 20% of causes. By creating a Pareto chart, teams can visualize which losses have the greatest impact and prioritize their improvement efforts accordingly.

How to use:

1. Collect data on all losses (downtime events, defects, speed reductions)
2. Categorize losses by type or cause
3. Calculate the frequency or duration of each category
4. Rank categories from highest to lowest
5. Create a bar chart with a cumulative percentage line

6. Focus improvement efforts on the "vital few" categories that account for most losses

13.8.2.2 Why-Why Analysis (5 Whys)

Why-Why Analysis is a simple but powerful technique for drilling down from a symptom to its root cause by repeatedly asking "Why?" The goal is to go beyond surface-level explanations to find the fundamental cause that, if addressed, will prevent recurrence.

Example:

Level	Question	Answer
Problem	Machine stopped	Overload tripped
Why 1?	Why did the overload trip?	Motor drew excessive current
Why 2?	Why excessive current?	Bearing was seized
Why 3?	Why was bearing seized?	Insufficient lubrication
Why 4?	Why insufficient lubrication?	Lubrication schedule not followed
Why 5?	Why not followed?	No visual indicator when lubrication is due
Root Cause		Missing visual management for lubrication schedule

Tips for effective Why-Why Analysis:

- Stay focused on one causal chain at a time
- Base answers on facts, not assumptions—verify at the Gemba
- Don't stop at "human error"—ask why the error was possible
- The number 5 is a guideline, not a rule—continue until you reach a controllable root cause

13.8.2.3 PM Analysis

PM Analysis (Physical Mechanism Analysis or Phenomenon-Mechanism Analysis) is a systematic method for analyzing **chronic losses**—problems that occur repeatedly despite multiple repair attempts. It examines the physical principles behind equipment failures.

The PM Analysis Process:

1. **Clarify the phenomenon**: Describe exactly what happens, when, and under what conditions
2. **Conduct physical analysis**: Understand the physical principles involved (friction, heat, vibration, etc.)

3. **Identify conditions that produce the phenomenon**: List all conditions required for the problem to occur
4. **Investigate equipment, materials, and methods**: Examine each condition against the 4Ms (Machine, Material, Method, Man)
5. **Plan and implement countermeasures**: Address the conditions that allow the problem to occur
6. **Verify results**: Confirm the chronic loss has been eliminated

PM Analysis is particularly valuable when Why-Why Analysis doesn't resolve a recurring problem.

13.8.2.4 SMED (Single-Minute Exchange of Die)

SMED is a systematic approach to reducing changeover or setup time—the time between the last good part of one product and the first good part of the next. The goal is to complete changeovers in under 10 minutes (single-digit minutes).

The SMED Process:

SMED: Reducing Changeover Time

1. **Step 1** – Separate; Internal & External
2. **Step 2** – Convert; Internal to External
3. **Step 3** – Streamline; All Elements

Key Concepts:

- **Internal setup**: Tasks that can only be done when the machine is stopped (changing the die itself)
- **External setup**: Tasks that can be done while the machine is running (preparing tools, preheating)

Typical SMED Improvements:

Technique	Example
Prepare in advance	Stage tools and materials before stopping the machine
Use quick-release clamps	Replace bolts with cam locks or hydraulic clamps
Eliminate adjustments	Use fixtures, stops, and standardized settings
Parallel operations	Two people working simultaneously on different tasks
Standardize components	Use common bolt sizes, connection types

13.8.2.5 Poka-Yoke (Error-Proofing)

Poka-Yoke ( ) means "mistake-proofing" or "error-proofing." These are devices or methods that either prevent errors from occurring or detect them immediately when they do occur.

Types of Poka-Yoke:

Type	Function	Example
Prevention	Makes errors impossible	Asymmetric connector that only fits one way
Detection	Identifies errors immediately	Sensor that stops the line if a part is missing
Warning	Alerts operator to potential error	Light or buzzer when a step is skipped

Poka-Yoke Design Principles:

1. **Design out the possibility of error** — The best Poka-Yoke makes the error impossible
2. **Detect at the source** — Catch errors where they occur, not downstream
3. **Stop and fix immediately** — Don't pass defects to the next process
4. **Keep it simple** — Low-cost, reliable solutions are better than complex ones

Examples in Manufacturing:

- **Guide pins** that prevent a part from being loaded incorrectly
- **Counters** that verify the correct number of components were installed
- **Checklists with physical verification** (e.g., removing a flag for each completed step)
- **Limit switches** that confirm proper positioning before the cycle starts
- **Color coding** to match parts with their correct locations

Question: How do you decide which tool to use?

Tool Selection Guide:

- **Pareto Analysis**: Use first to identify which loss to focus on
- **Why-Why Analysis**: Use for sporadic failures with clear cause-effect relationships
- **PM Analysis**: Use for chronic losses that keep recurring despite repairs
- **SMED**: Use specifically for setup/changeover losses (Loss #2)
- **Poka-Yoke**: Use to prevent defects and human errors (Losses #5, #6)

Often, multiple tools are used together—Pareto to select the problem, Why-Why or PM Analysis to find the root cause, and Poka-Yoke to prevent recurrence.

13.8.2.6 Summary of Loss Elimination Tools

Table 13.11: Tools for Loss Elimination

Tool	Purpose	Application
Pareto Analysis	Identify biggest losses to target	Prioritize improvement efforts
Why-Why Analysis	Find root cause of failures	Breakdown analysis, defect inves
PM Analysis	Analyze chronic losses systematically	Complex, recurring problems
SMED	Reduce setup/changeover time	Availability improvement (Loss
Poka-Yoke	Error-proof the process	Quality improvement (Losses #

13.9 TPM Implementation

13.9.1 The 12 Steps of TPM Implementation

Table 13.12: 12 Steps of TPM Implementation

Phase	Step	Activity	Duration
Preparation	1	Announce TPM introduction by top management	1 month
Preparation	2	Education and campaign to introduce TPM	3-6 months
Preparation	3	Create TPM promotion organization	1-2 months
Preparation	4	Establish basic TPM policies and goals	1-2 months
Preparation	5	Formulate master plan for TPM development	1-2 months
Implementation	6	Hold TPM kick-off	1 day
Implementation	7	Improve effectiveness of each piece of equipment	6-12 months
Implementation	8	Develop autonomous maintenance program	12-24 months
Implementation	9	Develop planned maintenance program	12-24 months
Implementation	10	Conduct training to improve skills	Ongoing
Stabilization	11	Develop early equipment management program	12-24 months
Stabilization	12	Perfect TPM implementation and raise levels	Ongoing

13.9.2 Success Factors

Table 13.13: TPM Success Factors

Factor	Why Critical
Top Management Commitment	Resources, culture change, sustained focus

****Training and Education****	Skills to identify problems, perform maintenance
****Small Group Activities****	Operator involvement, team problem-solving
****Visual Management****	Make abnormalities obvious, track progress
****Patience and Persistence****	TPM takes 3-5 years to fully implement

---

13.10 Case Study: TPM Implementation

13.10.1 Background

Company: Automotive component manufacturer

Equipment: CNC machining center

Initial OEE: 52%

13.10.2 Problem Analysis

```
# Initial State
cat("=== INITIAL OEE BREAKDOWN ===\n\n")

availability_initial <- 78  # %
performance_initial <- 82   # %
quality_initial <- 81       # %

oee_initial <- (availability_initial/100) * (performance_initial/100) * (quality_initial/100) * 1
cat("Availability:", availability_initial, "%\n")
cat("Performance:", performance_initial, "%\n")
cat("Quality:", quality_initial, "%\n")
cat("OEE:", round(oee_initial, 1), "%\n\n")

# Loss breakdown
cat("=== TOP LOSSES IDENTIFIED ===\n")
cat("1. Equipment breakdowns: 8% of time\n")
cat("2. Setup/changeover: 7% of time\n")
cat("3. Minor stops: 12% of time\n")
cat("4. Defects: 4% of production\n")
```

```
=== INITIAL OEE BREAKDOWN ===

Availability: 78 %
Performance: 82 %
Quality: 81 %
OEE: 51.8 %
```

```
=== TOP LOSSES IDENTIFIED ===
1. Equipment breakdowns: 8% of time
2. Setup/changeover: 7% of time
3. Minor stops: 12% of time
4. Defects: 4% of production
```

13.10.3 Improvements Implemented

Loss	Root Cause	Countermeasure	Result
Breakdowns	Lack of basic maintenance	Autonomous maintenance program	8% → 3%
Setup time	No standard procedure	SMED implementation	35 min → 12 min
Minor stops	Chip buildup in sensors	Improved chip management	12% → 4%
Defects	Tool wear not detected	In-process gauging	4% → 1.5%

13.10.4 Results After 18 Months

```
# After TPM Implementation
cat("=== FINAL OEE BREAKDOWN ===\n\n")

availability_final <- 91    # %
performance_final <- 94     # %
quality_final <- 98.5       # %

oee_final <- (availability_final/100) * (performance_final/100) * (quality_final/100) *
cat("Availability:", availability_final, "% (was", availability_initial, "%)\n")
cat("Performance:", performance_final, "% (was", performance_initial, "%)\n")
cat("Quality:", quality_final, "% (was", quality_initial, "%)\n")
cat("OEE:", round(oee_final, 1), "% (was", round(oee_initial, 1), "%)\n\n")

improvement <- oee_final - oee_initial
cat("=== IMPROVEMENT ===\n")
cat("OEE Improvement:", round(improvement, 1), "percentage points\n")
cat("Relative Improvement:", round((improvement/oee_initial)*100, 0), "%\n")
```

```
=== FINAL OEE BREAKDOWN ===

Availability: 91 % (was 78 %)
Performance: 94 % (was 82 %)
Quality: 98.5 % (was 81 %)
OEE: 84.3 % (was 51.8 %)
```

```
=== IMPROVEMENT ===
OEE Improvement: 32.4 percentage points
Relative Improvement: 63 %
```

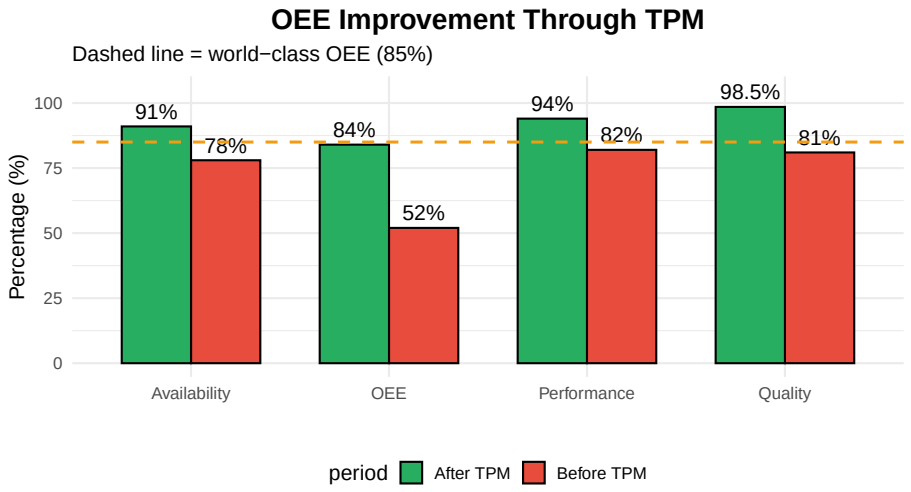


Figure 13.10: OEE Improvement: Before and After TPM

13.11 Summary

Table 13.14: TPM Key Concepts Summary

Topic	Key Points
TPM Definition	Total involvement, productive focus, maintenance excellence; zero breakdowns/defec
Eight Pillars	AM, PM, Quality, Focused Improvement, Early Equipment, Training, Safety, Office
OEE	OEE = Availability × Performance × Quality; World class > 85%
Six Big Losses	Equipment failure, setup, minor stops, speed loss, defects, startup rejects
Autonomous Maintenance	Operators maintain own equipment; 7 steps from cleaning to full ownership
Planned Maintenance	Preventive → Predictive → Proactive; use P-F curve for optimal timing

---

13.12 Review Questions

13.12.1 Conceptual Questions

Question 1: What does "Total" mean in Total Productive Maintenance?

"Total" has three dimensions:

1. **Total participation:** Everyone participates, from operators to top management
2. **Total equipment effectiveness:** Aim for maximum OEE (zero losses)
3. **Total system:** Covers the entire equipment lifecycle, from design to disposal

Key insight: TPM is not just a maintenance program — it's a company-wide philosophy where everyone takes responsibility for equipment care.

Question 2: Why is OEE calculated as a product rather than a sum?

OEE uses multiplication because losses are sequential and compound:

1. **Availability** determines how much time is available
2. **Performance** determines how much of that time is productive
3. **Quality** determines how much of that production is good

Example: - If each factor is 90%: OEE = $0.9 \times 0.9 \times 0.9 = 72.9\%$ (not 270% or 90%) - This reflects reality: you can only achieve quality on parts that were produced (performance) during time the machine ran (availability)

Why this matters: - Small improvements in all three factors compound - 5% improvement in each: $0.85 \times 0.85 \times 0.85 = 61\% \rightarrow 0.90 \times 0.90 \times 0.90 = 73\%$ - Focus on the weakest factor first for biggest impact

Question 3: What is the difference between autonomous maintenance and planned maintenance?

Aspect	Autonomous Maintenance	Planned Maintenance
Who	Operators	Maintenance technicians
What	Basic care (clean, inspect, lubricate)	Complex repairs, overhauls
When	Daily, during operation	Scheduled intervals
Skills	Basic equipment knowledge	Technical expertise
Goal	Prevent deterioration	Restore/improve equipment

They complement each other: - AM catches problems early (first line of defense) - PM provides specialized skills when needed - Together they achieve zero breakdowns

13.12.2 Calculation Questions

Question 4: Calculate OEE given the following data.

Given:

- Shift length: 480 minutes
- Breaks: 30 minutes
- Changeover: 45 minutes
- Breakdown: 20 minutes
- Ideal cycle time: 30 seconds (0.5 min)
- Total parts produced: 580
- Rejected parts: 23

Question 4: OEE

- Availability: _____ % *(to 1 decimal place)*
- Performance: _____ % *(to 1 decimal place)*
- Quality: _____ % *(to 1 decimal place)*
- OEE: _____ % *(to 1 decimal place)*

Show the worked solution

Solution:

```
Planned Production Time = 480 - 30 = 450 minutes
Run Time = 450 - 45 - 20 = 385 minutes

Availability = 385 / 450 = 85.6%

Theoretical Output = 385 / 0.5 = 770 parts
Performance = 580 / 770 = 75.3%

Quality = (580 - 23) / 580 = 557 / 580 = 96.0%

OEE = 85.6% × 75.3% × 96.0% = 61.9%
```

Analysis: Performance is the biggest opportunity for improvement (75.3% vs. targets of 95%). Investigate minor stops and speed losses.

Question 5: A machine has OEE of 65% with Availability 90%, Performance 80%, Quality 90%. Which factor should be prioritized?

Analysis:

Current state: - Availability: 90% (Gap to world class: 5%) - Performance: 80% (Gap to world class: 15%) - Quality: 90% (Gap to world class: 9%)

Performance should be prioritized because:

1. **Biggest gap:** 15 percentage points below world class (95%)

2. **Biggest OEE impact:** Improving performance 10% → OEE = 90% × 90% × 90% = 72.9%
3. **Often overlooked:** Speed losses and minor stops are frequently accepted as normal

Investigation areas for performance: - Minor stops (jams, blockages, sensor trips) - Reduced speed operation - Idling and waiting - Operator pace

Question 6: If current OEE is 60% and target is 85%, what improvement is needed in each component assuming equal improvement?

Question 6: equal improvement

- Value each of the three factors must reach: _____ % *(to 1 decimal place)*

Show the worked solution

Solution:

Current: 60% = A × P × Q

If we improve each equally, we need: - Target: 85% = A' × P' × Q' - If A' = P' = Q' = x, then $x^3 = 0.85$ - x = $\sqrt[3]{0.85} = 0.947 = 94.7\%$

Each component needs to reach approximately 95%

Check: 0.947 × 0.947 × 0.947 = 0.849 ≈ 85%

If current factors are A=80%, P=85%, Q=88%: - Availability needs: +15 points (80% → 95%) - Performance needs: +10 points (85% → 95%) - Quality needs: +7 points (88% → 95%)

Prioritize by gap size: Availability first, then Performance, then Quality.

13.12.3 Application Questions

Question 7: An operator notices a small oil leak on a machine. Under traditional maintenance vs. TPM, what happens?

Traditional Maintenance Approach: 1. Operator continues running machine 2. May or may not report to maintenance 3. If reported, added to backlog 4. Maintenance addresses when convenient (days/weeks) 5. Leak worsens, potential bearing damage 6. Eventually leads to breakdown

TPM Approach: 1. Operator notices during routine inspection 2. Tags the abnormality immediately 3. Determines if operator can fix (blue tag) or needs maintenance (red tag) 4. If simple: operator tightens fitting or adds sealant 5. If complex: maintenance prioritizes based on severity 6. Root cause investigated to prevent recurrence 7. Finding shared in team meeting

Key Differences: - **Ownership:** Operator takes responsibility vs. "not my job" - **Speed:** Immediate action vs. delayed response - **Prevention:** Fixed

before failure vs. reactive repair - **Cost:** $10 seal vs. $5,000 bearing replacement + downtime

Question 8: List five items an operator should check during daily autonomous maintenance.

Five Essential Daily AM Checks:

1. **Cleanliness:** Remove chips, debris, dust; clean safety windows and sensors
   - *Why:* Contamination causes wear, sensor malfunctions, quality issues
2. **Lubrication:** Check oil levels, grease points, coolant level
   - *Why:* Inadequate lubrication causes friction, heat, accelerated wear
3. **Fasteners:** Check for loose bolts, guards, covers
   - *Why:* Loose fasteners cause vibration, safety hazards, misalignment
4. **Leaks:** Look for oil, coolant, air, hydraulic leaks
   - *Why:* Leaks indicate seal failure, waste resources, create hazards
5. **Abnormal sounds/vibration:** Listen and feel during startup
   - *Why:* Changes indicate developing problems (bearings, alignment)

Bonus items: - Safety guards in place - Emergency stops functional - Pressure/temperature gauges in normal range - No unusual smells (burning, chemical)

Question 9: Explain how SMED relates to OEE improvement.

SMED (Single Minute Exchange of Die) directly improves Availability:

Loss #2: Setup and Adjustment is an availability loss captured in OEE.

How SMED helps:

1. **Reduces changeover time:**
   - Before SMED: 45-minute changeover
   - After SMED: 10-minute changeover
   - Gain: 35 minutes per changeover
2. **Impact on OEE:**
   - If 2 changeovers per shift: 70 minutes saved
   - On 450-minute shift: Availability improves 15%+
3. **Secondary benefits:**
   - More changeovers become practical
   - Smaller batch sizes possible
   - Flexibility increased
   - Less WIP inventory

SMED Process: 1. Separate internal (machine stopped) from external (while running) setup 2. Convert internal to external where possible 3. Streamline remaining internal operations 4. Practice and standardize

Example: Die change on press - External: Stage next die, prepare tools, pre-heat - Internal: Remove old die, install new die, adjust - Target: All possible work done while machine runs

---

13.13 References

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Chapter 14

Process Failure Mode and Effects Analysis (PFMEA)

14.1 Learning Objectives

By the end of this chapter, you will be able to:

- Explain the purpose and benefits of PFMEA in manufacturing
- Distinguish between DFMEA (Design) and PFMEA (Process)
- Apply the 10-step PFMEA methodology to a manufacturing process
- Correctly rate Severity, Occurrence, and Detection using standardized scales
- Calculate Risk Priority Numbers (RPN) and prioritize actions
- Develop recommended actions to reduce process risks
- Create and maintain PFMEA documentation as a living document

> "An ounce of prevention is worth a pound of cure." — Benjamin Franklin

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14.2 Introduction to PFMEA

14.2.1 What is PFMEA?

Process Failure Mode and Effects Analysis (PFMEA) is a systematic, proactive method for evaluating a manufacturing or business process to identify where and how it might fail, and to assess the relative impact of different failures.

The goal is to identify and prioritize actions that will prevent defects before they occur.

The PFMEA Mindset: Prevention Over Detection

1. **REACTIVE APPROACH** – Detect → Fix → Repeat; ; Find defects after they occur
2. **PFMEA APPROACH** – Predict → Prevent → Verify; ; Anticipate failures before they occur

14.2.2 Brief History of FMEA

Table 14.1: Evolution of FMEA

Year	Development	Significance
1949	US Military develops FMEA (MIL-P-1629)	First formal FMEA me
1960s	NASA adopts FMEA for Apollo program	Proven reliability in lif
1970s	Automotive industry begins using FMEA	Ford, GM, Chrysler ad
1980s	AIAG publishes first FMEA manual	Industry-wide consiste
1990s	QS-9000 requires FMEA for automotive suppliers	FMEA becomes manda
2000s	FMEA expands to healthcare, service industries	Methodology proves u
2019	AIAG-VDA FMEA Handbook published (current standard)	Harmonized global sta

14.2.3 DFMEA vs. PFMEA

Two main types of FMEA are used in product development:

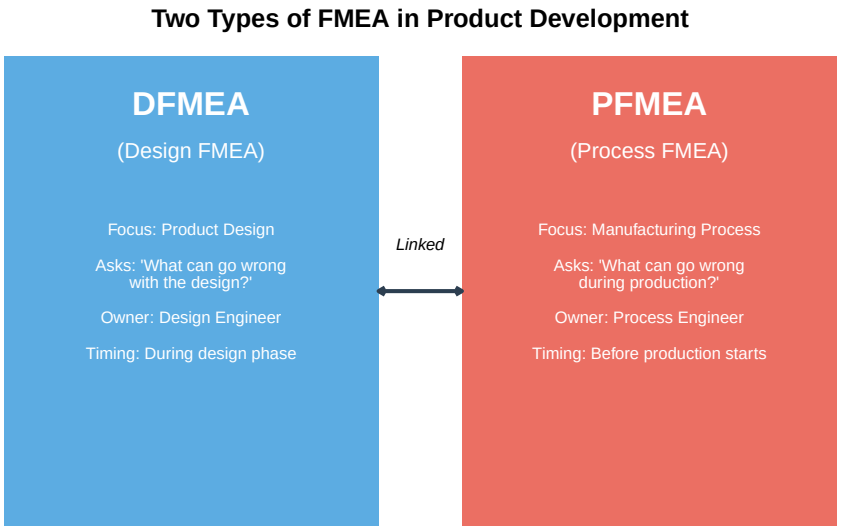


Figure 14.1: DFMEA vs PFMEA: Different Focus, Same Methodology

Table 14.2: DFMEA vs PFMEA Comparison

Aspect	DFMEA (Design)	PFMEA (Process)
Primary Question	Will the design meet requirements?	Will the process produce conforming parts?
Failure Mode Example	Shaft diameter too small	Shaft turned undersize
Effect Example	Premature bearing failure	Part fails incoming inspection
Cause Example	Inadequate stress analysis	Tool wear not monitored
Control Example	FEA simulation, prototype testing	In-process gauging, SPC
Responsible Team	Design engineering, R&D	Manufacturing, quality, production

Question: When should PFMEA be conducted?

Ideal Timing for PFMEA:

1. **New Process Development:** Before production begins, during process design
2. **New Product Introduction:** When a new product will use existing processes
3. **Process Changes:** Whenever significant changes are made to equipment, methods, or materials
4. **Quality Issues:** When field failures or customer complaints indicate process problems
5. **Regular Review:** Periodically (annually) to ensure continued relevance

Key Principle: The earlier PFMEA is conducted, the lower the cost of implementing improvements. Changes during design cost 10x less than changes during production, and 100x less than changes after customer delivery.

14.2.4 Why PFMEA Matters

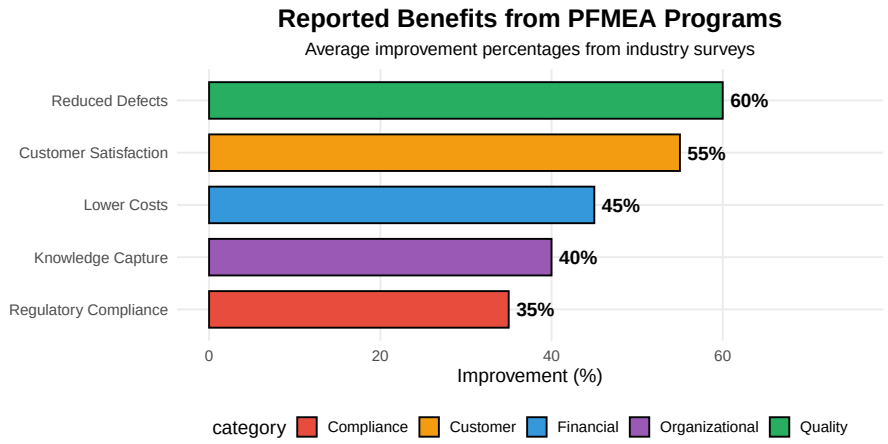


Figure 14.2: Benefits of PFMEA Implementation

14.3 PFMEA Fundamentals

14.3.1 Core Objectives

The fundamental objective of PFMEA is **prevention over detection** — identifying and eliminating potential failures before they occur, rather than finding defects after production.

Table 14.3: Core Objectives of PFMEA

Objective	Description	Outcon
****Identify Failure Modes****	Systematically identify all ways the process can fail	Compre
****Assess Risk****	Evaluate severity, likelihood, and detectability of each failure	Quantifi
****Prioritize Actions****	Focus resources on highest-risk failure modes	Action p
****Document Knowledge****	Capture institutional knowledge in a structured format	Living d

****Drive Improvement****	Implement and verify effectiveness of countermeasures	Measurable reductio

14.3.2 The PFMEA Team

PFMEA is most effective when conducted by a **cross-functional team**. No single person has all the knowledge needed to identify every potential failure.

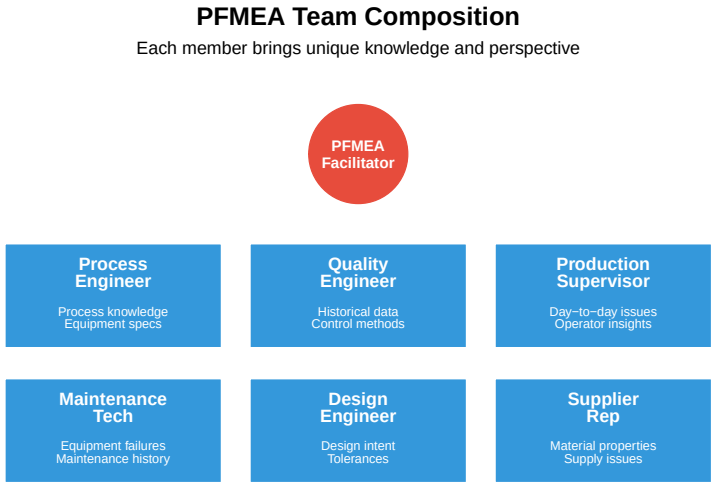


Figure 14.3: PFMEA Cross-Functional Team

Question: Why is a cross-functional team essential?

A cross-functional team is essential because:

1. **Diverse Knowledge:** No single person understands all aspects of the process
2. **Different Perspectives:** Quality sees defects, production sees constraints, maintenance sees failures
3. **Buy-in:** People support what they help create
4. **Historical Knowledge:** Operators often know about problems never formally documented
5. **System Thinking:** Process interactions are visible when multiple functions participate

Common Mistake: Having one engineer complete PFMEA alone as a "paperwork exercise." This misses critical failure modes and produces an ineffective document.

Best Practice: Minimum 4-6 team members, including at least one person with hands-on process experience.

14.3.3 Key Terminology

Understanding PFMEA terminology is essential for effective analysis:

Table 14.4: PFMEA Key Terminology

Term	Definition
****Process Function****	What the process step is supposed to do
****Failure Mode****	How the process can fail to perform its function
****Failure Effect****	The consequence of the failure on the customer or next operatio
****Severity (S)****	Rating (1-10) of how serious the failure effect is
****Failure Cause****	Why the failure mode might occur
****Occurrence (O)****	Rating (1-10) of how likely the cause is to happen
****Current Controls****	What currently prevents the cause or detects the failure
****Detection (D)****	Rating (1-10) of how likely current controls will detect the failur
****RPN****	Risk Priority Number = S × O × D
****Action Priority (AP)****	Alternative prioritization method (AIAG-VDA)

14.3.3.1 Try it yourself: mode, effect or cause?

Mixing up these three is the most common mistake on a PFMEA form. The failure mode is what goes wrong at the operation, the effect is what the customer or the next operation experiences, and the cause is why it happened. Sort these entries from an injection molding and a welding PFMEA.

Failure mode, effect or cause?

Decide what each entry is.

Categories: Failure mode / Effect / Cause

1. Short shot: the part is not completely filled. _____
2. Weld nugget undersized. _____
3. Hole drilled off location. _____
4. The part will not snap into the mating housing at the customer's assembly line. _____
5. The joint separates in service and the seat comes loose. _____
6. The customer's line stops and the lot is returned. _____
7. Melt temperature set too low. _____
8. Electrode tips worn beyond their change interval. _____
9. Locating pin worn, so the part sits crooked in the fixture. _____

Answers: 1 Failure mode; 2 Failure mode; 3 Failure mode; 4 Effect; 5 Effect; 6 Effect; 7 Cause; 8 Cause; 9 Cause.

14.3.4 The PFMEA Form

The PFMEA is documented in a structured form. Here is a simplified template:

Table 14.5: PFMEA Form Template (Simplified)

Process Step	Function	Failure Mode	Effect	S	Cause	O	Current Controls
Example:	Drill hole	Hole undersize	Assembly interference	7	Drill wear	5	Visual inspection

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14.4 The PFMEA Methodology: 10 Steps

PFMEA follows a systematic 10-step methodology. Each step builds on the previous one.

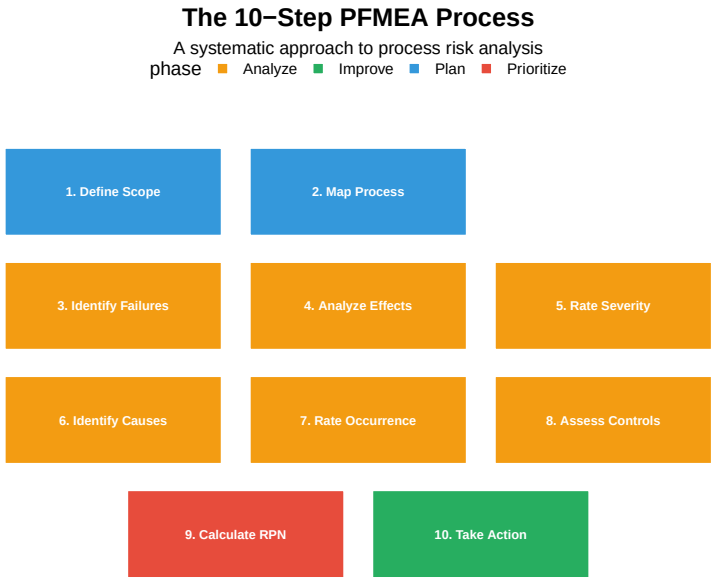


Figure 14.4: The 10-Step PFMEA Methodology

14.4.1 Step 1: Define Scope and Process Boundaries

Before starting the analysis, clearly define what is included and excluded.

Table 14.6: Step 1: Defining PFMEA Scope

Element	Description	Example: Injection M
****Process Name****	Clear name identifying the process	Plastic Part Injection Mo
****Start Boundary****	Where does our analysis begin?	Raw material loading int
****End Boundary****	Where does our analysis end?	Part removed from mold,
****Included Operations****	All operations within scope	Material prep, injection,
****Excluded Operations****	Operations analyzed separately or by others	Mold design (covered by
****Key Assumptions****	Conditions assumed to be true	Material meets incoming

14.4.2 Step 2: Process Mapping and Identifying Functions

Create a detailed process flow diagram and identify the function of each step.

Process Flow Diagram with Functions

1. **1. Load Material** (Function: Supply correct material)
2. **2. Heat & Plasticize** (Function: Melt to spec temp)
3. **3. Inject into Mold** (Function: Fill mold completely)
4. **4. Cool & Solidify** (Function: Solidify without defects)
5. **5. Eject Part** (Function: Remove without damage)

14.4.3 Step 3: Identifying Potential Failure Modes

For each process function, ask: **"How can this step fail to perform its intended function?"**

Table 14.7: Step 3: Identifying Failure Modes

Process Function	Potential Failure Mode	Category
Load correct material	Wrong material loaded	Wrong item
Load correct material	Material contaminated	Degraded item
Load correct material	Insufficient material	Missing item
Melt to specification temperature	Temperature too low	Below specification
Melt to specification temperature	Temperature too high	Above specification
Fill mold completely	Short shot (incomplete fill)	Incomplete operation
Fill mold completely	Flash (overfill)	Excessive operation
Fill mold completely	Air entrapment	Unintended result

Common failure mode categories: - **No operation** (step doesn't happen) - **Partial operation** (incomplete) - **Degraded operation** (below standard) - **Excessive operation** (too much) - **Wrong operation** (incorrect action) - **Intermittent operation** (inconsistent)

Interactive Exercise: Identify Failure Modes

Process Step: Heat treat steel parts at 850°C for 2 hours

Function: Achieve required hardness (58-62 HRC)

Identify at least 5 potential failure modes:

Answers: 1. Temperature too low → Parts soft (below hardness spec) 2. Temperature too high → Parts brittle/cracked 3. Time too short → Incomplete hardening 4. Time too long → Over-hardening, distortion 5. Uneven heating → Inconsistent hardness across part 6. Wrong atmosphere → Surface decarburization 7. Quench delay too long → Soft spots

14.4.4 Step 4: Analyzing Failure Effects

For each failure mode, determine: **"What is the consequence if this failure occurs?"**

Consider effects at multiple levels: 1. **Local effect:** Impact on immediate operation 2. **Next operation effect:** Impact on downstream process 3. **End customer effect:** Impact on final user

Table 14.8: Step 4: Analyzing Effects at Multiple Levels

Failure Mode	Effect Level	Effect Description
Short shot	Local	Part is incomplete/missing features
Short shot	Next Operation	Part rejected at inspection, line stoppage
Short shot	End Customer	If shipped: product malfunction, safety hazard
Flash	Local	Excess material on part edges
Flash	Next Operation	Secondary trimming required, increased cycle time
Flash	End Customer	If shipped: poor fit, appearance defect

14.4.5 Step 5: Determining Severity Ratings

Severity (S) rates the seriousness of the effect on a scale of 1-10.

Table 14.9: Severity Rating Scale (S)

Rating	Effect	Criteria
10	**Hazardous - without warning**	**Safety hazard; noncompliance with regulations; r**

9	**Hazardous - with warning**	**Safety hazard; noncompliance with reg**
8	Very High	Product inoperable; loss of primary function
7	High	Product operable but at reduced performanc
6	Moderate	Product operable; comfort/convenience affec
5	Low	Product operable; reduced comfort; noticed
4	Very Low	Product operable; slightly reduced comfort;
3	Minor	Fit/finish nonconforming; noticed by discrim
2	Very Minor	Fit/finish nonconforming; noticed only by tr
1	None	No discernible effect

Important: Severity is determined by the **effect**, not the failure mode. Severity can only be reduced by changing the design — process changes typically cannot reduce severity.

14.4.6 Step 6: Identifying Potential Causes

For each failure mode, identify all possible causes: **"Why might this failure occur?"**

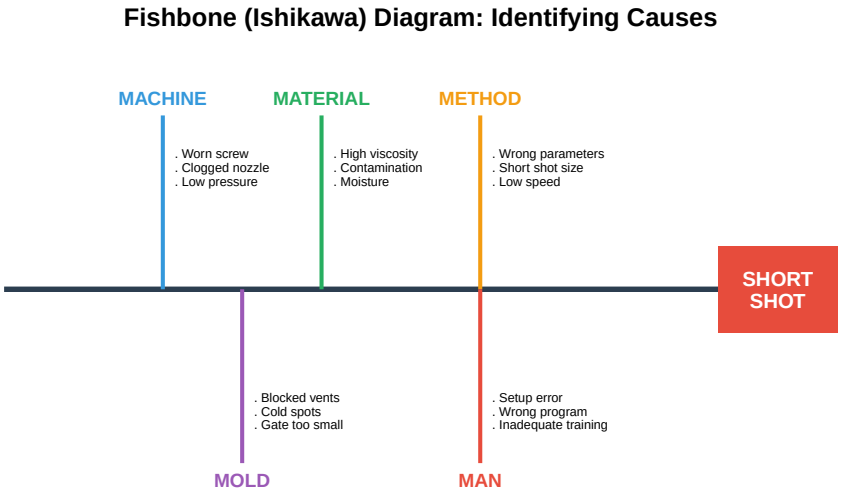


Figure 14.5: Fishbone Diagram: Causes of Short Shot

14.4.7 Step 7: Rating Occurrence

Occurrence (O) rates the likelihood that the cause will occur on a scale of 1-10.

Table 14.10: Occurrence Rating Scale (O)

Rating	Likelihood	Criteria	Approximate Rate	Cpk Equivalent
10	Very High	Failure is almost inevitable	100 per 1000 (10%)	< 0.33
9	Very High	Failures occur very frequently	50 per 1000 (5%)	0.33
8	High	Failures occur frequently	20 per 1000 (2%)	0.51
7	High	Failures occur on regular basis	10 per 1000 (1%)	0.67
6	Moderate	Occasional failures	5 per 1000 (0.5%)	0.83
5	Moderate	Occasional failures expected	2 per 1000 (0.2%)	1.00
4	Low	Relatively few failures	1 per 1000 (0.1%)	1.17
3	Low	Few failures	0.5 per 1000 (500 ppm)	1.33
2	Remote	Failure unlikely	0.1 per 1000 (100 ppm)	1.50
1	Remote	Failure nearly impossible	0.01 per 1000 (10 ppm)	1.67

14.4.8 Step 8: Current Controls and Detection Ratings

Identify current controls and rate their ability to detect the failure or cause.

Types of Process Controls:

Table 14.11: Types of Process Controls

Control Type	Purpose	Examples
****Prevention Controls****	Prevent the cause from occurring	Error-proofing (poka-yoke), interlo
****Detection Controls****	Detect the failure mode or cause after it occurs	Inspection, testing, gauging, SPC

Detection Rating Scale:

Table 14.12: Detection Rating Scale (D)

Rating	Detection	Criteria	Example Co
10	Absolute Uncertainty	No current control; cannot detect or is not analyzed	No inspection
9	Very Remote	Control will probably not detect	Indirect meas
8	Remote	Control has poor chance of detection	Visual inspect
7	Very Low	Control has low chance of detection	Manual 100%
6	Low	Control may detect	Variable gaug
5	Moderate	Control has moderate chance of detection	SPC charting
4	Moderately High	Control has moderately high chance of detection	In-station gau
3	High	Control has high chance of detection	Automated 10

2	Very High	Control almost certain to detect	M
1	Almost Certain	Control will detect; error-proofed	Er

Important: Detection ratings are often **harder to reduce** than occurrence ratings. Adding inspection doesn't prevent defects — it only catches them. Prevention is always better than detection.

Question: Why are detection ratings often harder to improve?

Detection ratings are harder to reduce because:

1. **Detection doesn't prevent:** The defect has already been made; you're just trying to find it
2. **Human inspection is unreliable:** Typical effectiveness is only 80-85% for visual inspection
3. **100% inspection is expensive:** Automated inspection requires significant investment
4. **Some defects are hard to detect:** Internal defects, intermittent problems, latent failures
5. **Sampling has limitations:** Even good sampling plans miss some defects

Better approach: Focus on reducing occurrence through: - Error-proofing (poka-yoke) - Process capability improvement - Preventive maintenance - Better training

14.4.9 Step 9: Calculating RPN and Prioritization

The **Risk Priority Number (RPN)** combines all three ratings:

$$\text{RPN} = \text{Severity (S)} \times \text{Occurrence (O)} \times \text{Detection (D)}$$

Teaching demo — try different Severity, Occurrence, and Detection ratings.

Show the live R code behind this card

```
#| label: rpn-calculation
# Example RPN Calculations

# Failure Mode 1: Short shot
S1 <- 7    # Severity: Product inoperable (assembly)
O1 <- 5    # Occurrence: Occasional
D1 <- 6    # Detection: Moderate (visual inspection)

RPN1 <- S1 * O1 * D1

# Failure Mode 2: Flash
S2 <- 4    # Severity: Appearance issue
O2 <- 6    # Occurrence: Regular
```

```
D2 <- 3    # Detection: High (easy to see)

RPN2 <- S2 * O2 * D2

# Failure Mode 3: Contamination
S3 <- 9    # Severity: Safety concern (medical device)
O3 <- 3    # Occurrence: Low
D3 <- 7    # Detection: Low (internal contamination)

RPN3 <- S3 * O3 * D3

# Comparison
modes <- c("Short shot", "Flash", "Contamination")
rpn <- c(RPN1, RPN2, RPN3)
ranking <- order(-rpn)

cat(sprintf("Highest priority: %s (RPN %g)\n", modes[ranking[1]], rpn[ranking[1]]))
cat("Priority order by RPN:\n")
for (k in seq_along(ranking)) {
  j <- ranking[k]
  value <- if (rpn[j] > 200) sprintf("!!%g!!", rpn[j]) else sprintf("%g", rpn[j])
  cat(sprintf("  %g. %s  RPN %s\n", k, modes[j], value))
}

par(mar = c(5, 5, 3, 1))
barplot(rpn, names.arg = modes, col = ifelse(rpn > 200, "#C0392B", "#2980B9"), ylim = c(0, 1000),
        ylab = "RPN", main = "RPN by failure mode (red = above 200)")
```

RPN Action Thresholds (Traditional):

Table 14.13: RPN Action Thresholds

RPN Range	Risk Level	Action Required
1-50	Low	Monitor; no immediate action
51-100	Moderate	Consider action if resources available
101-200	Significant	Action required within planning cycle
201-500	High	Immediate action required
501-1000	Critical	Stop production; immediate corrective action

Interactive Exercise: Calculate RPN

Calculate the RPN for each failure mode and rank them:

Failure Mode	Severity	Occurrence	Detection	RPN
A: Dimension out of spec	6	4	5	?
B: Surface scratch	3	7	2	?
C: Missing component	8	2	8	?
D: Wrong material	9	3	6	?

Answers: - A: $6 \times 4 \times 5 =$ **120** - B: $3 \times 7 \times 2 =$ **42** - C: $8 \times 2 \times 8 =$ **128** - D: $9 \times 3 \times 6 =$ **162**

Priority Order: D (162) > C (128) > A (120) > B (42)

Question: Should we always address the highest RPN first? See limitations discussion below.

14.4.10 Try it yourself: calculate and rank

A team has rated five failure modes on a brake caliper assembly line. Calculate the RPN for each, then rank them from 1 (highest RPN, first to be worked on) to 5 (lowest).

Calculate RPN and rank the failure modes

Fill in the RPN and its rank (1 = highest RPN).

Failure mode	S	O	D	RPN	Rank
Brake line fitting under-torqued	10	2	3		
Part label misprinted	3	6	6		
Cosmetic scratch on housing	2	8	8		
Porosity in bracket weld	7	4	4		
Wrong fastener installed	8	3	5		

Show the worked solution, and what is wrong with it

Failure mode	S	O	D	RPN	Rank
Cosmetic scratch on housing	2	8	8	128	1
Wrong fastener installed	8	3	5	120	2
Porosity in bracket weld	7	4	4	112	3
Part label misprinted	3	6	6	108	4
Brake line fitting under-torqued	10	2	3	60	5

Your arithmetic is right, and the ranking it produces should bother you. A cosmetic scratch that no customer will be hurt by is at the top of the list. The under-torqued brake line fitting, the only failure here with a severity of 10, is at the bottom, because it is rare and usually caught.

Ranking by RPN alone treats a point of severity as interchangeable with a point of detection. It is not. The next two sections explain the problem and the Action Priority method that was introduced to deal with it.

14.4.11 Limitations of RPN

RPN has known limitations that must be understood:

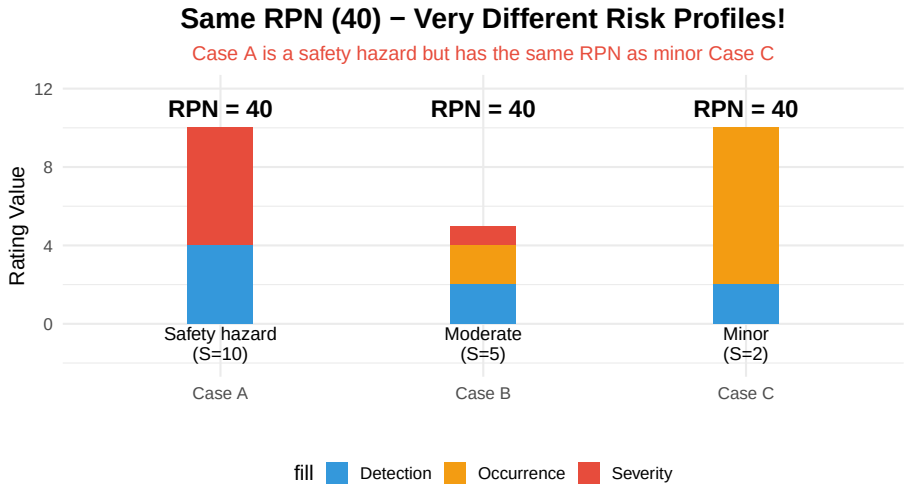


Figure 14.6: RPN Limitation Example

Key RPN Limitations:

1. **Same RPN, different risk:** $10\times1\times4 = 40$, but $2\times10\times2 = 40$
2. **Severity often overlooked:** High severity should always trigger action
3. **Arbitrary thresholds:** Why is 100 the magic number?
4. **Non-linear scale:** Difference between 2 and 3 is not same as 8 and 9

14.4.12 Action Priority (AP) Method — AIAG-VDA Alternative

The 2019 AIAG-VDA FMEA Handbook introduced the **Action Priority (AP)** method as an alternative to RPN.

Table 14.15: Action Priority (AP) Method

Priority	Criteria	Action Required
HIGH	Severity = 9 or 10 (any O, any D)	Must take action to improve preven
HIGH	Severity = 8 AND Occurrence 4	Must take action

MEDIUM	Severity = 7 AND Occurrence 5	Should take action
MEDIUM	Severity = 5-6 AND Occurrence 6 AND Detection 5	Should take action
LOW	All other combinations	May take action; docun

14.4.13 Step 10: Recommended Actions and Follow-Up

For high-priority items, develop specific recommended actions.

Table 14.16: Step 10: Recommended Ac

Failure Mode	Current RPN	Recommended Action	Respon
Short shot	**210**	Install shot size monitoring with automatic reject	Process
Short shot	**210**	Implement preventive maintenance for nozzle cleaning	Maintena
Short shot	**210**	Add mold temperature monitoring with alarm	Quality

---

14.5 Try it yourself: order the ten steps

The ten steps of the PFMEA method build on one another: you cannot rate what you have not yet identified. Put them in order.

The ten steps of a PFMEA

Number these in the correct order:

- _____ Analyze the effects of each failure
- _____ Calculate RPN and prioritize
- _____ Define the scope and process boundaries
- _____ Identify potential causes
- _____ Identify potential failure modes
- _____ List current controls and rate detection
- _____ Map the process and identify functions
- _____ Rate occurrence
- _____ Rate severity
- _____ Recommend actions and follow up

Answers, in the order listed: 4, 9, 1, 6, 3, 8, 2, 7, 5, 10.

14.6 Complete PFMEA Example: Injection Molding

Let's work through a complete PFMEA for an injection molding process step by step.

Table 14.17: Complete PFMEA Example: Injection Molding Proces

Step	Function	Failure Mode	Effect	S	Cause
3. Inject	Fill mold completely	Short shot	Part scrapped, line stoppage	7	Insufficient shot size
3. Inject	Fill mold completely	Flash	Secondary trim required	4	Excessive pressure
3. Inject	Fill mold completely	Air entrapment	Weak area, visual defect	6	Blocked vent
4. Cool	Solidify uniformly	Warpage	Dimensional rejection	7	Uneven cooling
4. Cool	Solidify uniformly	Sink marks	Visual defect	5	Thick section, short pa
5. Eject	Remove without damage	Stuck in mold	Part damage, mold damage	8	Insufficient draft angle

Your turn — change the baseline or improved S/O/D ratings below.

Show the live R code behind this calculator

```
#| label: rpn-improvement
# Improvement Analysis for Short Shot (Highest RPN = 210)

S_current <- 7  # Severity (cannot change - inherent to failure)
O_current <- 5  # Occurrence
D_current <- 6  # Detection
RPN_current <- S_current * O_current * D_current

# Option 1: Add automatic shot weight monitoring (improves Detection)
D_option1 <- 2
RPN_option1 <- S_current * O_current * D_option1

# Option 2: Add screw position monitoring with feedback control (improves Occurrence)
O_option2 <- 2
RPN_option2 <- S_current * O_option2 * D_current

# Option 3: Both improvements combined
RPN_option3 <- S_current * O_option2 * D_option1

cat(sprintf("Best option cuts RPN by %.0f%% (%d -> %d)\n",
            (1 - RPN_option3 / RPN_current) * 100, RPN_current, RPN_option3))
cat(sprintf("  Option 1: Detection only   RPN=%d (-%.0f%%)\n", RPN_option1, (1 - RPN_option1 / RP
cat(sprintf("  Option 2: Occurrence only  RPN=%d (-%.0f%%)\n", RPN_option2, (1 - RPN_option2 / RP
cat(sprintf("  Option 3: Both combined    RPN=%d (-%.0f%%)\n", RPN_option3, (1 - RPN_option3 / RP
```

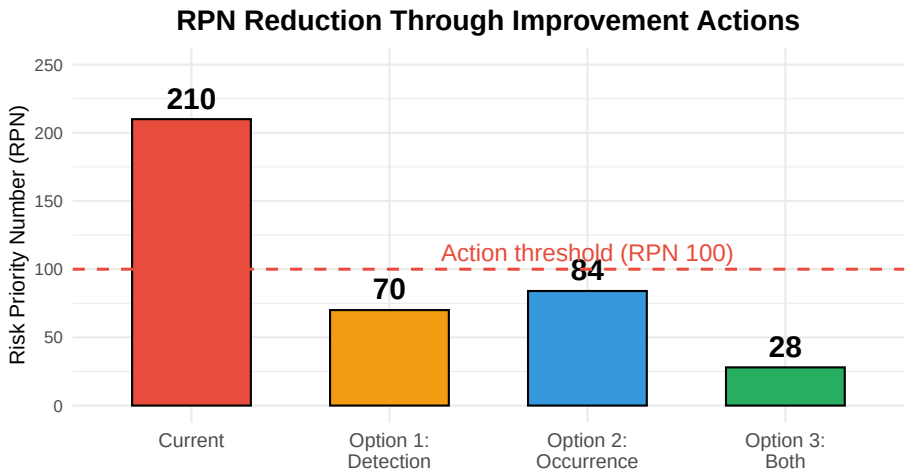


Figure 14.7: RPN Improvement Comparison

14.7 Industry-Specific PFMEA Examples

14.7.1 Example: Welding Process

Table 14.18: PFMEA Example: Welding Process

Failure Mode	Effect	S	Cause
Incomplete fusion	Weak joint, potential crack	9	Low heat input, contamination
Porosity	Leak path, strength reduction	8	Shielding gas issue, moisture
Undercut	Stress concentration, fatigue failure	7	Excessive current, wrong angle
Spatter	Appearance, cleanup required	3	Wrong parameters, contaminated
Distortion	Dimensional non-conformance	6	Excessive heat, improper sequenc

14.7.2 Example: Heat Treatment Process

Table 14.19: PFMEA Example: Heat Treatment Process

Failure Mode	Effect	S	Cause
Under-hardened	Premature wear, field failure	8	Low temp, short time
Over-hardened/Brittle	Part cracks under load	9	High temp, long time
Surface decarburization	Soft surface, wear	7	Wrong atmosphere

Distortion/Warpage	Dimensional rejection	6	Uneven heating, rapid quench	5	CMM s
Quench cracking	Scrap, safety hazard	10	Thermal shock, design	2	Visual

14.7.3 Example: Food Processing (Pasteurization)

Table 14.20: PFMEA Example: Pasteurization Proce

Failure Mode	Effect	S	Cause	O	C
Under-processing	Pathogen survival, food safety	10	Equipment malfunction, calibration	2	C
Over-processing	Nutrient loss, off-flavor	5	Control system error	3	T
Temperature deviation	Inconsistent kill, shelf life	8	Sensor drift, flow variation	3	C
Time deviation	Process variation	6	Timer malfunction	3	T
Cross-contamination	Product recall, illness	10	Gasket leak, CIP failure	2	C

Interactive Exercise: Create a PFMEA

Create a simple PFMEA for a coffee brewing process in a commercial setting.

Process Steps: 1. Fill water reservoir 2. Load coffee grounds 3. Brew coffee 4. Dispense into cup

Identify at least 4 failure modes with S, O, D ratings and RPN.

Example Answer:

Step	Failure Mode	Effect	S	Cause	O	Control	D	RPN
1. Fill water	Wrong water level	Weak/strong coffee	4	No level indicator	5	Visual check	5	100
2. Load grounds	Wrong amount	Weak/strong coffee	4	No measuring	6	Visual check	6	144
3. Brew	Temperature too low	Under-extracted	6	Heating element worn	3	Temp display	4	72
4. Dispense	Overflow	Burn hazard, mess	8	Cup sensor failure	2	Cup sensor	4	64

Priority: Wrong amount (144) > Water level (100) > Temperature (72) > Overflow (64)

14.8 Integration with Other Quality Tools

PFMEA doesn't exist in isolation — it connects to other quality tools and systems.

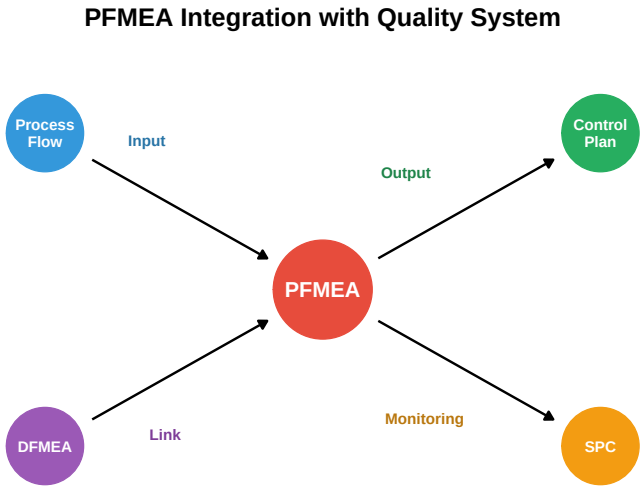


Figure 14.8: PFMEA Integration with Quality Tools

14.8.1 From PFMEA to Control Plan

The PFMEA directly feeds the **Control Plan** — a document that specifies what controls are needed at each process step.

Table 14.21: PFMEA to Control Plan Linkage

From PFMEA	To Control Plan
Process Step	Operation/Step
Failure Mode	Characteristic to Control
Severity	Classification (Critical/Significant)
Cause	Process Parameter
Current Control	Control Method
Detection	Reaction Plan

14.9 Common Pitfalls and Best Practices

14.9.1 Common Pitfalls to Avoid

Table 14.22: Common PFMEA Pitfalls and Solutions

Pitfall	Problem	Solution
Paperwork Exercise	PFMEA done only to satisfy customer requirement	Conduct meaningful a
One-Person Show	Engineer completes alone without team input	Require cross-functio
Optimistic Detection	Overestimating effectiveness of current controls	Validate detection eff
Static Document	Never updated after process changes	Review and update a
RPN Obsession	Focusing only on RPN numbers, missing high-severity items	Also consider severity
Vague Failure Modes	Failure modes too general to be actionable	Be specific: 'hole und

14.9.2 Best Practices

Table 14.23: PFMEA Best Practices

Practice	Description
Start Early	Begin PFMEA during process design, not after problems occur
Include Operators	Operators know failures that never get documented
Use Historical Data	Use warranty data, scrap reports, customer complaints
Be Specific	Specific failure modes lead to specific actions
Follow Up	Verify that actions were implemented and effective
Living Document	Review and update regularly (minimum annually)

14.10 Case Study: Preventing a Major Quality Issue

14.10.1 Background

Company: Automotive brake component manufacturer

Product: Brake caliper mounting bracket

Situation: New manufacturing process being implemented for a high-volume contract

14.10.2 PFMEA Analysis

During the PFMEA session, the team identified a critical failure mode:

Table 14.24: Case Study: Critical Failure Mode Identified

Element	Analysis
Process Step	Drill mounting holes
Function	Create holes at 45.0 ±0.1mm spacing
Failure Mode	Hole spacing out of tolerance
Effect	Caliper misalignment, brake performance affected
Severity	**9 (safety-related)**
Cause	Fixture wear, thermal expansion
Occurrence	4 (occasional)
Current Control	CMM check 5 per shift
Detection	5 (moderate sampling)
RPN	**180**

14.10.3 Actions Taken

Based on the PFMEA, the team implemented:

1. **In-process gauging:** Automated measurement after every part (D: 5 → 2)
2. **Fixture monitoring:** Temperature compensation and wear tracking (O: 4 → 2)
3. **Enhanced fixture design:** Carbide wear surfaces (O: 4 → 2)

Your turn — change the before/after ratings below.

Show the live R code behind this calculator

```
#| label: case-study-improvement
# Before improvements
S_before <- 9
O_before <- 4
D_before <- 5
RPN_before <- S_before * O_before * D_before

# After improvements
S_after <- 9    # Severity unchanged (inherent)
O_after <- 2    # Reduced through fixture improvements
D_after <- 2    # Reduced through 100% gauging
RPN_after <- S_after * O_after * D_after

cat(sprintf("RPN reduced by %.0f%% (%d -> %d)\n", (1 - RPN_after / RPN_before) * 100, I
```

```
cat(sprintf("  Before RPN           %d\n", RPN_before))
cat(sprintf("  After RPN            %d\n", RPN_after))
```

14.10.4 Results

- **Zero field failures** related to hole spacing after 18 months of production
- **Estimated avoided cost:** $2.4 million in potential recall and warranty
- **Investment:** $85,000 for gauging and fixture upgrades
- **ROI:** 28:1

Key Lesson: The PFMEA investment of $85,000 prevented a potential $2.4 million problem. This is the power of proactive risk management.

14.11 Summary

Table 14.25: PFMEA Key Concepts Summary

Topic	Key Points
PFMEA Purpose	Prevent defects before they occur; proactive vs. reactive quality
Key Ratings	Severity (effect), Occurrence (likelihood), Detection (control effectiveness)
RPN Calculation	RPN = S × O × D; Range 1-1000; Higher = More risk
Prioritization	Address high RPN first, but always act on high severity (9-10)
Action Priority	AIAG-VDA method: High/Medium/Low based on S-O-D combination
Best Practice	Cross-functional team, living document, follow-up on actions

---

14.12 Review Questions

14.12.1 Conceptual Questions

Question 1: What is the difference between a failure mode and a failure effect?

Failure Mode: How the process fails to perform its intended function - Example: “Hole drilled undersize” - Example: “Temperature too low” - Example: “Missing component”

Failure Effect: The consequence of the failure mode - Example: “Part rejected at assembly” (effect of undersize hole) - Example: “Product not properly cured” (effect of low temperature) - Example: “Product doesn’t function” (effect of missing component)

Key Relationship: One failure mode can have multiple effects at different levels (local, downstream, customer).

Question 2: Why should PFMEA be conducted by a cross-functional team?

Reasons for Cross-Functional Team:

1. **Diverse expertise:** No single person understands all aspects
2. **Multiple perspectives:** Quality sees defects differently than production
3. **Historical knowledge:** Operators know undocumented problems
4. **Buy-in:** People support what they help create
5. **System thinking:** Process interactions become visible
6. **Better solutions:** Different functions propose different improvements

Typical team composition: - Process Engineer (owner) - Quality Engineer - Production Supervisor/Operator - Maintenance Technician - Design Engineer - Supplier Representative (when applicable)

Question 3: Explain why detection ratings are often harder to reduce than occurrence ratings.

Detection is harder to reduce because:

1. **Detection doesn't prevent:** Defect already exists; just trying to find it
2. **Human inspection is unreliable:** 80-85% effectiveness typical
3. **100% inspection is expensive:** Requires automation investment
4. **Some defects are hidden:** Internal defects, intermittent failures
5. **Sampling limitations:** Cannot catch all defects with sampling

Better approach: Focus on occurrence reduction: - Error-proofing (poka-yoke) - Process capability improvement - Preventive maintenance - Better training

Example: Reducing drill wear (occurrence) is more effective than adding more inspection (detection).

Question 4: At what point in process development should PFMEA ideally be conducted?

Ideal Timing: During process design, before production begins

Why early is better: - Changes cost 1x during design - Changes cost 10x during production - Changes cost 100x after customer delivery

PFMEA Timing Points: 1. **New process development:** Before equipment is ordered 2. **New product introduction:** During process planning 3. **Process changes:** Before implementing changes 4. **Quality issues:** When problems indicate process risk 5. **Regular review:** Annually at minimum

Key Principle: The earlier PFMEA is conducted, the more opportunity exists to design out failures rather than inspect them out.

14.12.2 Calculation Questions

Question 5: Given S=8, O=6, D=4, calculate RPN. If improved controls reduce detection to 2, what is the new RPN?

Question 5: RPN

- Initial RPN: _____ *(to 0 decimal places)*
- RPN after detection improves to 2: _____ *(to 0 decimal places)*

Show the worked solution

Initial Calculation:

```
RPN = S × O × D
RPN = 8 × 6 × 4 = 192
```

After Detection Improvement:

```
New RPN = 8 × 6 × 2 = 96
```

Analysis: - Original RPN: 192 (Significant risk, action required) - New RPN: 96 (Moderate risk) - Reduction: 50%

Note: Severity (8) remains unchanged because it's determined by the effect, not by process controls. Only occurrence and detection can be reduced through process improvements.

Question 6: Two failure modes have RPNs of 120 and 150. Should the higher RPN always be addressed first?

Answer: Not necessarily!

Consider the following scenarios:

Scenario A: - Failure Mode 1: RPN = 150 (S=3, O=10, D=5) — Minor cosmetic issue - Failure Mode 2: RPN = 120 (S=10, O=3, D=4) — Safety hazard

Priority: Address Failure Mode 2 first despite lower RPN because: - Severity 10 indicates safety hazard - High-severity items should always be addressed regardless of RPN - The AP (Action Priority) method would classify this as HIGH priority

When to prioritize lower RPN: - Higher severity (especially 9-10) - Regulatory or safety implications - Customer-critical characteristic - Easier/faster to implement

Key Lesson: RPN is a prioritization tool, not the only decision criterion. Use engineering judgment.

14.12.3 Application Questions

Question 7: For a metal stamping operation, identify at least five potential failure modes and their likely effects.

Metal Stamping PFMEA:

#	Failure Mode	Effect
1	Part not fully formed	Dimensional rejection, functional failure
2	Burrs on edges	Injury hazard, assembly interference
3	Cracks/splits	Structural failure, scrap
4	Wrong material thickness	Dimensional/strength issues
5	Mislocated features	Assembly issues, rejection
6	Surface scratches	Appearance rejection, corrosion initiation
7	Springback out of tolerance	Dimensional rejection
8	Die mark/impression	Appearance defect

Potential Causes for "Part not fully formed": - Insufficient press tonnage - Die wear - Material too hard - Lubrication insufficient - Press speed too fast

Question 8: A process has RPN of 240 (S=8, O=6, D=5). Propose three improvement strategies and calculate resulting RPN.

Current State: S=8, O=6, D=5, RPN=240

Strategy 1: Improve Detection - Add automated inspection (100% gauging) - New D = 2 - New RPN = $8 \times 6 \times 2 = \mathbf{96}$ (60% reduction)

Strategy 2: Improve Occurrence - Add error-proofing (poka-yoke) to prevent cause - New O = 2 - New RPN = $8 \times 2 \times 5 = \mathbf{80}$ (67% reduction)

Strategy 3: Combined Approach - Error-proofing (O $\rightarrow$ 3) + Better detection (D $\rightarrow$ 3) - New RPN = $8 \times 3 \times 3 = \mathbf{72}$ (70% reduction)

Recommendation: Strategy 2 (error-proofing) is typically preferred because: - Prevents defects from occurring - More sustainable long-term - Reduces inspection costs - Better for customer confidence

14.12.4 Critical Thinking Questions

Question 9: Discuss the limitations of using RPN as the sole prioritization method.

RPN Limitations:

1. **Equal weighting problem:**
   - 10×1×4 = 40 (safety hazard, rare)
   - 2×10×2 = 40 (minor, frequent)
   - Same RPN, very different risk!
2. **Severity often overlooked:**
   - High-severity items (9-10) should always be addressed
   - RPN can hide safety-critical issues
3. **Non-linear scales:**
   - Difference between S=2 and S=3 is not same as S=8 and S=9
   - Multiplication assumes linear relationship
4. **Arbitrary thresholds:**
   - Why is RPN=100 the "action" threshold?
   - No scientific basis for specific cutoffs
5. **Gaming the system:**
   - Teams may manipulate ratings to avoid action
   - Subjectivity in ratings

Better Approaches: - Action Priority (AP) method - Always act on S 9 regardless of RPN - Consider multiple criteria (cost, feasibility, timing) - Use risk matrix visualization

Question 10: How would you handle a situation where team members disagree on severity ratings?

Approach to Rating Disagreements:

1. Clarify the failure effect: - Ensure everyone understands the same effect - Be specific about which customer sees which impact

2. Reference the rating criteria: - Use the standardized scale consistently - Point to specific criteria, not opinions

3. Consider worst-case scenario: - PFMEA should consider worst reasonable case - "Could this effect happen?" vs "Will it happen?"

4. Gather data: - Review historical data if available - What have similar failures caused in the past?

5. When in doubt, go higher: - Conservative approach is appropriate for safety - Can always reduce later with evidence

6. Document the discussion: - Record the rationale for the rating - Note if there was disagreement and why decision was made

7. Get external input: - Bring in customer perspective if needed - Consult subject matter experts

Final Rule: If a safety or regulatory issue is possible, severity should be 9 or 10 regardless of probability.

14.13 References

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Chapter 15

Lean Manufacturing

15.1 Learning Objectives

By the end of this chapter, you will be able to:

- Define Lean Manufacturing and explain its origins
- Identify and describe the 3 M's of Lean (Muda, Mura, Muri)
- Recognize and eliminate the seven types of waste
- Apply the 5S methodology to workplace organization
- Understand Just-in-Time (JIT) principles and Kanban systems
- Calculate Takt time and apply line balancing concepts
- Implement setup reduction techniques (SMED)

> "Perfection is not attainable. But if we chase perfection, we can catch excellence." — Vince Lombardi

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15.2 Introduction to Lean Manufacturing

Lean Manufacturing is the systematic elimination of waste. The term can be traced to Jim Womack, Daniel Jones, and Daniel Roos' book *The Machine that Changed the World* (1991), which examined the Toyota Production System.

What does "Lean" actually mean?

The term "Lean" refers to cutting "fat" from production activities — eliminating anything that doesn't add value to the customer. Think of it like a lean athlete: no excess weight, maximum efficiency, peak performance.

15.2.1 Definition of Lean

Table 15.1: Lean Manufacturing Targets (Womack, Jones, Roos 1990)

Metric	Lean Target
Human effort in factory	Half the hours
Defects in finished product	Half the defects
Engineering effort	One-third the hours
Factory space for same output	Half the space
In-process inventories	A tenth or less

15.3 Evolution of Manufacturing Systems

Manufacturing has evolved through three major paradigms. Understanding this evolution helps us appreciate why Lean emerged as the dominant philosophy.

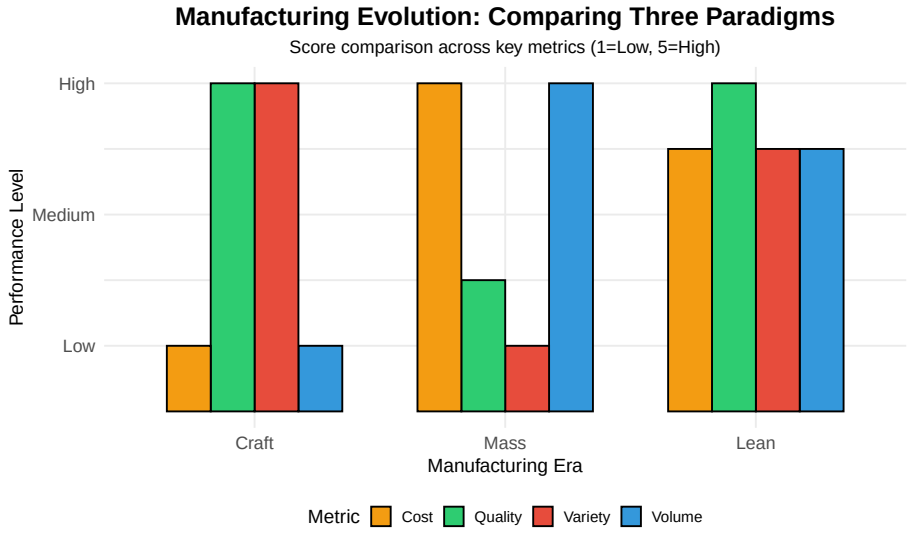


Figure 15.1: Evolution of Manufacturing Systems

15.3.1 Comparison of Manufacturing Systems

Table 15.2: Evolution of Manufacturing Systems

Characteristic	Craft Manufacturing	Mass Manufacturing	Le
Time Period	Late 1800s	1920s (Ford)	197
Process	Built on blocks, workers walk around	Assembly line	Cel
Workforce	Craftsmen with pride	Low skilled, simplistic jobs	Hig
Components	Hand-crafted, hand-fitted	Interchangeable parts	Int
Quality	Excellent	Lower	Exc
Cost	Very expensive	Affordable	Co
Volume	Few produced	Billions - identical	Hig
Flexibility	Very high	Very low	Hig

Discussion: Why did Mass Manufacturing dominate for so long?

Mass manufacturing dominated because:

1. **Economies of scale** - Higher volume = lower unit cost
2. **Predictable demand** - Post-war boom created stable markets
3. **Limited competition** - Few global competitors
4. **Consumer acceptance** - "Any color as long as it's black" (Henry Ford)

What changed? - Global competition (especially from Japan) - Customers demanded variety and quality - Product lifecycles shortened - Technology enabled flexibility

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15.4 The Toyota Production System (TPS)

The most influential lean manufacturing model is the **Toyota Production System (TPS)**, developed by Taiichi Ohno and Shigeo Shingo at Toyota.

15.4.1 Historical Context

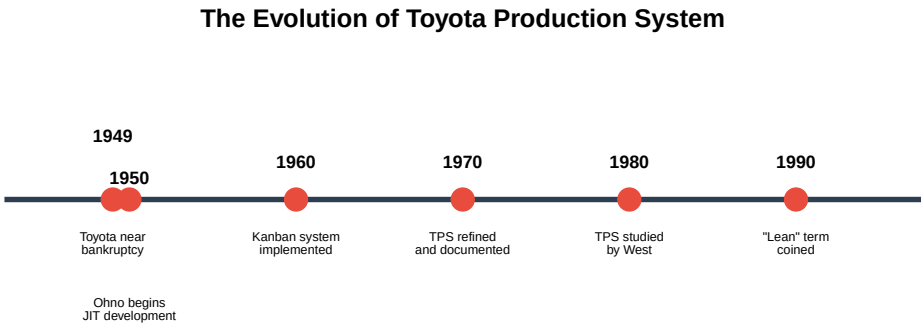


Figure 15.2: Toyota's Journey to Lean Excellence

In 1949, Toyota was on the brink of bankruptcy. Ford's car production was at least **8 times more efficient** than Toyota's.

> **Kiichiro Toyoda's Challenge:** "To achieve the same rate of production as the United States in three years."

Taiichi Ohno accepted this challenge. Inspired by American supermarkets, he developed the Just-in-Time method.

15.4.2 Two Pillars of TPS

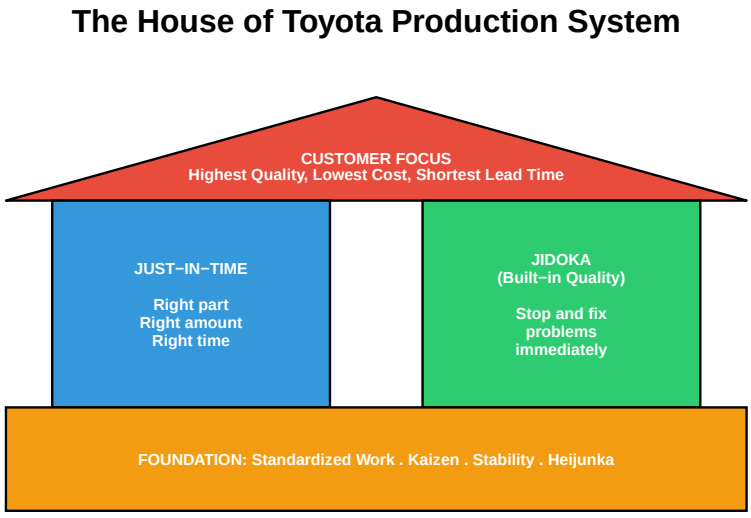


Figure 15.3: The Two Pillars of Toyota Production System

1. Just-in-Time (JIT): Produce only what is needed, when it is needed, in the amount needed.

2. Jidoka (Built-in Quality): Automation with a human touch — stop production immediately when a defect is detected.

15.5 The 3 M's of Lean

Lean manufacturing focuses on eliminating three types of inefficiency, known as the **3 M's**:

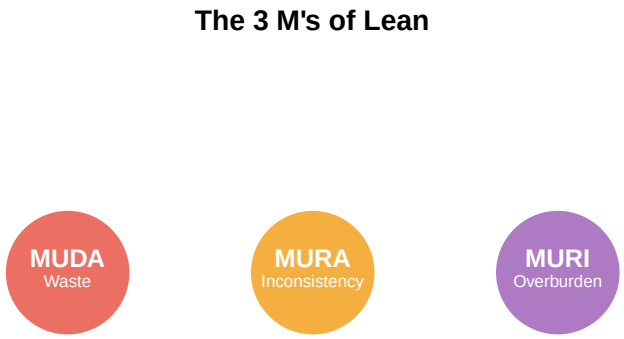


Figure 15.4: The 3 M's of Lean Manufacturing

Table 15.3: The 3 M's Explained

Japanese	English	Description
Muda ( )	Waste	Activities that consume resources without a
Mura ( )	Inconsistency/Unevenness	Variation and lack of uniformity in processe
Muri ( )	Overburden/Unreasonableness	Overburdening people or equipment beyond

---

15.6 The Seven Types of Waste (MUDA)

Shigeo Shingo identified seven categories of waste common to manufacturing. These are activities that add cost without adding value.

> "Waste is everything that is not absolutely essential." — Hiroyuki Hirano

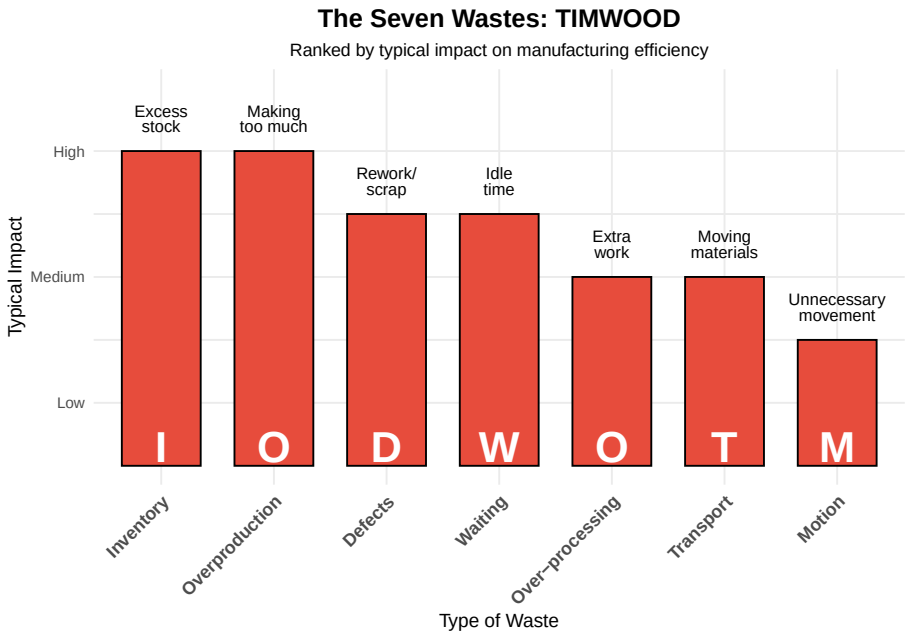


Figure 15.5: The Seven Types of Waste (TIMWOOD)

15.6.1 Detailed Breakdown of the Seven Wastes

Table 15.4: The Seven Wastes: Detailed

#	Waste Type	Description	Examples
1	**Overproduction**	Producing more than needed, earlier than needed	Building to fo
2	**Inventory**	Excess raw materials, WIP, or finished goods	Safety stock,
3	**Waiting**	Idle time waiting for materials, information, or equipment	Machine dow
4	**Transportation**	Unnecessary movement of materials between processes	Long distance
5	**Over-processing**	Doing more work than the customer requires	Tighter tolera
6	**Motion**	Unnecessary movement of people	Reaching, ben
7	**Defects**	Products that don't meet specifications	Scrap, rework

Memory Aid: TIMWOOD

Use **TIMWOOD** to remember the seven wastes:

- **T**ransportation
- **I**nventory
- **M**otion
- **W**aiting

- **O**verproduction
- **O**ver-processing
- **D**efects

Some people also use **DOWNTIME** (adding "Non-utilized talent" as an 8th waste).

Interactive Exercise: Identify the Waste

Scenario 1: A worker walks 50 feet to get a tool, uses it for 30 seconds, then walks back.

Answer: This is **Motion** waste. The solution would be to place tools at the point of use (5S).

Scenario 2: A machine produces 1000 parts, but only 800 are needed this week.

Answer: This is **Overproduction** waste. The solution would be to implement pull production based on actual demand.

Scenario 3: Parts sit in a queue for 3 days before being processed at the next station.

Answer: This is both **Inventory** and **Waiting** waste. The solution would be to balance the line and implement one-piece flow.

15.7 Try it yourself: spot the waste

Eight things you might see on a gemba walk. Name the waste in each one.

Which of the seven wastes?

Name the waste in each observation.

Categories: Transport / Inventory / Motion / Waiting / Overproduction / Over-processing / Defects

1. A forklift carries pallets 200 m from the press shop to assembly and returns empty. _____
2. Three weeks of stamped brackets sit on racks beside the weld cell. _____
3. An assembler walks to a shared toolbox for a torque wrench on every cycle. _____
4. An operator bends to floor level to pick each part out of a deep bin. _____
5. The paint line stands idle for 40 minutes because the upstream washer is being changed over. _____

6. The press keeps running to stay busy and makes 500 more panels than this week's orders. _____
7. A hidden bracket surface is polished to a mirror finish that the drawing does not ask for. _____
8. One housing in every twenty is reworked because of flash on the parting line. _____

Answers: 1 Transport; 2 Inventory; 3 Motion; 4 Motion; 5 Waiting; 6 Overproduction; 7 Over-processing; 8 Defects.

15.8 The 5S Methodology

5S is a workplace organization methodology that creates a clean, efficient, and safe working environment.



Figure 15.6: The 5S Methodology

Table 15.5: The 5S Steps Explained

Step	Japanese	English	Action	Key Question
1S	**Seiri**	Sort	Remove unnecessary items from the workplace	Do we need this?
2S	**Seiton**	Set in Order	Organize items so they are easy to find and use	Where does this belong?
3S	**Seiso**	Shine	Clean the workplace regularly	Is everything clean?
4S	**Seiketsu**	Standardize	Create standards for maintaining 1S-3S	How do we keep it this w

5S	**Shitsuke**	Sustain	Build discipline to maintain standards	Are we follow

15.8.1 Visual Factory

> "The ability to understand the status of a production area in **5 minutes or less** by simple observation without use of computers or speaking to anyone."

A well-implemented 5S creates a **visual factory** where problems are immediately apparent.

15.8.2 Try it yourself: put the 5S steps in order

5S only holds if the steps are done in sequence. Put them in order.

The five S's, first to last

Number these in the correct order:

- _____ Set in Order: a place for everything that remains
- _____ Shine: clean, and inspect while cleaning
- _____ Sort: remove what is not needed
- _____ Standardize: make the first three routine
- _____ Sustain: audit and keep it going

Answers, in the order listed: 2, 3, 1, 4, 5.

15.9 Just-in-Time (JIT) Production

15.9.1 The JIT Philosophy

> "Deliver the right material, in the exact quantity, with perfect quality, in the right place, just before it is needed." — Ohno and Shingo

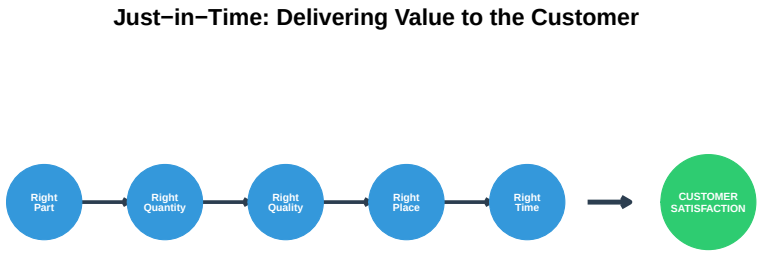


Figure 15.7: Just-in-Time: The Right Everything

15.9.2 Push vs. Pull Systems

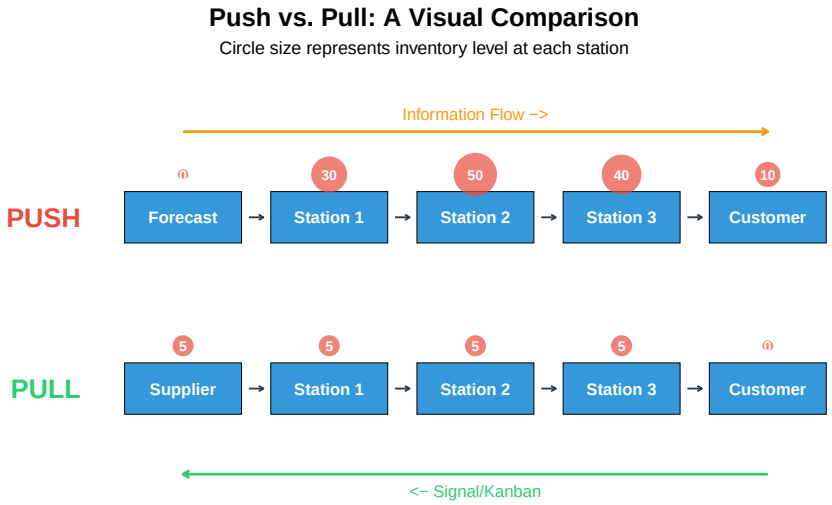


Figure 15.8: Push vs. Pull Production Systems

Table 15.6: Push vs. Pull System Comparison

Aspect	Push System	Pull System
Production trigger	Forecast/Schedule	Actual demand/Kanban signal
Information flow	Forward (upstream to downstream)	Backward (downstream to upstream)
Inventory levels	High (buffers at each station)	Low (minimal buffers)

Flexibility	Low	High
Problem visibility	Problems hidden by inventory	Problems immediately visible
Customer responsiveness	Slow	Fast

15.9.3 Try it yourself: push against pull

This is the dice game many lean courses play with poker chips, run for 200 days in a second. Four stations work in series. Each day every station can process between 1 and 6 parts, decided by a roll of a die, so the stations are balanced on average but never on the day.

The **push** line releases four new parts every morning, whatever is happening downstream. The **pull** line releases a part only when the total in the line is below a cap, which is what a kanban limit does. Both lines get the same dice.

Show the live R code behind this simulation

```
#| label: push-pull-sim
# The dice game: four stations in series, 200 days
wip_cap <- 12    # pull line: most parts allowed in the line at once
seed <- 1        # run number: same number, same dice

set.seed(seed)
n_days <- 200
n_stations <- 4
dice <- matrix(sample(1:6, n_days * n_stations, replace = TRUE), nrow = n_days)

run_line <- function(limit) {
  queue <- integer(n_stations)   # parts waiting at each station
  shipped <- 0
  wip <- numeric(n_days)
  for (day in seq_len(n_days)) {
    # Release work: push sends 4 a day regardless; pull tops the line up to its limit
    queue[1] <- queue[1] + if (is.finite(limit)) max(0, limit - sum(queue)) else 4
    for (s in n_stations:1) {
      moved <- min(queue[s], dice[day, s])
      queue[s] <- queue[s] - moved
      if (s < n_stations) queue[s + 1] <- queue[s + 1] + moved else shipped <- shipped
    }
    wip[day] <- sum(queue)
  }
  list(out = shipped / n_days, wip = wip)
}

push <- run_line(Inf)
```

```
pull <- run_line(wip_cap)
push_out <- push$out; push_wip <- mean(push$wip)
pull_out <- pull$out; pull_wip <- mean(pull$wip)

par(mar = c(4.5, 5, 2, 1), cex.main = 1.3, cex.lab = 1.2, cex.axis = 1.1)
plot(push$wip, type = "l", lwd = 3, col = "#e74c3c", las = 1, ylim = c(0, max(push$wip)),
     xlab = "Day", ylab = "Parts in the line", main = "Inventory in the line, day by day")
lines(pull$wip, lwd = 3, col = "#2980b9")
legend("topleft", legend = c("Push", "Pull"), col = c("#e74c3c", "#2980b9"), lwd = 3, bty = "n",

cat(sprintf("Pull: %.2f parts/day with %.0f parts in the line\n", pull_out, pull_wip))
cat(sprintf("  Push output            %.2f parts/day\n", push_out))
cat(sprintf("  Push inventory         %.0f parts on average, and still rising\n", push_wip))
cat(sprintf("  Pull output            %.0f%% of push\n", pull_out / push_out * 100))
cat(sprintf("  Pull inventory         %.0f%% of push\n", pull_wip / push_wip * 100))
cat(sprintf("  Pull lead time         %.1f days (inventory / output)\n", pull_wip / pull_out))
cat(sprintf("  Push lead time         %.1f days\n", push_wip / push_out))
```

Two things to notice. The push line's inventory never settles: it releases four a day into a line that ships a little over three, and the difference piles up. And a limit that is too tight starves the line, so the skill is in choosing a limit, not in having one.

15.9.4 Kanban System

Kanban ( ) is Japanese for "signboard" or "billboard." It's a scheduling system for lean and JIT production.

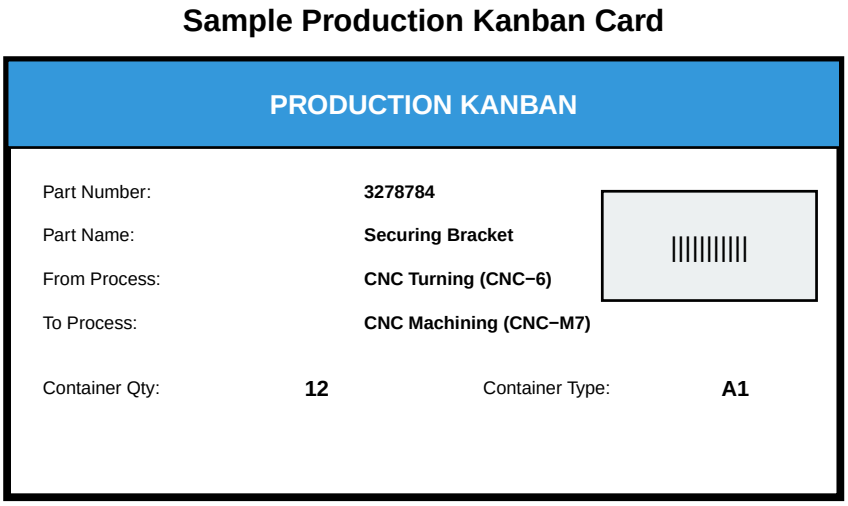


Figure 15.9: Example Kanban Card

15.10 Key Time Metrics

Understanding time metrics is essential for lean manufacturing analysis and improvement.

15.10.1 Definitions

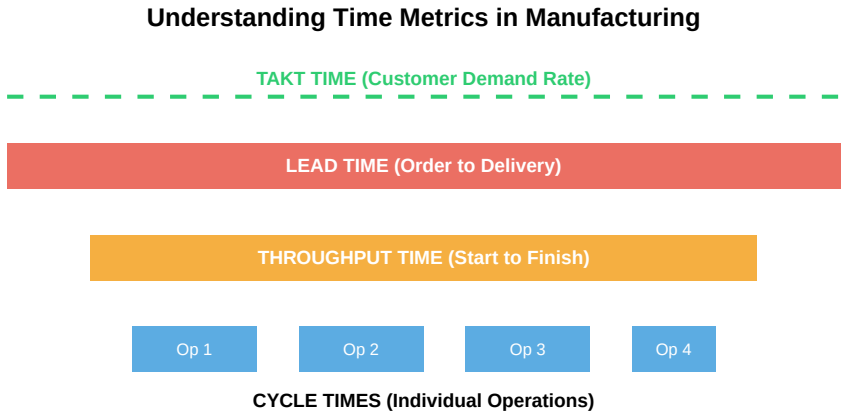


Figure 15.10: Relationship Between Time Metrics

15.10.2 Formulas and Calculations

Takt Time — The pace of production to meet customer demand:

$$\text{Takt Time} = \frac{\text{Available Production Time}}{\text{Customer Demand}}$$

Cycle Time — Time to complete one unit at a workstation:

$$\text{Cycle Time} = \frac{\text{Net Operating Time}}{\text{Units Produced}}$$

Throughput Time — Total time from start to finish:

$$\text{Throughput Time} = \text{Processing} + \text{Inspection} + \text{Queue} + \text{Move Time}$$

15.10.3 Interactive Example: Calculating Takt Time

Teaching demo — drag the shift length, the breaks and the demand and watch the takt time move.

Show the live R code behind this calculator

```
#| label: takt-calculation
# Given data
available_time_per_day <- 480     # 8 hours = 480 minutes
breaks_and_meetings <- 60         # 60 minutes for breaks/meetings
customer_demand <- 240            # 240 units per day

# Calculate net available time
net_available_time <- available_time_per_day - breaks_and_meetings

# Calculate Takt Time
takt_time <- net_available_time / customer_demand

cat(sprintf("Takt time: %.2f min per unit (%.1f sec)\n", takt_time, takt_time * 60))
cat(sprintf("  Net available time   %g min/day\n", net_available_time))

par(mar = c(5, 5, 3, 1))
demand_axis <- seq(10, 2000, by = 10)
plot(demand_axis, net_available_time / demand_axis * 60, type = "l", lwd = 2, col = "#2E86C1",
     xlim = c(10, 2000), ylim = c(0, 600), xlab = "Customer demand (units/day)",
     ylab = "Takt time (seconds per unit)", main = "Takt time against demand")
points(customer_demand, min(takt_time * 60, 600), pch = 19, cex = 2, col = "#C0392B")
```

Practice Problem: Calculate Your Takt Time

Given: - Shift length: 10 hours - Two 15-minute breaks and one 30-minute lunch - Customer requires 300 units per shift

Calculate: 1. Net available production time 2. Takt time in minutes 3. Takt time in seconds

Solution:

```
Net time = (10 × 60) − (15 + 15 + 30) = 600 − 60 = 540 minutes
Takt time = 540 / 300 = 1.8 minutes = 108 seconds per unit
```

15.11 Setup Reduction (SMED)

SMED (Single Minute Exchange of Dies) is a methodology developed by Shigeo Shingo to dramatically reduce changeover time.

15.11.1 The SMED Concept

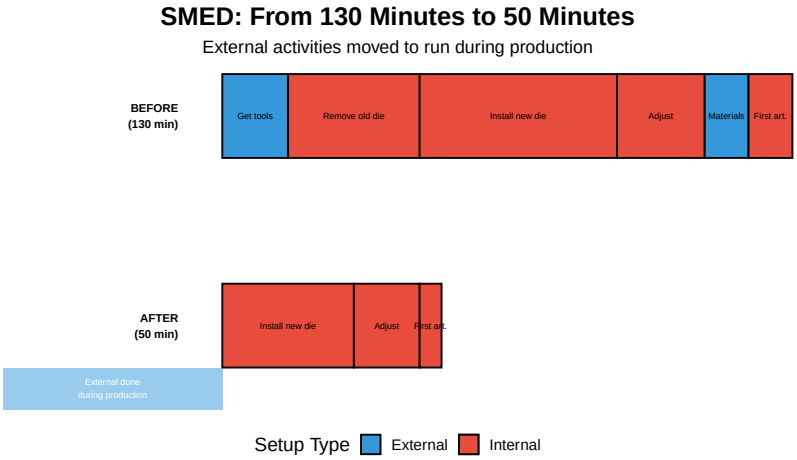


Figure 15.11: SMED: Internal vs External Setup

15.11.2 SMED Methodology

Table 15.7: The Four Steps of

Step	Action	Description
1	**Document current process**	Video the entire changeover, identify all steps
2	**Separate internal and external**	Internal = machine must be stopped; External =
3	**Convert internal to external**	Find ways to perform 'internal' tasks while machi

4	**Streamline all operations**	Reduce time for remaining internal tasks through better metho

Case Study: Press Setup Reduction

Problem: Presses were idle more than 50% of the time due to setup. - Average production run: 1-2 days - Setup time: 1-2 days

Root Cause Analysis: - Using standard bolt fasteners with open-end wrenches - No pre-staging of tools or dies - Sequential operations that could be parallel

Solutions Implemented:

Change	Time Saved
Socket sets + power tools	10%
Quick-release clamps	50%
Pre-staging materials	20%
Parallel operations	20%

Result: Setup time reduced from 1-2 days to 4-6 hours, with a target of under 1 hour.

15.12 Kaizen: Continuous Improvement

Kaizen ( ) means "change for the better" — the philosophy of continuous, incremental improvement.

15.12.1 Kaizen Principles



Figure 15.12: The Kaizen Continuous Improvement Cycle

15.12.2 Key Kaizen Concepts

- **Everyone participates** — from CEO to frontline workers
- **Small, incremental changes** — not major overhauls
- **Focus on process** — blame the process, not the person
- **Data-driven decisions** — measure before and after
- **Standardize improvements** — lock in gains

15.13 Value Stream Mapping (VSM)

Value Stream Mapping is a lean tool for analyzing the flow of materials and information required to bring a product to the customer.

15.13.1 VSM Symbols

Table 15.8: Key Metrics in Value Stream Maps

Metric	Description	Typical Location
Process Time (PT)	Time to actually process one unit	In process box
Lead Time (LT)	Total time from raw material to finished good	Timeline at bottom
Changeover Time (C/O)	Time to switch between products	In process box
Uptime	Percentage of time equipment is available	In process box
Takt Time	Required pace to meet demand	Noted on map
WIP Inventory	Units waiting between processes	Triangle symbol between processe

15.13.2 Try it yourself: read a value stream map

A supplier makes steering brackets in three steps and ships **500 brackets a day**. The current-state map shows the process time at each step and the inventory waiting in front of it. Convert each inventory triangle into days of demand, then total the timeline.

Days of inventory = pieces waiting ÷ daily demand. Lead time is the sum of those days; process time is the sum of the times in the boxes.

Current-state map: steering bracket

Fill in the days of inventory ahead of each step, then the two totals.

Box	Values	Leads to
Stamping	Process time = 2 s; Inventory ahead = 4 000 pcs; Days = _____	Welding
Welding	Process time = 40 s; Inventory ahead = 1 500 pcs; Days = _____	Assembly
Assembly	Process time = 60 s; Inventory ahead = 2 000 pcs; Days = _____	Shipping
Shipping	Inventory ahead = 1 000 pcs; Days = _____	-

- Total lead time: _____ days *(to 0 decimal places)*
- Total process (value-added) time: _____ sec *(to 0 decimal places)*

Show the worked solution

Step	Process time (s)	Inventory ahead (pcs)	Days of inventory
Stamping	2	4 000	8
Welding	40	1 500	3
Assembly	60	2 000	4
Shipping	-	1 000	2

Lead time = 8 + 3 + 4 + 2 = **17 days**. Process time = 2 + 40 + 60 = **102 seconds**.

A bracket spends 17 days in the plant and is worked on for under two minutes. That gap is why a value stream map puts the two numbers side by side: the improvement is in the inventory between the boxes, not in making any one process faster.

15.14 Summary

Table 15.10: Lean Manufacturing: Key C

Concept	Definition
Lean Manufacturing	Systematic elimination of waste
3 M's	Muda (waste), Mura (inconsistency), Muri (overburden)
7 Wastes	TIMWOOD: Transport, Inventory, Motion, Waiting, Overproduction, O
5S	Sort, Set in Order, Shine, Standardize, Sustain
JIT	Right part, right quantity, right quality, right place, right time
Kanban	Visual signal system for pull production
SMED	Single Minute Exchange of Dies - rapid changeover
Kaizen	Continuous incremental improvement

15.15 Review Questions

Question 1: What are the seven types of waste, and which is often considered the most impactful?

The seven types of waste (TIMWOOD) are: 1. **T**ransportation 2. **I**nventory 3. **M**otion 4. **W**aiting 5. **O**verproduction 6. **O**ver-processing 7. **D**efects

Inventory is often considered the most impactful because it hides other problems, ties up capital, and can become obsolete.

Question 2: Explain the difference between Push and Pull production systems.

Push System: - Production based on forecasts - Materials pushed through the system - Results in high inventory - Problems hidden by buffer stock

Pull System: - Production triggered by actual demand - Materials pulled by downstream operations - Minimal inventory - Problems immediately visible

Question 3: Calculate the Takt time for a facility with 450 minutes of available time and demand of 150 units.

Question 3: takt time

- Takt time: _____ min/unit *(to 2 decimal places)*
- Takt time: _____ sec/unit *(to 0 decimal places)*

Show the worked solution

$$\text{Takt Time} = \frac{450 \text{ minutes}}{150 \text{ units}} = 3 \text{ minutes per unit}$$

This means we need to produce one unit every 3 minutes (180 seconds) to meet customer demand.

Question 4: What is the difference between internal and external setup in SMED?

Internal Setup: Activities that can ONLY be performed when the machine is stopped (e.g., physically removing the old die).

External Setup: Activities that CAN be performed while the machine is still running (e.g., getting tools, pre-staging the next die, preparing materials).

The key to SMED is converting as many internal activities to external as possible.

15.16 References

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Part II

Process Engineering II

Chapter 16

Quality Management Systems

16.1 Learning Objectives

After completing this chapter, you will be able to:

1. Define quality management systems and explain their purpose in manufacturing
2. Describe the structure and requirements of ISO 9001:2015
3. Identify industry-specific quality standards (IATF 16949, AS9100, ISO 22000)
4. Explain the documentation hierarchy and control requirements
5. Conduct internal audits using systematic methodologies
6. Implement effective corrective and preventive action (CAPA) processes
7. Understand management review and continuous improvement requirements
8. Integrate quality tools (SPC, PFMEA, MSA) within a QMS framework

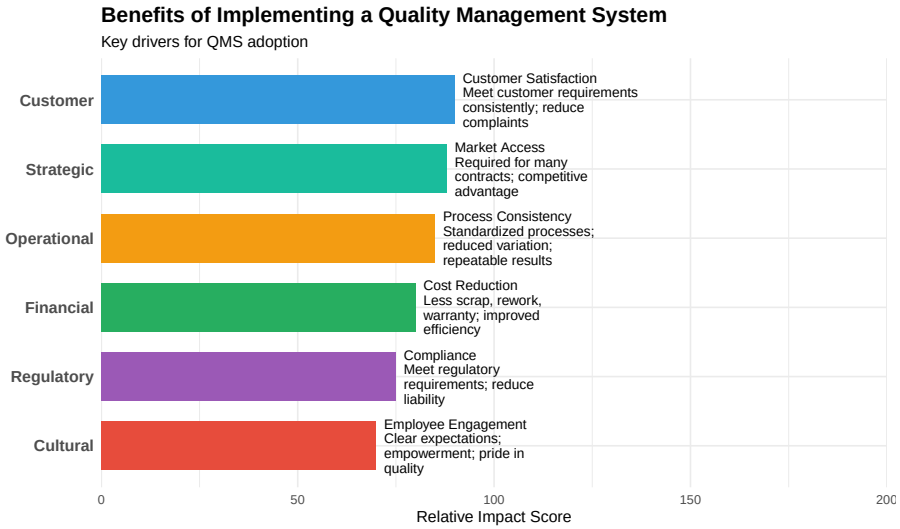
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16.2 Introduction to Quality Management Systems

A **Quality Management System (QMS)** is a formalized system that documents processes, procedures, and responsibilities for achieving quality policies and objectives. It coordinates and directs an organization's activities to meet

customer and regulatory requirements while continuously improving effectiveness.

16.2.1 Why Implement a QMS?



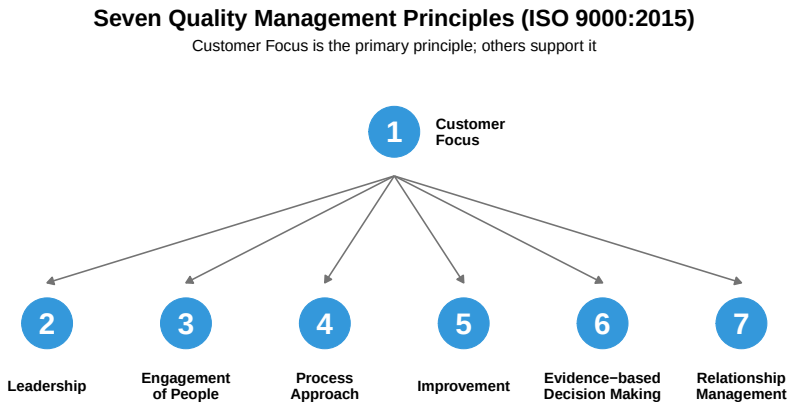
16.2.2 The Evolution of Quality Management

Table 16.1: Evolution of Quality Management

Era	Approach	Focus
Pre-1900s	Craftsmanship	Individual skill; master craftsmen
1900-1940s	Inspection	Sort good from bad; end-of-line inspection
1940-1960s	Statistical Quality Control	Control charts; sampling; Shewhart, Deming
1960-1980s	Quality Assurance	Prevention; systems approach; documented procedure
1980-2000s	Total Quality Management	Company-wide quality; customer focus; continuous im
2000s-Present	Integrated Management	Risk-based thinking; integration with business strateg

16.2.3 Quality Management Principles

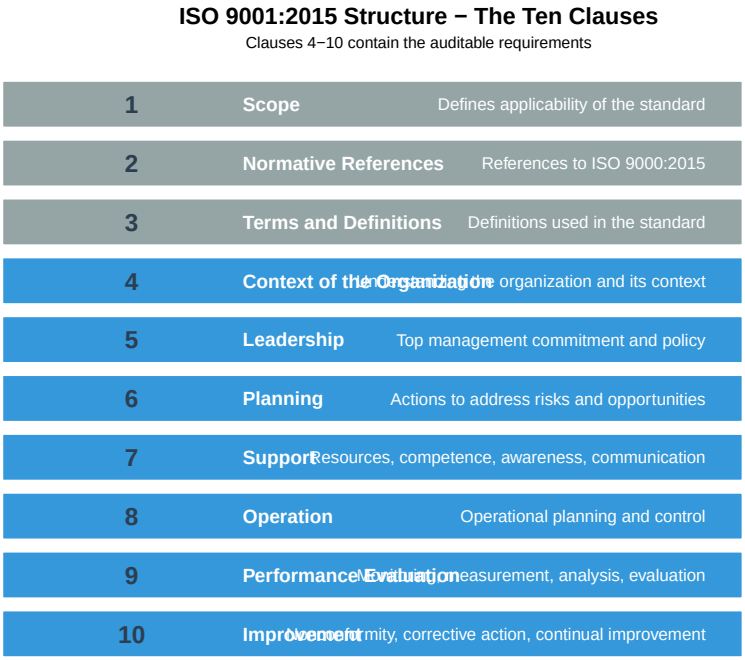
ISO 9000 defines seven quality management principles that form the foundation of all QMS standards:



16.3 ISO 9001:2015 Structure

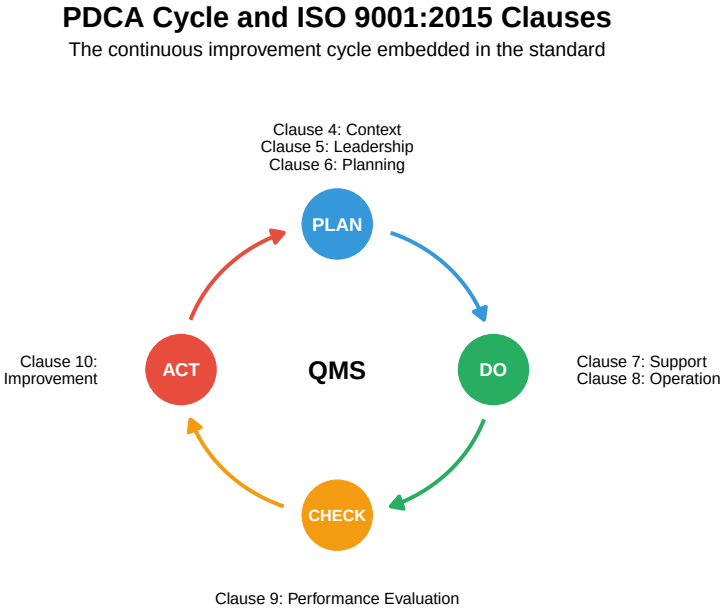
ISO 9001:2015 is the international standard for quality management systems. It uses the High-Level Structure (HLS) common to all ISO management system standards.

16.3.1 The Ten Clauses



16.3.2 The Process Approach and PDCA

ISO 9001:2015 emphasizes the **process approach** combined with the **Plan-Do-Check-Act (PDCA)** cycle:



16.3.3 Risk-Based Thinking

ISO 9001:2015 introduced **risk-based thinking** throughout the QMS:

Table 16.2: Risk-Based Thinking in ISO 9001:2015

Clause	Requirement	Risk_Application
4.1	Context of the Organization	Identify internal/external issues that could affect QMS ou
4.2	Interested Parties	Understand needs/expectations that could impact quality
6.1	Actions to Address Risks and Opportunities	Plan actions to address risks and opportunities; integrate
8.1	Operational Planning and Control	Control processes considering risks identified in planning
9.1	Monitoring and Measurement	Monitor effectiveness of actions taken to address risks
10.2	Nonconformity and Corrective Action	Analyze nonconformities; update risks/opportunities as ne

Understanding Risk-Based Thinking

Risk-based thinking doesn't require formal risk management (like FMEA for every process), but rather:

1. **Consider risks** when designing and implementing the QMS
2. **Preventive action** is built into the system, not a separate activity
3. **Proportional response** - more rigorous controls for higher-risk processes
4. **Opportunities** are also considered (not just negative risks)

Example: A food manufacturer identifies "allergen cross-contamination" as a significant risk. Risk-based thinking leads them to: - Dedicated production lines for allergen-containing products - Enhanced cleaning verification procedures - Staff training on allergen awareness - Increased inspection frequency after changeovers

The extent of controls is proportional to the risk level.

16.4 Industry-Specific Quality Standards

Different industries have developed standards that build upon ISO 9001 with additional sector-specific requirements.

16.4.1 Comparison of Major Standards



16.4.2 IATF 16949:2016 (Automotive)

IATF 16949 is the automotive quality standard, required by most major OEMs (Ford, GM, Toyota, VW, etc.).

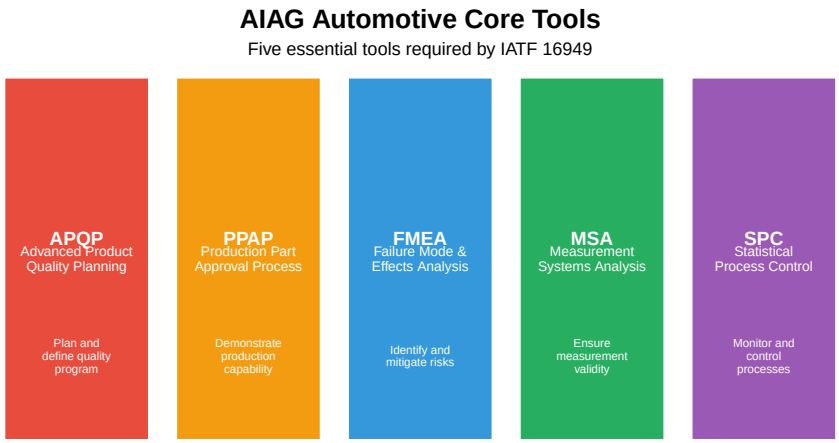
Table 16.3: IATF 16949 Additional Requirements Beyond ISO 9001

Requirement Area	ISO 9001	IATF 16949 Addition
Product Safety	Implied	Explicit requirements for product safety pro
APQP/PPAP	Not specified	APQP phases required; PPAP submission n
Control Plans	Not specified	Control plans required for all parts; specific

MSA	General requirement	MSA studies required per AIAG MSA manual
SPC	General requirement	SPC for all special characteristics
Supplier Quality	Basic purchasing	Supplier development; second-party audits; IATF certifi
Warranty Management	Complaint handling	NTF analysis; warranty data analysis; field return analy
Manufacturing Feasibility	Review of requirements	Documented manufacturing feasibility review

16.4.3 Automotive Core Tools

IATF 16949 requires the use of **five core tools**:



16.4.3.1 APQP: Advanced Product Quality Planning

APQP is a structured framework for developing products that ensures customer requirements are met through careful planning and defined milestones.

APQP Five Phases

1. **1. Plan & Define** – Voice of customer; Benchmarking; Product assumptions
2. **2. Product Design** – Design FMEA; DVP&R; Prototype build
3. **3. Process Design** – Process flow; Process FMEA; Control plan
4. **4. Product & Process Validation** – Production trial run; MSA & SPC studies; PPAP submission
5. **5. Feedback & Corrective Action** – Reduced variation; Customer satisfaction; Lessons learned

Example: New Brake Caliper Development

A Tier 1 supplier developing a new brake caliper for an OEM would follow APQP:

- **Phase 1**: Gather customer requirements (stopping distance, weight limits, environmental conditions), benchmark competitor products

- **Phase 2**: Design the caliper, create Design FMEA (what if piston seal fails?), build prototypes for testing
- **Phase 3**: Design manufacturing process, create Process FMEA, develop control plan for machining and assembly
- **Phase 4**: Run production trial, validate Cpk > 1.67 for critical dimensions, submit PPAP package
- **Phase 5**: Monitor warranty returns, implement kaizen improvements, document lessons for next program

16.4.3.2 PPAP: Production Part Approval Process

PPAP provides evidence that the supplier understands customer requirements and can consistently produce conforming parts during actual production runs.

Table 16.4: PPAP 18 Elements Required for Submission

Element	Description
1. Design Records	Part drawings, CAD files, specifications
2. Engineering Change Documents	Documentation of any design changes since original ap
3. Customer Engineering Approval	Evidence of customer sign-off on design
4. Design FMEA	Risk analysis of product design failure modes
5. Process Flow Diagram	Visual representation of manufacturing steps
6. Process FMEA	Risk analysis of process failure modes
7. Control Plan	Document describing controls for each process step
8. MSA Studies	Gage R&R studies proving measurement system adequ
9. Dimensional Results	Actual measurements vs. specifications for all dimensi
10. Material/Performance Test Results	Test results for material properties and performance r
11. Initial Process Studies (SPC)	Cpk/Ppk studies demonstrating process capability
12. Qualified Lab Documentation	Scope and accreditation of testing laboratories
13. Appearance Approval Report	Required for parts with color, grain, or surface require
14. Sample Production Parts	Parts from production run for customer evaluation
15. Master Sample	Retained reference sample for future comparison
16. Checking Aids	Fixtures, gages, or templates used to inspect parts
17. Customer-Specific Requirements	OEM-specific requirements (Ford, GM, Toyota, etc.)
18. Part Submission Warrant (PSW)	Summary document with supplier certification

PPAP Submission Levels:

Level	Requirements	When Used
1	PSW only	Low-risk parts, trusted suppliers
2	PSW + samples + limited data	Standard submission

Level	Requirements	When Used
3	PSW + samples + complete data	Default level; new parts/suppliers
4	PSW + other per customer direction	Customer-specified requirements
5	PSW + samples + complete data at supplier site	Full documentation review on-site

Example: Fuel Injector PPAP Submission

A supplier submitting PPAP for a new fuel injector would include:

- **Dimensional results**: 100% inspection of 300 parts showing all 47 dimensions in spec
- **MSA study**: Gage R&R $< 10\%$ for critical spray angle measurement
- **Process capability**: Cpk = 1.89 for nozzle hole diameter (spec requires Cpk 1.67)
- **Material certs**: Steel chemistry and hardness test results
- **Performance tests**: Flow rate, spray pattern, durability (1 million cycles)
- **PSW**: Signed warrant certifying all requirements met

16.4.3.3 FMEA: Failure Mode and Effects Analysis

FMEA is a systematic method for identifying potential failures, their causes, and effects, then prioritizing actions to reduce risk. Two types are used in automotive:

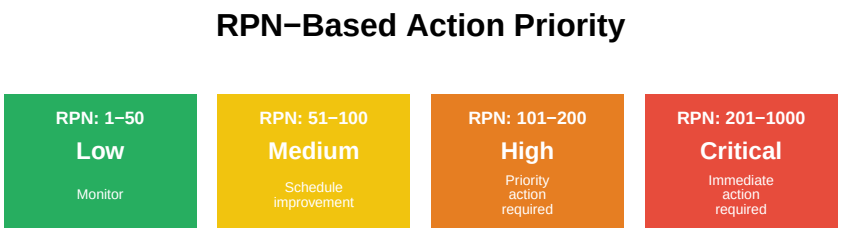
- **Design FMEA (DFMEA)**: Analyzes potential failures in product design
- **Process FMEA (PFMEA)**: Analyzes potential failures in manufacturing processes

Table 16.5: Process FMEA Structure and Example

Column	Description	Example
Process Step	The operation being analyzed	Torque fastener to 25 Nm
Potential Failure Mode	How the process could fail to meet requirements	Under-torque / Over-torq
Potential Effect(s)	Impact on customer if failure occurs	Joint loosens in service; w
Severity (S)	1-10 rating of effect seriousness (10 = safety/regulatory)	10 (safety critical)
Potential Cause(s)	Root cause of the failure mode	Worn tool, wrong program
Occurrence (O)	1-10 rating of cause likelihood (10 = very frequent)	4 (occasional occurrence)
Current Controls	Existing prevention or detection controls	Torque monitoring, error-p
Detection (D)	1-10 rating of detection ability (10 = cannot detect)	3 (high detection capabilit
RPN	S × O × D = Risk Priority Number	10 × 4 × 3 = 120

Recommended Actions	Actions to reduce S, O, or D	Add DC torq

Action Priority Based on RPN:



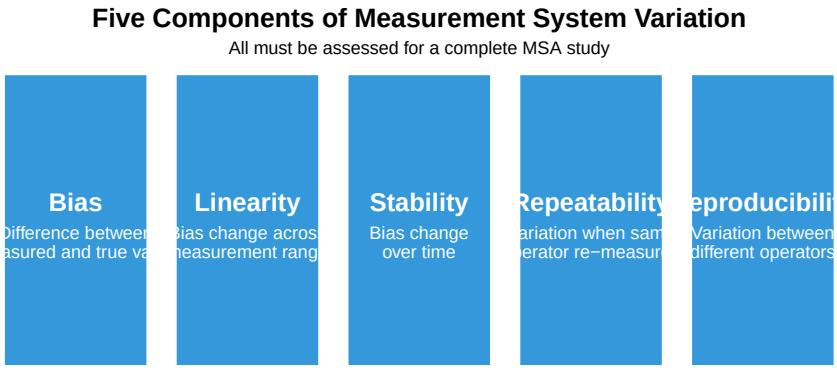
Example: Engine Block Machining PFMEA

Step	Failure Mode	Effect	S	Cause	O	Control	D	RPN	Action
Bore cylinder	Oversize bore	Piston slap, oil consumption	7	Tool wear	5	SPC chart, tool life monitor	4	140	Add in-process bore gage
Hone surface	Insufficient crosshatch	Poor ring seal, oil burning	8	Wrong hone stones	3	Visual inspection	6	144	Add surface roughness measurement
Wash block	Residual chips	Engine seizure	9	Blocked nozzle	4	Cleanliness spec check	5	180	Add particle counter, automated flush verification

16.4.3.4 MSA: Measurement Systems Analysis

MSA ensures that measurement systems provide reliable data for decision-making. Poor measurement systems can cause good parts to be rejected (false

rejects) or bad parts to be accepted (false accepts).



Gage R&R Study Acceptance Criteria:

%GRR	Interpretation	Action
$< 10\%$	Excellent	Acceptable for all applications
10-30%	Acceptable	May be acceptable depending on application, cost of gage, cost of repairs
$> 30\%$	Unacceptable	Measurement system must be improved; cannot make reliable decisions

Number of Distinct Categories (ndc): Should be 5 to adequately discriminate between parts.

Example: Gage R&R Study for Shaft Diameter

A supplier conducts a Gage R&R study for a micrometer used to measure shaft diameter (spec: 25.00 ± 0.05 mm):

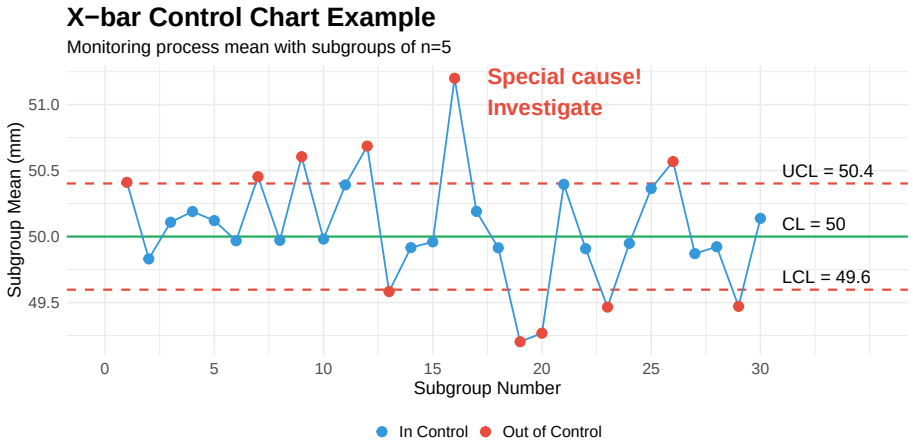
Study Setup: - 3 operators measure 10 parts, 3 times each = 90 measurements - Parts selected to represent the range of production variation

Results: | Source | Variance | % Contribution | |:——-|:———|:————| | Part-to-Part | 0.000156 | 87.2% | | Repeatability | 0.000018 | 10.1% | | Reproducibility | 0.000005 | 2.7% | | **Total Gage R&R** | **0.000023** | **12.8%** |

Conclusion: %GRR = 12.8% is acceptable (between 10-30%). ndc = 7 (5 required). The measurement system is adequate for this application.

16.4.3.5 SPC: Statistical Process Control

SPC uses control charts to monitor process stability and capability, distinguishing between common cause variation (inherent to the process) and special cause variation (assignable causes requiring investigation).



Common Control Chart Types:

Chart	Data Type	Monitors	When to Use
X-bar & R	Variable, subgroups	Mean and range	Most common; subgroups of 2-10
X-bar & S	Variable, subgroups	Mean and std dev	Subgroups > 10
I-MR	Variable, individuals	Individual values and moving range	Slow processes, destructive testing
p-chart	Attribute	Proportion defective	Variable sample sizes
np-chart	Attribute	Number defective	Fixed sample sizes
c-chart	Attribute	Count of defects	Fixed opportunity area
u-chart	Attribute	Defects per unit	Variable opportunity area

Process Capability Indices:

$$C_p = \frac{USL - LSL}{6\sigma}$$

$$C_{pk} = \min\left(\frac{USL - \bar{x}}{3\sigma}, \frac{\bar{x} - LSL}{3\sigma}\right)$$

Cpk Value	Interpretation	PPM Defective
< 1.00	Not capable	$> 2{,}700$
1.00 - 1.33	Marginally capable	64 - 2,700
1.33 - 1.67	Capable	0.6 - 64
1.67	Highly capable (IATF requirement)	< 0.6

Example: SPC for Injection Molded Connector

A supplier monitors critical dimensions on an injection molded electrical connector:

Process Setup: - Critical dimension: Pin spacing = 2.54 ± 0.08 mm - Sample 5 parts every 30 minutes - X-bar and R charts maintained at the press

Results over one shift: - Process mean (X-double-bar) = 2.539 mm - Average range (R-bar) = 0.024 mm - Estimated = R-bar / d2 = 0.024 / 2.326 = 0.0103 mm

Capability: - Cp = (2.62 - 2.46) / (6 × 0.0103) = 2.59 - Cpk = min[(2.62 - 2.539)/(3 × 0.0103), (2.539 - 2.46)/(3 × 0.0103)] = min[2.62, 2.56] = **2.56**

Conclusion: Cpk = 2.56 exceeds IATF requirement of 1.67. Process is highly capable.

16.4.4 AS9100D (Aerospace/Defense)

AS9100D adds aerospace-specific requirements to ISO 9001:

Table 16.6: AS9100D Key Additional Requirements

Area	Requirement
Configuration Management	Control product configuration throughout lifecycle; change management
First Article Inspection	FAI per AS9102; documented verification of first production parts
Counterfeit Parts Prevention	Controls to prevent counterfeit parts entering supply chain
Special Processes	NADCAP accreditation for special processes (welding, heat treat, NDT)
Product Safety	Product safety requirements; reporting of unsafe conditions
Risk Management	Explicit risk management process (often using AS/NZS 4360 or similar)
Human Factors	Consideration of human factors in process design

16.4.5 ISO 22000:2018 (Food Safety)

ISO 22000 integrates quality management with **HACCP** (Hazard Analysis Critical Control Points):

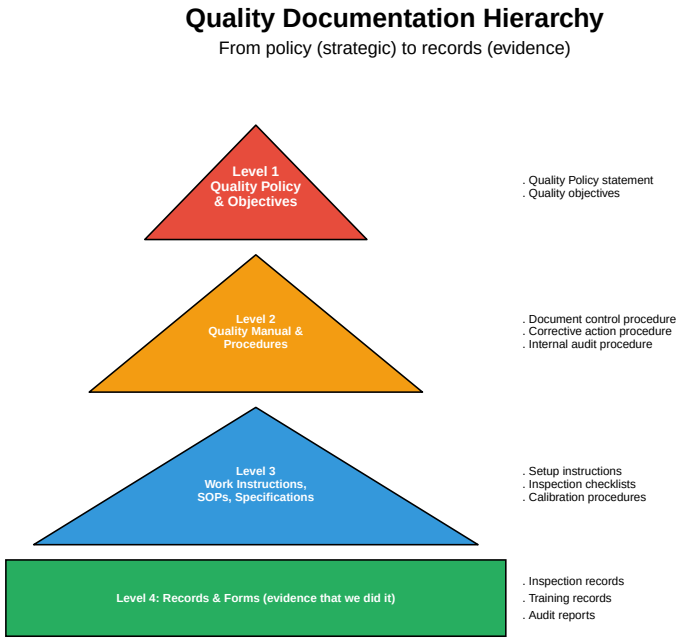
Table 16.7: ISO 22000 Food Safety M

Element	Description
Prerequisite Programs (PRPs)	Basic hygiene conditions: cleaning, pest control, personnel hygie
Operational PRPs (OPRPs)	PRPs essential to control specific identified hazards
HACCP Plan	Critical Control Points (CCPs) for significant hazards; critical li
Traceability	One-step-back, one-step-forward traceability; recall procedures
Emergency Preparedness	Procedures for food safety emergencies; recall/withdrawal
Validation and Verification	Validate control measures; verify HACCP plan effectiveness

16.5 Documentation Requirements

A QMS requires documented information to ensure consistency, provide evidence, and enable improvement.

16.5.1 Documentation Hierarchy



16.5.2 Document Control Requirements

Table 16.8: Document Control Requirements (ISO 9001 Clause 7.5)

Requirement	Description	Implementation
Approval	Documents reviewed and approved for adequacy before issue	Approval signature
Review & Update	Reviewed, updated as necessary, and re-approved	Periodic review sch
Identification	Identified with title, date, revision, author	Document numberi
Availability	Relevant versions available at points of use	Controlled copies a
Protection	Protected from loss, damage, unauthorized changes	Backup systems; ac
Obsolete Control	Obsolete documents identified and prevented from unintended use	Stamp 'OBSOLET
External Documents	External documents identified and distribution controlled	List of external doc
Record Retention	Records retained for specified periods; protected; retrievable	Retention matrix; s

16.5.3 Documented Information Required by ISO 9001:2015

Table 16.9: Documented Information Required by ISO 9001:2015

Type	Clause	Requirement
Documents (shall be maintained)		
Documents (shall be maintained)	4.3	Scope of the QMS
Documents (shall be maintained)	5.2	Quality policy
Documents (shall be maintained)	6.2	Quality objectives
Documents (shall be maintained)	7.1.5	Monitoring and measuring resources (calibration)
Documents (shall be maintained)	7.2	Evidence of competence
Documents (shall be maintained)	8.1	Evidence of conformity to processes
Documents (shall be maintained)	8.5.1	Controlled conditions for production
Records (shall be retained)		
Records (shall be retained)	7.1.5.1	Calibration/verification results
Records (shall be retained)	7.2	Training records
Records (shall be retained)	8.2.3.2	Results of review of requirements
Records (shall be retained)	8.3.3	Design and development inputs
Records (shall be retained)	8.3.4	Design and development controls
Records (shall be retained)	8.3.5	Design and development outputs
Records (shall be retained)	8.3.6	Design and development changes
Records (shall be retained)	8.5.2	Traceability requirements
Records (shall be retained)	8.6	Release of products/services
Records (shall be retained)	9.1.1	Results of monitoring and measurement

16.6 Internal Auditing

Internal audits are systematic, independent evaluations of the QMS to determine if it conforms to requirements and is effectively implemented.

16.6.1 The Audit Process

Internal Audit Process Flow

1. **1. Planning** – Schedule audits; assign auditors
2. **2. Preparation** – Review documents; prepare checklist
3. **3. Opening Meeting** – Explain purpose, scope, methods
4. **4. Audit Execution** – Collect evidence; interview; observe
5. **5. Closing Meeting** – Present findings; discuss
6. **6. Reporting** – Issue audit report
7. **7. Follow-up** – Verify corrective actions

16.6.2 Audit Checklist Example

Table 16.10: Sample Audit Checklist: Clause 7.1.5 Mo
Resources

Clause	Requirement	Question
7.1.5	Monitoring and measuring resources suitable for the type of activities	How do you determ
7.1.5	Resources maintained to ensure continued fitness for purpose	What maintenance
7.1.5	Documented information on fitness for purpose retained	Can you show me c
7.1.5	Measuring equipment identified to determine status	How can I tell if thi
7.1.5.2	Measurement traceability when required	What standards are

16.6.3 Types of Audit Findings

Finding Type	Definition
Major Nonconformity	Absence or total breakdown of a system; significant risk to produ
Minor Nonconformity	Single lapse or isolated incident; system exists but not fully effec
Observation	Weakness that could lead to nonconformity if not addressed
Opportunity for Improvement	Suggestion for enhancement; not a nonconformity

16.6.3.1 Try it yourself: grade the finding

You are closing out an internal audit. Grade each of these findings.

How serious is the finding?

Grade each audit finding.

Categories: Major Nonconformity / Minor Nonconformity / Opportunity for Improvement

1. There is no calibration system at all; none of the gauges on the floor has a record. _____
2. Nonconforming product was shipped to the customer because the final inspection step was skipped for a week. _____
3. No internal audits have been carried out for two years. _____
4. One of forty training records sampled is missing a signature. _____
5. One torque wrench out of thirty is a week past its calibration due date. _____
6. An obsolete revision of one work instruction is found at a single station. _____
7. The requirement is met, but the paper checklist could be replaced by a tablet form to save time. _____
8. Gauges are stored correctly, but a shadow board would make a missing one easier to spot. _____

Answers: 1 Major Nonconformity; 2 Major Nonconformity; 3 Major Nonconformity; 4 Minor Nonconformity; 5 Minor Nonconformity; 6 Minor Nonconformity; 7 Opportunity for Improvement; 8 Opportunity for Improvement.

16.6.4 Auditor Competence

Table 16.12: Internal Auditor Competency Requirements

Competency	Description	How_Developed
Knowledge of Standards	Understanding of ISO 9001 and applicable industry standards	Training courses; se
Audit Techniques	Planning, interviewing, evidence collection, reporting	Auditor training; sh
Industry Knowledge	Understanding of processes, products, and industry practices	Work experience; p
Communication Skills	Active listening, clear questioning, professional writing	Practice; feedback;
Objectivity	Independence from area being audited; impartial	Audit program desi
Professional Judgment	Ability to evaluate significance of findings	Experience; mentor

---

16.7 Corrective and Preventive Action (CAPA)

Corrective action eliminates the cause of a detected nonconformity to prevent recurrence. **Preventive action** eliminates the cause of a potential nonconformity to prevent occurrence.

16.7.1 The CAPA Process

CAPA Process Flow

1. **1. Identification** – Identify the nonconformity or potential issue
2. **2. Evaluation** – Assess significance; containment if needed
3. **3. Investigation** – Gather data and evidence
4. **4. Root Cause** – Determine root cause(s) using RCA tools
5. **5. Action Plan** – Define corrective / preventive actions
6. **6. Implementation** – Execute action plan; assign responsibility
7. **7. Verification** – Verify actions completed and effective
8. **8. Closure** – Document and close; share lessons learned

16.7.1.1 Step 1: Identification

The CAPA process begins when a nonconformity or potential problem is identified. Sources include:

Table 16.13: Common CAPA Initiation Sources

Source	Description	Priority
Customer Complaints	Product returns, warranty claims, customer feedback	High
Internal Audits	Nonconformities found during scheduled internal audits	Medium
Process Monitoring	SPC out-of-control conditions, trend analysis	Medium
Inspection/Testing	Failed inspections, test failures, rejected lots	High
Supplier Issues	Incoming inspection failures, supplier audit findings	Medium
Employee Reports	Near-misses, safety concerns, improvement suggestions	Low-Medium
Management Review	Issues identified during management review meetings	Medium
External Audits	Findings from certification body or customer audits	High

16.7.1.2 Step 2: Evaluation

Evaluate the significance of the issue to determine the appropriate response level:

Evaluation Criteria:

Factor	Questions to Consider
Safety	Could this harm people or the environment?
Customer Impact	How many customers affected? Severity of impact?
Regulatory	Does this violate regulations or standards?
Financial	What is the cost of the problem? Potential liability?
Frequency	Is this a one-time event or recurring pattern?
Detection	How was it found? Could others exist undetected?

Containment Actions - Immediate actions to limit the impact while the investigation proceeds:

- Quarantine suspect product
- Sort and inspect inventory
- Notify affected customers
- Implement temporary controls
- Stop shipments if necessary

Example: Evaluation of Dimensional Nonconformity

A batch of machined shafts is found with diameter 0.02 mm oversize:

- **Safety**: Low - not a safety-critical dimension
- **Customer Impact**: Medium - may cause assembly issues
- **Regulatory**: None
- **Financial**: 500 parts × \$15/part = \$7,500 potential scrap
- **Containment**: Quarantine batch, 100% inspection of last 3 lots, notify customer

Decision: Proceed with formal CAPA (cost exceeds \$5,000 threshold)

16.7.1.3 Step 3: Investigation

Gather all relevant data and evidence to understand what happened:

Investigation Activities:

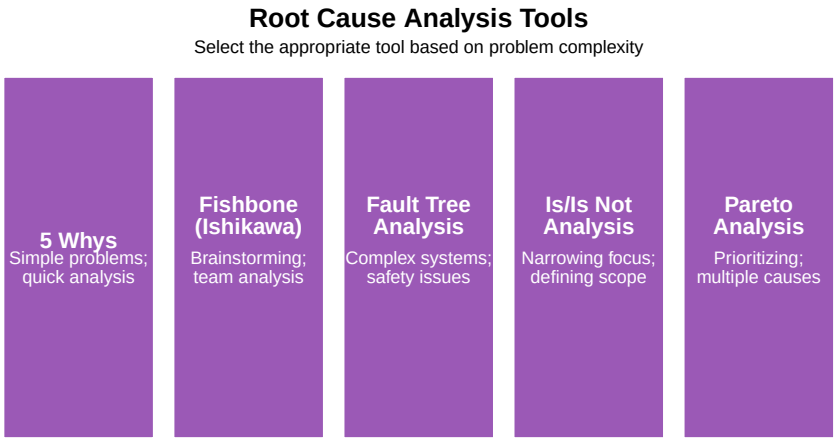
- **Review documentation**: Work instructions, control plans, inspection records
- **Examine physical evidence**: Failed parts, raw materials, tooling
- **Interview personnel**: Operators, inspectors, supervisors
- **Analyze data**: SPC charts, measurement data, process parameters
- **Timeline reconstruction**: When did the problem start? What changed?

Table 16.14: Data Collection for CAPA Investigation

Data Type	Examples	Collection Method
Process Data	Temperatures, pressures, speeds, feeds, cycle times	Data historians, SPC software, l
Material Data	Lot numbers, certificates, incoming inspection results	Traceability system, supplier rec
Equipment Data	Maintenance records, calibration status, tool wear	CMMS, calibration database
Personnel Data	Training records, shift assignments, experience level	HR system, training matrix
Environmental Data	Temperature, humidity, contamination levels	Environmental monitoring syste

16.7.1.4 Step 4: Root Cause Analysis

Determine the fundamental cause(s) of the problem using structured analysis tools:



The 5 Whys Example:

Level	Question	Answer
Problem	Shaft diameter is oversize	-
Why 1?	Why is the diameter oversize?	Tool is worn
Why 2?	Why is the tool worn?	Tool wasn't changed at scheduled interval
Why 3?	Why wasn't it changed?	Operator didn't know it was due
Why 4?	Why didn't operator know?	Tool life counter was reset incorrectly
Why 5?	Why was it reset incorrectly?	No verification step in setup procedure
Root Cause	**Setup procedure lacks tool counter verification step**	

The 6M Categories (Fishbone Diagram):

- **Man** (People): Training, skills, fatigue, attention
- **Machine**: Equipment condition, capability, maintenance
- **Material**: Raw material quality, specifications, storage
- **Method**: Procedures, work instructions, standards
- **Measurement**: Gages, calibration, measurement technique
- **Mother Nature** (Environment): Temperature, humidity, contamination

16.7.1.5 Step 5: Action Plan

Define specific corrective and/or preventive actions to address the root cause:

Table 16.15: Types of CAPA Actions

Action_Type	Definition	Example
Correction	Fix the immediate problem (symptom treatment)	Rework or scrap the nonconforming pa
Corrective Action	Eliminate the root cause to prevent recurrence	Revise setup procedure to include tool
Preventive Action	Eliminate potential causes before problems occur	Add tool counter verification to ALL n

SMART Action Planning:

- **S**pecific: Clearly defined action (not vague)
- **M**easurable: Can verify completion and effectiveness
- **A**ssignable: Single owner responsible
- **R**ealistic: Achievable with available resources
- **T**ime-bound: Due date specified

Example: SMART Corrective Action

Poor: "Improve the setup procedure"

SMART: "Quality Engineer (J. Smith) will revise Work Instruction WI-2345 to add Step 4.7: 'Verify tool life counter matches Tool Setup Sheet before starting production. Record counter reading on Setup Checklist Form F-102.' Revision to be completed by April 15, training completed by April 22."

16.7.1.6 Step 6: Implementation

Execute the action plan with clear ownership and tracking:

Implementation Requirements:

- Assign responsible person for each action
- Provide necessary resources (time, budget, authority)
- Communicate changes to affected personnel
- Update documentation (procedures, work instructions, forms)
- Train personnel on new requirements
- Implement changes in a controlled manner

Change Control Considerations:

Change Type	Approval Required	Documentation
Document revision	Document owner, Quality	Revision history, training records
Process change	Process owner, Engineering, Quality	Change request, validation if needed
Equipment change	Maintenance, Engineering, Quality	Equipment qualification, capability study

Change Type	Approval Required	Documentation
Product change	Engineering, Quality, Customer (if required)	ECN, customer approval, PPAP if applicable

16.7.1.7 Step 7: Verification

Verify that actions were completed AND effective:

Completion Verification: - Was the action actually implemented? - Is documentation updated and approved? - Are personnel trained? - Is the change being followed?

Effectiveness Verification: - Has the problem stopped recurring? - Do metrics show improvement? - Are there any unintended consequences?

Table 16.16: CAPA Effectiveness Verification Methods

Method	Description
Monitoring Period	Track for recurrence over defined period (e.g., 90 days, 3 production runs)
Statistical Analysis	Compare defect rates before/after; statistical significance testing
Audit Verification	Include in next internal audit; verify implementation
Customer Feedback	Monitor complaints related to this issue
Process Capability	Cpk study to confirm process improvement

16.7.1.8 Step 8: Closure

Complete documentation and share lessons learned:

Closure Requirements:

- All actions verified complete
- Effectiveness confirmed (no recurrence)
- Documentation updated and approved
- CAPA record completed with all evidence
- Management approval for closure

Lessons Learned:

- What was the root cause?
- Could this happen elsewhere? (Horizontal deployment)
- Should procedures/training be updated system-wide?
- Are there similar products/processes at risk?
- What can we do to prevent similar issues?

Example: Horizontal Deployment

The shaft diameter CAPA revealed that setup procedures lacked tool counter verification.

Horizontal deployment: Quality reviewed ALL 47 CNC machine setup procedures and found 12 others with the same gap. All procedures were revised, and the finding was shared at the monthly quality meeting as a lesson learned.

16.7.2 CAPA Effectiveness Criteria

This worked example is fixed as a reference — read through it once, then try the *editable* example further on in this chapter for yourself.

```
# CAPA Effectiveness Tracking
set.seed(42)

# Simulate CAPA data for a year
months <- c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
            "Jul", "Aug", "Sep", "Oct", "Nov", "Dec")

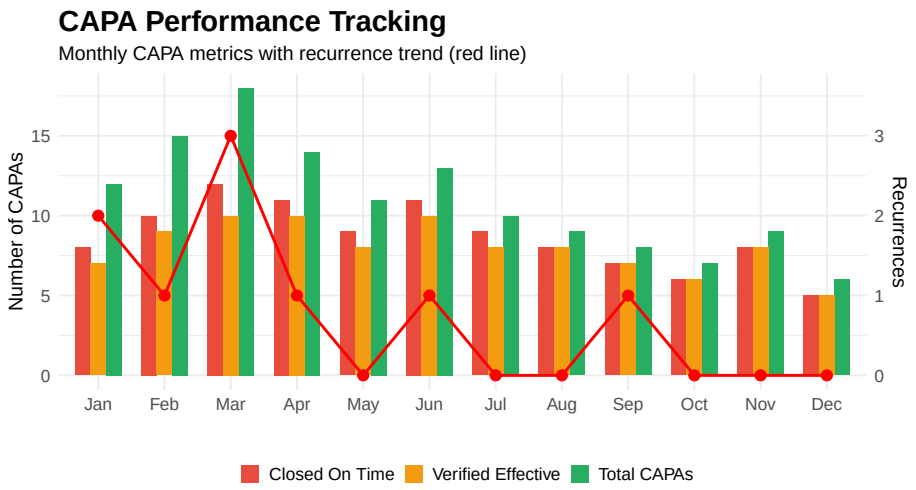
capa_data <- data.frame(
  Month = factor(months, levels = months),
  Total_CAPAs = c(12, 15, 18, 14, 11, 13, 10, 9, 8, 7, 9, 6),
  Closed_On_Time = c(8, 10, 12, 11, 9, 11, 9, 8, 7, 6, 8, 5),
  Effective = c(7, 9, 10, 10, 8, 10, 8, 8, 7, 6, 8, 5),
  Recurrence = c(2, 1, 3, 1, 0, 1, 0, 0, 1, 0, 0, 0)
)

# Calculate metrics
capa_data$On_Time_Rate <- round(capa_data$Closed_On_Time / capa_data$Total_CAPAs * 100, 1)
capa_data$Effectiveness_Rate <- round(capa_data$Effective / capa_data$Closed_On_Time * 100, 1)

cat("CAPA Performance Metrics Summary:\n")
cat("                  \n")
cat("Total CAPAs initiated:", sum(capa_data$Total_CAPAs), "\n")
cat("Closed on time:", sum(capa_data$Closed_On_Time), "(",
    round(sum(capa_data$Closed_On_Time)/sum(capa_data$Total_CAPAs)*100, 1), "%)\n")
cat("Verified effective:", sum(capa_data$Effective), "(",
    round(sum(capa_data$Effective)/sum(capa_data$Closed_On_Time)*100, 1), "%)\n")
cat("Recurrences:", sum(capa_data$Recurrence), "\n")
```

```
CAPA Performance Metrics Summary:

Total CAPAs initiated: 132
Closed on time: 104 ( 78.8 %)
Verified effective: 96 ( 92.3 %)
Recurrences: 9
```



16.8 Management Review

Management review is a formal evaluation of the QMS by top management to ensure its continuing suitability, adequacy, and effectiveness.

16.8.1 Management Review Inputs

Table 16.17: Managem

Required Input	Details to Review
Previous Review Actions	Status of actions from previous management reviews
Changes in Issues/Context	Changes in external/internal issues; interested party requirements
Quality Performance	Customer satisfaction; quality objectives achievement; process perfo
Resource Adequacy	Adequacy of resources for QMS maintenance and improvement
Risk/Opportunity Actions	Effectiveness of actions taken to address risks and opportunities
Improvement Opportunities	Opportunities for improvement identified from various sources

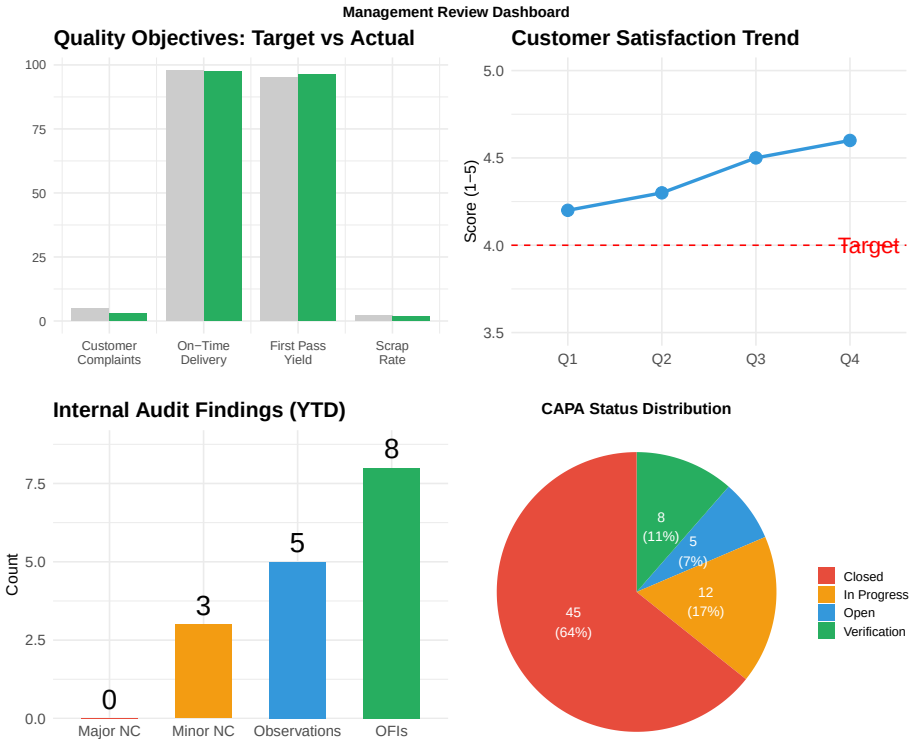
16.8.2 Management Review Outputs

Table 16.18: Management Review Outputs

Required Output	ISO Requirement	Examples
Improvement Decisions	Decisions and actions related to improvement opportunities	Launch ka
QMS Changes	Any need for changes to the QMS	Revise cor

Resource Needs	Any resource needs	Hire additional quality

16.8.3 Sample Management Review Dashboard



16.9 Continuous Improvement

Continuous improvement is a fundamental principle of quality management - the ongoing effort to improve products, services, and processes.

16.9.1 Improvement Methodologies

Table 16.19: Continuous Improvement Methodolog

Methodology	Origin	Best_For	Steps
PDCA	Deming/Shewhart	General improvement cycle; process changes	Plan-Do-Check-A
DMAIC	Six Sigma	Data-driven problem solving; reducing variation	Define-Measure-A
Kaizen	Toyota/Lean	Small, incremental improvements; team-based	Identify waste →

A3 Problem Solving	Toyota	Visual problem solving; root cause analysis	Back
8D	Ford	Customer complaints; complex problems	D0-I

16.9.2 Key Performance Indicators (KPIs)

Table 16.20: Common Quality Management KPIs

Category	KPI	Formula
Quality Metrics		
Quality	First Pass Yield (FPY)	(Good units first time / Total units) × 100%
Quality	Customer Complaints (PPM)	(Complaints / Units shipped) × 1,000,000
Quality	Internal Reject Rate	(Rejected units / Total inspected) × 100%
Quality	Cost of Quality	Prevention + Appraisal + Internal Failure + External
Delivery Metrics		
Delivery	On-Time Delivery (OTD)	(Orders delivered on time / Total orders) × 100%
Delivery	Lead Time	Order date to delivery date
Delivery	Order Fulfillment Rate	(Complete orders shipped / Total orders) × 100%
Cost Metrics		
Cost	Scrap Rate	(Scrap value / Total production value) × 100%
Cost	Rework Cost	Labor + Material for rework
Cost	Warranty Cost	Warranty claims cost / Revenue
Safety Metrics		
Safety	Recordable Incident Rate	(Recordable incidents × 200,000) / Hours worked
Safety	Near Miss Reports	Count of near miss reports

16.9.3 Cost of Quality (COQ)

```
# Cost of Quality Analysis
coq_data <- data.frame(
  Category = c("Prevention", "Appraisal", "Internal Failure", "External Failure"),
  Description = c("Quality planning, training, process control",
                  "Inspection, testing, audits, calibration",
                  "Scrap, rework, reinspection, downtime",
                  "Warranty, returns, complaints, recalls"),
  Cost = c(50000, 75000, 120000, 180000)
)

total_coq <- sum(coq_data$Cost)
revenue <- 10000000  # $10M revenue

coq_data$Percent_of_COQ <- round(coq_data$Cost / total_coq * 100, 1)
coq_data$Percent_of_Revenue <- round(coq_data$Cost / revenue * 100, 2)
```

```
cat("Cost of Quality Analysis:\n")
cat("                \n")
print(coq_data[, c("Category", "Cost", "Percent_of_COQ")])
cat("                \n")
cat("Total Cost of Quality: $", format(total_coq, big.mark = " ", scientific = FALSE), "\n")
cat("COQ as % of Revenue:", round(total_coq/revenue*100, 1), "%\n\n")

# Analysis
conformance <- coq_data$Cost[1] + coq_data$Cost[2]
nonconformance <- coq_data$Cost[3] + coq_data$Cost[4]

cat("Conformance Costs (Prevention + Appraisal): $", format(conformance, big.mark = " ", scientif
    "(", round(conformance/total_coq*100, 1), "%)\n")
cat("Nonconformance Costs (Failures): $", format(nonconformance, big.mark = " ", scientific = FAI
    "(", round(nonconformance/total_coq*100, 1), "%)\n")
cat("\nRecommendation: Increase prevention spending to reduce failure costs\n")
```

```
Cost of Quality Analysis:

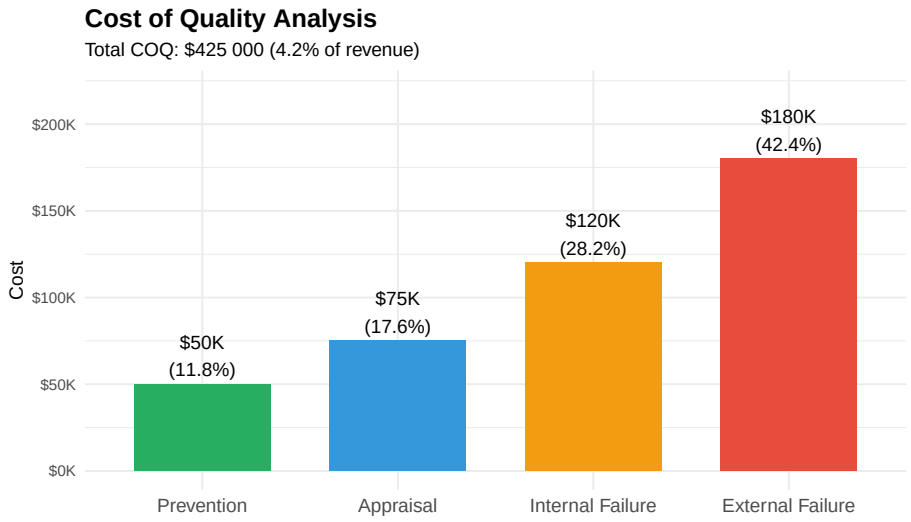
          Category   Cost Percent_of_COQ
1       Prevention  50000           11.8
2        Appraisal  75000           17.6
3 Internal Failure 120000           28.2
4 External Failure 180000           42.4

Total Cost of Quality: $ 425 000
COQ as % of Revenue: 4.2 %

Conformance Costs (Prevention + Appraisal): $ 125 000 ( 29.4 %)
Nonconformance Costs (Failures): $ 300 000 ( 70.6 %)

Recommendation: Increase prevention spending to reduce failure costs
```

```
Warning: Removed 1 row containing missing values or values outside the scale range
(`geom_text()`).
Removed 1 row containing missing values or values outside the scale range
(`geom_text()`).
```



16.9.4 Try it yourself: classify the cost

Eight lines from a plant's accounts. Decide which Cost of Quality category each belongs in.

Which Cost of Quality category?

Pick the Cost of Quality category for each line.

Categories: Prevention / Appraisal / Internal Failure / External Failure

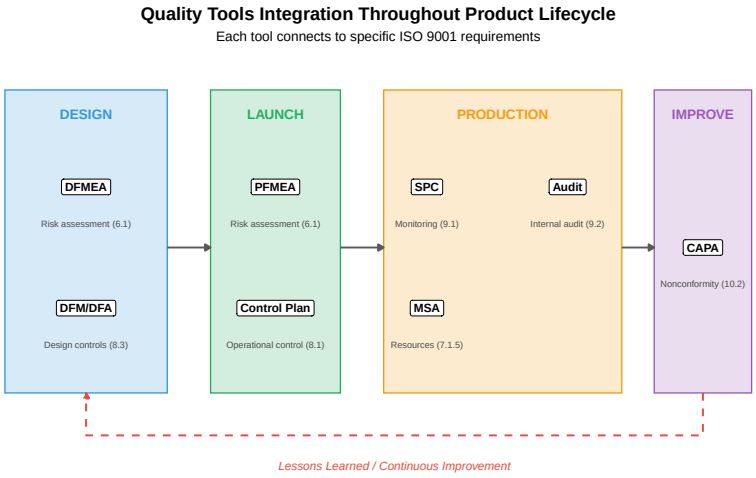
1. SPC training for 20 operators. _____
2. Engineering time spent on the PFMEA for a new product. _____
3. Calibration of the CMM and the shop-floor gauges. _____
4. Wages of the final inspectors. _____
5. Castings scrapped for porosity found at machining. _____
6. Overtime to rework a lot that failed final test. _____
7. Warranty claims on gearboxes that failed in the field. _____
8. Premium freight and a sorting crew sent to a customer's plant after a bad shipment. _____

Answers: 1 Prevention; 2 Prevention; 3 Appraisal; 4 Appraisal; 5 Internal Failure; 6 Internal Failure; 7 External Failure; 8 External Failure.

16.10 Integration with Quality Tools

A mature QMS integrates various quality tools throughout the organization.

16.10.1 Quality Tools Integration Map



16.10.2 Linking Tools to QMS Requirements

Table 16.21: Quality Tools and QMS Integration

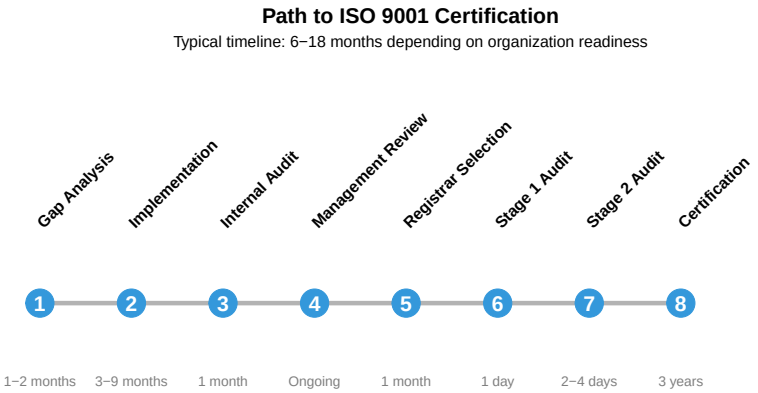
Quality Tool	ISO 9001 Clause	Purpose	Output Feed
PFMEA	6.1, 8.1	Identify and mitigate process risks before production	Control Plan
Control Plan	8.1, 8.5.1	Define controls for special characteristics	Work Instruc
SPC	9.1.1	Monitor process performance; detect special causes	CAPA trigge
MSA	7.1.5	Ensure measurement system is adequate	SPC validity
Internal Audit	9.2	Verify QMS conformance and effectiveness	CAPA trigge
CAPA	10.2	Eliminate causes of nonconformities	Process impr
Management Review	9.3	Evaluate QMS suitability and effectiveness	Strategic dec

---

16.11 Certification Process

Organizations can seek **third-party certification** to demonstrate QMS conformance to customers and stakeholders.

16.11.1 The Certification Journey



16.11.2 Certification Audit Types

Table 16.22: Third-Party Certi

Audit Type	Purpose	Focus Area
Stage 1 (Documentation)	Review QMS documentation; verify readiness for Stage 2	Quality ma
Stage 2 (Certification)	Evaluate implementation and effectiveness of QMS	All clauses
Surveillance	Verify continued conformance (annual)	Selected pr
Recertification	Full re-audit at end of 3-year cycle	Complete

---

16.12 Video Resources

16.12.1 ISO 9001:2015 Overview

16.12.2 Internal Auditing Best Practices

16.13 Summary

A Quality Management System provides the framework for consistently meeting customer requirements and improving organizational performance:

1. **ISO 9001:2015** provides the foundational requirements using risk-based thinking and PDCA

2. **Industry standards** (IATF 16949, AS9100, ISO 22000) add sector-specific requirements
3. **Documentation** establishes the hierarchy from policy to procedures to records
4. **Internal audits** verify conformance and identify improvement opportunities
5. **CAPA** eliminates root causes of nonconformities and prevents recurrence
6. **Management review** ensures QMS suitability and drives strategic improvement
7. **Continuous improvement** is embedded through KPIs, COQ analysis, and improvement methodologies
8. **Quality tools** (PFMEA, SPC, MSA) integrate throughout the product lifecycle

"Quality is not an act, it is a habit." — Aristotle

16.14 Review Questions

Question 1: Explain the seven quality management principles and how they relate to each other.

Answer:

The seven quality management principles form an interconnected foundation:

1. **Customer Focus** (Central principle)
   - Understanding and meeting customer needs is the primary purpose
   - All other principles support this goal
2. **Leadership**
   - Leaders establish unity of purpose and direction
   - Create conditions for people to achieve quality objectives
   - Enables all other principles
3. **Engagement of People**
   - Competent, empowered people at all levels
   - Essential for effective processes and customer focus
4. **Process Approach**
   - Activities managed as interrelated processes
   - Enables consistent, predictable results
5. **Improvement**
   - Ongoing focus on enhancing performance
   - Essential for maintaining customer satisfaction
6. **Evidence-based Decision Making**
   - Decisions based on analysis of data
   - Reduces uncertainty; improves decision quality
7. **Relationship Management**

- Managing relationships with interested parties (suppliers, partners)
- Optimizes their impact on performance

Relationships: - Customer Focus drives the need for all others - Leadership enables and supports all principles - Engagement of People is essential for Process Approach and Improvement - Evidence-based Decision Making supports Improvement - Relationship Management extends quality beyond the organization

Question 2: Compare ISO 9001, IATF 16949, and AS9100. When would each be required?

Answer:

Standard	Base	Industry	Additional Focus
ISO 9001:2015	Standalone	General	Quality fundamentals
IATF 16949:2016	ISO 9001 + automotive	Automotive	PPAP, APQP, Core Tools
AS9100D	ISO 9001 + aerospace	Aerospace/Defense	Configuration mgmt, FAI, counterfeit prevention

When Required:

ISO 9001: - Any organization wanting to demonstrate quality capability - Customer or contract requirement for QMS certification - Basis for all other standards - Suitable for service, manufacturing, any industry

IATF 16949: - Suppliers to automotive OEMs (Ford, GM, Toyota, VW, etc.) - Required for production parts/materials - Required throughout automotive supply chain - Includes Customer-Specific Requirements (CSRs)

AS9100: - Suppliers to aerospace/defense contractors - Boeing, Airbus, Lockheed Martin, military programs - Required for flight-critical components - OASIS database registration required

Key Differences: - IATF 16949: Emphasizes PPAP, APQP, manufacturing process control - AS9100: Emphasizes traceability, configuration control, first article inspection - Both build on ISO 9001 with sector-specific additions

Question 3: Describe the documentation hierarchy in a QMS. Give an example of each level.

Answer:

Level 1: Quality Policy and Objectives - Strategic direction for quality - Signed by top management - Communicated to all employees - *Example:* "ABC

Company is committed to delivering defect-free products that meet customer requirements through continuous improvement of our processes and people."

Level 2: Quality Manual and Procedures - Describes WHAT is done and WHO is responsible - System-level documents - *Example:* Document Control Procedure (QP-001) - Purpose: Control creation, approval, distribution of documents - Scope: All QMS documents - Responsibilities: Document Control Coordinator, Department Managers - Process flow for document changes

Level 3: Work Instructions and Specifications - Describes HOW tasks are performed - Detailed step-by-step instructions - *Example:* Work Instruction for CNC Lathe Setup (WI-MAC-015) - Step 1: Verify program number matches work order - Step 2: Install correct tooling per setup sheet - Step 3: Set tool offsets using touch-off procedure - Step 4: Run first piece and verify dimensions

Level 4: Records and Forms - Evidence that activities were performed - Completed forms, inspection results, logs - *Example:* First Article Inspection Report (FM-QC-008) - Part number, revision, date - Dimensional measurements vs. specifications - Inspector signature - Disposition (Accept/Reject)

Question 4: What are the key elements of an effective internal audit program?

Answer:

1. Audit Program Planning - Annual audit schedule covering all QMS processes - Risk-based frequency (higher risk = more frequent) - Documented audit program procedure - Resource allocation (trained auditors, time)

2. Auditor Competence - Trained in audit techniques - Knowledge of standards (ISO 9001, industry-specific) - Understanding of processes being audited - Independence from area being audited

3. Audit Preparation - Review of relevant documents (procedures, previous audits) - Prepared checklist linked to requirements - Notification to auditee - Defined scope and objectives

4. Audit Execution - Opening meeting (purpose, scope, schedule) - Evidence collection (documents, records, interviews, observation) - Objective evaluation against criteria - Note taking and evidence documentation

5. Reporting - Clear finding statements (nonconformities, observations) - Reference to requirement violated - Objective evidence cited - Closing meeting to present findings

6. Follow-up - Corrective action requests issued - Root cause analysis required - Verification of corrective action implementation - Verification of effectiveness

7. Continuous Improvement of Audit Program - Evaluate audit program effectiveness - Update based on organizational changes - Calibrate auditors periodically - Incorporate lessons learned

Question 5: Explain the difference between corrective action and preventive action. Provide an example of each.

Answer:

Corrective Action: - Eliminates the cause of a *detected* nonconformity - Reactive - responds to something that has already happened - Goal: Prevent recurrence of the specific problem

Example: - **Nonconformity:** Customer received shipment with wrong part numbers (3 incidents in past month) - **Root Cause:** Shipping labels printed from outdated part number list - **Corrective Action:** 1. Update master part number list 2. Link shipping system directly to ERP for current data 3. Add verification step comparing label to packing list - **Verification:** No recurrence after 3 months

Preventive Action: - Eliminates the cause of a *potential* nonconformity - Proactive - addresses something that hasn't happened yet - Goal: Prevent occurrence before it happens

Example: - **Potential Nonconformity:** New CNC machine may produce out-of-spec parts due to operator unfamiliarity - **Risk Identified Through:** PFMEA analysis during new equipment installation - **Preventive Action:** 1. Develop comprehensive training program before machine goes live 2. Create detailed setup procedures with photos 3. Implement first-article inspection for first 30 days 4. Assign experienced mentor to new operators - **Verification:** Track first-pass yield during initial production period

Key Difference: - Corrective = Fix what went wrong - Preventive = Stop what might go wrong

Note: ISO 9001:2015 doesn't explicitly require "preventive action" as a separate process. Instead, risk-based thinking throughout the QMS serves the preventive function.

Question 6: Calculate and analyze the Cost of Quality for the following data. What recommendations would you make?

Category	Cost
Quality planning	$35,000
Training	$25,000
Incoming inspection	$40,000
In-process inspection	$55,000
Final inspection	$30,000
Scrap	$85,000
Rework	$65,000
Warranty claims	$120,000
Customer returns	$45,000

Annual revenue: $8,000,000

Answer:

```
# Cost of Quality Analysis
prevention <- 35000 + 25000  # Planning + Training
appraisal <- 40000 + 55000 + 30000  # Inspections
internal_failure <- 85000 + 65000  # Scrap + Rework
external_failure <- 120000 + 45000  # Warranty + Returns

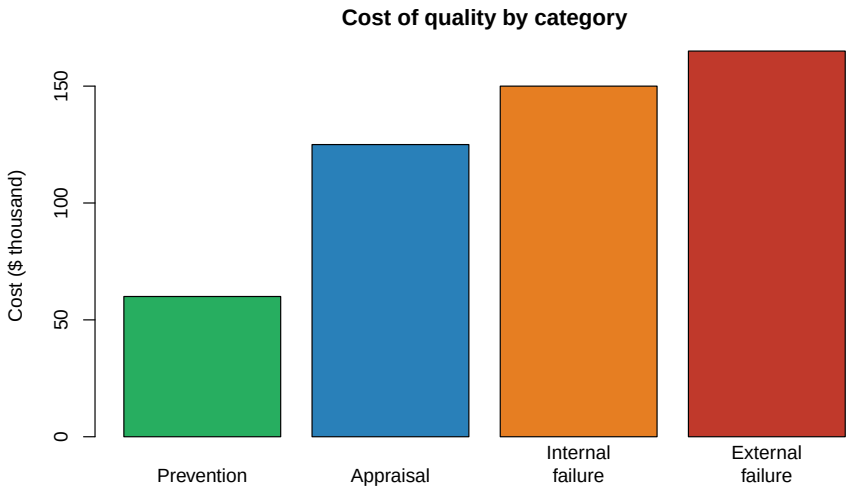
total_coq <- prevention + appraisal + internal_failure + external_failure
revenue <- 8000000

conformance <- prevention + appraisal
nonconformance <- internal_failure + external_failure

money <- function(x) format(x, big.mark = " ", scientific = FALSE)

cat(sprintf("Total cost of quality: $%s (%.1f%% of revenue)\n", money(total_coq), total_coq / rev
cat(sprintf("  Prevention  $%s (%.1f%%)\n", money(prevention), prevention / total_coq * 100))
cat(sprintf("  Appraisal  $%s (%.1f%%)\n", money(appraisal), appraisal / total_coq * 100))
cat(sprintf("  Internal failure  $%s (%.1f%%)\n", money(internal_failure), internal_failure / tot
cat(sprintf("  External failure  $%s (%.1f%%)\n", money(external_failure), external_failure / tot
cat(sprintf("  Conformance costs  $%s (%.1f%%)\n", money(conformance), conformance / total_coq *
cat(sprintf("  Nonconformance costs  $%s (%.1f%%)\n", money(nonconformance), nonconformance / tot

par(mar = c(5, 5, 3, 1))
barplot(c(prevention, appraisal, internal_failure, external_failure) / 1000,
        names.arg = c("Prevention", "Appraisal", "Internal\nfailure", "External\nfailure"),
        col = c("#27AE60", "#2980B9", "#E67E22", "#C0392B"),
        ylab = "Cost ($ thousand)", main = "Cost of quality by category")
```




```
Total cost of quality: $500 000 (6.2% of revenue)
  Prevention  $60 000 (12.0%)
  Appraisal  $125 000 (25.0%)
  Internal failure  $150 000 (30.0%)
  External failure  $165 000 (33.0%)
  Conformance costs  $185 000 (37.0%)
  Nonconformance costs  $315 000 (63.0%)
```

Analysis: - COQ at 6.25% of revenue is HIGH (target <3%) - External failure costs are the largest category (33%) - Nonconformance costs (63%) greatly exceed conformance (37%) - Prevention is only 12% of COQ - very low

Recommendations: 1. **Increase Prevention Investment:** - Implement SPC on critical processes to reduce defects - Enhance operator training on quality standards - Apply PFMEA to identify and control failure modes

2. **Reduce External Failures:**
   - Analyze warranty data for root causes
   - Improve final inspection effectiveness
   - Implement containment for known issues
3. **Optimize Appraisal:**
   - Use SPC to reduce inspection where processes are capable
   - Implement risk-based inspection (more for critical, less for proven)
4. **Target COQ Reduction:**
   - Short-term: Reduce external failures by 50% ($82,500 savings)
   - Medium-term: Reduce internal failures by 30% ($45,000 savings)
   - Long-term: Achieve COQ <3% of revenue ($240,000)

Question 7: What is risk-based thinking in ISO 9001:2015? How is it different

from formal risk management?

Answer:

Risk-Based Thinking in ISO 9001:2015:

Risk-based thinking is a way of considering risks and opportunities when designing, implementing, and operating the QMS. It:

- Applies throughout the standard (not a separate clause)
- Is proportional to potential impact on conformity
- Makes preventive action part of routine planning
- Considers both risks (negative) and opportunities (positive)

Where Risk-Based Thinking Applies:

Clause	Application
4.1 Context	Identify external/internal issues affecting QMS
4.2 Interested Parties	Understand requirements that could impact quality
6.1 Planning	Plan actions to address risks and opportunities
8.1 Operations	Consider risks in operational planning
9.1 Performance	Monitor effectiveness of risk actions
10.2 Corrective Action	Update risks/opportunities based on nonconformities

How It Differs from Formal Risk Management:

Aspect	Risk-Based Thinking	Formal Risk Management
Requirement	Implicit throughout ISO 9001	Explicit (ISO 31000)
Documentation	No formal risk register required	Documented risk register
Methodology	No specific method required	Defined methodology (FMEA, etc.)
Quantification	Not required	Often quantitative
Scope	QMS processes	Enterprise-wide

Practical Implementation:

Organizations should: 1. Consider risks when designing processes 2. Build controls proportional to risk level 3. Monitor and review effectiveness 4. Update as context changes

ISO 9001 does NOT require: - Formal risk management process - Risk register - Specific risk assessment methodology - Documented risk assessments (unless organization decides to)

However, many organizations do use tools like PFMEA to fulfill risk-based thinking requirements, especially in automotive (IATF 16949 requires it).

Question 8: Describe what management review is and what should be included in the inputs and outputs.

Answer:

What is Management Review?

Management review is a formal, periodic evaluation of the QMS by top management to: - Ensure continuing suitability (appropriate for the organization) - Ensure adequacy (sufficient to meet requirements) - Ensure effectiveness (achieving intended results) - Ensure alignment with strategic direction - Identify improvement opportunities

Frequency: At least annually; many organizations conduct quarterly reviews

Required Inputs (ISO 9001 Clause 9.3.2):

Input	What to Review
Status of previous actions	Actions from prior reviews - completed? effective?
Changes in external/internal issues	Market changes, regulations, technology, organizational changes
QMS performance and effectiveness	Customer satisfaction, quality objectives, process performance, nonconformities, audit results, supplier performance
Resource adequacy	People, infrastructure, environment, knowledge
Effectiveness of risk actions	Did actions address risks/opportunities?
Improvement opportunities	Suggestions, benchmarking, new technologies

Required Outputs (ISO 9001 Clause 9.3.3):

Output	Examples
Improvement decisions	Launch specific improvement projects, address recurring issues
QMS changes	Revise quality objectives, update processes, modify documentation
Resource needs	Budget approvals, hiring, equipment purchases, training

Best Practices: - Use a standard agenda aligned with required inputs - Prepare data in advance (dashboards, trend charts) - Document decisions and action

items with owners and due dates - Follow up on action completion - Keep minutes as quality records

Evidence Required: - Meeting minutes or management review report - Attendance record (top management participation) - Action items with status tracking - Data/reports presented

---

16.15 References

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7. Hoyle, D. (2017). *ISO 9000 Quality Systems Handbook* (7th ed.). Routledge.
8. AIAG. (2010). *Advanced Product Quality Planning and Control Plan (APQP)* (2nd ed.). Automotive Industry Action Group.
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Chapter 17

Advanced Measurement Systems Analysis

17.1 Learning Objectives

After completing this chapter, you will be able to:

1. Conduct a full crossed Gauge R&R study with the ANOVA method and extract the variance components, %GRR and ndc
2. Explain why ANOVA finds problems (operator-by-part interaction) that the Range method cannot
3. Run a Type 1 gauge study and judge a gauge with the capability indices C_g and C_{gk}
4. Choose between crossed, nested and Type 1 designs for a given measurement problem
5. Evaluate an attribute measurement system with kappa, effectiveness, miss rate and false-alarm rate
6. Build a measurement uncertainty budget and use it to compute the test uncertainty ratio and a guard-banded acceptance limit
7. Quantify how measurement error distorts control charts and capability indices

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17.2 From Screening to a Full Study

In the introductory MSA chapter you measured the pieces of a measurement system (bias, linearity, stability, repeatability and reproducibility) and ran a Gauge R&R with the **Range method**. That method is quick, and it is the right tool for a first look at a gauge in the lab. It is not the tool a customer

expects to see in a PPAP submission or a defence quality plan, and it breaks down in five situations:

Situation	Why the Range method is not enough	What this chapter uses
A customer or auditor wants the study "per AIAG MSA 4th edition"	The Range method is a screening tool and tends to overstate the Gauge R&R	ANOVA Gauge R&R
One operator seems to measure only certain parts differently	The Range method cannot see a part-by-operator interaction	ANOVA with the interaction term
A new gauge arrives and has not been used by anyone yet	There are no operators to compare, only the gauge	Type 1 study (C_g, C_{gk})
The inspection is pass/fail, or the test destroys the part	There is no continuous reading to take a range of, or no second measurement of the same part	Attribute agreement, nested Gauge R&R
A calibration lab or defence contract wants a stated "plus or minus" on a reading	%GRR says nothing about the size of the error in the measurement units	Uncertainty budget

Industry connection: in automotive supply the ANOVA Gauge R&R and the attribute study are standard PPAP evidence, and the Type 1 study is the usual first check on a new gauge (VDA 5 and many OEM specific requirements). In defence and aerospace work the uncertainty budget is what the calibration certificate and the measurement procedure must state. In food processing the same ideas apply to checkweighers, metal detectors and seal-integrity checks.

17.3 Full Gauge R&R with ANOVA

17.3.1 ANOVA Method (Full Study)

ANOVA (Analysis of Variance) is a statistical technique that partitions the total observed variation in a dataset into distinct components, each attributable to a specific source. In the context of Gauge R&R, ANOVA separates the total measurement variation into contributions from:

- **Parts** (the actual differences between items being measured)
- **Operators** (systematic differences between people taking measurements)
- **Part × Operator interaction** (whether certain operators measure certain parts differently)
- **Repeatability** (random error when the same operator measures the same part multiple times)

17.3.1.1 Why Use ANOVA for Gauge R&R?

The fundamental question in any measurement system analysis is: *"How much of the variation I observe comes from my measurement system versus the actual differences between parts?"* ANOVA answers this question by statistically decomposing variance—it calculates how much each source contributes to what we observe.

Consider the alternative: if we simply looked at the spread of all measurements, we wouldn't know whether a wide spread indicates highly variable parts (good—our gauge can detect differences) or a poor measurement system (bad—we're adding noise). ANOVA untangles these effects mathematically.

17.3.1.2 Why ANOVA is the Preferred Method

While the Range Method provides a quick estimate, ANOVA offers several critical advantages for Gauge R&R studies:

Aspect	Range Method	ANOVA Method
Interaction effects	Cannot detect	Identifies Part × Operator interactions

Aspect	Range Method	ANOVA Method
Statistical rigor	Approximation using range-based estimates	Exact variance component estimation
Hypothesis testing	Not available	Provides F-tests and p-values for each source
Accuracy	Tends to overestimate GRR	More precise variance estimates
Information richness	Basic EV/AV breakdown	Full decomposition with confidence intervals
Industry acceptance	Screening only	Required for formal MSA studies (AIAG)

The ability to detect **interaction effects** is particularly important. If Operator A consistently measures Part 3 higher than Operator B does (but they agree on other parts), this indicates a training issue or technique difference that the Range Method would miss entirely. ANOVA flags this interaction, enabling targeted corrective action.

17.3.1.3 How ANOVA Works in Gauge R&R

ANOVA uses the **F-statistic** to test whether variation between groups (e.g., between operators) is significantly larger than variation within groups (repeatability). The mathematical basis is:

$$F = \frac{\text{Mean Square Between Groups}}{\text{Mean Square Within Groups}} = \frac{MS_{factor}}{MS_{error}}$$

A large F-value indicates that the factor (Part, Operator, or their interaction) contributes meaningfully to the observed variation. The p-value tells us the probability of seeing such an F-value if the factor had no real effect.

From the ANOVA table's Mean Squares, we extract **variance components** (σ^2) for each source, then combine them to calculate the standard metrics: %EV (repeatability), %AV (reproducibility), %GRR, and the number of distinct categories (ndc).

```
# Full Gauge R&R using ANOVA method
# 10 parts, 3 operators, 3 trials

set.seed(42)

# Generate realistic GRR data
n_parts <- 10
n_operators <- 3
n_trials <- 3
```

```r
# Part true values (most of the variation)
part_effects <- rnorm(n_parts, 0, 0.05)

# Operator effects (some variation)
operator_effects <- c(-0.008, 0.003, 0.005)

# Create dataset
grr_data <- expand.grid(
  Part = 1:n_parts,
  Operator = paste0("Op", 1:n_operators),
  Trial = 1:n_trials
)

# Add measurements
grr_data$Measurement <- 10 +
  part_effects[grr_data$Part] +
  operator_effects[as.numeric(factor(grr_data$Operator))] +
  rnorm(nrow(grr_data), 0, 0.006)  # Repeatability error

# Convert to factors
grr_data$Part <- factor(grr_data$Part)
grr_data$Operator <- factor(grr_data$Operator)

# Fit ANOVA model
anova_model <- aov(Measurement ~ Part + Operator + Part:Operator, data = grr_data)

# Extract variance components
anova_table <- anova(anova_model)
print(anova_table)
```

```
Analysis of Variance Table

Response: Measurement
              Df   Sum Sq   Mean Sq  F value    Pr(>F)
Part           9 0.136635 0.0151816 343.6163 < 2.2e-16 ***
Operator       2 0.003430 0.0017152  38.8223  1.52e-11 ***
Part:Operator 18 0.000430 0.0000239   0.5401    0.9263
Residuals     60 0.002651 0.0000442
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Extract Mean Squares from ANOVA
MS_part <- anova_table["Part", "Mean Sq"]
MS_operator <- anova_table["Operator", "Mean Sq"]
MS_interaction <- anova_table["Part:Operator", "Mean Sq"]
```

```
MS_error <- anova_table["Residuals", "Mean Sq"]

# Calculate variance components
n_p <- 10  # number of parts
n_o <- 3   # number of operators
n_r <- 3   # number of trials

# Repeatability variance
var_repeatability <- MS_error

# Interaction variance
var_interaction <- max(0, (MS_interaction - MS_error) / n_r)

# Operator variance
var_operator <- max(0, (MS_operator - MS_interaction) / (n_p * n_r))

# Part variance
var_part <- max(0, (MS_part - MS_interaction) / (n_o * n_r))

# Reproducibility = Operator + Interaction
var_reproducibility <- var_operator + var_interaction

# Total Gauge R&R
var_GRR <- var_repeatability + var_reproducibility

# Total variation
var_total <- var_part + var_GRR

# Convert to standard deviations
sigma_repeatability <- sqrt(var_repeatability)
sigma_reproducibility <- sqrt(var_reproducibility)
sigma_GRR <- sqrt(var_GRR)
sigma_part <- sqrt(var_part)
sigma_total <- sqrt(var_total)

# Calculate study variation (5.15 sigma for 99% of distribution)
SV_repeatability <- 5.15 * sigma_repeatability
SV_reproducibility <- 5.15 * sigma_reproducibility
SV_GRR <- 5.15 * sigma_GRR
SV_part <- 5.15 * sigma_part
SV_total <- 5.15 * sigma_total

cat("\n=== ANOVA Gauge R&R Results ===\n\n")
cat("Variance Components:\n")
cat("  Repeatability:", round(var_repeatability, 8), "\n")
```

```
cat("  Reproducibility:", round(var_reproducibility, 8), "\n")
cat("  Part-to-Part:", round(var_part, 8), "\n")
cat("  Total:", round(var_total, 8), "\n")

cat("\nStudy Variation (5.15 ):\n")
cat("  Repeatability:", round(SV_repeatability, 5), "\n")
cat("  Reproducibility:", round(SV_reproducibility, 5), "\n")
cat("  Gauge R&R:", round(SV_GRR, 5), "\n")
cat("  Part-to-Part:", round(SV_part, 5), "\n")
cat("  Total:", round(SV_total, 5), "\n")
```

```
=== ANOVA Gauge R&R Results ===

Variance Components:
  Repeatability: 4.418e-05
  Reproducibility: 5.638e-05
  Part-to-Part: 0.0016842
  Total: 0.00178476

Study Variation (5.15 ):
  Repeatability: 0.03423
  Reproducibility: 0.03867
  Gauge R&R: 0.05164
  Part-to-Part: 0.21135
  Total: 0.21757
```

17.3.2 Calculating %GRR and Acceptance Criteria

```
# Calculate %GRR using both methods
tolerance <- 0.30  # Total tolerance

# Method 1: % of Tolerance (%GRR_Tolerance)
pct_GRR_tol <- (SV_GRR / tolerance) * 100

# Method 2: % of Total Variation (%GRR_TV)
pct_GRR_tv <- (sigma_GRR / sigma_total) * 100

# Number of Distinct Categories (ndc)
ndc <- 1.41 * (sigma_part / sigma_GRR)

cat("=== Gauge R&R Acceptance Criteria ===\n\n")
cat("%GRR (of Tolerance):", round(pct_GRR_tol, 1), "%\n")
cat("%GRR (of Total Variation):", round(pct_GRR_tv, 1), "%\n")
cat("Number of Distinct Categories (ndc):", round(ndc, 1), "\n\n")
```

```
# Acceptance criteria
cat("AIAG Acceptance Guidelines:\n")
cat("                          \n")
if(pct_GRR_tol < 10) {
  cat("%GRR < 10%: ACCEPTABLE - Measurement system is acceptable\n")
} else if(pct_GRR_tol < 30) {
  cat("%GRR 10-30%: MARGINAL - May be acceptable based on application\n")
} else {
  cat("%GRR > 30%: UNACCEPTABLE - Measurement system needs improvement\n")
}

if(ndc >= 5) {
  cat("ndc  5: ACCEPTABLE - Adequate discrimination\n")
} else if(ndc >= 2) {
  cat("ndc 2-4: MARGINAL - Limited discrimination ability\n")
} else {
  cat("ndc < 2: UNACCEPTABLE - Cannot distinguish between parts\n")
}
```

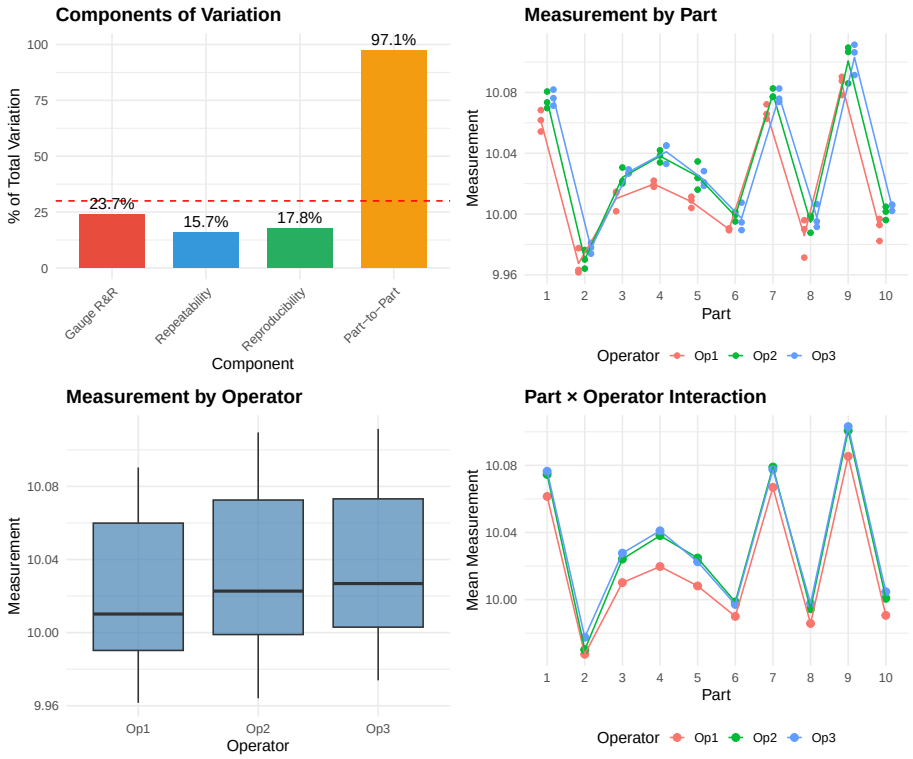
```
=== Gauge R&R Acceptance Criteria ===

%GRR (of Tolerance): 17.2 %
%GRR (of Total Variation): 23.7 %
Number of Distinct Categories (ndc): 5.8

AIAG Acceptance Guidelines:

%GRR 10-30%: MARGINAL - May be acceptable based on application
ndc  5: ACCEPTABLE - Adequate discrimination
```

17.3.3 Visual Analysis of Gauge R&R



17.4 Crossed, Nested and Destructive Studies

The ANOVA study above is a **crossed** design: every operator measures every part, several times. That only works when a part survives the measurement. Many of the tests that matter most do not leave a part to re-measure:

- a tensile test pulls the coupon apart,
- a weld peel or burst test destroys the joint,
- a food-safety plate count uses up the sample,
- a hardness indent on a thin section cannot be repeated on the same spot.

In these cases the study has to assume that parts from the same batch are close enough to be treated as "the same part". The batch is then **nested** inside the operator: each operator tests their own set of samples from a homogeneous batch, and nothing is shared across operators.

Design	Who measures what	Needs	Separates	Typical use
Crossed	Every operator measures every part, several times	A part that can be re-measured	Repeatability, reproducibility, part-by-operator interaction	Calipers, micrometers, CMM, gauges
Nested	Each operator measures their own parts, once each	Batches that are as uniform as possible	Repeatability, reproducibility (no interaction)	Tensile, burst, peel, microbiology
Type 1	One operator, one master, one gauge	A master of known value	Bias and repeatability only	Qualifying a new gauge

The analysis is the same ANOVA idea with a different model: the variation *within* a batch is repeatability, the variation *between* operator averages is reproducibility, and anything that is left over from batch-to-batch differences is mixed into the repeatability. That is why a nested study usually **overstates** the measurement error, and why the batch should be as uniform as you can make it.

Discussion: your team must qualify a destructive peel test on a sealed food pouch. What would you do to make the nested study fair?

Good practice:

- Take all samples from one production run and from consecutive pouches, so the seal quality is as similar as possible.
- Give each operator a randomised set of pouches, not the first ten off the line.
- Use the same test fixture, peel speed and conditioning time for every operator.
- Compare the result with the process: if the pouches in one run already differ by about as much as the measurement error, the study cannot separate the two. Say so in the report instead of quoting a number that looks more precise than it is.

17.5 Type 1 Gauge Study: C_g and C_{gk}

A Gauge R&R needs operators, parts and a lot of time. Before any of that, a quicker question is worth asking: *is the gauge itself good enough for this tolerance?* The **Type 1 study** answers it with one master part (a gauge block or a certified setting master) and one gauge. The gauge measures the master about 25 to 50 times, by the same operator under the same conditions, and the readings are compared with the master's known value.

Two indices come out of it, built the same way as the process capability indices you met in SPC:

$$C_g = \frac{0.2\,T}{6\,s_g} \qquad\qquad C_{gk} = \frac{0.1\,T - \left|\bar{x} - x_{ref}\right|}{3\,s_g}$$

where T is the tolerance width (USL − LSL), s_g is the standard deviation of the repeated readings, $\bar{x}$ is their average and x_{ref} is the master's reference value. The gauge is accepted when both are at least **1.33**.

Read the formulas the way you read C_p and C_{pk}:

- C_g asks only about **spread**. The gauge is allowed to use 20% of the tolerance ($0.2\,T$) for its own $6\,s_g$ scatter.
- C_{gk} asks about spread **and bias together**. The average has to sit close enough to the reference that $3\,s_g$ still fits inside $\pm 0.1\,T$.
- A big gap between C_g and C_{gk} means the gauge is precise but biased. That is cured by adjusting or recalibrating the gauge, not by buying a better one.

```
# Type 1 study: 25 readings of a 25.000 mm master, part tolerance 0.30 mm (+/-0.15)
set.seed(1717)
x_ref     <- 25.000
tolerance <- 0.30
readings  <- rnorm(25, mean = 25.008, sd = 0.006)

x_bar <- mean(readings)
s_g   <- sd(readings)
bias  <- x_bar - x_ref

Cg  <- 0.2 * tolerance / (6 * s_g)
Cgk <- (0.1 * tolerance - abs(bias)) / (3 * s_g)

cat("=== Type 1 Gauge Study ===\n\n")
cat(sprintf("Average reading:      %.4f mm\n", x_bar))
cat(sprintf("Bias:                 %.4f mm\n", bias))
cat(sprintf("Repeatability (s_g):  %.4f mm\n\n", s_g))
cat(sprintf("Cg  = %.2f   (spread only)\n", Cg))
```

```
cat(sprintf("Cgk = %.2f   (spread and bias)\n\n", Cgk))
cat(ifelse(Cgk >= 1.33, "Gauge is capable for this tolerance\n",
           "Gauge is NOT capable: Cgk is below 1.33\n"))

type1 <- data.frame(n = 1:25, reading = readings)
ggplot(type1, aes(x = n, y = reading)) +
  annotate("rect", xmin = -Inf, xmax = Inf, ymin = x_ref - 0.1 * tolerance,
           ymax = x_ref + 0.1 * tolerance, fill = "#D4EDDA", alpha = 0.6) +
  geom_hline(yintercept = x_ref, color = "#27AE60", linewidth = 1) +
  geom_hline(yintercept = x_bar, color = "#C0392B", linewidth = 1, linetype = "dashed") +
  geom_point(color = "steelblue", size = 3) +
  annotate("text", x = 26, y = x_ref - 0.1 * tolerance, label = "-0.1 T", hjust = 0,
           size = 4 * diagram_text_scale) +
  annotate("text", x = 26, y = x_ref + 0.1 * tolerance, label = "+0.1 T", hjust = 0,
           size = 4 * diagram_text_scale) +
  annotate("text", x = 26, y = x_ref - 0.002, label = "Reference", hjust = 0,
           color = "#27AE60", fontface = "bold", size = 4 * diagram_text_scale) +
  annotate("text", x = 26, y = x_bar + 0.002, label = "Average", hjust = 0,
           color = "#C0392B", fontface = "bold", size = 4 * diagram_text_scale) +
  scale_x_continuous(limits = c(0, 31)) +
  coord_cartesian(ylim = x_ref + c(-0.035, 0.035)) +
  labs(title = "Type 1 Study: Precise but Biased",
       subtitle = "The green band is the +/-0.1 T zone the gauge may use",
       x = "Reading number", y = "Reading (mm)") +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 12 * chart_text_scale))
```

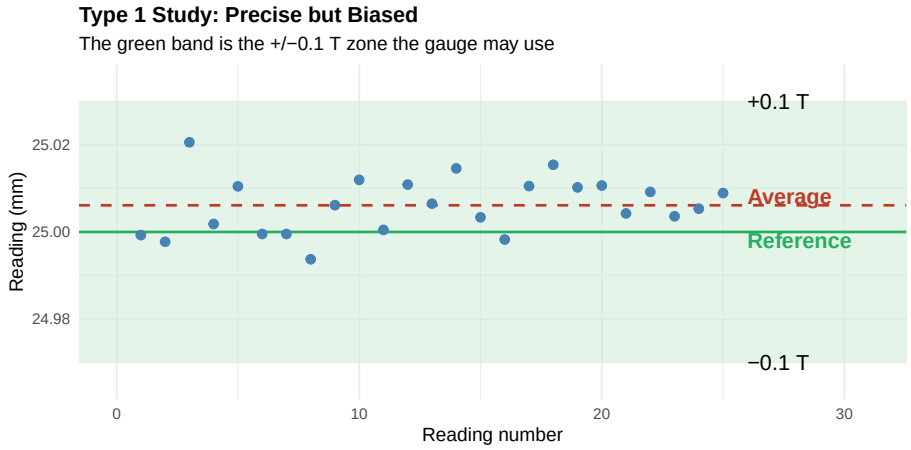


Figure 17.1: Type 1 study: 25 readings of a 25.000 mm master against the +/-0.1T band

```
=== Type 1 Gauge Study ===

Average reading:      25.0061 mm
Bias:                 0.0061 mm
Repeatability (s_g):  0.0065 mm

Cg  = 1.54   (spread only)
Cgk = 1.23   (spread and bias)

Gauge is NOT capable: Cgk is below 1.33
```

The scatter is small, so C_g is comfortable, but the whole cloud sits above the reference and C_{gk} falls below 1.33. This is a gauge that needs adjusting, not replacing.

Your turn : the gauge below is both biased and noisy. Change the average and the spread of the readings and watch which index moves.

Show the live R code behind this calculator

The card above re-runs this exact webR code every time you change an input, substituting your values for the variables at the top. Edit the code directly and press **Run Code** to experiment further.

```
#| label: type1-gauge
reference_value <- 25.000
tolerance <- 0.30
mean_reading <- 25.012
s_g <- 0.012

bias <- mean_reading - reference_value
Cg  <- 0.2 * tolerance / (6 * s_g)
Cgk <- (0.1 * tolerance - abs(bias)) / (3 * s_g)

cat(sprintf("Cgk: %.2f (%s)\n", Cgk, ifelse(Cgk >= 1.33, "capable", "!!not capable!!")))
cat(sprintf("  Cg  %.2f\n", Cg))
cat(sprintf("  Bias  %.4f mm\n", bias))
cat(sprintf("  Spread allowed for Cg = 1.33  %.4f mm\n", 0.2 * tolerance / (6 * 1.33)))
cat("  Target  Cg and Cgk at least 1.33\n")

par(mar = c(5, 5, 3, 1))
bp <- barplot(c(Cg, Cgk), names.arg = c("Cg", "Cgk"), col = c("#2E86C1", "#F39C12"),
              ylim = c(0, max(2, Cg, Cgk) * 1.1), ylab = "Index", main = "Gauge capability")
abline(h = 1.33, lty = 2, lwd = 2, col = "#C0392B")
```

Notice that moving the average back onto the reference raises C_{gk} only up to the value of C_g (with zero bias the two are equal). To go beyond that, the spread has to come down.

17.5.1 Try it yourself: a Type 1 study on a new gauge

A bore gauge is checked against a 10.000 mm ring master before it goes on the line. The part tolerance is 0.20 mm. Twenty readings gave an average of 10.006 mm and a standard deviation of 0.004 mm.

Type 1 study: bore gauge

- Bias (xbar - reference): _____ mm *(to 4 decimal places)*
- Cg: _____ *(to 2 decimal places)*
- Cgk: _____ *(to 2 decimal places)*

Show the worked solution

- Bias = 10.006 - 10.000 = **0.006 mm**
- $C_g = 0.2 \times 0.20/(6 \times 0.004) =$ **1.67**
- $C_{gk} = (0.1 \times 0.20 - 0.006)/(3 \times 0.004) =$ **1.17**

Both indices are not both above 1.33, so the gauge **is not yet capable** for this tolerance. The gap between C_g and C_{gk} comes from the 0.006 mm bias, so the first corrective action is to re-zero the gauge on the master.

17.6 Attribute Measurement Systems

For pass/fail or categorical measurements (visual inspection, go/no-go gauges), we use **Attribute MSA** or **Attribute Agreement Analysis**.

17.6.1 Kappa Statistic

When evaluating attribute measurement systems (pass/fail, accept/reject, visual inspection), you might think that **percent agreement** would be sufficient—if two inspectors agree 90% of the time, isn't that good enough? The problem is that some agreement happens purely by chance, and percent agreement doesn't account for this.

17.6.1.1 Why Percent Agreement Can Be Misleading

Consider a visual inspection where 95% of parts are good. If an inspector simply guesses "Accept" for every part (without even looking), they would be "correct" 95% of the time. Two such inspectors would agree 90.5% of the time by pure chance:

$$P_e = P(\text{both say Accept}) + P(\text{both say Reject}) = 0.95^2 + 0.05^2 = 0.905$$

This is why we need **Cohen's Kappa** (κ)—it measures agreement *beyond what we'd expect from chance alone.*

17.6.1.2 The Kappa Formula

$$\kappa = \frac{P_o - P_e}{1 - P_e}$$

Where:

- P_o = Observed proportion of agreement (how often they actually agree)
- P_e = Expected proportion of agreement by chance (how often they'd agree if guessing randomly based on their individual tendencies)
- $1 - P_e$ = Maximum possible agreement beyond chance

Interpreting the formula: The numerator $(P_o - P_e)$ represents *actual agreement minus chance agreement*—the "real" agreement attributable to consistent measurement. The denominator $(1 - P_e)$ normalizes this to a 0–1 scale by dividing by the maximum possible improvement over chance.

Kappa Value	Meaning
$\kappa = 1$	Perfect agreement (everyone agrees, and not by chance)
$\kappa = 0$	Agreement equals chance—no better than random guessing
$\kappa < 0$	Agreement is *worse* than chance (systematic disagreement)

17.6.1.3 When to Use Kappa

Use Kappa when evaluating:

- **Inspector vs. Reference**: Does the operator match the known correct answer? (Most important for MSA)
- **Inspector vs. Inspector**: Do two operators classify the same parts the same way?
- **Inspector vs. Self (Repeatability)**: Does the same operator give the same answer when re-inspecting the same part?

For a robust attribute MSA, you should calculate Kappa for all three comparisons. High agreement with other operators but low agreement with the reference standard indicates *consistent but incorrect* inspection—a training problem.

Table 17.1: Kappa Statistic Interpretation Guide

Kappa	Interpretation	Action
< 0	Less than chance agreement	Major issues - do not use
0.00 - 0.20	Slight agreement	Major improvement needed
0.21 - 0.40	Fair agreement	Significant improvement needed

0.41 - 0.60	Moderate agreement	Some improvement needed
0.61 - 0.80	Substantial agreement	Acceptable for most applications
0.81 - 1.00	Almost perfect agreement	Excellent - acceptable

17.6.2 Attribute Agreement Study Example

```
# Attribute Agreement Analysis
# 3 operators inspect 30 samples, 2 trials each
# Reference standard (known good/bad) available

set.seed(123)
n_samples <- 30
n_operators <- 3
n_trials <- 2

# Reference classification (10 bad, 20 good)
reference <- c(rep("Reject", 10), rep("Accept", 20))

# Simulate operator decisions (some disagreement)
simulate_decision <- function(ref, accuracy) {
  sapply(ref, function(r) {
    if(runif(1) < accuracy) r else ifelse(r == "Accept", "Reject", "Accept")
  })
}

# Operators with different accuracy levels
op1_t1 <- simulate_decision(reference, 0.92)
op1_t2 <- simulate_decision(reference, 0.92)
op2_t1 <- simulate_decision(reference, 0.88)
op2_t2 <- simulate_decision(reference, 0.88)
op3_t1 <- simulate_decision(reference, 0.95)
op3_t2 <- simulate_decision(reference, 0.95)

# Calculate agreement metrics

# 1. Within-operator agreement (repeatability)
within_op1 <- mean(op1_t1 == op1_t2)
within_op2 <- mean(op2_t1 == op2_t2)
within_op3 <- mean(op3_t1 == op3_t2)

# 2. Each operator vs. reference
op1_vs_ref <- mean(op1_t1 == reference & op1_t2 == reference)
op2_vs_ref <- mean(op2_t1 == reference & op2_t2 == reference)
op3_vs_ref <- mean(op3_t1 == reference & op3_t2 == reference)
```

```
# 3. All operators agree (both trials)
all_agree_t1 <- mean(op1_t1 == op2_t1 & op2_t1 == op3_t1)
all_agree_both <- mean(op1_t1 == op1_t2 & op1_t1 == op2_t1 & op2_t1 == op2_t2 &
                       op2_t1 == op3_t1 & op3_t1 == op3_t2)

cat("=== Attribute Agreement Analysis ===\n\n")
cat("Within-Operator Agreement (Repeatability):\n")
cat("  Operator 1:", round(within_op1 * 100, 1), "%\n")
cat("  Operator 2:", round(within_op2 * 100, 1), "%\n")
cat("  Operator 3:", round(within_op3 * 100, 1), "%\n\n")

cat("Each Operator vs. Reference (both trials correct):\n")
cat("  Operator 1:", round(op1_vs_ref * 100, 1), "%\n")
cat("  Operator 2:", round(op2_vs_ref * 100, 1), "%\n")
cat("  Operator 3:", round(op3_vs_ref * 100, 1), "%\n\n")

cat("All Operators Agreement:\n")
cat("  All agree (one trial):", round(all_agree_t1 * 100, 1), "%\n")
cat("  All agree (both trials):", round(all_agree_both * 100, 1), "%\n")
```

```
=== Attribute Agreement Analysis ===

Within-Operator Agreement (Repeatability):
  Operator 1: 83.3 %
  Operator 2: 76.7 %
  Operator 3: 93.3 %

Each Operator vs. Reference (both trials correct):
  Operator 1: 83.3 %
  Operator 2: 73.3 %
  Operator 3: 93.3 %

All Operators Agreement:
  All agree (one trial): 70 %
  All agree (both trials): 53.3 %
```

17.6.3 Calculating Kappa

```
# Calculate Kappa for Operator 1 vs Reference
# Create confusion matrix for trial 1
actual <- factor(reference, levels = c("Accept", "Reject"))
predicted <- factor(op1_t1, levels = c("Accept", "Reject"))

confusion <- table(Predicted = predicted, Actual = actual)
```

```
print(confusion)

# Calculate proportions
n <- sum(confusion)
p_o <- sum(diag(confusion)) / n  # Observed agreement

# Expected agreement by chance
p_accept <- sum(confusion[1,]) / n * sum(confusion[,1]) / n
p_reject <- sum(confusion[2,]) / n * sum(confusion[,2]) / n
p_e <- p_accept + p_reject

# Kappa
kappa <- (p_o - p_e) / (1 - p_e)

cat("\nKappa Calculation (Operator 1 vs Reference):\n")
cat("Observed agreement (Po):", round(p_o, 3), "\n")
cat("Expected agreement (Pe):", round(p_e, 3), "\n")
cat("Kappa:", round(kappa, 3), "\n")

# Interpretation
if(kappa >= 0.81) {
  cat("Interpretation: Almost perfect agreement\n")
} else if(kappa >= 0.61) {
  cat("Interpretation: Substantial agreement\n")
} else if(kappa >= 0.41) {
  cat("Interpretation: Moderate agreement\n")
} else {
  cat("Interpretation: Fair or less agreement - needs improvement\n")
}
```

```
         Actual
Predicted Accept Reject
   Accept     17      1
   Reject      3      9

Kappa Calculation (Operator 1 vs Reference):
Observed agreement (Po): 0.867
Expected agreement (Pe): 0.533
Kappa: 0.714
Interpretation: Substantial agreement
```

17.6.4 Effectiveness, Miss Rate and False-Alarm Rate

Kappa tells you whether inspectors agree beyond chance. A customer usually asks two plainer questions: *how often does a bad part get through?* and *how often is a good part thrown away?* The AIAG attribute study answers them

with three rates, all counted against the **reference** decision of each part:

$$\text{Effectiveness} = \frac{\text{correct decisions}}{\text{total opportunities}} \qquad \text{Miss rate} = \frac{\text{bad parts accepted}}{\text{opportunities on bad parts}} \qquad \text{False-alarm rate} = \frac{\text{good par}}{\text{opportunities}}$$

An "opportunity" is one part judged once by one appraiser, so 20 bad parts judged by 3 appraisers in 3 trials is 180 opportunities on bad parts.

Result	Effectiveness	Miss rate	False-alarm rate
Acceptable	90% or more	2% or less	5% or less
Marginal	80% to 90%	2% to 5%	5% to 10%
Unacceptable	below 80%	above 5%	above 10%

The two error rates carry very different costs. A **miss** ships a defect to the customer: in an airbag inflator weld or a food metal-detector check it is the failure that matters. A **false alarm** costs scrap, rework and re-inspection. A system can have a good overall effectiveness and still be unacceptable because of its miss rate, which is why all three are reported.

Your turn : the vision system below checks a seal on a food pouch. Change the counts and watch the three rates.

Show the live R code behind this calculator

The card above re-runs this exact webR code every time you change an input, substituting your values for the variables at the top. Edit the code directly and press **Run Code** to experiment further.

```
#| label: attribute-rates
bad_opps <- 60
good_opps <- 120
missed <- 5
false_alarms <- 9

total_opps <- bad_opps + good_opps
effectiveness <- (total_opps - missed - false_alarms) / total_opps * 100
miss_rate <- missed / bad_opps * 100
false_alarm_rate <- false_alarms / good_opps * 100

band <- function(x, ok, marginal) ifelse(x <= ok, "acceptable", ifelse(x <= marginal, "marginal",

cat(sprintf("Miss rate: %.1f%% (%s)\n", miss_rate, band(miss_rate, 2, 5)))
cat(sprintf("  False-alarm rate  %.1f%% (%s)\n", false_alarm_rate, band(false_alarm_rate, 5, 10))
cat(sprintf("  Effectiveness  %.1f%% (%s)\n", effectiveness,
            ifelse(effectiveness >= 90, "acceptable", ifelse(effectiveness >= 80, "marginal", "!!
```

```
par(mar = c(5, 5, 3, 1))
barplot(c(miss_rate, false_alarm_rate), names.arg = c("Miss rate", "False-alarm rate")
        col = c("#C0392B", "#F39C12"), ylab = "Percent of opportunities",
        ylim = c(0, max(12, miss_rate, false_alarm_rate) * 1.1), main = "Error rates ag
abline(h = c(2, 5), lty = 2, col = c("#C0392B", "#F39C12"))
```

With 60 opportunities on bad parts, one missed defect is already 1.7%, and two is 3.3%. A study this small cannot show a miss rate below 2% with any margin, which is why customers ask for larger samples when the consequence of a miss is severe.

17.6.5 Attribute MSA Best Practices

Table 17.2: Best Practices for Attribute MSA Studies

Practice	Description
Use boundary samples	Include samples at the accept/reject boundary where decisions are h
Include clear accept/reject	Include obvious accept and obvious reject samples as controls
Blind testing	Operators should not know which samples are repeated or reference
Multiple trials	Minimum 2-3 trials per operator to assess within-operator repeatabil
Reference standard	Must have known correct answers to assess accuracy, not just agreem
Adequate sample size	Minimum 30-50 samples recommended for statistical validity

---

17.7 Measurement Uncertainty

%GRR compares the measurement error with the *tolerance* or the *process*. It never says how big the error is in millimetres. When a calibration lab certifies a gauge block, or a defence contract states that a dimension must be verified to a given uncertainty, the answer is an **uncertainty budget** built the way the *Guide to the Expression of Uncertainty in Measurement* (GUM) describes.

17.7.1 Building a Budget

Every source of error becomes a **standard uncertainty**, a one-standard-deviation value in the unit of the measurement. How you get it depends on what you know about the source:

Source	Type	What you know	Standard uncertainty u
Repeatability	A (from data)	Standard deviation s of repeated readings	$u = s$ (or $s/\sqrt{n}$ for the average of n readings)
Calibration certificate	B (from a document)	Expanded uncertainty U at coverage factor k	$u = U/k$ (usually $k = 2$)
Resolution	B	Reading is rounded to a step d; any value in $\pm d/2$ is equally likely	$u = d/(2\sqrt{3})$
Temperature	B	Part and gauge may be off 20 °C by $\pm\Delta T$; expansion coefficient α	$u = \alpha\, L\, \Delta T/\sqrt{3}$

The standard uncertainties are independent, so they add as **variances** (root-sum-of-squares), exactly like the standard deviations in the Gauge R&R:

$$u_c = \sqrt{u_1^2 + u_2^2 + \cdots + u_n^2} \qquad U = k \cdot u_c \quad (k = 2 \text{ for about } 95\% \text{ coverage})$$

Two checks follow from the budget:

- **Test uncertainty ratio (TUR)** $= T/(2U)$, the tolerance width divided by the width of the uncertainty band. The common rule is **4:1** or better.
- **Guard banding.** If the uncertainty is not negligible, a reading just inside the specification limit may belong to a part that is actually outside. Tightening the acceptance limits by U (acceptance zone $= LSL + U$ to $USL - U$) makes the chance of accepting a bad part small, at the price of rejecting some good ones.

```
# Uncertainty budget: 0-25 mm micrometer measuring a 25 mm shaft, tolerance 0.10 mm (+/-0.05)
tolerance  <- 0.10
alpha      <- 11.5e-6   # steel, per degree C
length_mm  <- 25
delta_T    <- 2         # +/- degrees C from 20 C

budget <- data.frame(
  Source = c("Resolution (0.001 mm)", "Calibration certificate (U = 0.002, k = 2)",
             "Repeatability (s = 0.0015)", "Temperature (+/-2 C)"),
  Type = c("B", "B", "A", "B"),
```

```
  u = c(0.001 / (2 * sqrt(3)),
        0.002 / 2,
        0.0015,
        alpha * length_mm * delta_T / sqrt(3))
)
budget$Share <- budget$u^2 / sum(budget$u^2) * 100

u_c <- sqrt(sum(budget$u^2))
U   <- 2 * u_c
TUR <- tolerance / (2 * U)

cat("=== Uncertainty Budget ===\n\n")
print(transform(budget, u = round(u, 5), Share = round(Share, 1)), row.names = FALSE)
cat(sprintf("\nCombined standard uncertainty u_c = %.5f mm\n", u_c))
cat(sprintf("Expanded uncertainty U (k = 2)    = %.5f mm\n", U))
cat(sprintf("Test uncertainty ratio T / 2U     = %.1f : 1\n", TUR))
cat(sprintf("Acceptance zone (guard banded)    = %.3f to %.3f mm from nominal\n", -tole

budget$Source <- factor(budget$Source, levels = budget$Source[order(budget$Share)])
ggplot(budget, aes(x = Source, y = Share, fill = Type)) +
  geom_col(width = 0.7) +
  geom_text(aes(label = paste0(round(Share), "%")), hjust = -0.15, size = 4 * diagram_t
  coord_flip(ylim = c(0, 100)) +
  scale_x_discrete(labels = scales::label_wrap(28)) +
  scale_fill_manual(values = c(A = "#2E86C1", B = "#F39C12")) +
  labs(title = "Where the Uncertainty Comes From",
       subtitle = "Share of the combined variance, type A (data) and type B (documents)
       x = NULL, y = "Share of combined variance (%)", fill = "Type") +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 12 * chart_text_scale))
```

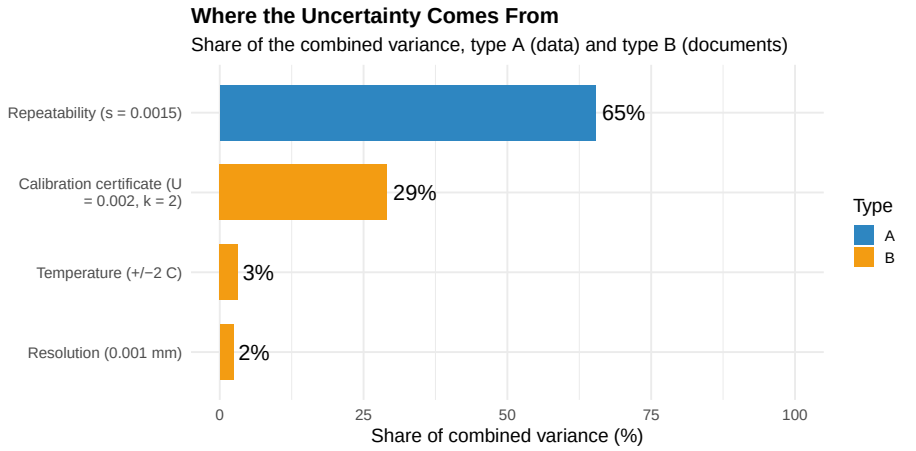


Figure 17.2: Contribution of each source to the combined uncertainty of a micrometer measurement

```
=== Uncertainty Budget ===

                                  Source Type       u Share
                     Resolution (0.001 mm)    B 0.00029   2.4
 Calibration certificate (U = 0.002, k = 2)    B 0.00100  29.0
                 Repeatability (s = 0.0015)    A 0.00150  65.3
                       Temperature (+/-2 C)    B 0.00033   3.2

Combined standard uncertainty u_c = 0.00186 mm
Expanded uncertainty U (k = 2)    = 0.00371 mm
Test uncertainty ratio T / 2U     = 13.5 : 1
Acceptance zone (guard banded)    = -0.046 to 0.046 mm from nominal
```

Because the contributions add as squares, the largest one dominates: here the calibration certificate and the repeatability account for almost all of the variance, and a gauge with finer resolution would barely change the answer. A budget tells you *which* source to attack before you spend money.

Your turn : the budget below is for a digital caliper checking a 25 mm feature with a total tolerance of 0.20 mm.

Show the live R code behind this calculator

The card above re-runs this exact webR code every time you change an input, substituting your values for the variables at the top. Edit the code directly and press **Run Code** to experiment further.

```
#| label: uncertainty-calc
tolerance <- 0.20
resolution <- 0.01
```

```
cal_U <- 0.03
repeat_s <- 0.02
delta_T <- 3

length_mm <- 25
alpha <- 11.5e-6

u_res  <- resolution / (2 * sqrt(3))
u_cal  <- cal_U / 2
u_rep  <- repeat_s
u_temp <- alpha * length_mm * delta_T / sqrt(3)
u_all  <- c(u_res, u_cal, u_rep, u_temp)

u_c <- sqrt(sum(u_all^2))
U   <- 2 * u_c
TUR <- tolerance / (2 * U)

cat(sprintf("Test uncertainty ratio: %.1f : 1 (%s)\n", TUR, ifelse(TUR >= 4, "meets 4:
cat(sprintf("  Combined standard uncertainty  %.4f mm\n", u_c))
cat(sprintf("  Expanded uncertainty U (k = 2)  %.4f mm\n", U))
cat(sprintf("  Guard-banded acceptance zone  +/- %.4f mm\n", max(tolerance / 2 - U, 0)

par(mar = c(5, 8, 3, 1))
barplot(rev(u_all^2 / sum(u_all^2) * 100), horiz = TRUE, las = 1, cex.names = 0.8,
        names.arg = rev(c("Resolution", "Calibration", "Repeatability", "Temperature")
        col = "#2E86C1", xlab = "Share of combined variance (%)", main = "Largest sour
```

17.7.2 Try it yourself: an uncertainty budget by hand

A 0.01 mm caliper checks a feature with a total tolerance of 0.30 mm. Its calibration certificate gives U = 0.02 mm at k = 2, and the repeatability standard deviation from a short study is 0.012 mm. Ignore temperature.

Uncertainty budget: digital caliper

- Resolution, standard uncertainty: _____ mm *(to 4 decimal places)*
- Calibration, standard uncertainty: _____ mm *(to 4 decimal places)*
- Combined standard uncertainty: _____ mm *(to 4 decimal places)*
- Expanded uncertainty U (k = 2): _____ mm *(to 4 decimal places)*
- Test uncertainty ratio, T / 2U: _____ *(to 2 decimal places)*

Show the worked solution

- Resolution: $u = 0.01/(2\sqrt{3}) =$ **0.0029 mm**
- Calibration: $u = 0.02/2 =$ **0.0100 mm**
- Combined: $u_c = \sqrt{0.0029^2 + 0.0100^2 + 0.012^2} =$ **0.0159 mm**

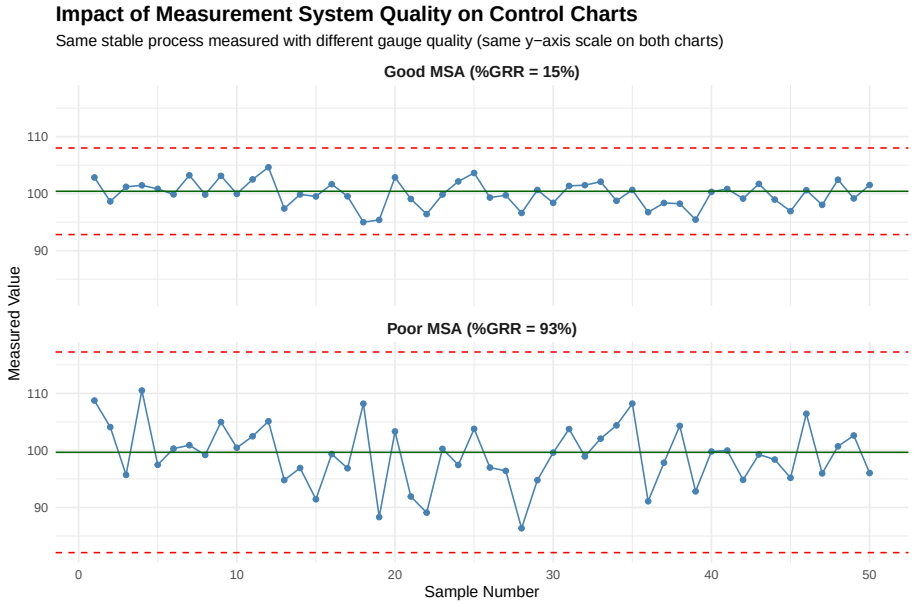
- Expanded: $U = 2 \times 0.0159 = \mathbf{0.0318\ mm}$
- $TUR = 0.30/(2 \times 0.0318) = \mathbf{4.72 : 1}$

The ratio is above 4:1, so the caliper is acceptable for this tolerance. The repeatability (57% of the combined variance) is the source to attack first.

17.8 MSA and SPC Relationship

Poor measurement systems directly impact SPC effectiveness.

17.8.1 Effect on Control Charts



17.8.2 Effect on Capability Studies

Teaching demo — drag the gauge quality and the process sigma, and watch the observed Cpk fall as the measurement error grows.

Show the live R code behind this calculator

```
#| label: msa-capability-effect
# True process parameters
true_mean <- 100
true_sigma <- 2.0
pct_grr_good <- 10   # %GRR of the good gauge, as a share of the tolerance
pct_grr_poor <- 50   # %GRR of the poor gauge
```

```
USL <- 106
LSL <- 94
tolerance <- USL - LSL

# True capability
true_Cpk <- min((USL - true_mean), (true_mean - LSL)) / (3 * true_sigma)

# With good measurement system
grr_good <- pct_grr_good / 100 * tolerance / 5.15
observed_sigma_good <- sqrt(true_sigma^2 + grr_good^2)
observed_Cpk_good <- min((USL - true_mean), (true_mean - LSL)) / (3 * observed_sigma_g

# With poor measurement system
grr_poor <- pct_grr_poor / 100 * tolerance / 5.15
observed_sigma_poor <- sqrt(true_sigma^2 + grr_poor^2)
observed_Cpk_poor <- min((USL - true_mean), (true_mean - LSL)) / (3 * observed_sigma_p

cat(sprintf("A capable process (true Cpk = %.2f) can appear incapable under poor MSA\n"
cat(sprintf("  True Cpk                  %.2f\n", true_Cpk))
cat(sprintf("  Observed Cpk (%g%% GRR)   %.2f\n", pct_grr_good, observed_Cpk_good))
cat(sprintf("  Observed Cpk (%g%% GRR)   %.2f\n", pct_grr_poor, observed_Cpk_poor))

par(mar = c(5, 5, 3, 1))
barplot(c(`True` = true_Cpk, `Good gauge` = observed_Cpk_good, `Poor gauge` = observed_
        col = c("#27AE60", "#2E86C1", "#C0392B"), ylim = c(0, 5), ylab = "Cpk", main =
abline(h = 1.33, lty = 2, lwd = 2)
```

17.9 Choosing the Right Study

By now there are five tools in the box. Each answers a different question, and picking the wrong one produces a number that looks official but means nothing.

Question	Study
Is this new gauge good enough for the tolerance before anyone uses it?	Type 1 (C_g, C_{gk})
How much of the variation comes from the gauge and the people using it?	Crossed Gauge R&R (ANOVA)
The test destroys the part. How do we qualify it?	Nested Gauge R&R
Do inspectors agree with each other and with the reference on pass/fail calls?	Attribute agreement (kappa, effectiveness, miss rate)

Question	Study
How large is the measurement error in the unit of measurement?	Uncertainty budget

17.9.1 Try it yourself: which study fits?

Which study answers the question?

Categories: Type 1 / Crossed Gauge R&R / Nested Gauge R&R / Attribute agreement / Uncertainty budget

1. A new bore gauge arrives and quality wants to approve it against a 10.000 mm ring master before it goes on the line. _____
2. Ten crankshaft journals are each measured three times by each of three operators on the same micrometer, in random order. _____
3. A weld peel test destroys each coupon, so every operator tests their own coupons cut from the same sheet. _____
4. Three inspectors accept or reject 50 sealed pouches, and the lab has already classified every pouch. _____
5. A defence customer asks for the CMM's reading of a bracket hole to be stated with a plus-or-minus value at k = 2. _____
6. A torque-wrench checker needs to be put into service and has never been used by anyone, so no operators exist to compare yet. _____
7. A checkweigher rejects under-filled packs, and a customer asks how often it passes a short pack. _____

Answers: 1 Type 1; 2 Crossed Gauge R&R; 3 Nested Gauge R&R; 4 Attribute agreement; 5 Uncertainty budget; 6 Type 1; 7 Attribute agreement.

17.10 Summary

1. **Range method is a screen, ANOVA is the study.** ANOVA gives exact variance components, tests for part-by-operator interaction and is what customers expect.
2. **Type 1 first.** C_g and C_{gk} qualify a gauge alone; accept at 1.33 or better, and read a large gap between them as bias.
3. **Pick the design to fit the part.** Crossed when parts survive, nested when they do not, Type 1 when there are no operators yet.
4. **Attribute systems have three rates, not one.** Report effectiveness, miss rate and false-alarm rate against the reference, and weight the miss rate when escapes are costly.
5. **An uncertainty budget adds variances.** The largest source dominates, so rank them before spending money; check the TUR (4:1 or better) and guard-band when it is not.

6. **Measurement error distorts SPC.** It widens the control limits and lowers the apparent capability, so fix the gauge before you chase the process.

"A number without an uncertainty is only an opinion."

17.11 Review Questions

Question 1: A visual inspection has the following results: Kappa vs. reference = 0.72, within-operator agreement = 88%, between-operator agreement = 75%. Evaluate this attribute measurement system.

Answer:

Results Interpretation:

1. **Kappa vs. Reference = 0.72**
   - Falls in "Substantial agreement" range (0.61-0.80)
   - Operators are correctly classifying parts most of the time
   - Acceptable for many applications, but improvement possible
2. **Within-Operator Agreement = 88%**
   - Operators are fairly consistent with themselves
   - 12% of the time, same operator gives different result on same part
   - Should target >90% for critical inspections
3. **Between-Operator Agreement = 75%**
   - Operators disagree on 25% of parts
   - This is a significant reproducibility issue
   - Major source of inconsistency in product quality

Assessment: MARGINAL - Improvement Needed

Recommended Actions:

1. **Standardize criteria**: Create clear visual standards with boundary samples
   - Photos of accept/reject borderline cases
   - Written descriptions of defect criteria
2. **Training**:
   - Calibration session with all operators and reference samples
   - Identify operators with lower agreement for targeted training
3. **Improve conditions**:
   - Check lighting adequacy and consistency
   - Ensure proper viewing distance and angle
   - Reduce fatigue with appropriate break schedules

4. **Consider automation**: For critical inspections, automated vision systems may provide better consistency
5. **Re-study after improvements** to verify effectiveness

Question 2: Given the following ANOVA table from a Gauge R&R study, calculate the variance components and %GRR.

Source	DF	SS	MS
Part	9	0.2850	0.03167
Operator	2	0.0025	0.00125
Part×Operator	18	0.0090	0.00050
Repeatability	60	0.0180	0.00030
Total	89	0.3145	

Study design: 10 parts, 3 operators, 3 trials

Answer:

```
# Given values from ANOVA table
MS_part <- 0.03167
MS_operator <- 0.00125
MS_interaction <- 0.00050
MS_error <- 0.00030

n_parts <- 10
n_operators <- 3
n_trials <- 3

# Variance components
var_repeatability <- MS_error
var_interaction <- max(0, (MS_interaction - MS_error) / n_trials)
var_operator <- max(0, (MS_operator - MS_interaction) / (n_parts * n_trials))
var_part <- max(0, (MS_part - MS_interaction) / (n_operators * n_trials))

# Reproducibility = Operator + Interaction
var_reproducibility <- var_operator + var_interaction

# Total Gauge R&R
var_GRR <- var_repeatability + var_reproducibility

# Total variation
var_total <- var_part + var_GRR

# Calculate %GRR (of total variation)
sigma_GRR <- sqrt(var_GRR)
```

```
sigma_total <- sqrt(var_total)
pct_GRR <- (sigma_GRR / sigma_total) * 100

# ndc
sigma_part <- sqrt(var_part)
ndc <- 1.41 * (sigma_part / sigma_GRR)

if(pct_GRR < 10) {
  assessment <- "ACCEPTABLE (< 10%)"
} else if(pct_GRR < 30) {
  assessment <- "MARGINAL (10-30%)"
} else {
  assessment <- "UNACCEPTABLE (> 30%)"
}

cat(sprintf("%%GRR (of total variation): %.1f%%\n", pct_GRR))
cat(sprintf("  Assessment  %s\n", assessment))
cat(sprintf("  Number of distinct categories (ndc)  %.1f\n", ndc))
cat(sprintf("  Var(Repeatability)  %.6f\n", var_repeatability))
cat(sprintf("  Var(Interaction)  %.6f\n", var_interaction))
cat(sprintf("  Var(Operator)  %.6f\n", var_operator))
cat(sprintf("  Var(Part)  %.6f\n", var_part))
cat(sprintf("  Var(Reproducibility)  %.6f\n", var_reproducibility))
cat(sprintf("  Var(GRR)  %.6f\n", var_GRR))
cat(sprintf("  Var(Total)  %.6f\n", var_total))
```

```
%GRR (of total variation): 31.9%
  Assessment  UNACCEPTABLE (> 30%)
  Number of distinct categories (ndc)  4.2
  Var(Repeatability)  0.000300
  Var(Interaction)  0.000067
  Var(Operator)  0.000025
  Var(Part)  0.003463
  Var(Reproducibility)  0.000092
  Var(GRR)  0.000392
  Var(Total)  0.003855
```

Question 3: A Type 1 study gives $C_g = 2.10$ and $C_{gk} = 0.95$. What is wrong with the gauge, and what do you do?

Answer:

C_g is well above 1.33, so the **spread** of the gauge is small enough for the tolerance. C_{gk} is below 1.33, and since C_{gk} differs from C_g only through the bias term, the gauge is **precise but biased**: its average sits too far from the master.

Action: re-zero or recalibrate the gauge (or apply a documented correction), then repeat the Type 1 study. There is no need to replace the gauge.

Question 4: Why can a nested Gauge R&R overstate the measurement error compared with a crossed study?

Answer:

In a nested study each operator tests different samples, so any difference between batches or samples lands inside the apparent repeatability and reproducibility. A crossed study measures the *same* part several times, so the part-to-part difference is removed and only the measurement system remains.

To keep the overstatement small, take the samples from a single homogeneous batch (consecutive items from one run), randomise the allocation to operators, and say in the report that the result is an upper bound on the measurement error.

Question 5: Compare the two ways a gauge can fail a tolerance check: a bad C_g and a bad TUR. What does each tell you?

Answer:

C_g and C_{gk} come from repeated readings of a *master* and describe the gauge's scatter and bias against the tolerance. The TUR comes from an *uncertainty budget* and also includes the sources that repeated readings of one master do not reveal: the calibration certificate, the resolution rounding and the temperature. A gauge can pass a Type 1 study and still have a poor TUR because its calibration uncertainty is large. Qualify with both when the specification allows little room.

17.12 References

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Chapter 18

Advanced Statistical Process Control

18.1 Learning Objectives

After completing this chapter, you will be able to:

1. Explain the difference between Phase I (setting up a chart) and Phase II (monitoring), and form rational subgroups
2. Construct and interpret X-bar/S charts, and say when S is the better choice than R
3. Construct and interpret np and u charts, including u charts with unequal sample sizes
4. Calculate and compare Cp/Cpk with Pp/Ppk and read the gap between them
5. Estimate capability for non-normal data with the percentile method, and judge how much a Cpk from a small sample can be trusted
6. Construct CUSUM and EWMA charts to detect small, sustained shifts, and compare charts by average run length
7. Choose a chart for short-run production and for autocorrelated data

---

18.2 Beyond the Basic Charts

The introductory SPC chapter gave you the working set: X-bar/R and I-MR for measurements, p and c for counts, the Western Electric rules, and Cp/Cpk. Those charts run most of the factory floor. They also carry assumptions that real processes break regularly:

Assumption behind the basic charts	When it fails	What this chapter adds
Subgroups contain only common-cause variation	Parts from two machines, cavities or shifts are mixed in one subgroup	Rational subgrouping
A range is a good enough spread estimate	Subgroups of 10 or more	X-bar/S chart
One defect rate and one sample size	Counts per lot, lots of different size	np and u charts
The process is stable, so Cpk describes it	Tool wear and drift make the long-term spread larger than the short-term	Pp/Ppk and the gap
Data are normal	Runout, flatness, surface roughness, contamination counts	Percentile capability
A big shift is the thing to catch	A 0.5 to 1.0 sigma drift is costly but never crosses a 3-sigma limit	CUSUM and EWMA
Many parts of one kind are made	Job shops make 50 parts of ten different kinds	Short-run charts

Industry connection: an automotive Tier 1 supplier submitting a PPAP needs to state Ppk for the initial run (stability not yet proven) and Cpk for ongoing production. A defence or aerospace supplier machining small lots of many part numbers uses short-run charts. A food plant filling bags uses X-bar/S on checkweigher data. A pharmaceutical or precision-casting line uses EWMA or CUSUM to catch slow drift before a limit is reached.

18.3 Setting Up Charts Properly

18.3.1 Phase I and Phase II

A control chart is used in two quite different ways:

	Phase I: set up	Phase II: monitor
Question	Is the process stable, and what are its natural limits?	Has the process changed?
Data	20 to 25 subgroups collected for the purpose	New subgroups as they are produced
Limits	*Trial limits* computed from the same data they are plotted against	Fixed limits carried forward from Phase I
When a point is out	Find the assignable cause. If it is found and fixed, remove those points and recompute	Signal: investigate and act
Danger	Removing points only because they are out, without a cause, makes the process look better than it is	Recomputing limits every shift hides drift

Never recompute the limits in Phase II unless there is a documented, deliberate change to the process. A chart whose limits chase the data cannot signal a shift.

18.3.2 Rational Subgrouping

The control limits of an X-bar chart come from the variation *within* subgroups. The chart can therefore detect only differences *between* subgroups. A **rational subgroup** is chosen so that the variation within it is only the common-cause variation, and any special cause shows up between subgroups. Consecutive parts from one machine, one cavity and one setup are the usual choice.

The two charts below use the same two machines. The only difference is how the subgroups are formed.

```
# Two machines feed one line: A runs at 50.0 mm, B at 51.0 mm, same scatter
set.seed(2024)
n_sub <- 20; n <- 4; A2 <- 0.729
make_parts <- function(mean_mm, k) rnorm(k, mean_mm, 0.25)

# (1) Mixed: each subgroup takes two parts from A and two from B
mixed <- t(replicate(n_sub, c(make_parts(50, 2), make_parts(51, 2))))
```

```
# (2) Rational: each subgroup is four consecutive parts from ONE machine (A, B, A, B .
by_machine <- t(sapply(1:n_sub, function(i) make_parts(ifelse(i %% 2 == 1, 50, 51), n))

xbar_chart <- function(m, label) {
  xb <- rowMeans(m)
  r  <- apply(m, 1, function(v) max(v) - min(v))
  cl <- mean(xb)
  data.frame(design = label, subgroup = seq_len(nrow(m)), xbar = xb,
             cl = cl, ucl = cl + A2 * mean(r), lcl = cl - A2 * mean(r))
}
charts <- rbind(xbar_chart(mixed, "Mixed subgroups (2 from A, 2 from B)"),
                xbar_chart(by_machine, "One machine per subgroup"))
charts$signal <- charts$xbar > charts$ucl | charts$xbar < charts$lcl

cat("=== Effect of subgroup design ===\n\n")
for (d in unique(charts$design)) {
  s <- charts[charts$design == d, ]
  cat(sprintf("%-38s limits %.2f to %.2f mm (width %.2f), %d of %d points out\n",
              d, s$lcl[1], s$ucl[1], s$ucl[1] - s$lcl[1], sum(s$signal), nrow(s)))
}

ggplot(charts, aes(x = subgroup, y = xbar)) +
  geom_line(aes(y = ucl), color = "#C0392B", linetype = "dashed") +
  geom_line(aes(y = lcl), color = "#C0392B", linetype = "dashed") +
  geom_line(aes(y = cl), color = "#27AE60") +
  geom_line(color = "grey50") +
  geom_point(aes(color = signal), size = 3) +
  scale_color_manual(values = c("FALSE" = "steelblue", "TRUE" = "#C0392B"), guide = "no
  facet_wrap(~design, ncol = 1) +
  labs(title = "Subgroup Design Decides What the Chart Can See",
       x = "Subgroup", y = "X-bar (mm)") +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 12 * chart_text_scale),
        strip.text = element_text(face = "bold", size = 10 * chart_text_scale))
```

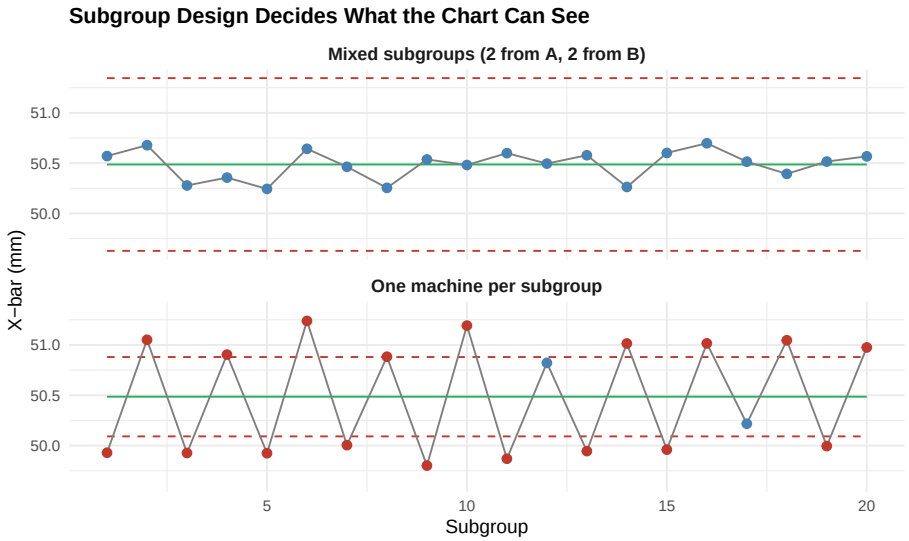


Figure 18.1: The same two machines charted with mixed subgroups and with one machine per subgroup

```
=== Effect of subgroup design ===

Mixed subgroups (2 from A, 2 from B)   limits 49.63 to 51.35 mm (width 1.72), 0 of 20 points out
One machine per subgroup               limits 50.09 to 50.88 mm (width 0.79), 18 of 20 points out
```

With mixed subgroups the machine difference sits *inside* every subgroup. The range is inflated, the limits are wide, and the X-bar chart looks calm: the chart has hidden a one-millimetre difference between two machines. With one machine per subgroup the within-subgroup variation is the true short-term scatter, the limits are tight, and nearly every point now signals. The process has not changed. The chart has finally been asked a question it can answer.

Discussion: a 4-cavity mould fills four parts per shot. Is "the four parts of one shot" a rational subgroup?

Usually not for an X-bar chart. The four cavities differ systematically (flow length, cooling), so the range of a shot mostly measures cavity-to-cavity difference. The limits widen and cavity problems stay hidden.

Better options: chart each cavity separately (the subgroup is consecutive shots from one cavity), or chart the shot average on an I-MR chart and monitor the cavity differences on a separate chart. Which one you choose depends on whether cavity differences are something the process can control or a fixed feature of the tool.

18.4 Charts for Variables

18.4.1 X-bar/S Chart

The range is quick to calculate, but it uses only the two extreme readings. In a subgroup of 10 or more, the standard deviation s uses all of them and estimates the spread better. The rule of thumb is **X-bar/R for $n \le 9$, X-bar/S for $n \ge 10$**, and also whenever the calculation is done by software anyway.

$$\begin{aligned} \text{X-bar chart:} \quad & UCL, LCL = \bar{\bar{x}} \pm A_3\,\bar{s} \\ \text{S chart:} \quad & UCL = B_4\,\bar{s}, \quad LCL = B_3\,\bar{s} \\ \text{Estimate of } \sigma: \quad & \hat{\sigma} = \bar{s}/c_4 \end{aligned}$$

Table 18.1: Control Chart Constants for X-bar/S Charts

n	c4	A3	B3	B4
2	0.7979	2.659	0.000	3.267
3	0.8862	1.954	0.000	2.568
4	0.9213	1.628	0.000	2.266
5	0.9400	1.427	0.000	2.089
6	0.9515	1.287	0.030	1.970
7	0.9594	1.182	0.118	1.882
8	0.9650	1.099	0.185	1.815
9	0.9693	1.032	0.239	1.761
10	0.9727	0.975	0.284	1.716

Example: a filling line checks 10 bags every 30 minutes (target 500 g).

```
set.seed(77)
n <- 10; k <- 25
A3 <- 0.975; B3 <- 0.284; B4 <- 1.716; c4 <- 0.9727   # constants for n = 10

weights <- matrix(rnorm(n * k, mean = 500, sd = 3.5), nrow = k)
weights[22:25, ] <- weights[22:25, ] + 4    # fill head drifts upward in the last four

xbar <- rowMeans(weights)
s    <- apply(weights, 1, sd)
xbarbar <- mean(xbar); sbar <- mean(s)

limits <- list(x_ucl = xbarbar + A3 * sbar, x_lcl = xbarbar - A3 * sbar,
               s_ucl = B4 * sbar, s_lcl = B3 * sbar)

cat("=== X-bar/S chart, bag fill weight (g) ===\n\n")
cat(sprintf("Grand mean  %.2f g,  average s  %.2f g,  estimated sigma  %.2f g\n", xbarb
```

```
cat(sprintf("X-bar chart  LCL %.2f   CL %.2f   UCL %.2f\n", limits$x_lcl, xbarbar, limits$x_ucl))
cat(sprintf("S chart      LCL %.2f   CL %.2f   UCL %.2f\n", limits$s_lcl, sbar, limits$s_ucl))
cat("Subgroups beyond the X-bar limits:", which(xbar > limits$x_ucl | xbar < limits$x_lcl), "\n")

charts <- rbind(
  data.frame(chart = "X-bar chart (g)", i = 1:k, y = xbar, cl = xbarbar, ucl = limits$x_ucl, lcl
  data.frame(chart = "S chart (g)", i = 1:k, y = s, cl = sbar, ucl = limits$s_ucl, lcl = limits$s
charts$out <- charts$y > charts$ucl | charts$y < charts$lcl

ggplot(charts, aes(x = i, y = y)) +
  geom_hline(aes(yintercept = ucl), color = "#C0392B", linetype = "dashed") +
  geom_hline(aes(yintercept = lcl), color = "#C0392B", linetype = "dashed") +
  geom_hline(aes(yintercept = cl), color = "#27AE60") +
  geom_line(color = "grey50") +
  geom_point(aes(color = out), size = 3) +
  scale_color_manual(values = c("FALSE" = "steelblue", "TRUE" = "#C0392B"), guide = "none") +
  facet_wrap(~chart, ncol = 1, scales = "free_y") +
  labs(title = "X-bar/S Chart: Fill Head Drift in the Last Four Subgroups", x = "Subgroup", y = N
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 12 * chart_text_scale),
        strip.text = element_text(face = "bold", size = 10 * chart_text_scale))
```

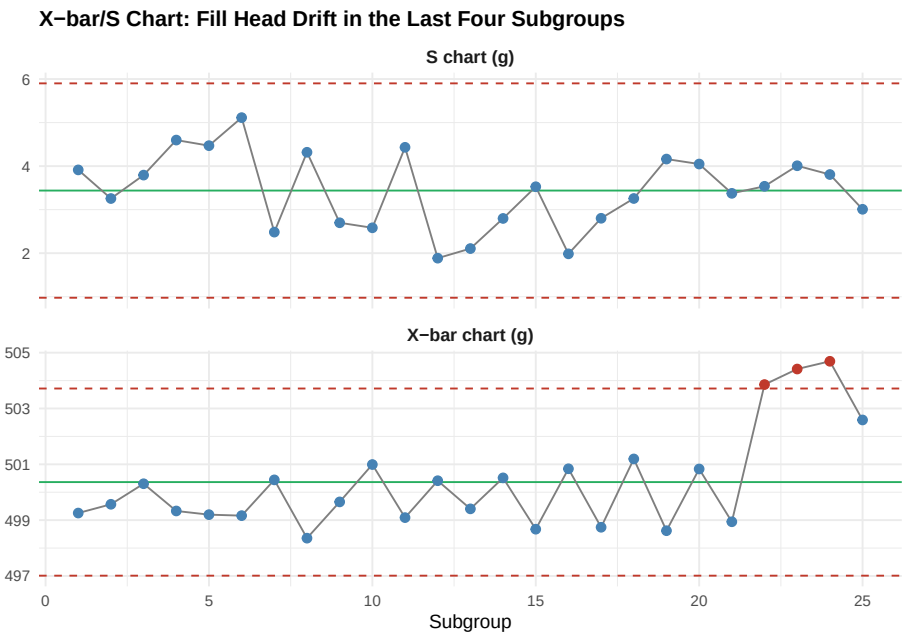


Figure 18.2: X-bar/S chart of bag fill weight, 25 subgroups of 10

```
=== X-bar/S chart, bag fill weight (g) ===

Grand mean  500.36 g,  average s  3.44 g,  estimated sigma  3.53 g
X-bar chart  LCL 497.01   CL 500.36   UCL 503.72
S chart      LCL 0.98   CL 3.44   UCL 5.90
Subgroups beyond the X-bar limits: 22 23 24
```

Your turn : change the subgroup size and see how much the X-bar limits tighten. This is the price of a larger sample.

Show the live R code behind this calculator

The card above re-runs this exact webR code every time you change an input, substituting your values for the variables at the top. Edit the code directly and press **Run Code** to experiment further.

```
#| label: xbar-s-limits
grand_mean <- 500
s_bar <- 3.5
n <- 3

# Constants for n = 2, 3, ..., 10
c4_tab <- c(0.7979, 0.8862, 0.9213, 0.9400, 0.9515, 0.9594, 0.9650, 0.9693, 0.9727)
A3_tab <- c(2.659, 1.954, 1.628, 1.427, 1.287, 1.182, 1.099, 1.032, 0.975)
B3_tab <- c(0, 0, 0, 0, 0.030, 0.118, 0.185, 0.239, 0.284)
B4_tab <- c(3.267, 2.568, 2.266, 2.089, 1.970, 1.882, 1.815, 1.761, 1.716)

A3 <- A3_tab[n - 1]
x_ucl <- grand_mean + A3 * s_bar
x_lcl <- grand_mean - A3 * s_bar
s_ucl <- B4_tab[n - 1] * s_bar
s_lcl <- B3_tab[n - 1] * s_bar
xbar_width <- x_ucl - x_lcl

cat(sprintf("X-bar chart width: %.2f g (%.1f to %.1f)\n", xbar_width, x_lcl, x_ucl))
cat(sprintf("  Constant A3  %.3f\n", A3))
cat(sprintf("  S chart LCL  %.2f g\n", s_lcl))
cat(sprintf("  S chart UCL  %.2f g\n", s_ucl))
cat(sprintf("  Estimated sigma  %.2f g\n", s_bar / c4_tab[n - 1]))

par(mar = c(5, 5, 3, 1))
plot(2:10, 2 * A3_tab * s_bar, type = "b", pch = 19, xlab = "Subgroup size n",
     ylab = "X-bar chart width (g)", main = "Wider subgroups, tighter limits")
points(n, xbar_width, pch = 19, cex = 2.2, col = "#C0392B")
```

The limits narrow quickly at first (from $n = 2$ to $n = 5$ the width nearly halves) and slowly after $n = 6$. That is why subgroups of 4 or 5 are the common choice, and a larger subgroup is only worth the extra measuring effort when the chart

needs to catch a small shift.

18.5 Attribute Charts for Counts

The p chart and c chart in the introductory chapter cover the two most common cases. Two companions complete the set.

Chart	Plots	Sample size	Centre line and limits
np	Number of defective units	Constant, n	$n\bar{p} \pm 3\sqrt{n\bar{p}(1-\bar{p})}$
u	Defects per unit	Varies from sample to sample, n_i	$\bar{u} \pm 3\sqrt{\bar{u}/n_i}$, with $\bar{u} = \sum c_i / \sum n_i$

The np chart is the p chart in disguise, plotted as a count instead of a fraction: it is preferred on the shop floor because operators prefer to plot "7 bad pieces" than "0.047". The u chart is the c chart for the case where the **area of opportunity changes**: boards per lot, metres of weld per shift, square metres of paint per day. Its limits widen and narrow with the sample size, just like the p chart's.

```
set.seed(11)
lots <- 20
boards  <- sample(30:60, lots, replace = TRUE)          # boards inspected per lot
defects <- rpois(lots, lambda = 1.8 * boards)            # in-control rate: 1.8 defects per board
defects[15] <- rpois(1, lambda = 3.6 * boards[15])       # lot 15: a bad solder-paste batch

u     <- defects / boards
u_bar <- sum(defects) / sum(boards)
ucl   <- u_bar + 3 * sqrt(u_bar / boards)
lcl   <- pmax(0, u_bar - 3 * sqrt(u_bar / boards))
u_data <- data.frame(lot = 1:lots, boards, u, ucl, lcl, out = u > ucl | u < lcl)

cat(sprintf("u-bar = %.3f defects per board\n", u_bar))
cat("Lots outside the limits:", u_data$lot[u_data$out], "\n")
cat(sprintf("Limit width ranges from +/- %.2f (60 boards) to +/- %.2f (30 boards)\n",
            3 * sqrt(u_bar / 60), 3 * sqrt(u_bar / 30)))

ggplot(u_data, aes(x = lot, y = u)) +
  geom_line(aes(y = ucl), color = "#C0392B", linetype = "dashed") +
  geom_line(aes(y = lcl), color = "#C0392B", linetype = "dashed") +
```

```
  geom_hline(yintercept = u_bar, color = "#27AE60") +
  geom_line(color = "grey50") +
  geom_point(aes(color = out), size = 3) +
  scale_color_manual(values = c("FALSE" = "steelblue", "TRUE" = "#C0392B"), guide = "no
  labs(title = "u Chart: Defects per Board with Unequal Lot Sizes",
       x = "Lot", y = "Defects per board") +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 12 * chart_text_scale))
```

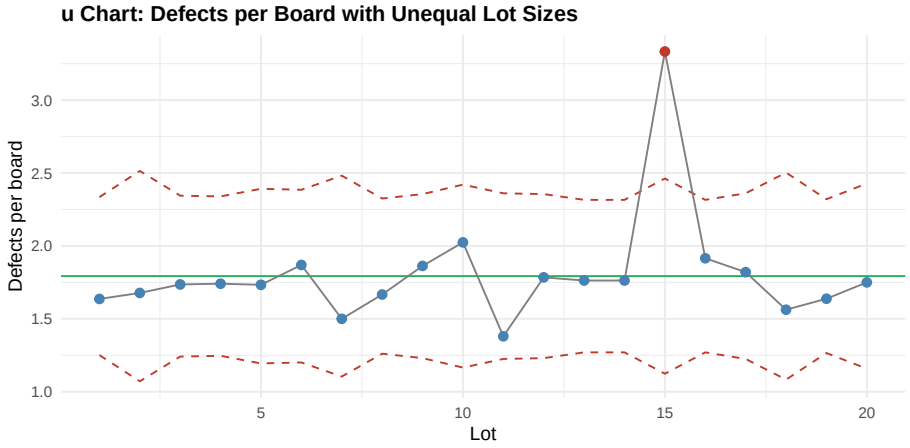


Figure 18.3: u chart of solder defects per board, with lot sizes between 30 and 60 boards

```
u-bar = 1.793 defects per board
Lots outside the limits: 15
Limit width ranges from +/- 0.52 (60 boards) to +/- 0.73 (30 boards)
```

18.5.1 Try it yourself: an np chart

A stamping cell checks 150 brackets every shift for a burr. Over the last 25 shifts, the average fraction defective was $\bar{p} = 0.04$.

np chart: burrs on stamped brackets

- Centre line, n x p-bar: _____ *(to 2 decimal places)*
- UCL: _____ *(to 1 decimal place)*
- LCL (enter 0 if the formula gives a negative number): _____ *(to 1 decimal place)*

Show the worked solution

- Centre line: $n\bar{p} = 150 \times 0.04 =$ **6.0**
- Spread: $\sqrt{6.0 \times (1 - 0.04)} = 2.400$

- UCL $= 6.0 + 3 \times 2.400 =$ **13.2**
- LCL $= 6.0 - 3 \times 2.400 = -1.2$, which is negative, so **LCL = 0**

A shift with 14 or more defective brackets signals. With an average of 6 per shift, the chart can only detect a rise, never a fall, which is why the zero lower limit matters.

18.6 Process Capability in Depth

18.6.1 Within-Subgroup and Overall Spread: Cp/Cpk and Pp/Ppk

Every capability index divides a specification width by a spread. The four indices differ only in which spread they use:

Index	Spread used	Estimated from	Describes
Cp, Cpk	$\hat{\sigma}_{within}$, the short-term spread	$\bar{R}/d_2$ or $\bar{s}/c_4$	What the process could do if it were centred and stable
Pp, Ppk	$\hat{\sigma}_{overall}$, the long-term spread	The standard deviation s of **all** the data	What the process actually delivered, drift and all

If the process is stable, the two spreads agree and the indices match. If the process drifts between subgroups (tool wear, temperature, material lots), the overall spread grows while the within-subgroup spread stays small. The **gap between Cpk and Ppk is therefore a measure of how unstable the process is**.

```
# 25 subgroups of 5 shaft diameters (mm); tool wear moves the process mean upward
set.seed(314)
LSL <- 24.85; USL <- 25.15
d2 <- 2.326    # n = 5

drift   <- seq(24.99, 25.05, length.out = 25)
shafts  <- t(sapply(drift, function(m) rnorm(5, mean = m, sd = 0.020)))

x_all <- as.vector(shafts)
r_bar <- mean(apply(shafts, 1, function(v) max(v) - min(v)))
```

```r
sigma_within  <- r_bar / d2
sigma_overall <- sd(x_all)
mu <- mean(x_all)

Cp  <- (USL - LSL) / (6 * sigma_within)
Cpk <- min(USL - mu, mu - LSL) / (3 * sigma_within)
Pp  <- (USL - LSL) / (6 * sigma_overall)
Ppk <- min(USL - mu, mu - LSL) / (3 * sigma_overall)

cat("=== Capability versus performance ===\n\n")
cat(sprintf("Mean %.4f mm,  sigma within %.4f mm,  sigma overall %.4f mm\n\n", mu, sigm
cat(sprintf("Cp  = %.2f    Pp  = %.2f\n", Cp, Pp))
cat(sprintf("Cpk = %.2f    Ppk = %.2f\n\n", Cpk, Ppk))
cat(sprintf("Ppk is %.0f%% below Cpk: %s\n", (1 - Ppk / Cpk) * 100,
            ifelse(Ppk / Cpk < 0.85, "the process is drifting between subgroups", "the
```

```
=== Capability versus performance ===

Mean 25.0188 mm,  sigma within 0.0195 mm,  sigma overall 0.0279 mm

Cp  = 2.56    Pp  = 1.79
Cpk = 2.24    Ppk = 1.57

Ppk is 30% below Cpk: the process is drifting between subgroups
```

Question: When should I quote Cpk and when Ppk?

Cp/Cpk (process capability):

- Uses within-subgroup variation (σ estimated from $\bar{R}/d_2$)
- Represents the *potential* capability if the process is in control
- Use for ongoing monitoring, once the process is shown to be stable
- Requires the process to be in statistical control

Pp/Ppk (process performance):

- Uses the overall standard deviation s of all the data
- Represents the *actual* performance, including all variation
- Use for the initial run (a PPAP submission asks for it) or when stability is unknown
- Includes both common-cause and special-cause variation

Rule of thumb:

- If Pp is close to Cp and Ppk is close to Cpk, the process is stable
- If Pp is much smaller than Cp, or Ppk much smaller than Cpk, the process has between-subgroup variation (drift, shifts, mixed sources)

Your turn : the process below has a short-term spread of 0.020 mm but an

overall spread of 0.035 mm, and it runs slightly off-centre.

Show the live R code behind this calculator

The card above re-runs this exact webR code every time you change an input, substituting your values for the variables at the top. Edit the code directly and press **Run Code** to experiment further.

```
#| label: cpk-ppk-calc
LSL <- 24.85
USL <- 25.15
process_mean <- 25.04
sigma_within <- 0.020
sigma_overall <- 0.035

nearest <- min(USL - process_mean, process_mean - LSL)
Cp  <- (USL - LSL) / (6 * sigma_within)
Cpk <- nearest / (3 * sigma_within)
Pp  <- (USL - LSL) / (6 * sigma_overall)
Ppk <- nearest / (3 * sigma_overall)

cat(sprintf("Ppk: %.2f (%s)\n", Ppk, ifelse(Ppk >= 1.33, "capable", "!!below 1.33!!")))
cat(sprintf("  Cpk  %.2f\n", Cpk))
cat(sprintf("  Pp  %.2f\n", Pp))
cat(sprintf("  Cp  %.2f\n", Cp))
cat(sprintf("  Distance to nearest limit  %.3f mm\n", nearest))

par(mar = c(5, 5, 3, 1))
barplot(c(Cp, Cpk, Pp, Ppk), names.arg = c("Cp", "Cpk", "Pp", "Ppk"),
        col = c("#2E86C1", "#2E86C1", "#F39C12", "#F39C12"), ylab = "Index",
        ylim = c(0, max(2, Cp, Pp) * 1.1), main = "Short-term (blue) vs long-term (orange)")
abline(h = 1.33, lty = 2, lwd = 2, col = "#C0392B")
```

18.6.2 Capability When the Data Are Not Normal

All four indices turn a spread into a defect rate through the normal curve. Many measured characteristics are not normal. Runout, flatness, surface roughness, concentricity and the count of particles all have a hard lower limit at zero and a long tail on the high side. Using the normal formula on such data usually **flatters** the process, because the tail it ignores is exactly where the parts fail.

The **percentile method** replaces $\pm 3\sigma$ around the mean with the actual 0.135% and 99.865% points of the data:

$$P_{pk} = \min\left(\frac{USL - \tilde{x}}{P_{99.865} - \tilde{x}},\ \frac{\tilde{x} - LSL}{\tilde{x} - P_{0.135}}\right)$$

where $\tilde{x}$ is the median. For a one-sided characteristic such as runout, only the upper term applies.

```
# Total indicated runout (mm), upper limit only: USL = 0.100 mm
set.seed(808)
USL <- 0.100
runout <- rlnorm(400, meanlog = log(0.040), sdlog = 0.40)

mu <- mean(runout); s <- sd(runout); med <- median(runout)
Ppk_normal <- (USL - mu) / (3 * s)

p99865 <- quantile(runout, 0.99865, names = FALSE)
Ppk_pct <- (USL - med) / (p99865 - med)

cat("=== Runout capability (USL = 0.100 mm) ===\n\n")
cat(sprintf("Mean %.4f, median %.4f, standard deviation %.4f mm\n", mu, med, s))
cat(sprintf("Ppk assuming normal data:   %.2f   (predicts %.2f%% over the limit)\n",
            Ppk_normal, 100 * (1 - pnorm((USL - mu) / s))))
cat(sprintf("Ppk by percentile method:   %.2f\n", Ppk_pct))
cat(sprintf("Parts actually over the limit: %.2f%%\n", 100 * mean(runout > USL)))

ggplot(data.frame(runout), aes(x = runout)) +
  geom_histogram(aes(y = after_stat(density)), bins = 30, fill = "steelblue", alpha = 
  stat_function(fun = dnorm, args = list(mean = mu, sd = s), color = "#E67E22", linewi
  geom_vline(xintercept = USL, color = "#C0392B", linetype = "dashed", linewidth = 1) 
  annotate("text", x = USL, y = Inf, label = "USL", hjust = -0.2, vjust = 1.6,
           color = "#C0392B", fontface = "bold", size = 4 * diagram_text_scale) +
  labs(title = "Skewed Data Fooled by a Normal Curve",
       subtitle = "Orange line: the normal curve with the same mean and standard deviat
       x = "Runout (mm)", y = "Density") +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 12 * chart_text_scale))
```

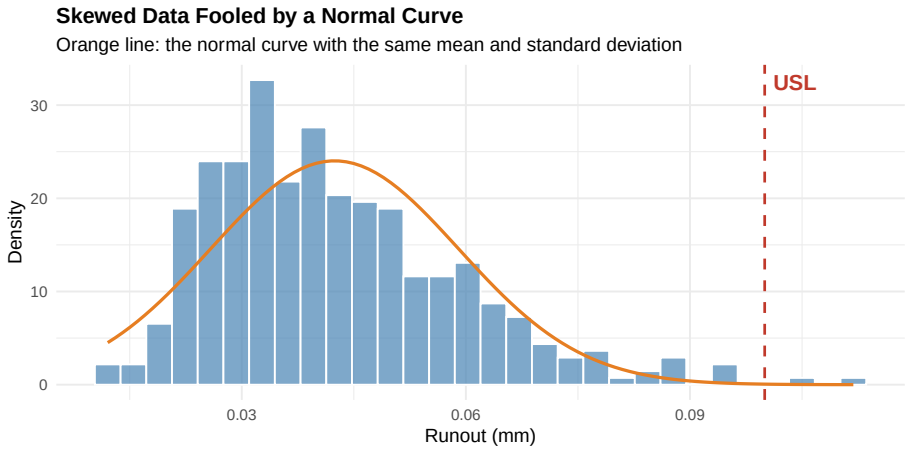


Figure 18.4: Shaft runout is skewed: the normal formula and the percentile method disagree

```
=== Runout capability (USL = 0.100 mm) ===

Mean 0.0424, median 0.0397, standard deviation 0.0166 mm
Ppk assuming normal data:   1.16   (predicts 0.03% over the limit)
Ppk by percentile method:   0.88
Parts actually over the limit: 0.50%
```

The orange curve spills below zero and under-predicts the tail on the high side, so the normal-based Ppk looks acceptable while a visible share of parts exceed the limit. Software offers a transformation (Box-Cox, Johnson) or a fitted distribution (Weibull, lognormal) as alternatives; the percentile method is the one you can calculate by hand and explain to a customer.

18.6.3 How Much Can You Trust a Cpk?

A Cpk is an estimate from a sample, so it has an uncertainty of its own. An approximate 95% confidence interval is

$$\hat{C}_{pk} \pm 1.96\sqrt{\frac{1}{9n} + \frac{\hat{C}_{pk}^2}{2(n-1)}}$$

```
cpk_hat <- 1.33
sample_sizes <- c(30, 50, 100, 200, 500)
half_width <- 1.96 * sqrt(1 / (9 * sample_sizes) + cpk_hat^2 / (2 * (sample_sizes - 1)))

ci <- data.frame(`Sample size` = sample_sizes,
                 `Lower bound` = round(cpk_hat - half_width, 2),
```

```
                `Upper bound` = round(cpk_hat + half_width, 2),
                check.names = FALSE)
ci %>%
  kable(caption = "95% confidence interval for an observed Cpk of 1.33") %>%
  kable_styling(bootstrap_options = c("striped", "hover"), full_width = FALSE)
```

Table 18.2: 95% confidence interval for an observed Cpk of 1.33

Sample size	Lower bound	Upper bound
30	0.97	1.69
50	1.05	1.61
100	1.13	1.53
200	1.19	1.47
500	1.24	1.42

An observed Cpk of 1.33 from 30 parts is consistent with a true Cpk anywhere from about 1.0 to 1.7. That is why customers specify the sample size of a capability study (typically 100 or more), and sometimes ask for a *lower confidence bound* of 1.33 rather than a point estimate of it.

18.7 Detecting Small Shifts: CUSUM and EWMA

A Shewhart chart judges each point on its own. A process that has moved by one half or one standard deviation produces points that are mostly inside the limits, and the Western Electric run rules take a long time to confirm it. Two charts solve this by **remembering the past**.

CUSUM (cumulative sum). Add up the deviations from target, ignoring any that are within a slack of k:

$$C_i^+ = \max\left(0,\; x_i - (\mu_0 + k) + C_{i-1}^+\right) \qquad C_i^- = \max\left(0,\; (\mu_0 - k) - x_i + C_{i-1}^-\right)$$

The chart signals when either sum passes the decision interval h. Common choices are $k = 0.5\sigma$ and $h = 4\sigma$ or 5σ.

EWMA (exponentially weighted moving average). Each plotted point blends the new reading with the previous plotted value:

$$z_i = \lambda x_i + (1-\lambda) z_{i-1}, \qquad z_0 = \mu_0 \qquad\qquad UCL, LCL = \mu_0 \pm L\sigma\sqrt{\frac{\lambda}{2-\lambda}\left[1-(1-\lambda)^{2i}\right]}$$

The weight λ sets the memory: a small λ (0.05 to 0.2) averages over a long history and catches small shifts; $\lambda = 1$ is a Shewhart chart. Typical settings are $\lambda = 0.2$ and $L = 2.7$ to 3.

```
set.seed(1)
mu0 <- 10; sigma <- 1; n_obs <- 40; shift_at <- 21; shift <- 1
x <- rnorm(n_obs, mu0, sigma)
x[shift_at:n_obs] <- x[shift_at:n_obs] + shift     # a 1-sigma shift starts at reading 21

# Individuals chart: +/- 3 sigma
i_sig <- which(abs(x - mu0) > 3 * sigma)

# CUSUM with k = 0.5 sigma and h = 4 sigma
k <- 0.5 * sigma; h <- 4 * sigma
c_up <- c_dn <- numeric(n_obs); up <- dn <- 0
for (i in seq_len(n_obs)) {
  up <- max(0, up + x[i] - (mu0 + k))
  dn <- max(0, dn + (mu0 - k) - x[i])
  c_up[i] <- up; c_dn[i] <- dn
}
c_sig <- which(c_up > h | c_dn > h)

# EWMA with lambda = 0.2 and L = 2.7
lambda <- 0.2; L <- 2.7
z <- numeric(n_obs); prev <- mu0
for (i in seq_len(n_obs)) { prev <- lambda * x[i] + (1 - lambda) * prev; z[i] <- prev }
ew_w  <- L * sigma * sqrt(lambda / (2 - lambda) * (1 - (1 - lambda)^(2 * seq_len(n_obs))))
e_sig <- which(abs(z - mu0) > ew_w)

first <- function(s) if (length(s)) s[1] else NA
cat("=== First signal (the shift starts at reading 21) ===\n\n")
cat("Individuals chart: ", ifelse(is.na(first(i_sig)), "no signal", paste("reading", first(i_sig)
cat("CUSUM:             ", ifelse(is.na(first(c_sig)), "no signal", paste("reading", first(c_sig)
cat("EWMA:              ", ifelse(is.na(first(e_sig)), "no signal", paste("reading", first(e_sig)

panels <- rbind(
  data.frame(chart = "Individuals chart", i = 1:n_obs, y = x, lo = mu0 - 3 * sigma, hi = mu0 + 3
  data.frame(chart = "CUSUM (upper sum above 0, lower sum below 0)", i = 1:n_obs, y = c_up - c_dn
  data.frame(chart = "EWMA (lambda = 0.2, L = 2.7)", i = 1:n_obs, y = z, lo = mu0 - ew_w, hi = mu
panels$chart <- factor(panels$chart, levels = unique(panels$chart))
panels$out <- panels$y > panels$hi | panels$y < panels$lo

ggplot(panels, aes(x = i, y = y)) +
  geom_vline(xintercept = shift_at - 0.5, color = "grey40", linetype = "dotted", linewidth = 1) +
  geom_line(aes(y = hi), color = "#C0392B", linetype = "dashed") +
  geom_line(aes(y = lo), color = "#C0392B", linetype = "dashed") +
```

```
  geom_hline(aes(yintercept = cl), color = "#27AE60") +
  geom_line(color = "grey50") +
  geom_point(aes(color = out), size = 2.5) +
  scale_color_manual(values = c("FALSE" = "steelblue", "TRUE" = "#C0392B"), guide = "no
  facet_wrap(~chart, ncol = 1, scales = "free_y") +
  labs(title = "A 1-Sigma Shift: Only the Memory Charts Notice",
       subtitle = "Dotted line: where the shift starts",
       x = "Reading", y = NULL) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 12 * chart_text_scale),
        strip.text = element_text(face = "bold", size = 10 * chart_text_scale))
```

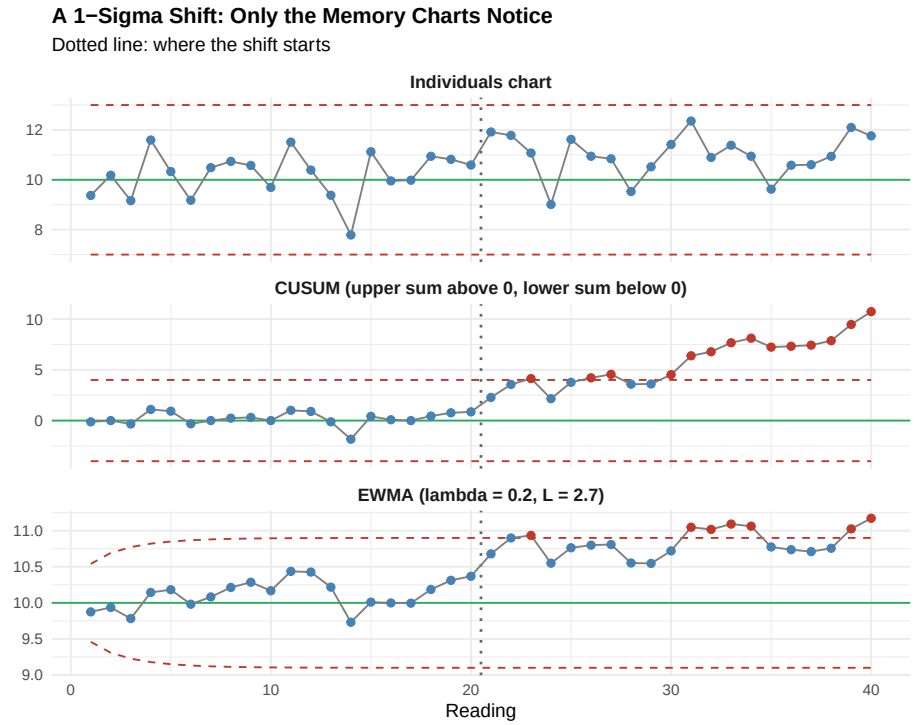


Figure 18.5: The same 40 readings on an individuals chart, a CUSUM and an EWMA; the mean shifts by 1 sigma after reading 20

```
=== First signal (the shift starts at reading 21) ===

Individuals chart:  no signal
CUSUM:              reading 23
EWMA:               reading 23
```

The individuals chart sees nothing: every shifted reading is still well inside the

3-sigma limits. The CUSUM and EWMA both react within a few readings of the shift, because each new reading above target adds to the evidence instead of being judged alone.

18.7.1 Average Run Length

Charts are compared by their **average run length (ARL)**: the average number of points plotted before the chart signals. We want a long ARL when the process is on target (few false alarms) and a short ARL after a shift. A 3-sigma Shewhart chart has an in-control ARL of about 370. For the same false-alarm rate, the memory charts are much faster at small shifts.

Shift size	Shewhart (3 sigma)	CUSUM ($k = 0.5$, $h = 4$)	EWMA ($\lambda = 0.2$, $L = 2.7$)
0.5 sigma	about 150	about 28	about 28
1.0 sigma	about 42	about 8	about 8
2.0 sigma	about 6	about 3	about 3
3.0 sigma	about 2	about 2	about 2

For large shifts all three charts perform about the same, and a Shewhart chart is simpler to run and to explain. The usual practice is to run an X-bar or I chart alongside a CUSUM or EWMA, so that large and small shifts are both covered.

Your turn : the simulation below estimates the ARL for a 0.5-sigma shift from 1000 simulated runs, for the three charts. It uses the same random numbers for every chart, so the differences are the charts and not luck.

Show the live R code behind this calculator

The card above re-runs this exact webR code every time you change an input, substituting your values for the variables at the top. Edit the code directly and press **Run Code** to experiment further.

```
#| label: arl-sim
shift <- 0.5
lambda <- 0.5
L <- 2.7
seed <- 1

set.seed(seed)
runs <- 1000
```

```
max_n <- 500
ewma <- rep(0, runs); cu_up <- rep(0, runs); cu_dn <- rep(0, runs)
rl_shewhart <- rl_cusum <- rl_ewma <- rep(max_n, runs)
found_s <- found_c <- found_e <- rep(FALSE, runs)

for (t in 1:max_n) {
  x <- rnorm(runs, mean = shift, sd = 1)          # shifted process, in units of sigma
  ewma  <- lambda * x + (1 - lambda) * ewma
  cu_up <- pmax(0, cu_up + x - 0.5)
  cu_dn <- pmax(0, cu_dn - x - 0.5)
  width <- L * sqrt(lambda / (2 - lambda) * (1 - (1 - lambda)^(2 * t)))
  hit_s <- !found_s & abs(x) > 3
  hit_c <- !found_c & (cu_up > 4 | cu_dn > 4)
  hit_e <- !found_e & abs(ewma) > width
  rl_shewhart[hit_s] <- t; rl_cusum[hit_c] <- t; rl_ewma[hit_e] <- t
  found_s <- found_s | hit_s; found_c <- found_c | hit_c; found_e <- found_e | hit_e
}

Shewhart_arl <- mean(rl_shewhart)
CUSUM_arl <- mean(rl_cusum)
EWMA_arl <- mean(rl_ewma)

cat(sprintf("EWMA average run length: %.1f readings\n", EWMA_arl))
cat(sprintf("  CUSUM  %.1f readings\n", CUSUM_arl))
cat(sprintf("  Shewhart (3 sigma)  %.1f readings\n", Shewhart_arl))
cat(sprintf("  Shift  %g sigma, %d simulated runs\n", shift, runs))

par(mar = c(5, 5, 3, 1))
barplot(c(Shewhart_arl, CUSUM_arl, EWMA_arl), names.arg = c("Shewhart", "CUSUM", "EWMA
        col = c("#95A5A6", "#2E86C1", "#F39C12"), ylab = "Average run length (readings
        main = "Readings until the chart signals")
```

Keep one thing in mind: shrinking L makes every EWMA faster, but it also makes it signal on a process that has not moved. The comparison is only fair at a fixed false-alarm rate, which is what the goal above enforces by keeping L at 2.6 or more.

18.7.2 Try it yourself: a CUSUM by hand

A filling process targets 10.0 g with $\sigma = 1$. Use $k = 0.5$ (so the slack is 0.5) and the decision interval $h = 4$. The table shows eight readings. Fill in the upper CUSUM C^+ for each, then find the reading at which the chart first signals.

Upper CUSUM, reading by reading

Target 10.0, k = 0.5, h = 4. Fill in C+ for each reading, then the reading

number of the first signal.

Reading	x	C+
1	10.40	
2	9.80	
3	10.90	
4	11.20	
5	10.60	
6	11.40	
7	11.10	
8	12.00	

- First reading at which C+ exceeds h = 4: _____ *(to 0 decimal places)*

Show the worked solution

Table 18.3: Upper CUSUM, target 10.0, k = 0.5

Reading	x	x - 10.5	C+
1	10.4	-0.1	0.0
2	9.8	-0.7	0.0
3	10.9	0.4	0.4
4	11.2	0.7	1.1
5	10.6	0.1	1.2
6	11.4	0.9	2.1
7	11.1	0.6	2.7
8	12.0	1.5	4.2

The sum stays at 0 while the readings sit near target, then grows by the excess over 10.5 on each high reading. It first exceeds h = 4 at **reading 8** (C+ = 4.2). A Shewhart chart with 3-sigma limits at 13.0 would not have signalled yet: the highest reading is 12.0.

18.8 Special Situations

18.8.1 Short-Run Production

A job shop machines 50 pieces of one part number, then changes over to another. Each part number has its own mean and its own spread, so there are never enough points on one chart to set limits. The solution is to plot a **standardised** value that makes all part numbers comparable:

- **Deviation from nominal** (a "DNOM" chart): plot $x-$nominal. It works when all part numbers have a similar spread.
- **Z chart:** plot $(x - \text{target})/\sigma_{part}$, with limits at ± 3. It also removes differences in spread, but needs a believable σ for each part number.

Whichever is used, the chart is plotted continuously across changeovers, so a trend or shift shows up even though the part number has changed.

18.8.2 Autocorrelated Data

Control limits assume that successive readings are independent. A furnace temperature, a bath concentration or the level of a tank changes slowly, so each reading looks like the one before it. A Shewhart chart on such data signals far too often. Options:

- Sample less often, until the readings are independent.
- Chart the *residuals* of a time-series model rather than the readings.
- Use an EWMA of the data with limits adjusted for the autocorrelation, or a "moving centre line" chart.

Check by plotting x_i against x_{i-1}: a visible slope means the independence assumption is broken.

18.8.3 Over-Adjustment

Deming's funnel experiment from the introductory chapter has a second lesson here: if a CUSUM or EWMA chart is allowed to trigger an adjustment *every time* it moves, the adjustments themselves add variation. A memory chart should signal an investigation first. Automatic compensation belongs to the process controller, not the operator chart.

18.9 Choosing the Chart

Question	Chart
Subgroups of 10 or more measurements	X-bar/S
Constant sample size, counting defective units	np
Varying number of units, counting defects per unit	u
Small drift matters (under about 1.5 sigma)	CUSUM or EWMA
Many part numbers, short runs	Z or DNOM chart
Initial run, stability unknown	Report Pp/Ppk and a chart together

18.9.1 Try it yourself: which chart fits?

Which chart for each situation?

Categories: X-bar / S / np chart / u chart / CUSUM or EWMA / Z or DNOM chart

1. A fill station weighs 12 packs every 15 minutes, and the supervisor wants to see standard deviation as well as the average. _____
2. Every shift, 200 sealed pouches are tested for leaks, and the count of leakers is plotted. _____
3. A paint line inspects a different number of car doors each day and counts the paint flaws found. _____
4. A heat-treat furnace drifts by a few tenths of a degree per week and nobody wants to wait for a 3-sigma point. _____
5. A CNC cell runs 30 different brackets in lots of 25, each with its own nominal. _____
6. A bottling plant takes 15 bottles from each batch of caps and wants a chart as sensitive as possible to a loose capping head. _____
7. A solder-paste printer's output is judged by a drift of 0.5 sigma in deposit volume that is costly if it persists. _____
8. The number of defective connectors in each lot of 500 is recorded. _____

Answers: 1 X-bar / S; 2 np chart; 3 u chart; 4 CUSUM or EWMA; 5 Z or DNOM chart; 6 X-bar / S; 7 CUSUM or EWMA; 8 np chart.

18.10 Summary

1. **Phase I sets the limits, Phase II uses them.** Do not recompute limits to follow the data, and do not delete a point without an assignable cause.
2. **Subgrouping is a design decision.** Mixed sources inside a subgroup hide the very differences the chart should find.
3. **Use S for large subgroups, np and u for counts.** The u chart handles unequal sample sizes; its limits move with the sample.
4. **Cpk is potential, Ppk is performance.** A large gap between them says the process is not stable.
5. **Normal formulas flatter skewed data.** Check the shape, use the percentile method, and state the sample size behind any capability figure.
6. **CUSUM and EWMA catch small shifts.** Keep a Shewhart chart for large ones and compare charts by ARL.
7. **Short-run and autocorrelated processes need adapted charts**, not forced onto the standard ones.

> "A control chart that nobody understands will eventually be ignored, and one that is over-tuned will be distrusted. Use the simplest chart that sees the problem you have."

18.11 Review Questions

Question 1: A process has Cpk = 1.60 and Ppk = 0.95. What does the gap tell you, and what would you check first?

Answer:

The short-term spread is small (Cpk is excellent) but the long-term performance is poor. The difference between σ_{within} and $\sigma_{overall}$ is **between-subgroup variation**: something is changing the process from subgroup to subgroup.

Check first:

1. Plot the X-bar chart: is there a trend (tool wear) or a step (new material lot, new shift)?
2. Look at the data by machine, cavity, shift and material lot: a mixture of sources produces the same pattern.
3. Look at the subgrouping: if the subgroup is too short, slow drift is hidden inside it.

Fix the cause of the shift before spending money on a tighter machine: the machine is already capable, but the process is not stable.

Question 2: Why does a chart on subgroups that each contain one part from each of two machines look "better" than a chart with one machine per subgroup, and which one should you trust?

Answer:

With mixed subgroups the machine difference sits inside every subgroup. The within-subgroup range is inflated, so the limits are wide enough to hide the difference, and the X-bar chart looks calm. With one machine per subgroup the range reflects only the short-term scatter, the limits are tight, and the machine difference appears as a pattern between subgroups.

Trust the second chart. The first one looks better only because it is unable to ask the right question.

Question 3: Give two situations where you would add an EWMA or CUSUM to an X-bar chart, and one where the X-bar chart alone is enough.

Answer:

Add a memory chart: tool wear that moves the dimension by a fraction of a sigma per day; a slow drift in a bath concentration or a furnace; any process where a half-sigma shift costs real money (rework, scrap, warranty).

X-bar alone is enough: a process where the failure modes are abrupt (a broken tool, a wrong setup, a missing component) and large. A Shewhart chart

catches a shift of 2 sigma or more within a few subgroups and is easier to run and explain to operators.

Question 4: A capability study of 30 parts gives Cpk = 1.45. The customer requires 1.33. Is the requirement met?

Answer:

The point estimate is above 1.33, but with only 30 parts the 95% confidence interval for Cpk is roughly 1.45 $\pm$ 0.40, so the true value could be near 1.05. Unless the customer specifies a point estimate, the evidence is **not conclusive**. Collect a larger sample (100 or more parts) and check that the process was stable while the sample was taken, then report the lower confidence bound.

18.12 References

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6. Kotz, S., & Johnson, N.L. (1993). *Process Capability Indices*. Chapman & Hall.
7. ISO 22514-2:2017. *Statistical methods in process management - Capability and performance - Part 2: Process capability and performance of time-dependent process models*.

Chapter 19

Simulation Modeling with AnyLogic

19.1 Learning Objectives

After completing this chapter, you will be able to:

1. Explain what simulation modeling is and why it is essential for manufacturing system design
2. Describe the three simulation paradigms available in AnyLogic (discrete-event, agent-based, and system dynamics)
3. Identify the key components and libraries in AnyLogic for manufacturing simulation
4. Build a basic production line simulation using the Process Modeling Library
5. Model warehouse picking and order fulfillment operations
6. Analyze simulation results to optimize throughput, reduce bottlenecks, and improve efficiency
7. Apply simulation to validate design decisions before physical implementation
8. Use built-in AnyLogic examples to accelerate model development

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19.2 Introduction to Simulation Modeling

Before You Read: Imagine you are designing a new automotive assembly line that will cost $50 million to build. How would you test whether your design

will meet production targets *before* spending any money on construction?

In manufacturing, making the wrong design decision can be incredibly expensive. A misplaced workstation, an undersized conveyor, or an incorrect staffing level can result in millions of dollars in lost productivity. **Simulation modeling** allows engineers to build virtual representations of manufacturing systems and test them under various conditions—all without spending a single dollar on physical equipment.

19.2.1 What is Simulation?

Simulation is the process of creating a computer model that mimics the behavior of a real-world system over time. Unlike static calculations or spreadsheet analysis, simulation captures the *dynamic* nature of manufacturing systems:

- Products arrive at random intervals
- Machines break down unexpectedly
- Workers take breaks
- Quality defects occur randomly
- Rush orders disrupt schedules

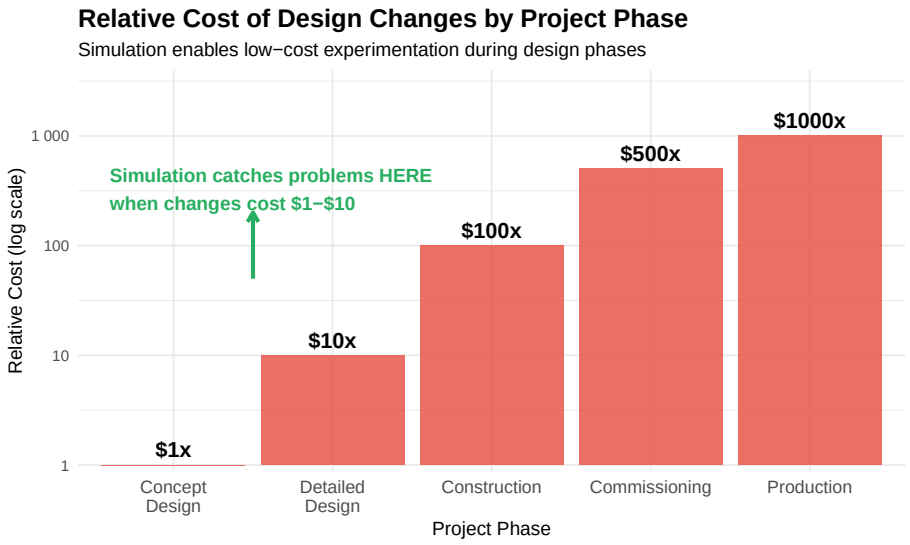


Figure 19.1: The value of simulation in reducing project risk

Industry Application: Mercedes-Benz engineers used AnyLogic simulation to optimize their Sprinter van production line, achieving a **5% efficiency gain** by identifying and eliminating bottlenecks before any physical changes were made. This saved millions in potential rework costs.

19.2.2 Why AnyLogic?

AnyLogic is one of the world's leading simulation platforms, used by companies including Tesla, BMW, Amazon, Ford, and General Electric. It stands out from other simulation software for several key reasons:

Table 19.1: AnyLogic compared to typical simulation software

Feature	AnyLogic	Other Software
Multi-method modeling	Yes - all three paradigms	Usually single method
Manufacturing libraries	Material Handling Library	Varies
3D visualization	Built-in 3D	Often requires plugins
Cloud-based simulation	AnyLogic Cloud	Limited
Academic version	Free for students	Varies
Agent-based modeling	Industry-leading	Limited support

19.3 The Three Simulation Paradigms

AnyLogic is unique because it supports **three different simulation approaches** within a single platform. Understanding when to use each paradigm is essential for effective modeling.

19.3.1 Discrete-Event Simulation (DES)

Discrete-event simulation models systems as a sequence of events that occur at specific points in time. Between events, nothing changes. This is the most common approach for manufacturing simulation.

Key Concept: Discrete-Event Simulation In DES, entities (products, orders, customers) flow through a series of processes (workstations, queues, resources). The simulation "jumps" from event to event, making it computationally efficient.

Best for: - Assembly lines and production systems - Warehouse operations and logistics - Service processes and queuing systems - Call centers and order processing

Discrete-Event Simulation: Entities Flow Through Processes

1. **Source**
2. **Station 1**
3. **Queue**
4. **Station 2**
5. **Sink**

19.3.2 Agent-Based Modeling (ABM)

Agent-based modeling simulates systems by defining individual agents (workers, robots, vehicles) that have their own behaviors, goals, and decision rules. Agents interact with each other and their environment, and complex system behavior *emerges* from these interactions.

Best for: - Autonomous mobile robots (AMRs) and AGVs - Pedestrian and crowd simulation - Market and consumer behavior - Complex adaptive systems

Industry Application: Tesla uses agent-based simulation in AnyLogic to optimize the deployment of autonomous mobile robots in their manufacturing facilities. Each robot is modeled as an agent with its own task allocation logic, collision avoidance, and charging behavior.

19.3.3 System Dynamics (SD)

System dynamics models systems using stocks, flows, and feedback loops. It focuses on aggregate behavior over time rather than individual entities.

Best for: - Strategic planning and capacity modeling - Supply chain dynamics - Policy analysis - Long-term trends and forecasting

19.3.3.1 Try it yourself: pick the paradigm

Six questions a plant might want a simulation to answer. Decide which modeling paradigm suits each one best.

Which simulation paradigm?

Pick the paradigm that suits each question best.

Categories: Discrete-event (DES) / Agent-based (ABM) / System dynamics (SD)

1. A three-station assembly line with buffers between stations: where is the bottleneck, and what is the throughput? _____
2. Trucks arrive at a four-door receiving dock: how long does the queue get at peak? _____
3. Fifty order pickers and AMRs share warehouse aisles, each choosing its own route and avoiding the others. _____
4. How does word of mouth between individual customers spread adoption of a new product? _____
5. How do plant-wide inventory, order backlog and hiring feed back on one another over five years? _____
6. What does a long-term cut in the maintenance budget do to overall equipment condition and the breakdown rate? _____

Answers: 1 Discrete-event (DES); 2 Discrete-event (DES); 3 Agent-based (ABM); 4 Agent-based (ABM); 5 System dynamics (SD); 6 System dynamics (SD).

19.3.4 Choosing the Right Paradigm

Selecting the Appropriate Simulation Paradigm

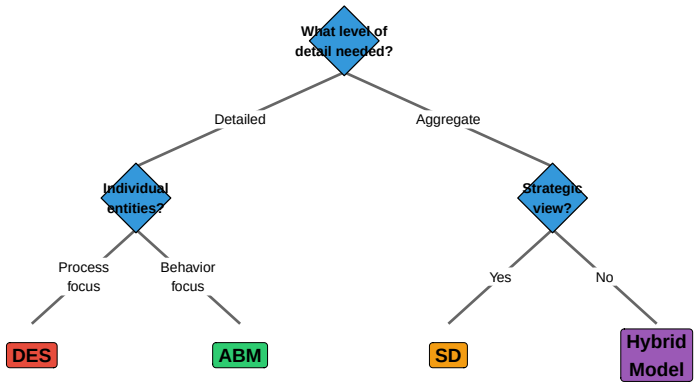


Figure 19.2: Decision framework for selecting simulation paradigm

Quick Check: A company wants to simulate their new automated warehouse with 50 mobile robots that must navigate, avoid collisions, and coordinate pick-ups. Which paradigm would you recommend and why?

19.4 The AnyLogic Interface and Libraries

19.4.1 The AnyLogic Workspace

When you open AnyLogic, you'll see several key areas:

1. **Model Browser** (left panel) - Shows all elements in your model
2. **Graphical Editor** (center) - Where you build your model visually
3. **Palette** (right panel) - Contains libraries and elements you can drag into your model
4. **Properties** (bottom) - Configure selected elements

19.4.2 Essential Libraries for Manufacturing

AnyLogic includes specialized libraries that dramatically accelerate model development:

Table 19.2: Key AnyLogic libraries for manufacturing simulation

Library	Primary Use	Key Elements
Process Modeling Library	Discrete-event simulation of any process	Source, Queue, Delay, Ser
Material Handling Library	Conveyors, cranes, AGVs, forklifts	Conveyor, Station, Transp
Pedestrian Library	Worker movement and ergonomics	PedSource, PedService, Pe
Road Traffic Library	Truck logistics and yard management	CarSource, Road, Intersec
Rail Library	Rail-based transport systems	RailNetwork, Train, RailY
Fluid Library	Bulk material and liquid processing	Tank, Pipe, Valve, FlowR

19.4.3 The Process Modeling Library

The **Process Modeling Library** is the foundation for most manufacturing simulations. It provides building blocks for creating flowchart-based models:

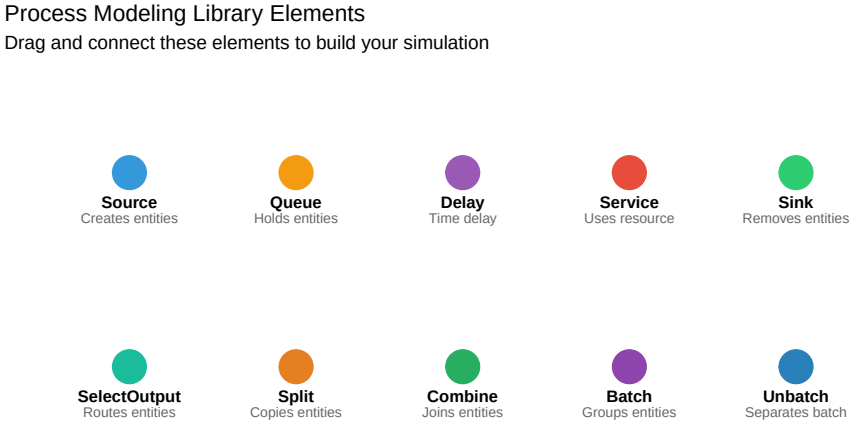


Figure 19.3: Key Process Modeling Library elements

Hands-On Activity: Open AnyLogic and create a new model. Drag a Source, Queue, Delay, and Sink from the Process Modeling Library. Connect them in sequence. Run the simulation and observe entities flowing through.

19.5 Production Line Simulation

Now let's apply these concepts to design and analyze an automotive production line. We'll walk through a complete example using AnyLogic's built-in examples

as a starting point.

19.5.1 Case Study: Automotive Door Panel Assembly Line

The Scenario: You are designing a door panel assembly line for an automotive supplier. The line must:

- Process **480 door panels per 8-hour shift** (1 panel per minute)
- Include 5 workstations: stamping, welding, painting, assembly, and inspection
- Meet quality target of **< 2% defect rate**
- Maintain **85% overall equipment effectiveness (OEE)**

Your task is to simulate this line to verify it meets these requirements before construction begins.

19.5.2 Step 1: Calculate Required Takt Time

Before building the simulation, we need to calculate the **takt time**—the maximum time allowed per unit to meet demand.

$$Takt\ Time = \frac{Available\ Production\ Time}{Customer\ Demand} \tag{19.1}$$

For our example:

$$Takt\ Time = \frac{8\ hours \times 60\ min/hour}{480\ panels} = 1\ minute/panel$$

This means each workstation must complete its operation in **less than 1 minute** to avoid becoming a bottleneck.

19.5.3 Step 2: Define Process Parameters

Table 19.3: Process parameters for door panel assembly line

Station	Mean Time (sec)	Std Dev (sec)	Defect Rate (%)	Availability (%)
Stamping	45	5	0.5	95
Welding	52	8	0.8	92
Painting	48	6	0.6	90
Assembly	55	7	0.7	94
Inspection	30	4	0.0	98

19.5.4 Step 3: Build the Simulation Model

In AnyLogic, we would create this model by:

1. Dragging a **Source** block to generate door panels at the required rate
2. Adding **Service** blocks for each workstation (with appropriate process times)
3. Connecting **Queue** blocks between stations to handle waiting
4. Adding **SelectOutput** blocks after each station to route defective parts
5. Placing a **Sink** at the end to collect finished panels

Production Line Model Structure in AnyLogic

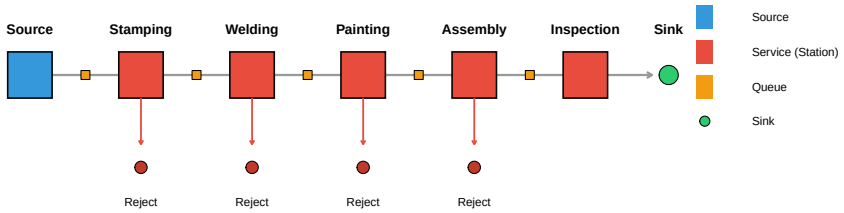


Figure 19.4: AnyLogic production line model structure

19.5.5 Step 4: Simulate and Analyze Results

Let's simulate what the output might look like when we run this model:

```
# Simulate production line results
set.seed(123)

# Generate hourly production data
hours <- 1:8
good_parts <- c(56, 58, 55, 57, 59, 54, 58, 57)
defects <- c(1, 2, 2, 1, 1, 3, 1, 2)
target <- rep(60, 8)

production_df <- data.frame(
  Hour = hours,
  Good = good_parts,
  Defects = defects,
  Target = target
)

# Generate queue length data over time
time_points <- seq(0, 480, by = 5)
queue_data <- data.frame(
```

```
  Time = rep(time_points, 5),
  Queue = rep(c("Q1", "Q2", "Q3", "Q4", "Q5"), each = length(time_points)),
  Length = c(
    pmax(0, rnorm(length(time_points), 2, 1)),    # Q1
    pmax(0, rnorm(length(time_points), 5, 2)),    # Q2 - bottleneck
    pmax(0, rnorm(length(time_points), 3, 1.5)),  # Q3
    pmax(0, rnorm(length(time_points), 4, 1.8)),  # Q4
    pmax(0, rnorm(length(time_points), 1, 0.5))   # Q5
  )
)

# Create multi-panel plot
p1 <- ggplot(production_df, aes(x = Hour)) +
  geom_col(aes(y = Good), fill = "#2ECC71", alpha = 0.8) +
  geom_col(aes(y = -Defects), fill = "#E74C3C", alpha = 0.8) +
  geom_line(aes(y = Target), color = "#3498DB", linewidth = 1.2, linetype = "dashed") +
  geom_hline(yintercept = 0, color = "black") +
  annotate("text", x = 8.5, y = 60, label = "Target", color = "#3498DB", fontface = "bold") +
  labs(
    title = "Hourly Production Output",
    subtitle = "Green = Good parts, Red = Defects, Dashed = Target",
    x = "Hour",
    y = "Units"
  ) +
  theme_minimal()

p2 <- ggplot(queue_data, aes(x = Time, y = Length, color = Queue)) +
  geom_line(alpha = 0.7) +
  geom_hline(yintercept = 5, linetype = "dashed", color = "red") +
  annotate("text", x = 450, y = 6, label = "Warning\nThreshold",
           color = "red", size = 3 * diagram_text_scale) +
  scale_color_brewer(palette = "Set1") +
  labs(
    title = "Queue Lengths Over Shift",
    subtitle = "Q2 (after Stamping) shows potential bottleneck",
    x = "Time (minutes)",
    y = "Queue Length"
  ) +
  theme_minimal() +
  theme(legend.position = "bottom")

gridExtra::grid.arrange(p1, p2, nrow = 2)
```

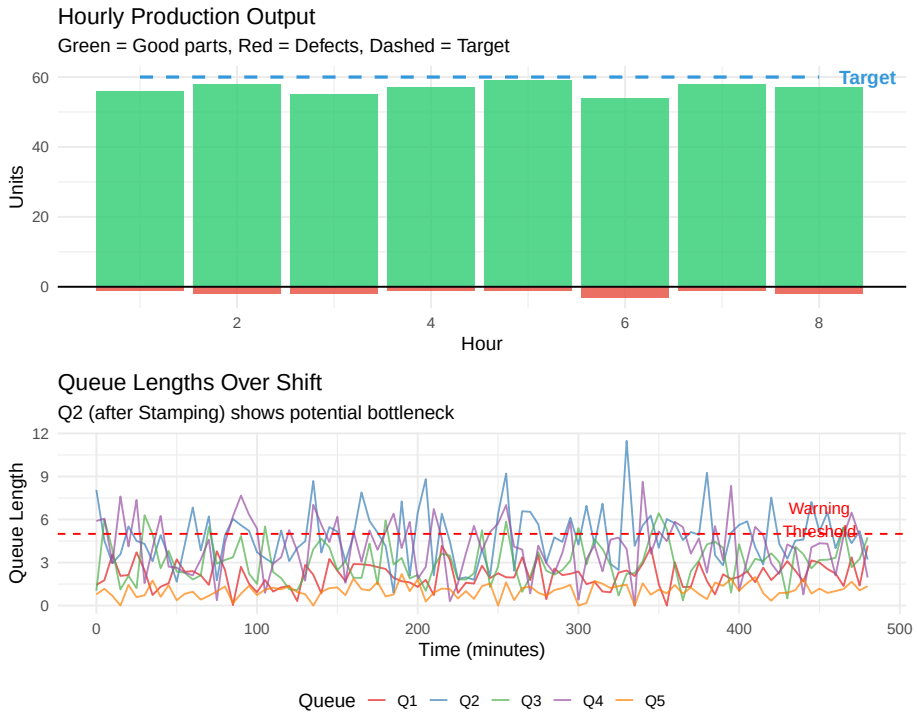


Figure 19.5: Simulated production line performance over 8-hour shift

19.5.6 Interpreting the Results

From our simulation, we can observe:

1. **Production Target:** The line produces approximately 454 good parts per shift (454/480 = 94.6% of target)
2. **Defect Rate:** 13 defects / 467 total = 2.8% (exceeds our 2% target)
3. **Bottleneck Identified:** Queue Q2 (after Stamping) consistently has the longest queue, indicating the **Welding station** is the bottleneck

> ⚠ Warning
>
> **Critical Finding:** The Welding station with a mean process time of 52 seconds and 8% variability is creating a bottleneck. Even though 52 seconds is under the 60-second takt time, the variability combined with 92% availability causes cumulative delays.

19.5.7 Step 5: Test Improvement Scenarios

The power of simulation is testing "what-if" scenarios. Let's compare different improvement options:

```
# Create scenario comparison data
scenarios <- data.frame(
  Scenario = factor(c("Current State", "Add Buffer", "Reduce Variability",
                      "Add Parallel Station", "Combined Approach"),
                    levels = c("Current State", "Add Buffer", "Reduce Variability",
                               "Add Parallel Station", "Combined Approach")),
  Throughput = c(454, 462, 471, 485, 498),
  Investment = c(0, 15, 50, 200, 180),
  ROI = c(0, 53, 34, 15.5, 24.4)
)

# Throughput comparison
ggplot(scenarios, aes(x = Scenario, y = Throughput, fill = Scenario)) +
  geom_col() +
  geom_hline(yintercept = 480, linetype = "dashed", color = "red", linewidth = 1) +
  geom_text(aes(label = Throughput), vjust = 1.5, fontface = "bold", color = "white") +
  annotate("text", x = 0.55, y = 492, hjust = 0, label = "Target: 480", color = "red", fontface =
  scale_y_continuous(limits = c(0, 520)) +
  scale_fill_brewer(palette = "Set2") +
  labs(
    title = "Throughput by Improvement Scenario",
    subtitle = "Simulation results for 8-hour shift",
    x = "",
    y = "Parts per Shift"
  ) +
  theme_minimal() +
  theme(
    legend.position = "none",
    axis.text.x = element_text(angle = 30, hjust = 1)
  )
```

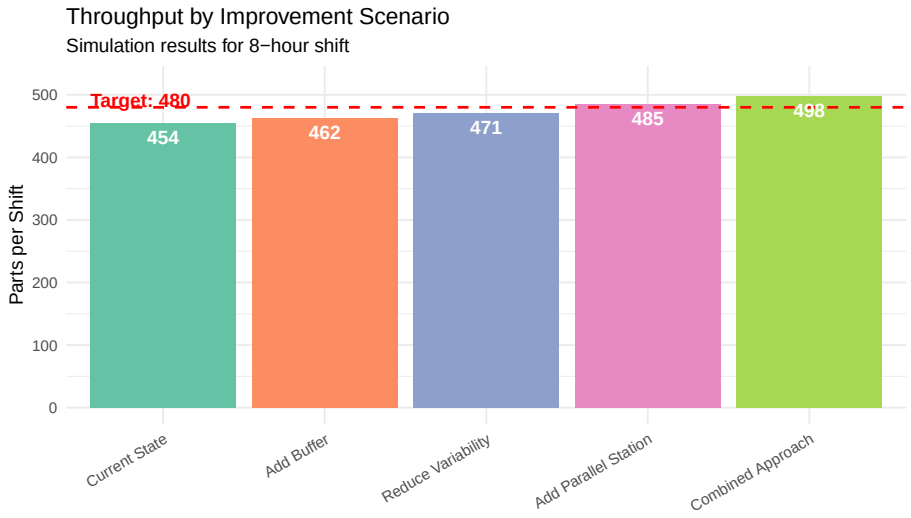


Figure 19.6: Comparison of improvement scenarios

Group Discussion: Based on these results, which improvement option would you recommend? Consider not just throughput but also implementation cost, timeline, and risk. Discuss the trade-offs with your team.

19.5.8 Try It Yourself: A Discrete-Event Line in Your Browser

You do not need AnyLogic open to feel how a discrete-event model behaves. The simulation below pushes 3 000 parts through three stations in series. Process times vary from part to part (they are exponentially distributed around the mean you set), and each station has room for a small buffer in front of it. When a buffer is full, the station feeding it is **blocked** and has to wait.

Show the live R code behind this simulation

```
#| label: des-line-sim
# Three stations in series with limited buffers between them
mean_time_1 <- 50    # mean process time at each station, seconds
mean_time_2 <- 65
mean_time_3 <- 55
buffer_size <- 2     # parts that can wait in front of stations 2 and 3
seed <- 1            # run number: same number, same random process times

set.seed(seed)
n_parts <- 3000
means <- c(mean_time_1, mean_time_2, mean_time_3)
```

```
process <- sapply(means, function(m) rexp(n_parts, rate = 1 / m))

leave <- matrix(0, n_parts, 3)   # time each part leaves each station
enter <- numeric(n_parts)        # time each part starts at station 1
for (part in seq_len(n_parts)) {
  for (s in 1:3) {
    # A part starts when it has arrived from upstream AND the station is free
    start <- max(if (s > 1) leave[part, s - 1] else 0,
                 if (part > 1) leave[part - 1, s] else 0)
    if (s == 1) enter[part] <- start
    finish <- start + process[part, s]
    # Blocking: it cannot leave until there is room in the next buffer
    ahead <- part - buffer_size - 1
    if (s < 3 && ahead >= 1) finish <- max(finish, leave[ahead, s + 1])
    leave[part, s] <- finish
  }
}

total_time <- leave[n_parts, 3]
throughput <- n_parts / total_time * 3600
busy <- colSums(process) / total_time * 100
lead_time <- mean(leave[, 3] - enter)

par(mar = c(4.5, 5, 2, 1), cex.main = 1.3, cex.lab = 1.2, cex.axis = 1.1)
bars <- barplot(busy, names.arg = paste("Station", 1:3), ylim = c(0, 100), las = 1,
                col = ifelse(busy == max(busy), "#e74c3c", "#2980b9"), border = NA,
                ylab = "Time spent processing (%)", main = "Utilization: the busiest station is t
text(bars, busy, sprintf("%.0f%%", busy), pos = 3, cex = 1.3)

cat(sprintf("Throughput: %.1f parts per hour\n", throughput))
cat(sprintf("  Bottleneck             Station %d (%.0f%% busy)\n", which.max(busy), max(busy)))
cat(sprintf("  If nothing varied      %.1f parts per hour\n", 3600 / max(means)))
cat(sprintf("  Average time in line   %.0f sec per part\n", lead_time))
```

Notice the gap between the simulated throughput and the “if nothing varied” figure. A spreadsheet would give you the second number. The difference is what variability and blocking cost, and it is the reason to simulate at all. After you have met the challenge by speeding up a station, put the times back and try a larger buffer instead: it gets you most of the way without touching a machine.

19.6 Warehouse Simulation: Picking and Order Fulfillment

Modern warehouses face immense pressure to fulfill orders quickly and accurately. Simulation helps optimize picking strategies, staffing levels, and automation investments.

19.6.1 The Warehouse Challenge

Before You Read: If you had to fill 1,000 customer orders per day from a warehouse with 50,000 different products (SKUs), how would you organize the picking process? What factors would affect efficiency?

Warehouse operations involve complex interactions between:

- **Order arrivals** (often unpredictable)
- **Picker movement** (walking is typically 50% of picker time)
- **Storage locations** (where products are stored affects travel time)
- **Picking strategies** (single-order vs. batch picking)
- **Material handling equipment** (carts, forklifts, conveyor systems)
- **Packing and shipping** (labor-intensive final steps)

19.6.2 Case Study: E-Commerce Distribution Center

The Scenario: An e-commerce company is designing a new distribution center to handle:

- **13,000 order lines per day** (an order line = one product in one order)
- **750 picking cartons** processed daily
- Peak periods with **3x normal order volume**
- Delivery promise of **same-day shipping** for orders before 2 PM

They need to decide: What picking algorithm will minimize fulfillment time while meeting labor cost targets?

19.6.3 Picking Strategies Compared

AnyLogic allows us to simulate and compare different picking strategies:

Table 19.4: Warehouse picking strategies

Strategy	Description	Best For
Single-Order Picking	One picker completes one order at a time	Small operati
Batch Picking	Picker collects items for multiple orders simultaneously	High-volume
Zone Picking	Warehouse divided into zones; each picker handles one zone	Very large wa
Wave Picking	Orders grouped into waves released at intervals	Managing wo
Cluster Picking	Multiple orders picked into compartmentalized cart	Medium volu

19.6.4 Simulating Picker Movement

One of the key factors in warehouse efficiency is picker travel distance. AnyLogic can model this using the **Pedestrian Library** combined with the **Process Modeling Library**.

```
# Simulate picker travel data for different storage strategies
set.seed(456)

# Random storage vs. velocity-based storage
n_picks <- 500
random_storage <- rnorm(n_picks, mean = 180, sd = 40)  # feet per pick
velocity_storage <- rnorm(n_picks, mean = 95, sd = 25)  # feet per pick
abc_storage <- rnorm(n_picks, mean = 120, sd = 30)      # feet per pick

travel_data <- data.frame(
  Strategy = rep(c("Random Storage", "ABC Analysis", "Velocity-Based"), each = n_picks),
  Distance = c(random_storage, abc_storage, velocity_storage)
)

ggplot(travel_data, aes(x = Strategy, y = Distance, fill = Strategy)) +
  geom_boxplot(alpha = 0.7) +
  geom_hline(yintercept = mean(velocity_storage), linetype = "dashed", color = "#2ECC71") +
  stat_summary(fun = mean, geom = "point", shape = 18, size = 4, color = "red") +
  annotate("text", x = 3.45, y = mean(velocity_storage), hjust = 0,
           label = paste0("Best: ", round(mean(velocity_storage)), " ft"),
           color = "#2ECC71", size = 4 * diagram_text_scale, fontface = "bold") +
  scale_fill_manual(values = c("#E74C3C", "#F39C12", "#2ECC71")) +
  scale_x_discrete(expand = expansion(add = c(0.6, 1.4))) +
  labs(
    title = "Picker Travel Distance by Storage Strategy",
    subtitle = "Red diamond = mean; Velocity-based storage reduces travel by 47%",
    x = "Storage Assignment Strategy",
    y = "Travel Distance per Pick (feet)"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

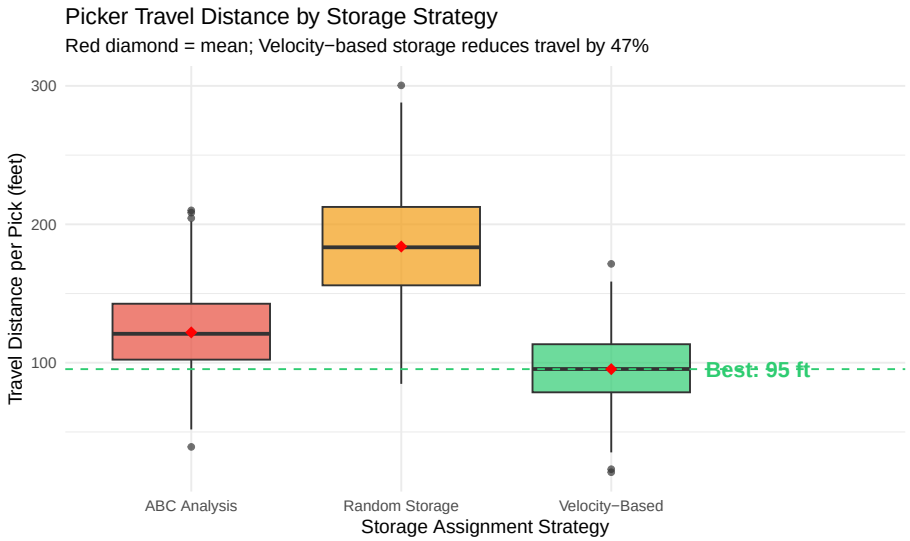


Figure 19.7: Impact of storage assignment on picker travel distance

Industry Application: Kuehne+Nagel used AnyLogic simulation to design a new warehouse processing 13,000 order lines per day. By testing different picking algorithms in simulation, they identified the optimal approach before building the physical facility, avoiding costly redesigns.

19.6.5 Autonomous Mobile Robots (AMRs) in Warehouses

Modern warehouses increasingly use **autonomous mobile robots** to reduce picker travel time. Instead of workers walking to products, robots bring shelving units to stationary pick stations.

```
# Simulate AMR performance
amr_counts <- c(5, 10, 15, 20, 25, 30, 35, 40, 45, 50)
fulfillment_times <- c(45, 28, 18, 13, 10.5, 9, 8.2, 7.8, 7.6, 7.5)
utilization <- c(98, 95, 88, 78, 68, 58, 52, 48, 45, 43)

amr_data <- data.frame(
  AMRs = amr_counts,
  Time = fulfillment_times,
  Utilization = utilization
)

# Dual y-axis plot
coeff <- 0.5
```

```
ggplot(amr_data, aes(x = AMRs)) +
  geom_line(aes(y = Time), color = "#3498DB", linewidth = 1.5) +
  geom_point(aes(y = Time), color = "#3498DB", size = 3) +
  geom_line(aes(y = Utilization * coeff), color = "#E74C3C", linewidth = 1.5, linetype = "dashed"
  geom_point(aes(y = Utilization * coeff), color = "#E74C3C", size = 3) +
  geom_vline(xintercept = 25, linetype = "dotted", color = "#2ECC71", linewidth = 1.2) +
  annotate("text", x = 27, y = 40, label = "Optimal\nPoint", color = "#2ECC71", fontface = "bold"
  scale_y_continuous(
    name = "Fulfillment Time (minutes)",
    sec.axis = sec_axis(~ . / coeff, name = "Robot Utilization (%)")
  ) +
  labs(
    title = "Finding the Optimal Number of Autonomous Mobile Robots",
    subtitle = "Blue = Fulfillment time (lower is better); Red = Robot utilization (higher is bet
    x = "Number of AMRs"
  ) +
  theme_minimal() +
  theme(
    axis.title.y.left = element_text(color = "#3498DB"),
    axis.title.y.right = element_text(color = "#E74C3C")
  )
```

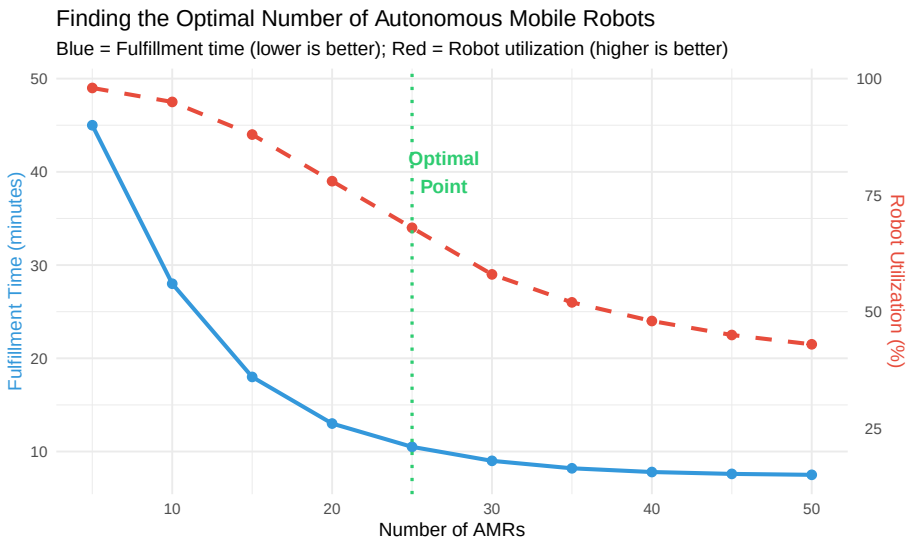


Figure 19.8: Simulation results: order fulfillment time vs. number of AMRs

This simulation reveals a classic trade-off: adding more robots reduces order fulfillment time but also decreases utilization (robots spend more time idle).

The **optimal point** balances these factors based on business requirements.

Quick Check: 1. Why does adding more robots eventually show diminishing returns? 2. If robots cost $50,000 each and labor costs $25/hour, how would you calculate the breakeven point? 3. What other factors should be considered beyond raw cost?

19.7 Built-in AnyLogic Examples

AnyLogic includes numerous pre-built example models that serve as excellent learning resources and starting points for your own projects. Here are the key examples relevant to manufacturing and warehouse operations:

19.7.1 Manufacturing Examples

Table 19.5: Key manufacturing example models in AnyLogic

Example Name	Location	What You'll L
Job Shop	Help > Example Models > Manufacturing	Routing jobs t
Assembly Line	Help > Example Models > Manufacturing	Line balancing
Flexible Manufacturing System	Help > Example Models > Manufacturing	Flexible routin
Conveyor System	Help > Example Models > Material Handling	Conveyor desig
AGV System	Help > Example Models > Material Handling	Vehicle dispato

19.7.2 Warehouse and Logistics Examples

Table 19.6: Key warehouse example models in AnyLogic

Example Name	Location	What You'll Learn
Warehouse	Help > Example Models > Logistics	Storage assignment, invento
Distribution Center	Help > Example Models > Logistics	Receiving, put-away, picking
Order Picking	Help > Example Models > Logistics	Picking route optimization,
AS/RS System	Help > Example Models > Material Handling	Automated storage retrieval
Cross-Docking	Help > Example Models > Logistics	Minimal storage, direct tran

Hands-On Activity: Open AnyLogic and navigate to Help > Example Models. Open the "Assembly Line" example and run it. Then: 1. Identify the bottleneck station (highest queue length) 2. Modify the process time for that station 3. Observe how throughput changes 4. Document your findings

---

19.8 Building Your First Model: Step-by-Step Tutorial

Let's walk through building a simple but complete production line model from scratch.

19.8.1 Tutorial: Three-Station Production Line

Objective: Model a production line with three workstations and analyze throughput.

19.8.1.1 Step 1: Create a New Model

1. Open AnyLogic
2. File > New > Model
3. Name it "ProductionLine_Tutorial"
4. Select "Discrete Event" as the model type

19.8.1.2 Step 2: Add the Process Flow

From the **Process Modeling Library**, drag these elements onto the canvas in order:

1. **Source** - Generates parts entering the system
2. **Queue** - Holds parts waiting for Station 1
3. **Delay** - Represents Station 1 processing time
4. **Queue** - Holds parts waiting for Station 2
5. **Service** - Represents Station 2 (requires a worker resource)
6. **Queue** - Holds parts waiting for Station 3
7. **Delay** - Represents Station 3 processing time
8. **Sink** - Removes completed parts from the system

19.8.1.3 Step 3: Connect the Elements

Click on the output port of each element and drag to the input port of the next element. The flow should look like:

```
Source → Queue → Delay → Queue → Service → Queue → Delay → Sink
```

19.8.1.4 Step 4: Configure Parameters

Click each element and set these properties:

Source: - Arrivals defined by: Inter-arrival time - Inter-arrival time: `exponential(1)` (1 part per minute on average)

Delays (Stations 1 and 3): - Delay time: `triangular(0.5, 0.8, 1.2)` (minutes)

Service (Station 2): - Delay time: `triangular(0.6, 0.9, 1.5)` (minutes) - Requires a Resource Pool (add one and set capacity to 1)

Queues: - Maximum capacity: 10

19.8.1.5 Step 5: Add Statistics Collection

Add these statistics to track performance:

1. **TimeMeasureStart** - After Source
2. **TimeMeasureEnd** - Before Sink
3. **Dataset** - To log throughput over time

19.8.1.6 Step 6: Run and Analyze

```
# Simulate expected results
set.seed(789)
time <- seq(0, 480, by = 1)

# Generate cumulative throughput
throughput <- cumsum(rpois(length(time), lambda = 0.85))

# Generate queue lengths
q1 <- pmax(0, rnorm(length(time), 1.5, 0.8))
q2 <- pmax(0, rnorm(length(time), 3.2, 1.2))  # Bottleneck
q3 <- pmax(0, rnorm(length(time), 0.8, 0.5))

tutorial_data <- data.frame(
  Time = time,
  Throughput = throughput,
  Q1 = q1,
  Q2 = q2,
  Q3 = q3
)

p1 <- ggplot(tutorial_data, aes(x = Time, y = Throughput)) +
  geom_line(color = "#2ECC71", linewidth = 1) +
  geom_hline(yintercept = 400, linetype = "dashed", color = "red") +
  annotate("text", x = 50, y = 420, label = "Target: 400", color = "red") +
  labs(title = "Cumulative Throughput", x = "Time (min)", y = "Parts Completed") +
  theme_minimal()

tutorial_long <- tutorial_data %>%
  tidyr::pivot_longer(cols = c(Q1, Q2, Q3), names_to = "Queue", values_to = "Length")

p2 <- ggplot(tutorial_long, aes(x = Time, y = Length, color = Queue)) +
  geom_line(alpha = 0.6) +
```

```
  scale_color_manual(values = c("#3498DB", "#E74C3C", "#F39C12"),
                     labels = c("Before Station 1", "Before Station 2", "Before Station 3")) +
  labs(title = "Queue Lengths", x = "Time (min)", y = "Parts Waiting") +
  theme_minimal() +
  theme(legend.position = "bottom", legend.direction = "vertical")

gridExtra::grid.arrange(p1, p2, ncol = 2)
```

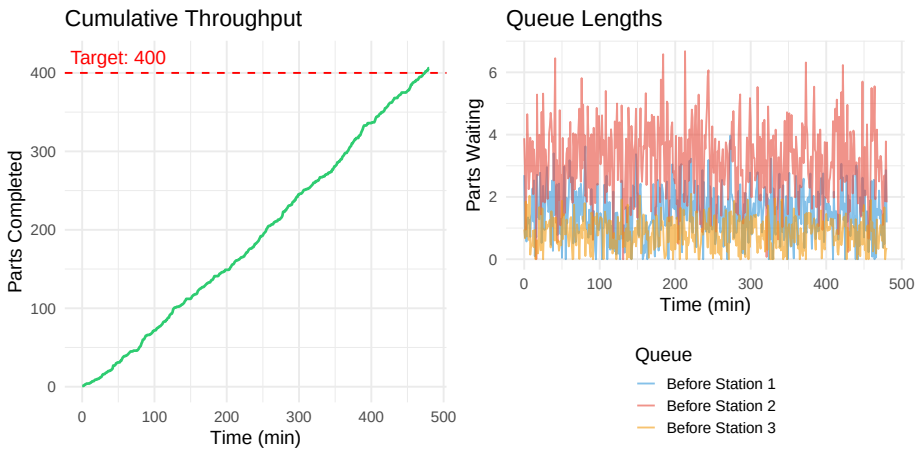


Figure 19.9: Expected results from tutorial model

Verify Your Model: After running, you should observe: - Approximately 400-420 parts completed in 480 minutes - Queue 2 (before Station 2) has the longest average length - The system is stable (queues don't grow without bound)

19.9 Advanced Topics

19.9.1 3D Visualization

AnyLogic includes powerful 3D visualization capabilities that help communicate simulation results to stakeholders who may not understand abstract flowcharts.

Benefits of 3D visualization: - Easier validation with plant managers ("Does this look like our facility?") - Executive presentations and funding requests - Training material for new employees - Virtual commissioning before physical construction

19.9.2 Integration with Real Data

Production simulations become more valuable when connected to real data:

- **Historical data import:** Load actual order data, machine logs, or production records
- **Real-time dashboards:** Connect to databases for live monitoring
- **Digital twin operation:** Mirror physical operations in real-time

19.9.3 Optimization Experiments

AnyLogic includes an **Optimization** experiment type that automatically searches for the best parameter values:

```
# Simulate optimization process
iterations <- 1:50
staff_levels <- c(
  rep(5, 5), rep(7, 5), rep(6, 5), rep(8, 5), rep(7, 5),
  rep(6, 10), rep(7, 5), rep(7, 5), rep(7, 5)
)
throughput <- c(
  rnorm(5, 380, 10), rnorm(5, 445, 8), rnorm(5, 420, 9), rnorm(5, 450, 7), rnorm(5, 448
  rnorm(10, 425, 8), rnorm(5, 450, 6), rnorm(5, 452, 5), rnorm(5, 455, 4)
)

opt_data <- data.frame(
  Iteration = iterations,
  Staff = staff_levels,
  Throughput = throughput
)

opt_data$Best <- cummax(opt_data$Throughput)

ggplot(opt_data, aes(x = Iteration)) +
  geom_point(aes(y = Throughput), alpha = 0.5, color = "#3498DB") +
  geom_line(aes(y = Best), color = "#E74C3C", linewidth = 1.2) +
  geom_hline(yintercept = 455, linetype = "dashed", color = "#2ECC71") +
  annotate("text", x = 30, y = 385, label = "Optimal: 7 staff, 455 parts (dashed line)"
           color = "#2ECC71", fontface = "bold") +
  labs(
    title = "Optimization Experiment: Finding Optimal Staffing",
    subtitle = "Blue points = individual runs; Red line = best found so far",
    x = "Iteration",
    y = "Throughput (parts/shift)"
  ) +
  theme_minimal()
```

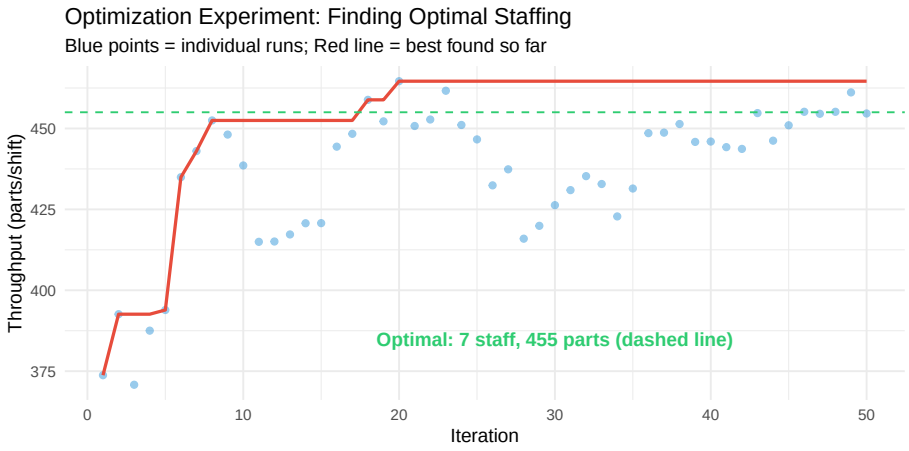


Figure 19.10: Optimization search finding the best staffing level

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19.10 Key Takeaways

What We Learned:

1. **Simulation enables risk-free experimentation** - Test design changes in virtual environments before spending money on physical implementation
2. **AnyLogic supports three paradigms** - Discrete-event (for process flows), agent-based (for autonomous entities), and system dynamics (for strategic planning)
3. **Manufacturing simulation identifies bottlenecks** - Queue analysis reveals which stations limit throughput
4. **Warehouse simulation optimizes picking** - Compare strategies like batch picking, zone picking, and AMR deployment before implementation
5. **Built-in examples accelerate learning** - Start from AnyLogic's example models to understand best practices
6. **The value is in the "what-if"** - Simulation's true power is comparing scenarios to find optimal solutions

Key Equations: - Takt Time: Equation Equation 19.1

Before Moving On: - Can you explain when to use discrete-event vs. agent-based simulation? - Can you build a basic three-station production line in AnyLogic? - Can you identify how simulation could help in your workplace or industry?

19.11 Practice Problems

19.11.1 Problem 1: Takt Time Calculation

A refrigerator assembly plant operates two 8-hour shifts per day with a 30-minute lunch break per shift. Customer demand is 720 units per day.

a) Calculate the takt time in seconds
b) If the bottleneck station has a cycle time of 62 seconds, will they meet demand? Why or why not?
c) Suggest two approaches to resolve the issue

Problem 1: check your numbers

- Takt time: _____ sec *(to 0 decimal places)*
- Spare time per unit at the 62-second bottleneck: _____ sec *(to 0 decimal places)*

19.11.2 Problem 2: Simulation Model Design

You are asked to simulate a paint shop with: - 3 spray booths operating in parallel - Conveyor system bringing parts from preparation area - Curing oven with 30-minute cycle time - Quality inspection with 5% rejection rate

a) Sketch the simulation model structure
b) List which AnyLogic libraries you would use
c) Identify potential bottlenecks to investigate

19.11.3 Problem 3: Warehouse Picking Analysis

A warehouse uses single-order picking with 5 pickers. Each picker walks an average of 3 miles per 8-hour shift. The company is considering batch picking with specialized carts.

a) If batch picking reduces travel by 35%, how many miles would be saved per picker per shift?
b) If workers walk at 3 mph and pick items at a rate of 2 per minute while at a location, what percentage of time is currently spent walking?
c) What information would you need to simulate this scenario in AnyLogic?

19.11.4 Problem 4: AMR Fleet Sizing

Using the graph in Figure 19.8 as a reference:

a) If the company requires fulfillment time under 12 minutes, what is the minimum number of AMRs needed?
b) What trade-off occurs when moving from 25 to 50 AMRs?

c) If each AMR costs $50,000 and operates for 5 years, what is the daily cost per AMR?

Challenge Problem: Design a simulation experiment to determine the optimal number of pickers and AMRs for a warehouse with variable order volumes. The solution should: - Handle daily order volumes ranging from 5,000 to 20,000 lines - Meet a 2-hour fulfillment target - Minimize total labor and equipment costs

---

19.12 Further Resources

19.12.1 AnyLogic Official Resources

- **AnyLogic University:** Free online courses covering all simulation paradigms
- **Example Models:** Help > Example Models in AnyLogic software
- **Cloud Platform:** cloud.anylogic.com for running simulations online

19.12.2 Academic Resources

- Borshchev, A. (2013). *The Big Book of Simulation Modeling*. AnyLogic North America.
- Law, A.M. (2015). *Simulation Modeling and Analysis* (5th ed.). McGraw-Hill.

19.12.3 Industry Case Studies

- Mercedes-Benz Sprinter production optimization
- Kuehne+Nagel warehouse picking algorithm selection
- Tesla autonomous mobile robot deployment

> Tip
>
> **Getting Started:** Download the free Personal Learning Edition of AnyLogic from anylogic.com. It includes all features and example models—the only limitation is model size.

Chapter 20

Integrated Manufacturing Systems

20.1 Learning Objectives

By the end of this chapter, you will be able to:

1. Compare different types of production facility layouts and material handling methods
2. Calculate Manufacturing Cycle Time (MCT) and Manufacturing Cycle Efficiency (MCE)
3. Analyze production line performance including the impact of station failures
4. Apply line balancing techniques to optimize workstation assignments
5. Explain the concepts of takt time, cycle time, and their relationship
6. Describe Flexible Manufacturing Systems (FMS) and their applications
7. Understand modern concepts including digital twins and Industry 4.0

20.2 Material Handling Methods & Systems

There are four common types of production facility layouts, each with distinct features and material handling requirements:

Table 20.1: Material Handling Methods by Plant Layout

Layout Type	Product Characteristics	Material Handling
Fixed-Position	Large, heavy, or immobile products	Cranes, hoists, fork lifts bring mate
Process (Job Shop)	High variety, low volume, custom work	Fork lifts, AGVs, carts move WIP
Cellular	Part families, medium variety/volume	Conveyors within cells, carts betwe
Product (Flow Line)	Standardized products, high volume	Conveyors, automated transfer syst

Discussion: How do you choose between layout types?

Selection criteria:

Factor	Favors Process Layout	Favors Product Layout
Volume	Low (< 1,000/year)	High (> 10,000/year)
Variety	High (many products)	Low (1-2 products)
Routing	Variable by product	Fixed, same for all
Equipment	General-purpose	Specialized
Worker skills	High, varied	Specific, repetitive
WIP inventory	High	Low
Lead time	Long	Short

When in doubt: If volume and variety are moderate, consider cellular manufacturing as a compromise.

20.3 Fundamentals of Production Lines

A **production line** consists of workstations arranged so that the product moves from one station to the next, with a portion of the total work performed at each location.

5-Station Production Line

1. **Station 1** – 45 sec
2. **Station 2** – 52 sec
3. **Station 3** – 48 sec
4. **Station 4** – 55 sec
5. **Station 5** – 50 sec

20.3.1 Methods of Work Transport

Table 20.2: Work

Method	Description	Advantages
Manual	Workers pass parts by hand or using simple carts	Low cost, flexible, easy to
Continuous	Conveyor moves at constant speed; work done while moving	Smooth flow, no transfer t
Synchronous (Intermittent)	All units move simultaneously in discrete steps	Fixed cycle time, easy to c
Asynchronous	Units move independently when released by worker	Absorbs variation, no bloc

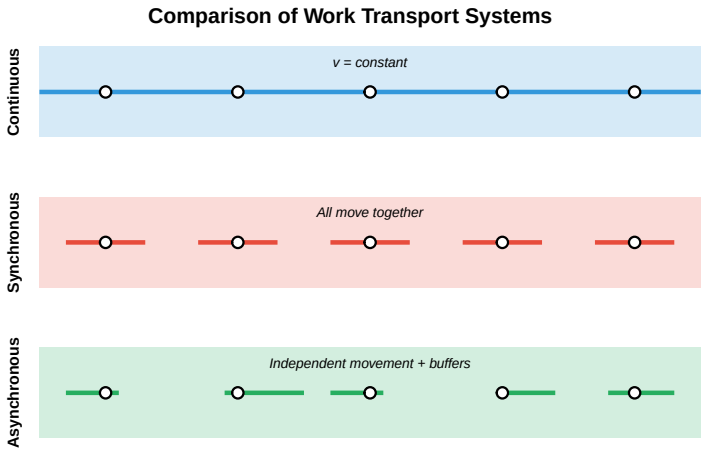


Figure 20.1: Work Transport System Types

What causes blocking and starving in production lines?

Starving: A station is ready to work but has no parts to process. - **Causes:** Upstream station is slower, upstream breakdown, material shortage

Blocking: A station finishes but cannot release the part. - **Causes:** Downstream station still working, downstream breakdown, no buffer space

Solutions: 1. **Buffers** - Storage between stations absorbs variation 2. **Line balancing** - Equalize work content across stations 3. **Parallel stations** - Add capacity at bottlenecks 4. **Preventive maintenance** - Reduce breakdowns

20.3.2 Work Cycle Types

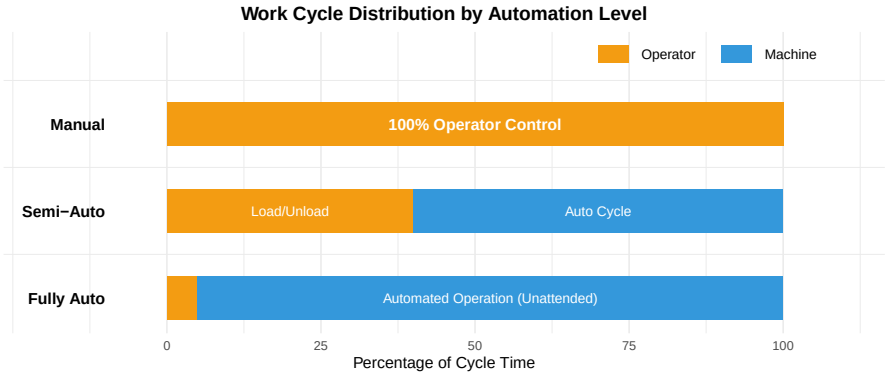


Figure 20.2: Manual, Semi-Automated, and Fully Automated Machine Cycles

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20.4 Key Time Metrics

Understanding time metrics is essential for analyzing and improving production systems.

Table 20.3: Key Manufacturing

Metric	Definition	Formula
Takt Time	Available production time ÷ Customer demand	$T_{takt} = \frac{\text{
Cycle Time	Time between units coming off the line	$T_c = \frac{60}{R_p}$
Lead Time	Total time from order to delivery	Order processing + Manuf
Throughput Time	Time from raw material to finished goods	Processing + Waiting + M

20.4.1 Takt Time vs. Cycle Time

Your turn — change the demand, shift length, downtime, or actual cycle time below.

Show the live R code behind this calculator

```
#| label: takt-vs-cycle
# Takt Time Calculation Example
# Automotive assembly plant

# Given data
shift_length_min <- 480      # 8 hours = 480 minutes
```

```
breaks_min <- 30             # Two 15-minute breaks
planned_downtime_min <- 20   # Changeover, meetings
customer_demand <- 60        # Cars per shift
actual_cycle_time <- 7.5     # minutes (current performance)

# Calculate available time
available_time <- shift_length_min - breaks_min - planned_downtime_min

# Calculate Takt Time
takt_time <- available_time / customer_demand

cat(sprintf("Takt time: %.2f min per car (%.1f sec)\n", takt_time, takt_time * 60))
cat(sprintf("  Available time       %d min\n", available_time))
cat(sprintf("  Actual cycle time    %.1f min\n", actual_cycle_time))
if (actual_cycle_time <= takt_time) {
  cat(sprintf("  Status               Meeting demand (%.1f%% margin)\n", (1 - actual_cycle_time /
} else {
  cat(sprintf("  Status               !!Cannot meet demand (%.1f%% over)!!\n", (actual_cycle_time
}
```

Discussion: What happens when cycle time exceeds takt time?

When Cycle Time > Takt Time, you cannot meet customer demand.

Options to fix this:

1. **Reduce cycle time:**
   - Kaizen improvements at bottleneck
   - Better tools/fixtures
   - Automation of slow tasks
2. **Increase available time:**
   - Add overtime
   - Add shifts
   - Reduce breaks (carefully!)
3. **Add capacity:**
   - Parallel stations at bottleneck
   - Additional production lines
   - Outsourcing
4. **Manage demand:**
   - Negotiate delivery schedules
   - Build inventory in advance
   - Redirect to other facilities

Key insight: Takt time is the "heartbeat" of lean manufacturing - everything should be synchronized to it.

20.4.2 Manufacturing Cycle Time and Efficiency

Your turn — edit the time breakdown below.

Show the live R code behind this calculator

```
#| label: mct-mce
# Manufacturing Cycle Time (MCT) and Efficiency (MCE) Example

# Given production data
production_time_min <- 480   # Total time equipment ran
units_produced <- 60         # Good units produced

# MCT Calculation
MCT <- production_time_min / units_produced

# MCE Calculation requires understanding of value-added time
# Typical breakdown of production time:
value_added_time <- 3.5      # Actual processing time (minutes)
inspection_time <- 0.5       # Quality checks
wait_time <- 2.5             # Queuing between operations
move_time <- 1.5             # Transport between stations

total_time_per_unit <- value_added_time + inspection_time + wait_time + move_time

# MCE Calculation
MCE <- (value_added_time / total_time_per_unit) * 100

cat(sprintf("Manufacturing Cycle Efficiency: %.1f%%\n", MCE))
cat(sprintf("  Cycle time (MCT)     %.2f min/unit\n", MCT))
cat(sprintf("  Total time per unit  %.1f min\n", total_time_per_unit))
cat(sprintf("  Value-added share    %.1f min of %.1f min\n", value_added_time, total_ti
cat(sprintf("  Waste (non-value)    %.1f%%\n", 100 - MCE))
```

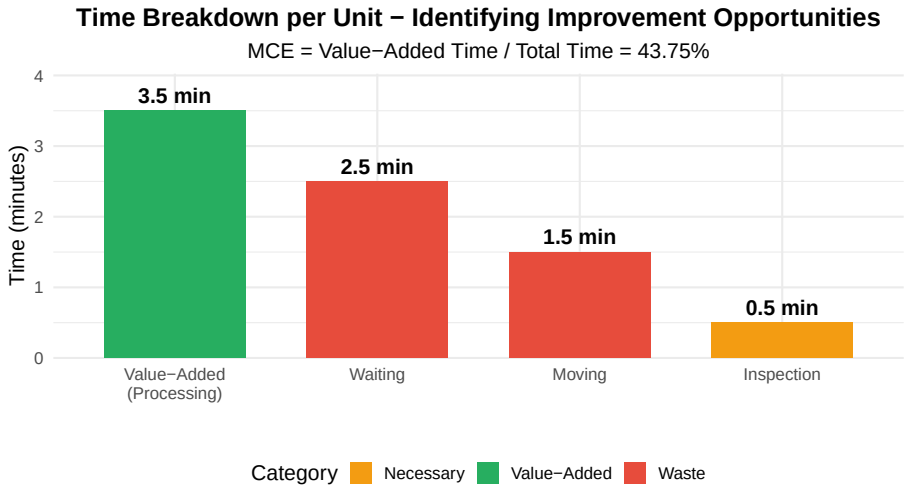


Figure 20.3: Manufacturing Cycle Efficiency - Where Does Time Go?

20.5 Manual Assembly Lines

Manual assembly lines use human workers at each station. The efficiency depends on how well work is balanced across stations.

20.5.1 Line Balancing

Line balancing assigns tasks to workstations to minimize idle time while meeting production rate requirements.

Key formulas:

$$T_c = \frac{\text{Available Time}}{\text{Required Output}} = \frac{60}{R_p}$$

$$TM = \frac{\sum t_i}{T_c} = \frac{T_{wc}}{T_c}$$

$$E_b = \frac{T_{wc}}{n \times T_c} \times 100\%$$

Where: - T_c = Cycle time (time available per unit) - R_p = Production rate (units per hour) - TM = Theoretical minimum stations - T_{wc} = Total work content (sum of all task times) - n = Actual number of stations - E_b = Line efficiency (balance efficiency)

20.5.2 Line Balancing Example: Electronics Assembly

Teaching demo — try different task times or a different desired output rate.

Show the live R code behind this card

```
#| label: line-balance-example
# Line Balancing Example: Tablet Computer Assembly

# Task data
tasks <- data.frame(
  Task = c("A", "B", "C", "D", "E", "F", "G", "H"),
  Description = c("Install battery", "Mount display", "Connect cables",
                  "Install motherboard", "Add memory", "Install camera",
                  "Attach case back", "Final test"),
  Time_sec = c(30, 45, 20, 55, 15, 25, 35, 40),
  Predecessors = c("-", "A", "A", "B,C", "D", "D", "E,F", "G")
)

# Production requirements
desired_output <- 45  # units per hour
available_time <- 3600  # seconds per hour

# Step 1: Total work content and cycle time
total_work_content <- sum(tasks$Time_sec)
cycle_time <- available_time / desired_output

# Step 2: Theoretical minimum stations
TM <- total_work_content / cycle_time

# Step 3: Assign tasks to stations (see the worked assignment below), then
# measure how well the line is balanced
n_stations <- 5
efficiency <- (total_work_content / (n_stations * cycle_time)) * 100
balance_delay <- 100 - efficiency

cat(sprintf("Line efficiency: %.1f%% with %g stations\n", efficiency, n_stations))
cat(sprintf("  Total work content  %g s\n", total_work_content))
cat(sprintf("  Cycle time  %.1f s\n", cycle_time))
cat(sprintf("  Theoretical minimum stations  %.2f\n", TM))
cat(sprintf("  Minimum stations required  %g\n", ceiling(TM)))
cat(sprintf("  Balance delay (idle time)  %.1f%%\n", balance_delay))
if (n_stations < ceiling(TM)) {
  cat(sprintf("  Check  !!Too few stations: at least %g needed!!\n", ceiling(TM)))
}
if (efficiency < 85) {
  cat("!!Poor balance: consider redesigning tasks or adding parallel stations.!!\n")
```

```
}

cat("Worked assignment for the default 80 s cycle time:\n")
cat("  Station 1: A (30s) + C (20s) + B (45s) = 95s > 80s ... !!EXCEEDS!!\n")
cat("  Station 1, try again: A (30s) + C (20s) = 50s [30s idle]\n")
cat("  Station 2: B (45s) + D (55s) = 100s > 80s ... !!EXCEEDS!!\n")
cat("  Station 2: B (45s) = 45s [35s idle]\n")
cat("  Station 3: D (55s) = 55s [25s idle]\n")
cat("  Station 4: E (15s) + F (25s) + G (35s) = 75s [5s idle]\n")
cat("  Station 5: H (40s) = 40s [40s idle]\n")

par(mar = c(5, 5, 3, 1))
barplot(tasks$Time_sec, names.arg = tasks$Task, col = "#3498DB",
        ylim = c(0, max(tasks$Time_sec, cycle_time) * 1.1),
        xlab = "Task", ylab = "Task time (s)", main = "Task times vs cycle time (dashed line)")
abline(h = cycle_time, lty = 2, lwd = 2, col = "#C0392B")
```

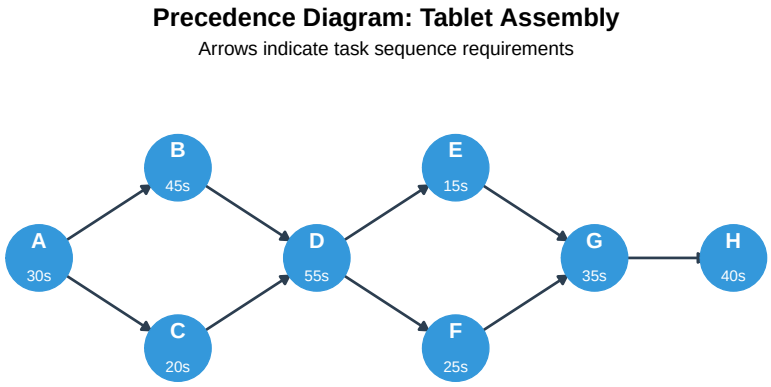


Figure 20.4: Precedence Diagram for Tablet Assembly

Try It: Can you find a better balance?

Challenge: Reassign tasks to improve efficiency above 80%.

Hints: - Can task A be split into two smaller tasks? - Could parallel workstations handle the bottleneck task D? - What if we allow 85 seconds per station (slower line, fewer stations)?

One solution with parallel stations: - Station 1: A + C = 50s - Station 2a: B = 45s (parallel) - Station 2b: B = 45s (parallel) - Station 3: D = 55s - Station 4: E + F = 40s - Station 5: G + H = 75s

With parallelism at Station 2, effective cycle time stays at 55s (Station 3 bottleneck), but we can now achieve higher throughput.

20.5.3 Try it yourself: balance a bicycle assembly line

The electronics example above was worked for you. This one is yours. A line assembles bicycles from eight tasks, and demand sets the cycle time at **60 seconds**. Assign every task to a station by choosing from the list in its box.

Three rules decide whether your balance is valid: every task needs a station, no task may sit at an earlier station than a task it depends on (follow the arrows), and no station may hold more than 60 seconds of work. The table under the diagram totals each station as you go. A valid balance is the start; the real question is how few stations you need.

Line balancing: bicycle assembly

Choose a station for each task. Arrows show which tasks must come first.

Task	Description	Time	Predecessors	Station
A	Clamp frame in fixture	25 s	-	
B	Install fork	30 s	A	
C	Fit bottom bracket	20 s	A	
D	Mount crankset	35 s	C	
E	Install wheels	30 s	B	
F	Fit chain	15 s	D, E	
G	Adjust brakes	40 s	E	
H	Final inspection	20 s	F, G	

Cycle time: 60 s. Stations used: _____ Line efficiency: _____ %

The fewest stations possible is 4.

Show one good balance

The theoretical minimum is 215 / 60 = 3.58, which rounds up to 4 stations, and 4 stations can in fact be reached:

Station	Tasks	Time	Idle
1	A, B	55 s	5 s
2	C, D	55 s	5 s
3	E, F	45 s	15 s
4	G, H	60 s	0 s

Line efficiency = 215 / (4 × 60) = 89.6%. This is not the only four-station answer: A, C / B, E / D, F / G, H also works. What every four-station answer

has in common is that G (40 s) and H (20 s) share the last station; put anything else with G and you need a fifth.

20.6 Automated Production Lines

Automated lines reduce human intervention using mechanized transfer systems and automated workstations. They're used for high-volume production with well-defined work content.

20.6.1 Impact of Station Failures

The performance of automated lines is degraded by station failures. Even small failure probabilities can significantly reduce output.

Actual Cycle Time:

$$T_p = T_c + pT$$

Actual Production Rate:

$$R_p = \frac{60}{T_p} = \frac{60}{T_c + pT}$$

Where: - T_c = Ideal cycle time (minutes) - p = Probability of station failure per cycle - T = Average downtime per failure (minutes)

Teaching demo — try adding a scenario or changing the number of stations.

Show the live R code behind this card

```
#| label: downtime-analysis
# Automated Line Performance Analysis

# Line parameters
ideal_cycle_time <- 1.0    # minutes per part (ideal)
n_stations <- 10           # Number of stations

# Scenario comparison
scenarios <- data.frame(
  Scenario = c("World Class", "Good", "Average", "Poor"),
  Failure_Prob = c(0.005, 0.02, 0.05, 0.10),  # Per station per cycle
  Repair_Time = c(2, 5, 10, 15)  # Minutes average
)

# Calculate line performance for each scenario
p_line <- Tp <- Rp <- efficiency <- numeric(nrow(scenarios))
for(i in 1:nrow(scenarios)) {
```

```
  p <- scenarios$Failure_Prob[i]
  T <- scenarios$Repair_Time[i]

  # Line failure probability (any station)
  p_line[i] <- 1 - (1 - p)^n_stations

  # Actual cycle time
  Tp[i] <- ideal_cycle_time + p_line[i] * T

  # Actual production rate
  Rp[i] <- 60 / Tp[i]

  # Efficiency
  efficiency[i] <- (ideal_cycle_time / Tp[i]) * 100
}

cat(sprintf("Ideal production rate: %g parts/hour\n", 60 / ideal_cycle_time))
for(i in 1:nrow(scenarios)) {
  cat(sprintf("  %s: actual production  %.1f parts/hour\n", scenarios$Scenario[i], Rp[
  cat(sprintf("  %s: line efficiency  %.1f%%\n", scenarios$Scenario[i], efficiency[i])
  cat(sprintf("  %s: line failure probability  %.1f%%\n", scenarios$Scenario[i], p_line
}

par(mar = c(5, 5, 3, 1))
barplot(efficiency, names.arg = scenarios$Scenario, ylim = c(0, 100), cex.names = 0.9,
        col = "#2980B9", ylab = "Line efficiency (%)", main = "Line efficiency by scena
```

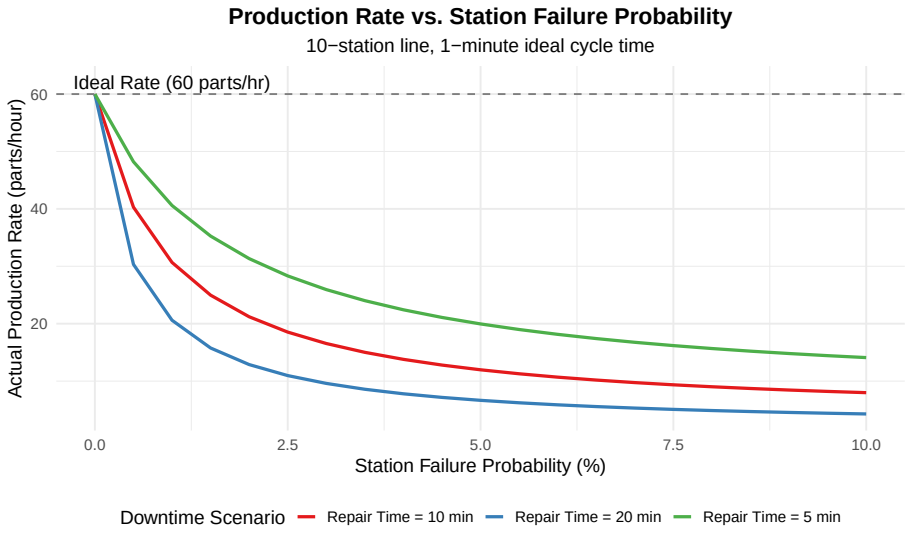


Figure 20.5: Impact of Station Failures on Production Rate

Discussion: Why is preventive maintenance critical for automated lines?

The math shows why: - At 5% failure probability with 10-minute repairs, you lose 25% of capacity - This costs more than the maintenance to prevent it

Cost comparison (example): - Lost production: 15 parts/hour × \$50/part × 8 hours = \$6,000/day - Preventive maintenance: \$500/day - **ROI of PM: 12:1**

Best practices: 1. Track failure data by station 2. Prioritize PM on high-failure stations 3. Stock critical spare parts 4. Train operators on quick changeover/repair 5. Consider redundant stations for bottlenecks

20.6.2 Buffer Sizing

Buffers between stations decouple operations and improve overall line availability.

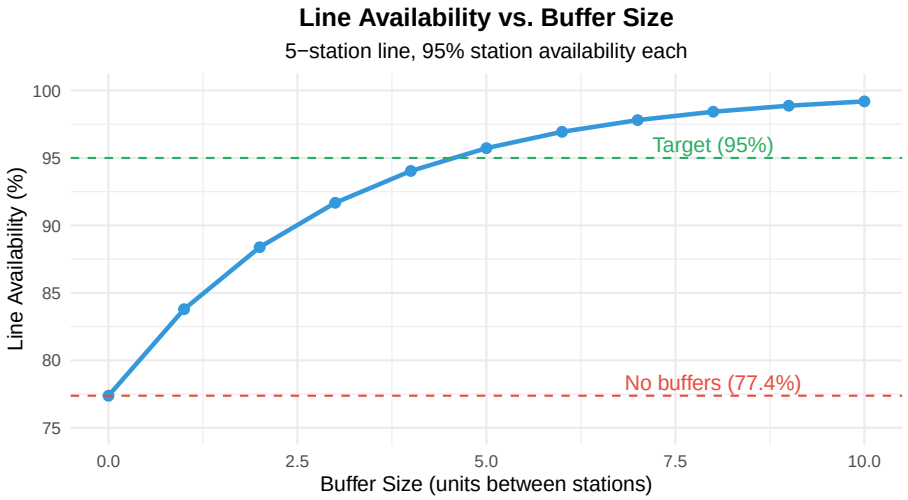


Figure 20.6: Effect of Buffers on Line Availability

20.6.3 Try it yourself: what does a buffer buy you?

The formulas above give the average. A simulation shows what happens cycle by cycle. This one runs a six-station automated line for 2 000 cycles. Each station that is working has a small chance of breaking down, and a breakdown lasts several cycles. With no buffers the stations are rigidly linked, so one stoppage stops them all. With buffers, a station can keep working from the stock in front of it and into the space behind it.

Change the buffer size and watch the efficiency. Then press **Run again** a few times: the answer moves, because breakdowns are random. In class, each team can enter its team number as the run number and compare.

Show the live R code behind this simulation

```
#| label: line-buffer-sim
# A serial production line with random breakdowns
n_stations <- 6       # stations in the line
fail_pct <- 4         # chance (%) that a working station breaks down in a cycle
repair_cycles <- 5    # cycles a breakdown lasts
buffer_size <- 0      # parts of storage between neighbouring stations (0 = rigid line)
seed <- 1             # run number: same number, same breakdowns

run_line <- function(buffer) {
  set.seed(seed)
  n_cycles <- 2000
  down <- integer(n_stations)      # cycles of repair left at each station
  stock <- integer(n_stations - 1) # parts waiting in each buffer
```

```
  made <- 0

  for (cycle in seq_len(n_cycles)) {
    down <- pmax(down - 1, 0)
    up <- down == 0
    breaks <- runif(n_stations) < fail_pct / 100   # would this station break if it ran?

    if (buffer == 0) {
      # Rigid line: every station is tied to its neighbours
      if (all(up)) {
        if (any(breaks)) down[breaks] <- repair_cycles else made <- made + 1
      }
    } else {
      for (i in n_stations:1) {    # last station first, so it frees space behind it
        has_part  <- i == 1 || stock[i - 1] > 0
        has_space <- i == n_stations || stock[i] < buffer
        if (up[i] && has_part && has_space) {
          if (breaks[i]) {
            down[i] <- repair_cycles
          } else {
            if (i > 1) stock[i - 1] <- stock[i - 1] - 1
            if (i < n_stations) stock[i] <- stock[i] + 1 else made <- made + 1
          }
        }
      }
    }
  }
  made / n_cycles * 100
}

efficiency <- run_line(buffer_size)

# The same line at a range of buffer sizes, for comparison
sizes <- c(0, 1, 2, 3, 5, 8, 12, 20, 30)
curve <- sapply(sizes, run_line)
par(mar = c(4.5, 5, 3.5, 1), cex.main = 1.3, cex.lab = 1.2, cex.axis = 1.1)
plot(sizes, curve, type = "b", lwd = 3, pch = 19, col = "#2c3e50", ylim = c(0, 100), las = 1,
     xlab = "Buffer between stations (parts)", ylab = "Line efficiency (%)",
     main = "Each extra part of buffer buys less than the one before")
points(buffer_size, efficiency, pch = 19, cex = 2.6, col = "#e74c3c")

cat(sprintf("Line efficiency: %.1f%%\n", efficiency))
cat(sprintf("  Parts made             %.0f in 2000 cycles\n", efficiency * 20))
cat(sprintf("  Buffer between stations  %d parts\n", buffer_size))
cat(sprintf("  Work-in-process allowed  up to %d parts\n", buffer_size * (n_stations - 1)))
```

20.7 Cellular Manufacturing

Cellular manufacturing applies Group Technology to organize production into cells that specialize in "families" of similar parts.

20.7.1 Part Families and Machine Cells

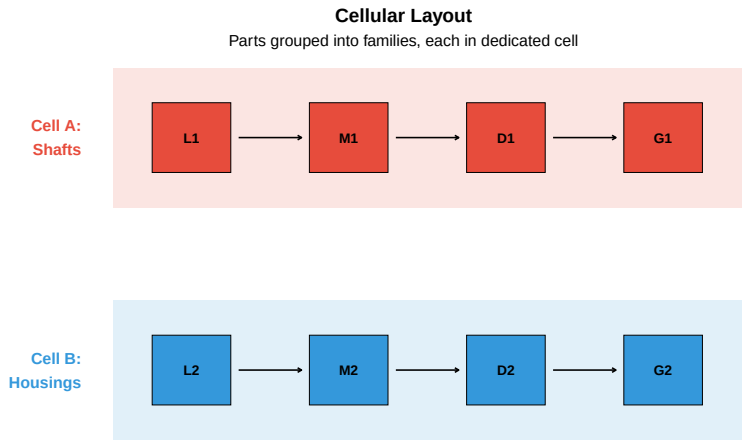


Figure 20.7: Cellular Manufacturing: From Process Layout to Cells

Table 20.4: Cellular Manufacturing Benefits vs. Traditional Process Layout

Metric	Traditional (Process)	Cellular	Typical Improver
Setup time	High - frequent changeovers	Low - dedicated to family	50-90% reduction
Work-in-process	High - batches waiting	Low - continuous flow	50-90% reduction
Lead time	Long - complex routing	Short - U-cell flow	50-80% reduction
Quality defects	Higher - multiple handoffs	Lower - cell ownership	30-50% reduction
Material handling	High - long distances	Low - within cell	40-70% reduction
Floor space	More - large aisles	Less - compact cells	20-50% reduction
Worker skills	Specialized by machine	Cross-trained on cell	Increased flexibility

20.8 Flexible Manufacturing Systems (FMS)

An **FMS** is a highly automated cell consisting of CNC machines interconnected by automated material handling, all controlled by a central computer.

20.8.1 FMS Components

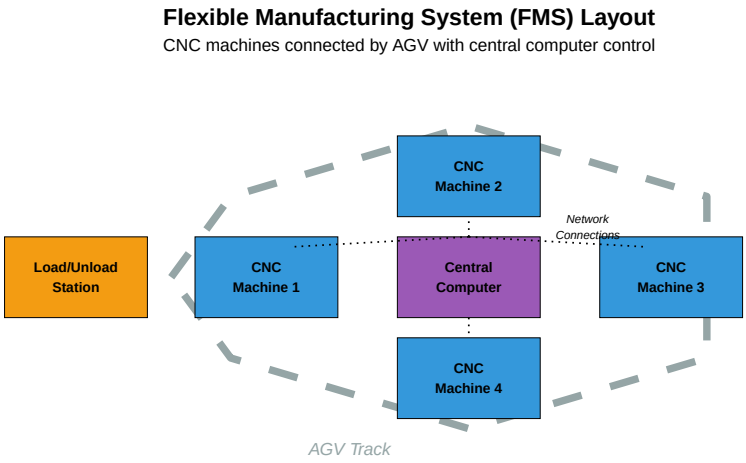


Figure 20.8: Flexible Manufacturing System Components

Table 20.5: FMS Characteristics and Applications

Characteristic	FMS Capability
Production Volume	Medium (2,000 - 100,000/year per part type)
Part Variety	Medium (4-100 different part types)
Batch Size	Small to medium (1-500 per batch)
Setup Time	Minimal (automatic pallet/fixture changes)
Typical Parts	Prismatic parts (housings, brackets, covers)
Investment	$5M - $50M+ depending on size
Best Suited For	Aerospace, automotive components, job shops with repeat parts

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20.9 Try it yourself: match the system to the job

Eight production situations. Decide which type of system from this chapter suits each one. Think about volume, variety, and how much of the work a person must do.

Which production system?

Pick the system that suits each situation.

Categories: Manual assembly line / Automated transfer line / Manufacturing cell / FMS

1. A dashboard needs wiring harnesses routed and clipped by hand; 400 vehicles a day. _____
2. Laptops are assembled in a fixed task sequence that is balanced to takt time. _____
3. One engine block design, 300,000 a year, machined at 20 stations in sequence. _____
4. Beverage cans are formed, washed, coated and printed at 2,000 a minute. _____
5. Twelve similar shaft part numbers are grouped as a family and made on a U-shaped group of machines. _____
6. Two cross-trained operators run five machines arranged in a U for one family of brackets. _____
7. CNC machining centres linked by an automated pallet system make 40 part numbers in any order, lights-out. _____
8. A central computer routes pallets between machines and changes part programs with almost no changeover loss. _____

Answers: 1 Manual assembly line; 2 Manual assembly line; 3 Automated transfer line; 4 Automated transfer line; 5 Manufacturing cell; 6 Manufacturing cell; 7 FMS; 8 FMS.

20.10 Modern Manufacturing Concepts

20.10.1 SMED (Single-Minute Exchange of Die)

SMED is a lean manufacturing technique to reduce changeover time to under 10 minutes (“single digit minutes”).

Table 20.6: SMED M

Step	Description
1. Separate	Distinguish internal setup (machine stopped) from external setup (machine runni
2. Convert	Convert as much internal setup to external setup as possible
3. Streamline	Reduce time for remaining internal and external activities

Case Study: Injection Molding Changeover

Before SMED: - Changeover time: 90 minutes - Machine stopped for entire process - One operator working alone

After SMED analysis:

Activity	Before	Type	After	Change
Find tools	10 min	Internal	0	External - tools pre-staged
Remove old mold	15 min	Internal	5 min	Quick-release clamps
Get new mold	10 min	Internal	0	External - mold pre-staged at machine
Install new mold	20 min	Internal	8 min	Standardized mold heights
Connect utilities	10 min	Internal	4 min	Quick-connect fittings
Adjust settings	15 min	Internal	3 min	Pre-programmed recipes
First article	10 min	Internal	5 min	Improved documentation

Results: - Total changeover: 25 minutes (72% reduction) - Additional capacity: +15 changeovers/week possible - Smaller batch sizes now economical

20.10.2 AGV and AMR Systems

Table 20.7: AGV vs. AMR Comparison

Feature	AGV (Automated Guided Vehicle)	AMR (Autonomous Mobile Robot)
Navigation	Fixed path (wires, magnets, tape)	Dynamic (SLAM, vision, sensors)
Flexibility	Low - follows predetermined routes	High - calculates optimal routes
Infrastructure	Requires installation (floor modifications)	Minimal (maps environment)
Cost	Lower vehicle cost, higher infrastructure	Higher vehicle cost, lower infrastructure
Path Changes	Requires physical modification	Software update only
Obstacle Handling	Stops and waits	Navigates around obstacles
Best Application	High-volume, repetitive routes	Variable routes, dynamic environments

20.10.3 Digital Twin

A **digital twin** is a virtual representation of a physical manufacturing system that is updated in real-time with sensor data.

Digital Twin Data Flow

1. **Physical System**
2. **Sensors & IoT**
3. **Digital Twin**
4. **Analytics & AI**
5. **Visualization**

Feedback for optimization

Digital Twin Applications: - Predictive maintenance (predict failures before they occur) - Process optimization (simulate changes before implementation) - Training (virtual commissioning and operator training) - Quality prediction (correlate parameters with outcomes)

20.11 Industry Applications

Table 20.8: Manufacturing System Applications by Ind

System Type	Automotive	Food Processing
Transfer Line	Engine block machining, cylinder head lines	Beverage bottling, canning lin
Manual Assembly	Final vehicle assembly, interior trim	Packaging, quality inspection
FMS	Transmission components, brake parts	Multi-product bakery equipme
Cellular	Subassemblies, wiring harnesses	Specialty product lines
AGV/AMR	Body shop material delivery, parts sequencing	Ingredient delivery, packaging

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20.12 Review Questions

Question 1: Takt Time Calculation

A plant operates 8-hour shifts with 30 minutes of breaks. Daily demand is 400 units. Calculate the takt time.

Question 1: takt time

- Takt time: _____ min/unit *(to 3 decimal places)*
- Takt time: _____ sec/unit *(to 1 decimal place)*

Show the worked solution

Solution:

Available time = 8 hours × 60 min - 30 min = 450 minutes

Takt time = 450 min / 400 units = **1.125 minutes per unit** = 67.5 seconds

If actual cycle time is 60 seconds, the plant has (67.5 - 60)/67.5 = **11% capacity margin**.

Question 2: Line Efficiency

A 6-station assembly line has a cycle time of 2 minutes. The task times at each station are: 1.8, 1.9, 2.0, 1.7, 1.6, and 1.5 minutes. Calculate the line efficiency.

Question 2: line efficiency

- Line efficiency: _____ % *(to 1 decimal place)*
- Balance delay: _____ % *(to 1 decimal place)*

Show the worked solution

Solution:

Total work content = 1.8 + 1.9 + 2.0 + 1.7 + 1.6 + 1.5 = 10.5 minutes

Line efficiency = (10.5) / (6 × 2.0) × 100% = 10.5/12 × 100% = **87.5%**

Balance delay (idle time) = 100% - 87.5% = **12.5%**

Question 3: Downtime Impact

An automated line has 8 stations, each with 3% failure probability per cycle and 8-minute average repair time. The ideal cycle time is 0.5 minutes. What is the actual production rate?

Question 3: downtime impact

- Line failure probability per cycle: _____ % *(to 1 decimal place)*
- Actual cycle time: _____ min *(to 2 decimal places)*
- Actual production rate: _____ parts/hr *(to 1 decimal place)*

Show the worked solution

Solution:

Line failure probability = 1 - (1 - 0.03)^8 = 1 - 0.97^8 = 1 - 0.784 = **21.6%**

Actual cycle time = 0.5 + (0.216 × 8) = 0.5 + 1.73 = **2.23 minutes**

Actual production rate = 60 / 2.23 = **26.9 parts/hour**

(vs. ideal of 120 parts/hour - only 22% of ideal capacity!)

This shows why reliability is critical in automated systems.

Question 4: FMS Selection

A company makes 15 different part numbers with annual volumes of 500-5,000 each. Parts are machined from aluminum and require milling, drilling, and tapping. Which system is most appropriate: dedicated transfer line, FMS, or CNC job shop?

Answer: FMS is most appropriate because:

- **Volume range (500-5,000)** is ideal for FMS
- **Part variety (15 types)** requires flexibility
- **Similar operations** (milling, drilling, tapping) can share equipment
- **Material (aluminum)** machines well on CNC

Transfer line would require too much volume per part. Job shop would have high setup times and low efficiency for these volumes.

Question 5: SMED Application

A stamping press has a 45-minute die change. Internal activities are 35 minutes, external are 10 minutes but currently done with machine stopped. How much can SMED reduce this?

Answer:

Step 1 - Separate: Do the 10 minutes of external work while press runs. - New changeover: 35 minutes (30% reduction)

Step 2 - Convert: Analyze the 35 minutes: - If 15 minutes can be converted to external (pre-staging, etc.): New changeover = 20 minutes

Step 3 - Streamline: Quick-release clamps, standardization might save another 5 minutes. - Final changeover: **15 minutes (67% reduction)**

This enables smaller batch sizes and more frequent changeovers.

20.13 Summary

Table 20.9: Chapter 20 Summary

Topic	Key Points
Production Lines	Manual, continuous, synchronous, or asynchronous transfer; each has tra
Time Metrics	Takt time = demand pace; Cycle time = actual pace; MCE measures va
Line Balancing	Assign tasks to minimize idle time; Efficiency = work content / (station
Automated Lines	Reliability is critical; Small failure probabilities cause large capacity loss
Cellular Manufacturing	Group Technology groups part families; Reduces setup, WIP, lead time
FMS	Automated cells for medium variety/volume; Computer-controlled flexib
Modern Concepts	SMED for quick changeover; AGV/AMR for material handling; Digital

20.13.1 Key Takeaways

1. **Match system to product** - Volume and variety determine optimal manufacturing system

2. **Takt time sets the pace** - Everything should be designed to meet customer demand rate
3. **Balance matters** - Unbalanced lines waste capacity through idle time
4. **Reliability is critical** - Small failures compound to large losses in automated systems
5. **Flexibility has value** - FMS and cellular systems enable quick response to change
6. **Technology evolves** - Digital twins and AMRs are transforming manufacturing

---

20.14 References

1. Groover, M.P. (2020). *Automation, Production Systems, and Computer-Integrated Manufacturing* (5th ed.). Pearson.
2. Liker, J.K. (2004). *The Toyota Way.* McGraw-Hill.
3. Shingo, S. (1985). *A Revolution in Manufacturing: The SMED System.* Productivity Press.
4. Black, J.T. (1991). *The Design of the Factory with a Future.* McGraw-Hill.

Chapter 21

Troubleshooting and Root Cause Analysis

21.1 Learning Objectives

After completing this chapter, you will be able to:

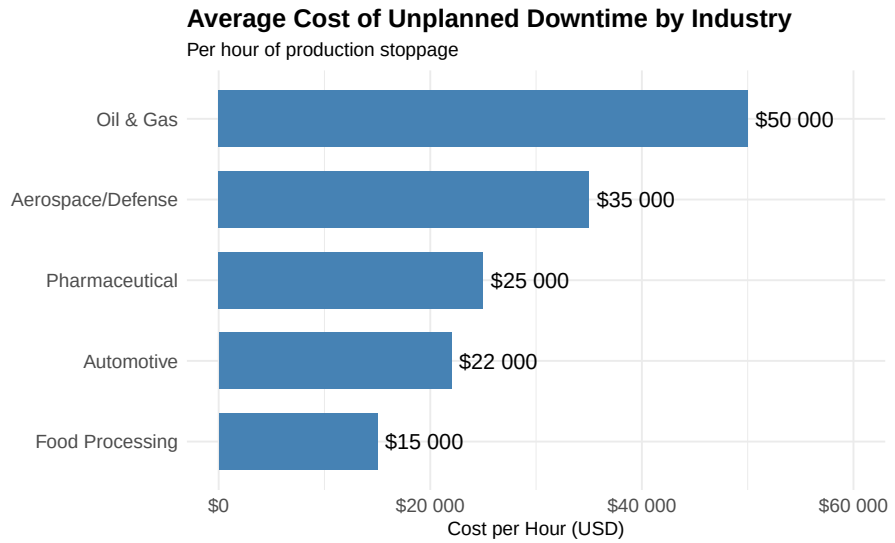
1. Apply systematic troubleshooting methodologies to diagnose equipment and process failures
2. Distinguish between symptoms, immediate causes, and root causes
3. Use the 5 Whys technique to drill down to fundamental causes
4. Construct and analyze Ishikawa (fishbone) diagrams
5. Apply Fault Tree Analysis (FTA) to complex failure scenarios
6. Use Pareto analysis to prioritize problem-solving efforts
7. Implement the 8D problem-solving methodology
8. Document findings and implement effective corrective actions
9. Prevent problem recurrence through systemic improvements

21.2 Introduction to Troubleshooting

Troubleshooting is the systematic process of identifying, diagnosing, and resolving problems in equipment, processes, or systems. For electromechanical technicians, effective troubleshooting is perhaps the most valuable skill you can develop.

21.2.1 The Cost of Downtime

Equipment failures and process problems have significant financial impacts:



21.2.2 Reactive vs. Proactive Problem Solving

Table 21.1: Comparison of Problem-Solving Approaches

Aspect	Reactive Approach	Proactive Approach
Timing	After failure occurs	Before or early in failure
Focus	Quick fix to restore operation	Root cause elimination
Stress Level	High - production pressure	Lower - planned approach
Cost	High - emergency repairs	Lower - scheduled work
Learning	Limited - rush to fix	High - thorough analysis
Documentation	Often skipped	Systematic and complete

Think About It: Why do organizations often default to reactive troubleshooting?

Common reasons include: - **Time pressure**: Production demands make it tempting to apply quick fixes - **Lack of training**: Technicians may not know RCA methodologies - **Reward systems**: Organizations may reward "firefighters" who quickly restore production - **Missing data**: Without good records, patterns are hard to identify - **Culture**: "We've always done it this way" mentality

The irony is that reactive troubleshooting actually costs more time in the long run due to recurring problems.

21.3 Systematic Troubleshooting Methodology

Effective troubleshooting follows a structured approach rather than random trial-and-error.

21.3.1 The DMAIC Troubleshooting Framework

Systematic Troubleshooting Process

1. **1. DEFINE** – Clearly state the problem
2. **2. MEASURE** – Gather data and observations
3. **3. ANALYZE** – Identify root cause
4. **4. IMPLEMENT** – Apply corrective action
5. **5. CONTROL** – Verify solution works
6. **6. DOCUMENT** – Record findings

21.3.2 Step 1: Define the Problem

A well-defined problem is half-solved. Use the **5W2H** method:

Table 21.2: 5W2H Problem Definition Framework

Question	What to Ask	Example Response
WHAT	What is the problem? What is happening vs. what should happen?	Motor overheating on c
WHERE	Where does the problem occur? Location, machine, station?	Discharge end of convey
WHEN	When did it start? When does it occur? Continuous or intermittent?	Started Monday, occurs
WHO	Who discovered it? Who is affected? Who was operating?	Operator noticed, maint
WHY	Why is this a problem? What is the impact?	Risk of motor failure, li
HOW	How was the problem detected? How is it manifesting?	Detected by high-temp
HOW MUCH	How often? How many units affected? What is the magnitude?	Occurs every day, 100%

21.3.3 Problem Statement Template

> **Good Problem Statement Format:** "[WHAT] is occurring at [WHERE] since [WHEN], affecting [WHO/WHAT], with a magnitude of [HOW MUCH]."

Example: > "Motor overheating (85°C vs. normal 70°C) is occurring at Conveyor Line 3 discharge motor since Monday morning, affecting packaging department production, with the condition occurring every shift after approximately 2 hours of operation."

Practice: Write a problem statement for this scenario

Scenario: An injection molding machine is producing parts with short shots (incomplete fill). The problem was noticed by the quality inspector during the

night shift. About 15% of parts are affected. The issue seems worse when running the larger 500g parts compared to the smaller 200g parts.

Sample Problem Statement: "Short shots (incomplete cavity fill) are occurring on Injection Molding Machine #4 since night shift on Tuesday, affecting 15% of parts produced, with the defect rate increasing significantly when running 500g parts compared to 200g parts."

21.3.4 Step 2: Gather Data and Observations

Before making any changes, collect information systematically:

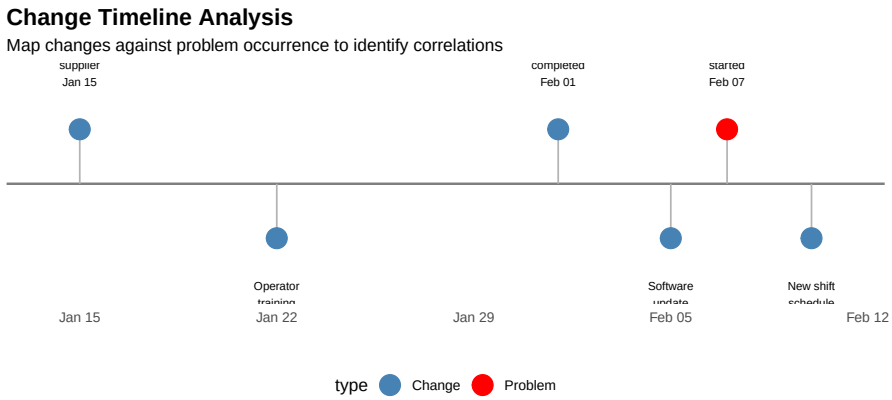
Table 21.3: Systematic Data Gathering Checklist

Data Source	What to Check	Tools/Metho
Visual Inspection	Physical condition, wear, contamination, alignment, damage	Flashlight, m
Operator Interview	What changed? When did it start? Any unusual events?	Open-ended
Machine Parameters	Temperatures, pressures, speeds, currents, positions	HMI screens,
Historical Records	Maintenance logs, previous repairs, similar issues	CMMS, logb
Product/Output	Defect patterns, measurements, reject rates	Calipers, gau
Environmental	Temperature, humidity, vibration, power quality	Thermomete

21.3.5 The "Change Analysis" Approach

Many problems are caused by changes. Ask:

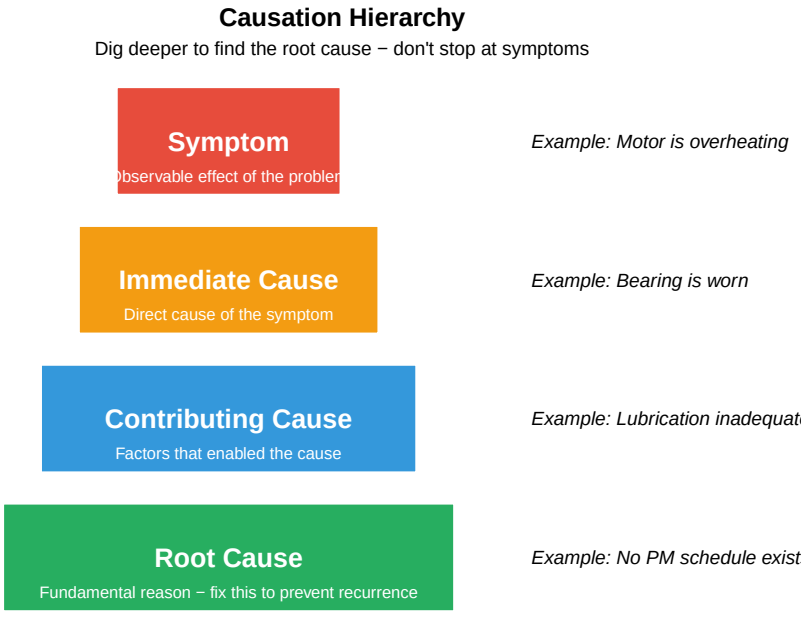
- What changed recently?
- New materials, suppliers, or batches?
- New operators or different shifts?
- Maintenance performed?
- Environmental changes (weather, season)?
- Software updates or parameter changes?
- New products or recipes running?



21.4 Understanding Root Cause vs. Symptoms

A critical concept in troubleshooting is distinguishing between different levels of causation.

21.4.1 The Causation Hierarchy



21.4.2 Why Finding Root Cause Matters

Table 21.4: Comparison of Fix

Fix Level	Action Taken	Immediate Result
Symptom	Add cooling fan to motor	Motor runs cooler tem
Immediate Cause	Replace worn bearing	Motor works for a whil
Root Cause	**Implement PM schedule with lubrication tasks**	**Problem eliminated**

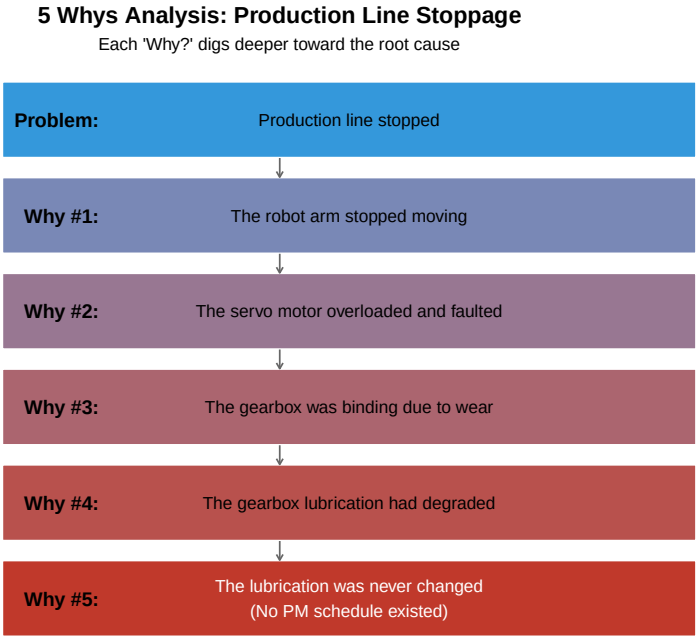
21.5 The 5 Whys Technique

The **5 Whys** is a simple but powerful technique developed by Sakichi Toyoda and used within Toyota's manufacturing methods.

21.5.1 How It Works

1. State the problem
2. Ask "Why?" to identify the cause
3. Ask "Why?" again about that cause
4. Continue asking "Why?" until you reach the root cause
5. Typically requires 5 iterations (sometimes more, sometimes less)

21.5.2 5 Whys Example: Production Line Stoppage



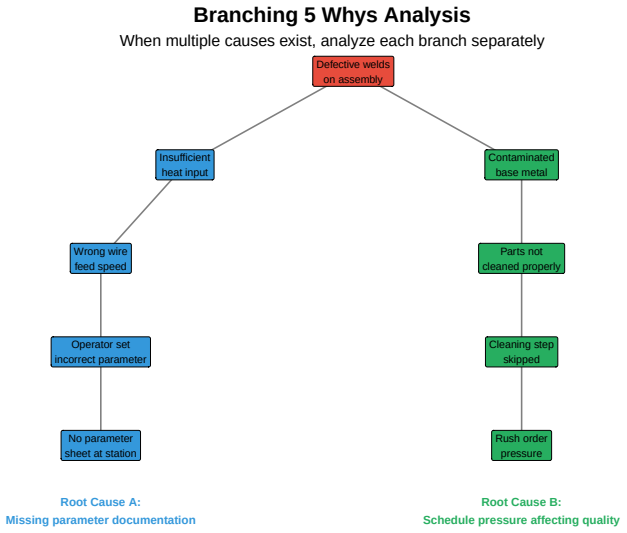
21.5.3 5 Whys Template

Table 21.5: 5 Whys Analysis Template

Step	Description	Typical Cause Type
Problem Statement	Clear, specific description of what happened	Symptom
Why #1	First-level cause (usually technical/physical)	Physical
Why #2	Why did that first cause occur?	Physical/Human
Why #3	Why did the second cause occur?	Human/Process
Why #4	Why did the third cause occur?	Process/System
Why #5	Why did the fourth cause occur?	System/Culture
Root Cause	**The fundamental cause that, if fixed, prevents recurrence**	**Root**

21.5.4 Multiple Branches in 5 Whys

Sometimes a problem has multiple causes. When you encounter this, branch your analysis:



Practice Exercise: Perform a 5 Whys Analysis

Scenario: A CNC machine produced 50 parts that were all 0.5mm undersize on a critical diameter.

Try to complete the 5 Whys analysis before looking at the sample answer below.

Sample Analysis:

1. **Problem:** CNC produced 50 undersize parts

2. **Why #1:** The tool offset was set incorrectly
3. **Why #2:** The operator entered the wrong offset value
4. **Why #3:** The operator misread the setup sheet
5. **Why #4:** The setup sheet had poor formatting and small print
6. **Why #5:** No standard format exists for setup documentation

Root Cause: Lack of standardized, clear setup documentation

Corrective Actions: - Create standardized setup sheet template with large, clear fonts - Add visual verification step (photo of correct setup) - Implement first-piece inspection requirement before production run

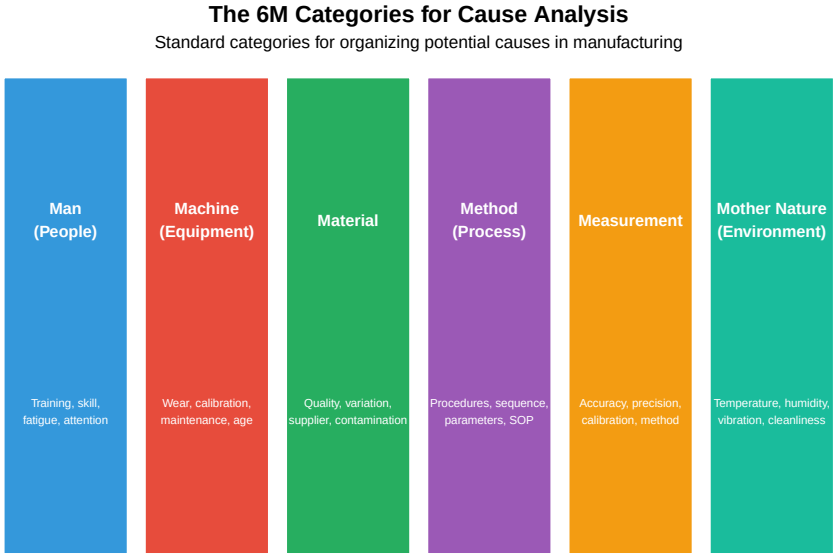
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21.6 Ishikawa (Fishbone) Diagram

The **Ishikawa diagram**, also called a **fishbone diagram** or **cause-and-effect diagram**, was developed by Kaoru Ishikawa in the 1960s. It provides a structured way to brainstorm and categorize potential causes.

21.6.1 The 6M Categories

Manufacturing problems are typically organized using the **6M** categories:



21.6.1.1 Try it yourself: hang the cause on the right bone

A team is brainstorming why a CNC lathe is producing out-of-tolerance diameters. Put each suggested cause on the correct bone of the fishbone.

Which of the 6M categories?

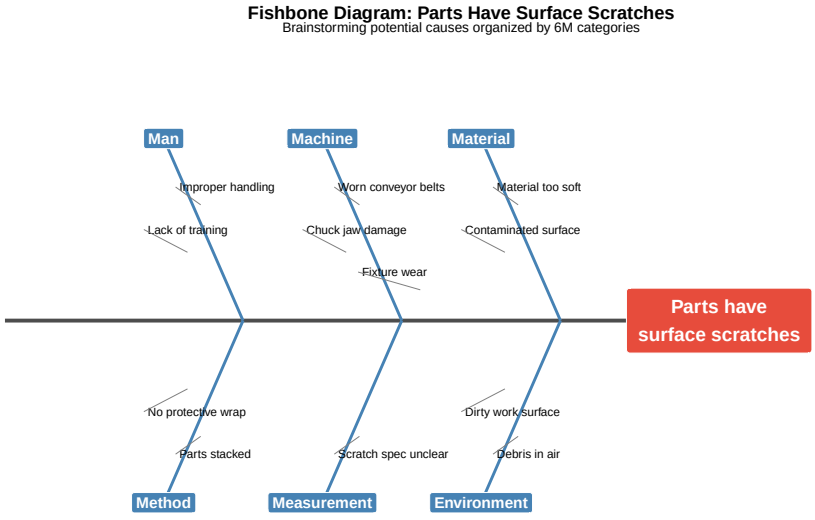
Put each cause on its bone.

Categories: Man / Machine / Material / Method / Measurement / Mother Nature

1. A newly hired operator has not yet been trained on tool offsets. _____
2. The spindle bearing has excessive play. _____
3. The chuck jaws are worn and no longer grip concentrically. _____
4. Bar stock from a new supplier is harder than specified. _____
5. The work instruction does not say how often to check the insert for wear. _____
6. The micrometer used at the machine is out of calibration. _____
7. The shop is 12 degrees C warmer in the afternoon and the parts grow before they are measured. _____

Answers: 1 Man; 2 Machine; 3 Machine; 4 Material; 5 Method; 6 Measurement; 7 Mother Nature.

21.6.2 Constructing a Fishbone Diagram



21.6.3 Fishbone Diagram Best Practices

Table 21.6: Best Practices for Fishbone Diagram Development

Guideline	Description
Team Approach	Include operators, engineers, maintenance, quality - dive
No Evaluation During Brainstorming	Capture all ideas first; evaluate feasibility later
Use Specific Language	Not 'machine problem' but 'bearing wear on spindle mot
Aim for 3-5 Causes per Category	Some categories may have more; don't force causes into
Circle the Most Likely	After brainstorming, identify 2-3 most probable causes t
Verify with Data	Don't assume - test hypotheses with data before implem

21.6.4 Video: How to Create a Fishbone Diagram

21.7 Fault Tree Analysis (FTA)

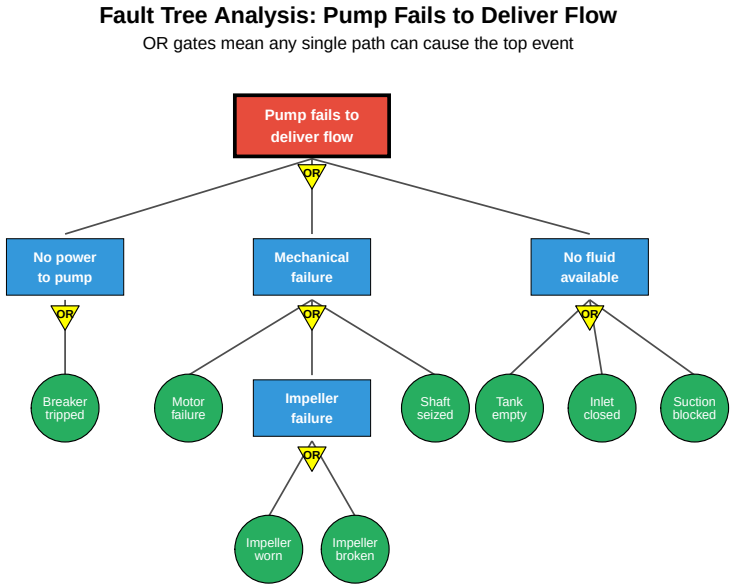
Fault Tree Analysis is a top-down, deductive analysis method that uses Boolean logic to analyze undesired events. Originally developed for the aerospace industry, FTA is particularly useful for complex systems with multiple potential failure paths.

21.7.1 FTA Symbols

Table 21.7: Fault Tree Analysis Symbols

Symbol	Name	Meaning
Rectangle	Intermediate Event	A fault event that results from combination of other events
Circle	Basic Event	A basic failure that cannot be broken down further
Diamond	Undeveloped Event	Event not analyzed further (outside scope or insufficient data)
House	Normal Event	An event expected to occur during normal operation
AND Gate	AND Gate	Output occurs only if ALL inputs occur
OR Gate	OR Gate	Output occurs if ANY input occurs

21.7.2 FTA Example: Pump Fails to Deliver Flow



21.7.3 Calculating Probability with FTA

FTA allows quantitative risk analysis by calculating the probability of the top event:

OR Gate: P(output) = 1 - [(1 - P_A) × (1 - P_B) × ...] *Simplified:* P(output) $P_A + P_B$ (when probabilities are small)

AND Gate: P(output) = $P_A \times P_B \times \ldots$

Your turn — change the basic-event probabilities below.

Show the live R code behind this calculator

```
#| label: fta-calculation
# Example: Calculate top event probability
# Basic event probabilities (annual failure rates)
P_breaker_trip <- 0.02
P_motor_fail <- 0.01
P_impeller_worn <- 0.03
P_impeller_break <- 0.005
P_shaft_seize <- 0.008
P_tank_empty <- 0.05
P_valve_closed <- 0.01
P_line_blocked <- 0.02
```

```
# Impeller failure (OR gate)
P_impeller_fail <- 1 - (1 - P_impeller_worn) * (1 - P_impeller_break)

# Mechanical failure (OR gate)
P_mechanical <- 1 - (1 - P_motor_fail) * (1 - P_impeller_fail) * (1 - P_shaft_seize)

# No fluid available (OR gate)
P_no_fluid <- 1 - (1 - P_tank_empty) * (1 - P_valve_closed) * (1 - P_line_blocked)

# Top event: Pump fails (OR gate)
P_top <- 1 - (1 - P_breaker_trip) * (1 - P_mechanical) * (1 - P_no_fluid)

cat(sprintf("P(pump fails to deliver flow): %.1f%% per year\n", P_top * 100))
cat(sprintf("  P(impeller failure)      %.4f\n", P_impeller_fail))
cat(sprintf("  P(mechanical failure)    %.4f\n", P_mechanical))
cat(sprintf("  P(no fluid available)    %.4f\n", P_no_fluid))
```

21.7.4 Try it yourself: work a fault tree by hand

A packaging line's infeed conveyor keeps stopping without warning. The team has drawn the fault tree below and estimated the probability of each basic event per shift. Work upward: fill in the probability of the two intermediate events, then the top event.

Remember the two gate rules. Through an **AND** gate every input must occur, so multiply the probabilities. Through an **OR** gate any one input is enough, so use $1 - (1 - \text{P1})(1 - \text{P2})\ldots$

Fault tree: conveyor stops unexpectedly

Fill in P for the three gates. Give four decimal places.

Box	Values	Inputs from below
Conveyor stops unexpectedly (OR gate)	P = _____	Drive fails, Control power lost, False jam signal
Drive fails (OR gate)	P = _____	Motor burns out, Gearbox seizes
Control power lost (AND gate)	P = _____	Mains supply fails, UPS fails
False jam signal (Basic event)	P = 0.020	-
Motor burns out (Basic event)	P = 0.010	-
Gearbox seizes (Basic event)	P = 0.005	-

Box	Values	Inputs from below
Mains supply fails (Basic event)	P = 0.050	-
UPS fails (Basic event)	P = 0.040	-

Show the worked solution

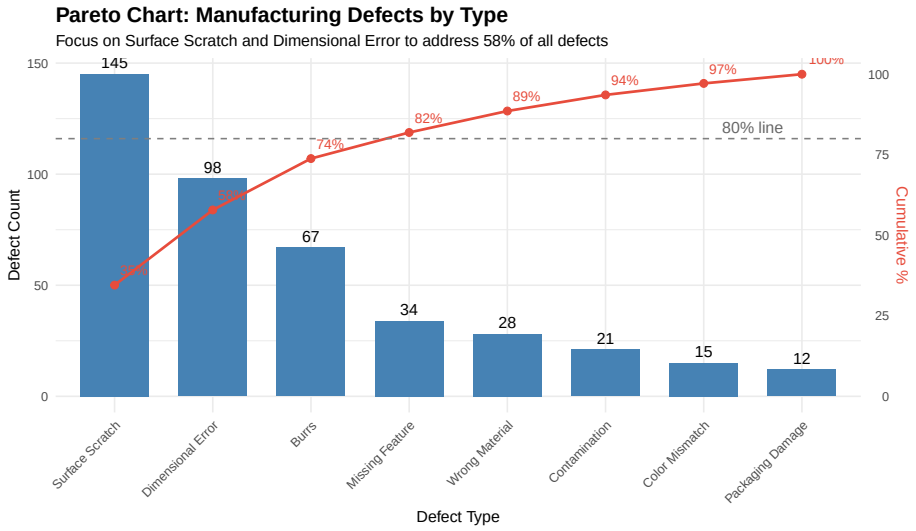
- **Drive fails** (OR): $1 - (1 - 0.010)(1 - 0.005) = \mathbf{0.0150}$
- **Control power lost** (AND): $0.050 \times 0.040 = \mathbf{0.0020}$
- **Conveyor stops** (OR): $1 - (1 - 0.0150)(1 - 0.0020)(1 - 0.020) = \mathbf{0.0366}$, about 3.7% of shifts

Look at where the risk comes from. The two power supplies are individually the least reliable items on the tree, yet because both must fail together they contribute almost nothing. The single false jam signal, with no redundancy, is the largest contributor. That is the argument for redundancy in one picture.

21.8 Pareto Analysis

Pareto Analysis is based on the Pareto Principle (80/20 rule): roughly 80% of effects come from 20% of causes. This helps prioritize problem-solving efforts.

21.8.1 Creating a Pareto Chart



21.8.2 Interpreting Pareto Analysis

Teaching demo — try different defect counts.

Show the live R code behind this card

```
#| label: pareto-interpretation
# Pareto analysis interpretation
defects <- data.frame(
  Defect = c("Surface Scratch", "Dimensional Error", "Burrs",
             "Missing Feature", "Wrong Material", "Contamination",
             "Color Mismatch", "Packaging Damage"),
  Count = c(145, 98, 67, 34, 28, 21, 15, 12)
)

defects <- defects[order(-defects$Count),]
defects$Cumulative_Pct <- cumsum(defects$Count) / sum(defects$Count) * 100

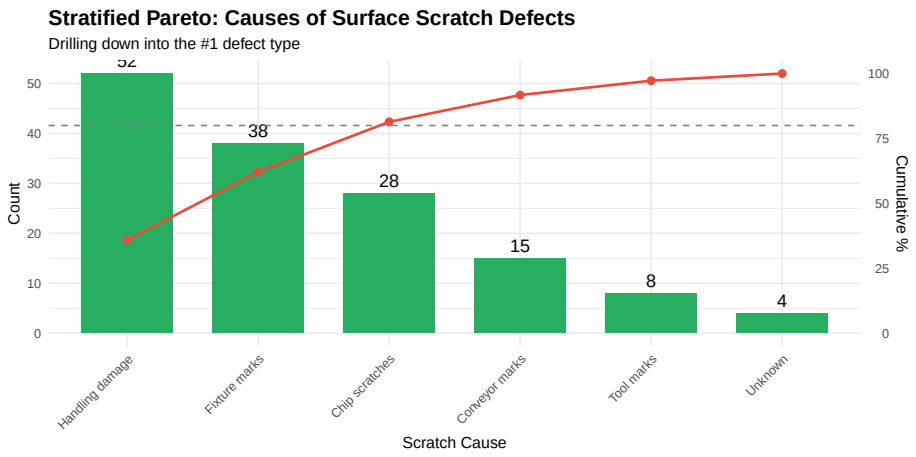
# Identify the "vital few" (those that contribute to 80%)
vital_few <- defects[defects$Cumulative_Pct <= 80 |
                       c(TRUE, defects$Cumulative_Pct[-nrow(defects)] < 80),]

cat(sprintf("Vital few: %g defect types address %.1f%% of all defects\n",
            nrow(vital_few), max(vital_few$Cumulative_Pct)))
cat(sprintf("  Total defects  %g\n", sum(defects$Count)))
for (i in seq_len(nrow(vital_few))) {
  cat(sprintf("  %s  %g (%.1f%% cumulative)\n", vital_few$Defect[i], vital_few$Count[i]
}

par(mar = c(5, 5, 3, 1))
barplot(defects$Count, names.arg = defects$Defect, cex.names = 0.55,
        col = ifelse(defects$Defect %in% vital_few$Defect, "#C0392B", "#95A5A6"),
        ylab = "Count", main = "Pareto chart (red = vital few)")
```

21.8.3 Stratified Pareto Analysis

Sometimes you need to break down categories further:



21.9 Is/Is Not Analysis

Is/Is Not Analysis is a structured comparison technique that helps narrow down the scope and identify distinguishing factors of a problem.

21.9.1 Is/Is Not Matrix

Table 21.8: Is/Is Not Analysis: Motor Overheating Problem

Dimension	Question	IS	IS NOT	W
WHAT	What object has the problem?	Motor on Line 3	Motors on Lines 1, 2, 4	L
WHAT	What is the defect/symptom?	Overheating (85°C)	Noise, vibration, or failure	T
WHERE	Where is the problem observed?	Packaging area, Station 4	Any other station	S
WHERE	Where on the object is the defect?	Motor housing, bearing end	Drive end, junction box	B
WHEN	When was it first observed?	Monday morning shift	Before weekend (Friday OK)	S
WHEN	When in the process/cycle?	After 2 hours of operation	Cold start or end of shift	H
EXTENT	How many units affected?	100% - every shift	Intermittent	C
EXTENT	Is it trending up/down/stable?	Stable - same every day	Getting worse over time	S

21.9.2 Key Insights from Is/Is Not Analysis

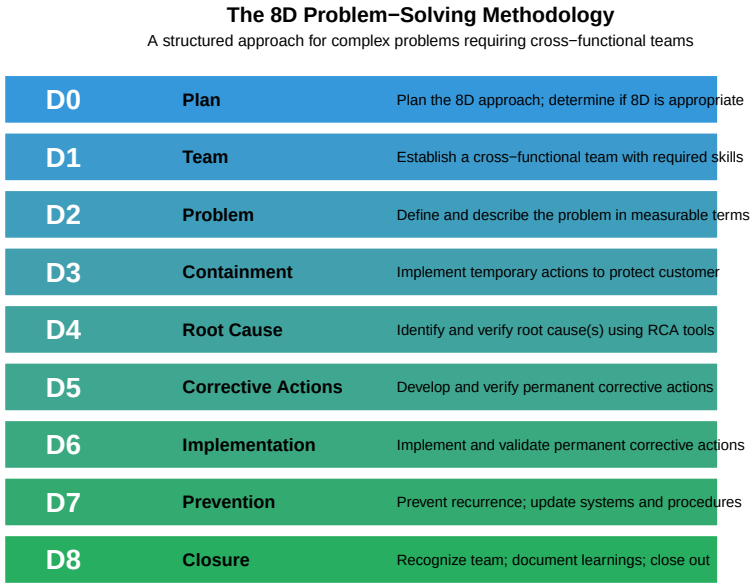
From the example above, we can identify:

1. **Line 3 specific** - Not a general problem across all lines
2. **Thermal, not mechanical** - Overheating but no noise/vibration yet
3. **After warm-up** - Heat accumulation issue
4. **Started after weekend** - Something changed

Investigation focus: What changed over the weekend? Check maintenance logs, any work done on Line 3, any environmental changes.

21.10 The 8D Problem-Solving Methodology

The **8D (Eight Disciplines)** methodology is a comprehensive problem-solving approach developed by Ford Motor Company. It's widely used in automotive and other industries.



21.10.1 Detailed 8D Steps

Table 21.9: 8D Methodology Detailed Breakdown

Discipline	Key Activities
D0: Plan	Emergency response actions, initial assessment, determine if 8D needed
D1: Team	Select team leader, identify required expertise, define roles
D2: Problem	5W2H, Is/Is Not, quantify impact, identify affected parts/lots
D3: Containment	Sort suspect material, increase inspection, alert downstream
D4: Root Cause	5 Whys, Fishbone, FTA, verify cause by testing
D5: Corrective Actions	Brainstorm solutions, verify effectiveness before full rollout
D6: Implementation	Execute action plan, validate results, monitor KPIs
D7: Prevention	Update FMEAs, procedures, training, share across organization

D8: Closure	Final documentation, lessons learned, team recognition	8D report,

21.10.1.1 Try it yourself: order the 8D disciplines

The 8D method runs from D0 to D8. Containment comes before root cause for a reason: the customer has to be protected while you investigate. Put the nine disciplines in order.

The 8D disciplines, D0 to D8

Number these in the correct order:

- _____ Choose the permanent corrective actions
- _____ Close out and recognize the team
- _____ Contain the problem to protect the customer
- _____ Describe the problem
- _____ Find and verify the root cause
- _____ Form the team
- _____ Implement and validate the corrective actions
- _____ Plan and prepare for the 8D
- _____ Prevent recurrence

Answers, in the order listed: 6, 9, 4, 3, 5, 2, 7, 1, 8.

21.10.2 D3: Containment Actions

Containment is crucial to protect the customer while you investigate:

Table 21.10: Types of Containment Actions

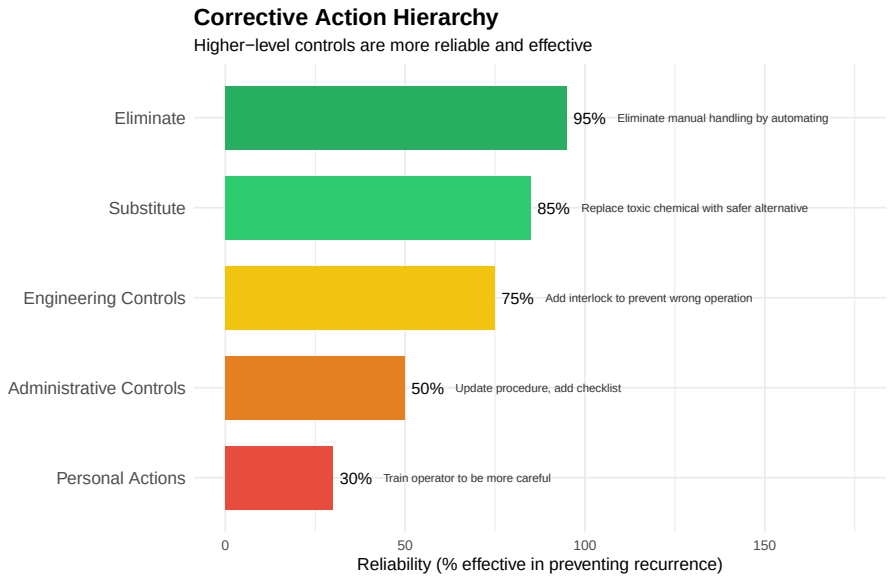
Type	Description	Example
Sort & Inspect	100% inspection of suspect inventory to separate good from bad	Inspect all parts from
Rework/Repair	Repair nonconforming units if possible and economical	Re-machine undersiz
Scrap	Dispose of units that cannot be repaired	Dispose of parts with
Hold/Quarantine	Prevent movement of suspect material until disposition	Tag and segregate al
Increase Monitoring	Add inspections, increase sampling, tighten control limits	Add 100% visual insp
Supplier Notification	Notify suppliers if incoming material is suspected	Request material cer
Customer Alert	Alert customers to check their inventory if product shipped	Contact distributor a

---

21.11 Corrective Action Implementation

Effective corrective actions address the root cause and prevent recurrence.

21.11.1 Types of Corrective Actions



21.11.2 SMART Corrective Actions

Corrective actions should be **SMART**:

Table 21.11: SMART Corrective Action Cri

	Meaning	Description	Poor Example	Good Ex
S	Specific	Clearly defined action, not vague	Improve quality	Install p
M	Measurable	Can verify completion and effectiveness	Train operators	Train all
A	Achievable	Realistic with available resources	Redesign entire product line	Add insp
R	Relevant	Directly addresses the root cause	Repaint the floor	Add chec
T	Time-bound	Has a target completion date	Do this sometime soon	Complet

---

21.12 Verification and Validation

Before closing out a problem, verify that your actions worked.

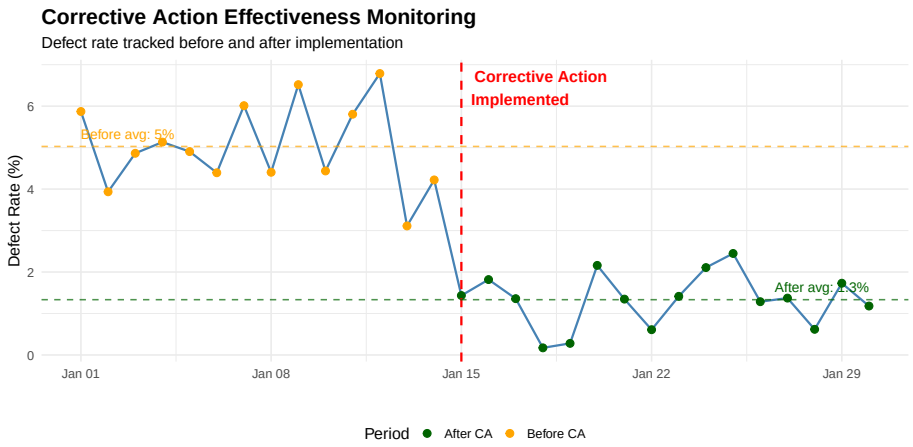
21.12.1 Verification vs. Validation

Table 21.12: Verification vs. Validation of Corrective Actions

Aspect	Verification	Validation
Definition	Did we implement the action correctly?	Did the action solve the problem?
Focus	Process compliance	Results and effectiveness
Question	Did we build the fix right?	Did we build the right fix?
Timing	During/after implementation	After implementation, during monitoring
Methods	Audits, inspections, checklists	Data analysis, trend charts, capability studies

21.12.2 Effectiveness Monitoring

```
Warning in scale_x_date(): A <numeric> value was passed to a Date scale.
i The value was converted to a <Date> object.
```




```r
# Calculate improvement
before_avg <- mean(defect_rate[1:14])
after_avg <- mean(defect_rate[15:30])
improvement <- (before_avg - after_avg) / before_avg * 100

cat("Before corrective action: ", round(before_avg, 2), "% defect rate\n")
cat("After corrective action:  ", round(after_avg, 2), "% defect rate\n")
cat("Improvement:              ", round(improvement, 1), "% reduction\n")
```

```
Before corrective action:  5.03 % defect rate
After corrective action:   1.33 % defect rate
Improvement:               73.5 % reduction
```

21.13 Case Study: Automotive Welding Defects

Let's work through a comprehensive troubleshooting case using multiple RCA tools.

21.13.1 The Problem

A Tier-1 automotive supplier is experiencing weld porosity defects on a critical structural component. Customer complaints have increased, and the plant has received a quality alert.

21.13.2 Initial Problem Definition (5W2H)

Table 21.13: Case Study Problem Definition

Question	Details
WHAT	Porosity (gas pockets) in MIG welds on crossmember subassembly
WHERE	Robot welding Cell 4, occurring at weld joint W-23 (corner weld)
WHEN	Started approximately 2 weeks ago, occurs on all shifts
WHO	Detected by customer during their incoming inspection
WHY	Customer rejecting parts; risk of production line stop
HOW	Visible voids on X-ray inspection; some visible to naked eye
HOW MUCH	Estimated 8% of parts affected (up from baseline of 0.5%)

21.13.3 5 Whys Analysis

Table 21.14: 5 Whys Analysis: Weld Porosity

Level	Finding
Problem	Porosity in welds at joint W-23
Why #1	Gas entrapment during solidification
Why #2	Inadequate shielding gas coverage
Why #3	Gas flow rate dropped below specification
Why #4	Flow meter was not calibrated; showing higher than actual
Why #5	**No calibration schedule existed for welding gas flow meters**

21.13.4 Fishbone Diagram Findings

After brainstorming with the cross-functional team:

Table 21.15: Fishbone Analysis Results with Verification

Category	Potential Cause	Likelihood	Verification Result
Machine	Gas flow meter out of calibration	HIGH	YES - Root cause
Machine	Worn contact tip affecting arc stability	Medium	No - tips OK
Material	Different wire lot from new supplier	Medium	No - tested OK
Material	Base metal surface contamination	Low	No - clean parts
Method	Robot path too fast for corner weld	Medium	Partial contributor
Method	Gas pre-flow time insufficient	HIGH	YES - Contributing
Man	Recent operator turnover, less experience	Low	No - all shifts affected
Environment	Seasonal humidity changes	Low	No - controlled

21.13.5 8D Report Summary

Table 21.16: 8D Report: Weld Porosity Case Study

D#	Discipline	Summary
D1	**Team**	Team: Weld Engineer (lead), Quality, Maintenance, Production Supervisor, Supplie
D2	**Problem**	8% porosity rate at W-23 joint since Jan 15; customer quality alert received Jan 28
D3	**Containment**	100% X-ray inspection of all crossmembers; sort and quarantine 3 days production
D4	**Root Cause**	Primary: Flow meter out of calibration (reading 25 CFH, actual 18 CFH). Contribu
D5	**Corrective Actions**	1. Recalibrate flow meter immediately; 2. Increase pre-flow time to 0.5s; 3. Add flo
D6	**Implementation**	Actions completed Feb 5; validation testing shows 0.1% porosity rate (below 0.5% t
D7	**Prevention**	Updated PM schedule to include all gas flow meters quarterly; added to PFMEA w
D8	**Closure**	Customer satisfied with response; 8D closed Feb 15; cost avoidance: $45,000 (avoide

---

21.14 Documentation and Knowledge Management

Proper documentation ensures learnings are captured and problems don't recur.

21.14.1 Key Documentation Elements

Table 21.17: RCA Documentation Checklist

Element	Contents	Benefit
Problem Description	5W2H, problem statement, impact/cost	Clear reference for s
Timeline	When discovered, key investigation dates, implementation dates	Understanding of re
Data Collected	Measurements, photos, test results, operator interviews	Evidence-based anal
Root Cause Analysis	5 Whys, fishbone diagrams, FTA as applicable	Documented logical
Corrective Actions	Actions taken, responsible parties, target/actual dates	Accountability, closu

Verification Results	Before/after data, statistical comparison, effectiveness %	Proof of
Lessons Learned	What we'd do differently, recommendations for similar situations	Continu
System Updates	PFMEA updates, procedure changes, training records	Prevent

21.14.2 Creating a Troubleshooting Knowledge Base

Build organizational memory by:

1. **Standardized templates** - Use consistent formats for all RCA documents
2. **Searchable database** - Store in CMMS, SharePoint, or dedicated quality system
3. **Categories and tags** - Equipment type, failure mode, root cause category
4. **Cross-references** - Link related issues and solutions
5. **Regular reviews** - Periodic analysis of trends and patterns

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21.15 Troubleshooting Tools Quick Reference

Table 21.18: RCA Tool Selection Guide

Tool	Best For	When to Use
5 Whys	Simple problems with linear cause chains	Quick investigations, st
Fishbone Diagram	Brainstorming multiple potential causes	Team problem-solving s
Fault Tree Analysis	Complex systems with multiple failure paths	Safety-critical systems,
Pareto Chart	Prioritizing which problems to tackle first	When facing multiple p
Is/Is Not	Narrowing down scope and identifying distinctions	When problem is hard t
8D	Customer complaints requiring formal response	Formal quality requirem
Change Analysis	Problems that started suddenly	When timing of problem

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21.16 Video Resources

21.16.1 Understanding 5 Whys

21.16.2 8D Problem Solving

21.17 Summary

Effective troubleshooting and root cause analysis require:

1. **Systematic approach** - Don't guess; follow a structured methodology
2. **Clear problem definition** - Use 5W2H to clearly define what you're solving
3. **Data-driven analysis** - Collect facts before jumping to conclusions
4. **Root cause focus** - Don't stop at symptoms; dig to fundamental causes
5. **Multiple tools** - Use 5 Whys, Fishbone, FTA, Pareto as appropriate
6. **Effective corrective actions** - Aim for elimination, not just mitigation
7. **Verification** - Confirm your fix actually works with data
8. **Documentation** - Capture learnings to prevent recurrence

Remember: The goal isn't just to fix today's problem - it's to prevent tomorrow's.

21.18 Review Questions

Question 1: What is the difference between a symptom, immediate cause, and root cause? Provide an example of each for a machine that is producing parts with excessive surface roughness.

Answer:

- **Symptom**: The observable effect of the problem
  - *Example: Parts have Ra surface roughness of 3.2 m when specification is 1.6 m*
- **Immediate cause**: The direct, technical cause of the symptom
  - *Example: Cutting tool is worn beyond its effective life*
- **Root cause**: The fundamental reason that, if eliminated, prevents recurrence
  - *Example: No tool life monitoring system exists; operators judge tool changes by intuition rather than data*

Fixing the root cause (implementing tool life tracking) prevents the problem from recurring, whereas just replacing the tool (addressing immediate cause) means it will happen again.

Question 2: A food processing plant is experiencing frequent jams on their packaging line. Perform a 5 Whys analysis given the following information: The jam occurs at the carton loading station, started last week, happens 4-5 times per shift, and the maintenance team found the timing of the pusher mechanism is off.

Answer:

1. **Problem**: Carton loading station jams 4-5 times per shift

2. **Why #1**: Cartons are not in position when pusher activates → Timing mismatch
3. **Why #2**: Pusher timing is off by approximately 0.3 seconds
4. **Why #3**: Timing adjustment was changed during last week's maintenance
5. **Why #4**: Technician didn't have correct specification for timing setting
6. **Why #5**: Maintenance procedure doesn't include timing specifications

Root Cause: Maintenance procedures lack critical parameter specifications

Corrective Actions: - Update maintenance procedure with correct timing specification (180ms ± 10ms) - Restore timing to correct setting immediately - Review other procedures for missing specifications - Add parameter verification checklist to PM activities

Question 3: Create a fishbone diagram outline (list the potential causes under each 6M category) for the problem "CNC machine producing parts 0.05mm oversize."

Answer:

MAN (People) - Operator entered wrong offset - Incorrect setup procedure followed - Inadequate training on new program - Fatigue/distraction during setup

MACHINE (Equipment) - Spindle thermal growth - Ballscrew wear/backlash - Tool holder runout - Axis positioning error

MATERIAL - Material hardness variation - Different lot from supplier - Material springback - Incorrect material grade

METHOD (Process) - Wrong tool offset entered - Program error in G-code - Incorrect cutting parameters - Missing roughing pass

MEASUREMENT - Gauge out of calibration - Wrong measurement technique - Temperature affecting measurement - Incorrect datum reference

MOTHER NATURE (Environment) - Shop temperature variation - Coolant temperature change - Vibration from nearby equipment - Humidity affecting gauges

Question 4: Given the following defect data from a sheet metal stamping operation, create a Pareto analysis and identify the vital few defects to focus on.

Defect Type	Count
Scratches	89
Burrs	156
Wrinkles	45
Splits/Cracks	28
Dimensional	112

Defect Type	Count
Surface Dents	67
Missing Features	18

Answer:

```
# Create and analyze the data
defects <- data.frame(
  Type = c("Burrs", "Dimensional", "Scratches", "Surface Dents",
           "Wrinkles", "Splits/Cracks", "Missing Features"),
  Count = c(156, 112, 89, 67, 45, 28, 18)
)

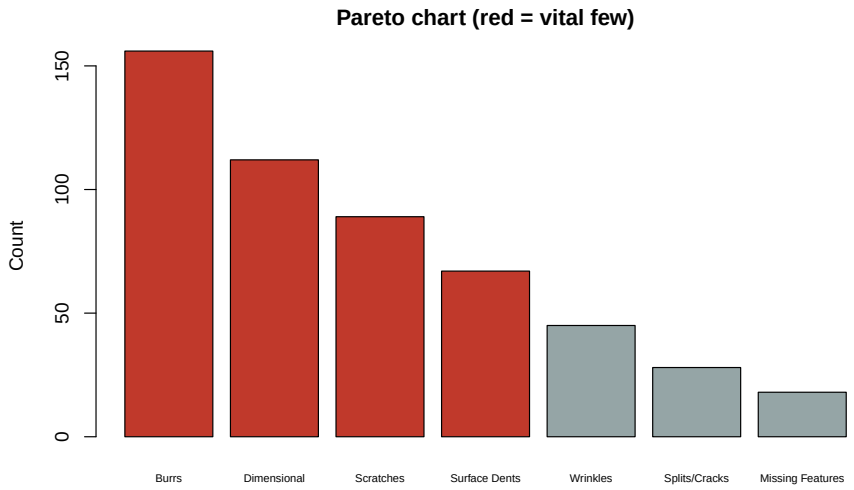
# Sort by count descending
defects <- defects[order(-defects$Count),]

# Calculate percentages
defects$Percent <- defects$Count / sum(defects$Count) * 100
defects$Cumulative <- cumsum(defects$Percent)

# Identify vital few
vital_few <- defects$Type[defects$Cumulative <= 80 |
                          c(TRUE, defects$Cumulative[-nrow(defects)] < 80)]

cat(sprintf("Vital few (about 80%%): %s\n", paste(vital_few, collapse = ", ")))
for (i in seq_len(nrow(defects))) {
  cat(sprintf("  %s  %g (%.1f%%, cumulative %.1f%%)\n", defects$Type[i], defects$Count[i],
              defects$Percent[i], defects$Cumulative[i]))
}

par(mar = c(5, 5, 3, 1))
barplot(defects$Count, names.arg = defects$Type, cex.names = 0.55,
        col = ifelse(defects$Type %in% vital_few, "#C0392B", "#95A5A6"),
        ylab = "Count", main = "Pareto chart (red = vital few)")
```




```
Vital few (about 80%): Burrs, Dimensional, Scratches, Surface Dents
  Burrs  156 (30.3%, cumulative 30.3%)
  Dimensional  112 (21.7%, cumulative 52.0%)
  Scratches  89 (17.3%, cumulative 69.3%)
  Surface Dents  67 (13.0%, cumulative 82.3%)
  Wrinkles  45 (8.7%, cumulative 91.1%)
  Splits/Cracks  28 (5.4%, cumulative 96.5%)
  Missing Features  18 (3.5%, cumulative 100.0%)
```

The **vital few** are Burrs, Dimensional errors, and Scratches - addressing these three defect types would address approximately 69% of all defects. Adding Surface Dents would bring the total to 82%.

Question 5: What are the key differences between verification and validation in the context of corrective actions?

Answer:

Aspect	Verification	Validation
Question asked	Did we implement the action correctly?	Did the action solve the problem?
Focus	Process compliance	Results/outcomes
Timing	During/immediately after implementation	Ongoing monitoring after implementation
Methods	Audits, checklists, inspections	Data analysis, trend charts, SPC

Aspect	Verification	Validation
Example	Confirm new sensor was installed per specification	Measure defect rate over 30 days to confirm reduction

Verification confirms the action was done right. **Validation** confirms we did the right action.

Both are required - an action can be implemented perfectly (verified) but still not solve the problem (not validated).

Question 6: An aerospace manufacturer has a requirement to use the 8D methodology for customer complaints. What happens in D3 (Containment) and why is it critical?

Answer:

D3 Containment Actions protect the customer while the root cause investigation continues. Key activities include:

1. **Immediate actions**:
   - Quarantine suspect inventory (tagged, segregated)
   - 100% inspection of suspect lots
   - Increase monitoring at subsequent operations
   - Sort parts at customer location if already shipped
2. **Documentation**:
   - Identify affected lot numbers, date codes, serial numbers
   - Quantify the scope of potential exposure
   - Document all containment actions taken
3. **Communication**:
   - Notify customer of actions being taken
   - Alert downstream operations
   - Inform suppliers if incoming material involved

Why critical: - Prevents additional defective product from reaching the customer - Limits the scope of the problem (cost, liability) - Demonstrates responsiveness to customer - Buys time for proper root cause investigation - In aerospace, safety-of-flight concerns make containment essential

Without effective containment, the problem continues while you investigate, increasing customer impact and costs exponentially.

Question 7: Why might a team's 5 Whys analysis stop too early? What are signs that you haven't reached the true root cause?

Answer:

Reasons teams stop too early: 1. **Blame arrives** - Analysis ends at "operator error" without asking why the error occurred 2. **Comfort zone** - Team

stops when they reach a cause they can easily fix 3. **Time pressure** - Rush to implement a fix, skipping deeper analysis 4. **Technical limit** - Team lacks expertise to go deeper into certain areas 5. **Defensiveness** - Reaching causes that implicate management or systems creates resistance

Signs you haven't reached root cause: 1. **Recurrence** - If the fix is "more training" or "be more careful," problem will return 2. **Human-based answer** - Root causes should typically be systems/processes, not individuals 3. **No prevention mechanism** - If fixing this cause doesn't prevent future occurrences, keep asking why 4. **Can still ask "why?"** - If there's a logical next level to investigate 5. **Administrative fix only** - If the solution is just a procedure or checklist with no engineering controls

Example of stopping too early: - Stopping at: "Operator set wrong parameter" - Should continue: Why? → No parameter sheet at workstation → Why? → No standard work documentation exists → Root cause: Missing standardized work process

The true root cause typically relates to systems, processes, or management decisions rather than individual actions.

Question 8: Describe how you would use Pareto analysis in a stratified manner to drill down into a quality problem.

Answer:

Stratified Pareto Analysis involves creating multiple levels of Pareto charts, progressively drilling down into the vital few categories.

Example: High scrap rate at a machining shop

Level 1: Scrap by defect type 1. Create Pareto of all defect types 2. Result: "Dimensional errors" is 45% of all scrap - investigate this first

Level 2: Dimensional errors by machine 1. Create Pareto of dimensional errors broken down by machine 2. Result: Machine #4 accounts for 60% of dimensional errors

Level 3: Machine #4 dimensional errors by feature 1. Create Pareto of which dimensions are failing on Machine #4 2. Result: Bore diameter is 70% of Machine #4 dimensional issues

Level 4: Bore diameter issues by shift/operator 1. Create Pareto by shift 2. Result: Evenly distributed - not operator-dependent (machine issue confirmed)

Final Analysis: Investigate boring operation on Machine #4 specifically - likely tool wear, spindle issue, or program error.

The stratified approach efficiently narrows focus from "we have a scrap problem" to "boring operation on Machine #4 has a systematic error."

Question 9: A defense contractor needs to analyze a safety-critical failure. Why might Fault Tree Analysis (FTA) be preferred over simpler methods like 5 Whys?

Answer:

FTA advantages for safety-critical analysis:

1. **Handles complexity**:
   - Safety-critical systems often have multiple redundancies and complex failure paths
   - FTA can model combinations of events that must occur together (AND gates) or alternatively (OR gates)
   - 5 Whys follows single linear path; FTA captures parallel paths
2. **Quantitative capability**:
   - FTA allows probability calculations for top event
   - Can identify minimum cut sets (smallest combinations that cause failure)
   - Supports reliability and safety integrity level (SIL) calculations
   - Defense/aerospace require quantitative risk assessment
3. **Documentation requirements**:
   - Regulatory bodies (FAA, DoD) often require FTA for critical systems
   - Provides auditable, traceable analysis
   - Standard symbols and methodology
4. **Identifies hidden failures**:
   - Can reveal single points of failure
   - Shows where redundancy is (or isn't) effective
   - Identifies common cause failures that defeat redundancy
5. **Design improvement**:
   - Visual representation helps identify where to add safeguards
   - Sensitivity analysis shows which basic events most affect risk
   - Supports design reviews and safety cases

Example: A missile guidance system failure - 5 Whys might identify one cause path - FTA would show all possible failure paths, calculate overall failure probability, and identify that two specific sensor failures AND a software error all occurring together could cause the top event, even though each alone is unlikely.

21.19 References

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6. Ishikawa, K. (1990). *Introduction to Quality Control*. 3A Corporation.

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Chapter 22

Industrial Automation Fundamentals

22.1 Learning Objectives

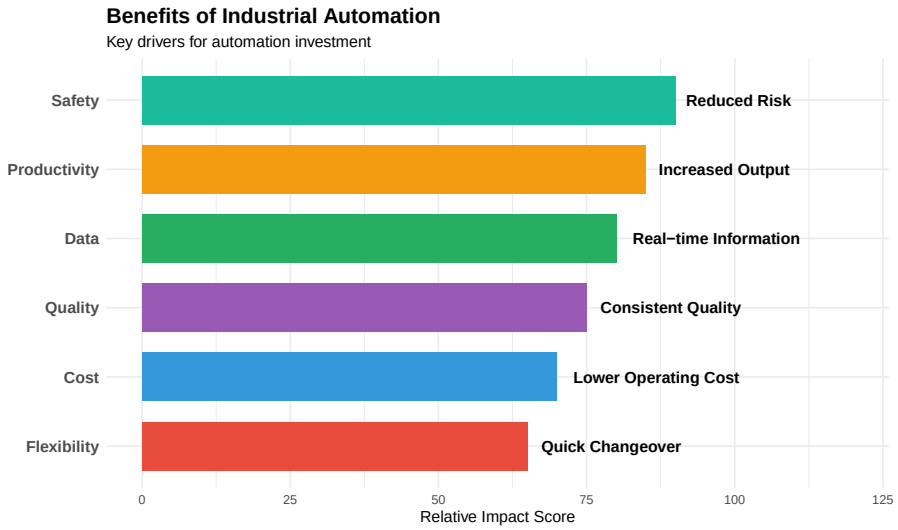
After completing this chapter, you will be able to:

1. Define industrial automation and explain its role in modern manufacturing
2. Distinguish between fixed, programmable, and flexible automation
3. Describe the architecture and function of Programmable Logic Controllers (PLCs)
4. Identify common industrial sensors and their applications
5. Explain motor control fundamentals including VFDs and servo systems
6. Understand industrial communication protocols and networking
7. Describe the role of HMI and SCADA in process monitoring
8. Apply safety principles to automated systems
9. Explain Industry 4.0 concepts and the Industrial Internet of Things (IIoT)

22.2 Introduction to Industrial Automation

Industrial automation is the use of control systems, machinery, and information technologies to handle processes and machinery in an industry, replacing human intervention where possible to increase efficiency, quality, and safety.

22.2.1 Why Automate?



22.2.2 Automation in Key Industries

Table 22.1: Automation Applications by Industry

Industry	Application	Technology	Benefit
Automotive			
Automotive	Robotic welding and assembly	6-axis robots, PLCs	Consistent w
Automotive	Automated paint systems	Conveyors, spray robots	Uniform coat
Automotive	Vision inspection	Machine vision, AI	100% inspect
Food & Beverage			
Food & Beverage	Filling and packaging lines	Servo drives, sensors	High speed,
Food & Beverage	Pasteurization control	PLCs, temperature control	Food safety,
Food & Beverage	Sorting and grading	Vision systems, conveyors	Quality sorti
Aerospace/Defense			
Aerospace/Defense	CNC precision machining	CNC, CMM integration	Tight toleran
Aerospace/Defense	Automated testing systems	Automated test equipment	Comprehensi
Aerospace/Defense	Clean room automation	Robotics, HEPA systems	Contaminati

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22.3 Types of Automation

Industrial automation systems can be classified based on their flexibility and programming capability.

22.3.1 Automation Classification

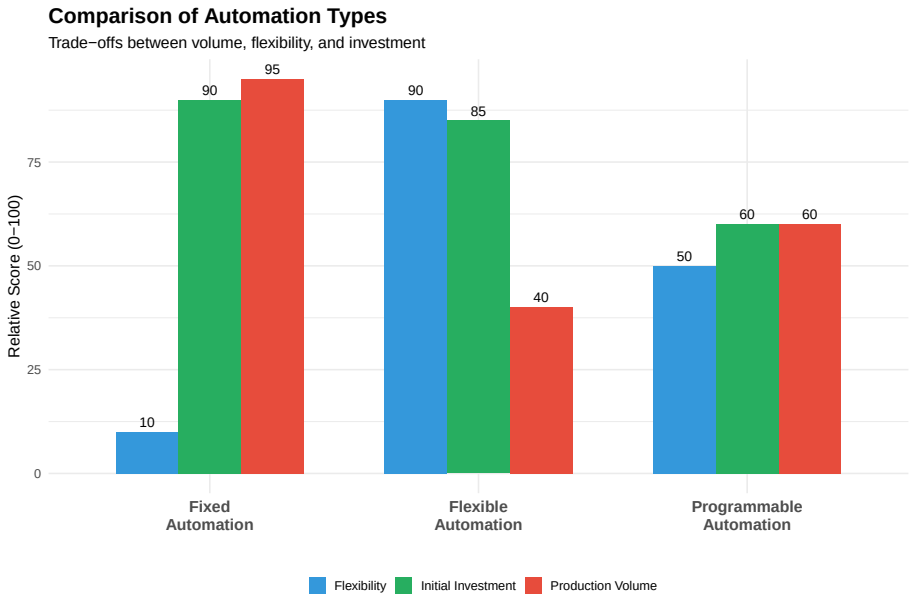
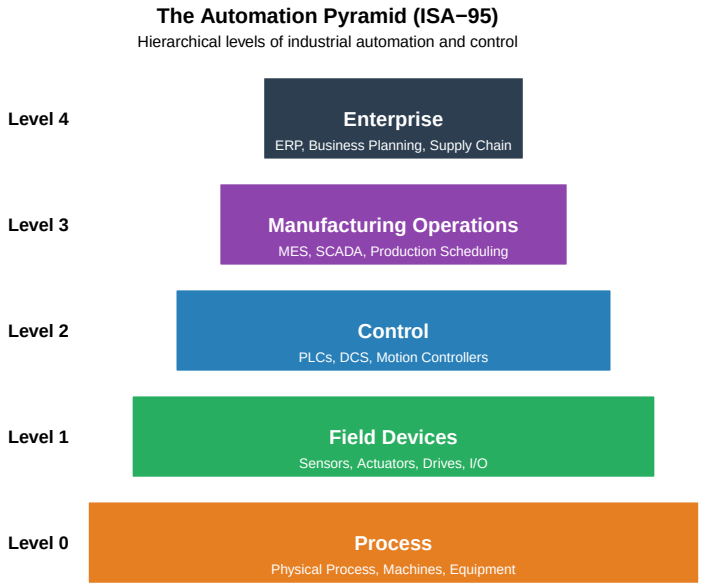


Table 22.2: Comparison of Automation Types

	Fixed Automation	Programmable Automation
Definition	Hard-wired, dedicated equipment for single product	Equipment can be reprogrammed f
Product Variety	Single product or very similar variants	Batches of different products
Production Volume	Very high (millions of units)	Medium to high
Changeover	Difficult, expensive, time-consuming	Requires reprogramming and setup
Initial Investment	Very high	High
Unit Cost	Very low per unit	Medium
Typical Equipment	Transfer lines, dedicated assembly machines	CNC machines, PLCs, industrial ro
Best For	Automotive components, fasteners, bottles	Batch manufacturing, job shops

22.3.2 The Automation Pyramid



Understanding the Pyramid Levels

Level 0 - Process: The actual physical equipment, machines, and processes being controlled. This includes conveyors, motors, valves, tanks, and the product being manufactured.

Level 1 - Field Devices: Sensors that measure process variables (temperature, pressure, flow, position) and actuators that affect the process (motors, valves, solenoids, drives).

Level 2 - Control: The "brain" of automation - PLCs, DCS, and motion controllers that execute control logic, process sensor inputs, and command actuators. Also includes HMI for operator interaction.

Level 3 - Manufacturing Operations: MES (Manufacturing Execution Systems), SCADA, production scheduling, quality management, and maintenance management. Bridges plant floor and business systems.

Level 4 - Enterprise: ERP systems, business intelligence, supply chain management, financial systems. Makes business decisions based on plant floor data.

Key Principle: Data flows up (process information), commands flow down (control directives).

22.3.3 Try it yourself: place it on the pyramid

Nine pieces of a plant's automation. Place each one on the correct level of the automation pyramid.

Which level of the automation pyramid?

Place each item on its level of the pyramid.

Categories: Level 0: Process / Level 1: Field Devices / Level 2: Control / Level 3: Manufacturing Operations / Level 4: Enterprise

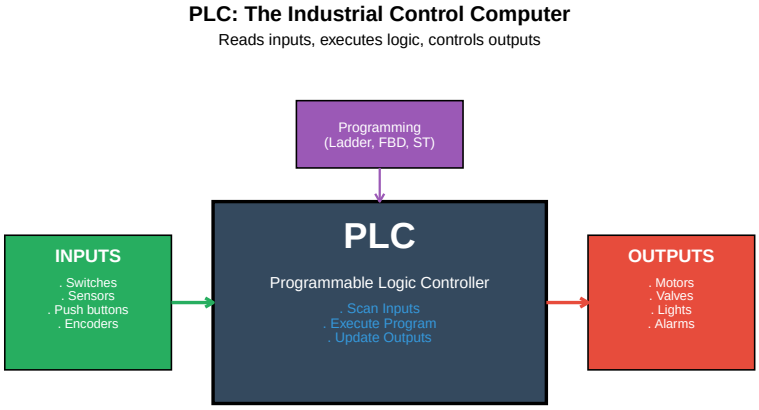
1. A 500-litre mixing tank and the product inside it. _____
2. A conveyor carrying cartons to the palletizer. _____
3. An inductive proximity sensor detecting a part at a station. _____
4. A solenoid valve that opens the filling nozzle. _____
5. A PLC running the ladder logic for a packaging machine. _____
6. A DCS controller holding reactor temperature at setpoint. _____
7. An MES releasing work orders to each line and tracking what was built. _____
8. A SCADA overview screen showing the status of every line in the plant. _____
9. The ERP system taking a customer order and planning material purchases. _____

Answers: 1 Level 0: Process; 2 Level 0: Process; 3 Level 1: Field Devices; 4 Level 1: Field Devices; 5 Level 2: Control; 6 Level 2: Control; 7 Level 3: Manufacturing Operations; 8 Level 3: Manufacturing Operations; 9 Level 4: Enterprise.

22.4 Programmable Logic Controllers (PLCs)

The **PLC (Programmable Logic Controller)** is the workhorse of industrial automation, providing reliable, real-time control of machines and processes.

22.4.1 What is a PLC?



22.4.2 PLC Hardware Components

Table 22.3: PLC Hardware Componer

Component	Function
CPU (Processor)	Executes the control program, manages memory, coordinates all modu
Power Supply	Converts AC power to DC voltages required by PLC components
Input Modules	Interface field devices (sensors, switches) to CPU; converts signals to d
Output Modules	Interface CPU to field devices (motors, valves); converts digital to pow
Communication Modules	Enable networking: Ethernet/IP, Profinet, Modbus, DeviceNet
Programming Device	Laptop/PC with programming software for creating and downloading

22.4.3 The PLC Scan Cycle

PLCs operate in a continuous scan cycle:

PLC Scan Cycle

1. **1. Read Inputs** – Read all input module states into the input image table (~1 ms)
2. **2. Execute Program** – Execute ladder logic / program using input image data (~5-20 ms)
3. **3. Update Outputs** – Write output image table values to physical outputs (~1 ms)
4. **4. Housekeeping** – Communications, diagnostics, self-testing (~1 ms)

The cycle repeats continuously (about 10-50 ms in total)

Your turn — change the scan time components below.

Show the live R code behind this calculator

```
#| label: scan-time-calc
# Scan Time Calculation Example
input_scan_time <- 1.2    # ms
program_execution <- 15.5  # ms (depends on program size)
output_update <- 1.0       # ms
housekeeping <- 2.3        # ms

total_scan_time <- input_scan_time + program_execution + output_update + housekeeping

cat(sprintf("Total scan time: %.1f ms (%.0f scans/sec)\n", total_scan_time, 1000 / total_scan_tim
cat(sprintf("  Input scan           %.1f ms\n", input_scan_time))
cat(sprintf("  Program execution    %.1f ms\n", program_execution))
cat(sprintf("  Output update        %.1f ms\n", output_update))
cat(sprintf("  Housekeeping         %.1f ms\n", housekeeping))
```

22.4.3.1 Try it yourself: order the scan cycle

A PLC repeats the same four steps on every scan. Put them in order, starting with the step that reads the field.

The PLC scan cycle

Number these in the correct order:

- _____ Housekeeping: communications and self-diagnostics
- _____ Input scan: read all inputs into memory
- _____ Output update: write the results to the outputs
- _____ Program execution: solve the logic, rung by rung

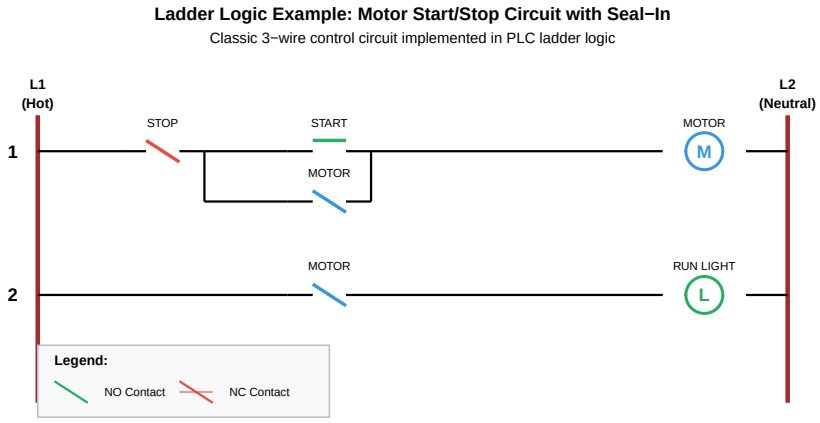
Answers, in the order listed: 4, 1, 3, 2.

22.4.4 PLC Programming Languages (IEC 61131-3)

Table 22.4: IEC 61131-3 PLC Programming Lang

Language	Type	Description
Ladder Diagram (LD)	Graphical	Resembles electrical relay circuits; most common in discrete
Function Block Diagram (FBD)	Graphical	Uses function blocks connected by lines; good for analog/pro
Structured Text (ST)	Textual	High-level language similar to Pascal; powerful for complex c
Instruction List (IL)	Textual	Low-level assembly-like language; rarely used today
Sequential Function Chart (SFC)	Graphical	Sequence/state-based programming; excellent for batch proce

22.4.5 Ladder Logic Example



How the Start/Stop Circuit Works

Rung 1 - Motor Control: 1. Power flows from L1 through the NC (Normally Closed) STOP button 2. If STOP is pressed, the circuit breaks and motor stops 3. START is NO (Normally Open) - pressing it allows power to flow to the MOTOR coil 4. When MOTOR energizes, its NO contact in the parallel branch closes 5. This "seals in" the circuit - motor stays running even after START is released 6. To stop, press STOP which breaks the seal-in circuit

Rung 2 - Run Indicator: 1. When MOTOR coil is energized, the MOTOR contact closes 2. This allows power to flow to the RUN LIGHT output 3. Light is ON whenever motor is running

This is called a "3-wire control" circuit - it provides: - Low voltage release protection (motor won't restart after power failure) - Maintained contact operation (don't need to hold START button)

22.4.5.1 Try it yourself: step through the seal-in rung

Be the PLC. The table gives the state of the two pushbuttons at each scan (1 = pressed) for the start/stop rung above. Work out whether the MOTOR output is ON (1) or OFF (0) at the end of every scan. The motor starts OFF.

On each scan the rung is true when STOP is not pressed **and** either START is pressed **or** the motor was already on from the previous scan.

Seal-in rung, scan by scan

Enter 1 (ON) or 0 (OFF) for the MOTOR output at the end of each scan.

Scan	START	STOP	MOTOR
1	0	0	
2	1	0	
3	0	0	
4	0	0	
5	0	1	
6	0	0	
7	1	1	
8	0	0	

Show the worked solution

Scan	START	STOP	MOTOR	Why
1	0	0	0	Nothing pressed, motor was off
2	1	0	1	START pressed: rung true
3	0	0	1	START released, but the MOTOR contact seals in
4	0	0	1	Still sealed in
5	0	1	0	STOP pressed: the rung is broken
6	0	0	0	The seal was broken, so the motor stays off
7	1	1	0	Both pressed: STOP is in series, so it wins
8	0	0	0	Motor was off and START is released

Scans 3 and 4 are the whole point of the seal-in branch: the operator has let go of START, and the motor's own contact keeps the rung true. Scan 7 shows why STOP is wired in series with everything else. Whatever else is happening, pressing STOP drops the motor out.

22.5 Industrial Sensors

Sensors are the "eyes and ears" of automation systems, providing feedback about the process to the control system.

22.5.1 Sensor Classification

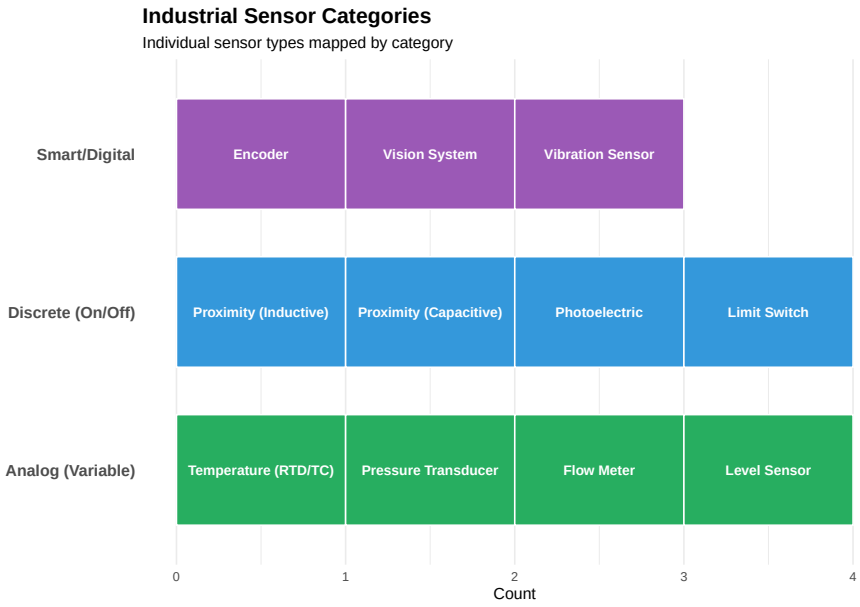
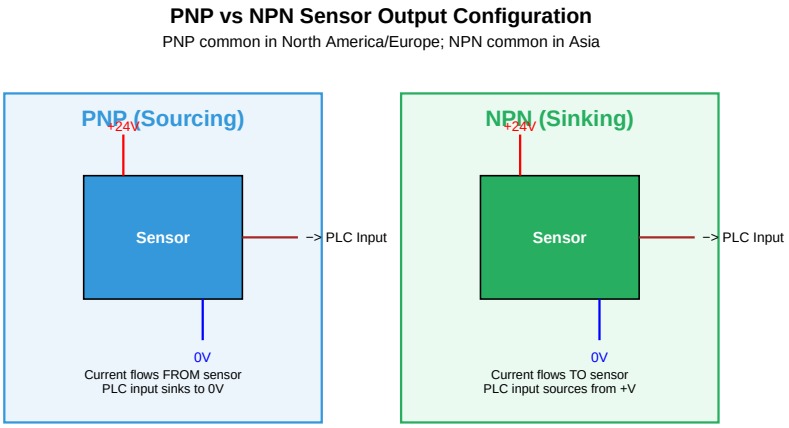


Table 22.6: Common Industrial Sensor Specifications

Sensor	Principle	Range	Adv
Inductive Proximity	Eddy current change in oscillating field	2-40mm typical	Du
Capacitive Proximity	Capacitance change with target approach	2-25mm typical	Det
Photoelectric	Light beam interrupted or reflected	0.1-30m	Lon
RTD (PT100)	Resistance changes with temperature	-200 to 850°C	Acc
Thermocouple	Voltage generated at junction of dissimilar metals	-200 to 2300°C	Wi
Pressure Transducer	Diaphragm deflection converted to electrical signal	0-10000 psi	Acc
Incremental Encoder	Optical/magnetic pulses per revolution	100-10000 PPR	Sim
Absolute Encoder	Unique code for each position	12-25 bit	Pos

22.5.2 Sensor Wiring: PNP vs NPN



22.6 Actuators and Motor Control

Actuators convert control signals into physical motion or action.

22.6.1 Types of Industrial Actuators

Table 22.7: Industrial Actuator Types and Applications

Category	Type	Application	Control
Electric			
Electric Motors	AC Induction Motor	Conveyors, pumps, fans, compressors	VFD for variable speed
Electric Motors	Servo Motor	Precise positioning, robotics, CNC	Servo drive with feedback
Electric Motors	Stepper Motor	Indexing, low-speed positioning	Stepper drive (open or close
Electric Motors	DC Motor	Battery vehicles, legacy systems	PWM, SCR drive
Pneumatic			
Pneumatic	Cylinder	Clamping, pushing, lifting	Solenoid valves, proportiona
Pneumatic	Rotary Actuator	Rotating grippers, indexing	Solenoid valves
Hydraulic			
Hydraulic	Cylinder	Heavy lifting, presses	Proportional/servo valves
Hydraulic	Motor	Heavy machinery, mobile equipment	Proportional/servo valves

22.6.2 Variable Frequency Drives (VFDs)

A **Variable Frequency Drive (VFD)** controls AC motor speed by varying the frequency and voltage of the power supplied to the motor.

Variable Frequency Drive (VFD) Block Diagram

1. **AC Input** – 60 Hz, fixed frequency
2. **Rectifier** – AC → DC
3. **DC Bus** – Filter & capacitors
4. **Inverter** – DC → AC (PWM)
5. **AC Motor** – Variable speed

Teaching demo — change the motor parameters or drag the frequency slider below.

Show the live R code behind this calculator

```
#| label: vfd-calculations
# VFD Speed and Torque Calculations

# Motor nameplate data
poles <- 4
rated_voltage <- 460      # V
rated_frequency <- 60     # Hz
rated_rpm <- 1750
selected_frequency <- 30  # Hz - frequency to check

slip_rpm <- (120 * rated_frequency / poles) - rated_rpm
sync_speed_rated <- 120 * rated_frequency / poles
vhz_ratio <- rated_voltage / rated_frequency

speed_at <- function(f) (120 * f / poles) - slip_rpm  # assumes constant slip (approxi

frequencies <- seq(5, 75, length.out = 150)
par(mar = c(4, 5, 2, 1), cex.main = 1.3, cex.lab = 1.2, cex.axis = 1.1)
plot(frequencies, speed_at(frequencies), type = "l", lwd = 4, col = "#2c3e50", xlim =
     xlab = "Frequency (Hz)", ylab = "Motor speed (RPM)",
     main = "Motor Speed vs. VFD Frequency", las = 1)
abline(v = selected_frequency, lty = 2, col = "#bdc3c7")
points(selected_frequency, speed_at(selected_frequency), pch = 19, cex = 2.2, col = "#

cat(sprintf("At %d Hz: %.0f RPM (voltage should be %.0f V)\n",
            selected_frequency, speed_at(selected_frequency), selected_frequency * vhz
cat(sprintf("  Synchronous speed @ rated Hz  %.0f RPM\n", sync_speed_rated))
cat(sprintf("  Slip                          %.0f RPM (%.1f%%)\n", slip_rpm, slip_rpm
cat(sprintf("  V/Hz ratio                    %.2f V/Hz\n", vhz_ratio))
```

22.6.3 VFD Benefits

Table 22.8: Variable Frequency Drive Benefits

Benefit	Description	Typical
Energy Savings	Match motor speed to load requirements; huge savings on fans/pumps	20-50%
Soft Start/Stop	Ramp up/down gradually; eliminates inrush current (6-8x normal)	Reduce
Speed Control	Precise speed control from 0-100%+ of base speed	N/A
Process Control	Maintain constant pressure, flow, or tension	Improv
Reduced Mechanical Stress	Reduced wear on belts, gears, couplings from smooth acceleration	Extend
Power Factor	VFD presents near-unity power factor to supply	Avoid

22.6.4 Servo Systems

Servo System: Closed-Loop Motion Control

1. **Motion Controller**
2. **Servo Drive**
3. **Servo Motor**
4. **Encoder**
5. **Load**

Position / velocity feedback from the encoder returns to the servo drive

Table 22.9: VFD vs. Servo System Comparison

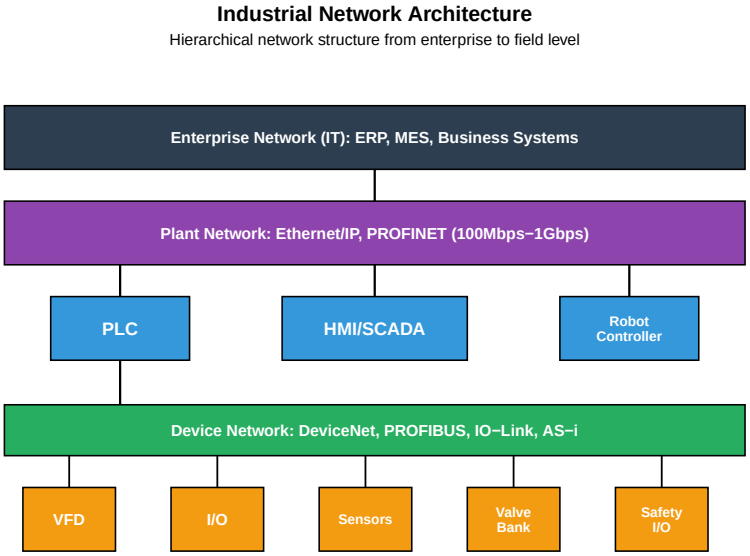
Characteristic	VFD + Induction Motor	Servo System
Control Type	Open loop (usually)	Closed loop (always)
Feedback	Optional encoder	High-resolution encoder required
Positioning Accuracy	±1-5% of speed	±0.01° or better
Dynamic Response	Moderate (100-500ms)	Fast (1-10ms)
Torque at Zero Speed	Limited (10-20%)	100% continuous
Cost	Lower ($500-5000)	Higher ($2000-20000)
Typical Application	Fans, pumps, conveyors	Robotics, CNC, packaging

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22.7 Industrial Communication Networks

Modern automation systems rely on communication networks to connect devices, share data, and enable coordinated control.

22.7.1 Network Architecture



22.7.2 Common Industrial Protocols

Table 22.10: Common Industrial Communication Protocols

Protocol	Developer	Medium	Speed	Application
Ethernet/IP	ODVA (Rockwell)	Ethernet	100Mbps/1Gbps	General automation, I/C
PROFINET	PI (Siemens)	Ethernet	100Mbps/1Gbps	Siemens ecosystem, moti
Modbus TCP	Modicon/Schneider	Ethernet	100Mbps	Simple, open, legacy syst
EtherCAT	Beckhoff	Ethernet	100Mbps	High-speed motion, prec
DeviceNet	ODVA	CAN-based	500kbps	I/O, drives (legacy)
PROFIBUS	PI (Siemens)	RS-485	12Mbps	I/O, drives (legacy)
IO-Link	IO-Link Consortium	Point-to-point	230.4kbps	Smart sensors
OPC UA	OPC Foundation	Ethernet/Any	Varies	IT/OT integration, IIoT

Protocol Selection Guide

Choose Ethernet/IP if: - Using Allen-Bradley/Rockwell equipment - Need general-purpose industrial Ethernet - Require seamless integration with IT networks

Choose PROFINET if: - Using Siemens equipment - Need deterministic communication for motion control - Require IRT (Isochronous Real-Time) performance

Choose EtherCAT if: - Need highest speed/lowest latency - High-precision motion control - Many axes of coordinated motion

Choose Modbus TCP if: - Simple, low-cost solution needed - Connecting legacy equipment - Open protocol preference (no licensing)

Choose OPC UA if: - Need IT/OT convergence - IIoT/Industry 4.0 implementation - Secure, platform-independent communication

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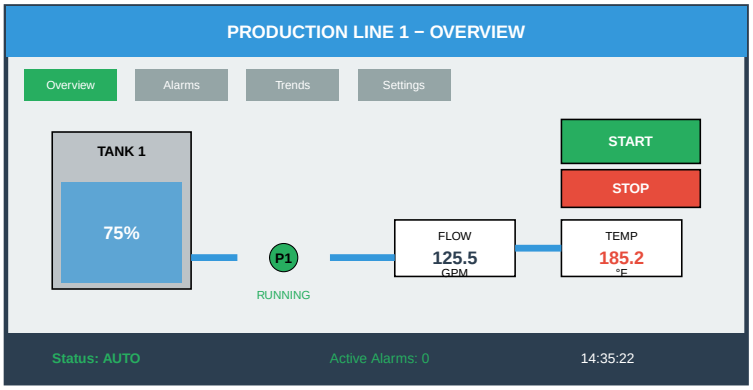
22.8 Human-Machine Interface (HMI) and SCADA

22.8.1 HMI Overview

An **HMI (Human-Machine Interface)** provides operators with a visual interface to monitor and control automated processes.

HMI Screen Example

Typical operator interface showing process status and controls



22.8.2 HMI Design Best Practices

Table 22.11: HMI Design Best Practices

Principle	Best Practice
Situational Awareness	Show abnormal conditions prominently; operator should know status at a glance
Alarm Management	Prioritize alarms (critical/warning/info); avoid alarm floods; require acknowledgment
Navigation	Consistent layout; max 3 clicks to any screen; clear hierarchy
Color Usage	Use color sparingly for meaning; gray for normal; avoid red/green for critical info (colo

Data Display	Show trends, not just values; use appropriate precision; include units
Controls	Confirm destructive actions; use interlocks; provide feedback

22.8.3 SCADA Systems

SCADA (Supervisory Control and Data Acquisition) systems provide centralized monitoring and control across multiple locations or processes.

Table 22.12: SCADA System Components

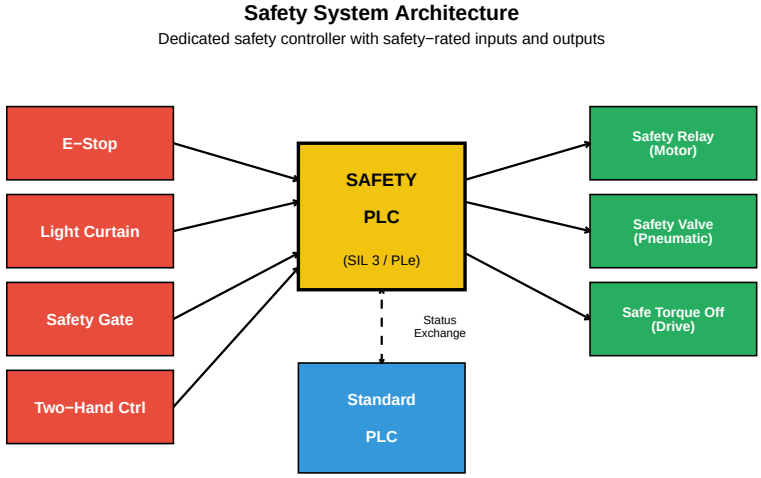
Component	Function
MTU (Master Terminal Unit)	Central server running SCADA software; processes data, execute
RTU (Remote Terminal Unit)	Field device that collects data from sensors and sends to MTU
Communication Network	Links MTU to RTUs; can be radio, cellular, satellite, fiber
HMI/Workstations	Operator interface for monitoring and control
Historian Database	Stores historical process data for trending and analysis
Alarm Server	Manages alarm generation, notification, and logging

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22.9 Safety in Automation

Safety is paramount in automated systems. Safety systems protect personnel and equipment from harm.

22.9.1 Safety System Architecture



22.9.2 Safety Integrity Levels (SIL)

Table 22.13: Safety Integrity Levels (IEC 61508)

SIL Level	PFD (avg)	Risk Reduction Factor	Risk Level	Example
SIL 1	10^{-2} to $<10^{-1}$	10-100	Minor injury possible; first line of defense	Warning
SIL 2	10^{-3} to $<10^{-2}$	100-1,000	Serious injury possible; general industrial	Machine
SIL 3	10^{-3} to $<10^{-3}$	1,000-10,000	Death or severe injury; process industry standard	Emergen
SIL 4	10^{-4} to $<10^{-4}$	10,000-100,000	Catastrophic; nuclear, aviation	Nuclear

---

22.10 Industry 4.0 and IIoT

Industry 4.0 represents the fourth industrial revolution, characterized by the integration of digital technologies into manufacturing.

22.10.1 Industry 4.0 Technologies



22.10.2 IIoT Architecture

Table 22.14: IIoT Architecture Layers

Layer	Components	Function
Edge/Device Layer	Sensors, actuators, PLCs, gateways, edge devices	Collect data, lo
Network Layer	Industrial Ethernet, Wi-Fi, 5G, LoRaWAN, MQTT	Secure data tra
Platform Layer	Cloud/on-premise servers, databases, data lakes	Store, process,
Application Layer	Analytics dashboards, AI/ML models, business applications	Derive insights

22.10.3 Predictive Maintenance Example

```
# Simulating predictive maintenance with vibration data
set.seed(42)

# Generate 30 days of vibration readings
days <- 1:30
baseline_vibration <- 2.5  # mm/s RMS (normal)

# Simulate gradual bearing degradation
vibration <- baseline_vibration +
  0.05 * days +           # Gradual increase
  0.9 * exp((days - 25)/5) * (days > 20) +  # Accelerating failure
  rnorm(30, 0, 0.2)       # Normal variation

# Alert thresholds
warning_threshold <- 4.0
alarm_threshold <- 6.0

# Find when thresholds crossed
warning_day <- min(which(vibration > warning_threshold))
alarm_day <- min(which(vibration > alarm_threshold))

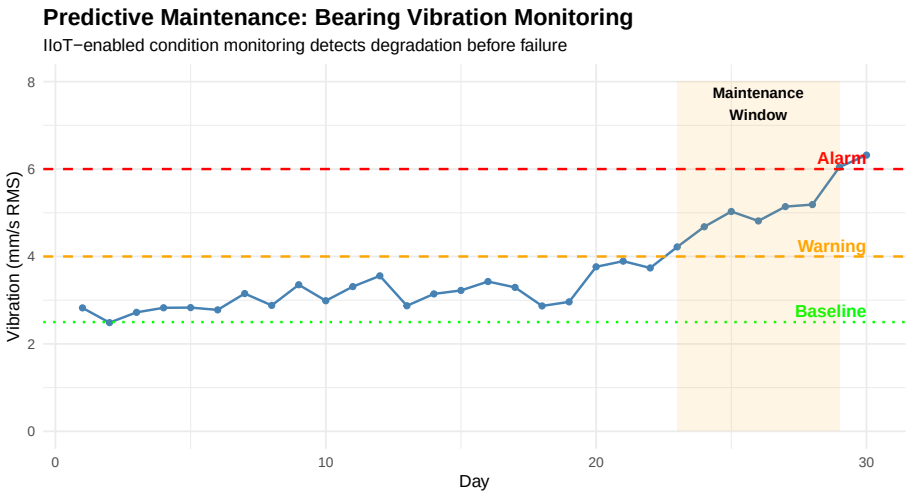
cat("Predictive Maintenance Analysis:\n")
cat("                  \n")
cat("Baseline vibration:", baseline_vibration, "mm/s RMS\n")
cat("Warning threshold:", warning_threshold, "mm/s RMS\n")
cat("Alarm threshold:", alarm_threshold, "mm/s RMS\n\n")
cat("Warning triggered on day:", warning_day, "\n")
cat("Alarm triggered on day:", alarm_day, "\n")
cat("Days of warning before alarm:", alarm_day - warning_day, "\n")
cat("\nRecommendation: Schedule bearing replacement before day", alarm_day, "\n")
```

```
Predictive Maintenance Analysis:

Baseline vibration: 2.5 mm/s RMS
Warning threshold: 4 mm/s RMS
Alarm threshold: 6 mm/s RMS
```

```
Warning triggered on day: 23
Alarm triggered on day: 29
Days of warning before alarm: 6

Recommendation: Schedule bearing replacement before day 29
```



22.11 Video Resources

22.11.1 Introduction to PLCs

22.11.2 Industrial Networks Explained

22.12 Summary

Industrial automation integrates multiple technologies to improve manufacturing efficiency, quality, and safety:

1. **Automation types** range from fixed (high-volume, single product) to flexible (low-volume, high variety)
2. **PLCs** are the workhorses of industrial control, using scan cycles to process inputs and control outputs
3. **Sensors** provide feedback on process conditions (discrete, analog, and smart)
4. **Motor control** through VFDs and servo systems enables precise speed and position control

5. **Industrial networks** connect devices using protocols like Ethernet/IP, PROFINET, and Modbus
6. **HMI and SCADA** provide operator interface for monitoring and control
7. **Safety systems** use dedicated controllers and devices to protect personnel
8. **Industry 4.0** brings IIoT, cloud computing, and AI to enable smart manufacturing

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22.13 Review Questions

Question 1: Compare and contrast fixed, programmable, and flexible automation. Give an example application for each.

Answer:

Aspect	Fixed	Programmable	Flexible
Product variety	Single product	Batches of different products	High variety, mixed production
Volume	Very high	Medium-high	Low-medium
Changeover	Difficult, expensive	Hours to days	Minutes
Investment	Very high	High	Very high
Flexibility	None	Limited	High

Examples:

1. **Fixed Automation**: Automotive transfer line producing engine blocks
   - Same machining operations repeated millions of times
   - Dedicated stations optimized for specific operations
   - Changing product would require complete retooling
2. **Programmable Automation**: CNC machining job shop
   - Different parts programmed and run in batches
   - Significant setup time between batches
   - Equipment reprogrammable but not instantaneous
3. **Flexible Automation**: Robotic welding cell in aerospace
   - Different assemblies run with quick changeover
   - Robot program changes automatically based on part ID
   - Same cell handles multiple product variants

Question 2: Explain the PLC scan cycle and why scan time is important.

Answer:

The PLC Scan Cycle consists of four main phases executed continuously:

1. **Input Scan** (~1ms): Read all physical inputs and store values in the input image table

2. **Program Execution** (~5-20ms): Execute the user program (ladder logic, etc.) using input image data
3. **Output Update** (~1ms): Write output image table values to physical outputs
4. **Housekeeping** (~1-3ms): Communications, diagnostics, self-testing

Total scan time = Sum of all phases, typically 10-50ms

Why Scan Time Matters:

1. **Response Time**: An input change isn't acted upon until the next scan completes. If scan time is 50ms, worst-case response time is 50ms.

2. **Safety**: Fast-moving machinery requires short scan times for safety functions. A 100ms scan time with a motor at 1800 RPM means 3 shaft revolutions between scans.

3. **Application Limits**:

   - Motion control may need <1ms scan times (special motion PLCs)
   - High-speed counting needs special modules
   - Safety PLCs have guaranteed maximum scan times

4. **Program Size Impact**: Larger programs take longer to execute, increasing scan time. Optimization may be needed for time-critical applications.

5. **Communication Delays**: Network updates typically happen once per scan, affecting data freshness.

Question 3: What are the key differences between PNP and NPN sensor outputs? When would you use each?

Answer:

PNP (Sourcing): - Sensor output sources current TO the load - When activated, output connects to +V - Load is connected between sensor output and 0V - Current flows: +V → Sensor → Output → Load → 0V - Common in North America and Europe

NPN (Sinking): - Sensor output sinks current FROM the load - When activated, output connects to 0V (ground) - Load is connected between sensor output and +V - Current flows: +V → Load → Output → Sensor → 0V - Common in Asia (Japan)

Selection Criteria:

1. **PLC Input Type**:
   - Sinking PLC inputs work with PNP (sourcing) sensors
   - Sourcing PLC inputs work with NPN (sinking) sensors
2. **Regional Standards**:
   - North America: PNP is standard
   - Europe: PNP is standard
   - Japan/Asia: NPN is common

3. **Safety Considerations**:
   - PNP: A ground fault can cause false ON signal
   - NPN: A ground fault typically causes safe-off condition
   - For safety applications, consider sensor design carefully
4. **Existing Infrastructure**:
   - Match new sensors to existing system type
   - Converting between types requires different PLC input cards

Question 4: A VFD is controlling a 10 HP, 4-pole motor rated at 1750 RPM and 460V at 60Hz. Calculate the motor speed at 30Hz and 45Hz. What voltage should the VFD supply at these frequencies?

Answer:

```
# Motor parameters
poles <- 4
rated_voltage <- 460  # V
rated_frequency <- 60  # Hz
rated_rpm <- 1750

# Calculate synchronous speed and slip at rated conditions
sync_speed_60 <- 120 * rated_frequency / poles
slip_rpm <- sync_speed_60 - rated_rpm
slip_percent <- (slip_rpm / sync_speed_60) * 100

# V/Hz ratio
vhz_ratio <- rated_voltage / rated_frequency

# Calculate at 30Hz
freq_30 <- 30
sync_30 <- 120 * freq_30 / poles
speed_30 <- sync_30 - slip_rpm  # Assuming constant slip in RPM
voltage_30 <- freq_30 * vhz_ratio

# Calculate at 45Hz
freq_45 <- 45
sync_45 <- 120 * freq_45 / poles
speed_45 <- sync_45 - slip_rpm
voltage_45 <- freq_45 * vhz_ratio

cat(sprintf("Motor speed: %g RPM at %g Hz, %g RPM at %g Hz\n", speed_30, freq_30, speed
cat(sprintf("  Synchronous speed at %g Hz  %g RPM\n", rated_frequency, sync_speed_60))
cat(sprintf("  Slip at rated speed  %g RPM (%.1f%%)\n", slip_rpm, slip_percent))
cat(sprintf("  V/Hz ratio  %.2f V/Hz\n", vhz_ratio))
cat(sprintf("  At %g Hz: synchronous speed  %g RPM\n", freq_30, sync_30))
cat(sprintf("  At %g Hz: motor speed (approx)  %g RPM\n", freq_30, speed_30))
cat(sprintf("  At %g Hz: VFD output voltage  %g V\n", freq_30, voltage_30))
```

```
cat(sprintf("  At %g Hz: synchronous speed  %g RPM\n", freq_45, sync_45))
cat(sprintf("  At %g Hz: motor speed (approx)  %g RPM\n", freq_45, speed_45))
cat(sprintf("  At %g Hz: VFD output voltage  %g V\n", freq_45, voltage_45))
```

```
Motor speed: 850 RPM at 30 Hz, 1300 RPM at 45 Hz
  Synchronous speed at 60 Hz  1800 RPM
  Slip at rated speed  50 RPM (2.8%)
  V/Hz ratio  7.67 V/Hz
  At 30 Hz: synchronous speed  900 RPM
  At 30 Hz: motor speed (approx)  850 RPM
  At 30 Hz: VFD output voltage  230 V
  At 45 Hz: synchronous speed  1350 RPM
  At 45 Hz: motor speed (approx)  1300 RPM
  At 45 Hz: VFD output voltage  345 V
```

Note: The calculation assumes constant slip in RPM, which is an approximation. Actual slip varies somewhat with load and frequency. The V/Hz ratio is maintained constant to provide constant torque capability below base speed.

Question 5: Compare Ethernet/IP and PROFINET. When would you choose each?

Answer:

Feature	Ethernet/IP	PROFINET
Developer	ODVA (Rockwell Automation)	PROFIBUS International (Siemens)
Base Protocol	CIP over TCP/UDP	Based on standard Ethernet
Real-time	CIP Motion for motion control	IRT (Isochronous Real-Time)
Determinism	Standard: non-deterministic; CIP Sync: deterministic	RT: soft real-time; IRT: hard real-time
Typical Cycle	2-10ms (standard); <1ms (CIP Motion)	1-10ms (RT); <1ms (IRT)
Ecosystem	Allen-Bradley, DeviceNet-heritage	Siemens, PROFIBUS-heritage

Choose Ethernet/IP when: - Using Allen-Bradley/Rockwell PLCs - Integrating with DeviceNet legacy systems - Standard industrial Ethernet needs - North American installations (common) - Need CIP protocol compatibility

Choose PROFINET when: - Using Siemens PLCs (S7-1200, S7-1500) - Migrating from PROFIBUS installations - Need IRT for high-performance motion - European installations (common) - Require I-Device functionality

General Guidance: - Match the protocol to your PLC vendor's ecosystem - Both are capable industrial Ethernet solutions - Consider existing infrastructure and expertise - For motion, evaluate specific timing requirements

Question 6: What is the difference between HMI and SCADA? Where would each be used?

Answer:

HMI (Human-Machine Interface): - Local operator interface for a single machine or process - Typically a touchscreen panel mounted on or near equipment - Communicates directly with one or few PLCs - Provides real-time control and monitoring - Limited data storage (trends, alarms) - Examples: Allen-Bradley PanelView, Siemens Comfort Panel

SCADA (Supervisory Control and Data Acquisition): - Centralized system monitoring multiple processes/locations - Server-based architecture with multiple client workstations - Communicates with many PLCs/RTUs across large distances - Supervisory control - high-level commands, not direct control - Extensive historical data collection and analysis - Examples: Wonderware, Ignition, FactoryTalk View SE

Where Each is Used:

Application	HMI	SCADA
Single CNC machine		
Packaging line		
Entire factory floor		
Water treatment plant	(local)	(central)
Oil pipeline network		
Building automation	(per zone)	(campus-wide)
Power grid		

Key Distinction: HMI is typically embedded/local for one process; SCADA is a system supervising many processes across the enterprise.

Question 7: Explain what SIL 3 means and give an example of a SIL 3 application.

Answer:

SIL 3 (Safety Integrity Level 3) is defined in IEC 61508 and represents:

- **PFD (Probability of Failure on Demand):** 10^{-3} to 10^{-4} (0.01% to 0.1%)
- **Risk Reduction Factor:** 1,000 to 10,000
- **Availability:** 99.9% to 99.99%

What SIL 3 Means: - The safety function will fail to operate when needed less than 1 in 1,000 demands - Requires redundant architecture (typically 2oo3 or 1oo2D) - Requires certified safety-rated components - Requires rigorous design, testing, and validation processes - Requires regular proof testing and maintenance

SIL 3 Requirements: - Hardware fault tolerance 1 (single fault won't cause dangerous failure) - Safe Failure Fraction > 90% (for Type A components) - Systematic capability rating of SC3 - Extensive documentation and lifecycle management

SIL 3 Application Examples:

1. **Emergency Shutdown System (ESD)** in oil refinery
   - Detects dangerous conditions (high pressure, fire)
   - Shuts down process and isolates fuel sources
   - Failure could result in explosion and fatalities
2. **Burner Management System (BMS)**
   - Controls furnace/boiler ignition sequence
   - Monitors flame presence and fuel/air ratio
   - Prevents furnace explosions
3. **High Integrity Pressure Protection System (HIPPS)**
   - Prevents overpressure in pipelines
   - Isolates high-pressure source before relief valve capacity exceeded
   - Protects against catastrophic pipeline rupture

Question 8: Describe three benefits of implementing IIoT in a manufacturing facility and give a specific example of each.

Answer:

1. Predictive Maintenance

Benefit: Detect equipment degradation before failure, enabling planned maintenance during scheduled downtime rather than emergency repairs.

Example: Vibration sensors on a conveyor gearbox transmit data to cloud analytics. Machine learning algorithms detect bearing wear signature 3 weeks before expected failure. Maintenance schedules replacement during next planned shutdown, avoiding $50,000 in lost production from unplanned downtime.

2. Real-Time Production Visibility

Benefit: Instant access to production metrics, OEE, and quality data from anywhere, enabling faster decision-making.

Example: Dashboard shows real-time OEE for all production lines on plant manager's tablet. Sudden drop in Line 3 performance triggers alert. Investigation reveals material feed issue; corrected within 30 minutes instead of waiting for end-of-shift report.

3. Energy Optimization

Benefit: Monitor energy consumption at machine level, identify waste, and optimize based on production schedule.

Example: Power meters on each machine report to central system. Analysis reveals HVAC in unused areas runs at full capacity during weekends. Automated scheduling reduces weekend HVAC operation, saving 15% on energy bills ($30,000/year).

Additional Benefits:

4. **Quality Traceability** - Every part traced through production with complete process data
5. **Remote Monitoring** - Engineers can diagnose issues without traveling to site
6. **Supply Chain Integration** - Real-time inventory and production data shared with suppliers

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22.14 References

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Chapter 23

Labs

This chapter is a **catalogue of labs**. It holds more labs than any one term can deliver: an instructor picks the labs that suit their course, their own background and the equipment available. Labs are identified by **name, not by number** — no lab depends on where it sits in this chapter. Where labs *do* depend on one another, the catalogue below and each lab's *Prep work* say so.

The chapter is organised by **topic**. Each topic is a sub-chapter, and each lab is a section within its topic. The last sub-chapter, Teacher preparation, is a planning aid for the instructor: which labs must be delivered together, how to schedule the materials-testing rotation, and the room, equipment, software and shopping lists for every lab.

Catalogue of labs

Topic	Lab	Where / tools	Format	Notes
Robotics and Programming Logic	Logic Thinking for Robot Control: Rocksi	Browser (Rocksi)	Individual	No install, no login
	Robotic Work-Cell Design in RoboDK for Web	Browser (RoboDK), screen recorder	Individual, take-home	Uses the takt time from the Lean lab

Topic	Lab	Where / tools	Format	Notes
Process Design and Simulation	Engineering Design Process Documentation	Office, Project, Visio	Individual	Précitech case study
	Manufacturing Facility Design & Simulation (AnyLogic)	AnyLogic PLE, Office	Individual	Builds on the Design Documentation lab
Lean Manufacturing	Lean: Value Stream Mapping, Takt and Pull	Browser (draw.io, Sheets, AnyLogic Cloud)	Individual	
Ergonomics	Ergonomics in Automated Manufacturing (REBA)	REBA worksheet, task videos	Individual	
Metrology and Calibration	Linear and Angular Measuring Instruments	Metrology lab	Instructor's choice	
	Groove Angle Measurement (V-Block)	Metrology lab	Instructor's choice	
	Taper Angle Measurement with a Sine Bar	Metrology lab	Instructor's choice	
	Micrometer and Caliper Calibration	Metrology lab	Instructor's choice	
Statistical Quality Control and Measurement Systems	Statistics with Excel for Quality Control	Computer, Excel	Individual	
	Variable Control Charts: $\bar{X}$, R and S	Metrology lab, Excel	Instructor's choice	

Topic	Lab	Where / tools	Format	Notes
	Attribute Control Charts: p and np	Bench, Excel	Instructor's choice	
	Process Capability: Cp, Cpk and Distributions	Metrology lab, Excel	Instructor's choice	
	Repeatability and Reproducibility (Gauge R&R)	Metrology lab, Excel	Individual (swap calipers with a partner)	
	SPC and Gauge R&R: the Virtual Caliper	This page (webR)	Individual	No install
Materials Testing	Hardness Testing: Rockwell and Brinell	Materials lab	Team, **3-week rotation**	**Deliver all three as a set**
	Tensile and Microhardness Testing of Case-Hardened Steel	Materials lab	Team, **3-week rotation**	**Deliver all three as a set**
	Charpy Impact Testing and the Ductile-to-Brittle Transition	Materials lab	Team, **3-week rotation**	**Deliver all three as a set; needs dry ice**
Virtual Welding	GMAW Flat Surfacing: Single Pass	VRTEX 360+ simulator	Individual	Start with this one
	GMAW Flat Surfacing: Multi-Pass	VRTEX 360+ simulator	Individual	
	GMAW Groove Weld: Single Pass	VRTEX 360+ simulator	Individual	

Topic	Lab	Where / tools	Format	Notes
	GMAW Groove Weld: Multi-Pass	VRTEX 360+ simulator	Individual	
	GMAW T and Lap Welds	VRTEX 360+ simulator	Individual	
	GMAW Vertical T Welds	VRTEX 360+ simulator	Individual	
Machine Learning: Predictive Maintenance	Machine Learning, Week 1: AI Literacy and Prompt Engineering	Browser, LLM chat	Individual	**Deliver all six as a set, in order**
	Machine Learning, Week 2: Manufacturing Data and SPC	Google Colab	Individual	**Deliver all six as a set, in order**
	Machine Learning, Week 3: The C-MAPSS Dataset	Google Colab	Individual	**Deliver all six as a set, in order**
	Machine Learning, Week 4: Building the Predictive-Maintenance Model	Google Colab	Individual	**Deliver all six as a set, in order**
	Machine Learning, Week 5: Deploying the App	GitHub, Streamlit Cloud	Individual	**Deliver all six as a set, in order**
	Machine Learning, Week 6: Capstone Presentations	Projector, deployed app	Individual	**Deliver all six as a set, in order**

Topic	Lab	Where / tools	Format	Notes
Maintenance and Root-Cause Analysis	TPM/OEE, Root Cause and a New Dataset	Browser (Sheets, draw.io), Colab	Individual	Best after the machine-learning set

"Instructor's choice" means the source handout does not fix the group size — these are bench exercises that work equally well individually or in pairs.

How to use this chapter

Every lab is broken into the **same fixed parts**, and each part has its own colour so you always know which part of the lab you are reading:

Colour	Part	What it is
Dark blue	**Overview**	What the lab is about and why you do it
Teal	**Learning outcomes**	What you should be able to do afterwards
Orange	**Prep work**	Readings, videos and set-up to finish *before* the lab period
Blue	**Materials & equipment**	What to bring / what is provided

! Where the step-by-step instructions live

The **detailed instructions, the tasks, the deliverables and the marking scheme for each lab are delivered on FOL** as a **FOL quiz** you complete *during the lab period*: it walks you through each task in order, collects your written description of what you did, and is where you upload the files you create (Rocksi stations, AnyLogic models, notebooks, model files, screenshots and screen recordings). For the hands-on labs — metrology, quality control, materials testing and virtual welding — the FOL quiz walks you through each task, collects your measurements and answers, and is where you upload the **lab worksheet or report template** you filled in. This chapter does not reproduce the procedure or the rubric; open the FOL item for your lab and work through it top to bottom.

Unless a lab says otherwise, work is individual. The materials-testing rotation runs in teams and says so, and even there **each student**

writes and submits their own report. The hands-on Gauge R&R lab is individual too: you take your own readings, and the only thing you share with a partner is a swap of calipers and, afterwards, of readings.

! Précitech Components — the running case

Many labs are set at **Précitech Components**, a Tier-1 automotive supplier in Granby, Québec, whose CNC turning cells make transmission shaft blanks and whose new shaft-kit assembly cell you design, simulate, balance, assess and finally program a robot for (see the case study in the *Engineering Design Process Documentation* lab). The metrology, quality-control, materials and welding labs use lab hardware instead and are not tied to the narrative.

How the labs fit together:

- **The Virtual Caliper precedes ML Week 2.** That is deliberately the arc Chapter Chapter 11 is built on: *do the SPC yourself first, then direct an LLM to do it.*
- **Facility Design & Simulation follows Engineering Design Process Documentation** — you turn your own layout drawing into a simulation.
- **Robotic Work-Cell Design checks its cycle time against the takt time you compute in the Lean lab.** It also bookends *Logic Thinking*: block-logic puzzles at the start of the course, a real industrial arm at the end.
- **TPM/OEE, Root Cause and a New Dataset follows the machine-learning set** because it reuses the whole ML pipeline on an unseen, differently shaped dataset — which only works once students have the skill.
- **The metrology labs underpin the quality-control labs** — a control chart or a Gauge R&R study is only as good as the measurement behind it.

Not in this catalogue: the popsicle-stick bridge lab, the lean paper-cutting simulation, and the automation/functional-safety lab series. The old handouts still exist in the `LabInstructions/` folder and can be reinstated; the older **instrumentation-trainer** stream (B1033 benches) is summarised in Appendix: instrumentation-trainer lab stream.

Each lab is due at the end of its scheduled lab period; the late penalty is 25 % per lab session. A take-home lab (Robotic Work-Cell Design) is released about three weeks before its due date and includes a short screen recording. Marks may be deducted for incomplete prelab work, missing safety training, not bringing required materials or protective equipment, not cleaning your station, or unprofessional conduct in the lab room. The course schedule in `index.qmd` shows the plan the current offering follows; follow FOL if it diverges.

23.1 Robotics and Programming Logic

Two labs that bookend a course: **block-based programming logic** on a browser robot simulator, and the design and programming of a work cell for a **real 6-axis industrial arm** in RoboDK. Neither needs an install. They pair with Chapter 3 (Industrial Robotics).

23.1.1 Logic Thinking for Robot Control: Rocksi

An introduction to programming logic — **sequencing**, **loops** and **conditional/branching logic** — and to programming a robot arm, using one free browser simulator: **Rocksi** (a browser, block-based simulator for a collaborative robot arm — Scratch-style blocks, no install and no login). You build the logic concepts *and* a pick-and-place sequence in the same tool, and the focus is on *optimising* the number of blocks.

(Earlier versions of this lab also used LightBot; it needs a Flash-era plug-in that is not available on college machines, so the whole lab now runs in Rocksi.)

AI use: not permitted — the outcome *is* your own logic. The Robotic Work-Cell Design lab returns to a real 6-axis arm in RoboDK for Web.

Learning outcomes

After this lab you should be able to:

- Explain why programming logic matters for reliable, safe, repeatable automation.
- Use sequences, loops and conditional/branching blocks to solve a control problem with the fewest blocks.
- Describe the role of a teach pendant (jogging, teaching points, deadman/E-stop, reduced-speed teach mode, I/O testing).
- Program a simple pick-and-place-style sequence on a block-based robot simulator and simulate a digital input/output.

Prep work

- Read the Programming Logic handout (importance of programming logic; loops; conditionals; teach pendants).
- Watch: *Rocksi robot simulator: Meet Sawyer.*

Rocksi robot simulator: Meet Sawyer: https://www.youtube.com/watch?v=3THTZFnrxtc

- Open **Rocksi** and confirm it loads and runs on your device. Try dragging one move block onto the workspace and pressing run.

Materials & equipment

- Computer with internet access — **one per student** (all labs are individual)
- Access to **Rocksi** block-based robot-arm simulator (browser, free, no login, no plug-in)
- The **Snipping Tool** (Windows) for screenshots you will upload
- The **FOL quiz for this lab** — it carries every task, in order, and is where you type your answers and upload your Rocksi program and screenshots

> **i** Instructions & submission
>
> Work through the **FOL quiz for this lab** during the lab period. It covers sequencing, loops, conditional/branching logic, a pick-and-place sequence, simulating a digital I/O, and an optimisation challenge; for each task you describe what you did and attach a screenshot, and at the end you upload your saved Rocksi program.

23.1.2 Robotic Work-Cell Design in RoboDK for Web

A **take-home** lab: released about three weeks before it is due, done individually, with a short screen recording as part of the submission. You design and program the Précitech kitting cell's pick-and-place robot in **RoboDK for Web** — a free, no-install, in-browser robot simulator with 600+ real arms. It bookends Logic Thinking: block-logic puzzles at the start, a real 6-axis industrial arm at the end. You close the loop on the course by checking the robot's cycle time against the **takt time you computed in the Lean lab**.

AI use: permitted with disclosure. The station layout, the program and the cycle times must be your own work.

Learning outcomes

After this lab you should be able to:

- Jog a 6-axis robot in joint, linear and tool coordinates and recognise singularities and joint-limit stops.
- Lay out a work cell and verify that every pick and place point lies inside the robot's reach envelope.
- Program a pick-and-place cycle with targets and gripper I/O and read the simulated cycle time.
- Compare a simulated cycle time against a required takt time and optimise the program to close the gap.
- Compute a minimum safeguarding distance from robot stopping performance and select an appropriate safeguard.

Prep work

- Read Chapter 3 (industrial robotics — coordinate systems, the work envelope, singularities) and Chapter 12 (machine guarding — the minimum-distance / safeguarding-distance formula and the safeguard hierarchy).
- Have your **takt calculation from the Lean lab** to hand — you need the takt time for 250 kits/day.
- Open RoboDK for Web and confirm it loads and runs a sample station in your browser. No account or licence key is required.
- Confirm your screen-recording tool works (the same tool the course plan's take-home lab uses).

Materials & equipment

- Personal computer with internet access and a browser (this is a take-home lab — no lab PC required)
- RoboDK for Web — no install, no licence key
- A screen-recording tool — the built-in recorder is enough (Windows **Xbox Game Bar**, Win+G; macOS **Screenshot**, Shift+Cmd+5) or free OBS Studio
- Your takt time from the Lean lab; the Précitech case study (250 kits/day, one 8-hour shift)
- The **FOL quiz for this lab** — the five tasks in order, and where you upload the report, the RoboDK station file and the 3–5 min screen recording

i Instructions & submission

This is a **take-home** lab: released about three weeks before it is due, done on your own computer. Work through the **FOL quiz for this lab** — it runs the five tasks in order (jog and observe, build the kitting cell, program the pick-and-place, cycle time vs the Lean-lab takt, and the safeguarding distance), asks you to describe each design decision, and collects your report, the RoboDK station file and a 3–5 minute screen recording talking through one decision. The layout, the program and the cycle times must be your own work; disclose any AI help.

23.2 Process Design and Simulation

The engineering documentation a new production cell needs before it is built, and a discrete-event simulation of that cell — both set at Précitech Components. The simulation lab builds on the documentation lab, so deliver them in that order. They pair with Chapters 2, 8 and 16.

23.2.1 Engineering Design Process Documentation

You produce the engineering-documentation package that a real project needs to move a design into production — specification sheets, a bill of materials, a task list / process flow, a project schedule and Gantt chart, a production-facility layout drawing, and a health-safety-environment (HSE) risk assessment — for the **Précitech Components shaft-kit assembly cell** described in the case study below. The same cell is simulated in Facility Design & Simulation, so the documents you build here feed directly into the next lab. Tools: Microsoft Visio/Word, Project and Excel.

AI use: not permitted for any part of the report.

ℹ Case study — Précitech Components: new shaft-kit assembly & packing cell

Précitech Components is a Tier-1 automotive supplier in Granby, Québec. Its CNC turning cells produce transmission **shaft blanks** (AISI 4140 steel, Ø 25.000 mm ± 0.050 mm) for a major OEM customer. The OEM now wants the shafts delivered as a **ready-to-install kit** rather than as loose blanks, so Précitech is adding a **manual kitting, sub-assembly and packing cell** immediately downstream of the CNC turning cells.
One shaft kit contains: 1 machined shaft blank, 2 tapered roller bearings, 1 retaining ring, 1 lip seal, 1 dowel pin, 1 desiccant sachet — placed in a moulded tray inside a labelled carton. The exact part numbers, quantities and shaft dimensions are on the assembly drawing (410-2365, Rev B) and shaft detail drawing (205-5478, Rev B) — you need both for this lab.
Target output: 250 kits/day on one 8-hour shift.
Process outline: finished blanks arrive from the CNC cells on a trolley; bought-in components (bearings, rings, seals, pins) are held in a small storage rack that is replenished from the main warehouse when stock falls to a reorder level; an operator picks the components for one kit, presses the two bearings onto the shaft, fits the retaining ring and lip seal, inspects the assembly against a checklist, places it in a tray, closes and labels the carton, and stages it for the shipping dock.
Plant manager **Sophie Tran** has asked you to produce the engineering documentation package for this cell **before it is built**.
This is the same plant used in the machine-learning labs (see Machine Learning), which build a predictive-maintenance proof of concept for Précitech's CNC turning cells.

Learning outcomes

After this lab you should be able to:

- Explain terms such as HSE, Basic Engineering Design Data (BEDD), Front-End Engineering Design (FEED), fast-tracking, project schedule and General Arrangement (GA) drawing.
- Produce a specification sheet and a bill of materials for a manufactured product.
- Build a task list / process-flow chart with durations and turn it into a project schedule and Gantt chart in Microsoft Project.
- Draft a production-facility layout drawing showing product in/out, stations, storage, and material and people flow, and mark starvation / blockage risk points.
- Compare a highly automated (turntable) cell against a manual U-cell and justify a configuration choice for a given volume, timeline and labour situation.
- Complete an HSE risk assessment for a manual assembly cell.

Prep work

- Read the **case study** above and Unit 2 content on the basics of process engineering and design.
- Review the assembly drawing (410-2365, Rev B) and the shaft detail drawing (205-5478, Rev B) — you need the part numbers, quantities, materials and key dimensions from these for your spec sheet and BOM. Purchased components are sourced from **McMaster-Carr**.
- Skim the Design Documentation supplement for the document templates and the detail-design / engineering process-flow diagram (ignore the popsicle-bridge framing — the document types are the same).
- Confirm you can open Microsoft Visio/Word, Project and Excel on a lab PC.

Materials & equipment

- Lab PC with **Microsoft Word / Visio**, **Microsoft Project**, **Microsoft Excel**
- Assembly drawing 410-2365, Rev B and shaft drawing 205-5478, Rev B — the part numbers, quantities and dimensions you need
- Templates from FOL:
  - `Bill_of_Materials_Template_for_Excel.xlsx` — pre-filled with the kit's part numbers, descriptions and quantities from the assembly drawing; you fill in unit, source and supplier
  - `Gantt_chart_template.xlsx` — open it in Microsoft Project (File > Open) and use Project's Import Wizard to turn it into a schedule, then Save As `.mpp`; the template's "Read me first" sheet walks you through it
  - `Risk_Assessment_Template.docx` (HSE risk assessment)
- The Précitech shaft-kit case study (above) — the single source of requirements

- The **FOL quiz for this lab** — every phase in order, and where you upload your spec sheets, BOM `.xlsx`, flow chart, Project `.mpp`, layout drawing and risk assessment

i The automation decision

Précitech's plant manager has not committed to a cell configuration yet — that choice is yours to make and defend as part of your facility layout. Two realistic options are on the table for a 250-kit/day, single-shift cell:
Option A — highly automated: rotary turntable, 6–7 stations. An indexing turntable carries the shaft through fixed stations (press bearing 1, press bearing 2, fit retaining ring, fit seal, insert dowel pin + desiccant, inspect, place in tray / close carton), with **one operator** at a single load/unload station. Everything else is tooled and sensored.
Option B — manual U-cell. A standard U-shaped layout (Chapter 8) with **one operator per station**, parts and trays moving between stations on roller conveyors — matching the *manual* kitting cell Précitech originally scoped.

	Option A — automated turntable	Option B — manual U-cell
Capital cost	High — custom turntable + tooled/sensored stations	Low — conveyor + hand tools/fixtures
Time to stand up	Long — design, build and debug before day 1	Short — matches the "before it is built" schedule pressure
Labour	1 operator/shift	Up to 6–7 operators/shift
Cycle-time consistency	High — paced by the index, not by people	Lower — depends on operator skill and fatigue
Flexibility to rebalance	Low — station count and index time are fixed	High — move people, cross-train, absorb absences
Changeover to a kit variant	Retool multiple stations at once	Operators adapt; fixtures swap faster
Failure mode	Single point of failure — one jammed station stops the whole cell	A blocked station can sometimes be buffered around
Ergonomic risk	Concentrated at one manual station	Spread across every station (Ergonomics)

Maintenance skill needed	PLC/servo/automation technicians (Chapter 22)	Standard mechanical/tooling upkeep

Neither option is "correct." Your facility-layout task in the FOL quiz asks you to pick one, sketch it, and justify the choice with at least two pros and two cons — from the table above or your own reasoning — tied to Précitech's actual volume, timeline and labour situation.

i Instructions & submission

Work through the **FOL quiz for this lab** during the lab period. It takes you through all five phases — specification sheet and BOM (with a prepared checklist to work from), task list / process flow, project schedule and Gantt chart, facility layout drawing (including the automation decision above), and HSE risk assessment — asks you to describe each one, and collects the electronic files. The task-list and Gantt-chart examples in the quiz are **starting points only** — part of the mark is finding and adding the steps they're missing. Keep your layout drawing: you refine and reuse it in Facility Design & Simulation. AI tools may not be used for any part of this lab.

23.2.2 Manufacturing Facility Design & Simulation (AnyLogic)

You turn the Précitech kitting-cell layout from the Engineering Design Process Documentation lab into an electronic drawing, then use **AnyLogic Personal Learning Edition** to run a discrete-event simulation of the cell's order-fulfilment behaviour. By changing order and replenishment parameters and reading the statistics screens, you see how component-stock policy drives waiting time, picking (kitting) time and throughput.

AI use: not permitted for any part of the report.

Learning outcomes

After this lab you should be able to:

- Produce an electronic facility layout showing stations, storage, flows, manual vs. automated stations, and starvation / blockage risk points.
- Run an AnyLogic model, change input parameters, and capture the statistics output.
- Interpret order statistics (preorder waiting time, picking time, full processing time, order states) and goods statistics (unavailable goods, stored reserved / unreserved goods, storing time) and explain how they are linked.

- Explain how simulation supports facility-design decisions such as where to hold buffer stock and how often to replenish.

Prep work

- Re-read the Précitech case study and review Unit 3 content on integrated manufacturing systems.
- Read the Facility Simulation supplement (order statistics, goods statistics, and the relationships between them).
- Watch the AnyLogic walk-through video (link on FOL).
- Install **AnyLogic Personal Learning Edition** (free) on your machine *before* the lab and confirm it launches.
- Bring your layout drawing, spec sheets, schedule, BOM and flow documents.

Materials & equipment

- Lab PC with **AnyLogic Personal Learning Edition** and **Microsoft Visio/Word/Project/Excel**
- The AnyLogic order-fulfilment / warehouse demo model referenced in the video — *confirm the exact model and stage it on FOL / the lab PCs*
- Your Engineering Design Process Documentation lab documents for the Précitech kitting cell
- The **FOL quiz for this lab** — the facility-drawing checklist, the four AnyLogic parameter sets to run, and where you upload your drawing, your saved AnyLogic model and the labelled statistics screenshots

i Instructions & submission

Work through the **FOL quiz for this lab** during the lab period. Phase 1 walks you through recreating your Engineering Design Process Documentation kitting-cell layout electronically against a checklist; Phase 2 gives you four AnyLogic parameter sets to run, each for three simulated minutes, and asks for a labelled Statistics screenshot and a written observation/analysis tied to the case study for each. You upload the drawing, the AnyLogic model file and the screenshots in the quiz. AI tools may not be used for any part of this lab.

23.3 Lean Manufacturing

Map, size and redesign a production flow. This lab pairs with Chapters 17 and 18, and its takt time is used again in the robot work-cell lab.

23.3.1 Lean: Value Stream Mapping, Takt and Pull

You map the Précitech shaft line as it runs today, size it against customer demand, and redesign the flow. Working in **draw.io** (browser) you build a current-state **value stream map**; in **Google Sheets** you compute **takt time**, an operator balance chart and **process cycle efficiency**; and on **AnyLogic Cloud** (browser, free public tier) you compare push, pull and a kanban-capped supermarket. You finish with a future-state map carrying one SMED and one pull improvement. Everything runs in a browser — nothing to install.

AI use: permitted with disclosure, but **not** for the map itself or the takt / PCE calculations — those must be your own work. Name any tool you used and what for.

Learning outcomes

After this lab you should be able to:

- Draw a current-state value stream map from a station data table, with a material-flow line, an information-flow line and a lead-time ladder.
- Compute takt time from customer demand and compare it against station cycle times on an operator balance chart to find the bottleneck.
- Classify observed waste against the seven wastes and compute process cycle efficiency (value-added time ÷ lead time).
- Explain how push, pull and a kanban-capped supermarket differ in WIP and lead time, using simulation output as evidence.
- Propose a future state and justify the projected lead-time reduction.

Prep work

- Read Chapter 15 (Lean) on takt time, the seven wastes, pull systems, kanban and SMED, and the Chapter 20 material on line balancing.
- Create a free **draw.io** document ("Device" storage is fine, no account needed) and skim the built-in *Value Stream* shape library.
- Confirm **AnyLogic Cloud** loads and that you can run a public model and edit its input parameters in the browser.
- Download `vsm_station_data.csv` from FOL — the station table for the Précitech shaft line (cycle time, changeover, uptime, WIP, batch size per station).

Materials & equipment

- Computer with internet access — one per student (all labs are individual)
- **draw.io** (browser) — current- and future-state maps
- **Google Sheets** (browser) — takt, balance chart, PCE
- **AnyLogic Cloud** (browser, free public tier) — push/pull model
- `vsm_station_data.csv` from FOL
- The Précitech case study — demand is 250 kits/day on one 8-hour shift

- The **FOL quiz for this lab** — the five tasks in order, and where you upload the draw.io exports, the Sheets calculations and the AnyLogic Cloud results

> **i** Instructions & submission
>
> Work through the **FOL quiz for this lab** during the lab period. It runs the five tasks in order — current-state VSM, takt and the operator balance chart, the seven-wastes classification and PCE, push vs pull vs supermarket on AnyLogic Cloud, and the future-state map — asks you to describe and interpret each, and collects the draw.io exports, the Sheets file and the simulation-results table. The map and the takt/PCE arithmetic must be your own work; disclose any AI tool you used for the parts where it is permitted.

23.4 Ergonomics

Assess a workplace task for musculoskeletal risk. This lab pairs with Chapter 4.

23.4.1 Ergonomics in Automated Manufacturing (REBA)

You assess a workplace task for musculoskeletal risk using the **REBA (Rapid Entire Body Assessment)** worksheet. (The FOL quiz assigns you one of three sample tasks.)

AI use: permitted with disclosure, not for the REBA scoring itself.

Learning outcomes

After this lab you should be able to:

- Explain why ergonomics matters in an automated manufacturing facility and name common ergonomic risks and their effects on workers.
- Score a task with the REBA worksheet and interpret the result.
- Identify the sources of error and limitations of a manual ergonomic assessment.

Prep work

- Review the introduction-to-ergonomics and REBA material (video links in the handout, Procedure topics 1–2).
- Skim the REBA step-by-step guide and the REBA Employee Assessment Worksheet.

Materials & equipment

- **REBA Employee Assessment Worksheet** (printed, one per student)
- Computer with internet access
- Sample tasks from an automated manufacturing facility (videos / photo sets), one assigned to you in the FOL quiz:
  - Task 1 — lifting boxes repeatedly
  - Task 2 — computer-workstation setup
  - Task 3 — floor cleaning with a buffer
- The **FOL quiz for this lab** — the intro videos, the assigned task, the REBA steps, and where you upload your completed REBA worksheet (scan or photo)

i Instructions & submission

Work through the **FOL quiz for this lab** during the lab period. It plays the intro videos, assigns you one of the three sample tasks, and walks you through scoring it on the REBA worksheet. You upload the completed REBA worksheet and any screenshots in the quiz. Disclose any AI tool you used.

23.5 Metrology and Calibration

Dimensional measurement fundamentals: reading precision instruments, measuring angles *indirectly* with gauge blocks and rollers, and calibrating an instrument against a traceable standard. These hands-on labs sit underneath the SPC and MSA chapters — a control chart or a Gauge R&R study is only as good as the measurement behind it (Chapter 5). They are independent of each other and can be delivered in any order; the instruments lab is the natural first one.

23.5.1 Linear and Angular Measuring Instruments

Before you can trust a measurement you have to understand the instrument that made it. You study three linear instruments — the **vernier caliper**, the **vernier height gauge** and the **vernier micrometer** — and one angular reference, the **gauge (slip) block**. For each instrument you illustrate it, label its parts, state the units of its scale(s), write the procedure for reading it and derive the formula for its **least count**; you then work two least-count problems and answer three questions on gauge-block grades, sets and wringing.

AI use: not permitted for the least-count calculations. Disclose any AI tool you used for background research on the instruments.

Learning outcomes

After this lab you should be able to:

- Identify the main parts of a vernier caliper, a vernier height gauge and a vernier micrometer, and state the unit of each scale.
- Read each instrument and explain, step by step, how the reading is built up from the main scale and the vernier or thimble scale.
- Calculate the least count of a vernier caliper and of a vernier micrometer from the way their scales are graduated.
- Explain what a gauge-block grade means, what each grade is used for, and how blocks are *wrung* together to build a stack.

Prep work

- Read the parts of Chapter 5 (MSA) on resolution, accuracy and calibration.
- Review how a vernier scale works: a sliding scale whose divisions are slightly shorter than the main-scale divisions, so the one line that aligns gives the fraction of a division.
- Bring a phone or camera — you photograph the instruments on your bench.

Materials & equipment

- Vernier caliper, vernier height gauge and vernier (or digital) micrometer
- A gauge-block (slip-gauge) set — rectangular, square and square-with-hole blocks if the set has them
- The FOL quiz for this lab (and the lab assignment sheet on FOL)

> **i** Instructions & submission
>
> Complete the assignment sheet from FOL: one entry per instrument (illustration, labelled parts, scale unit(s), reading procedure, least-count formula), the two least-count problems, and the three gauge-block questions. Submit it in the FOL quiz. The **FOL quiz for this lab** walks you through these steps in order and is where you record your results and upload your file.

23.5.2 Groove Angle Measurement (V-Block)

You measure the included angle of a V-groove *without* an angle-measuring instrument. Two cylindrical rollers of different, precisely known diameters are placed in the groove one at a time; with the V-block on a surface plate you measure how high the top of each roller sits with a **vernier height gauge**. The groove angle then follows from the two roller diameters and the two heights by trigonometry. You repeat the whole procedure on four samples.

AI use: not permitted — the measurements and calculations must be your own.

Learning outcomes

After this lab you should be able to:

- Explain the principle of measuring an angle indirectly from two roller diameters and two heights.
- Set up a vernier height gauge on a surface plate (zero at the plate, scriber just touching the roller) and read it.
- Calculate the V-groove angle from your measurements and comment on how well your repeats agree.

Prep work

- Review right-triangle trigonometry and inverse sine.
- Review how to zero and read a vernier height gauge (see Linear and Angular Measuring Instruments).
- Read the precautions on the FOL sheet: the rollers must not move while the height is read, the scriber edge should only *just* touch the roller, and the roller diameter is measured at its middle.

Materials & equipment

- V-block and a clean, dust-free surface plate
- Vernier height gauge and vernier caliper
- Two rollers (gauge pins) of different diameters
- The V-groove angle sheet from FOL (Table 1: roller diameter, roller height and calculated angle for each of four samples)

i Instructions & submission

Measure the two roller diameters with the caliper; place the smaller roller in the groove and measure its height above the plate; repeat for the larger roller; calculate the angle with the equation on the sheet and record it in Table 1; then repeat the whole procedure for the remaining samples. Submit the completed table in the FOL quiz. The **FOL quiz for this lab** walks you through these steps in order and is where you record your results and upload your file.

23.5.3 Taper Angle Measurement with a Sine Bar

A **sine bar** turns an angle measurement into a length measurement. The bar, a stack of gauge blocks and the surface plate form a right-angled triangle: the bar is the hypotenuse L and the gauge-block stack is the side h opposite the angle, so $\sin\theta = h/L$. You build a gauge-block stack under one roller of the sine

bar until the top surface of an unknown **taper** is parallel to the surface plate — confirmed by sliding a **dial indicator** along it and reading zero deviation — and calculate the taper angle. You measure three tapers.

AI use: not permitted — the measurements and calculations must be your own.

Learning outcomes

After this lab you should be able to:

- Explain the sine principle and why the two rollers must be equal in diameter, parallel, and a precisely known distance apart.
- Wring gauge blocks into a stack and use a dial indicator to prove that a surface is parallel to the plate.
- Calculate the half-angle $\theta = \sin^{-1}(h/L)$ and the whole taper angle 2θ, and compare it with the nominal taper angle.

Prep work

- Review right-triangle trigonometry.
- Review gauge-block wringing (see Linear and Angular Measuring Instruments).
- Note that the dial-indicator reading tells you *which way* to adjust the stack: if the indicator deflects, add or remove gauge blocks and try again.

Materials & equipment

- Surface plate, sine bar, dial indicator on a stand, gauge-block (slip-gauge) set
- Three tapers of different angles
- The sine-bar sheet from FOL (table: taper, gauge-block height h, bar length L, half-angle , whole angle 2 , comparison with the actual taper angle)

> **i** Instructions & submission
>
> Follow the procedure on the FOL sheet for each assigned taper, record h, L, and 2 in the table, and compare with the actual taper angle. Submit the completed sheet in the FOL quiz. The **FOL quiz for this lab** walks you through these steps in order and is where you record your results and upload your file.

23.5.4 Micrometer and Caliper Calibration

Calibration compares an instrument against a *traceable standard* — in dimensional metrology, a gauge block whose deviation from nominal is known

and traceable to the international standard. You calibrate a micrometer and a caliper: clean and inspect the instrument, let it come to ambient temperature, set zero, then check it against gauge-block stacks at ten points spread over its range, record the deviation at each point and report the instrument's accuracy (the deviation over its range) against the manufacturer's published accuracy. This is the first step of any measurement-systems analysis, and the reference values it produces are what *bias* and *linearity* are measured against.

AI use: not permitted — the readings and the report must be your own.

Learning outcomes

After this lab you should be able to:

- Explain calibration, traceability and why calibration is done at defined intervals.
- Carry out a calibration check on a micrometer and on a caliper following a standard procedure (clean, inspect, stabilise, zero, check at each size).
- Record and interpret the deviation at each size and decide whether the instrument is within its published accuracy.
- Complete a calibration report identifying the instrument, the standard used and the findings.

Prep work

- Read Chapter 5 (MSA) — calibration, bias and linearity.
- Review gauge-block wringing (see Linear and Angular Measuring Instruments).
- Read the calibration procedure on the FOL sheet before the lab: the check sizes for the micrometer and for the caliper (which depend on its range) are listed there.

Materials & equipment

- A micrometer and a caliper to calibrate (record make, model, serial number, range, resolution and published accuracy)
- A gauge-block set with its serial number, and clean lint-free wipes
- Cleaning fluid (methyl alcohol) — keep it off your skin
- The calibration report forms from FOL (one for the micrometer, one for the caliper)

i Instructions & submission

Complete one calibration report for the micrometer and one for the caliper: instrument identification, the reading and deviation at each check size, the accuracy over the range, the standard used, calibration notes and your observations and findings. Submit both reports in the FOL quiz. The

FOL quiz for this lab walks you through these steps in order and is where you record your results and upload your file.

23.6 Statistical Quality Control and Measurement Systems

The laboratory companion to Chapter 9 (SPC) and Chapter 5 (MSA). Five of the six labs are Excel worksheets: you collect real data — measuring roller bearings, inspecting beads, measuring bar stock — and then do the statistics yourself in the worksheet. The sixth, the Virtual Caliper, runs entirely on this page. Any lab can be delivered on its own; a sensible order is *Statistics with Excel* → *Variable Control Charts* → *Attribute Control Charts* → *Process Capability*, with the two measurement-system labs (hands-on Gauge R&R and the Virtual Caliper) alongside or after them. The *Virtual Caliper* and the hands-on *Gauge R&R* lab cover overlapping ground — deliver one or both.

23.6.1 Statistics with Excel for Quality Control

Statistics is the language of quality control, and Excel is the tool most quality technicians actually use. You learn the Excel functions for the basic statistical parameters — average, median, mode, population and sample variance and standard deviation, count, minimum, maximum and range — and apply them to three small data sets: the processing times of 12 hot-rolled-steel samples (compute the statistics and interpret the difference between mean, median and mode), the depth of cut versus tool wear of a milling operation (20 observations, plotted as a labelled scatterplot), and 50 call-centre waiting times (built into a frequency table and a histogram, then read for the mean, the mode/median and whether the shape looks like a normal curve).

AI use: not permitted — the calculations and graphs must be your own.

Learning outcomes

After this lab you should be able to:

- Use Excel's statistical functions to compute measures of centre and spread, and explain when to use the population and the sample versions.
- Interpret the difference between mean, median and mode.
- Build a frequency table and histogram in Excel and judge whether a distribution looks normal.
- Plot and label a scatterplot to look for a relationship between two process variables.

Prep work

- Read the introductory sections of Chapter 9 (SPC) on variation, mean, standard deviation and histograms.
- Confirm you can use basic Excel formulas and insert a chart.

Materials & equipment

- Computer with Microsoft Excel
- The statistics worksheet from FOL (the data sets, function list and answer cells are in the file)

> **i** Instructions & submission
>
> Work through the worksheet in order: the function warm-up, the processing-time statistics with your interpretation, the scatterplot, and the histogram with the four short-answer questions. Save the workbook and submit it in the FOL quiz. The **FOL quiz for this lab** walks you through these steps in order and is where you record your results and upload your file.

23.6.2 Variable Control Charts: X, R and S

You measure the length of **100 roller bearings** — in ten subgroups of ten — and use the data to build an **X chart**, an **R chart** and an **S chart** in Excel. You compute the subgroup averages, ranges and standard deviations, the grand average, the average range and average standard deviation, and the control limits from the tabulated constants; then you plot the charts and evaluate them against the out-of-control rules on the worksheet. (The bearings are tapered — you measure length, *not* diameter.)

AI use: not permitted — the measurements, calculations and charts must be your own.

Learning outcomes

After this lab you should be able to:

- Organise measurements into rational subgroups and compute subgroup statistics.
- Calculate centre lines and control limits for X, R and S charts using the control-chart constants.
- Plot the charts and apply the out-of-control rules to decide whether the process is in statistical control.
- Explain the difference between what the X chart and the R/S charts monitor.

Prep work

- Read Chapter 9 (SPC) — X/R charts, control limits and the out-of-control rules (X/S charts are in Chapter 18).
- Have the control-chart constants table to hand (the FOL sheet gives the constants for subgroups of ten; the full table is on FOL).
- Review how to read a micrometer or caliper to the resolution the sheet specifies.

Materials & equipment

- 100 roller bearings (the same batch) and a micrometer or caliper
- Computer with Microsoft Excel
- The variable-control-charts worksheet from FOL and the control-chart constants table

> **i** Instructions & submission
>
> Measure the bearings in ten subgroups of ten and enter them in the worksheet, complete the X, R and S calculations, plot the three charts, and evaluate the charts against the out-of-control rules. Save the workbook and submit it in the FOL quiz. The **FOL quiz for this lab** walks you through these steps in order and is where you record your results and upload your file.

23.6.3 Attribute Control Charts: p and np

Not every quality characteristic is a measurement — many are pass/fail. You inspect a production run of coloured beads: in each of ten samples of 50 beads you count the defectives (the instructor tells you which colour is "defective" for your sample), then compute the proportion defective and build a **p chart** and an **np chart** with their control limits. You judge whether the process is in control and say what should be done next.

AI use: not permitted for the counts, calculations and charts. Your written answers must be your own.

Learning outcomes

After this lab you should be able to:

- Explain what a p chart and an np chart monitor and when to use each.
- Calculate the centre line and control limits for both charts (setting a negative lower limit to zero).
- Decide whether the process is in statistical control and justify the next step.

- Name a complementary tool (for example a Pareto chart) and say why it helps.

Prep work

- Read Chapter 9 (SPC) — attribute data and p charts (np charts are in Chapter 18).
- Review the difference between a *defect* and a *defective*.

Materials & equipment

- A bag of beads of two or more colours per bench (enough for ten samples of 50)
- Computer with Microsoft Excel
- The attribute-charts worksheet from FOL

> **i** Instructions & submission
>
> Draw and inspect ten samples of 50 beads, record the number defective in each, complete the p-chart and np-chart calculations, plot both charts and answer the questions on the worksheet. Save the workbook and submit it in the FOL quiz. The **FOL quiz for this lab** walks you through these steps in order and is where you record your results and upload your file.

23.6.4 Process Capability: Cp, Cpk and Distributions

A process can be in control and still not be **capable** — its natural spread may be wider than the specification. You measure the length of **50 roller bearings** with a micrometer, plot the data to see what distribution it follows, compute the average, the average range and the standard deviation (both directly and as the average range divided by d_2), and calculate **Cp** and **Cpk** against the specification limits on the worksheet. A capable process has an index of at least 1.33.

AI use: not permitted — the measurements and calculations must be your own.

Learning outcomes

After this lab you should be able to:

- Plot a data set as a bar chart and scatter plot and describe its distribution.
- Estimate process standard deviation from the average range and from the raw data.
- Calculate Cp and Cpk and decide whether the process is capable.
- Explain why Cpk, not Cp, is the right index when the process is off-centre.

- Describe the steps of a real capability study — why the process must be stable and the parts must come straight from production.

Prep work

- Read Chapter 9 (SPC) — process capability, Cp and Cpk.
- Read the **Preparing a Capability Study** handout (FOL): the checklist a quality department follows before running a study — review history, size the sample, confirm the process is stable and the gauges are calibrated, quarantine and number the parts, and do not discard defective parts.

Materials & equipment

- 50 roller bearings (the same batch) and a micrometer
- Computer with Microsoft Excel
- The capability worksheet from FOL, and the *Preparing a Capability Study* handout

> **i** Instructions & submission
>
> Measure the 50 bearings and enter them in the worksheet, plot the distribution, compute the average, average range and standard deviation, then Cp and Cpk, and state whether the process is capable. Save the workbook and submit it in the FOL quiz. The **FOL quiz for this lab** walks you through these steps in order and is where you record your results and upload your file.

23.6.5 Repeatability and Reproducibility (Gauge R&R)

A measurement system has two sources of variation you can study directly: **repeatability** (one person measuring the same item repeatedly) and **reproducibility** (different people measuring the same item). Here the "instrument" is a **caliper**, and calipers differ too, so a difference between two people might be the people or might be their calipers. This lab pulls the two apart. You have your **own numbered caliper** and measure the **diameter and the length** of **ten numbered steel bars** that were sawn to length, twice each, in a randomised order. Then you **swap calipers with a partner** and measure all ten bars twice more with *their* caliper. Afterwards you and your partner exchange readings, so each of you holds a complete 2 operators $\times$ 2 calipers study.

Before you measure anything you check both calipers against a **gauge block** and record each one's bias. You then work out, for each characteristic, repeatability (EV), the **operator effect** (AV), the **caliper effect** (CV), R&R, part variation, total variation, %GRR, %GRR of the tolerance and the number of distinct categories, and compare the two characteristics: is the same caliper equally good at measuring a diameter and a length? This is the hands-on, physical

version of the study you can also run in the browser in SPC and Gauge R&R: the Virtual Caliper.

The lab is **individual**: you take your own readings and write your own report. The swap partner only lends you a caliper and, at the end, a copy of their readings.

AI use: not permitted — the measurements and calculations must be your own.

Learning outcomes

After this lab you should be able to:

- Explain repeatability, reproducibility and bias, and how each is estimated.
- Check a caliper against a gauge block and calculate its bias.
- Explain why a study in which each operator uses only their own instrument cannot separate operator from instrument, and how swapping instruments does.
- Collect a balanced, randomised Gauge R&R data set for two characteristics.
- Calculate repeatability, operator effect, caliper effect, R&R, part variation, total variation, %GRR, %GRR of tolerance and ndc with the average-and-range method.
- Judge whether a measurement system is adequate for each characteristic, explain why the verdicts can differ, and say whether operator or caliper matters more.

Prep work

- Read Chapter 5 (MSA) — bias, repeatability and reproducibility, the range method, and the resolution "rule of ten".
- Review how to read a caliper to its resolution (see Linear and Angular Measuring Instruments).
- Know who your swap partner is (assigned by your instructor).

Materials & equipment

- Your own numbered caliper (note its serial number), and your partner's for the second half
- A certified gauge block, to check the calipers
- Ten numbered steel bars, sawn to length (the set your instructor gives you)
- Lint-free wipes for the caliper jaws and the gauge block
- Computer with Microsoft Excel
- The Gauge R&R data sheet from FOL (setup, caliper check, randomised run order, diameter and length sheets, summary)

Instructions & submission

With your partner, check both calipers against the gauge block and record each bias. Measure the diameter and the length of all ten bars twice with your own caliper, in the run order on the data sheet, then swap calipers and measure them twice more with your partner's. Do not look at your earlier readings or your partner's while you measure. When you have both finished, copy your partner's readings into your sheet. Calculate the Gauge R&R results for the diameter and for the length, judge each against the acceptance bands, and decide whether the operator or the caliper matters more. Save the workbook and submit it to the FOL quiz. The **FOL quiz for this lab** walks you through these steps in order and is where you record your results and upload your file.

23.6.6 SPC and Gauge R&R: the Virtual Caliper

Before you ask an LLM to do statistical process control in Python (that is Machine Learning, Week 2), you do it yourself, by hand, as the measurement system. This lab runs entirely inside this page: **webR cells** render a **virtual vernier caliper** at a hidden, seeded true value, you read the scale and type the value, and the readings you produce carry *genuine human reading error against known ground truth* — so bias, linearity, repeatability and reproducibility are all computable, which a paper Gauge R&R cannot do. You then build an X/R chart and compute Cpk against the Précitech shaft spec of Ø 25.000 ± 0.050 mm.

AI use: not permitted for the readings or the calculations — *you* are the measurement system being studied. You may use an LLM to check your written reflection only, with disclosure.

Learning outcomes

After this lab you should be able to:

- Read a vernier scale to its resolution and record a blinded, randomised measurement set.
- Run a range-method Gauge R&R: compute EV, AV, GRR, PV, TV, %GRR and the number of distinct categories (ndc), and judge gauge acceptability.
- Estimate gauge bias and linearity against certified reference values.
- Build an X/R control chart, apply the Western Electric rules, and compute Cpk against a two-sided specification.
- Explain how an inadequate measurement system distorts a capability index.

Prep work

- Read Chapter 9 (SPC — X/R charts, the Western Electric rules, Cpk) and Chapter 5 (MSA — bias, linearity, repeatability, reproducibility, the range method, ndc).
- Skim the worked webR cells in Chapter 5: `bias-calculation` and `range-method` — you reuse that method here.
- Nothing to install and no account to create. This lab needs **no external service** — if the book page loads, the lab works.
- You run the study **solo**: you play all three operator roles yourself, as three independent measurement rounds of the same 10 parts. Do the three rounds in separate sittings so each is a genuine re-read, not a copy of the last.

Materials & equipment

- One computer with this book page open (a second screen helps — read the caliper on one, record on the other)
- The webR cells in this section (the caliper, the Gauge R&R calculator, the X/R + Cpk cell)
- A spreadsheet or paper grid to hold 90 readings (10 parts × 3 rounds × 3 trials) before you type them into the calculator cell
- The Précitech case study: shaft blanks, AISI 4140, Ø 25.000 ± 0.050 mm
- The **FOL quiz for this lab** — the six tasks in order, and where you paste your reading grid and the cell outputs and upload screenshots of the X/R chart

The lab cells

> **i** Instructions & submission
>
> Work through the **FOL quiz for this lab** during the lab period. It runs the six tasks in order — calibrate your eye, the 90-reading measurement run (three rounds of 10 parts × 3 trials), the range-method Gauge R&R, bias and linearity, the X/R chart and Cpk, and the reflection — and asks you to paste your reading grid and every cell output and to upload a screenshot of the X/R chart. The readings and the calculations must be your own; you may use an LLM to check the written reflection only, with disclosure.

The quiz drives the webR cells below. They share one page environment and run **top to bottom**: run the **setup cell first** and leave it run; every later cell depends on the objects it creates. If a later cell errors with "object not found", re-run the setup cell. Base R graphics only — do not add `library(ggplot2)`; it is slow and fragile in webR and not needed here. Read the virtual caliper to its **0.05 mm** resolution: the last main-scale mark the vernier zero has passed,

plus the vernier mark that best lines up with a main-scale mark. When you play the three operator rounds, read blind — do not look at your previous round's numbers.

```
#| label: caliper-setup
#   Virtual caliper setup - RUN THIS FIRST
# Defines the virtual caliper and the hidden true values. No third-party
# packages: base R graphics only.

set.seed(3027)

# Certified reference value of each of the 10 study parts (what a CMM reports),
# chosen to span the process range around the Ø 25.000 mm nominal.
caliper_ref <- round(sort(runif(10, 24.90, 25.10)), 3)

# The caliper under study has a small, slightly size-dependent bias (mm).
caliper_bias <- 0.03 + 0.05 * (caliper_ref - 25.00)

# What the caliper *shows* for a given part and trial: reference + gauge bias +
# a small amount of mechanism/technique slop. Seeded per (part, trial) so the
# same call always draws the same scene.
caliper_shown <- function(part, trial) {
  set.seed(1000L * part + trial)
  caliper_ref[part] + caliper_bias[part] + rnorm(1, 0, 0.015)
}

# Draw a zoomed vernier caliper (resolution 0.05 mm) at the shown value.
draw_caliper <- function(part, trial) {
  stopifnot(part >= 1, part <= 10, trial >= 1, trial <= 3)
  shown <- caliper_shown(part, trial)
  lo <- floor(shown) - 4
  hi <- lo + 24
  op <- par(mar = c(1, 1, 3, 1)); on.exit(par(op))
  plot(NA, xlim = c(lo, hi), ylim = c(3.5, 9), axes = FALSE, xlab = "", ylab = "",
       main = sprintf("draw_caliper(part = %d, trial = %d)", part, trial))
  # fixed main scale (mm), ticks every 1 mm, labels every 5 mm
  for (m in lo:hi) {
    big <- (m %% 5 == 0)
    lines(c(m, m), c(7, if (big) 6.1 else 6.6), lwd = if (big) 2 else 1)
    if (big) text(m, 7.5, labels = m, cex = 0.9)
  }
  lines(c(lo, hi), c(7, 7), lwd = 2)
  text(lo, 8.6, "main scale - millimetres", pos = 4, cex = 0.9, font = 3)
  # sliding vernier: 20 divisions spanning 19 mm  ->  each 0.95 mm, resolution 0.05
  v0 <- shown
  for (k in 0:20) {
```

```
    xk <- v0 + k * 0.95
    big <- (k %% 5 == 0)
    lines(c(xk, xk), c(5.9, if (big) 5.0 else 5.5), lwd = if (big) 2 else 1)
    if (big) text(xk, 4.7, labels = sprintf("%.2f", k * 0.05), cex = 0.8)
  }
  lines(c(v0, v0 + 19), c(5.9, 5.9), lwd = 2)
  polygon(c(v0 - 0.22, v0 + 0.22, v0), c(6.0, 6.0, 6.5), col = "black")
  text(v0, 4.2, "vernier 0", cex = 0.8, font = 3)
  invisible(NULL)
}

# Randomised part order, one per operator (A, B, C).
operator_order <- lapply(1:3, function(i) { set.seed(70 + i); sample(1:10) })
names(operator_order) <- c("A", "B", "C")

cat("Setup complete.\n\n")
cat("Randomised part order:\n")
for (nm in names(operator_order))
  cat("  Operator", nm, ":", paste(operator_order[[nm]], collapse = " "), "\n")
cat("\nUse draw_caliper(part, trial) with part 1..10 and trial 1..3.\n")
```

```
#| label: caliper-demo
# Practice reads - set demo to 1, 2 or 3, run, write your reading, then check.
demo <- 1

practice_parts <- c(2, 6, 9)
p <- practice_parts[demo]
draw_caliper(part = p, trial = 1)
cat("Write your reading (mm) before running the next line.\n")
cat("Correct reading to 0.05 mm:",
    format(round(caliper_shown(p, 1) / 0.05) * 0.05, nsmall = 2), "mm\n")
```

```
#| label: caliper-grr
#   Range-method Gauge R&R - 10 parts, 3 operator rounds, 3 trials
# Replace the numbers below with YOUR 90 readings. Each round's matrix is
# 10 rows (parts 1..10) x 3 columns (trials 1..3). You produce all three rounds
# yourself (op_A = round 1, op_B = round 2, op_C = round 3).

op_A <- matrix(c(
#  trial1  trial2  trial3
   25.00,  25.05,  25.00,   # part 1
   24.95,  24.95,  25.00,   # part 2
   24.90,  24.95,  24.90,   # part 3
   25.10,  25.05,  25.10,   # part 4
   24.85,  24.90,  24.85,   # part 5
```

```
   25.20,  25.15,  25.20,   # part 6
   24.80,  24.85,  24.80,   # part 7
   25.15,  25.15,  25.10,   # part 8
   24.75,  24.80,  24.80,   # part 9
   25.25,  25.20,  25.25    # part 10
), ncol = 3, byrow = TRUE)

op_B <- op_A + 0.01   # <-- replace with your round-2 readings
op_C <- op_A - 0.01   # <-- replace with your round-3 readings

trials    <- 3
operators <- 3
parts     <- 10

# AIAG average-and-range constants
K1 <- 0.5908                # trials = 3   (repeatability)
K2 <- 0.5231                # operators = 3 (reproducibility)
K3 <- 0.3146                # parts = 10   (part variation)

ops <- list(A = op_A, B = op_B, C = op_C)

# Repeatability (EV): mean within-operator, within-part range
part_op_range <- sapply(ops, function(M) apply(M, 1, function(x) max(x) - min(x)))
Rbar <- mean(part_op_range)
EV   <- Rbar * K1

# Reproducibility (AV): range of the operator grand means, corrected for EV
op_means <- sapply(ops, mean)
Xdiff    <- max(op_means) - min(op_means)
AV       <- sqrt(max(0, (Xdiff * K2)^2 - (EV^2 / (parts * trials))))

# Gauge R&R
GRR <- sqrt(EV^2 + AV^2)

# Part variation (PV): range of the part averages across all operators & trials
part_means <- rowMeans(sapply(ops, function(M) rowMeans(M)))
Rp <- max(part_means) - min(part_means)
PV <- Rp * K3

TV   <- sqrt(GRR^2 + PV^2)
pGRR <- 100 * GRR / TV
ndc  <- floor(1.41 * PV / GRR)

cat(sprintf("EV  (repeatability)   = %.4f mm\n", EV))
cat(sprintf("AV  (reproducibility) = %.4f mm\n", AV))
```

```
cat(sprintf("GRR                    = %.4f mm\n", GRR))
cat(sprintf("PV  (part variation)   = %.4f mm\n", PV))
cat(sprintf("TV  (total variation)  = %.4f mm\n", TV))
cat(sprintf("%%GRR                   = %.1f %%\n", pGRR))
cat(sprintf("ndc                    = %d\n", ndc))
cat("\nAIAG: %GRR < 10 acceptable | 10-30 marginal | > 30 unacceptable; ndc >= 5.\n")
```

```
#| label: caliper-bias
#   Bias and linearity vs certified reference values
# Uses op_A / op_B / op_C from the caliper-grr cell and caliper_ref from the setup cell.

ops <- list(A = op_A, B = op_B, C = op_C)
obs_part_mean <- rowMeans(sapply(ops, function(M) rowMeans(M)))   # 10 part means
bias_by_part  <- obs_part_mean - caliper_ref

cat("Part | reference | observed mean | bias (mm)\n")
for (i in 1:10)
  cat(sprintf("%4d | %9.3f | %13.3f | %+7.3f\n",
              i, caliper_ref[i], obs_part_mean[i], bias_by_part[i]))

cat(sprintf("\nOverall bias = %+.4f mm\n", mean(bias_by_part)))

lin <- lm(bias_by_part ~ caliper_ref)
slope <- coef(lin)[2]
cat(sprintf("Linearity slope = %+.4f mm of bias per mm of size\n", slope))
cat(sprintf("Linearity over the %.2f mm study range = %.4f mm\n",
            diff(range(caliper_ref)), abs(slope) * diff(range(caliper_ref))))
```

```
#| label: caliper-spc
#   X-bar / R chart and Cpk - seeded 25-subgroup production dataset
# Ø 25.000 +/- 0.050 mm. Same data for every group.

set.seed(250)
n_sub <- 25; n <- 5
USL <- 25.050; LSL <- 24.950; nominal <- 25.000

mu_series <- rep(25.002, n_sub)
mu_series[12:16] <- mu_series[12:16] + 0.030   # a real special-cause shift
x <- t(sapply(mu_series, function(m) rnorm(n, m, 0.011)))

xbar <- rowMeans(x)
R    <- apply(x, 1, function(v) max(v) - min(v))
xbarbar <- mean(xbar); Rbar <- mean(R)

A2 <- 0.577; D3 <- 0; D4 <- 2.114; d2 <- 2.326   # n = 5
```

```
UCL_x <- xbarbar + A2 * Rbar; LCL_x <- xbarbar - A2 * Rbar
UCL_R <- D4 * Rbar;           LCL_R <- D3 * Rbar

par(mfrow = c(2, 1), mar = c(4, 4, 2, 1))
plot(xbar, type = "b", pch = 19, ylim = range(xbar, UCL_x, LCL_x),
     xlab = "Subgroup", ylab = "X-bar (mm)", main = "X-bar chart")
abline(h = c(UCL_x, xbarbar, LCL_x), lty = c(2, 1, 2), col = c("red", "black", "red"))
plot(R, type = "b", pch = 19, ylim = range(0, R, UCL_R),
     xlab = "Subgroup", ylab = "R (mm)", main = "R chart")
abline(h = c(UCL_R, Rbar, LCL_R), lty = c(2, 1, 2), col = c("red", "black", "red"))
par(mfrow = c(1, 1))

sigma_hat <- Rbar / d2
Cp  <- (USL - LSL) / (6 * sigma_hat)
Cpk <- min(USL - xbarbar, xbarbar - LSL) / (3 * sigma_hat)

cat(sprintf("X-bar-bar = %.4f mm   R-bar = %.4f mm   sigma_hat = %.4f mm\n",
            xbarbar, Rbar, sigma_hat))
cat(sprintf("X-bar control limits: [%.4f, %.4f]\n", LCL_x, UCL_x))
cat(sprintf("Cp = %.2f   Cpk = %.2f   (automotive Tier-1 target: Cpk >= 1.33)\n", Cp, C
cat("\nBeyond-3-sigma subgroups:",
    paste(which(xbar > UCL_x | xbar < LCL_x), collapse = " "), "\n")
cat("Apply the remaining Western Electric rules by eye on the chart above.\n")
```

23.7 Materials Testing

Three destructive and hardness tests that show how a steel's heat treatment and processing change its strength, hardness and toughness: **hardness testing**, **tensile and microhardness testing of case-hardened steel**, and **Charpy impact testing**. **These three labs run together as a rotation over three consecutive weeks.** The class is split into three groups; each week every group is at a different one of the three labs, and each group starts at a different lab, so by the end of week three every student has completed all three (see Materials-testing rotation). Your instructor posts your group and your starting lab on FOL.

> **!** Safety in the materials laboratory
>
> Wear **safety glasses and safety footwear** — you will not be admitted without them. Do not handle dry ice with bare hands: use insulated gloves and tongs. Keep alcohol off your skin (it is absorbed through it), keep etchant off your skin, and stay clear of the pendulum's swing path in the

Charpy lab.

Testing is done in teams so that everyone can see the test, but **each student writes and submits their own report** on the template provided on FOL.

23.7.1 Hardness Testing: Rockwell and Brinell

Hardness is a material's resistance to indentation, and it is the quickest test of what a heat treatment has done to a steel. You measure the hardness of three steel specimens — **annealed**, **quenched (hardened)** and **normalized** — with two methods: the **Rockwell** tester (choosing the right scale, minor load and major load) and the **Brinell** tester (measuring the indentation diameter and converting it to a Brinell number). You average repeated readings, convert between scales, calculate the percentage deviation and the reproducibility of each method, and explain why the three heat treatments give different hardness.

AI use: not permitted for the readings and calculations. Your written answers must be your own.

Learning outcomes

After this lab you should be able to:

- Carry out Rockwell and Brinell hardness tests correctly, including scale and load selection.
- Convert between Rockwell and Brinell values using a conversion chart.
- Compare the reproducibility of the two methods for each specimen.
- Explain annealing, quenching and normalizing and relate each to the hardness you measured.

Prep work

- Read your notes on the heat treatment of steel — annealing, quenching and normalizing — and the difference between the Rockwell B and C scales.
- Review the definition of range (maximum minus minimum), which you use as your measure of reproducibility.
- Download the report template from FOL (*Hardness Testing*).

Materials & equipment

- Rockwell hardness tester (with ball and diamond indenters) and Brinell hardness tester with a measuring microscope
- Annealed, quenched and normalized steel specimens
- Rockwell–Brinell conversion chart (or an online source)
- Safety glasses and safety footwear
- The hardness report template from FOL

Instructions & submission

Test each specimen with both methods, record the readings in Tables 1 and 2 of the template, complete the calculations, answer the questions and write the conclusions on how closely the two methods agree and how reproducible each is. Submit your own report in the FOL quiz. The **FOL quiz for this lab** walks you through these steps in order and is where you record your results and upload your file.

23.7.2 Tensile and Microhardness Testing of Case-Hardened Steel

A case-hardened part is hard on the outside and tough in the core. You examine the microstructure of the **case** and the **core** of a case-hardened steel (AISI 1566) and run a **Vickers microhardness traverse** from the surface inwards in 0.5 mm steps, converting each indentation diameter to a Vickers and then a Rockwell number. You plot hardness against depth and read off the **effective case depth** (where hardness falls to HRC 50) and the **true case depth** (HRC 40). You then test tensile specimens, plot their stress–strain curves and use them to decide which material is the most brittle, the most ductile and the toughest, and you calculate stress and strain for each.

AI use: not permitted for the readings, calculations and graphs. Your written conclusions must be your own.

Learning outcomes

After this lab you should be able to:

- Explain what case hardening does and how the microstructure of case and core differ.
- Perform a Vickers microhardness traverse, convert the results to Rockwell numbers and plot a hardness profile.
- Determine the effective and true case depths from the profile.
- Explain what a tensile test measures — yield strength, ultimate tensile strength, ductility — and calculate stress and strain from load and extension.
- Compare materials from their stress–strain curves: brittle versus ductile, and toughness as the area under the curve.

Prep work

- Read the **Tensile Testing and Microhardness Testing of Case-Hardened Material** slides from FOL — loading, stress, strain, the stress–strain curve, what the tensile test reveals, and how Vickers microhardness works.

- Review the formulas for engineering stress (load ÷ original area) and strain (extension ÷ original gauge length).
- Download the report template from FOL (*Tensile and MicroHardness Test*).

Materials & equipment

- Case-hardened AISI 1566 specimen (mounted and polished) and a metallurgical microscope
- Vickers microhardness tester
- Tensile testing machine with data acquisition, and the tensile specimens
- Vickers-to-Rockwell conversion chart
- Safety glasses and safety footwear
- The tensile and microhardness report template from FOL

> **i** Instructions & submission
>
> Record the metallography sketches, complete the hardness table (depth, indentation diameter, Vickers, Rockwell), plot the two hardness graphs, state the effective and true case depths, then complete the tensile results and conclusions in the template. Submit your own report in the FOL quiz. The **FOL quiz for this lab** walks you through these steps in order and is where you record your results and upload your file.

23.7.3 Charpy Impact Testing and the Ductile-to-Brittle Transition

Some steels are tough at room temperature and shatter when cold — the failure mode behind brittle-fracture disasters in cold-weather structures. You break notched **Charpy** bars of **AISI 1018 hot-rolled** and **cold-rolled** steel in a pendulum impact tester at temperatures from well below freezing to above boiling-water heat, recording the energy absorbed at each. Low temperatures come from a bath of alcohol and **dry ice**; high temperatures from heated water. You plot impact toughness against temperature for each steel, read off the **ductile-to-brittle transition temperature (DBTT)**, and compare the fracture surfaces (fine granular for brittle fracture, necking for ductile).

AI use: not permitted for the readings, calculations and graphs. Your written conclusions must be your own.

Learning outcomes

After this lab you should be able to:

- Explain what the Charpy value measures and how rate of loading, temperature and a notch affect toughness.

- Conduct Charpy tests safely at controlled temperatures, transferring the specimen and striking it within the time limit.
- Plot impact toughness against temperature and determine the DBTT for each steel.
- Describe and explain brittle and ductile fracture surfaces.
- Explain which crystal structure is susceptible to low-temperature brittleness and why the two steels differ.

Prep work

- Read the **Material Toughness, Charpy Impact Test, and the Ductile to Brittle Temperature Transition** slides from FOL — toughness, strain rate, notch effect, the Charpy test and the DBTT.
- Read the safety rules for the pendulum tester and for dry ice before the lab.
- Download the report template from FOL (*Charpy Impact Test*).

Materials & equipment

- Charpy pendulum impact tester
- Notched AISI 1018 hot-rolled and cold-rolled Charpy specimens
- Temperature baths: alcohol and **dry ice** for the low temperatures (about −60, −40, −20 and 0 °C), room temperature, and heated water for the high temperatures (about 60 and 100 °C); a thermometer and tongs
- Insulated gloves, safety glasses and safety footwear
- The Charpy report template from FOL

i Instructions & submission

Hold each specimen at its test temperature for several minutes so the whole bar is at temperature, and break it **within five seconds** of removing it from the bath. Record the energy absorbed in the results table, plot impact toughness against temperature for both steels, sketch or photograph the two fracture surfaces, answer the questions, state the DBTT for each steel and write your conclusions. Submit your own report in the FOL quiz. The **FOL quiz for this lab** walks you through these steps in order and is where you record your results and upload your file.

23.8 Virtual Welding

Six progressive labs in **GMAW** (gas metal arc welding, often called MIG) on the **Lincoln Electric VRTEX 360+** virtual-reality welding simulator. There is no arc, no fume and no wasted metal: you practise on a simulated coupon while the machine tracks your gun position, angle and travel speed and **scores every**

weld. The process is the one automated and robotic welding cells replicate, so these labs give you the operator's view of what a welding cell has to get right. In every lab you set the machine up **from a Welding Procedure Specification (WPS)** — the document that fixes the shielding gas and flow rate, wire feed speed, voltage and polarity for the joint — weld the coupon, and save your best scored result. The labs share one routine (start-up, login, set-up screens, monitor screens), so the first lab teaches it in detail and the later labs reuse it. They build in difficulty in the order below, so it is best to deliver them in that order, although any one can be delivered alone if students are given the start-up routine first.

! Working on the simulator

- Log in with your username in the form **student ID + family name + first letter of your given name** (the first lab uses your first name). Every scored weld is saved under this name.
- Set-up values come **only from the WPS** for the joint you are welding. Wire feed speed and voltage have tolerances (for example ±5 in/min and ±1 V); a reading outside tolerance triggers an "Incorrect Settings" prompt.
- Practise as often as you like. **Save your best result to your flash drive and upload it in the FOL quiz.**

23.8.1 GMAW Flat Surfacing: Single Pass

Your first lab on the simulator. You learn the machine — power-up, the coloured function buttons, login, the set-up and monitor screens — then lay a **single-pass stringer bead** on a flat 1/4 in mild-steel plate with GMAW short-circuit transfer. The WPS supplies the settings: 75/25 argon–CO shielding gas at 25–35 cfh, wire feed speed 250 in/min, 18 V, DC electrode positive (DCEP), with a contact-tip-to-work distance of 3/8 in and a travel speed of 15 in/min.

AI use: not applicable — the work is hands-on in the simulator.

Learning outcomes

After this lab you should be able to:

- Start up the VRTEX 360+, log in, and navigate the set-up and monitor screens.
- Read a WPS and set the shielding gas, flow rate, wire feed speed, voltage and polarity from it.
- Lay a single-pass stringer bead and interpret the score.
- Save your best result to a flash drive.

Prep work

- Read the **VRTEX 360 start-up** guide from FOL (power-up, login, gas and flow, wire feed speed and voltage, placing the coupon).
- Find this lab's WPS on FOL, or in the WPS booklet at the simulator using the index at the front (look up the identification number in the top right of the WPS).
- Bring a flash drive.

Materials & equipment

- Lincoln VRTEX 360+ simulator
- The WPS for this weld (FOL or the WPS booklet at the simulator)
- The FOL quiz for this lab (and the lab assignment sheet on FOL)
- A USB flash drive

> **i** Instructions & submission
>
> Complete the lab assignment sheet — the start-up questions (the coloured buttons, units, saving a weld, the monitor views) and the set-up values you took from the WPS — then practise and save your best single-pass weld. Upload the saved weld to the FOL quiz. The **FOL quiz for this lab** walks you through these steps in order and is where you record your results and upload your file.

23.8.2 GMAW Flat Surfacing: Multi-Pass

You repeat the start-up routine and set the machine up from the WPS again, then lay a **multi-pass** weld on the flat 1/4 in plate. Each pass has to be placed relative to the one before it, so consistency of speed, work angle and stick-out matters more than in a single pass.

AI use: not applicable — the work is hands-on in the simulator.

Learning outcomes

After this lab you should be able to:

- Set up the simulator confidently from a WPS with minimal prompting.
- Lay overlapping multi-pass beads on a flat surface and compare the score of each pass.
- Identify from the monitor screens what you changed between passes and whether it improved the score.

Prep work

- Review the start-up routine and your notes and best score from GMAW Flat Surfacing: Single Pass.

- Find this lab's WPS on FOL or in the WPS booklet.
- Bring a flash drive.

Materials & equipment

- Lincoln VRTEX 360+ simulator
- The WPS for this weld
- The FOL quiz for this lab (and the lab assignment sheet on FOL)
- A USB flash drive

> **i** Instructions & submission
>
> Complete the lab assignment sheet, then practise the multi-pass weld as many times as you like and save your best result. Upload it in the FOL quiz. The **FOL quiz for this lab** walks you through these steps in order and is where you record your results and upload your file.

23.8.3 GMAW Groove Weld: Single Pass

You move from surfacing a plate to a **groove joint**: a butt joint in 3/8 in A36 plate, welded in the flat position (1G), with a backing strip, a 1/4 in root opening and a 45° included groove angle. Unlike the surfacing labs, you change the machine's default selection to match the WPS — 75/25 argon–CO , wire feed speed 350 in/min, 20 V, DCEP — and you lay a single pass in the groove.

AI use: not applicable — the work is hands-on in the simulator.

Learning outcomes

After this lab you should be able to:

- Select the weld type and joint on the simulator and set it up from the groove WPS.
- Explain how a groove joint differs from a surfacing weld.
- Complete a single-pass groove weld and interpret the score.

Prep work

- Review the start-up routine from GMAW Flat Surfacing: Single Pass.
- Read the groove-weld WPS (FOL or the WPS booklet): note the joint design, root opening and groove angle as well as the machine settings.
- Bring a flash drive.

Materials & equipment

- Lincoln VRTEX 360+ simulator
- The groove-weld WPS

- The FOL quiz for this lab (and the lab assignment sheet on FOL)
- A USB flash drive

> **i** Instructions & submission
>
> Set up the machine from the WPS (the sheet asks for the gas and flow rate, wire feed speed, voltage and polarity), practise the single-pass groove weld and save your best result. Upload it in the FOL quiz. The **FOL quiz for this lab** walks you through these steps in order and is where you record your results and upload your file.

23.8.4 GMAW Groove Weld: Multi-Pass

A thick joint cannot be filled in one pass. You weld the same groove joint with a **multi-pass** sequence — the sheet has room to score up to **seven passes** — and finish with a virtual **bend test**: if the weld is sound you earn a **bend-test certificate**.

AI use: not applicable — the work is hands-on in the simulator.

Learning outcomes

After this lab you should be able to:

- Plan and execute a multi-pass weld sequence in a groove joint.
- Record the score of each pass and see how the score changes as the joint fills.
- Explain what a bend test checks and what a pass or fail tells you about weld quality.

Prep work

- Review GMAW Groove Weld: Single Pass.
- Read the WPS for the multi-pass groove weld.
- Bring a flash drive.

Materials & equipment

- Lincoln VRTEX 360+ simulator
- The groove-weld WPS
- The FOL quiz for this lab (and the lab assignment sheet on FOL)
- A USB flash drive

Instructions & submission

Set up the machine from the WPS, weld and record the score for each pass, run the bend test and note whether you earned the certificate. Save your best result and upload it in the FOL quiz. The **FOL quiz for this lab** walks you through these steps in order and is where you record your results and upload your file.

23.8.5 GMAW T and Lap Welds

Fillet welds join two plates that overlap or meet at a right angle. In three parts you weld a **T joint** in thin (10 gauge) sheet with a single pass, a **lap joint** in 10 gauge with a single pass, and a **T joint in 1/4 in plate with three passes**, all in the horizontal fillet position. Each part has its own WPS, so you re-set the machine for each, and each finishes with a bend test.

AI use: not applicable — the work is hands-on in the simulator.

Learning outcomes

After this lab you should be able to:

- Distinguish T, lap and groove joints and the weld each calls for.
- Switch the machine between joints and thicknesses by working from the right WPS each time.
- Complete single-pass and multi-pass fillet welds, score them and run the bend test.

Prep work

- Review the start-up routine and the multi-pass groove lab.
- Locate all three WPSs (T single-pass, lap single-pass, T multi-pass) before the lab.
- Bring a flash drive.

Materials & equipment

- Lincoln VRTEX 360+ simulator
- The three WPSs for Parts A–C
- The FOL quiz for this lab (and the lab assignment sheet on FOL)
- A USB flash drive

Instructions & submission

Complete Parts A, B and C on the assignment sheet — for each, the set-up values from the WPS, the score of each pass and the bend-test result.

Save your best result of each weld and upload them in the FOL quiz. The **FOL quiz for this lab** walks you through these steps in order and is where you record your results and upload your file.

23.8.6 GMAW Vertical T Welds

The hardest position on the simulator. You weld **T joints in the vertical position** in two parts — a single pass in 10 gauge sheet, then a second T weld of up to three passes — each set up from its own WPS. Working against gravity, you must control the puddle with travel speed and gun angle much more carefully than on a flat plate.

AI use: not applicable — the work is hands-on in the simulator.

Learning outcomes

After this lab you should be able to:

- Explain why vertical welding is harder to control than flat welding.
- Set up and weld vertical T joints from the WPS and interpret the score.
- Complete the bend test and explain the result.

Prep work

- Review GMAW T and Lap Welds.
- Locate the vertical-T WPSs before the lab and note the direction of travel the WPS specifies.
- Bring a flash drive.

Materials & equipment

- Lincoln VRTEX 360+ simulator
- The vertical-T WPSs
- The FOL quiz for this lab (and the lab assignment sheet on FOL)
- A USB flash drive

i Instructions & submission

Complete Parts A and B on the assignment sheet — the set-up values from each WPS, the score of each pass and the bend-test result — then save your best result and upload it in the FOL quiz. The **FOL quiz for this lab** walks you through these steps in order and is where you record your results and upload your file.

23.9 Machine Learning: Predictive Maintenance

> **!** Deliver as a set — do not cherry-pick
>
> These six labs are **one six-week project**. Each week's output is the next week's input, and the capstone rubric spans four of the six, so they are delivered together, in order, or not at all.

These six labs are the laboratory component of Chapter Chapter 11. Each week has two hours of classroom theory (Chapter Chapter 11) followed by a two-hour lab, and the six sessions build on each other: by the end of Week 5 you will have a working predictive-maintenance web application at a public URL, and Week 6 is the live demo of it.

Each lab's tasks, deliverables and rubric are in that lab's **FOL quiz** (see How to use this chapter); the files you upload in the quiz are:

Module week	Lab	Focus	Files you upload in the FOL quiz
1	AI Literacy and Prompt Engineering	AI literacy: prompt engineering, hallucination hunting, Python code generation	`prompts.md` + `spc_starter.py`
2	Manufacturing Data and SPC	Manufacturing data: cleaning, SPC, Cpk on real plant data	Completed `week2_lab.ipynb`
3	The C-MAPSS Dataset	Capstone kickoff: understanding the C-MAPSS dataset	Notebook with sensor plots and reflection
4	Building the Predictive-Maintenance Model	Capstone build: feature engineering, model training, evaluation	Notebook + `rul_model.pkl` + `feature_cols.pkl`
5	Deploying the App	Capstone deploy: Streamlit app on the public internet	Public URL submitted via FOL

Module week	Lab	Focus	Files you upload in the FOL quiz
6	Capstone Presentations	Capstone presentations: live demo to the class	Presentation (no file submission)

Tools used across the module:

- **Google Colab** — free, no local installation required
- **Claude.ai** or **ChatGPT** (free tier) — for the Week 1–2 exercises
- **GitHub** (free account) — required from Week 5; optional earlier
- **Streamlit Community Cloud** (free) — required for the Week 5 deployment

A note on using AI assistants in these labs: you are explicitly encouraged to use LLMs to help you write, debug, and understand code throughout this module. However, you must be able to explain every line you submit. If the instructor asks "why does this `groupby().transform()` reset the rolling window at each engine boundary?" and you cannot answer, that line does not count as your work — regardless of whether it runs correctly.

> **i** The course narrative — Précitech Components
>
> **You are a new process engineer at Précitech Components**, a Tier-1 automotive supplier in Granby, Québec (the same plant documented in the Engineering Design Process Documentation and Facility Design & Simulation labs). The plant runs CNC turning cells producing transmission shaft blanks for a major OEM customer; your cell produces roughly 250 parts per day.
> Your plant manager, **Sophie Tran**, has seen what GE and Toyota are doing with predictive maintenance and wants to know whether Précitech can benefit too — without leaking IP to public tools, without trusting unverified outputs, and without a million-dollar infrastructure investment. She has assigned you to:
>
> 1. Learn enough about AI tools to use them safely and productively for daily engineering work (Weeks 1–2).
> 2. Build a proof-of-concept predictive-maintenance system using publicly available data and free tools (Weeks 3–5).
> 3. Present your findings and the deployed application to plant leadership (Week 6).
>
> By the end of Week 5 you will have a working web application at a public URL. It will not yet work on Précitech's actual machines — you used a NASA simulation dataset, not plant data — but it will demonstrate the

methodology, code quality and user interface well enough for the plant manager to decide on next steps.

23.9.1 Machine Learning, Week 1: AI Literacy and Prompt Engineering

The first lab of the machine-learning module. You compare how an LLM responds to poorly vs. well-structured prompts, deliberately trigger and document a hallucination, and use an LLM to generate the Python code you will build on in Machine Learning, Week 2. Set in the Précitech narrative above.

AI use: required — you are assessed on directing an LLM well, not on writing Python.

Learning outcomes

- How an LLM responds to poorly structured vs. well-structured prompts.
- How to detect and document an LLM hallucination.
- How to generate Python code using an LLM and evaluate whether the output is trustworthy.

Prep work

- Read Chapter Chapter 11, **Week 1** (the AI landscape; how LLMs work; why they fail; IP risk; the R-C-T-C-F prompt pattern).
- Create a free account on Claude.ai or chat.openai.com (free tier is fine).
- Create a text file `prompts.md` in your course folder — you add to it throughout the session and submit it at the end.

Materials & equipment

- Computer with internet access — one per student (individual lab)
- An LLM chat account (Claude.ai or ChatGPT, free tier)
- `prompts.md` (you create it)
- No dataset or local install needed this week
- The **FOL quiz for this lab** — the four exercises in order, and where you paste your prompts and upload `prompts.md` and `spc_starter.py`

i Instructions & submission

Work through the **FOL quiz for this lab** during the lab period. It runs the four exercises in order — bad prompt → good prompt with the R-C-T-C-F pattern, FMEA summarisation at three audiences, the hallucination hunt, and LLM-assisted Python starter code — asks you to paste each prompt and response and to say what changed, and collects `prompts.md`

and `spc_starter.py`. AI use is required; you must be able to explain every line of code you submit.

23.9.2 Machine Learning, Week 2: Manufacturing Data and SPC

You run your Week 1 Python on real (messy) plant data — 90 days of production from Précitech CNC Cell 3, with intentional data-quality problems embedded — find and document each problem without silently deleting data, then build an X-bar control chart and compute Cpk. This is the SPC you may have done by hand in the Virtual Caliper lab, now directed through an LLM.

AI use: required — you direct an LLM to write the Python and must be able to explain every line.

Learning outcomes

- How LLM-generated code behaves on real, messy industrial data.
- How to find and document data-quality problems without silent data deletion.
- How to build an X-bar control chart in Python and identify out-of-control events.

Prep work

- Read Chapter Chapter 11, **Week 2** (where plant data comes from; why industrial data is messy; statistical process control; where SPC fails).
- Have your `spc_starter.py` from Week 1 ready.
- Open Google Colab and check you can upload files.

Materials & equipment

- Computer with internet access; Google Colab
- `spc_data.csv` (90 days of CNC Cell 3 data, with embedded problems) — from FOL
- `week2_lab.ipynb` — pre-structured lab notebook, from FOL
- Your `spc_starter.py` from Week 1
- The **FOL quiz for this lab** — the four tasks in order, and where you upload the completed `week2_lab.ipynb` and paste your quality log and Cpk

i Instructions & submission

Work through the **FOL quiz for this lab** during the lab period. It runs the four tasks in order — run your Week 1 code on the real data, find

and fix the five data-quality issues with a traceable quality log, build the X-bar chart with `sigma / sqrt(n)` limits, and the reflection — asks you to describe what you did and what you found at each step, and collects the executed `week2_lab.ipynb`. You direct an LLM to write the Python and must be able to explain every line.

23.9.3 Machine Learning, Week 3: The C-MAPSS Dataset

The capstone kickoff. You load NASA's C-MAPSS turbofan run-to-failure data, compute the supervised-learning label (Remaining Useful Life, RUL) from the raw records, and identify which of the 21 sensors actually carry predictive information.

AI use: required — you direct an LLM to write the Python and must be able to explain every line.

Learning outcomes

- How to load and explore industrial time-series run-to-failure data.
- How to compute a supervised-learning label (RUL) from raw run-to-failure records.
- How to identify which sensors carry predictive information.

Prep work

- Read Chapter Chapter 11, **Week 3** (the economics of maintenance; the P-F curve; the C-MAPSS dataset; the ML problem; the fatal mistake of row-wise splitting; Random Forest intuition; evaluation).
- Get `train_FD001.txt` from the instructor's FOL folder (or the Kaggle mirror, free account).
- Open Google Colab.

Materials & equipment

- Computer with internet access; Google Colab
- `train_FD001.txt` — NASA C-MAPSS FD001 training set, from FOL
- `predictive_maintenance.ipynb` — capstone notebook, pre-structured with starter code for Weeks 3–5, from FOL
- The **FOL quiz for this lab** — the five tasks in order, and where you upload the executed `predictive_maintenance.ipynb` and the 7×3 sensor plot

Instructions & submission

Work through the **FOL quiz for this lab** during the lab period. It runs the five tasks in order — load and label the data, visualise one engine's life on a 7×3 grid, identify the constant sensors, correlate the rest with RUL, and the reflection — asks you to describe your findings and classifications, and collects the executed `predictive_maintenance.ipynb` and the sensor plot. This is the capstone kickoff; you keep working in the same notebook in Weeks 4 and 5.

23.9.4 Machine Learning, Week 4: Building the Predictive-Maintenance Model

The capstone build. You engineer rolling-window features per engine, perform a correct engine-wise train/validation split, train a Random Forest to predict RUL, evaluate it against a naive baseline, and save the model for deployment.

AI use: required — you direct an LLM to write the Python and must be able to explain every line.

Learning outcomes

- How to engineer rolling-window features per engine without data leakage.
- How to perform a correct engine-wise train/validation split and verify it is clean.
- How to train, evaluate and compare a Random Forest against a naive baseline.
- How to save a trained model for deployment.

Prep work

- Read Chapter Chapter 11, **Week 4** (feature engineering for time-series; Random Forest internals; evaluation done right).
- Have your `predictive_maintenance.ipynb` from Week 3 with the data loaded, RUL computed, and the useful-sensor list established.

Materials & equipment

- Computer with internet access; Google Colab
- Your `predictive_maintenance.ipynb` from Week 3 (continue in the same notebook)
- Python packages available in a standard Colab environment (`pandas`, `numpy`, `scikit-learn`, `matplotlib`, `joblib`)
- The **FOL quiz for this lab** — the seven tasks in order, and where you upload the executed `predictive_maintenance.ipynb`, `rul_model.pkl` and `feature_cols.pkl`

Instructions & submission

Work through the **FOL quiz for this lab** during the lab period. It runs the seven tasks in order — rolling features per engine, clip RUL at 125, the engine-wise train/validation split with the leak-check assert, train the Random Forest, evaluate against the naive baseline, feature importance, and save the model — asks you to describe each step and report your numbers, and collects the executed notebook plus `rul_model.pkl` and `feature_cols.pkl`. You must be able to explain every line.

23.9.5 Machine Learning, Week 5: Deploying the App

You wrap the trained model in a Streamlit web application, push it to GitHub, and deploy it to Streamlit Community Cloud so it is live at a public URL — the proof-of-concept deliverable for Sophie Tran. If your instructor has assigned the take-home Robotic Work-Cell Design lab, keep an eye on its due date while you build.

AI use: required — you direct an LLM to write the Python and must be able to explain every line.

Learning outcomes

- How to wrap a trained ML model in a user-facing web application.
- How to deploy that application to the public internet for free.
- The difference between "works on my machine" and "deployed and accessible to others".

Prep work

- Read Chapter Chapter 11, **Week 5** (the OT/IT boundary; dashboards and operators; model drift and the production AI lifecycle).
- Create a free **GitHub** account and a new **public** repo named `rul-predictor`.
- Create a **Streamlit Community Cloud** account using your GitHub login.
- Have `rul_model.pkl` and `feature_cols.pkl` from Week 4 on your machine.

Materials & equipment

- Computer with internet access
- GitHub account + `rul-predictor` public repo
- Streamlit Community Cloud account
- `streamlit_app.py` starter template (with `TODO` markers), from FOL
- `rul_model.pkl`, `feature_cols.pkl` from Week 4

- A sample one-engine CSV (`sample_engine.csv`) — last 30 cycles of engine #1
- The **FOL quiz for this lab** — the five tasks in order, and where you submit the public Streamlit URL, the GitHub repo URL and a screenshot of a live prediction

> **Instructions & submission**
>
> Work through the **FOL quiz for this lab** during the lab period. It runs the five tasks in order — run the Streamlit app locally, push to GitHub, deploy to Streamlit Community Cloud, polish the app, and test another student's deployed URL — asks you to describe what you did and what broke, and collects your public Streamlit URL, your GitHub repo URL and a screenshot of a live prediction. If the take-home Robotic Work-Cell Design lab is assigned, check its due date.

23.9.6 Machine Learning, Week 6: Capstone Presentations

Each student demos their deployed predictive-maintenance app live to the class and defends one technical decision — the "present to plant leadership" step of the Précitech assignment. This session carries the capstone rubric for Weeks 3–6.

AI use: required across the module — but the presentation and defence must be your own.

Learning outcomes

- Present a working technical deliverable to a non-specialist audience.
- Explain and defend a modelling decision under questioning.
- State the limitations of your own model honestly and specifically.

Prep work

- Read Chapter Chapter 11, **Week 6** (AI ethics in manufacturing; the broader AI landscape; where this is going).
- Confirm your app loads from its public URL and the sample CSV produces a sensible prediction.
- Prepare an 8–10 minute walk-through (no slides required — the app is the presentation) and be ready for 3–5 minutes of questions.

Materials & equipment

- Computer + projector/display; internet access
- Your deployed Streamlit app (public URL) and `sample_engine.csv`

- Your Week 4 notebook (for the RMSE / feature-importance figures if asked)
- The **FOL quiz for this lab** — the demo checklist and the technical-decision and limitation questions you answer in writing after presenting

> **i** Instructions & submission
>
> The **FOL quiz for this lab** is where you record the capstone presentation session: after your 8–10 minute live demo you answer, in writing, which one technical decision you explained and how you defended it, and one specific limitation of your model. Your app URL and GitHub repo (submitted in Week 5) must be live at presentation time. This session carries the capstone rubric for Weeks 3–6.

23.10 Maintenance and Root-Cause Analysis

Bring OEE, the Six Big Losses, root-cause analysis and machine learning together on one failing cell. It pairs with Chapters 12 and 19, and reuses the machine-learning pipeline, so it works best after that set.

23.10.1 TPM/OEE, Root Cause and a New Dataset

You bring the whole term together on one problem: a Précitech CNC turning cell that is not making its numbers. You compute **OEE** from a shift log and split the losses into the **Six Big Losses**; you Pareto the failure modes; you run a **5-Why and Ishikawa** root-cause analysis; and then you **direct an LLM** to build a failure classifier on a *new* dataset — the **AI4I 2020 Predictive Maintenance Dataset** from the UCI repository (CC BY 4.0, 10 000 rows, 5 labelled failure modes). This is *classification* on a differently-shaped dataset, so the C-MAPSS recipe from the capstone does not carry over — the point is to show you can transfer the skill, not copy the code. Tools: **Google Sheets**, **draw.io**, **Google Colab**.

AI use: required for the classifier task; permitted with disclosure elsewhere. As in the ML module, you must be able to explain every line.

Learning outcomes

After this lab you should be able to:

- Compute OEE (availability × performance × quality) from a shift log and attribute the losses to the Six Big Losses.
- Build a Pareto chart of failure modes and identify the vital few.
- Run a 5-Why and an Ishikawa (fishbone) analysis using measured process data as evidence.

- Direct an LLM to build and evaluate a classifier on an unfamiliar, class-imbalanced dataset, and choose metrics that survive the imbalance.
- Write the D4 (root cause) and D5 (corrective action) sections of an 8D report, tying model evidence back to the fishbone.

Prep work

- Read Chapter 13 (TPM, OEE, the Six Big Losses) and Chapter 21 (troubleshooting, 5-Why, Ishikawa, the 8D report).
- Skim your capstone notebook from the machine-learning module — you reuse the *workflow* (load, explore, split, train, evaluate) but not the code.
- Open the AI4I 2020 dataset page and confirm you can download the CSV in the browser. Columns: air temperature, process temperature, rotational speed, torque, tool wear, a machine-failure flag, and five failure-mode flags — **TWF** (tool wear), **HDF** (heat dissipation), **PWF** (power), **OSF** (overstrain), **RNF** (random).
- Download `precitech_shift_log.csv` from FOL (one week of shift records for the CNC cell: planned time, downtime events, ideal cycle time, count, scrap).

Materials & equipment

- Computer with internet access
- **Google Sheets** — OEE calculation and the Pareto chart
- **draw.io** — the Ishikawa diagram
- **Google Colab** — the classifier
- `precitech_shift_log.csv` from FOL
- **AI4I 2020** CSV, downloaded in-browser from UCI
- Chapter 13 OEE definitions and Chapter 21 8D template
- The **FOL quiz for this lab** — the five tasks in order, and where you upload the Sheets OEE/Pareto file, the draw.io Ishikawa export and the Colab notebook link

i Instructions & submission

Work through the **FOL quiz for this lab** during the lab period. It runs the five tasks in order — OEE and the Six Big Losses from the shift log, the failure-mode Pareto on the AI4I data, the 5-Why and Ishikawa with measured feature evidence, directing an LLM to build the failure classifier, and the 8D D4/D5 sections — asks you to describe your method and results, and collects the Sheets file, the draw.io export and the Colab notebook link. The classifier task requires an LLM; disclose AI use elsewhere and be able to explain every line.

23.11 Teacher preparation

Everything the instructor needs to prep the term: how to choose and schedule labs from the catalogue (including the labs that must be delivered as a set), then the room, equipment and software per lab, a consolidated shopping list, a software/accounts checklist, and a room-booking checklist. Items marked [**confirm**] are not stated in the source handouts.

> The full term-prep material for the current offering — quick-start, term grid, grade-item checklist, paste-ready Course Plan notes, the AI-use policy, answer keys, rubrics and the per-lab failure modes for the labs it covers — lives in the **instructor handbook** (`instructor/instructor-guide.qmd`, renders to `instructor/instructor-guide.pdf` for FOL). The matrix below is the quick per-lab view.

23.11.1 Choosing and scheduling labs

This chapter holds more labs than one term can deliver, so **pick labs to fit your course, your own background and your equipment.** Most labs stand alone. Three groups of labs do not, and they are the constraints to plan around:

> ! Labs that must be delivered together
>
> - **The machine-learning labs are a set — they cannot be cherry-picked.** The six labs form one six-week predictive-maintenance project: each week's output is the next week's input (`prompts.md` and `spc_starter.py`, then the cleaned SPC notebook, then the C-MAPSS notebook, then the trained model files, then the deployed app, then the presentation of that app), and the capstone rubric spans four of the six. Deliver all six, in order, in six consecutive weeks (each week pairs two hours of Chapter Chapter 11 theory with the two-hour lab) — or leave the module out.
> - **The three materials labs are a 3-week rotation.** Deliver them over three consecutive weeks: split the class into three groups and have each group work through all three labs, each group starting at a different one — see Materials-testing rotation. All three labs run in the same room *every* week, so you cannot deliver just one or two of them. **The Charpy lab needs dry ice, and the lab technician needs two weeks' notice to order it for the three weeks.**

Dependencies between other labs (deliver the first before the second):

Lab	Needs first
Manufacturing Facility Design & Simulation (AnyLogic)	Engineering Design Process Documentation (its layout drawing becomes the simulation)
Robotic Work-Cell Design in RoboDK for Web	Lean: Value Stream Mapping, Takt and Pull (supplies the takt time)
Machine Learning, Week 2	Machine Learning, Week 1 — and, ideally, the Virtual Caliper (the SPC done by hand)
TPM/OEE, Root Cause and a New Dataset	The machine-learning set (reuses the whole pipeline)
Any Virtual Welding lab after the first	GMAW Flat Surfacing: Single Pass (teaches the start-up routine) — *recommended*, not strictly required

Other planning notes

- **Take-home lab.** Release Robotic Work-Cell Design about three weeks before it is due, and reserve 15 minutes in a lab period to demo the tool.
- **Statistical labs.** Deliver *Statistics with Excel* first if students are not yet comfortable with Excel; the Virtual Caliper and hands-on Gauge R&R overlap, so choose one or both.
- **Simulator and bench capacity.** Virtual welding and the bench labs are limited by the number of simulators and instruments; plan rotations accordingly [**confirm counts**].
- The default plan for the current offering, lab by lab and week by week, is in the course schedule in `index.qmd`.

23.11.2 Materials-testing rotation

Split the class into **three groups** (A, B, C). Each group starts at a different lab and moves on every week, so all three stations run every week and every group completes all three labs:

	Week 1	Week 2	Week 3
Group A	Hardness Testing	Tensile and Microhardness	Charpy Impact
Group B	Tensile and Microhardness	Charpy Impact	Hardness Testing
Group C	Charpy Impact	Hardness Testing	Tensile and Microhardness

- **Dry ice (Charpy lab).** One group is at the Charpy station each of the three weeks, so dry ice is needed in **all three weeks**. Give the lab

technician the three lab dates and **at least two weeks' notice** to order it for the full three-week block; agree the quantity per session with them [**confirm**]. Dry ice sublimates, so it is normally collected on the day of the lab [**confirm with the technician**].

- **Staff the Charpy station properly.** Cold-bath handling, dry ice and a swinging pendulum make it the highest-risk station; the technician or instructor should be at it for the whole period.
- **Each group needs its own specimens.** Count specimens per station per week (see the shopping list).
- Post the group list and the rotation on FOL before week 1 so students know which lab they start with and can do that lab's prep.
- Tests run in teams, but each student **submits an individual report** on the FOL template.

23.11.3 Per-lab prep matrix

Lab	Room	Equipment (reusable)	Software / online	Consumables & printing
Logic Thinking for Robot Control: Rocksi	Computer lab, 1 PC/student [**confirm**]	PCs with internet	**Rocksi** (browser, no plug-in); prelab YouTube `3THTZFnrxtc`	None (all instructions and submission in the FOL quiz)
Robotic Work-Cell Design in RoboDK for Web *(take-home)*	**None — done from home.** Reserve 15 min in a lab period to demo the tool	Student's own computer	**RoboDK for Web** (`web.robodk.com/simulation`) — no install, no licence key; a screen-recording tool	None; provide a `.rdk` station file and an expected cycle-time range
Engineering Design Process Documentation	Computer lab with full Office [**confirm**]	PCs	**Microsoft Word/Visio**, **Microsoft Project**, **Microsoft Excel**	None; post the Précitech case study + re-upload FOL templates (run `python R/gen_lab_templates.py`): BOM `.xlsx`, Gantt `.xlsx` (Project-importable), HSE `.docx`

Lab	Room	Equipment (reusable)	Software / online	Consumables & printing
Manufacturing Facility Design & Simulation (AnyLogic)	Computer lab **[confirm]**	PCs (check AnyLogic PLE model-size limit)	**AnyLogic Personal Learning Edition**, Microsoft Visio/Project/Excel; AnyLogic order-fulfilment demo model **[confirm which]** + prelab video	None
Lean: Value Stream Mapping, Takt and Pull	Computer lab with internet	PCs	**draw.io** (browser), **Google Sheets**, **AnyLogic Cloud** (free public tier) — all browser, no install	None; generate + stage `vsm_station_data.csv` (run `Rscript R/make_lab_data.R`); build/fork the AnyLogic Cloud push–pull model and expose its input parameters
Ergonomics in Automated Manufacturing (REBA)	Computer lab with internet **[confirm]**	PCs	Browser; ergonomics + REBA videos	Print **REBA worksheets** + **REBA cheat sheet** (per student); stage task clips/photos (Lifting Boxes, Workstation, Cleaning) on FOL

Lab	Room	Equipment (reusable)	Software / online	Consumables & printing
Linear and Angular Measuring Instruments	Metrology lab	Vernier calipers, vernier height gauges, micrometers, gauge-block set [**confirm counts**]	—	Post the assignment sheet on FOL
Groove Angle Measurement (V-Block)	Metrology lab	Surface plates, V-blocks, vernier height gauges, calipers, two rollers of different diameters	—	Post the sheet; check the angle equation is legible (see instructor notes)
Taper Angle Measurement with a Sine Bar	Metrology lab	Surface plates, sine bar, dial indicator + stand, gauge-block set, three tapers	—	Post the sheet
Micrometer and Caliper Calibration	Metrology lab	Micrometer and caliper to calibrate, gauge-block set with its serial/certificate	—	Methyl alcohol, lint-free wipes; print the calibration report forms
Statistics with Excel for Quality Control	Computer lab with Excel	PCs	**Microsoft Excel**	Post the worksheet

Lab	Room	Equipment (reusable)	Software / online	Consumables & printing
Variable Control Charts: $\bar{X}$, R and S	Metrology lab + PCs	100 roller bearings per class/bench **[confirm]**, micrometers or calipers	**Microsoft Excel**	Post the worksheet + the control-chart constants table
Attribute Control Charts: p and np	Bench + PCs	Bead bags, enough for ten samples of 50	**Microsoft Excel**	Post the worksheet; assign the "defective" colour per student
Process Capability: Cp, Cpk and Distributions	Metrology lab + PCs	50 roller bearings, micrometers	**Microsoft Excel**	Post the worksheet + the *Preparing a Capability Study* handout
Repeatability and Reproducibility (Gauge R&R)	Metrology lab + PCs	One serial-numbered caliper per student, a certified gauge block, 10 numbered sawn steel bars per pair (a pool of 10–30)	**Microsoft Excel**	Stage `Gauge_RR_Bar_Stock_Data_Sheet.xl` (from `R/gen_grr_sheet.py`); enter the bars' real tolerances; assign swap partners and each pair's ten bars
SPC and Gauge R&R: the Virtual Caliper	Computer lab, 1 PC/student	PCs with internet (to load the book page)	**This book's webR cells only — no external site, no account, no dataset.** Just confirm the rendered `23-Labs.html` page and its webR cells load	None; print the answer key from `Rscript R/make_lab_data.R` for marking

Lab	Room	Equipment (reusable)	Software / online	Consumables & printing
Hardness Testing: Rockwell and Brinell	Materials lab	Rockwell tester, Brinell tester + measuring microscope, annealed / quenched / normalized specimens	—	Conversion chart; post the report template
Tensile and Microhardness Testing of Case-Hardened Steel	Materials lab	Tensile testing machine with data acquisition, Vickers microhardness tester, metallography kit (mounting, grinding, polishing, microscope), mounted case-hardened AISI 1566 specimens	—	Nital etchant, polishing consumables **[confirm]**; students bring a memory stick for the tensile data **[confirm]**; post the report template
Charpy Impact Testing and the Ductile-to-Brittle Transition	Materials lab	Charpy pendulum tester, notched AISI 1018 hot- and cold-rolled specimens, insulated cold-bath container, hot-water bath, thermometers, tongs, insulated gloves	—	**Dry ice — order 2 weeks ahead for the 3-week block**; alcohol for the cold bath; post the report template

Lab	Room	Equipment (reusable)	Software / online	Consumables & printing
Virtual Welding — all six labs	Welding-simulator lab	Lincoln VRTEX 360+ units, WPS booklets **[confirm number of units]**	—	USB flash drives; post the start-up guide, the WPSs and the six assignment sheets on FOL
Machine Learning, Week 1: AI Literacy	Computer lab, 1 PC/student	PCs with internet	**Claude.ai** or **ChatGPT** (free tier); Google Colab	None; post FMEA table + blank `prompts.md` on FOL
Machine Learning, Week 2: Data & SPC	Computer lab	PCs with internet	**Google Colab**; Python (pandas, numpy, matplotlib)	None; stage `spc_data.csv` + `week2_lab.ipynb` on FOL
Machine Learning, Week 3: C-MAPSS	Computer lab	PCs with internet	**Google Colab**; scikit-learn stack	None; stage `train_FD001.txt` + `predictive_maintenance.ipynb` on FOL
Machine Learning, Week 4: Building the Model	Computer lab	PCs with internet	**Google Colab**; scikit-learn, joblib	None; have a reference solution notebook (expected RMSE 21)
Machine Learning, Week 5: Deploying the App	Computer lab	PCs with internet	**GitHub** + **Streamlit Community Cloud** accounts; Google Colab	None; stage `streamlit_app.py` + tested `requirements.txt` on FOL
Machine Learning, Week 6: Capstone Presentations	Room with projector/display + internet **[confirm]**	Projector; PCs	Student public Streamlit URLs	None; presentation schedule

Lab	Room	Equipment (reusable)	Software / online	Consumables & printing
TPM/OEE, Root Cause and a New Dataset	Computer lab with internet	PCs	**Google Sheets**, **draw.io**, **Google Colab**; **AI4I 2020** CSV from `archive.ics.uci.edu`	None; generate + stage `precitech_shift_log.csv` (run `Rscript R/make_lab_data.R`); host an AI4I 2020 copy on FOL as a fallback

23.11.4 Instructor notes by lab

Marking guidance and lab-specific prep, moved here from the individual lab sub-chapters. For the software labs the tasks, deliverables and rubric each lab is marked against live in that lab's **FOL quiz**; for the hands-on labs they are on the FOL worksheet or report template. The full answer keys and per-lab failure modes for the labs in the current offering are in the instructor handbook.

23.11.4.1 Logic Thinking for Robot Control: Rocksi

- **Prep:** confirm Rocksi is reachable through the lab network/firewall *the day before* — it is the single point of failure for this lab. Have an offline fallback (screenshots / a slide walk-through) ready.
- **LightBot has been dropped:** it needs a Flash-era plug-in that is not available on college machines. The whole lab now runs in Rocksi — sequencing, loops, conditional/branching blocks, a pick-and-place sequence and a simulated I/O.
- Individual lab — one machine per student; no group submission.
- **Marking:** award the loop / conditional marks for the *explanation*, not just a working program. For the optimisation challenge, ask for the block count as evidence.

23.11.4.2 Robotic Work-Cell Design in RoboDK for Web

- **Release three weeks before the due date** (in the default plan: week 12, alongside Week 3 of the machine-learning set, due at the end of week 15). This mirrors the course plan's existing take-home lab pattern (released three weeks out, video submission).
- **Prep:** the day before release, run through Tasks 1–3 in RoboDK for Web yourself on the college network — confirm `web.robodk.com` is reachable, the library loads, and a station can be saved and re-loaded.

Have a reference station file and an expected cycle-time range (from the `R/make_lab_data.R` answer key) ready for marking.
- No lab room is needed — this is done from home — but reserve 15 minutes in a lab period to demo the tool and walk through the brief.
- **Marking:** the takt comparison in Task 4 only works if the student's takt from the Lean lab was correct; if it was not, mark Task 4 on the *method* against a takt you supply. Watch for cycle times that are physically implausible (joint speeds cranked past limits) — that is where the "your own work" line is tested.

23.11.4.3 Engineering Design Process Documentation

- **Prep:** verify Microsoft **Project** and **Visio** licences are live on every lab PC (the parts most likely to be missing from a standard Office install). Run `python R/gen_lab_templates.py` to (re)generate the three FOL templates into `labs/templates/` and re-upload them each term; confirm the Gantt template's Import Wizard walk-through still matches your Project version, and save a native `.mpp` copy once you've verified it. Post the case study on FOL as its own one-page handout.
- **Continuity:** this lab and Facility Design & Simulation share the Précitech kitting cell. Tell students up front that the layout drawing they produce here becomes the model they build in the AnyLogic lab.
- **Marking:** look for traceability — the spec sheet, BOM, task list, schedule and layout should all describe the *same* cell and the *same* kit. Check the starvation / blockage annotations on the layout are present and sensible; they are the link to the AnyLogic lab.

23.11.4.4 Manufacturing Facility Design & Simulation (AnyLogic)

- **Prep:** AnyLogic PLE install is the biggest failure point — push it to lab PCs ahead of the term and test a full model run. PLE has a model-size limit; confirm the demo model stays under it. Have the exact model file on FOL.
- Give the Phase 1 (facility-drawing) sign-off some rigour — students who skip it produce screenshots with no layout to tie them to. Check the layout matches their Design Documentation submission.
- **Marking:** the drawing carries ~40 % of the mark; analysis should reference the *specific* statistic that moved *and* the case-study consequence (e.g. "lower reorder level → more unavailable goods → longer preorder waiting time → kitting station idle, shipments late").

23.11.4.5 Lean: Value Stream Mapping, Takt and Pull

- **Prep:** generate `vsm_station_data.csv` with `Rscript R/make_lab_data.R` and stage it on FOL. Publish the answer key from the same script — expected takt, bottleneck station, PCE and the push-vs-pull WIP/lead-time ordering.

- Build or fork the AnyLogic Cloud push/pull model once and confirm the input parameters students need (control policy, WIP cap) are exposed on the public run screen. Test it through the college firewall the day before.
- Confirm draw.io and cloud.anylogic.com are reachable on the lab network (see the software checklist below).
- **Marking:** the map and the takt/PCE arithmetic must be the student's own (AI not permitted for those). Check the lead-time ladder actually totals to the numbers quoted — a ladder that does not add up is the most common error.
- This is the takt reference for **Robotic Work-Cell Design** — students compare the RoboDK cycle time against the takt they compute here, so make sure the takt value is right.

23.11.4.6 Ergonomics in Automated Manufacturing (REBA)

- **Print:** REBA worksheets + the REBA cheat sheet, one per student.
- Stage the three task video/photo sets on FOL (`Lifting Boxes.pdf`, `Workstation.pdf`, `Cleaning.pdf` and the matching clips).
- Individual lab — the FOL quiz assigns each student one of the three tasks; spread the assignments roughly evenly across the class.
- **Marking:** REBA scoring is where students lose marks — check the trunk/neck/leg and arm/wrist sub-scores and the coupling/activity adjustments, not just the final number.

23.11.4.7 Linear and Angular Measuring Instruments

- **Prep:** post the assignment sheet; lay out one of each instrument per bench so students can photograph and label real instruments rather than copy pictures.
- **Post the corrected assignment sheet** (`Linear and Angular Measurements - fixed.docx` in `LabInstructions/Quality_Labs/fixed/`). The original caliper problem called the main scale's 1 mm division a "least count of 0.1 mm", which made the question ambiguous; both least-count problems are now worded unambiguously. **Answers:** caliper = 1 mm − 0.9 mm = **0.1 mm**; micrometer = 1 mm ÷ 50 = **0.02 mm** without the vernier and 0.02 mm ÷ 5 = **0.004 mm** with it.
- **Marking:** the derivation of the least-count formula, not just the number.

23.11.4.8 Groove Angle Measurement (V-Block)

- **Prep:** confirm each roller pair is precisely measured and the roller diameters differ enough that both rollers touch only the sloping sides of the groove (as the sheet requires). Check the V-blocks and surface plates are clean.
- **Answer key:** the sheet's equation (a Word equation object) is $\alpha = 2\sin^{-1}\left[\dfrac{d_2 - d_1}{2(h_2 - h_1) - (d_2 - d_1)}\right]$, with d_1, d_2 the roller diameters and h_1,

h_2 the heights from the plate to the top of each roller.
- **Marking:** compare the repeats across the four samples; large scatter usually means the roller moved or the scriber pressed on it.

23.11.4.9 Taper Angle Measurement with a Sine Bar

- **Prep:** confirm the length L of each sine bar (roller centre distance) and post it, and check the gauge-block set has fine enough steps to reach parallelism.
- Walk students through wringing once (see the Instruments lab).
- **Marking:** the sheet takes as the taper's *half*-angle and reports 2 as the whole angle — check students follow that definition and compare 2 with the tapers' stated angles.

23.11.4.10 Micrometer and Caliper Calibration

- **Prep:** the source sheet's check sizes are **inch** sizes (micrometer 0.106 to 0.954 in in nine steps; 6 in or 8 in caliper in ten steps of range/10). If your instruments are metric, substitute an equivalent sequence and update the report forms.
- The gauge-block set must carry a serial number and a certificate: the report asks for the standard's identification, and traceability is the point of the lab.
- Let instruments and blocks stabilise to room temperature before the lab starts; cold blocks from a cabinet give false deviations.
- **Marking:** the deviation at each size, the accuracy over the range compared with the published accuracy, and an actual finding (in / out of tolerance) rather than a restatement of the numbers.

23.11.4.11 Statistics with Excel for Quality Control

- **Prep:** post the worksheet; check the students' Excel version supports `VAR.P`, `VAR.S`, `STDEV.P` and `STDEV.S`.
- **Marking:** the sheet carries its own mark allocations (interpretation of mean/median/mode, range/variance/standard deviation, the labelled scatterplot, the histogram and the four questions). Check the histogram bins match the ones given and that the "normal curve?" answer is argued from the shape.

23.11.4.12 Variable Control Charts: X̄, R and S

- **Prep:** you need a batch of 100 roller bearings (or an equivalent part) and one measuring instrument per student or pair. The worksheet reads **length**; the bearings are tapered, so tell students *not* to measure diameter. Post the control-chart constants table (the constants for subgroups of ten are already in the sheet: A2 = 0.308, B3 = 0.284, B4 = 1.716, D3 = 0.223, D4 = 1.777).

- **Post the corrected worksheet** (`Variable Control Charts - fixed.xlsx` in `LabInstructions/Quality_Labs/fixed/`): its out-of-control rules now match the Western Electric rules in Chapter 9 (rules 1–4), plus two extra pattern rules (fifteen points within 1 ; eight points none within 1).
- **Marking:** the subgroup statistics, the limits, the three charts and a justified in/out-of-control call.

23.11.4.13 Attribute Control Charts: p and np

- **Prep:** put together bead bags with a known mix of colours, and tell each student (or bench) a different "defective" colour so results cannot be copied.
- **Marking:** the p-chart limits use the sample size $n = 50$; a negative lower limit is set to 0. Check the reasoning on "what to do next" and on the complementary tool (a Pareto chart is the expected answer).

23.11.4.14 Process Capability: Cp, Cpk and Distributions

- **Post the corrected worksheet** (`Process Capability - fixed.xlsx` in `LabInstructions/Quality_Labs/fixed/`). The original Cp formula lacked brackets (it now reads (USL − LSL) / (6)) and told students to use Cpk "if the distribution is not normal" — Cpk is the index for an **off-centre** process, and the corrected sheet says so.
- The specification on the sheet is USL = 0.5008 and LSL = 0.4992 (inches). Confirm it suits the bearings and micrometer you supply.
- **Marking:** the estimate two ways (R/d and STDEV.P), Cp, Cpk, and the capable / not-capable decision against 1.33.

23.11.4.15 Repeatability and Reproducibility (Gauge R&R)

- **Design:** an **individual** lab built on a **swap pair**. Each student is an operator with their own serial-numbered caliper. Two students swap calipers, so the study is fully crossed: 2 operators × 2 calipers × 10 bars × 2 trials, for the diameter and the length. Without the swap each operator is confounded with their caliper (the operator and caliper effects cannot be told apart); with it, the operator difference is read with the caliper held constant and the caliper difference with the operator held constant. Each student takes 80 readings (10 bars × 2 trials × 2 calipers × 2 characteristics) and copies their partner's 80 into their own sheet, so **the only thing shared is the readings** — the calculations, graphs and answers are the student's own.
- **Prep — equipment:** one caliper per student, each with its own serial number (the sheet records it); one certified gauge block (choose a size near the bar diameter if you have a choice); ten numbered bars per pair, drawn from your pool of 10–30. With 30 bars, three pairs can measure at once; with fewer, stagger the pairs so the same bars are never needed twice at

the same moment. Give partners the **same ten bars**, entered in the same (ascending) order on both sheets, or the rows will not line up. Stamp or engrave the numbers — labels that fall off ruin the study.

- **Prep — pairs and numbers:** assign swap partners, and give every student a **student number from 1 to 30**, different from their partner's. The number selects the student's random run order on the sheet (a look-up in the *Order table* sheet), so students beside each other are rarely after the same bar. With an **odd** class, let the extra student pair with someone from a finished pair, who then repeats only the swap half of the study using the extra student's caliper.
- **Prep — workbook:** run `python R/gen_grr_sheet.py` and stage `labs/templates/Gauge_RR_Bar_Stock_Data_Sheet.xlsx` on FOL. **Replace the placeholder tolerances on the Setup sheet** (Ø1.000 ± 0.005 in and length 2.000 ± 0.020 in) with the real specification for your bars; %GRR of tolerance depends on them. The sheet is in **inches** and assumes a **0.0005 in** caliper resolution (typical of a digital caliper) — if yours read 0.001 in (dial or vernier), change the resolution cell; it sets the caliper-check limit (two resolution steps). A metric caliper needs the readings and tolerances entered in the same unit throughout. The run orders are fixed random sequences; change the seed in `run_orders()` to regenerate them.
- **Operational definitions — agree these before the students start:** diameter at the mid-length of the bar, jaws square to the axis, one reading, no rotating the bar between trials; length between the two sawn faces, with the jaws placed at the centre of each face (sawn faces are rough and not perfectly square, which is a real source of variation). Students measure both dimensions each time they pick a bar up, in the run order, and must not look at their earlier readings or their partner's until both have finished.
- **Gauge block:** one block means a queue — partners check their two calipers together at the start of the period, one at a time. Clean the jaws and the block, handle the block by its edges, return it to its case. The check covers one size only; use it to discuss what it does not tell you (linearity, the bar's length).
- **The method and its limits:** repeatability EV comes from the average range of the four conditions; the operator effect AV and caliper effect CV each come from the difference between the two levels ($d^* = 1.41$), corrected for repeat scatter over the 40 readings behind each average. Two levels each is a crude basis — a small real difference is magnified by the 5.15/1.41 factor — so treat AV and CV as estimates. If you want firmer numbers, **pool the class**: collect every student's readings and run a class-level study with many operators and calipers (or an ANOVA, as in Chapter 17). The sheet also reports an operator-by-caliper **interaction** contrast against about 2 of repeat scatter: if it is noticeable, the operator effect depends on the caliper and the split is only approximate.
- **Timing (indicative — not tested):** caliper check about 10 minutes for the two calipers; 80 readings per student take roughly 45–60 minutes

including the swap. Plan a second period for the calculations if the first is short.

- **Answer key (simulated):** `Gauge RR Bar Stock - EXAMPLE (instructor).xlsx` in `LabInstructions/Quality_Labs/fixed/` holds **simulated** readings (not measurements), in inches, with tolerances ± 0.005 in and ± 0.020 in, a 1.000 in gauge block, a caliper bias of 0 and −0.0007 in, and a partner technique offset of +0.0015 in on length. Its formulas were checked against an independent calculation. Diameter: EV 0.0021, AV 0.0006, CV 0.0029, GRR 0.0036, PV 0.0043, TV 0.0056 in, **%GRR 65 %**, %tolerance 36 %, ndc = 1, largest component the caliper. Length: EV 0.0059, AV 0.0050, CV 0.0025, GRR 0.0082, PV 0.0541, TV 0.0547 in, **%GRR 14.9 %**, %tolerance 20 %, ndc = 9. The caliper difference recovered from the bars (+0.0008 in on diameter, +0.0007 in on length) matches the gauge-block difference (+0.0007 in); the operator difference on length (−0.0014 in) matches the built-in offset. Real student results will differ.
- **What to expect (a teaching expectation, not a measured result):** sawn lengths vary a lot from bar to bar, while drawn bar stock has almost the same diameter along its stock; so the part variation is large for length and small for diameter. The same caliper therefore tends to look better on length than on diameter when judged against total variation — a good moment to compare %GRR of total variation with %GRR of tolerance, and to ask what the 0.0005 in resolution has to do with it.
- **Constants (5.15 convention):** EV = 5.15 R / 1.128 (d , 2 trials); AV = $\sqrt{[(\text{Xdiff,operator} \times 5.15 / 1.41)^2 - \text{EV}^2 / 40]}$; CV = $\sqrt{[(\text{Xdiff,caliper} \times 5.15 / 1.41)^2 - \text{EV}^2 / 40]}$, with d * = 1.41 for two levels and 40 = 10 bars × 2 trials × 2; PV = 5.15 Rp / 3.18 (d * for 10 parts); GRR = $\sqrt{(\text{EV}^2 + \text{AV}^2 + \text{CV}^2)}$. Ratios such as %GRR do not depend on the convention.
- **Marking:** the check of both calipers and their biases, the balanced data set (80 readings per characteristic, 40 of them the partner's) and how independence was kept, each calculation step (accept answers that follow from the student's own R and Xdiff), the verdicts against the acceptance bands, and the operator-or-caliper conclusion against the gauge-block difference. **Common slips:** readings typed in the wrong bar's row, or the partner's columns pasted into the wrong place (their C:F go into the student's G:J); partners using different bars or a different order; results quoted as %GRR of tolerance when total variation was asked (or the reverse); a negative value under the AV or CV root not set to zero; ndc rounded instead of truncated; placeholder tolerances left in; readings in the wrong unit.

23.11.4.16 SPC and Gauge R&R: the Virtual Caliper

- **Prep:** none beyond opening the rendered page and checking the webR cells load. This lab has **no external dependency** — no site, no account, no dataset file.
- Run `Rscript R/make_lab_data.R` to print the Virtual Caliper answer

key: the `caliper_ref` values, the expected overall bias (+0.03 mm), and the expected %GRR / ndc / Cpk ranges. The caliper and SPC data are generated inside the webR cells, so there is no file to stage.
- Individual lab — each student plays all three operator rounds themselves, in separate sittings. A reading grid where the three rounds are identical (or every trial matches) was not read blind.
- **Marking:** check that %GRR and ndc are consistent with the EV/AV/PV reported.
- This lab is the "do it yourself first" half of the arc into Machine Learning, Week 2 (SPC in Python, directed by an LLM). Point that forward explicitly.

23.11.4.17 Hardness Testing: Rockwell and Brinell

- **Prep:** stage annealed, quenched and normalized specimens and check the two testers are calibrated with test blocks. Post the report template.
- Safety glasses and safety footwear are required; keep students clear of the indenter when loading.
- **Marking:** the average and % deviation, the reproducibility (max − min) for each method, and a reasoned explanation of why the hardened specimen gives a narrower spread than the annealed one. For the online-conversion questions, Rockwell C 51 and Brinell 500 are roughly equivalent, as are Rockwell B 85 and Brinell 165 — check against your conversion chart.

23.11.4.18 Tensile and Microhardness Testing of Case-Hardened Steel

- **Prep:** mounted, polished case-hardened AISI 1566 specimens; 2% nital etchant and polishing consumables if students etch (keep alcohol and etchant off skin, ventilate); tensile specimens of **two or more** different materials so that the most-brittle / most-ductile / toughest comparison has something to compare.
- Tell students to bring a memory stick if the tensile machine's data are saved to file [**confirm**].
- The template defines the effective case depth at HRC 50 and the true case depth at HRC 40 — check that matches the convention you teach.
- **Marking:** the hardness table and the two graphs, the case depths read from them, and stress and strain calculated with the working shown.

23.11.4.19 Charpy Impact Testing and the Ductile-to-Brittle Transition

- **Order dry ice two weeks ahead** through the technician, for all three weeks of the rotation.
- **Prep:** notched AISI 1018 hot-rolled and cold-rolled specimens (the template has seven temperature rows: about −60, −40, −20, 0 °C, room

temperature, 60 and 100 °C). Set up the alcohol/dry-ice bath and the hot-water bath before the period so specimens can equilibrate for several minutes.

- **Safety:** insulated gloves for dry ice, never sealed in a container, keep the room ventilated. Before each swing, make sure everyone is clear on both sides of the pendulum and out of its striking distance; the specimen is centred on the anvil with the notch facing away from the striking edge. The break must be made **within five seconds** of leaving the bath.
- **Marking:** the plotted curves, a DBTT read from each curve, the fracture-surface descriptions (fine granular vs. necking), and the answers on the Charpy value and BCC crystal structure.

23.11.4.20 Virtual Welding (all six labs)

- **Prep:** power up the VRTEX 360+ units before the lab (they need two or more minutes to start), check the WPS booklets are at the simulators, and post the start-up guide, the WPSs and the six assignment sheets on FOL. Students need a flash drive for their best weld.
- **The FOL quiz for each welding lab replaces the source PDF assignment sheets**, which are not editable and contain copy-paste slips (the quizzes carry the correct titles and steps): the flat-surfacing multi-pass sheet is titled "Single Pass"; the groove single-pass sheet is headed "Lab Assignment 2" and describes flat surfacing; the T-and-lap file is named for lab 4 but headed lab 5; the vertical-T sheet's Part B is headed "Lap Single Pass" but selects **T 10GA** and scores three passes. The single-pass and multi-pass groove labs cite the **same WPS**, whose table lists two passes while the multi-pass sheet has seven score lines. The WPS pages for the flat multi-pass and vertical-T welds were not among the supplied files — confirm they are in the booklet.
- **Marking:** compare the set-up values on the sheet with the WPS (within the stated tolerances), check the saved weld belongs to the student's login, and record the score and any bend-test result.

23.11.4.21 Machine Learning, Week 1: AI Literacy and Prompt Engineering

- **Prep:** confirm Claude.ai / ChatGPT and colab.research.google.com are reachable on the lab network. Post the FMEA table (Chapter Chapter 11, Week 1) and a blank `prompts.md` on FOL.
- No dataset needed this week; `spc_data.csv` is staged for Week 2.
- **Marking:** Exercise 3 is the point of the week — require genuine verification (a screenshot of the empty ASTM search, a Google Scholar "no results"), not just "I think this is made up".
- The capstone (Weeks 3–6) is assessed together against the capstone rubric held in FOL and the instructor handbook.

23.11.4.22 Machine Learning, Week 2: Manufacturing Data and SPC

- **Prep:** stage `spc_data.csv` and `week2_lab.ipynb` on FOL. Keep a key of the embedded problems (row indices, correct fixes) for marking.
- The most common error is using the standard deviation of the daily means for the control limits instead of `sigma_individual / sqrt(n)` — check this explicitly.
- Follow up on a `spc_starter.py` line from Week 1: ask a student to explain it.
- **Marking:** deduct for silent row deletion even if the final chart looks right — the traceable quality log is the learning objective.

23.11.4.23 Machine Learning, Week 3: The C-MAPSS Dataset

- **Prep:** stage `train_FD001.txt` and `predictive_maintenance.ipynb` on FOL (host the dataset yourself so it does not depend on Kaggle sign-in during the lab). Confirm Colab is reachable.
- The "sawtooth" reflection question (Task 5.1) sets up the engine-wise split in Week 4 — make sure students actually answer it.
- **If you are delivering the take-home Robotic Work-Cell Design lab, release it this week** (three weeks before it is due, in the default plan).
- **Marking:** rolled into the capstone rubric; here just verify the notebook runs end to end and the reflection is attempted.

23.11.4.24 Machine Learning, Week 4: Building the Model

- **Prep:** none beyond Week 3 — same notebook, same data. Have a reference solution notebook with the expected RMSE (~21 cycles) and feature-importance chart.
- **The teaching moment:** circulate during the evaluation task and look for RMSE < 10 — that is the row-wise-split data leak. Make students diagnose it themselves via the split assert.
- **Marking:** rolled into the capstone rubric.

23.11.4.25 Machine Learning, Week 5: Deploying the App

- **Prep:** stage `streamlit_app.py` and a known-good `requirements.txt` on FOL. Walk through one full deploy yourself the week before — Streamlit Cloud and GitHub UIs change. Confirm github.com and streamlit.io are reachable on the lab network.
- Have students do the GitHub / Streamlit account creation as prep, not in the lab — it eats 20 min otherwise.
- **Marking:** rolled into the capstone rubric; the deliverable is a URL that loads and predicts.

23.11.4.26 Machine Learning, Week 6: Capstone Presentations

- **Prep:** build the presentation schedule; test the projector with a couple of student URLs beforehand; have your own reference app ready to show if a student's deployment is down.
- **Stretch goals** continue in TPM/OEE, Root Cause and a New Dataset, which reuses this pipeline on the AI4I 2020 dataset: XGBoost comparison, SHAP values, FD002 generalisation, prediction intervals.
- The Weeks 3–6 capstone rubric (100 points: working model 30, code quality 20, Streamlit app 25, documentation & honesty 15, presentation 10) and its weighting are maintained in FOL and the instructor handbook; publish them before the module starts.

23.11.4.27 TPM/OEE, Root Cause and a New Dataset

- **Prep:** generate `precitech_shift_log.csv` with `Rscript R/make_lab_data.R` and stage it on FOL; publish the expected OEE and Pareto order from the answer key the script prints. Host a copy of the AI4I 2020 CSV on FOL as a fallback in case UCI is unreachable from the lab.
- Confirm `archive.ics.uci.edu`, draw.io and Colab are reachable on the lab network the day before.
- Individual lab.
- **Marking:** Task 4 is the point of the lab. A student who reports high accuracy and stops has missed it — full marks require *noticing* the imbalance and switching to precision/recall (or a resampling / class-weight fix). Reused C-MAPSS code that does not fit tabular data is a red flag that the transfer did not happen.

23.11.5 Consolidated shopping list (buy / restock each term)

Tick only the labs you are delivering. The software-only labs (Rocksi, RoboDK, AnyLogic, Lean, ML, TPM) use **no physical consumables**; the bridge and lean-simulation labs that did are not in the catalogue. If either is reinstated from `LabInstructions/`, restore its shopping list from the previous revision of this chapter (git history) or the original handouts.

Consumables — order ahead

- ☐ **Dry ice for the Charpy lab — needed in all three weeks of the materials-testing rotation. The lab technician needs two weeks' notice to order it.** Agree the quantity per session with the technician.
- ☐ Alcohol for the Charpy cold bath; methyl alcohol and lint-free wipes for instrument and gauge-block cleaning (calibration lab)
- ☐ Notched AISI 1018 hot-rolled and cold-rolled Charpy specimens — seven temperatures × two steels per test set, times the number of test sets

- ☐ Tensile specimens (two or more materials), mounted case-hardened AISI 1566 specimens, annealed / quenched / normalized hardness specimens
- ☐ Nital (2 %) etchant, polishing consumables and mounting resin, if the metallography portion is run **[confirm]**
- ☐ Roller bearings — 100 for the variable-charts lab and 50 for the process-capability lab, per class or bench **[confirm]**
- ☐ Bead bags in two or more colours, enough for ten samples of 50 per student or bench (attribute-charts lab)
- ☐ USB flash drives for the Virtual Welding labs

Printing (per term)

- ☐ REBA worksheets + REBA cheat sheet (one per student)
- ☐ Précitech case-study handout + document templates
- ☐ FMEA table for Machine Learning, Week 1 (from Chapter Chapter 11, Week 1)
- ☐ Answer key for the Virtual Caliper (caliper reference values, expected %GRR / ndc / Cpk) — for marking only
- ☐ Metrology and statistical-quality-control assignment sheets / worksheets, the control-chart constants table, and the *Preparing a Capability Study* handout
- ☐ Materials-testing report templates (hardness, tensile and microhardness, Charpy) and the rotation schedule
- ☐ Calibration report forms (micrometer, caliper)

Reusable — check stock

- ☐ Projector / display for the capstone presentations
- ☐ Vernier calipers, height gauges and micrometers; gauge-block sets (check serial numbers and certificates); surface plates, V-blocks, sine bar, dial indicators, rollers, tapers
- ☐ Rockwell and Brinell testers, Vickers microhardness tester, tensile testing machine, Charpy pendulum tester, metallography equipment; thermometers, tongs, insulated gloves, safety glasses
- ☐ Lincoln VRTEX 360+ simulators and WPS booklets

23.11.6 Software, licences & accounts — set up before the term

- ☐ **AnyLogic Personal Learning Edition** installed on every Facility Design & Simulation PC; run a full test of the demo model (watch the PLE size limit)
- ☐ **Microsoft Project** and **Microsoft Visio** licences live on the Design Documentation and Facility Design PCs (most likely gap in a standard Office image)
- ☐ **Microsoft Excel** / **Word** available on the Design Documentation, Facility Design and statistical-quality-control PCs

- ☐ **Rocksi** confirmed reachable through the lab network/firewall (Logic Thinking); offline fallback prepared. *(LightBot dropped — needs a plug-in the college machines do not have.)*
- ☐ **draw.io** reachable on the lab network (Lean; TPM/OEE)
- ☐ **AnyLogic Cloud** reachable, and the push–pull model published with editable input parameters (Lean)
- ☐ **Google Sheets** reachable (Lean; TPM/OEE)
- ☐ **UCI ML Repository** reachable for the AI4I 2020 download, with a FOL fallback copy staged (TPM/OEE)
- ☐ **RoboDK for Web** reachable; run the first three tasks through the college firewall before releasing the lab
- ☐ **The Virtual Caliper needs no external service** — only confirm the rendered book page and its webR cells load in the lab browsers
- ☐ **Google Colab** reachable on the lab network (ML Weeks 2–5; TPM/OEE)
- ☐ **Claude.ai / ChatGPT** reachable on the lab network (ML Weeks 1–2; TPM/OEE)
- ☐ **GitHub** and **Streamlit Community Cloud** reachable (ML Week 5); students create accounts as prep
- ☐ VRTEX 360+ units start and log in; WPS booklets present; this term's WPSs posted on FOL
- ☐ FOL document templates generated and re-uploaded — run `python R/gen_lab_templates.py`: `Bill_of_Materials_Template_for_Excel.xlsx`, `Gantt_chart_template.xlsx`, `Risk_Assessment_Template.docx` (Design Documentation); `week2_lab.ipynb` (ML Week 2); `predictive_maintenance.ipynb` (ML Week 3, reused in Weeks 4–5); `streamlit_app.py` + `requirements.txt` (ML Week 5)
- ☐ Lab data generated and staged on FOL — run `Rscript R/make_lab_data.R`: `vsm_station_data.csv` (Lean), `spc_data.csv` (ML Week 2), `precitech_shift_log.csv` (TPM/OEE)
- ☐ `train_FD001.txt` — the real NASA C-MAPSS FD001 dataset — hosted on FOL (ML Week 3; do not fabricate this one, it is real turbofan sensor data)
- ☐ Task clips / photo sets staged on FOL for Ergonomics (Lifting Boxes, Workstation, Cleaning)
- ☐ Prelab video links checked for Logic Thinking, Facility Design and Ergonomics
- ☐ Grade items verified in FOL against the instructor handbook's grade-item checklist for the labs you deliver

23.11.7 Room bookings

- ☐ **Computer lab** with internet + full Microsoft Office — for the software labs. The Design Documentation and Facility Design labs also need Project, Visio and AnyLogic PLE; the browser-based labs (Rocksi, Lean, Virtual Caliper, ML, TPM/OEE) need only a browser and internet
- ☐ **Metrology lab** with surface plates and the measuring instruments — the

metrology labs and the hands-on statistical-quality-control labs (the latter also need PCs with Excel, or a computer lab booked alongside)
- ☐ **Materials lab** (safety glasses and safety footwear required) — three consecutive weeks, the same slot each week, for the materials-testing rotation
- ☐ **Welding-simulator lab** with the VRTEX 360+ units — for the Virtual Welding labs
- ☐ **Room with a projector/display** for the machine-learning capstone presentations [**confirm — the computer lab may suffice**]
- ☐ **No room for Robotic Work-Cell Design** — it is a take-home lab; only a 15-minute demo slot in a lab period is needed

23.11.8 Appendix: instrumentation-trainer lab stream

An older set of labs for this course code was written for the **B1033** trainer benches (author "MZhang"). They are **not** part of the current lab set, but the handouts are in the `LabInstructions/` folder and are summarised here in case the course runs them:

Alt lab	Title	Room	Equipment	Software
Alt — Pressure/level	Pressure Measurement for Level	B1033	Trainer: power-supply module, pump/tank unit, RTD temperature transmitter, pressure transmitter, float/reed switch, 2 × DMM, banana leads	—

Alt lab	Title	Room	Equipment	Software
Alt — Motor control	Motor Control & Intro to VFDs	B1033	3-phase motor, FVNR (full-voltage non-reversing) starters, control-relay/MCR wiring, E-stop, pilot lights, pushbuttons; VFD	—
Alt — PLC	PLC Introduction with Siemens	B1033	Siemens PLC trainer, field devices (pushbuttons, selector switches, pilot lights, cylinders with proxy sensors), potentiometer 0–10 VDC, DMM	**Siemens TIA Portal**; reference video `YOvQhvjaeqY`
Alt — Flow rate	Flow Rate Measurement	B1033	Trainer: pump/tank unit, paddle-wheel flow transmitter, rotameter (LPM/GPM), RTD temperature transmitter, 2 × DMM, hoses/banana leads	—

Prep for the alt stream: book **B1033**; confirm every trainer bench has a working

power-supply module, pump/tank unit and the two DMMs; confirm **Siemens TIA Portal** licences for the PLC lab; print each handout's instructor sign-off sheet (4 sign-offs per lab).

Bibliography